



Article No. 712

# Chemistry<sup>0625</sup>

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## Paper-2

Classified/Topical MCQ'S  
Updated Till – 2022

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- All variant
  - Mark scheme included
  - Questions order new to old
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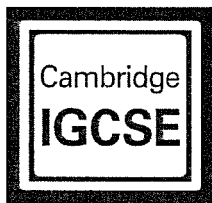
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## Unit 1

# THE PARTICULATE NATURE OF MATTER

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## Unit 1: The Particulate nature of matter

### 1. O/N 22/P22/Q1

The rate of diffusion of three gases, ammonia, carbon dioxide and methane, is measured. What is the order of the rate of diffusion of the gases from slowest to fastest?

- A  $\text{CO}_2 \rightarrow \text{NH}_3 \rightarrow \text{CH}_4$     B  $\text{CO}_2 \rightarrow \text{CH}_4 \rightarrow \text{NH}_3$     C  $\text{CH}_4 \rightarrow \text{NH}_3 \rightarrow \text{CO}_2$     D  $\text{NH}_3 \rightarrow \text{CH}_4 \rightarrow \text{CO}_2$

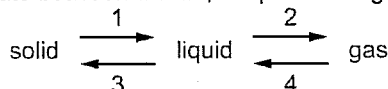
### 2. O/N 22/P22/Q2

Which description of Brownian motion is correct?

- A random movement of particles due to bombardment by larger particles  
B random movement of particles due to bombardment by smaller particles  
C random movement of particles from a high concentration to a low concentration  
D random movement of particles from a low concentration to a high concentration

### 3. O/N 22/P21/Q1

The diagram shows the changes of state between a solid, a liquid and a gas.



In which changes of state is energy being given out?

- A 1 and 2    B 1 and 4    C 2 and 3    D 3 and 4

### 4. O/N 22/P23/Q1

Which gas diffuses the most slowly?

- A  $\text{CH}_4$     B  $\text{CO}_2$     C  $\text{H}_2$     D  $\text{NH}_3$

### 5. M/J 22/P22/Q1, M/J 22/P21/Q1, M/J 22/P23/Q1, O/N 19/P23/Q1

Which two gases will diffuse at the same rate, at the same temperature?

- A carbon monoxide and carbon dioxide    C chlorine and fluorine  
B carbon monoxide and nitrogen    D nitrogen and oxygen

### 6. M/J 22/P22/Q12, M/J 22/P21/Q12, M/J 22/P23/Q12, O/N 19/P22/Q15

Which row identifies a chemical change and a physical change?

|   | chemical change             | physical change             |
|---|-----------------------------|-----------------------------|
| A | boiling ethanol             | burning ethanol             |
| B | burning ethanol             | evaporating ethanol         |
| C | dissolving ethanol in water | burning ethanol             |
| D | evaporating ethanol         | dissolving ethanol in water |

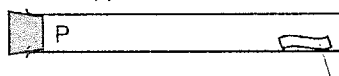
### 7. MAR 22/P22/Q1

Which gas has the fastest rate of diffusion?

- A  $\text{H}_2$     B  $\text{CH}_4$     C  $\text{CO}_2$     D  $\text{SO}_2$

### 8. M/J 21/P22/Q1, M/J 21/P21/Q1, M/J 21/P23/Q1, M/J 18/P22/Q1

A gas is released at point P in the apparatus shown.



damp universal indicator paper

Which gas turns the damp universal indicator paper red most quickly?

- A ammonia,  $\text{NH}_3$     C hydrogen chloride,  $\text{HCl}$   
B chlorine,  $\text{Cl}_2$     D sulfur dioxide,  $\text{SO}_2$



**13. M/J 21/P23/Q12**

Gas syringe X contains 100 cm<sup>3</sup> of hydrogen bromide gas, HBr.

Gas syringe Y contains 100 cm<sup>3</sup> of carbon dioxide gas. The volume of each gas is measured at room temperature and pressure.

Which statement is correct?

- A The mass of HBr is less than the mass of CO<sub>2</sub>.  
 B The number of molecules of HBr equals the number of molecules of CO<sub>2</sub>.  
 C The gas in syringe X contains more atoms than the gas in syringe Y.  
 D The number of moles of HBr is more than the number of moles of CO<sub>2</sub>.

**14. M/J 21/P23/Q14**

When sulfur is heated it undergoes a .....1..... change as it melts.

Further heating causes the sulfur to undergo a .....2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

|   | 1        | 2        |
|---|----------|----------|
| A | chemical | chemical |
| B | chemical | physical |
| C | physical | chemical |
| D | physical | physical |

**15. O/N 20/P22/Q1, O/N 20/P21/Q1, O/N 20/P23/Q1**

Which gas has the slowest rate of diffusion?

- A H<sub>2</sub>                                      B NH<sub>3</sub>                                      C CH<sub>4</sub>                                      D CO<sub>2</sub>

**16. M/J 20/P22/Q1, M/J 20/P21/Q1, M/J 20/P23/Q1**

A mixture of ice and water is left to stand and the ice melts.

Which row describes what happens as the ice is melting?

|   | temperature of mixture | energy changes                                |
|---|------------------------|-----------------------------------------------|
| A | increases              | average kinetic energy of particles increases |
| B | increases              | energy is used to overcome attractive forces  |
| C | stays the same         | average kinetic energy of particles increases |
| D | stays the same         | energy is used to overcome attractive forces  |

**17. M/J 20/P22/Q14**

Which list contains only chemical changes?

- A melting, evaporating, dissolving                                      C neutralisation, polymerisation, combustion  
 B rusting, freezing, subliming                                      D boiling, condensing, distillation

**18. M/J 20/P23/Q15**

A chemical reaction occurs when the reacting particles collide. Which reaction conditions would produce the greatest rate of particle collisions?

|   | concentration of acid | reaction temperature |
|---|-----------------------|----------------------|
| A | decrease              | decrease             |
| B | no change             | increase             |
| C | increase              | increase             |
| D | increase              | no change            |

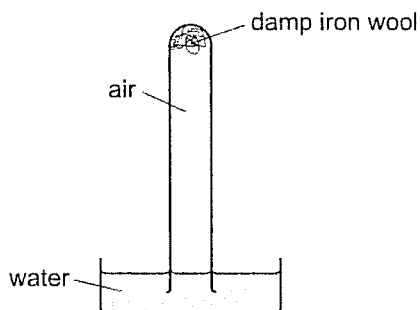
**19. O/N 19/P22/Q1**

The rate of diffusion of a gas depends on its molecular mass and the temperature.  
Which combination of molecular mass and temperature gives the slowest rate of diffusion?

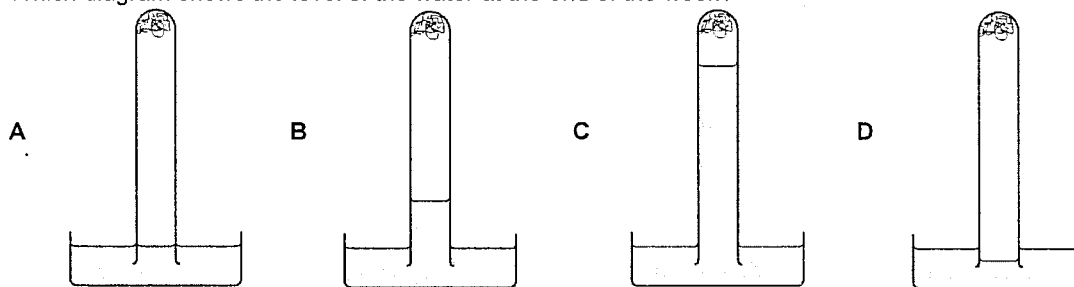
|   | molecular mass | temperature |
|---|----------------|-------------|
| A | high           | high        |
| B | high           | low         |
| C | low            | high        |
| D | low            | low         |

**20. O/N 19/P22/Q32, O/N 19/P21/Q32, O/N 19/P23/Q32**

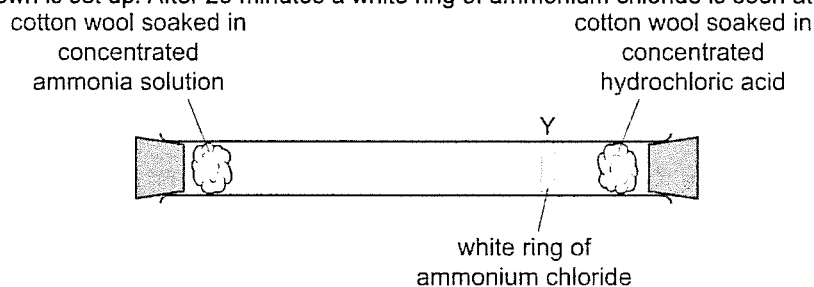
The apparatus shown is set up and left for a week.



Which diagram shows the level of the water at the end of the week?

**21. M/J 19/P22/Q1**

The apparatus shown is set up. After 20 minutes a white ring of ammonium chloride is seen at position Y.



Which statement about the molecules of ammonia and hydrogen chloride is correct?

- A Molecules in ammonia have a larger  $M_r$  than molecules of hydrogen chloride and so they move more slowly.
- B Molecules in ammonia have a larger  $M_r$  than molecules of hydrogen chloride and so they move more quickly.
- C Molecules in ammonia have a smaller  $M_r$  than molecules of hydrogen chloride and so they move more slowly.
- D Molecules in ammonia have a smaller  $M_r$  than molecules of hydrogen chloride and so they move more quickly.

**22. O/N 19/P21/Q1**

Samples of four gases are released in a room at the same time.

The gases are carbon dioxide,  $\text{CO}_2$ , hydrogen chloride,  $\text{HCl}$ , hydrogen sulfide,  $\text{H}_2\text{S}$ , and nitrogen dioxide,  $\text{NO}_2$ . Which gas diffuses fastest?

- A carbon dioxide      B hydrogen chloride      C hydrogen sulfide      D nitrogen dioxide

**23. O/N 19/P21/Q15**

Which is a chemical change?

- A boiling water      B cooking an egg      C dissolving sugar      D melting ice cubes

**24. M/J 19/P21/Q1**

Which statement explains why ammonia gas,  $\text{NH}_3$ , diffuses at a faster rate than hydrogen chloride gas,  $\text{HCl}$ ?

- A Ammonia expands to occupy all of the space available.  
B Ammonia has a smaller relative molecular mass than hydrogen chloride.  
C Ammonia is an alkali and hydrogen chloride is an acid.  
D Ammonia molecules diffuse in all directions at the same time.

**25. MAR 19/P22/Q1**

Pure water boils at  $100^\circ\text{C}$ .

What happens to the water particles when water boils?

- A They gain energy and move further apart.      C They lose energy and move further apart.  
B They gain energy and stay close together.      D They lose energy and stay close together.

**26. O/N 18/P22/Q1**

Oxygen and fluorine are gaseous elements next to each other in the Periodic Table.

Under the same conditions of temperature and pressure, oxygen diffuses .....1..... than fluorine because its .....2..... is less than that of fluorine.

Which words correctly complete gaps 1 and 2?

|   | 1      | 2              |
|---|--------|----------------|
| A | faster | molecular mass |
| B | faster | reactivity     |
| C | slower | molecular mass |
| D | slower | reactivity     |

**27. O/N 18/P21/Q1**

When smoke particles are observed with a microscope they are seen to move around randomly. This is called Brownian motion.

What causes Brownian motion?

- A diffusion of the smoke particles  
B molecules in the air hitting the smoke particles  
C sublimation of the smoke particles  
D the smoke particles hitting the walls of the container

**28. O/N 18/P23/Q1**

Gases are separated from liquid air by fractional distillation. The boiling points of four gases are shown.

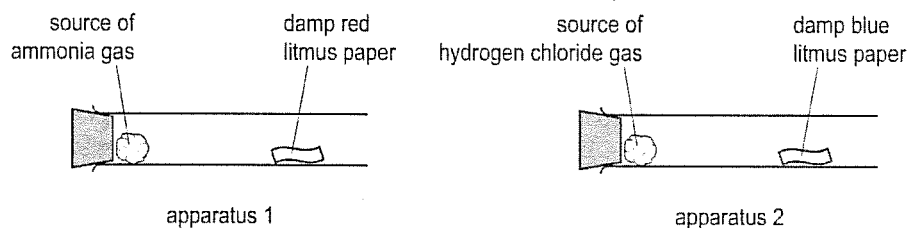
Which gas is both monatomic and a liquid at  $-200^\circ\text{C}$ ?

|   | gas      | boiling point / $^\circ\text{C}$ |
|---|----------|----------------------------------|
| A | argon    | -186                             |
| B | helium   | -269                             |
| C | neon     | -246                             |
| D | nitrogen | -196                             |

29. M/J 18/P21/Q1

A student investigated the diffusion of ammonia gas,  $\text{NH}_3$ , and hydrogen chloride gas,  $\text{HCl}$ .

Two sets of apparatus were set up as shown at room temperature and pressure.



The damp red litmus paper in apparatus 1 changed colour after 30 seconds.

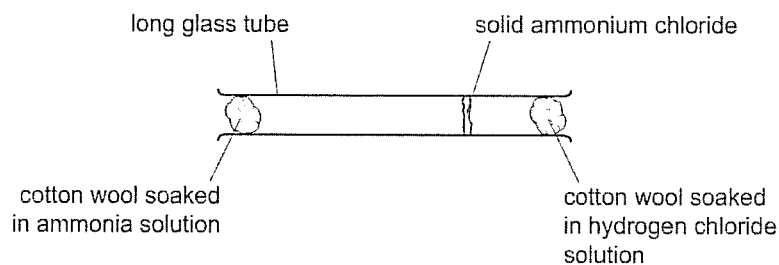
How long does it take for the damp blue litmus paper to change colour in apparatus 2?

- A 64 seconds  
B 30 seconds  
C 21 seconds  
D The blue litmus paper would not change colour.

30. M/J 18/P23/Q1

Ammonia gas is reacted with hydrogen chloride gas using the apparatus shown.

Solid ammonium chloride is produced.



Which statement explains why the solid ammonium chloride is formed nearer to the hydrogen chloride?

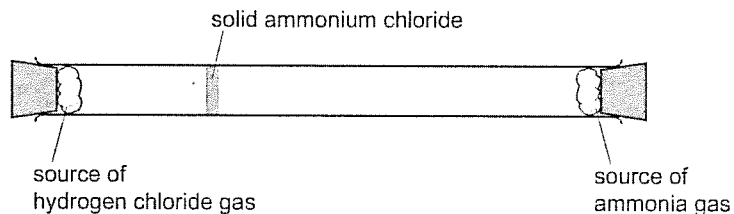
- A** Ammonia solution is a base and hydrogen chloride solution is an acid.  
**B** Ammonia molecules diffuse more slowly than hydrogen chloride molecules.  
**C** Hydrogen chloride has a greater molecular mass than ammonia.  
**D** Hydrogen chloride moves by Brownian motion.

31. MAR 18/P22/Q1

Hydrogen chloride gas,  $\text{HCl}$ , reacts with ammonia gas,  $\text{NH}_3$ , to form solid ammonium chloride.

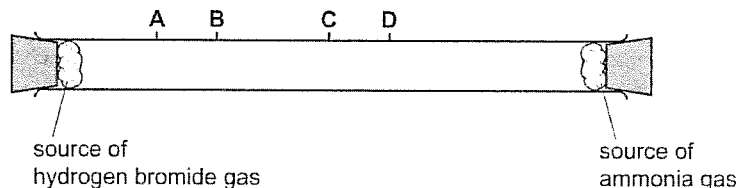
The apparatus is set up as shown.

After a few minutes, solid ammonium chloride forms where the two gases meet.



The experiment is repeated using hydrogen bromide, HBr, in place of hydrogen chloride.

How far along the tube does the solid ammonium bromide form?



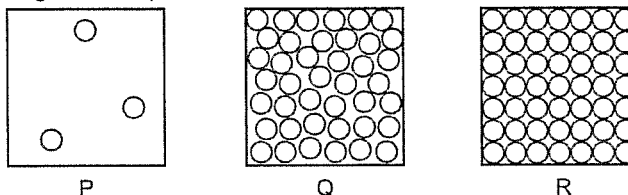
**32. MAR 18/P22/Q2**

Substance L melts at  $-7^{\circ}\text{C}$  and is a brown liquid at room temperature.  
Which temperature is the boiling point of pure L?

- A  $-77^{\circ}\text{C}$       B  $-7^{\circ}\text{C}$  to  $+7^{\circ}\text{C}$       C  $59^{\circ}\text{C}$       D  $107^{\circ}\text{C}$  to  $117^{\circ}\text{C}$

**33. O/N 17/P22/Q1**

The diagram shows the arrangement of particles in the three states of matter.



Solid carbon dioxide (dry ice) sublimates to gaseous carbon dioxide.  
Which row describes the initial and final states?

|   | initial state | final state |
|---|---------------|-------------|
| A | P             | R           |
| B | Q             | P           |
| C | R             | P           |
| D | R             | Q           |

**34. O/N 17/P21/Q1**

Which process causes the greatest increase in the distance between particles?

- A condensation      B freezing      C melting      D sublimation

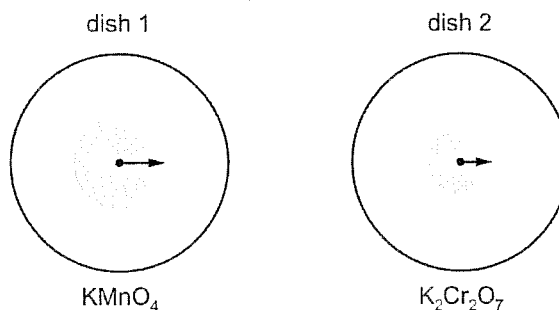
**35. O/N 17/P23/Q1**

Which statement describes sublimation?

- A Particles moving slowly past each other speed up and move further apart.  
B Particles vibrating next to each other become mobile and move slowly past each other.  
C Particles vibrating next to each other start to move rapidly and move further apart.  
D Rapidly moving particles slow down and move closer together.

**36. M/J 17/P22/Q1, M/J 17/P21/Q1, M/J 17/P23/Q1**

Small crystals of purple  $\text{KMnO}_4$  ( $M_r = 158$ ) and orange  $\text{K}_2\text{Cr}_2\text{O}_7$  ( $M_r = 294$ ) were placed at the centres of separate petri dishes filled with agar jelly. They were left to stand under the same physical conditions. After some time, the colour of each substance had spread out as shown.



The lengths of the arrows indicate the relative distances travelled by particles of each substance.  
Which statement is correct?

- A Diffusion is faster in dish 1 because the mass of the particles is greater.  
B Diffusion is faster in dish 2 because the mass of the particles is greater.  
C Diffusion is slower in dish 1 because the mass of the particles is smaller.  
D Diffusion is slower in dish 2 because the mass of the particles is greater.



**37. M/J 17/P23/Q2**

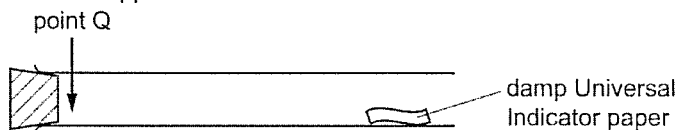
A compound, X, has a melting point of  $71^{\circ}\text{C}$  and a boiling point of  $375^{\circ}\text{C}$ .

Which statement about X is correct?

- A It is a liquid at  $52^{\circ}\text{C}$  and a gas at  $175^{\circ}\text{C}$ .      C It is a liquid at  $75^{\circ}\text{C}$  and a gas at  $350^{\circ}\text{C}$ .  
 B It is a liquid at  $69^{\circ}\text{C}$  and a gas at  $380^{\circ}\text{C}$ .      D It is a liquid at  $80^{\circ}\text{C}$  and a gas at  $400^{\circ}\text{C}$ .

**38. MAR 17/P22/Q1**

A gas is released at point Q in the apparatus shown.



Which gas changes the colour of the damp Universal Indicator paper most quickly?

|   | gas            | relative molecular mass |
|---|----------------|-------------------------|
| A | ammonia        | 17                      |
| B | carbon dioxide | 44                      |
| C | chlorine       | 71                      |
| D | hydrogen       | 2                       |

**39. O/N 16/P22/Q1, O/N 16/P21/Q1, O/N 16/P23/Q1, O/N 12/P13/Q1**

'Particles moving **very slowly** from an area of higher concentration to an area of lower concentration.'

Which process is being described?

- A a liquid being frozen      C a substance diffusing through a liquid  
 B a solid melting      D a substance diffusing through the air

**40. M/J 16/P22/Q1**

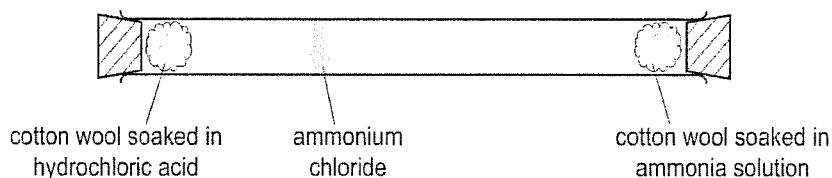
The particles of a substance gain energy and change from a regular ordered structure to a disordered structure with large distances between the particles.

Which change of state is described?

- A boiling      B evaporation      C melting      D sublimation

**41. M/J 16/P23/Q1**

The diagram shows an experiment to demonstrate diffusion.

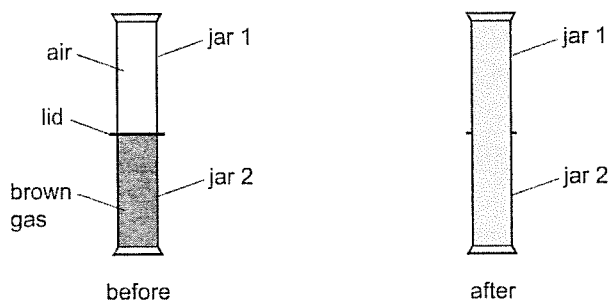


Which statement explains why the ring of ammonium chloride appears as shown?

- A Ammonia solution only produces a gas which moves until it meets the hydrochloric acid.  
 B Both solutions produce a gas, but ammonia moves quicker than hydrogen chloride because it is lighter.  
 C Hydrochloric acid produces hydrogen chloride which stays at one end of the tube until the ammonia reaches it.  
 D The two solutions run along the tube until they meet.

**42. MAR 16/P22/Q1**

Two gas jars are set up as shown.



The lid is removed and the gas jars are left to stand. After some time the contents of both gas jars are brown. Which process causes this to happen?

- A** condensation      **B** diffusion      **C** evaporation      **D** filtration

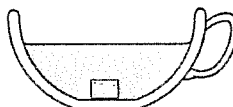
**43. O/N 15/P13/Q1**

In which process do particles move closer together but remain in motion?

- A** condensation      **B** diffusion      **C** evaporation      **D** freezing

**44. M/J 15/P12/Q1**

The diagram shows a sugar lump in a cup of tea.

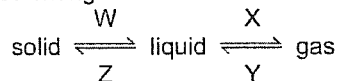


Which two processes must happen to spread the sugar evenly in the tea?

|          | first process | second process |
|----------|---------------|----------------|
| <b>A</b> | diffusion     | dissolving     |
| <b>B</b> | dissolving    | diffusion      |
| <b>C</b> | dissolving    | melting        |
| <b>D</b> | melting       | diffusion      |

**45. M/J 15/P11/Q1**

The changes that occur when a substance changes state are shown below.



Which process, W, X, Y or Z, is occurring in the following four situations?

- Butter melts on a warm day.
- Water condenses on a cold surface.
- The volume of liquid ethanol in an open beaker reduces.
- Ice forms inside a freezer.

|          | 1 | 2 | 3 | 4 |
|----------|---|---|---|---|
| <b>A</b> | W | X | Y | Z |
| <b>B</b> | W | Y | X | Z |
| <b>C</b> | X | Y | Z | W |
| <b>D</b> | X | Z | Y | W |

**46. M/J 15/P13/Q1**

A sugar cube is dropped into a hot cup of tea.

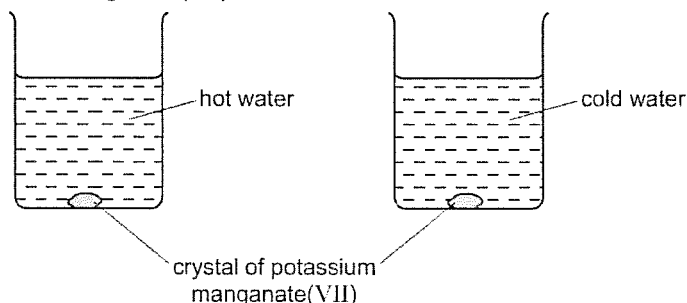
The tea is not stirred.

Which statement explains why the tea becomes sweet?

- A The heated water molecules penetrate the sugar cube.
- B The hot tea causes the sugar to melt.
- C The sugar cube dissolves and its molecules diffuse.
- D The sugar molecules get hot and evaporate.

**47. MAR 15/P12/Q1**

A crystal of purple potassium manganate(VII) was added to each of the beakers shown in the diagram.



One beaker contained hot water and the other beaker contained cold water.

In both beakers the purple colour of the potassium manganate(VII) spreads out.

Which result and explanation are correct?

|          | result                              | explanation                                   |
|----------|-------------------------------------|-----------------------------------------------|
| <b>A</b> | colour spreads faster in cold water | particles move faster at a higher temperature |
| <b>B</b> | colour spreads faster in cold water | particles move slower at a higher temperature |
| <b>C</b> | colour spreads faster in hot water  | particles move faster at a higher temperature |
| <b>D</b> | colour spreads faster in hot water  | particles move slower at a higher temperature |

**48. O/N 14/P12/Q2, O/N 14/P11/Q1**

Which statement is an example of diffusion?

- A A kitchen towel soaks up some spilt milk.
- B Ice cream melts in a warm room.
- C Pollen from flowers is blown by the wind.
- D The smell of cooking spreads through a house.

**49. O/N 14/P13/Q1**

A few drops of perfume were spilt on the floor. A few minutes later the perfume could be smelt a few metres away.

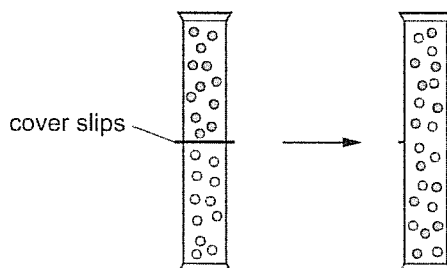
Which two processes had taken place?

- A distillation and condensation
- B distillation and diffusion
- C evaporation and condensation
- D evaporation and diffusion

**50. M/J 14/P12/Q1**

Two gas jars each contain a different gas. The gas jars are connected and the cover slips are removed.

The diagram shows what happens to the particles of the gases.

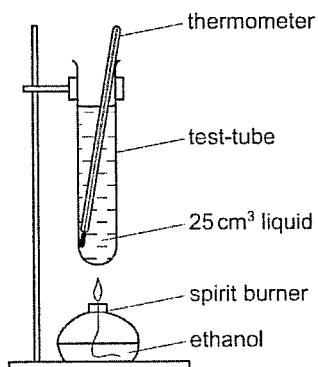


Which process has occurred?

- A chemical reaction
- B condensation
- C diffusion
- D evaporation

**51. M/J 14/P12/Q2**

A liquid is heated until it boils.

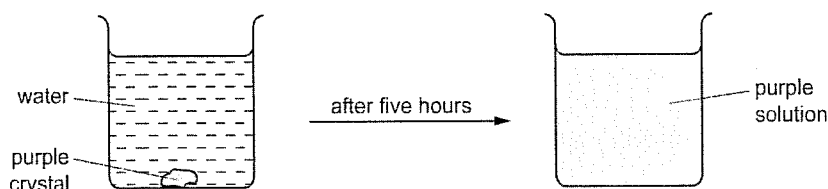


Which result shows that the liquid in the test-tube is pure water?

- A Condensation forms at the top of the test-tube.
- B Steam is produced.
- C The thermometer reads 100 °C.
- D There is nothing left behind in the test-tube.

**52. M/J 14/P11/Q1, M/J 14/P13/Q1**

The diagram shows the result of dropping a purple crystal into water.

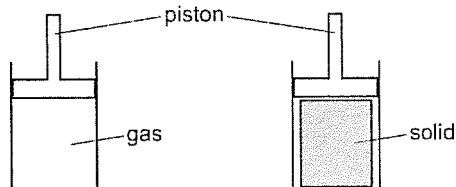


Which processes take place in this experiment?

|   | chemical reaction | diffusing | dissolving |
|---|-------------------|-----------|------------|
| A | ✓                 | ✓         | ✓          |
| B | ✓                 | x         | ✓          |
| C | x                 | x         | ✓          |
| D | x                 | ✓         | ✓          |

**53. O/N 13/P12/Q1, O/N 13/P11/Q1**

An attempt was made to compress a gas and a solid using the apparatus shown.



Which substance would be compressed and what is the reason for this?

|   | substance | reason                                 |
|---|-----------|----------------------------------------|
| A | gas       | the gas particles are close together   |
| B | gas       | the gas particles are far apart        |
| C | solid     | the solid particles are close together |
| D | solid     | the solid particles are far apart      |

**54. O/N 13/P13/Q1**

A gas jar of bromine vapour and a gas jar of air are set up as shown in diagram 1.

The glass slide is removed. Diagram 2 shows the appearance of the gas jars after one hour.

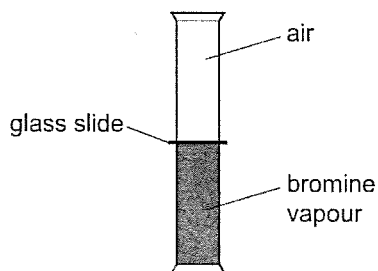


diagram 1

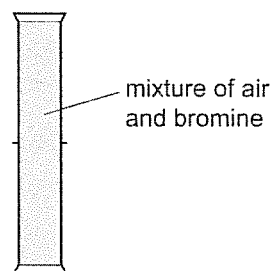


diagram 2

Which statement explains why the bromine and air mix together?

- A Bromine is denser than air.
- B Bromine is lighter than air.
- C Bromine molecules moved upwards and molecules in air moved downwards.
- D Molecules in bromine and air moved randomly.

**55. M/J 13/P12/Q1, M/J 13/P11/Q1, M/J 13/P13/Q1, M/J 10/P12/Q1, M/J 10/P11/Q1, M/J 10/P13/Q1**

The diagram shows a cup of tea.



Which row describes the water particles in the air above the cup compared with the water particles in the cup?

|   | moving faster | closer together |
|---|---------------|-----------------|
| A | ✓             | ✓               |
| B | ✓             | x               |
| C | x             | ✓               |
| D | x             | x               |

**Answers Section**

| Unit 1 |   |     |   |
|--------|---|-----|---|
| 1.     | A | 41. | B |
| 2.     | B | 42. | B |
| 3.     | D | 43. | A |
| 4.     | B | 44. | B |
| 5.     | B | 45. | B |
| 6.     | B | 46. | C |
| 7.     | A | 47. | C |
| 8.     | C | 48. | D |
| 9.     | D | 49. | D |
| 10.    | A | 50. | C |
| 11.    | B | 51. | C |
| 12.    | B | 52. | D |
| 13.    | B | 53. | B |
| 14.    | C | 54. | D |
| 15.    | D | 55. | B |
| 16.    | D |     |   |
| 17.    | C |     |   |
| 18.    | C |     |   |
| 19.    | B |     |   |
| 20.    | B |     |   |
| 21.    | D |     |   |
| 22.    | C |     |   |
| 23.    | B |     |   |
| 24.    | B |     |   |
| 25.    | A |     |   |
| 26.    | A |     |   |
| 27.    | B |     |   |
| 28.    | A |     |   |
| 29.    | A |     |   |
| 30.    | C |     |   |
| 31.    | A |     |   |
| 32.    | C |     |   |
| 33.    | C |     |   |
| 34.    | D |     |   |
| 35.    | C |     |   |
| 36.    | D |     |   |
| 37.    | D |     |   |
| 38.    | A |     |   |
| 39.    | C |     |   |
| 40.    | D |     |   |



## Unit 2

# EXPERIMENTAL TECHNIQUES

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IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

2.1 Measurement  
2.2 Purity

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## Unit 2: Experimental Techniques

### 2.1: Measurement

1. M/J 22/P22/Q2, M/J 22/P21/Q2, M/J 22/P23/Q2

A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm<sup>3</sup> of dilute sulfuric acid. Which apparatus is essential to complete the experiment?

- 1 stop-clock
- 2 measuring cylinder
- 3 thermometer
- 4 balance

A 1, 2 and 4                      B 1 and 2 only                      C 1 and 4 only                      D 2, 3 and 4

2. O/N 21/P22/Q2, O/N 21/P21/Q2

A student put exactly 25.00 cm<sup>3</sup> of dilute hydrochloric acid into a conical flask. The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture. Which apparatus does the student need to use?

- A balance, measuring cylinder, thermometer
- B balance, pipette, stopwatch
- C balance, pipette, thermometer
- D burette, pipette, thermometer

3. M/J 21/P21/Q3

Which statement about paper chromatography is correct?

- A A solvent is needed to dissolve the paper.
- B Paper chromatography separates mixtures of solvents.
- C The solvent should cover the baseline.
- D The baseline should be drawn in pencil.

4. MAR 21/P22/Q1

Which row about a change of state is correct?

|   | change of state | energy change  | process     |
|---|-----------------|----------------|-------------|
| A | solid → liquid  | heat given out | melting     |
| B | gas → liquid    | heat taken in  | evaporation |
| C | solid → gas     | heat taken in  | sublimation |
| D | liquid → solid  | heat given out | condensing  |

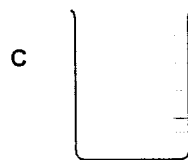
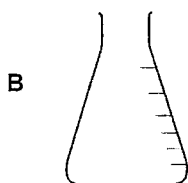
5. O/N 20/P21/Q3

Which piece of apparatus can only measure a single fixed volume?

- A 250 cm<sup>3</sup> beaker
- B 50 cm<sup>3</sup> burette
- C 100 cm<sup>3</sup> measuring cylinder
- D 25 cm<sup>3</sup> pipette

6. M/J 20/P22/Q2

Which piece of apparatus is used to measure 25.0 cm<sup>3</sup> of aqueous sodium hydroxide?





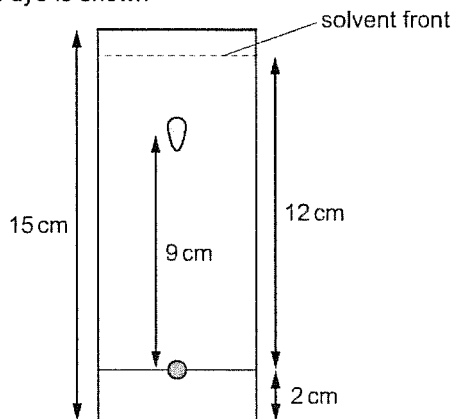
## 7. M/J 20/P21/Q2

Which piece of apparatus should be used to measure exactly  $21.4 \text{ cm}^3$  of water?

- A  $25 \text{ cm}^3$  beaker  
 B  $25 \text{ cm}^3$  pipette  
 C  $50 \text{ cm}^3$  burette  
 D  $50 \text{ cm}^3$  measuring cylinder

## 8. M/J 20/P21/Q3

The chromatogram for an unknown dye is shown



What is the  $R_f$  value of the dye?

- A 0.60  
 B 0.64  
 C 0.75  
 D 0.82

## 9. M/J 20/P23/Q2

Which piece of apparatus is used to measure  $13.7 \text{ cm}^3$  of dilute hydrochloric acid?

- A balance  
 B burette  
 C conical flask  
 D pipette

## 10. O/N 19/P22/Q2, O/N 19/P21/Q2

A student is asked to measure the time taken for  $0.4 \text{ g}$  of magnesium carbonate to react completely with  $25.0 \text{ cm}^3$  of dilute hydrochloric acid.

Which pieces of apparatus does the student need?

- A balance, stop-clock, pipette  
 B balance, stop-clock, thermometer  
 C balance, pipette, thermometer  
 D stop-clock, pipette, thermometer

## 11. O/N 19/P22/Q31

Which process is used to separate nitrogen and oxygen from air?

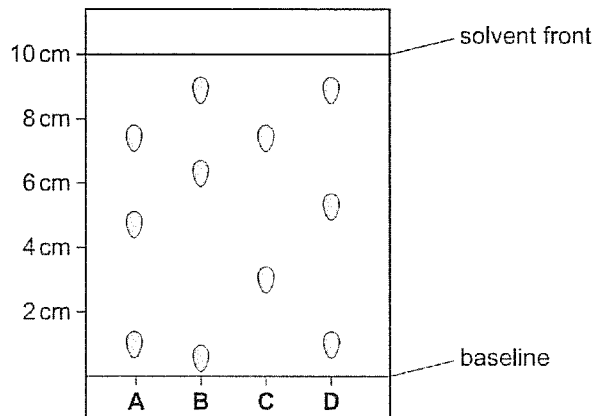
- A chromatography  
 B evaporation  
 C filtration  
 D fractional distillation

## 12. O/N 19/P21/Q3

Four different food colourings are analysed using chromatography.

The results are shown on the chromatogram. The diagram is not drawn to scale.

Which food colouring contains a component with an  $R_f$  value of 0.3?



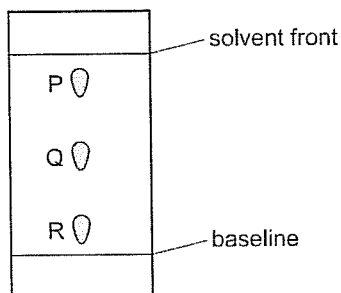
## 13. O/N 19/P21/Q31

Which physical property is used to separate the nitrogen and oxygen from air?

- A boiling point  
B density  
C electrical conductivity  
D molecular mass

## 14. O/N 19/P23/Q3

A substance is separated using chromatography.  
The chromatogram is shown.



Which statement is not correct?

- A P has a higher  $R_f$  value than Q.  
B P, Q and R are all soluble in the solvent.  
C R is the most soluble substance.  
D The  $R_f$  value of P is less than 1.

## 15. M/J 19/P22/Q2

A student measures  $25.00 \text{ cm}^3$  of dilute hydrochloric acid accurately.  
Which apparatus is most suitable?

- A beaker  
B measuring cylinder  
C burette  
D dropping pipette

## 16. M/J 19/P21/Q2

$2.00 \text{ g}$  of powdered calcium carbonate is added to  $50.0 \text{ cm}^3$  of hydrochloric acid.  
Which apparatus is used to measure the calcium carbonate and the hydrochloric acid?

|   | calcium carbonate | hydrochloric acid |
|---|-------------------|-------------------|
| A | balance           | burette           |
| B | balance           | thermometer       |
| C | pipette           | burette           |
| D | pipette           | thermometer       |

## 17. M/J 19/P23/Q2

Which piece of apparatus is used to measure  $24.8 \text{ cm}^3$  of gas produced during a reaction?

- A beaker  
B conical flask  
C measuring cylinder  
D pipette

## 18. M/J 17/P23/Q14

Which changes are physical changes?

- 1 melting ice to form water  
2 burning hydrogen to form water  
3 adding sodium to water  
4 boiling water to form steam  
A 1 and 2  
B 1 and 4  
C 2 and 3  
D 3 and 4

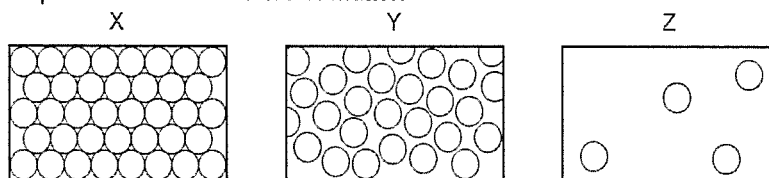
## 19. O/N 15/P12/Q1

Which change of state takes place during evaporation?

- A gas to liquid  
B liquid to gas  
C liquid to solid  
D solid to gas

**20. O/N 15/P11/Q1**

Diagrams X, Y and Z represent the three states of matter.

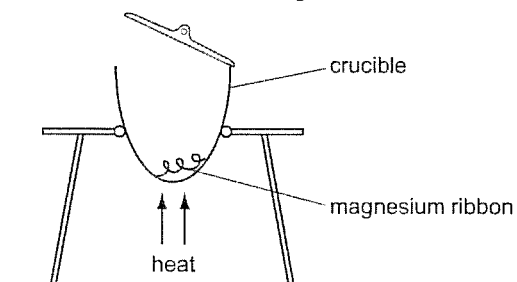


Which change occurs during boiling?

- A X to Y                      B Y to Z                      C Z to X                      D Z to Y

**21. M/J 13/P12/Q2**

The diagram shows an experiment to find the formula of magnesium oxide.



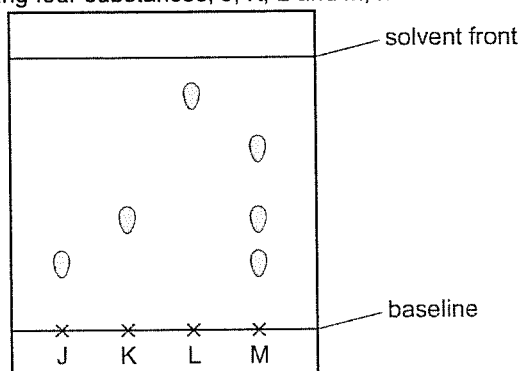
Which piece of apparatus would be needed in addition to those shown?

- A a balance                      C a spatula  
B a measuring cylinder                      D a thermometer

## 2.2: Purity

### 1. O/N 22/P22/Q3

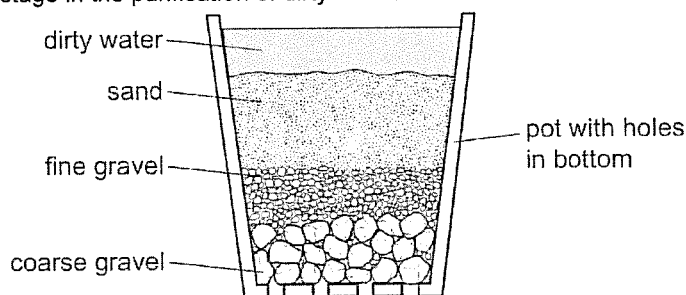
The chromatogram obtained using four substances, J, K, L and M, is shown.



Which statement about M is correct?

- A It is a mixture of J and K only.
  - B It is a pure substance.
  - C It is a mixture of J, K and L.
  - D It is a mixture of J, K and an unknown substance.
2. O/N 22/P22/Q28, O/N 22/P21/Q28, O/N 22/P23/Q28

The diagram shows a stage in the purification of dirty water.



Which process does this apparatus show?

- A chlorination
  - B condensation
  - C distillation
  - D filtration
3. O/N 22/P21/Q2
- A coloured dye is separated by chromatography.  
One component of the dye moves a distance of 13 cm and has an  $R_f$  value of 0.86.  
Which distance did the solvent front move?
- A 6.6 cm
  - B 11.9 cm
  - C 15.1 cm
  - D 21.6 cm

### 4. O/N 22/P21/Q3

A mixture contains salt, sand and sulfur.

Salt dissolves in water but not in xylene.

Sulfur dissolves in xylene but not in water.

Sand does not dissolve in water or xylene.

What is the order of the processes used to separate the salt, the sand and the sulfur from the mixture?

- A add water → filter → add xylene to the filtrate → filter
- B add water → filter → add xylene to the filtrate → filter
- C add water → filter → add water to the filtrate → filter
- D add xylene → filter → add xylene to the residue → filter

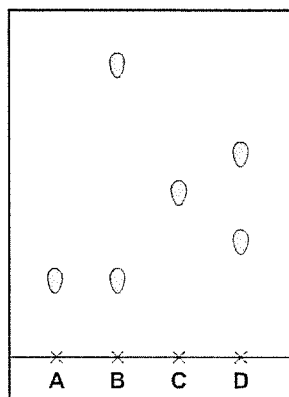
## 5. O/N 22/P21/Q19

Aqueous ammonium sulfate is made by reacting aqueous ammonia with dilute sulfuric acid.  
How is solid ammonium sulfate obtained from the resulting solution?

- A crystallisation      B distillation      C filtration      D solvent extraction

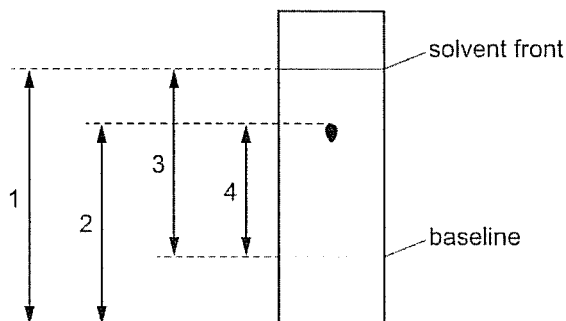
## 6. O/N 22/P23/Q2

The chromatogram from four different substances is shown.  
Which pure substance has the largest  $R_f$  value?



## 7. M/J 22/P22/Q3

A chromatogram of a single substance T is shown.

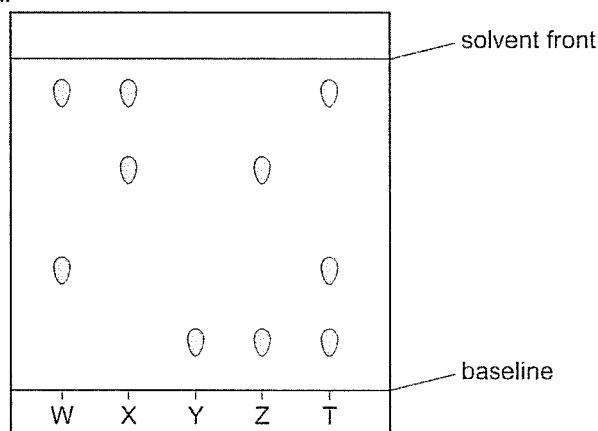


Which measurements are used to find the  $R_f$  value of T?

- A 1 and 2      B 1 and 4      C 2 and 3      D 3 and 4

## 8. M/J 22/P23/Q4

Paper chromatography is used to separate four different coloured inks, W, X, Y and Z, and an unknown ink T.  
The chromatogram is shown.

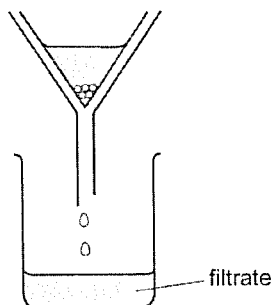


Which inks are present in ink T?

- A W and X      B W and Y      C X and Z      D Y and Z

**9. O/N 21/P22/Q3, O/N 21/P21/Q3, O/N 21/P23/Q3**

A student separates sugar from pieces of broken glass by dissolving the sugar in water and filtering off the broken glass.

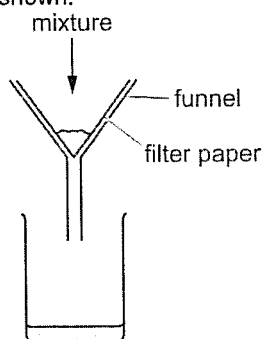


What is the filtrate?

- A broken glass only  
B broken glass and sugar solution  
C pure water  
D sugar solution

**10. M/J 21/P21/Q2**

A mixture is separated using the apparatus shown.



What is the mixture?

- A aqueous copper(II) sulfate and aqueous sodium chloride  
B aqueous copper(II) sulfate and copper  
C copper and sulfur  
D ethanol and ethanoic acid

**11. M/J 21/P21/Q19**

Copper(II) sulfate is prepared by adding excess copper(II) oxide to warm dilute sulfuric acid.

Which purification methods are used to obtain pure solid copper(II) sulfate from the reaction mixture?

- 1 crystallisation  
2 filtration  
3 chromatography  
4 distillation

- A 1 and 4  
B 1 and 2  
C 2 and 3  
D 3 and 4

**12. M/J 21/P22/Q2**

A mixture of colourless compounds is separated using chromatography.

Which type of reagent is used to detect these compounds after separation?

- A a dehydrating agent  
B a locating agent  
C an oxidising agent  
D a reducing agent

**13. M/J 21/P22/Q3, M/J 21/P23/Q3**

Which statement about paper chromatography is correct?

- A A solvent is needed to dissolve the paper.  
B Paper chromatography separates mixtures of solvents.  
C The solvent should cover the baseline.  
D The baseline should be drawn in pencil.

**14. MAR 21/P22/Q2**

Gases are separated from liquid air by fractional distillation.

The boiling points of four gases are shown.

Which gas is both monoatomic and a liquid at  $-200\text{ }^{\circ}\text{C}$ ?

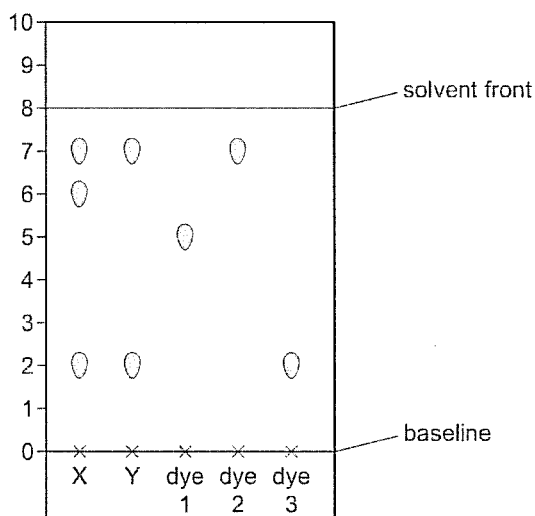
|          | gas      | boiling point/ $^{\circ}\text{C}$ |
|----------|----------|-----------------------------------|
| <b>A</b> | argon    | $-186$                            |
| <b>B</b> | helium   | $-269$                            |
| <b>C</b> | neon     | $-246$                            |
| <b>D</b> | nitrogen | $-196$                            |

**15. MAR 21/P22/Q3**

Two different food colourings, X and Y, are tested using chromatography.

Three pure dyes, 1, 2 and 3, are also tested.

The chromatogram is shown.



Which statements are correct?

- 1 X and Y both contain two or more dyes.
- 2 Dyes 2 and 3 are present in both X and Y.
- 3 The  $R_f$  of dye 1 is 0.625.

**A** 1 and 2 only      **B** 1 and 3 only      **C** 1, 2 and 3      **D** 2 and 3 only

**16. O/N 20/P21/Q2**

A mixture of colourless amino acids is separated using chromatography. The solvent used is propanol. The chromatogram is sprayed with a locating agent. Which row describes the purpose of the propanol and the locating agent?

|          | purpose of propanol                                    | purpose of locating agent                     |
|----------|--------------------------------------------------------|-----------------------------------------------|
| <b>A</b> | to make the individual amino acids visible             | to prevent the amino acids moving any further |
| <b>B</b> | to move the amino acids up the chromatography paper    | to make the individual amino acids visible    |
| <b>C</b> | to move the amino acids up the chromatography paper    | to prevent the amino acids moving any further |
| <b>D</b> | to prevent the amino acids moving too far up the paper | to make the individual amino acids visible    |

**17. O/N 20/P22/Q2**

A chromatography experiment is carried out to analyze the pigments present in four different types of leaf. The student carrying out the experiment forgot to complete his table of results, which is shown.

| plant leaf | number of pigments identified | colour of identified pigments | distance travelled by the solvent front (cm) | distance travelled from the origin by each pigment (cm) | $R_f$ value                                     |
|------------|-------------------------------|-------------------------------|----------------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| maple      | <b>F</b>                      | green / yellow                | 3.7                                          | green: 3.0<br>yellow: 3.1                               | green: 0.81<br>yellow: 0.83                     |
| laurel     | 2                             | green / yellow                | <b>G</b>                                     | green: 2.5<br>yellow: 2.5                               | green: 0.78<br>yellow: 0.78                     |
| lime       | 3                             | green / yellow / orange       | 3.5                                          | green: 2.9<br>yellow: 3.0<br>orange: 2.7                | green: 0.83<br>yellow: 0.86<br>yellow: 0.77     |
| ash        | 3                             | green / yellow / orange       | 3.5                                          | green: 2.8<br>yellow: 3.0<br>orange: 2.7                | green: 0.80<br>yellow: <b>H</b><br>orange: 0.77 |

Which row identifies the values of F, G and H?

|          | <b>F</b> | <b>G</b> | <b>H</b> |
|----------|----------|----------|----------|
| <b>A</b> | 2        | 3.2      | 0.80     |
| <b>B</b> | 3        | 3.5      | 0.83     |
| <b>C</b> | 2        | 3.2      | 0.86     |
| <b>D</b> | 3        | 3.5      | 0.78     |

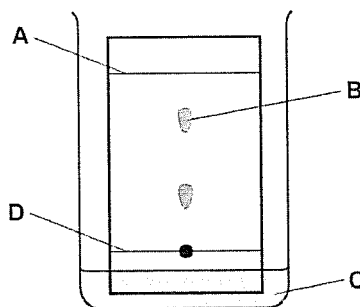
**18. O/N 20/P22/Q5**

Different methods of separation rely on substances having different properties. Which property does distillation make use of?

- A** boiling point  
**B** colour  
**C** particle size  
**D** solubility in different solvents

**19. O/N 20/P22/Q4, O/N 20/P21/Q4, O/N 20/P23/Q4**

In the chromatography experiment shown, which label represents the solvent front?

**20. O/N 20/P23/Q30**

Which process is used to separate oxygen from liquid air?

- A** chromatography  
**B** distillation  
**C** filtration  
**D** fractional distillation

**21. O/N 20/P23/Q2**

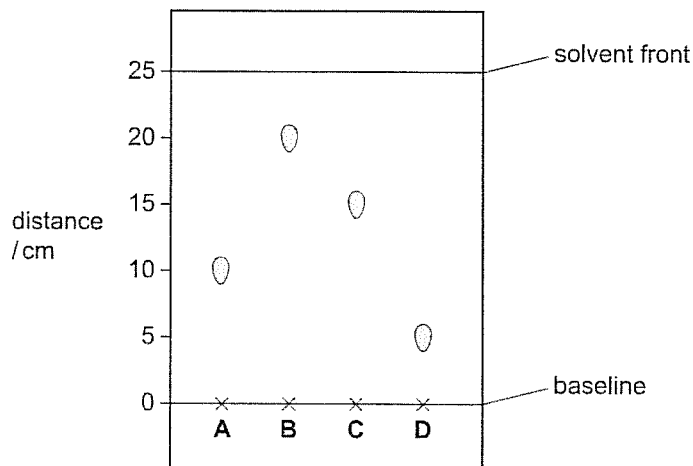
When a dark grey solid element is heated, it changes directly into a purple gas. Which word describes this change?

- A** boiling  
**B** evaporation  
**C** melting  
**D** sublimation



**22. M/J 20/P22/Q3**

Paper chromatography is used to determine the  $R_f$  values for four different food colourings. Which food colouring has an  $R_f$  value of 0.6?

**23. M/J 20/P21/Q7**

Which statement explains why methane has a lower boiling point than water?

- A Methane has weaker covalent bonds than water.
- B Methane has weaker attractive forces than water.
- C Methane molecules are heavier than water molecules.
- D Methane molecules have more bonds than water molecules.

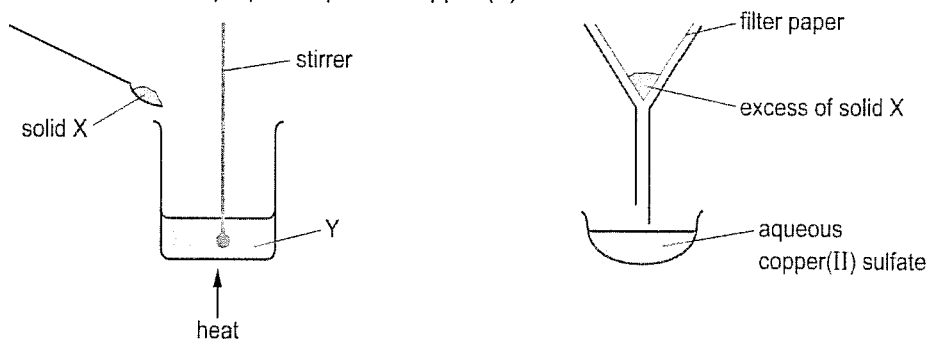
**24. M/J 20/P21/Q29**

Which statement shows that a liquid is pure water?

- A It boils at 100 °C.
- B It has a pH value of 7.
- C It turns blue cobalt (II) chloride pink.
- D It turns white copper (II) sulfate blue.

**25. M/J 20/P22/Q20, M/J 20/P21/Q20**

The apparatus shown is used to prepare aqueous copper (II) sulfate

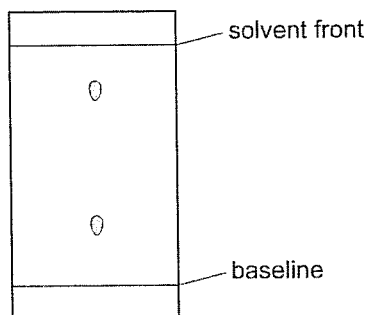


What are X and Y?

|   | X                   | Y                           |
|---|---------------------|-----------------------------|
| A | copper              | aqueous iron(II) sulfate    |
| B | copper(II) chloride | dilute sulfuric acid        |
| C | copper(II) oxide    | dilute sulfuric acid        |
| D | sulfur              | aqueous copper(II) chloride |

**26. M/J 20/P23/Q3**

Chromatography is carried out on a mixture of three substances. The chromatogram is sprayed with a locating agent. The result is shown.



What are possible reasons why the chromatogram shows only two spots?

- 1 One of the substances in the mixture is insoluble in the solvent.
- 2 The locating agent did not react with one of the substances in the mixture.
- 3 Two of the substances in the mixture have the same  $R_f$  values.
- 4 The  $R_f$  value of one of the substances is too small.

A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

**27. MAR 20/P22/Q2**

Which test is used to show that a sample of water is pure?

- A Evaporate the water to see if any solids remain.
- B Heat the water to check its boiling point.
- C Test with anhydrous cobalt (II) chloride.
- D Use universal indicator paper to check its pH

**28. MAR 20/P22/Q3**

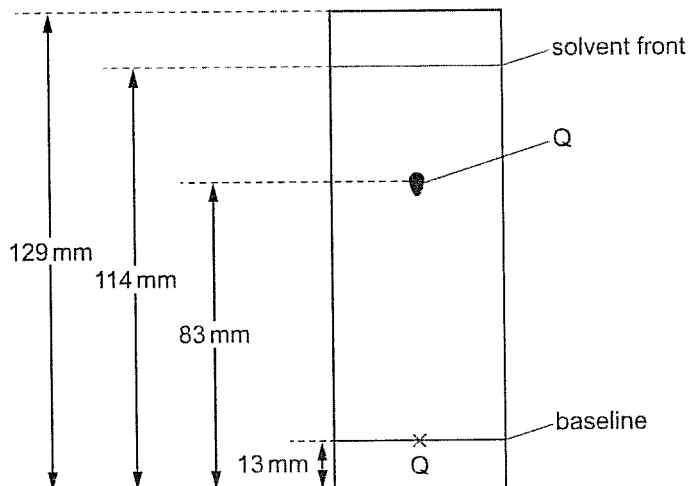
Chromatography is used to separate and identify the components in both colored and color less mixtures. For color less mixtures the chromatogram has to be treated with another chemical. What is the name of this type of chemical?

- A coloring agent              B display agent              C finding agent              D locating agent

**29. O/N 19/P22/Q3**

Substance Q was investigated using chromatography.

The chromatogram is shown. The diagram is not drawn to scale.



What is the  $R_f$  value of Q?

- A 0.60                      B 0.64                      C 0.69                      D 0.72

**30. O/N 19/P21/Q30**

River water contains soluble impurities, insoluble impurities and bacteria.

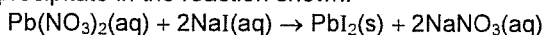
River water is made safe to drink by filtration and chlorination.

Which statement is correct?

- A Filtration removes bacteria and insoluble impurities, and chlorination removes soluble impurities.
- B Filtration removes insoluble impurities, and chlorination kills the bacteria.
- C Filtration removes soluble and insoluble impurities, and chlorination kills the bacteria.
- D Filtration removes soluble impurities and bacteria, and chlorination removes insoluble impurities.

**31. O/N 19/P23/Q22**

Lead(II) iodide is formed as a precipitate in the reaction shown.



Which method is used to separate the lead(II) iodide from the mixture?

- A crystallisation
- B distillation
- C evaporation
- D filtration

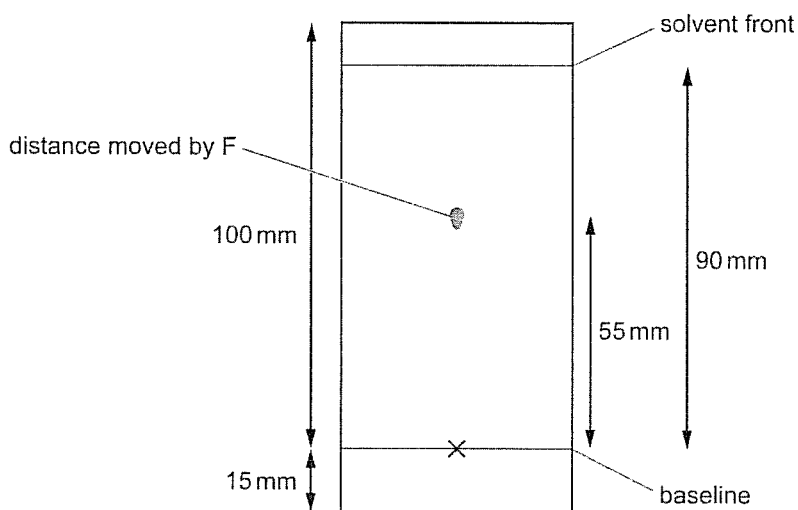
**32. O/N 19/P23/Q31**

How are oxygen and nitrogen separated from air?

- A chromatography
- B condensation and filtration
- C crystallisation
- D fractional distillation

**33. M/J 19/P21/Q3**

The measurements from a chromatography experiment using substance F are shown. The diagram is not drawn to scale.



What is the  $R_f$  value of F?

- A 0.55
- B 0.61
- C 0.90
- D 1.64

**34. M/J 19/P23/Q3**

$R_f$  values are used to identify unknown substances using paper chromatography.

Which statements about  $R_f$  values are correct?

- 1  $R_f$  values are always less than 1.0.
- 2  $R_f$  value = distance travelled by solvent ÷ distance travelled by unknown substance.
- 3 The higher the  $R_f$  value, the further the unknown substance travels.
- 4  $R_f$  values are not affected by the solubility of the unknown substance.

- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 3 and 4

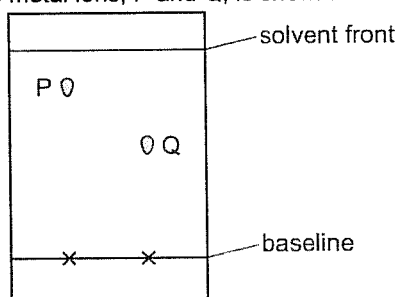
**35. MAR 19/P22/Q2**

Which method should be used to separate a mixture of two liquids?

- A crystallisation
- B electrolysis
- C filtration
- D fractional distillation

**36. M/J 19/P22/Q3**

The chromatogram of solutions of two metal ions, P and Q, is shown.



P is coloured. A locating agent is used to find the position of Q.

The  $R_f$  value of each solution is calculated.

P is a .....1..... element and has an  $R_f$  value .....2..... than that of Q.

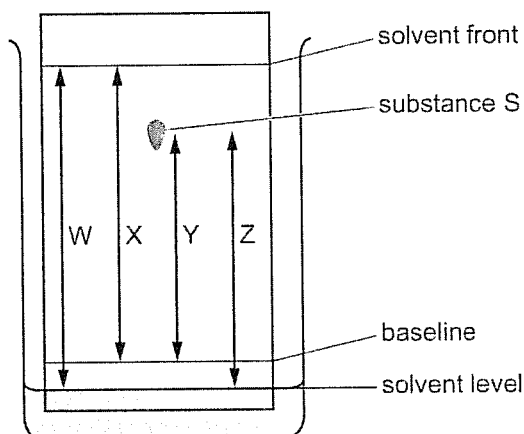
Which words complete gaps 1 and 2?

|          | 1              | 2       |
|----------|----------------|---------|
| <b>A</b> | non-transition | greater |
| <b>B</b> | non-transition | smaller |
| <b>C</b> | transition     | greater |
| <b>D</b> | transition     | smaller |

**37. MAR 19/P22/Q4**

The chromatogram of substance S is shown.

Some distances, W, X, Y and Z, are labelled on the diagram.



How is the  $R_f$  value of substance S calculated?

**A**  $\frac{X}{Y}$

**B**  $\frac{W}{Z}$

**C**  $\frac{Y}{X}$

**D**  $\frac{Y}{W}$

**38. O/N 18/P23/Q6**

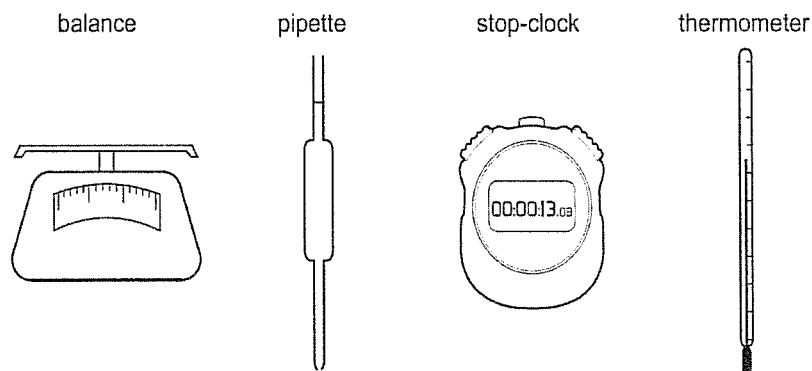
The formula of ammonia is  $\text{NH}_3$ .

Which statement about a molecule of ammonia is correct?

- A** The bonding in a molecule of ammonia is ionic.
- B** The nitrogen atom has a noble gas configuration, the hydrogen atoms do not.
- C** The nitrogen atom shares all of its electrons with hydrogen atoms.
- D** There are six shared electrons in a molecule of ammonia.

**39. O/N 18/P22/Q2, O/N 18/P21/Q2, O/N 18/P23/Q2**

The diagrams show four pieces of laboratory equipment.



Which equipment is essential to find out if dissolving a salt in water is an exothermic process?

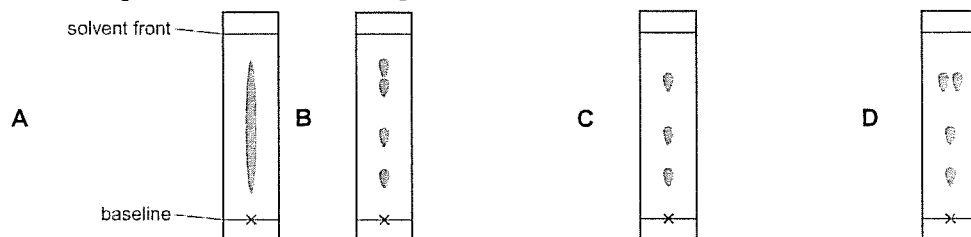
|          | balance | pipette | stop-clock | thermometer |
|----------|---------|---------|------------|-------------|
| <b>A</b> | x       | x       | x          | ✓           |
| <b>B</b> | ✓       | x       | x          | ✓           |
| <b>C</b> | x       | ✓       | x          | ✓           |
| <b>D</b> | ✓       | x       | ✓          | x           |

**40. M/J 18/P22/Q2**

A chromatography experiment was done to separate a mixture of four substances.

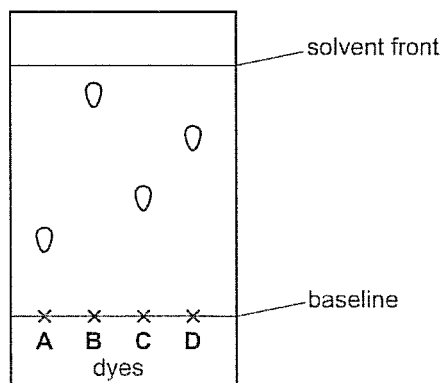
The  $R_f$  values measured for these substances were 0.3, 0.5, 0.8 and 0.8.

Which diagram shows the chromatogram obtained?

**41. M/J 18/P21/Q2**

Chromatography is a technique used to separate coloured dyes.

Which dye has an  $R_f$  value of 0.7?



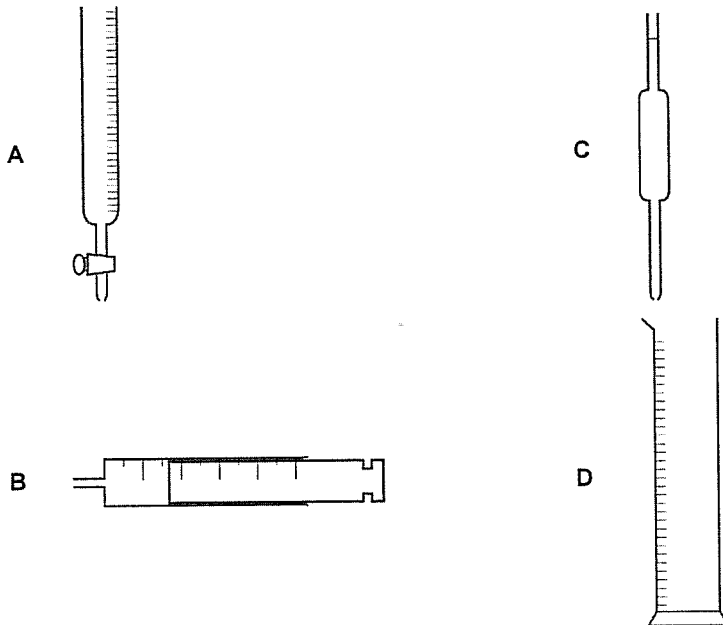
## 42. M/J 18/P22/Q3

Which piece of apparatus **cannot** be used to collect and measure the volume of gas produced in an experiment?

- A burette                      B gas syringe                      C measuring cylinder                      D pipette

## 43. M/J 18/P21/Q3

Which piece of apparatus is used to measure exactly 26.3 cm<sup>3</sup> of a liquid?



## 44. M/J 18/P23/Q2

Paper chromatography is done in the same way with three different mixtures of dyes. Each mixture contains at least one of the dyes W, X, Y and Z.

The  $R_f$  values of the dyes in the three mixtures are shown.

| dye | $R_f$ values from mixture 1 | $R_f$ values from mixture 2 | $R_f$ values from mixture 3 |
|-----|-----------------------------|-----------------------------|-----------------------------|
| W   | 0.15                        | 0.15                        | 0.15                        |
| X   | 0.00                        | 0.00                        | 0.00                        |
| Y   | 0.50                        | 0.50                        | 0.50                        |
| Z   | 0.00                        | 0.91                        | 0.91                        |

Which conclusion is correct?

- A Dye W is nearest the solvent front and is present only in mixture 1 and mixture 3.  
 B Dye X has travelled furthest up the chromatography paper.  
 C Dye Y is the only dye present in all three mixtures.  
 D Dye Z is nearest the solvent front and is found in only two of the mixtures.

## 45. O/N 17/P22/Q2

During an experiment a measurement is recorded in cm<sup>3</sup>.

Which apparatus is used?

- A balance                      B measuring cylinder                      C stopclock                      D thermometer

## 46. O/N 17/P22/Q3

A student carried out paper chromatography on a mixture of amino acids.

The student sprayed the dried chromatogram with a locating agent.

What is the function of the locating agent?

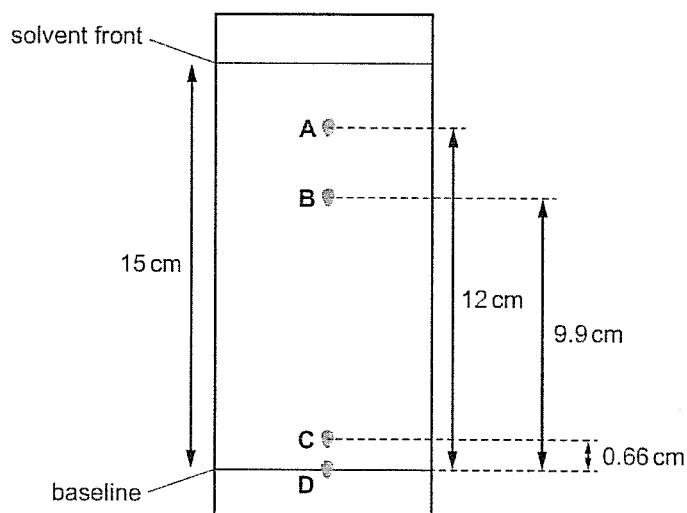
- A to dissolve the amino acids                      C to preserve the amino acids  
 B to form coloured spots with the amino acids                      D to stop the amino acids reacting

**47. MAR 18/P22/Q3**

Chromatography is done on a mixture containing a drug. The drug has an  $R_f$  value of 0.66.

The diagram is **not** drawn to scale.

Which spot on the chromatogram represents the drug?

**48. O/N 17/P21/Q3**

The results obtained from a chromatogram are shown.

|             | distance travelled / cm |
|-------------|-------------------------|
| solvent     | 5.0                     |
| substance X | 3.0                     |
| substance Y | 2.5                     |

Which row gives the  $R_f$  values of substance X and substance Y?

|   | $R_f$ (X) | $R_f$ (Y) |
|---|-----------|-----------|
| A | 0.5       | 0.6       |
| B | 0.6       | 0.5       |
| C | 1.6       | 2.0       |
| D | 2.0       | 1.6       |

**49. M/J 17/P22/Q2**

Impurities change the melting and boiling points of substances.

Sodium chloride is added to a sample of pure water.

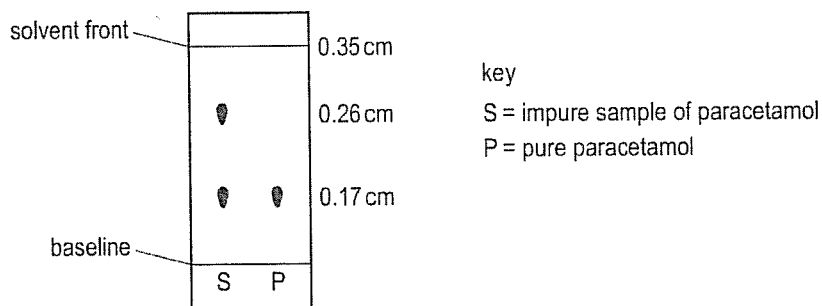
How does the addition of sodium chloride affect the melting point and boiling point of the water?

|   | melting point | boiling point |
|---|---------------|---------------|
| A | increases     | increases     |
| B | increases     | decreases     |
| C | decreases     | increases     |
| D | decreases     | decreases     |

## 50. O/N 17/P23/Q3

The painkiller paracetamol is synthesised from 4-aminophenol.

Chromatography was carried out on an impure sample of paracetamol. The results are shown (not drawn to scale).



The sample of paracetamol was contaminated with 4-aminophenol only.

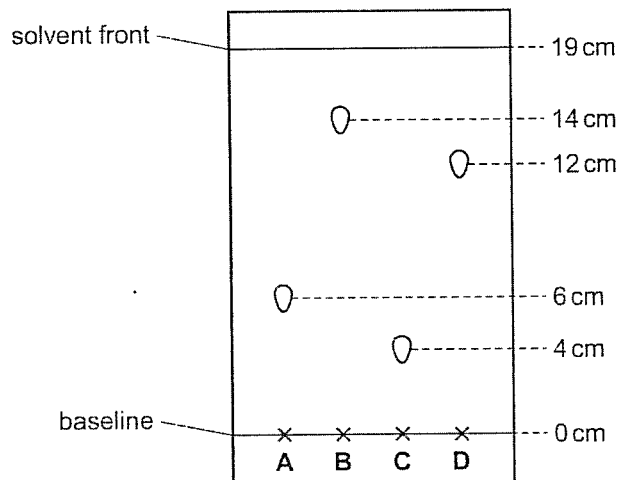
What is the  $R_f$  value of 4-aminophenol?

- A 0.49      B 0.65      C 0.74      D 1.35

## 51. M/J 17/P22/Q3

The diagram shows a chromatogram of four substances.

Which substance has an  $R_f$  value of approximately 0.32?



## 52. M/J 17/P21/Q2

Pure water has a boiling point of 100 °C and a freezing point of 0 °C.

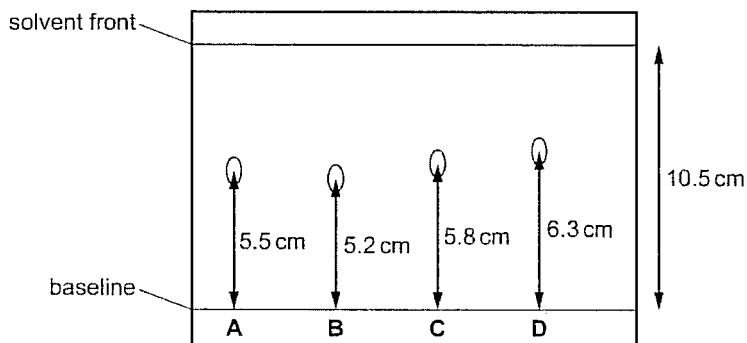
What is the boiling point and freezing point of a sample of aqueous sodium chloride?

|   | boiling point / °C | freezing point / °C |
|---|--------------------|---------------------|
| A | 98                 | -2                  |
| B | 98                 | 2                   |
| C | 102                | -2                  |
| D | 102                | 2                   |

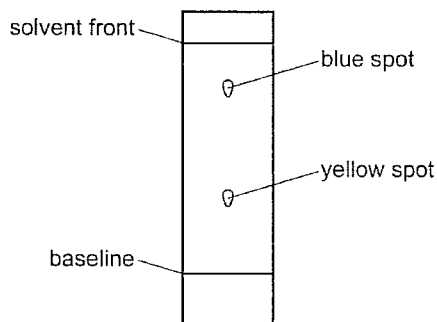


**53. M/J 17/P21/Q3**

A chromatogram obtained from the chromatography of four substances is shown. Which substance has an  $R_f$  value of 0.6?

**54. M/J 17/P23/Q3**

A student used chromatography to analyse a green food colouring. The chromatogram obtained is shown.

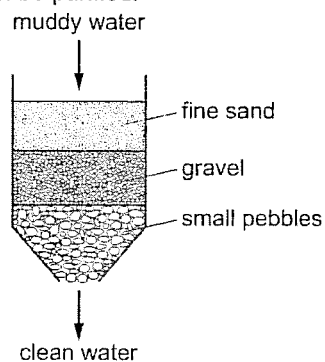


The table lists some yellow food dyes and their  $R_f$  values. Which yellow food dye does the green food colouring contain?

|   | yellow food dye  | $R_f$ value |
|---|------------------|-------------|
| A | Quinoline Yellow | 0.48        |
| B | Sunset Yellow    | 0.32        |
| C | tartrazine       | 0.69        |
| D | Yellow 2G        | 0.82        |

**55. MAR 17/P22/Q3**

The diagram shows how muddy water can be purified.



Which process for purifying the muddy water is shown?

- A crystallisation      B distillation      C filtration      D solvent extraction

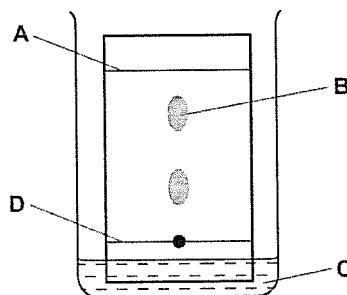
**56. O/N 16/P22/Q3**

A sample contains a mixture of powdered limestone (calcium carbonate), sugar and wax. What is the correct way to obtain a pure sample of sugar?

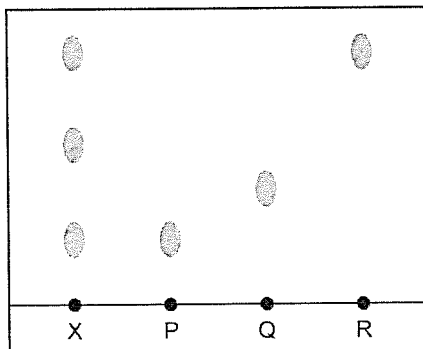
- A Dissolve the mixture in dilute hydrochloric acid, filter and wash the residue.
- B Dissolve the mixture in hexane, filter and evaporate the filtrate.
- C Dissolve the mixture in water, filter and evaporate the filtrate.
- D Dissolve the mixture in water, filter and wash the residue.

**57. M/J 16/P22/Q2**

In the chromatography experiment shown, which label represents the solvent front?

**58. M/J 16/P22/Q3**

X is a mixture of colourless compounds. The diagram shows a chromatogram of X and of three pure compounds, P, Q and R.

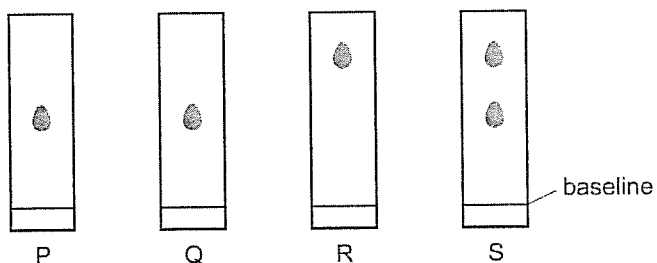


Which statement is **not** correct?

- A A locating agent was used to develop the chromatogram of X.
- B P and R could be present in X.
- C P and R have different solubilities in the solvent.
- D Q has a greater  $R_f$  value than R.

**59. M/J 16/P23/Q2**

Chromatography experiments are carried out on four substances, P, Q, R and S. The same solvent is used in each experiment. The resulting chromatograms are shown below.



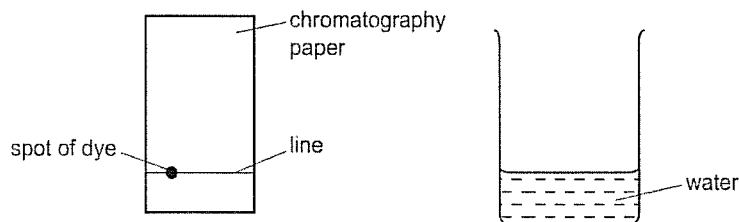
Which statement is **not** correct?

- A P and Q are pure substances.
- B P and R are different substances.
- C R and S are pure substances.
- D S is a mixture of substances.

**60. M/J 16/P21/Q2**

A sample of a dye is investigated by chromatography.

A line is drawn across a piece of chromatography paper and a spot of the dye is placed on it. The paper is placed in water.



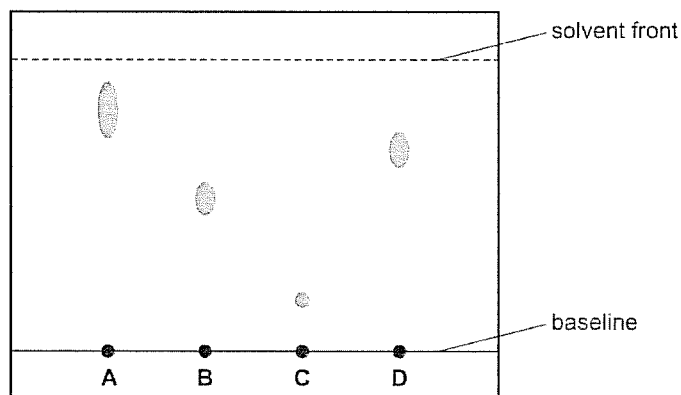
Which row is correct?

|          | what is used to draw the line | position of spot             |
|----------|-------------------------------|------------------------------|
| <b>A</b> | ink                           | above the level of the water |
| <b>B</b> | ink                           | below the level of the water |
| <b>C</b> | pencil                        | above the level of the water |
| <b>D</b> | pencil                        | below the level of the water |

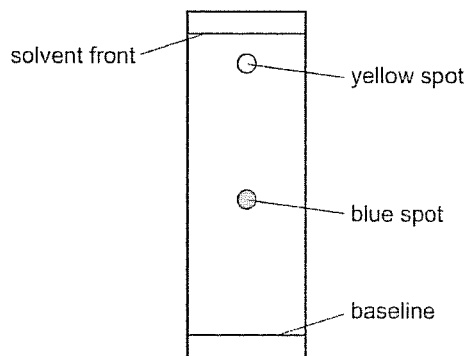
**61. M/J 16/P21/Q3**

The paper chromatogram below was obtained from four different dyes.

Which dye has an  $R_f$  value of 0.7?

**62. MAR 16/P22/Q3**

A sample of a green food colouring was separated into its component colours using paper chromatography. The results obtained are shown.



What is the  $R_f$  value of the blue spot?

**A** 0.45

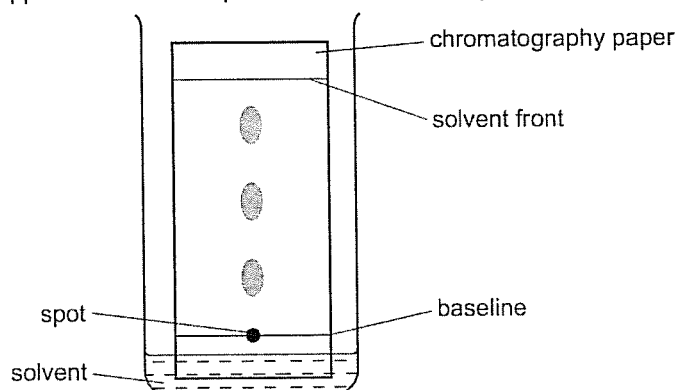
**B** 0.90

**C** 1.10

**D** 2.20

**63. M/J 16/P23/Q3**

The diagram shows the apparatus used to separate the different components of a mixture by chromatography.



Which statement about this experiment is correct?

- A A locating agent is used to find the position of the solvent front.  
 B The components to be separated must be soluble in the solvent.  
 C The baseline on which the spot of the mixture is placed is drawn in ink.  
 D The  $R_f$  value is calculated by  $\frac{\text{the distance travelled by the solvent front}}{\text{the distance travelled by the component}}$

**64. M/J 15/P13/Q2**

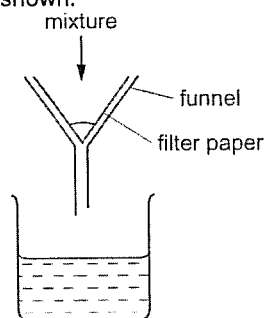
A blue solid, X, is soluble in water.

Which method is used to obtain pure solid X from an aqueous solution?

- A chromatography    B crystallisation    C filtration    D neutralisation

**65. O/N 14/P12/Q3, O/N 14/P11/Q2**

A mixture is separated using the apparatus shown.

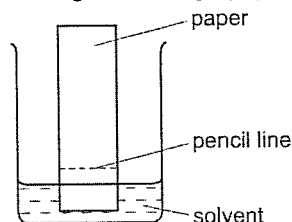


What is the mixture?

- A aqueous copper chloride and copper    C ethane and methane  
 B aqueous copper chloride and sodium chloride    D ethanol and water

**66. O/N 14/P13/Q3**

A student is investigating a coloured mixture using chromatography.



Where should he place the coloured mixture?

- A in the solvent    C just below the pencil line  
 B just above the pencil line    D on the pencil line

**67. M/J 14/P12/Q3**

Which two methods can be used to separate a salt from its solution in water?

- 1 crystallisation
- 2 decanting
- 3 distillation
- 4 filtration

A 1 and 2                      B 1 and 3                      C 2 and 3                      D 3 and 4

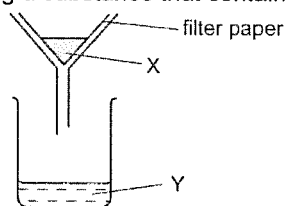
**68. M/J 14/P11/Q3, M/J 14/P13/Q2**

Alcohol and water are completely miscible. This means when mixed together they form only one liquid layer. Which method is used to separate alcohol from water?

- A crystallisation                      C fractional distillation  
B filtration                              D precipitation

**69. O/N 13/P13/Q2**

The diagram shows a method for separating a substance that contains X and Y.



Which types of substance can be separated as shown?

- A compounds                      B elements                      C mixtures                      D molecules

**70. O/N 13/P12/Q3, O/N 13/P11/Q3, O/N 13/P13/Q3**

Diagram 1 shows the paper chromatogram of substance X.

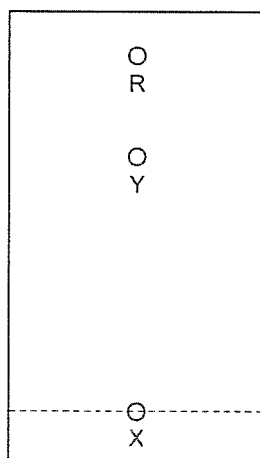


diagram 1

Diagram 2 shows the cooling curve for substance Y.

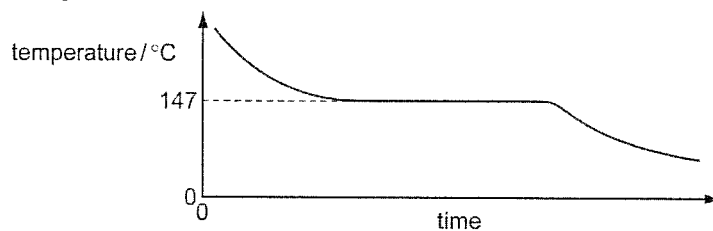


diagram 2

Which statement about X and Y is correct?

- A X is a mixture and Y is a pure substance.                      C X and Y are mixtures.  
B X is a pure substance and Y is a mixture.                      D X and Y are pure substances.

**71. M/J 13/P11/Q2, M/J 13/P13/Q2**

Crystals of sodium chloride were prepared by the following method.

- 1 25.0 cm<sup>3</sup> of dilute hydrochloric acid was accurately measured into a conical flask.
- 2 Aqueous sodium hydroxide was added until the solution was neutral. The volume of sodium hydroxide added was measured.
- 3 The solution was evaporated and the crystals washed with approximately 15 cm<sup>3</sup> of water.

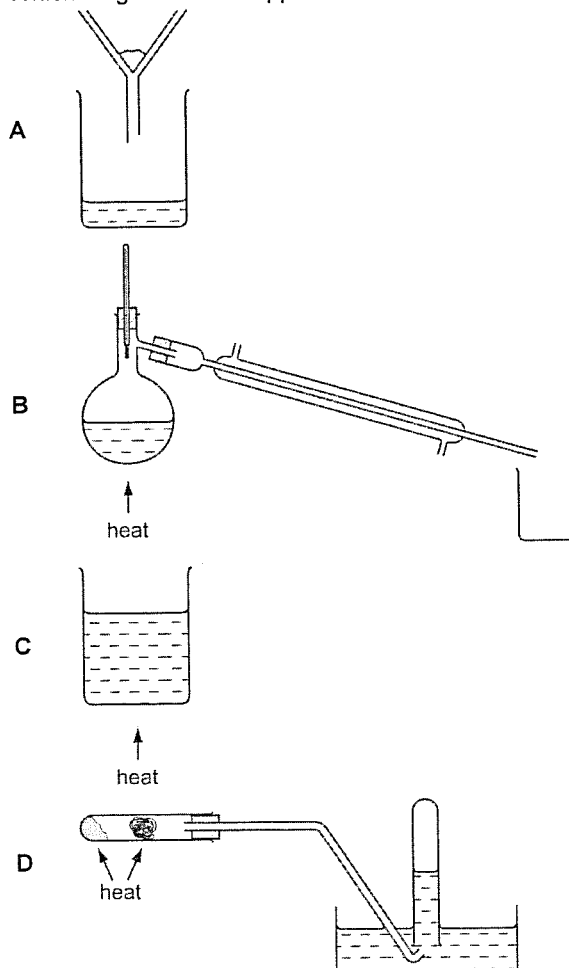
Which row shows the pieces of apparatus used to measure the 25.0 cm<sup>3</sup> of hydrochloric acid, the volume of aqueous sodium hydroxide and the 15 cm<sup>3</sup> of water?

|   | 25.0 cm <sup>3</sup> of hydrochloric acid accurately | the volume of aqueous sodium hydroxide added | 15 cm <sup>3</sup> of water approximately |
|---|------------------------------------------------------|----------------------------------------------|-------------------------------------------|
| A | burette                                              | pipette                                      | measuring cylinder                        |
| B | measuring cylinder                                   | burette                                      | pipette                                   |
| C | pipette                                              | burette                                      | measuring cylinder                        |
| D | pipette                                              | measuring cylinder                           | burette                                   |

**72. M/J 13/P12/Q3**

Methanol, CH<sub>3</sub>OH, and ethanol, C<sub>2</sub>H<sub>5</sub>OH, are miscible liquids.

Which diagram shows apparatus that is used to obtain methanol from a mixture of ethanol and methanol?

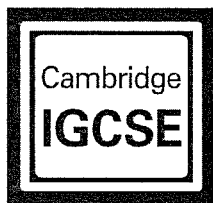


## Answers Section

| 2.1   | 2.2         |
|-------|-------------|
| 1. A  | 1. D 41. D  |
| 2. C  | 2. D 42. D  |
| 3. D  | 3. C 43. A  |
| 4. C  | 4. B 44. D  |
| 5. D  | 5. A 45. B  |
| 6. D  | 6. C 46. B  |
| 7. C  | 7. D 47. B  |
| 8. C  | 8. B 48. B  |
| 9. B  | 9. D 49. C  |
| 10. A | 10. B 50. C |
| 11. D | 11. B 51. A |
| 12. C | 12. B 52. C |
| 13. A | 13. D 53. D |
| 14. C | 14. A 54. B |
| 15. C | 15. C 55. C |
| 16. A | 16. B 56. C |
| 17. C | 17. C 57. A |
| 18. B | 18. A 58. D |
| 19. B | 19. A 59. C |
| 20. B | 20. D 60. C |
| 21. A | 21. D 61. D |
|       | 22. C 62. A |
|       | 23. B 63. B |
|       | 24. A 64. B |
|       | 25. C 65. A |
|       | 26. C 66. D |
|       | 27. B 67. B |
|       | 28. D 68. C |
|       | 29. C 69. C |
|       | 30. B 70. A |
|       | 31. D 71. C |
|       | 32. D 72. B |
|       | 33. B       |
|       | 34. B       |
|       | 35. D       |
|       | 36. C       |
|       | 37. C       |
|       | 38. D       |
|       | 39. A       |
|       | 40. C       |

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## Unit 3

# ATOMS, ELEMENTS AND COMPOUNDS

IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 3.1 Atomic structure and periodic table
- 3.2 Ions and ionic bonds
- 3.3 Molecules and covalent bonds
- 3.4 Macromolecules
- 3.5 Metallic bonding

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## Unit 3: Atoms, Elements And Compounds

### 3.1: Atomic structure and the periodic table

1. O/N 22/P22/Q4, O/N 22/P21/Q4, O/N 22/P23/Q5, M/J 16/P22/Q4, M/J 16/P21/Q4, M/J 16/P23/Q4

Which statements about isotopes of the same element are correct?

- 1 They are atoms which have the same chemical properties because they have the same number of electrons in their outer shell.
- 2 They are atoms which have the same number of electrons and neutrons but different numbers of protons.
- 3 They are atoms which have the same number of electrons and protons but different numbers of neutrons.

A 1 and 2                      B 1 and 3                      C 2 only                      D 3 only

2. O/N 22/P22/Q18

X and Y are oxides of two different elements.

- X reacts with water to produce aqueous solution Z.
- Z turns universal indicator paper blue.
- An aqueous solution of Y reacts with sodium carbonate to produce carbon dioxide gas.

Which statement is correct?

- A X and Y are both the oxides of metals.
- B X and Y are both the oxides of non-metals.
- C X is the oxide of a metal and Y is the oxide of a non-metal.
- D X is the oxide of a non-metal and Y is the oxide of a metal.

3. O/N 22/P23/Q4

Which two particles have the same electronic structure?

A C and O<sup>2-</sup>                      B F<sup>-</sup> and Na                      C K<sup>+</sup> and S<sup>2-</sup>                      D Mg and Na<sup>+</sup>

4. M/J 22/P22/Q4

X and Y are two different elements.

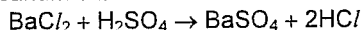
X and Y have the same number of nucleons.

Which statement about X and Y is correct?

- A They have the same physical properties.
- B Their atoms have the same number of electrons.
- C They are in different groups of the Periodic Table.
- D They have different relative masses.

5. M/J 22/P22/Q7, M/J 22/P21/Q7, M/J 22/P23/Q7

The equation for the reaction between barium chloride and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

|   | BaCl <sub>2</sub> | H <sub>2</sub> SO <sub>4</sub> | BaSO <sub>4</sub> | 2HCl |
|---|-------------------|--------------------------------|-------------------|------|
| A | (aq)              | (aq)                           | (s)               | (aq) |
| B | (aq)              | (l)                            | (s)               | (aq) |
| C | (l)               | (aq)                           | (s)               | (l)  |
| D | (aq)              | (l)                            | (aq)              | (l)  |

6. M/J 22/P21/Q4

Which substance should be pure for the intended use?

- A a drug for curing disease
- B limestone for iron extraction
- C petroleum for fractional distillation
- D water for washing a car



## 13. MAR 22/P22/Q28

The table gives some properties of an element.

|                                           |           |
|-------------------------------------------|-----------|
| melting point in °C                       | 3422      |
| appearance of the element                 | grey      |
| appearance of the chloride of the element | dark blue |
| density in g/cm <sup>3</sup>              | 19.2      |
| electrical conductivity when solid        | good      |

Which other property does this element have?

- A acts as a catalyst  
B brittle  
C forms an acidic oxide  
D highly reactive with water

## 14. O/N 21/P22/Q4

How many protons, neutrons and electrons are there in one atom of the isotope  $^{27}_{13}\text{A}$ ?

|   | protons | neutrons | electrons |
|---|---------|----------|-----------|
| A | 13      | 13       | 13        |
| B | 13      | 14       | 13        |
| C | 14      | 13       | 13        |
| D | 14      | 14       | 13        |

## 15. O/N 21/P21/Q9

The formula of an aluminium ion is  $\text{Al}^{3+}$ .

What is the formula of aluminium sulfate?

- A  $\text{Al}_2\text{SO}_4$   
B  $\text{Al}(\text{SO}_4)_2$   
C  $\text{Al}_2(\text{SO}_4)_3$   
D  $\text{Al}_3(\text{SO}_4)_2$

## 16. O/N 21/P23/Q4

The nucleus of a particular atom consists of nineteen particles.

Nine of them are positively charged and ten of them are uncharged.

Which statement about this nucleus is correct?

- A The nucleus has a nucleon number of nine.  
B The nucleus has a nucleon number of ten.  
C The nucleus has a proton number of nine.  
D The nucleus has a proton number of ten.

## 17. M/J 21/P22/Q4, M/J 21/P21/Q4, M/J 21/P23/Q4

Element X has 7 protons.

Element Y has 8 more protons than X.

Which statement about element Y is correct?

- A Y has more electron shells than X.  
B Y has more electrons in its outer shell than X.  
C Y is in a different group of the Periodic Table from X.  
D Y is in the same period of the Periodic Table as X.

## 18. M/J 21/P22/Q8

Some information about particles P, Q, R and S is shown.

|   | nucleon number | number of neutrons | number of electrons |
|---|----------------|--------------------|---------------------|
| P | 12             | 6                  | 6                   |
| Q | 24             | 12                 | 10                  |
| R | 16             | 8                  | 10                  |
| S | 14             | 8                  | 6                   |

Which two particles are isotopes of the same element?

- A P and Q  
B P and S  
C Q and R  
D R and S

**19. M/J 21/P23/Q7**

Which statement about isotopes of the same element is correct?

- A They have different numbers of electrons.      C They have different numbers of protons.  
 B They have different numbers of neutrons.      D They have the same mass number.

**20. MAR 21/P22/Q4**

Which statement about the atoms of all the isotopes of carbon is correct?

- A They are all radioactive.  
 B They have the same mass.  
 C They have the same number of neutrons.  
 D They have the same number of electrons in the outer shell.

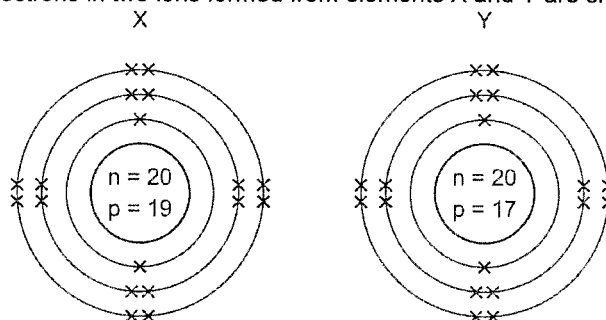
**21. O/N 20/P22/Q3**

Which statement about isotopes is correct?

- A They have different proton numbers.  
 B They have different chemical properties.  
 C They have the same nucleon number.  
 D They have the same number of electrons in their outer shell.

**22. O/N 20/P22/Q6, O/N 20/P21/Q6**

The arrangements of the electrons in two ions formed from elements X and Y are shown.



Which equation represents the reaction between elements X and Y?

- A  $X_2 + 2Y \rightarrow 2X^+ + 2Y^-$       C  $2X + Y_2 \rightarrow 2X^+ + 2Y^-$   
 B  $X_2 + 2Y \rightarrow 2X^- + 2Y^+$       D  $2X + Y_2 \rightarrow 2X^- + 2Y^+$

**23. O/N 20/P22/Q11, O/N 20/P21/Q12, O/N 20/P23/Q11**

The relative atomic mass of chlorine is 35.5. When calculating relative atomic mass, which particle is the mass of a chlorine atom compared to?

- A a neutron      C an atom of carbon-12  
 B a proton      D an atom of hydrogen-1

**24. O/N 20/P21/Q5**

The atomic structure of four particles are shown.

|   | electrons | protons | neutrons |
|---|-----------|---------|----------|
| P | 18        | 17      | 18       |
| Q | 18        | 17      | 20       |
| R | 17        | 17      | 18       |
| S | 17        | 17      | 20       |

Which particles have the same chemical properties?

- A P and R only      B P and S      C P, Q and R      D R and S

**25. O/N 20/P21/Q25**

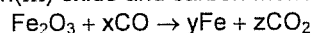
Elements P and Q have the same number of electron shells. Q has more electrons in its outer shell than P. Which statements are correct?

- 1 P and Q are in the same group of the Periodic Table.
- 2 P and Q are in the same period of the Periodic Table.
- 3 P has a greater tendency to form positive ions than Q.
- 4 The oxides of Q are more basic than those of P.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**26. O/N 20/P23/Q29**

The equation for the reaction between iron(III) oxide and carbon monoxide is shown.

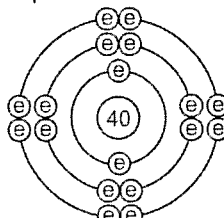


Which values of x, y and z balance the equation?

|   | x | y | z |
|---|---|---|---|
| A | 2 | 2 | 2 |
| B | 2 | 3 | 3 |
| C | 3 | 1 | 3 |
| D | 3 | 2 | 3 |

**27. M/J 20/P22/Q4**

The diagram shows the electronic structure of a particle with a nucleon number (mass number) of 40.



The table shows the suggestions that three students, 1, 2 and 3, made to identify the particle.

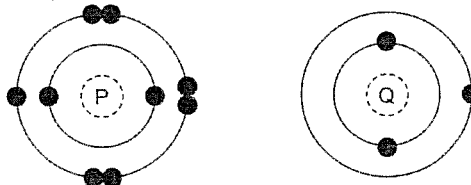
|          | student |    |                  |
|----------|---------|----|------------------|
|          | 1       | 2  | 3                |
| particle | Ar      | Cl | Ca <sup>2+</sup> |

Which students are correct?

A 1 and 2 only                      B 1 and 3 only                      C 2 and 3 only                      D 1, 2 and 3

**28. M/J 20/P21/Q5**

The electronic structures of two atoms, P and Q, are shown.



P and Q combine together to form a compound.

What is the type of bonding in the compound and what is the formula of the compound?

|   | type of bonding | formula         |
|---|-----------------|-----------------|
| A | ionic           | PQ              |
| B | ionic           | PQ <sub>2</sub> |
| C | covalent        | PQ <sub>2</sub> |
| D | covalent        | PQ              |

**29. M/J 20/P21/Q4**

The atomic number and nucleon number of a potassium atom are shown.

|                | potassium atom |
|----------------|----------------|
| atomic number  | 19             |
| nucleon number | 39             |

How many protons, neutrons and electrons are in a potassium ion,  $K^+$ ?

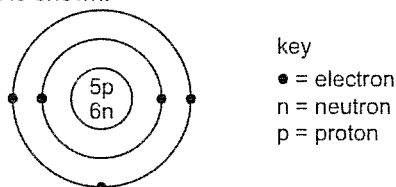
|          | protons | neutrons | electrons |
|----------|---------|----------|-----------|
| <b>A</b> | 19      | 20       | 18        |
| <b>B</b> | 19      | 20       | 20        |
| <b>C</b> | 20      | 19       | 18        |
| <b>D</b> | 20      | 19       | 19        |

**30. M/J 20/P21/Q14**

Which diagram represents a chemical change?

**31. M/J 20/P23/Q4**

The structure of an atom of element X is shown.



What is element X?

- A** boron                      **B** carbon                      **C** sodium                      **D** sulfur

**32. MAR 20/P22/Q5**

$^{14}_6\text{C}$  and  $^{12}_6\text{C}$  are isotopes of carbon. Which statement about these isotopes is correct?

- A**  $^{12}_6\text{C}$  is more reactive than  $^{14}_6\text{C}$  because the atoms have less mass.
- B**  $^{12}_6\text{C}$  is more reactive than  $^{14}_6\text{C}$  because the atoms have different numbers of neutrons.
- C** The reactions of  $^{12}_6\text{C}$  are similar to  $^{14}_6\text{C}$  because they have the same number of outer shell electrons.
- D** The reactions of  $^{12}_6\text{C}$  are similar to  $^{14}_6\text{C}$  because they have the same number of protons in the nucleus.

**33. O/N 19/P22/Q6**

Rubidium has two isotopes,  $^{85}_{37}\text{Rb}$  and  $^{87}_{37}\text{Rb}$ .

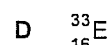
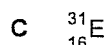
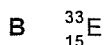
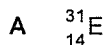
Which statement explains why both isotopes have the same chemical properties?

- A** They have the same number of protons.
- B** They have the same number of outer shell electrons.
- C** They have different numbers of neutrons.
- D** They have different mass numbers.





## 40. M/J 19/P22/Q4

What is an isotope of  ${}^{31}_{15}\text{E}$ ?

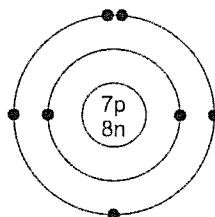
## 41. M/J 19/P21/Q4

Which statement about an atom of fluorine,  ${}^{19}_9\text{F}$ , is correct?

- A It contains more protons than neutrons.  
 B It contains a total of 28 protons, neutrons and electrons.  
 C Its isotopes contain different numbers of protons.  
 D Its nucleus contains 9 neutrons.

## 42. M/J 19/P23/Q4

The structure of an atom is shown.



key

● = electron

n = neutron

p = proton

Which element is the atom an isotope of?

A nitrogen

B oxygen

C phosphorus

D titanium

## 43. MAR 19/P22/Q5

Which row describes isotopes of the same element?

|   | number of protons | number of neutrons |
|---|-------------------|--------------------|
| A | different         | different          |
| B | different         | same               |
| C | same              | different          |
| D | same              | same               |

## 44. O/N 18/P22/Q3

How many neutrons are present in the atom  ${}^{45}_{21}\text{X}$ ?

A 21

B 24

C 45

D 66

## 45. O/N 18/P22/Q4

Two naturally occurring isotopes of oxygen are  ${}^{16}\text{O}$  and  ${}^{17}\text{O}$ .

Which statement is correct?

- A Both isotopes react with iron to form rust.      C Only  ${}^{16}\text{O}$  reacts with iron to form rust.  
 B Neither isotope reacts with iron to form rust.      D Only  ${}^{17}\text{O}$  reacts with iron to form rust.

## 46. O/N 18/P21/Q4

Which statement about the isotopes of an element is correct?

- A Their physical properties are different because they have different proton numbers.  
 B Their atomic masses are different because they have different numbers of electron shells.  
 C They have the same chemical properties because they have the same number of electrons in their outer shells.  
 D They have the same physical properties because they have the same number of neutrons in their nuclei.

## 47. O/N 18/P21/Q5

Which two molecules contain the same number of electrons?

A  $\text{Cl}_2$  and  $\text{SO}_2$ B  $\text{CH}_4$  and  $\text{H}_2\text{O}$ C  $\text{CO}$  and  $\text{NH}_3$ D  $\text{CO}_2$  and  $\text{HCl}$

**48. O/N 18/P23/Q3**

Which statement describes isotopes?

- A Isotopes of the same element have different electron arrangements.  
 B Isotopes of the same element have different nuclear charges.  
 C Isotopes of the same element have nuclei with masses that are the same.  
 D Isotopes of the same element have the same number of protons.

**49. O/N 18/P23/Q4**

X and Y are both atoms.

X and Y have the same chemical properties as each other.

Which row describes the atomic structures of X and Y?

|   | X       |          |           | Y       |          |           |
|---|---------|----------|-----------|---------|----------|-----------|
|   | protons | neutrons | electrons | protons | neutrons | electrons |
| A | 6       | 6        | 6         | 6       | 6        | 7         |
| B | 6       | 6        | 6         | 6       | 8        | 6         |
| C | 6       | 6        | 6         | 16      | 16       | 16        |
| D | 7       | 6        | 7         | 6       | 6        | 7         |

**50. M/J 18/P22/Q5**

Which pair shows particles with the same chemical properties?

- A  $^{23}_{11}\text{M}$  and  $^{23}_{11}\text{M}^+$       B  $^{23}_{11}\text{M}$  and  $^{24}_{11}\text{M}$       C  $^{23}_{11}\text{M}$  and  $^{23}_{12}\text{M}$       D  $^{24}_{11}\text{M}^+$  and  $^{24}_{12}\text{M}^+$

**51. M/J 18/P21/Q5**Chlorine exists as two common isotopes,  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ .

Information about these two isotopes is shown.

|                  | number of protons | number of neutrons | number of electron shells |
|------------------|-------------------|--------------------|---------------------------|
| $^{35}\text{Cl}$ | 17                | 18                 | 3                         |
| $^{37}\text{Cl}$ | 17                | 20                 | 3                         |

Which statement explains why the two isotopes are of the same element?

- A Both have the same number of electron shells.  
 B Both have the same number of protons.  
 C Both have 7 outer shell electrons.  
 D  $^{37}\text{Cl}$  has 2 more neutrons than  $^{35}\text{Cl}$ .

**52. M/J 18/P23/Q5**Iron has an atomic number of 26. It occurs as the isotopes  $^{54}\text{Fe}$ ,  $^{56}\text{Fe}$ ,  $^{57}\text{Fe}$  and  $^{58}\text{Fe}$ .

Which statement explains why these isotopes have the same chemical properties?

- A They have similar mass numbers.  
 B They have the same number of electrons in their outer shells.  
 C They have the same number of neutrons in their nuclei.  
 D They have the same number of protons in their nuclei.

**53. O/N 17/P22/Q5**

Why do isotopes of the same element have the same chemical properties?

- A They have the same nucleon number.  
 B They have the same number of electrons in the outer shell.  
 C They have the same number of neutrons in the nucleus.  
 D They have the same number of protons as neutrons.

**54. O/N 17/P23/Q5**

Carbon has three naturally occurring isotopes,  $^{12}\text{C}$ ,  $^{13}\text{C}$  and  $^{14}\text{C}$ .

Which statement explains why the isotopes have the same chemical properties?

- A They have the same number of electrons in the first shell.
- B They have the same number of electrons in the outer shell.
- C They have the same number of neutrons in the nucleus.
- D They have the same number of protons as neutrons.

**55. MAR 17/P22/Q4**

Which statement explains why isotopes of an element have the same chemical properties?

- A They have different numbers of neutrons.
- B They have the same number of electrons as protons.
- C They have the same number of electrons in the outer shell.
- D They have the same number of protons in the nucleus.

**56. O/N 16/P23/Q4**

An atom has three electron shells. There are three electrons in the outer shell.

How many protons and how many neutrons are in this atom?

|   | protons | neutrons |
|---|---------|----------|
| A | 13      | 14       |
| B | 13      | 27       |
| C | 14      | 13       |
| D | 21      | 24       |

**57. O/N 16/P22/Q4**

The table shows information about four different particles.

| particle        | proton number | nucleon number | number of protons | number of neutrons | number of electrons |
|-----------------|---------------|----------------|-------------------|--------------------|---------------------|
| Na              | 11            | 23             | 11                | W                  | 11                  |
| Na <sup>+</sup> | 11            | 23             | 11                | 12                 | X                   |
| O               | 8             | 16             | 8                 | Y                  | 8                   |
| O <sup>2-</sup> | 8             | 16             | 8                 | 8                  | Z                   |

What are the values of W, X, Y and Z?

|   | W  | X  | Y  | Z  |
|---|----|----|----|----|
| A | 11 | 10 | 10 | 8  |
| B | 11 | 11 | 8  | 10 |
| C | 12 | 10 | 8  | 10 |
| D | 12 | 11 | 10 | 8  |

**58. MAR 16/P22/Q4**

In which row are the substances correctly classified?

|   | element | compound | mixture |
|---|---------|----------|---------|
| A | brass   | sulfur   | water   |
| B | sulfur  | brass    | water   |
| C | sulfur  | water    | brass   |
| D | water   | sulfur   | brass   |

**59. O/N 16/P21/Q4**

The table shows information about atoms of three different elements.

| element   | proton number | nucleon number | number of protons | number of neutrons | number of electrons |
|-----------|---------------|----------------|-------------------|--------------------|---------------------|
| chlorine  | 17            | 35             | 17                | W                  | 17                  |
| chlorine  | 17            | X              | 17                | 19                 | 17                  |
| argon     | Y             | 40             | 18                | 22                 | 18                  |
| potassium | 19            | 39             | 19                | 20                 | Z                   |

What are the values of W, X, Y and Z?

|   | W  | X  | Y  | Z  |
|---|----|----|----|----|
| A | 18 | 35 | 18 | 19 |
| B | 18 | 36 | 18 | 19 |
| C | 19 | 35 | 19 | 18 |
| D | 19 | 36 | 19 | 18 |

**60. MAR 16/P22/Q7**

X and Y are isotopes of the same element.

Which statement is correct?

- A X and Y have atoms with different numbers of electron shells.
- B X and Y have atoms with the same nucleon number.
- C X and Y have atoms with the same number of outer shell electrons.
- D X and Y have different chemical properties.

**61. O/N 15/P12/Q3**

The table shows the nucleon number and the number of neutrons in one atom of isotopes W, X, Y and Z.

| isotope | nucleon number | number of neutrons |
|---------|----------------|--------------------|
| W       | 35             | 18                 |
| X       | 37             | 20                 |
| Y       | 39             | 20                 |
| Z       | 40             | 22                 |

Which statement about W, X, Y and Z is correct?

- A W and X are isotopes of the same element.
- B X and Y are isotopes of elements in the same group of the Periodic Table.
- C Y and Z are isotopes of elements in the same period of the Periodic Table.
- D Z has a higher proton number than Y.

**62. O/N 15/P12/Q5, O/N 15/P11/Q5, O/N 15/P13/Q5**

What do the nuclei of  ${}^1_1\text{H}$  hydrogen atoms contain?

- A electrons and neutrons
- B electrons and protons
- C neutrons only
- D protons only

**63. O/N 15/P11/Q3**

Which statement about atoms is correct?

- A Atoms contain protons and electrons in the nucleus.
- B Neutrons are negatively charged.
- C Protons are positively charged.
- D The nucleon number is the number of neutrons.

**64. M/J 15/P12/Q3**

The atomic structures of four particles are shown.

| particle | electrons | neutrons | protons |
|----------|-----------|----------|---------|
| W        | 8         | 9        | 8       |
| X        | 7         | 9        | 7       |
| Y        | 8         | 10       | 8       |
| Z        | 9         | 10       | 9       |

Which two particles are isotopes?

- A W and X      B W and Y      C X and Z      D Y and Z

**65. M/J 15/P12/Q4**

$Q^+$  is an ion of element Q.

What has the highest value in the ion?

- A the nucleon number      C the number of neutrons  
B the number of electrons      D the proton number

**66. M/J 15/P12/Q5, M/J 15/P11/Q5, M/J 15/P13/Q5, O/N 12/P12/Q5, O/N 12/P11/Q4**

Which statements comparing the properties of electrons, neutrons and protons are correct?

|   | neutrons and protons are both heavier than electrons | only electrons and neutrons are charged |
|---|------------------------------------------------------|-----------------------------------------|
| A | ✓                                                    | ✓                                       |
| B | ✓                                                    | x                                       |
| C | x                                                    | ✓                                       |
| D | x                                                    | x                                       |

**67. M/J 15/P11/Q3**

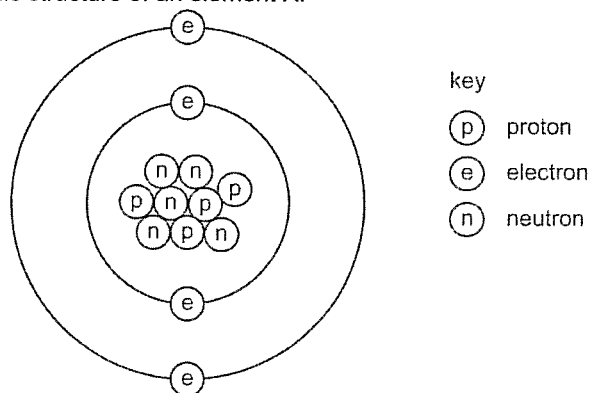
Atoms contain electrons, neutrons and protons.

What is the definition of nucleon number?

- A the number of neutrons in the nucleus of an atom  
B the number of protons in the nucleus of an atom  
C the total number of neutrons and protons in the nucleus of an atom  
D the total number of particles in an atom

**68. M/J 15/P11/Q4**

The diagram shows the atomic structure of an element X.



What is X?

- A aluminium      B beryllium      C boron      D fluorine

**69. M/J 15/P13/Q3**

Two atoms, X and Y, can be represented as shown.



Which statement is **not** correct?

- A X and Y are atoms of different elements.      C X and Y have different mass numbers.  
B X and Y are isotopes.      D X and Y have the same number of electrons.

**70. M/J 15/P13/Q4**

Two atoms have the same relative atomic mass but different chemical properties.  
Which row about the proton and neutron numbers of these atoms is correct?

|   | proton numbers | neutron numbers |
|---|----------------|-----------------|
| A | different      | different       |
| B | different      | same            |
| C | same           | different       |
| D | same           | same            |

**71. MAR 15/P12/Q4**

The table shows the numbers of particles present in the nuclei of four atoms or ions.

|   | protons | neutrons | electronic structure |
|---|---------|----------|----------------------|
| 1 | 18      | 22       | 2,8,8                |
| 2 | 19      | 20       | 2,8,8                |
| 3 | 19      | 21       | 2,8,8,1              |
| 4 | 20      | 20       | 2,8,8,2              |

Which two particles belong to the same element?

- A 1 and 2      B 1 and 4      C 2 and 3      D 2 and 4

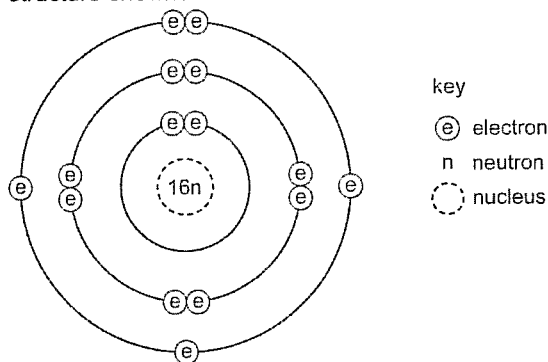
**72. O/N 14/P12/Q4, O/N 14/P11/Q4**

What is different for isotopes of the same element?

- A nucleon number      C number of electrons in the outer shell  
B number of electron shells      D proton number

**73. O/N 14/P12/Q8, O/N 14/P11/Q5, O/N 14/P13/Q5**

Which element has the atomic structure shown?



- A Al      B P      C S      D Si

**74. O/N 14/P13/Q4**

Which statement about a neutron is **not** correct?

- A It can be present in different numbers in atoms of the same element.  
B It has no electrical charge.  
C It is always found in the nucleus of an atom.  
D It weighs much less than a proton.

**75. O/N 14/P13/Q7**

Element X,  $^{19}_9\text{X}$ , forms a compound with element Y,  $^{39}_{19}\text{Y}$ .

Which statement describes the bonding in the compound formed?

- A X and Y share electrons. C Y gives away one electron to X.  
B X gives away one electron to Y. D Y gives away two electrons to X.

**76. O/N 14/P13/Q10**

An element, X, can be represented as  $^a_b\text{X}$ .

Which statement is correct?

- A The number of protons in an atom of X is a.  
B The exact position of X in the Periodic Table can be found from a.  
C The relative atomic mass of X is b.  
D The total number of electrons in one atom of X is b.

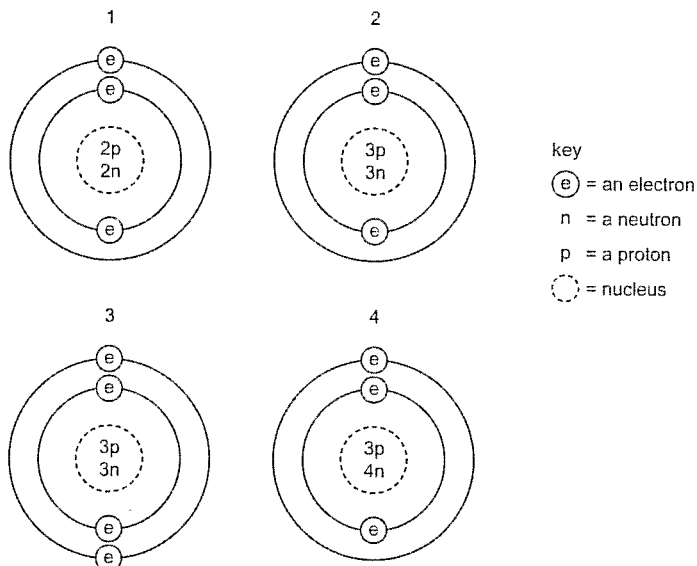
**77. M/J 14/P12/Q4**

Which statements about a phosphorus atom,  $^{31}_{15}\text{P}$ , are correct?

- 1 The nucleon number is 16.  
2 The number of outer electrons is 5.  
3 The proton number is 15.  
A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

**78. M/J 14/P12/Q5, M/J 14/P11/Q5, M/J 14/P13/Q6**

The diagrams show four particles.

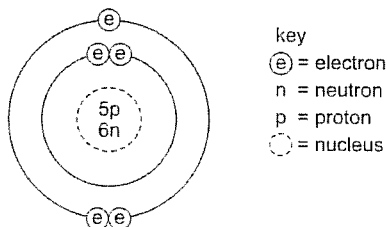


Which two diagrams show **atoms** that are isotopes of each other?

- A 1 and 2 B 1 and 3 C 2 and 3 D 2 and 4

**79. M/J 14/P11/Q4, M/J 14/P13/Q4**

The diagram shows the structure of an atom of element X.



What is X?

- A boron B carbon C sodium D sulfur

**80. O/N 13/P12/Q4, O/N 13/P11/Q5**

Which statements about a sodium atom,  $^{23}_{11}\text{Na}$ , are correct?

- 1 The number of protons and neutrons is the same.
- 2 The number of protons and electrons is the same.
- 3 The number of outer electrons is one.

A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

**81. O/N 13/P12/Q7, O/N 13/P11/Q4**

Element X has 7 protons.

Element Y has 8 more protons than X.

Which statement about element Y is correct?

- A Y has more electron shells than X.
- B Y has more electrons in its outer shell than X.
- C Y is in a different group of the Periodic Table from X.
- D Y is in the same period of the Periodic Table as X.

**82. O/N 13/P13/Q4**

The atomic structures of four atoms are shown.

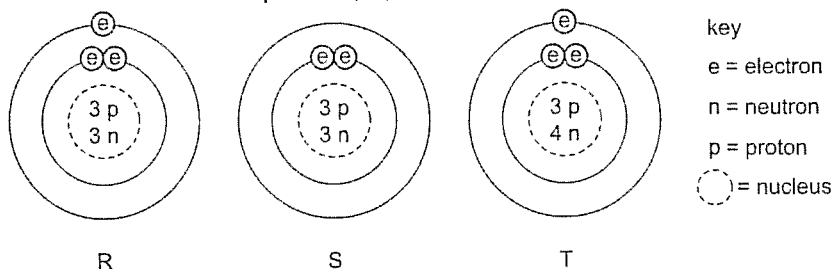
| atom | number of neutrons | number of protons | number of electrons |
|------|--------------------|-------------------|---------------------|
| W    | 6                  | 6                 | 6                   |
| X    | 7                  | 7                 | 7                   |
| Y    | 8                  | 6                 | 6                   |
| Z    | 8                  | 8                 | 8                   |

Which pair of atoms are isotopes?

- A W and X      B W and Y      C X and Y      D Y and Z

**83. O/N 13/P13/Q5**

The diagram shows the structure of three particles, R, S and T.



key  
e = electron  
n = neutron  
p = proton  
○ = nucleus

Which row describes these particles?

|   | ions    | isotopes |
|---|---------|----------|
| A | R       | S and T  |
| B | R and S | T        |
| C | S       | R and T  |
| D | T       | R and S  |

**84. M/J 13/P11/Q4, M/J 13/P13/Q5**

Element X is represented by  $^{27}_{13}\text{X}$ .

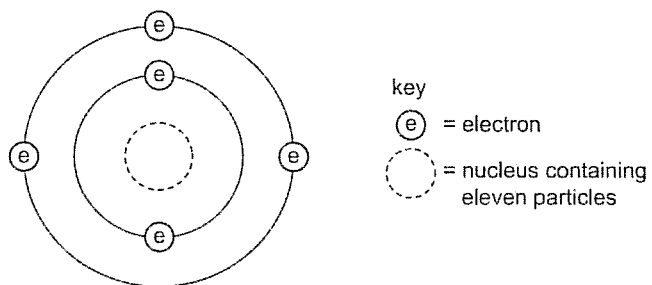
Which statement about element X is correct?

- A An atom of X contains 13 protons and 13 neutrons.
- B An atom of X contains 27 protons and 13 electrons.
- C X forms an ion by gaining electrons.
- D X is placed in Group III of the Periodic Table.



**85. M/J 13/P12/Q5**

The diagram shows an atom of an element.



How many protons and neutrons are in the nucleus of the atom and in which group and period of the Periodic Table is the element found?

|          | number of protons | number of neutrons | group number | period number |
|----------|-------------------|--------------------|--------------|---------------|
| <b>A</b> | 5                 | 6                  | 3            | 2             |
| <b>B</b> | 5                 | 11                 | 2            | 3             |
| <b>C</b> | 6                 | 5                  | 3            | 2             |
| <b>D</b> | 6                 | 11                 | 2            | 3             |

### 3.2: Ions and ionic bonds

1. O/N 22/P22/Q5

Which statement about solid magnesium oxide is correct?

- A It is a giant structure made up of magnesium and oxygen atoms bonded covalently.
- B It is an electrical conductor with mobile magnesium ions and oxygen ions.
- C Magnesium loses electrons and these electrons move freely through a lattice.
- D Oxygen ions and magnesium ions are attracted to each other in a giant lattice.

2. O/N 22/P21/Q6

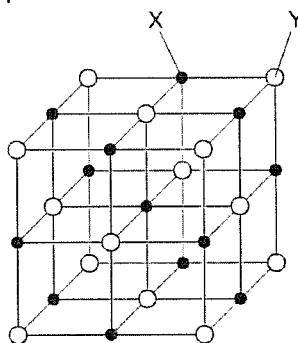
Which statements about potassium bromide are correct?

- 1 It has a high melting point.
- 2 It dissolves in water.
- 3 It conducts electricity when solid.

- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 3 only

3. O/N 22/P23/Q3

The structure of sodium chloride can be represented as shown.

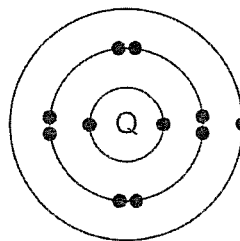
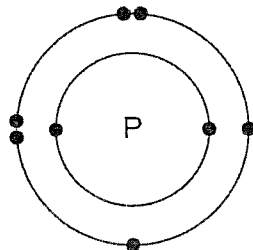


What are X and Y?

|   | X            | Y              |
|---|--------------|----------------|
| A | metal atom   | non-metal atom |
| B | negative ion | electron       |
| C | positive ion | negative ion   |
| D | positive ion | electron       |

4. MAR 22/P22/Q4

The electronic structures of atoms P and Q are shown.



P and Q form an ionic compound.

What is the formula of the compound?

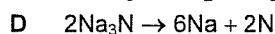
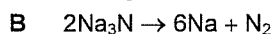
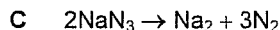
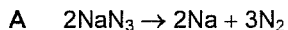
- A PQ
- B P<sub>2</sub>Q
- C P<sub>2</sub>Q<sub>3</sub>
- D PQ<sub>2</sub>

**5. MAR 22/P22/Q8**

Sodium nitride contains the nitride ion,  $\text{N}^{3-}$ .

Sodium nitride is unstable and decomposes into its elements.

What is the equation for the decomposition of sodium nitride?

**6. MAR 22/P22/Q10**

Caesium fluoride is an ionic compound.

Which statements about caesium fluoride are correct?

1 It conducts electricity when solid.

2 It has a high melting point.

3 It is soluble in water.

4 It is highly volatile.

A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

**7. O/N 21/P22/Q7**

Which statement describes the attractive forces between molecules?

A They are strong covalent bonds which hold molecules together.

B They are strong ionic bonds which hold molecules together.

C They are weak forces formed between covalently-bonded molecules.

D They are weak forces which hold ions together in a lattice.

**8. O/N 21/P23/Q9**

Aluminium oxide is an ionic compound containing  $\text{Al}^{3+}$  ions and  $\text{O}^{2-}$  ions.

Aluminium hydroxide is an ionic compound containing  $\text{Al}^{3+}$  ions and  $\text{OH}^-$  ions.

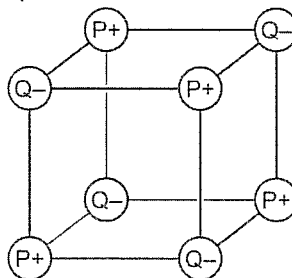
In which row are the formulae for aluminium oxide and aluminium hydroxide correct?

|   | aluminium oxide         | aluminium hydroxide      |
|---|-------------------------|--------------------------|
| A | $\text{Al}_2\text{O}_3$ | $\text{Al}(\text{OH})_3$ |
| B | $\text{Al}_3\text{O}_2$ | $\text{AlOH}_3$          |
| C | $\text{Al}_2\text{O}_3$ | $\text{AlOH}_3$          |
| D | $\text{Al}_3\text{O}_2$ | $\text{Al}(\text{OH})_3$ |

**9. M/J 21/P21/Q8**

Two elements, P and Q, are in the same period of the Periodic Table.

P and Q react together to form an ionic compound. Part of the lattice of this compound is shown.



Which statement is correct?

A An ion of P has more electrons than an ion of Q.

B Element P is non-metallic.

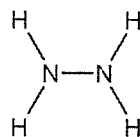
C P is to the left of Q in the Periodic Table.

D The formula of the compound is  $\text{P}_4\text{Q}_4$ .



**14. MAR 20/P22/Q6**

The molecular structure of hydrazine,  $\text{N}_2\text{H}_4$ , is shown.



Which description of the bonding in hydrazine is not correct?

- A Each nitrogen atom has a non-bonding pair of electrons.
- B Each nitrogen atom has four bonding pairs of electrons.
- C Each nitrogen atom shares one of its electrons with a nitrogen atom.
- D Each nitrogen atom shares two of its electrons with hydrogen atoms.

**15. MAR 20/P22/Q8**

The formulae of some ions are shown.

| positive ions    | negative ions      |
|------------------|--------------------|
| $\text{Al}^{3+}$ | $\text{Cl}^-$      |
| $\text{Fe}^{2+}$ | $\text{N}^{3-}$    |
| $\text{Mg}^{2+}$ | $\text{NO}_3^-$    |
| $\text{Na}^+$    | $\text{O}^{2-}$    |
| $\text{Zn}^{2+}$ | $\text{SO}_4^{2-}$ |

In which row is the formula not correct?

|   | compound         | formula                    |
|---|------------------|----------------------------|
| A | aluminium oxide  | $\text{Al}_2\text{O}_3$    |
| B | iron(II) nitride | $\text{Fe}_2\text{N}_3$    |
| C | sodium sulfate   | $\text{Na}_2\text{SO}_4$   |
| D | zinc nitrate     | $\text{Zn}(\text{NO}_3)_2$ |

**16. O/N 19/P22/Q4, O/N 19/P21/Q4, O/N 19/P23/Q4**

Which statement about an ionic compound is **not** correct?

- A It conducts electricity when dissolved in water.
- B It has a high melting point due to strong attractive forces between ions.
- C It has a regular lattice of oppositely charged ions in a 'sea of electrons'.
- D The ionic bonds are formed between metallic and non-metallic elements.

**17. O/N 19/P22/Q26, O/N 19/P21/Q26, O/N 19/P23/Q26**

Some properties of substance X are listed.

- It conducts electricity when molten.
- It has a high melting point.
- It burns in oxygen and the oxide dissolves in water to give a solution with pH 11.

What is X?

- |                       |                     |
|-----------------------|---------------------|
| A a covalent compound | C a metal           |
| B a macromolecule     | D an ionic compound |

**18. M/J 19/P22/Q6, M/J 19/P21/Q6, M/J 19/P23/Q6**

Which statement describes the structure of an ionic compound?

- A It is a giant lattice of oppositely charged ions.
- B It is a giant lattice of positive ions in a 'sea' of electrons.
- C It is a giant molecule of oppositely charged ions.
- D It is a simple molecule of oppositely charged ions.

**19. MAR 19/P22/Q6**

Which row describes the structure of the positive ion in sodium chloride?

|          | protons | electrons | neutrons |
|----------|---------|-----------|----------|
| <b>A</b> | 11      | 11        | 12       |
| <b>B</b> | 11      | 10        | 12       |
| <b>C</b> | 17      | 17        | 18       |
| <b>D</b> | 17      | 18        | 18       |

**20. O/N 18/P21/Q6**

Which statement describes the lattice structure of a metal?

- A** The lattice consists of alternating positive ions and negative ions.  
**B** The lattice consists of neutral atoms arranged in layers.  
**C** The lattice consists of positive ions in a 'sea of electrons'.  
**D** The lattice consists of neutral atoms in a 'sea of electrons'.

**21. MAR 18/P22/Q5**

The structure of copper is described as a lattice of positive ions in a 'sea of electrons'.

Which statements are correct?

- 1 Copper has a high melting point because of the strong electrostatic attraction between the positive ions and the 'sea of electrons'.  
 2 Copper is malleable because the layers of atoms in the lattice can slide over each other.  
 3 Copper atoms can be oxidised to form copper ions by losing electrons.

- A** 1, 2 and 3                      **B** 1 and 2 only                      **C** 1 and 3 only                      **D** 2 and 3 only

**22. M/J 17/P22/Q4**Which element does **not** form a stable ion with the same electronic structure as argon?

- A** aluminium                      **B** chlorine                      **C** phosphorus                      **D** potassium

**23. M/J 17/P21/Q4**

Sodium reacts with chlorine to form sodium chloride.

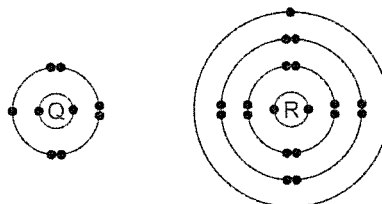
Which statements describe what happens to the sodium atoms in this reaction?

- 1 Sodium atoms form positive ions.  
 2 Sodium atoms form negative ions.  
 3 Sodium atoms gain electrons.  
 4 Sodium atoms lose electrons.

- A** 1 and 3                      **B** 1 and 4                      **C** 2 and 3                      **D** 2 and 4

**24. M/J 17/P23/Q4**

The electronic structures of atoms Q and R are shown.



Q and R form an ionic compound.

What is the formula of the compound?

- A**  $QR_7$                       **B**  $Q_2R_4$                       **C**  $QR$                       **D**  $Q_7R$

**25. O/N 16/P22/Q5**

In which ionic compound do the metal ion and the non-metal ion have the same electronic structure?

- A**  $CaO$                       **B**  $KBr$                       **C**  $MgO$                       **D**  $NaCl$

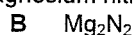
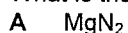
**26. M/J 16/P23/Q7**

Which substance exists as a lattice of positive ions in a 'sea of electrons'?

- A** liquid potassium chloride                      **C** solid magnesium  
**B** solid graphite                      **D** solid silicon(IV) oxide

**27. MAR 16/P22/Q9**

Magnesium nitride is formed when magnesium burns in air. Magnesium nitride is an ionic compound. What is the formula of magnesium nitride?

**28. M/J 15/P11/Q6**

Rescuers are drilling through fallen rock in order to rescue some men trapped in a cave. The drill needs lubricating from time to time.

The following statements were made about the materials used for the drill tip and the lubricant and the reasons for their use.

1 Diamond was used for the drill tip as it does not conduct electricity.

2 Diamond was used for the drill tip as it is very hard.

3 Graphite was used as the lubricant as it conducts electricity.

4 Graphite was used as the lubricant as it is soft and flaky.

Which statements are correct?

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4

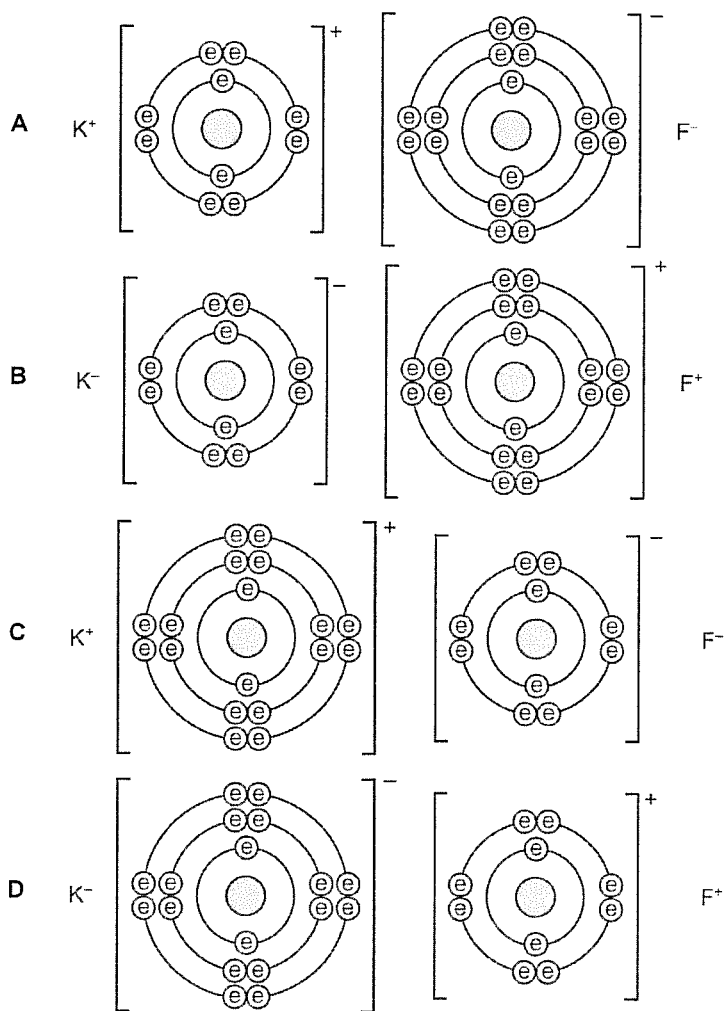
**29. O/N 15/P11/Q4**

Which diagram correctly shows the ions present in the compound potassium fluoride?

key

 = nucleus

 = electron



**30. O/N 15/P11/Q7**

Two atoms of magnesium, Mg, react with one molecule of oxygen, O<sub>2</sub>.  
What is the formula of the product?

- A MgO                      B MgO<sub>2</sub>                      C Mg<sub>2</sub>O                      D Mg<sub>2</sub>O<sub>2</sub>

**31. MAR 15/P12/Q3**

Which statement about bonding is **not** correct?

- A Carbon can form four single covalent bonds.  
B Chlorine atoms react to gain a noble gas electronic structure.  
C Covalent bonding involves losing and gaining electrons.  
D Hydrogen molecules have the formula H<sub>2</sub>.

**32. MAR 15/P12/Q5**

Which substance is an ionic compound?

|   | volatility | electrical conductivity<br>when molten | solubility in water |
|---|------------|----------------------------------------|---------------------|
| A | high       | good                                   | soluble             |
| B | high       | poor                                   | insoluble           |
| C | low        | good                                   | soluble             |
| D | low        | poor                                   | insoluble           |

**33. MAR 15/P12/Q6**

Covalent bonds are formed when electrons are .....1.....  
Most covalent compounds have .....2..... electrical conductivity.  
Which words correctly complete gaps 1 and 2?

|   | 1           | 2    |
|---|-------------|------|
| A | shared      | high |
| B | shared      | low  |
| C | transferred | high |
| D | transferred | low  |

**34. O/N 14/P12/Q6, O/N 14/P11/Q7**

Sodium chloride is an ionic solid.  
Which statement is **not** correct?

- A Ions are formed when atoms lose or gain electrons.  
B Ions in sodium chloride are strongly held together.  
C Ions with the same charge attract each other.  
D Sodium chloride solution can conduct electricity.

**35. O/N 14/P12/Q7, O/N 14/P11/Q8**

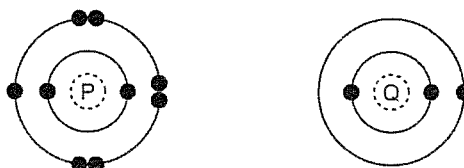
Caesium chloride and rubidium bromide are halide compounds of Group I elements.  
Caesium chloride has the formula .....1....., a relative formula mass .....2..... that of rubidium bromide and bonds that are .....3.....  
Which words correctly complete gaps 1, 2 and 3?

|   | 1    | 2              | 3        |
|---|------|----------------|----------|
| A | CaCl | different from | ionic    |
| B | CaCl | the same as    | covalent |
| C | CsCl | different from | ionic    |
| D | CsCl | the same as    | covalent |



**36. O/N 13/P13/Q7**

The electronic structures of atoms P and Q are shown.



P and Q react to form an ionic compound.  
What is the formula of the compound?

A  $\text{Q}_7\text{P}$

B  $\text{QP}$

C  $\text{QP}_3$

D  $\text{QP}_7$

**3.3: Molecules and covalent bonds****1. O/N 22/P22/Q6**

Which molecule contains only three shared pairs of electrons?

A  $\text{CH}_3\text{OH}$ B  $\text{Cl}_2$ C  $\text{H}_2\text{O}$ D  $\text{N}_2$ **2. O/N 22/P21/Q7**

Which substance has a similar structure to silicon(IV) oxide?

A carbon dioxide

B diamond

C graphite

D sodium oxide

**3. O/N 22/P23/Q6**What is the total number of shared electrons in a molecule of methanol,  $\text{CH}_3\text{OH}$ ?

A 4

B 5

C 8

D 10

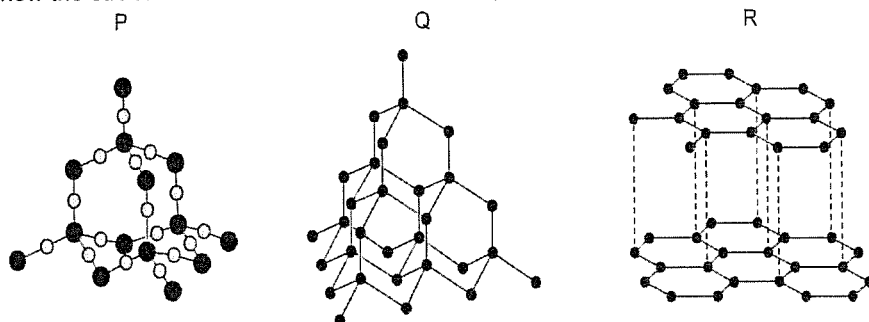
**4. O/N 22/P23/Q7**

Which row about the structures and uses of diamond and graphite is correct?

|   | structure                                | use                                    |
|---|------------------------------------------|----------------------------------------|
| A | diamond has a giant covalent structure   | diamond is used to make electrodes     |
| B | diamond has a simple covalent structure  | diamond is used to make cutting tools  |
| C | graphite has a giant covalent structure  | graphite is used as a lubricant        |
| D | graphite has a simple covalent structure | graphite is used to make cutting tools |

**5. M/J 22/P22/Q5**

The diagrams show the structures of three macromolecules P, Q and R.



What are P, Q and R?

|   | P                 | Q                 | R                 |
|---|-------------------|-------------------|-------------------|
| A | diamond           | silicon(IV) oxide | graphite          |
| B | graphite          | diamond           | silicon(IV) oxide |
| C | silicon(IV) oxide | diamond           | graphite          |
| D | silicon(IV) oxide | graphite          | diamond           |

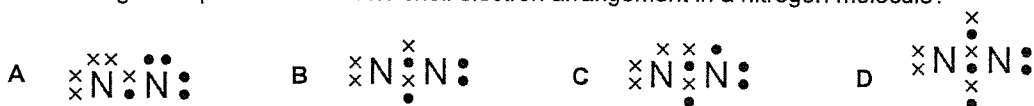
**6. M/J 22/P22/Q6**

Which dot-and-cross diagram shows the arrangement of outer shell electrons in a molecule of hydrogen chloride?



## 7. M/J 22/P21/Q6

Which diagram represents the outer-shell electron arrangement in a nitrogen molecule?



## 8. M/J 22/P23/Q3

Which statement describes the properties of both diamond and silicon(IV) oxide?

- A They are brittle, with a low melting point, and are insoluble in water.  
 B They are hard, with a high melting point, and are electrical insulators.  
 C They are malleable, with a high melting point, and are electrical conductors.  
 D They are soft, with a low melting point, and are electrical insulators.

## 9. M/J 22/P23/Q23

Selenium is an element in Group VI.

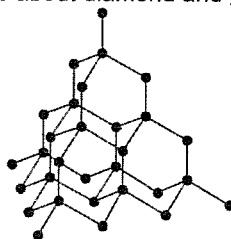
Group VI elements follow similar trends to Group VII elements.

Which statement about selenium is correct?

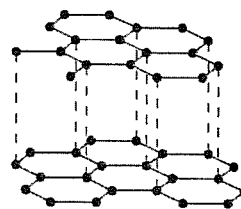
- A It has a higher density than sulfur.      C It has six electron shells.  
 B It has a lower melting point than sulfur.      D It is a monoatomic element.

## 10. MAR 22/P22/Q7

Which pair of statements about diamond and graphite is correct?



diamond



graphite

- A Diamond and graphite are both pure carbon. They are both macromolecules.  
 B Diamond and graphite can both be used as electrodes. Graphite is also used as a lubricant.  
 C Diamond has covalent bonds. Graphite has ionic bonds.  
 D Diamond is hard with a high melting point. Graphite is soft with a low melting point.

## 11. O/N 21/P22/Q6

Some properties of diamond are shown.

- 1 It is very hard.  
 2 Every atom forms four bonds.  
 3 It does not conduct electricity.

Which properties are also shown by silicon(IV) oxide?

- A 1 only      B 1 and 2      C 1 and 3      D 2 and 3

## 12. O/N 21/P21/Q7, O/N 21/P23/Q7

Which statement describes the attractive forces between molecules?

- A They are strong covalent bonds which hold molecules together.  
 B They are strong ionic bonds which hold molecules together.  
 C They are weak forces formed between covalently-bonded molecules.  
 D They are weak forces which hold ions together in a lattice.

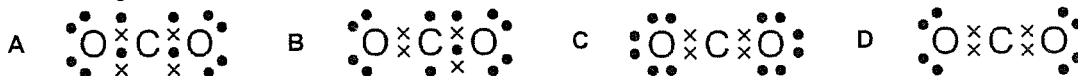
## 13. O/N 21/P21/Q35

Which molecule contains only single covalent bonds?

- A propane      B propanoic acid      C propene      D propyl propanoate

## 14. O/N 21/P23/Q8

Which diagram shows the outer electron arrangement in a molecule of carbon dioxide?



## 15. M/J 21/P22/Q5, M/J 21/P21/Q5, M/J 21/P23/Q5

A covalent molecule Q contains only six shared electrons.

What is Q?

- A ammonia,  $\text{NH}_3$       B chlorine,  $\text{Cl}_2$       C methane,  $\text{CH}_4$       D water,  $\text{H}_2\text{O}$

## 16. M/J 21/P22/Q6

Information about four substances E, F, G and H is shown.

|   | melting point/ $^{\circ}\text{C}$ | electrical conductivity     |
|---|-----------------------------------|-----------------------------|
| E | 1710                              | does not conduct when solid |
| F | 3500                              | conducts when solid         |
| G | 120                               | does not conduct            |
| H | 801                               | conducts when molten        |

E, F, G and H are graphite, poly(ethene), sodium chloride and silicon(IV) oxide but not in that order.  
What are E, F, G and H?

|   | E                 | F               | G                 | H                 |
|---|-------------------|-----------------|-------------------|-------------------|
| A | graphite          | poly(ethene)    | silicon(IV) oxide | sodium chloride   |
| B | sodium chloride   | graphite        | poly(ethene)      | silicon(IV) oxide |
| C | poly(ethene)      | sodium chloride | graphite          | silicon(IV) oxide |
| D | silicon(IV) oxide | graphite        | poly(ethene)      | sodium chloride   |

## 17. M/J 21/P22/Q9

Chlorine gas will react with iron metal.

Exactly 21.3 g of chlorine reacts with 11.2 g of iron.

How many iron atoms react with 30 molecules of chlorine?

- A 10      B 15      C 20      D 30

## 18. M/J 21/P21/Q24

Element R forms a covalent compound  $\text{R}_2\text{Si}$  with silicon.

Which row describes R?

|   | metallic or non-metallic character | group number in the Periodic Table |
|---|------------------------------------|------------------------------------|
| A | metallic                           | II                                 |
| B | metallic                           | VI                                 |
| C | non-metallic                       | II                                 |
| D | non-metallic                       | VI                                 |

## 19. M/J 21/P23/Q8

The element silicon has the same structure as diamond.

Which statement about silicon is correct?

- A Every silicon atom is bonded to three other atoms only.  
B Silicon has a high melting point.  
C Silicon is a good conductor of electricity.  
D Silicon is used as a lubricant.

**20. M/J 21/P23/Q20**

Information about element J is shown.

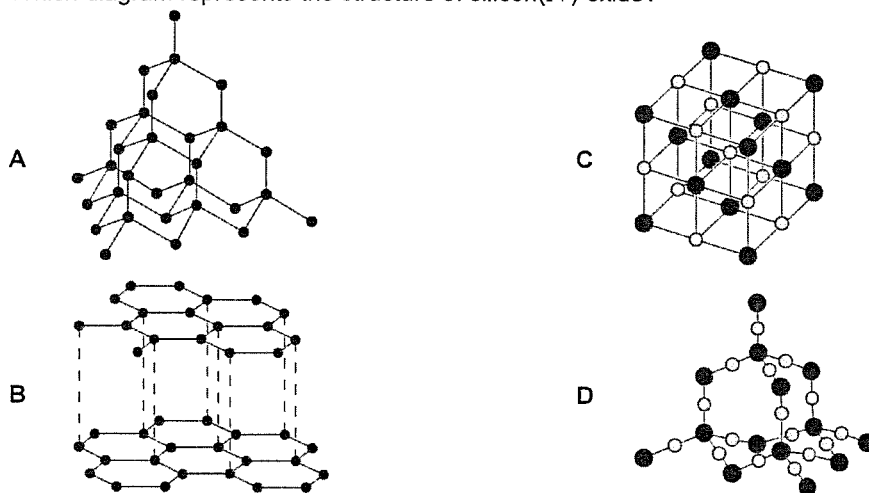
- Its atoms have four electrons in their outer shell.
- It is a non-metal.
- Its oxide has a macromolecular structure.
- It has a high melting point.

What is J?

- A beryllium                      B carbon                      C silicon                      D sulfur

**21. MAR 21/P22/Q5**

Which diagram represents the structure of silicon(IV) oxide?

**22. MAR 21/P22/Q8**

Magnesium oxide has a high melting point.

Carbon dioxide has a low melting point.

Which row identifies the attractive forces that are broken when these compounds are melted?

|   | magnesium oxide                      | carbon dioxide                     |
|---|--------------------------------------|------------------------------------|
| A | strong attractions between molecules | weak attractions between atoms     |
| B | strong attractions between molecules | weak attractions between molecules |
| C | strong attractions between ions      | weak attractions between atoms     |
| D | strong attractions between ions      | weak attractions between molecules |

**23. MAR 21/P22/Q22**

Metal X reacts with non-metal Y to form an ionic compound with the formula  $X_2Y$ .

Which statements are correct?

- 1 X is in Group I of the Periodic Table.
- 2 X is in Group II of the Periodic Table.
- 3 Y is in Group VI of the Periodic Table.
- 4 Y is in Group VII of the Periodic Table.

- A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**24. O/N 20/P22/Q10, O/N 20/P21/Q10, O/N 20/P23/Q10**

Which statement explains why graphite is used as a lubricant?

- A All bonds between the atoms are weak.  
 B It conducts electricity.  
 C It has a low melting point.  
 D Layers in the structure can slide over each other.

**25. MAR 21/P22/Q23**

The table gives some properties of Group IV elements.

| element   | density<br>g/cm <sup>3</sup> | boiling point<br>/°C |
|-----------|------------------------------|----------------------|
| carbon    | 2.2                          | 4827                 |
| silicon   |                              |                      |
| germanium | 5.3                          | 2830                 |
| tin       | 5.8                          | 2270                 |
| lead      | 11.3                         | 1755                 |

Which row describes the properties of silicon?

|          | density<br>g/cm <sup>3</sup> | boiling point<br>/°C |
|----------|------------------------------|----------------------|
| <b>A</b> | 2.3                          | 3 265                |
| <b>B</b> | 3.1                          | 1 997                |
| <b>C</b> | 6.2                          | 2 920                |
| <b>D</b> | 24.6                         | 11 682               |

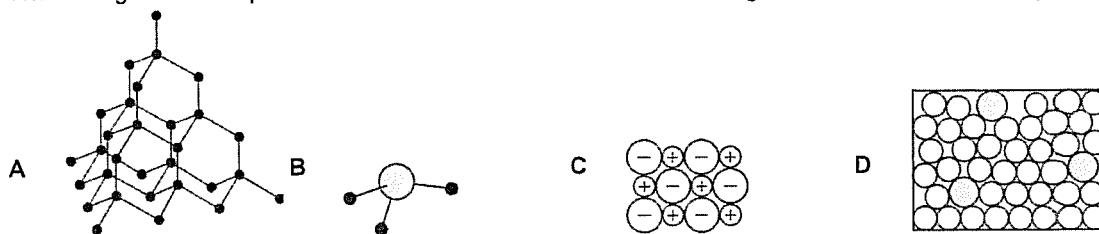
**26. O/N 20/P22/Q7**

Which row identifies compounds that contain single covalent bonds only, double covalent bonds only or both single and double covalent bonds?

|          | single covalent bonds<br>only | double covalent bonds<br>only | both single and<br>double covalent bonds |
|----------|-------------------------------|-------------------------------|------------------------------------------|
| <b>A</b> | C <sub>2</sub> H <sub>4</sub> | CH <sub>3</sub> OH            | CO <sub>2</sub>                          |
| <b>B</b> | CH <sub>3</sub> OH            | C <sub>2</sub> H <sub>4</sub> | CO <sub>2</sub>                          |
| <b>C</b> | CH <sub>3</sub> OH            | CO <sub>2</sub>               | C <sub>2</sub> H <sub>4</sub>            |
| <b>D</b> | CO <sub>2</sub>               | C <sub>2</sub> H <sub>4</sub> | CH <sub>3</sub> OH                       |

**27. O/N 20/P22/Q29**

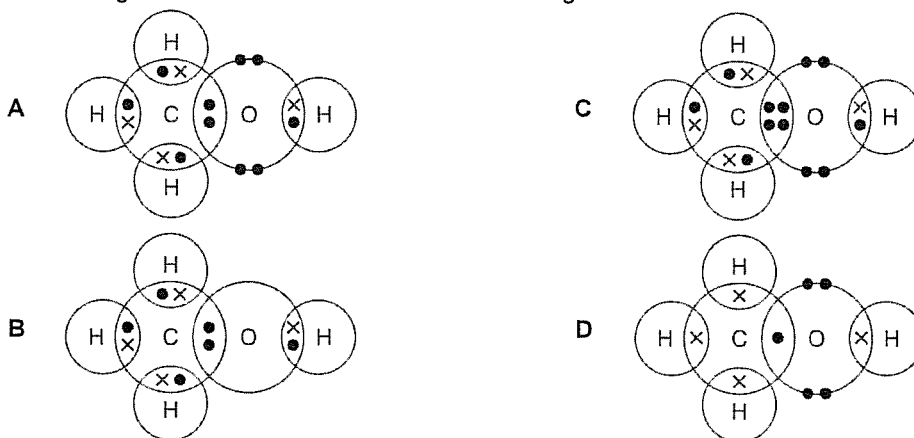
Which diagram best represents the structure of a substance that is a good conductor of electricity at 25 °C?

**28. O/N 20/P21/Q8**

Which statement about silicon dioxide, SiO<sub>2</sub>, is correct?

- A** It conducts electricity because it contains free electrons
- B** It is a macromolecule with four oxygen atoms bonded to each silicon atom.
- C** It is a simple covalent molecule.
- D** Its structure is similar to graphite.

## 29. O/N 20/P21/Q7

Which diagram shows the outer shell electron arrangement in a molecule of methanol,  $\text{CH}_3\text{OH}$ ?

## 30. O/N 20/P23/Q5

Molecules containing only non-metal atoms are covalently bonded.

The formulae of four covalently bonded molecules are given below:

- 1 nitrogen,  $\text{N}_2$
- 2 carbon dioxide,  $\text{CO}_2$
- 3 ethene,  $\text{C}_2\text{H}_4$
- 4 methanol,  $\text{CH}_3\text{OH}$

Which of the molecules contain double bonds?

- A 1 and 4                      B 2 and 3                      C 2 and 4                      D 4 only

## 31. O/N 19/P22/Q7

Which statement about the structure and properties of silicon(IV) oxide is **not** correct?

- A It has a giant structure similar to that of diamond.
- B It has a high melting point due to the strong attractive force between molecules.
- C There are strong covalent bonds between silicon and oxygen.
- D There are no free electrons, so silicon(IV) oxide does not conduct electricity.

## 32. O/N 19/P23/Q7

Silicon(IV) oxide is a covalently bonded compound.

Which statements are correct?

- 1 Silicon atoms form four single bonds in silicon(IV) oxide.
- 2 Oxygen atoms form two double bonds in silicon(IV) oxide.
- 3 Silicon(IV) oxide has a high melting point.
- 4 Silicon(IV) oxide contains one silicon atom and four oxygen atoms.

- A 1 and 2 only                      B 1 and 3 only                      C 2 and 3 only                      D 3 and 4 only

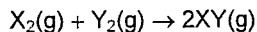
## 33. M/J 19/P22/Q5, M/J 19/P21/Q5, M/J 19/P23/Q5

Which row describes the formation of single covalent bonds in methane?

|   |                                                    |                                                                   |
|---|----------------------------------------------------|-------------------------------------------------------------------|
| A | atoms share a pair of electrons                    | both atoms gain a noble gas electronic structure                  |
| B | atoms share a pair of electrons                    | both atoms have the same number of electrons in their outer shell |
| C | electrons are transferred from one atom to another | both atoms gain a noble gas electronic structure                  |
| D | electrons are transferred from one atom to another | both atoms have the same number of electrons in their outer shell |

**34. M/J 19/P22/Q12**

Two elements, X and Y, react together to form a covalent molecule as shown.  
The reaction is exothermic.



The bond energies are shown in the table.

| bond | bond energy in kJ/mol |
|------|-----------------------|
| X–X  | 436                   |
| Y–Y  | 242                   |
| X–Y  | 431                   |

What is the energy change for the reaction?

- A +184 kJ/mol      B –184 kJ/mol      C +247 kJ/mol      D –247 kJ/mol

**35. MAR 19/P22/Q7**

Which statement about copper, diamond and silicon(IV) oxide is correct?

- A Copper and silicon(IV) oxide have similar electrical conductivity.  
B In diamond the carbon atoms are covalently bonded as flat sheets.  
C In silicon(IV) oxide the silicon and oxygen atoms are covalently bonded as flat sheets.  
D The structure of copper includes a lattice of positive ions.

**36. O/N 18/P22/Q5**

How many electrons are used to form covalent bonds in a molecule of methanol,  $\text{CH}_3\text{OH}$ ?

- A 5      B 6      C 8      D 10

**37. O/N 18/P22/Q6**

Potassium bromide and methanol are both compounds.

Their melting points are different.

Which row is correct?

|   | substance with the higher melting point | reason why the melting points are different                                                                   |
|---|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| A | methanol                                | the attractive forces between oppositely charged ions is greater than the attractive forces between molecules |
| B | methanol                                | the attractive forces between molecules is greater than the attractive forces between oppositely charged ions |
| C | potassium bromide                       | the attractive forces between oppositely charged ions is greater than the attractive forces between molecules |
| D | potassium bromide                       | the attractive forces between molecules is greater than the attractive forces between oppositely charged ions |

**38. O/N 18/P23/Q5**

Which covalent molecule contains two atoms bonded together by exactly four shared electrons?

- A  $\text{N}_2$       B  $\text{C}_3\text{H}_8$       C  $\text{CH}_3\text{OH}$       D  $\text{CH}_3\text{COOH}$

**39. MAR 18/P22/Q6**

Three statements about diamond, graphite and silicon(IV) oxide are listed.

- 1 Diamond and graphite both have giant covalent structures.
- 2 In silicon(IV) oxide, silicon and oxygen atoms are joined together by covalent bonds throughout the whole structure.
- 3 Diamond and silicon(IV) oxide have similar structures.

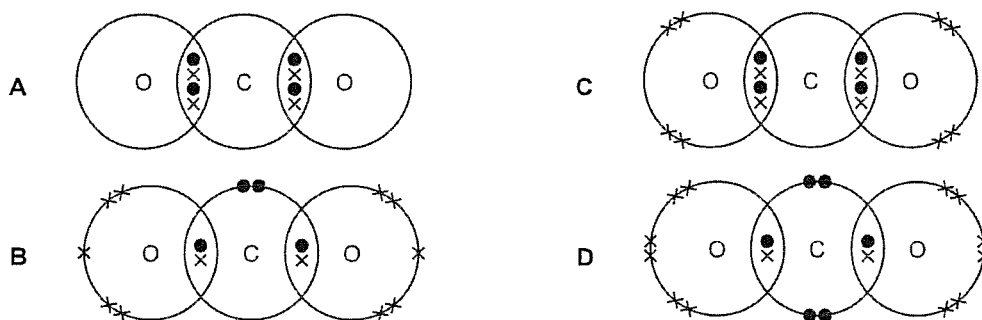
Which statements are correct?

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only



## 40. O/N 17/P22/Q6

Which dot-and-cross diagram shows the outer shell electron arrangement in a molecule of carbon dioxide?



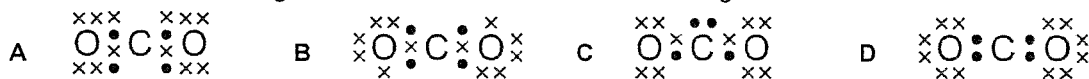
## 41. O/N 17/P21/Q6

Which dot-and-cross diagram shows the outer shell electron arrangement in a molecule of nitrogen?



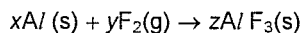
## 42. O/N 17/P23/Q6

Which dot-and-cross diagram shows the outer shell electron arrangement in a molecule of carbon dioxide?



## 43. M/J 17/P21/Q7

Aluminium reacts with fluorine.



Which values of x, y and z balance the equation?

|   | x | y | z |
|---|---|---|---|
| A | 1 | 2 | 1 |
| B | 2 | 3 | 2 |
| C | 3 | 2 | 3 |
| D | 4 | 3 | 4 |

## 44. O/N 16/P21/Q5

Metal P reacts with non-metal Q to form a compound.

Which process takes place and which type of compound is formed?

|   | process                               | type of compound |
|---|---------------------------------------|------------------|
| A | electrons are transferred from P to Q | covalent         |
| B | electrons are transferred from P to Q | ionic            |
| C | electrons are transferred from Q to P | covalent         |
| D | electrons are transferred from Q to P | ionic            |

**45. O/N 16/P23/Q5**

Ethanol is a liquid at room temperature and boils at 78 °C.

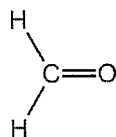
Sodium chloride is a solid at room temperature.

Which statement about the bonding in ethanol and sodium chloride is **not** correct?

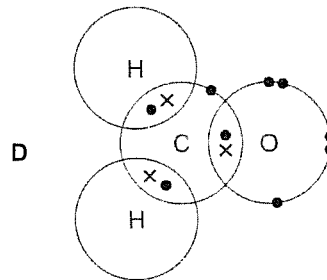
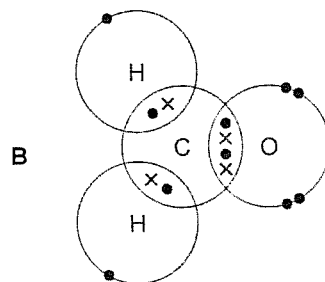
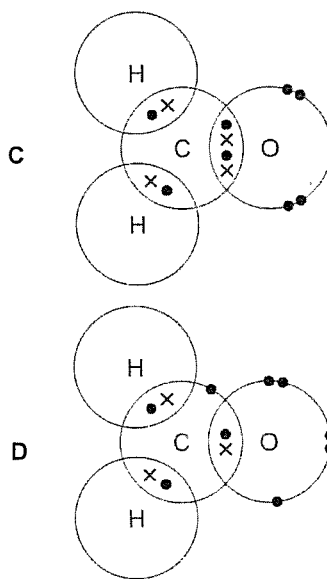
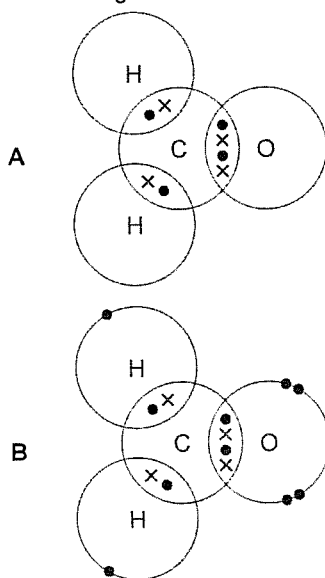
- A Each ethanol molecule is held together by weak covalent bonds.
- B The ethanol molecules are held together by weak attractive forces.
- C The sodium ions and chloride ions are held together by strong attractive forces.
- D The sodium ions and chloride ions are held together in a giant lattice.

**46. O/N 16/P22/Q6**

The structure of methanal is shown.



Which diagram shows the arrangement of outer shell electrons in a molecule of methanal?

**47. O/N 16/P23/Q6**

The molecules N<sub>2</sub>, C<sub>2</sub>H<sub>4</sub>, CO<sub>2</sub> and CH<sub>3</sub>OH all have covalent bonds.

These bonds consist of shared pairs of electrons.

Which row gives the total number of shared pairs of electrons in the molecules shown?

|          | molecule                      | total number of shared pairs of electrons |
|----------|-------------------------------|-------------------------------------------|
| <b>A</b> | N <sub>2</sub>                | 2                                         |
| <b>B</b> | C <sub>2</sub> H <sub>4</sub> | 6                                         |
| <b>C</b> | CO <sub>2</sub>               | 2                                         |
| <b>D</b> | CH <sub>3</sub> OH            | 4                                         |

**48. M/J 16/P22/Q5, M/J 16/P21/Q5, M/J 16/P23/Q5**

The table shows the electronic structure of four atoms.

| atom | electronic structure |
|------|----------------------|
| W    | 2,8,1                |
| X    | 2,8,4                |
| Y    | 2,8,7                |
| Z    | 2,8,8                |

Which two atoms combine to form a covalent compound?

- A W and X                      B W and Y                      C X and Y                      D X and Z

**49. M/J 16/P22/Q6, M/J 16/P21/Q6, M/J 16/P23/Q6**

Which statement describes the attractive forces between molecules (intermolecular forces)?

- A They are strong covalent bonds which hold molecules together.  
 B They are strong ionic bonds which hold molecules together.  
 C They are weak forces formed between covalently-bonded molecules.  
 D They are weak forces which hold ions together in a lattice.

**50. MAR 16/P22/Q5**

Which molecule contains only single covalent bonds?

- A  $\text{Cl}_2$                       B  $\text{CO}_2$                       C  $\text{N}_2$                       D  $\text{O}_2$

**51. O/N 15/P12/Q4**

Compound X melts at  $801^\circ\text{C}$  and is a good electrical conductor when dissolved in water.

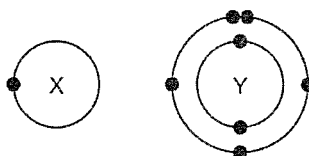
Compound Y boils at  $77^\circ\text{C}$ , is insoluble in water and is a non-conductor of electricity.

Which type of bonding is present in X and in Y?

|   | X        | Y        |
|---|----------|----------|
| A | covalent | covalent |
| B | covalent | ionic    |
| C | ionic    | covalent |
| D | ionic    | ionic    |

**52. O/N 15/P12/Q6, O/N 15/P11/Q6, O/N 15/P13/Q6**

The electronic structures of atoms X and Y are shown.



X and Y form a covalent compound.

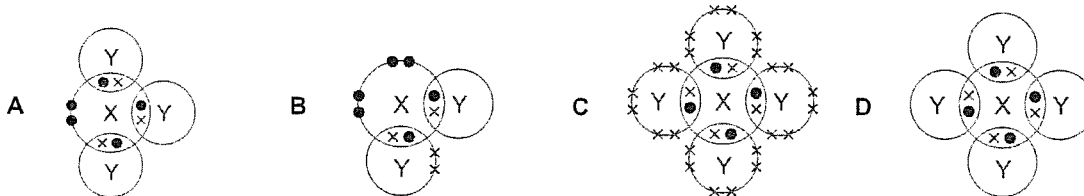
What is its formula?

- A  $\text{XY}_5$                       B  $\text{XY}_3$                       C  $\text{XY}$                       D  $\text{X}_3\text{Y}$

**53. O/N 15/P13/Q4**

In the following diagrams, X and Y are atoms of different elements.

Which diagram correctly shows the arrangement of outer electrons in a molecule of methane?



**54. M/J 15/P12/Q7, M/J 15/P11/Q7, M/J 15/P13/Q7, M/J 10/P12/Q10, M/J 10/P11/Q9, M/J 10/P13/Q10**

In which compounds are pairs of electrons shared between atoms?

- 1 methane  
2 lead bromide  
3 sodium chloride

A 1 only                      B 2 only                      C 1 and 3                      D 1, 2 and 3

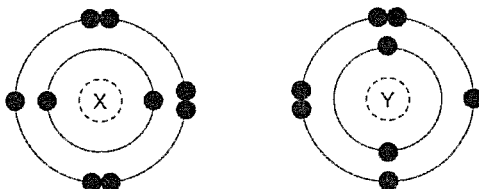
**55. M/J 15/P12/Q8**

A molecule, Z, contains two atoms of oxygen, six atoms of hydrogen and three atoms of carbon.  
What is the formula of Z?

- A  $\text{CH}_3\text{CH}_2\text{CHO}$                       B  $\text{CH}_3\text{COCH}_3$                       C  $\text{C}_2\text{H}_5\text{CO}_2\text{H}$                       D  $\text{C}_3\text{H}_6\text{CO}_2\text{H}$

**56. M/J 14/P12/Q7**

The electronic structures of two atoms, X and Y, are shown.



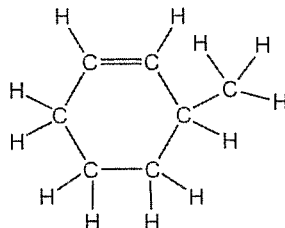
X and Y combine together to form a compound.

What is the type of bonding in the compound and what is the formula of the compound?

|          | type of bonding | formula              |
|----------|-----------------|----------------------|
| <b>A</b> | covalent        | $\text{X}_2\text{Y}$ |
| <b>B</b> | covalent        | $\text{XY}_2$        |
| <b>C</b> | ionic           | $\text{XY}_2$        |
| <b>D</b> | ionic           | $\text{X}_2\text{Y}$ |

**57. M/J 14/P12/Q8**

The structure of an organic compound, X, is shown.

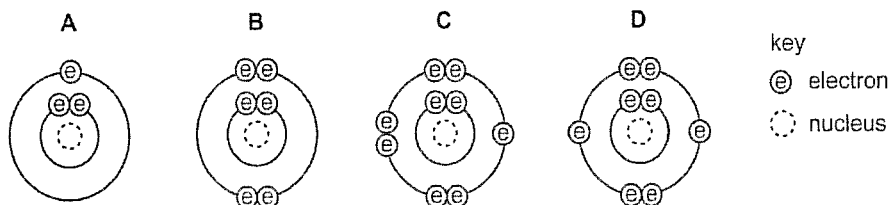


What is the molecular formula of X?

- A  $\text{C}_6\text{H}_9$                       B  $\text{C}_6\text{H}_{12}$                       C  $\text{C}_7\text{H}_{12}$                       D  $\text{C}_7\text{H}_{14}$

**58. O/N 13/P12/Q5, O/N 13/P11/Q7**

The diagrams show the electron arrangements in the atoms of four elements.

Which element does **not** form a covalent bond?**59. M/J 13/P12/Q6**

Electrons from each element are shared by both of the elements in a compound.

Which compound matches this description?

- A lead bromide                      B sodium chloride                      C water                      D zinc oxide

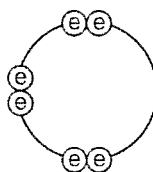
## 60. M/J 13/P11/Q6, M/J 13/P13/Q7

For which substance is the type of bonding **not** correct?

|   | substance         | type of bonding |          |          |
|---|-------------------|-----------------|----------|----------|
|   |                   | ionic           | covalent | metallic |
| A | chlorine          |                 | ✓        |          |
| B | potassium bromide | ✓               |          |          |
| C | sodium            |                 |          | ✓        |
| D | sodium chloride   |                 | ✓        |          |

## 61. M/J 13/P12/Q8, M/J 13/P11/Q7, M/J 13/P13/Q6, M/J 10/P12/Q11, M/J 10/P11/Q8, M/J 10/P13/Q11

Element X has six electrons in its outer shell.



key

ⓔ = electron

How could the element react?

- A by gaining two electrons to form a positive ion
- B by losing six electrons to form a negative ion
- C by sharing two electrons with two electrons from another element to form two covalent bonds
- D by sharing two electrons with two electrons from another element to form four covalent bonds

### 3.4: Macromolecules

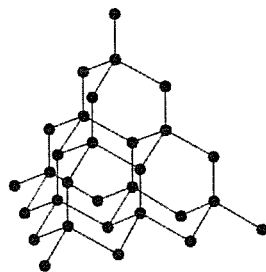
1. O/N 21/P22/Q8

Which substance is described as a macromolecule?

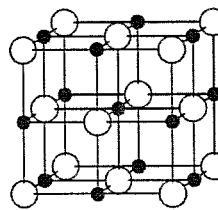
- A ammonia                      B graphite                      C iron                      D sodium chloride

2. M/J 21/P21/Q6

The arrangement of particles in each of two solids, S and T, are shown.



S



T

What are S and T?

|   | S        | T                 |
|---|----------|-------------------|
| A | diamond  | silicon(IV) oxide |
| B | diamond  | sodium chloride   |
| C | graphite | silicon(IV) oxide |
| D | graphite | sodium chloride   |

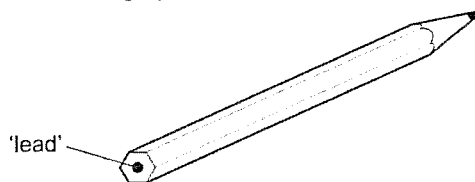
3. O/N 19/P21/Q7

How are the structures of diamond and silicon(IV) oxide similar?

- A Molecules of both diamond and silicon(IV) oxide are held together by weak attractive forces.  
 B They both contain atoms arranged in planes held together by weak bonds.  
 C They both contain ions that are free to move.  
 D The carbon in diamond and the silicon in silicon(IV) oxide each have four covalent bonds.

4. M/J 18/P22/Q4, M/J 18/P21/Q4, M/J 18/P23/Q4, M/J 14/P12/Q6, M/J 14/P11/Q6, M/J 14/P13/Q5

The 'lead' in a pencil is made of a mixture of graphite and clay.



When the percentage of graphite is increased, the pencil slides across the paper more easily.

Which statement explains this observation?

- A Graphite has a high melting point.                      C Graphite is a lubricant.  
 B Graphite is a form of carbon.                      D Graphite is a non-metal.

5. M/J 18/P22/Q6

Which substances have similar structures?

- A diamond and graphite                      C graphite and poly(ethene)  
 B diamond and silicon(IV) oxide                      D graphite and silicon(IV) oxide

6. M/J 18/P22/Q7, M/J 18/P21/Q6, M/J 18/P23/Q7

Which substance is **not** a macromolecule?

- A diamond                      B graphite                      C silicon(IV) oxide                      D sulfur

**7. M/J 18/P23/Q6**

How many silicon atoms are bonded to each oxygen atom in a crystal of silicon(IV) oxide?

- A 1                                      B 2                                      C 3                                      D 4

**8. O/N 17/P23/Q4**

Which compound is silicon(IV) oxide?

|   | melting point<br>/°C | good electrical<br>conductor when solid | good electrical<br>conductor when molten |
|---|----------------------|-----------------------------------------|------------------------------------------|
| A | -73                  | no                                      | no                                       |
| B | 801                  | no                                      | yes                                      |
| C | 1495                 | yes                                     | yes                                      |
| D | 1710                 | no                                      | no                                       |

**9. M/J 17/P22/Q5**

Graphite and diamond are both forms of the element carbon.

Which row shows the number of other carbon atoms that each carbon atom is covalently bonded to in graphite and diamond?

|   | graphite | diamond |
|---|----------|---------|
| A | 3        | 3       |
| B | 3        | 4       |
| C | 4        | 3       |
| D | 4        | 4       |

**10. M/J 17/P21/Q5**

Diamond is extremely hard and does not conduct electricity.

Which statement explains these properties?

- A It has a lattice of positive carbon ions in a 'sea of electrons'.  
 B It has delocalised electrons and each carbon atom forms three covalent bonds with other carbon atoms.  
 C It has no delocalised electrons and each carbon atom forms four covalent bonds with other carbon atoms.  
 D It has strong ionic bonds between each carbon atom.

**11. M/J 17/P23/Q5**

Which substance is a macromolecule?

- A ammonia                              B carbon dioxide                              C diamond                              D water

**12. MAR 17/P22/Q6**

Diamond and silicon(IV) oxide both have giant structures.

Which statements are correct?

- 1 Both substances are compounds.  
 2 There are strong covalent bonds in diamond.  
 3 Silicon(IV) oxide is bonded ionically.  
 4 Both substances have very high melting points.

- A 1 and 2                              B 2 and 3                              C 2 and 4                              D 3 and 4

**13. M/J 16/P23/Q25**

Some properties of substance X are listed.

- It conducts electricity when molten.
- It has a high melting point.
- It burns in oxygen and the product dissolves in water to give a solution with pH 11.

What is X?

- A a covalent compound                              C a metal  
 B a macromolecule                              D an ionic compound

**14. M/J 15/P12/Q6**

Graphite is a form of carbon.

Why can graphite be used as a lubricant?

- A Graphite contains delocalised electrons which move throughout the structure.
- B Graphite contains weak covalent bonds so the atoms move easily.
- C Graphite has a low melting point so it easily turns into a liquid.
- D Graphite has weak forces of attraction between layers so they can move.

**15. M/J 15/P13/Q6**

Diamond and graphite are both macromolecules.

Which statement is **not** correct?

- A Diamond and graphite contain carbon atoms only.
- B Diamond and graphite contain charged ions.
- C Diamond and graphite have high melting points.
- D The atoms in diamond and graphite are held together by covalent bonds.



**3.5: Metallic bonding****1. O/N 22/P22/Q7**

Which particles are present in the structure of metals?

- 1 positive ions
- 2 negative ions
- 3 shared pairs of electrons
- 4 mobile electrons

A 1 and 2                      B 1 and 4                      C 2 and 3                      D 2 and 4

**2. O/N 22/P21/Q5**

Which type of structure and bonding is present in an element that is malleable and conducts electricity?

- A covalent molecular
- B ionic lattice
- C covalent macromolecular
- D metallic lattice

**3. M/J 22/P21/Q5**

Metals and ionic compounds have similarities and differences in their structure and properties.  
Which row about metals and ionic compounds is correct?

|   | similarity                 | difference                                         |
|---|----------------------------|----------------------------------------------------|
| A | both contain positive ions | only ionic compounds contain anions                |
| B | both contain positive ions | ionic compounds conduct using a 'sea of electrons' |
| C | both are malleable         | only ionic compounds contain anions                |
| D | both are malleable         | ionic compounds conduct using a 'sea of electrons' |

**4. M/J 21/P21/Q7**

Which statement about metals is correct?

- A Metals conduct electricity when molten because negative ions are free to move.
- B Metals conduct electricity when solid because positive ions are free to move.
- C Metals are malleable because the bonds between the atoms are weak.
- D Metals are malleable because the layers of ions can slide over each other.

**5. M/J 21/P23/Q6**

Which statement explains why metals are malleable?

- A The atoms release electrons to become cations.
- B The electrons are free to move.
- C The electrons and the cations are attracted to each other.
- D The layers of ions can slide over each other.

**6. M/J 20/P21/Q6**

Which row contains a description of metallic bonding and a property that is explained by reference to metallic bonding?

|   | description of metallic bonding                  | property explained by metallic bonding                  |
|---|--------------------------------------------------|---------------------------------------------------------|
| A | a lattice of negative ions in a sea of electrons | a metal will react with an acid, producing hydrogen     |
| B | a lattice of negative ions in a sea of electrons | a piece of a metal can be moulded into different shapes |
| C | a lattice of positive ions in a sea of electrons | a metal will react with an acid, producing hydrogen     |
| D | a lattice of positive ions in a sea of electrons | a piece of a metal can be moulded into different shapes |

**7. M/J 20/P22/Q6**

Which statement about the structure of a metal explains why metals are malleable?

- A The electrons can move freely throughout the lattice.
- B The layers of metal ions can slide over each other.
- C The metal ions are positively charged.
- D There is a strong force of attraction between the metal ions and the electrons.

**8. MAR 20/P22/Q7**

Solid X has a high boiling point. Its structure has positive ions surrounded by a sea of electrons. Which other properties does solid X have?

- |                                       |                                         |
|---------------------------------------|-----------------------------------------|
| A brittle and an electrical conductor | C malleable and an electrical conductor |
| B brittle and an insulator            | D malleable and an insulator            |

**9. O/N 19/P21/Q8, O/N 19/P22/Q8, O/N 19/P23/Q8**

Which statement describes the structure of copper?

- A It has a lattice of negative ions in a 'sea of electrons'.
- B It has a lattice of negative ions in a 'sea of protons'.
- C It has a lattice of positive ions in a 'sea of electrons'.
- D It has a lattice of positive ions in a 'sea of protons'.

**10. M/J 18/P21/Q7**

Copper is a metallic element.

Which statements about copper are correct?

- 1 Copper is malleable because layers of ions are in fixed positions and cannot move.
  - 2 The structure of copper consists of negative ions in a lattice.
  - 3 Copper conducts electricity because electrons can move through the metal.
  - 4 Electrons hold copper ions together in a lattice by electrostatic attraction.
- A 1 and 2                      B 2, 3 and 4                      C 2 and 3 only                      D 3 and 4 only

**11. O/N 17/P21/Q4**

Two statements about silicon(IV) oxide are given.

- 1 It is a hard substance.
- 2 It has a macromolecular structure with strong covalent bonds.

Which is correct?

- A Both statements are correct and statement 2 explains statement 1.
- B Both statements are correct but statement 2 does not explain statement 1.
- C Statement 1 is correct but statement 2 is not correct.
- D Statement 2 is correct but statement 1 is not correct.

**12. M/J 17/P22/Q6**

Which statement describes metallic bonding?

- A The attraction between a lattice of negative ions and delocalised protons.
- B The attraction between a lattice of positive ions and delocalised electrons.
- C The attraction between delocalised protons and electrons.
- D The attraction between oppositely charged ions.

**13. M/J 17/P21/Q6**

Which statement about metals is **not** correct?

- A Metals are malleable because the metal ions can slide over one another.
- B Metals conduct electricity because electrons can move through the lattice.
- C Metals consist of a giant lattice of metal ions in a 'sea of electrons'.
- D Metals have high melting points because of the strong attraction between the metal ions.

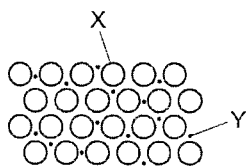
**14. MAR 17/P22/Q7**

Which statement about metals is correct?

- A Layers of positive ions can slide over each other making metals malleable.
- B Metallic bonding consists of a lattice of negative ions in a sea of delocalised electrons.
- C Metallic bonding consists of a lattice of positive ions in a sea of delocalised negative ions.
- D Metals conduct electricity because positive ions are free to move.

**15. M/J 17/P23/Q6**

The diagram shows metallic bonding.



Which labels are correct?

|   | X              | Y               |
|---|----------------|-----------------|
| A | atomic nucleus | outer electron  |
| B | metal atom     | mobile electron |
| C | metal ion      | mobile electron |
| D | positive ion   | negative ion    |

**16. O/N 16/P22/Q7**

Iron is a metal. Its structure consists of a giant lattice of positive ions in a 'sea of electrons'.

Which statements about solid iron are correct?

- 1 Iron conducts electricity because the electrons are free to move.
- 2 Iron conducts heat because the positive ions are free to move.
- 3 Iron has a high melting point due to the strong covalent bonds.
- 4 Iron is malleable because the layers of ions can slide over one another.

A 1 and 3

B 1 and 4

C 1 only

D 2, 3 and 4

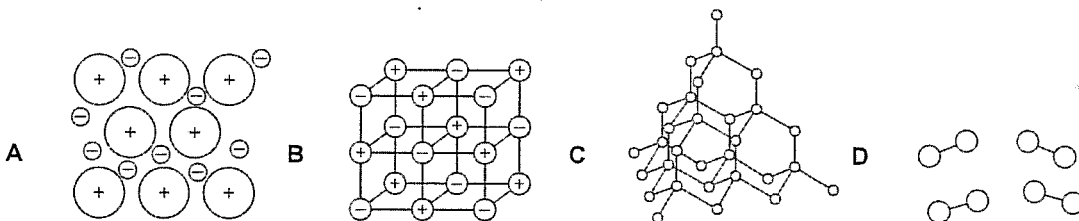
**17. O/N 16/P21/Q7**

X is a solid at room temperature.

X has a high melting point.

Solid X conducts electricity.

Which diagram shows how the particles are arranged in solid X?

**18. O/N 16/P23/Q7**

Metals are malleable.

Which statement explains why metals are malleable?

- A Metallic bonding is very strong.
- B Metals are good conductors of electricity.
- C Positive metal ions are arranged in a regular lattice structure.
- D The layers of positive metal ions can slide over each other.

**19. M/J 16/P22/Q7**

Metals consist of a lattice of positive ions in a 'sea of electrons'.

Why is aluminium malleable?

- A Its ions are attracted to the 'sea of electrons'.
- B Its ions are tightly packed together.
- C Its ions repel each other.
- D Its layers of ions can slide over each other.

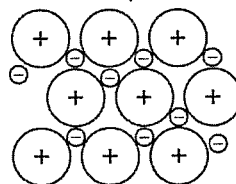
**20. MAR 16/P22/Q23**

In the Periodic Table, how does the metallic character of the elements vary from left to right across a period?

- A It decreases.
- B It increases.
- C It increases then decreases.
- D It stays the same.

**21. M/J 16/P21/Q7**

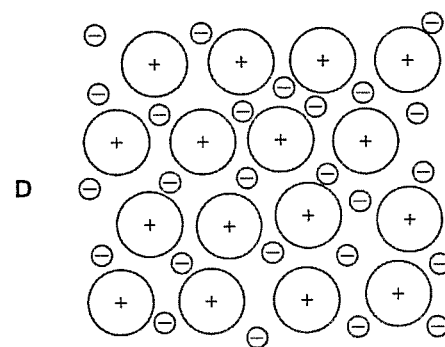
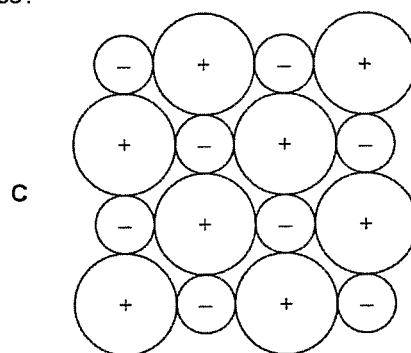
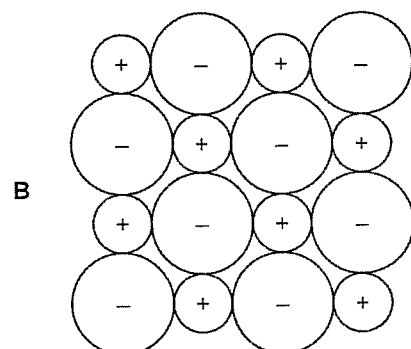
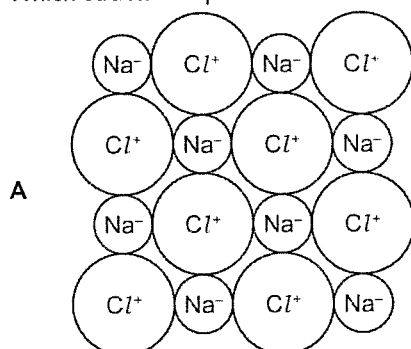
The diagram represents the general structure of a solid Z.



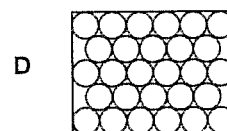
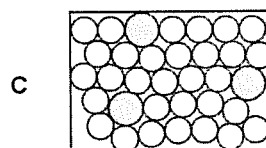
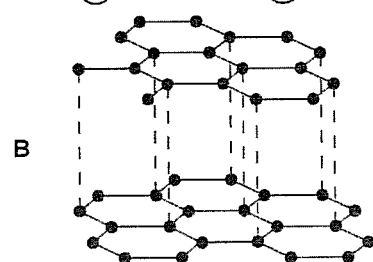
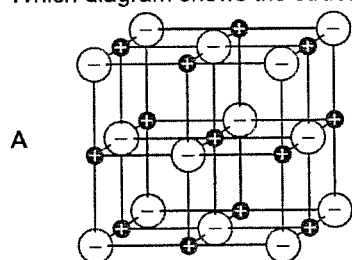
What is Z?

**A** aluminium**B** iodine**C** silicon dioxide**D** sulfur**22. MAR 16/P22/Q6**

Which structure represents the sodium chloride lattice?

**23. O/N 15/P11/Q23**

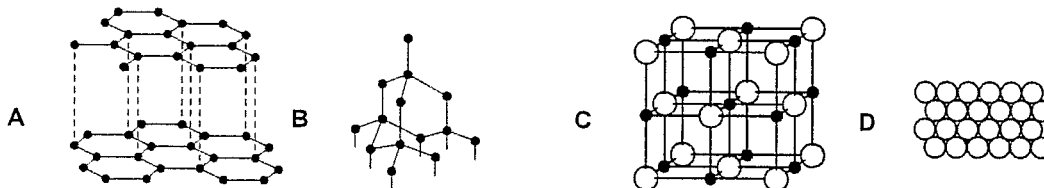
Which diagram shows the structure of an alloy?



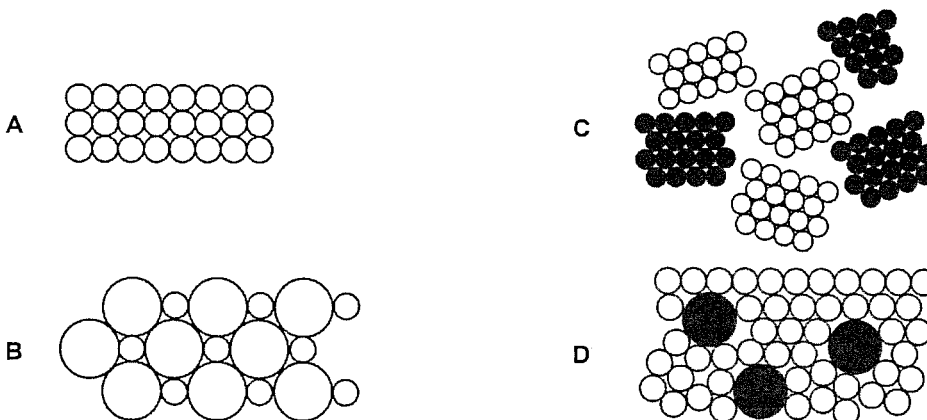
**24. O/N 14/P12/Q5, O/N 14/P11/Q6, O/N 14/P13/Q6**

Slate has a layered structure and can easily be split into thin sheets.

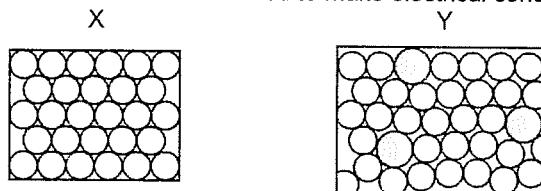
Which diagram shows a structure most like that of slate?

**25. M/J 15/P12/Q23**

Which diagram represents an alloy?

**26. M/J 15/P13/Q23**

The diagrams show the structure of two substances used to make electrical conductors.



Which statement correctly describes X and Y?

- A X is a pure metal and Y is a compound.  
B X is a pure metal and Y is an alloy.  
C X is a solid and Y is a liquid.  
D X is harder and stronger than Y.

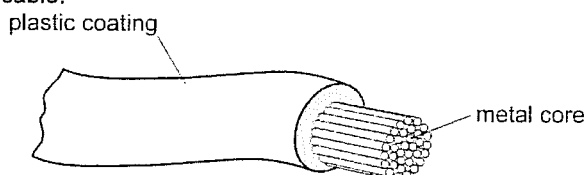
**27. M/J 14/P11/Q11, M/J 14/P13/Q11**

Which substance will **not** conduct electricity?

- A aluminium      B copper      C plastic      D steel

**28. M/J 13/P12,11/Q11, M/J 13/P13/Q10, O/N 09/P12/Q12, O/N 09/P11/Q14**

The diagram shows an electrical cable.



Which statement about the substances used is correct?

- A The coating is plastic because it conducts electricity well.  
B The core is copper because it conducts electricity well.  
C The core is copper because it is cheap and strong.  
D The core is iron because it is cheap and strong.

## Answers Section

| 3.1   |       |       |  | 3.2   | 3.3   |       |  | 3.4   | 3.5   |
|-------|-------|-------|--|-------|-------|-------|--|-------|-------|
| 1. B  | 41. B | 81. A |  | 1. D  | 1. D  | 41. B |  | 1. B  | 1. B  |
| 2. C  | 42. A | 82. B |  | 2. A  | 2. B  | 42. A |  | 2. B  | 2. D  |
| 3. C  | 43. C | 83. C |  | 3. C  | 3. D  | 43. B |  | 3. D  | 3. A  |
| 4. C  | 44. B | 84. D |  | 4. D  | 4. C  | 44. B |  | 4. C  | 4. D  |
| 5. A  | 45. A | 85. A |  | 5. B  | 5. C  | 45. A |  | 5. B  | 5. D  |
| 6. A  | 46. C |       |  | 6. C  | 6. A  | 46. C |  | 6. D  | 6. D  |
| 7. D  | 47. B |       |  | 7. C  | 7. D  | 47. B |  | 7. B  | 7. B  |
| 8. C  | 48. D |       |  | 8. A  | 8. B  | 48. C |  | 8. D  | 8. C  |
| 9. A  | 49. B |       |  | 9. C  | 9. A  | 49. C |  | 9. B  | 9. C  |
| 10. C | 50. B |       |  | 10. B | 10. A | 50. A |  | 10. C | 10. D |
| 11. D | 51. B |       |  | 11. C | 11. C | 51. C |  | 11. C | 11. A |
| 12. A | 52. B |       |  | 12. C | 12. C | 52. D |  | 12. C | 12. B |
| 13. A | 53. B |       |  | 13. A | 13. A | 53. D |  | 13. C | 13. D |
| 14. B | 54. B |       |  | 14. B | 14. A | 54. A |  | 14. D | 14. A |
| 15. C | 55. C |       |  | 15. B | 15. A | 55. C |  | 15. B | 15. C |
| 16. C | 56. A |       |  | 16. C | 16. D | 56. A |  |       | 16. B |
| 17. A | 57. C |       |  | 17. C | 17. C | 57. C |  |       | 17. A |
| 18. B | 58. C |       |  | 18. A | 18. D | 58. A |  |       | 18. D |
| 19. B | 59. B |       |  | 19. B | 19. B | 59. C |  |       | 19. D |
| 20. D | 60. C |       |  | 20. C | 20. C | 60. D |  |       | 20. A |
| 21. D | 61. A |       |  | 21. A | 21. D | 61. C |  |       | 21. A |
| 22. C | 62. D |       |  | 22. A | 22. D |       |  |       | 22. B |
| 23. C | 63. C |       |  | 23. B | 23. A |       |  |       | 23. C |
| 24. D | 64. B |       |  | 24. C | 24. D |       |  |       | 24. A |
| 25. C | 65. A |       |  | 25. C | 25. A |       |  |       | 25. D |
| 26. D | 66. B |       |  | 26. C | 26. C |       |  |       | 26. B |
| 27. B | 67. C |       |  | 27. D | 27. D |       |  |       | 27. C |
| 28. A | 68. B |       |  | 28. D | 28. B |       |  |       | 28. B |
| 29. A | 69. A |       |  | 29. C | 29. A |       |  |       |       |
| 30. D | 70. A |       |  | 30. A | 30. B |       |  |       |       |
| 31. A | 71. C |       |  | 31. C | 31. B |       |  |       |       |
| 32. C | 72. A |       |  | 32. C | 32. B |       |  |       |       |
| 33. B | 73. B |       |  | 33. B | 33. A |       |  |       |       |
| 34. C | 74. D |       |  | 34. C | 34. B |       |  |       |       |
| 35. A | 75. C |       |  | 35. C | 35. D |       |  |       |       |
| 36. C | 76. D |       |  | 36. B | 36. D |       |  |       |       |
| 37. B | 77. D |       |  |       | 37. C |       |  |       |       |
| 38. C | 78. D |       |  |       | 38. D |       |  |       |       |
| 39. B | 79. A |       |  |       | 39. A |       |  |       |       |
| 40. B | 80. D |       |  |       | 40. C |       |  |       |       |



## Unit 4

# STOICHIOMETRY

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IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 4.1 Stoichiometry
- 4.2 The mole concept

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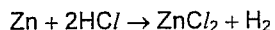
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## Unit 4: Stoichiometry

### 4.1: Stoichiometry

1. O/N 22/P21/Q9

4.55 g of zinc is reacted with 50 cm<sup>3</sup> of 2.25 mol/dm<sup>3</sup> dilute hydrochloric acid.  
The equation for the reaction is shown.

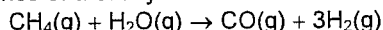


Which volume of hydrogen gas, at room temperature and pressure, is produced in the reaction?

- A 1.35 dm<sup>3</sup>      B 1.67 dm<sup>3</sup>      C 2.70 dm<sup>3</sup>      D 3.34 dm<sup>3</sup>

2. M/J 22/P22/Q8

Methane and steam react in the presence of a catalyst.



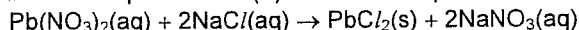
0.5 mol of methane reacts completely with 0.5 mol of steam.

What is the volume of carbon monoxide and hydrogen produced, measured at room temperature and pressure?

|   | volume<br>of CO / dm <sup>3</sup> | volume<br>of H <sub>2</sub> / dm <sup>3</sup> |
|---|-----------------------------------|-----------------------------------------------|
| A | 0.5                               | 1.5                                           |
| B | 1.0                               | 3.0                                           |
| C | 12.0                              | 12.0                                          |
| D | 12.0                              | 36.0                                          |

3. M/J 22/P21/Q9

The equation for the reaction between aqueous lead(II) nitrate and aqueous sodium chloride is shown.



If 100 cm<sup>3</sup> of aqueous lead(II) nitrate of concentration 0.1 mol / dm<sup>3</sup> is reacted with an excess of aqueous sodium chloride, which mass of lead(II) chloride is obtained?

- A 1.16 g      B 2.42 g      C 2.78 g      D 3.31 g

4. M/J 22/P23/Q8

A 0.5 g sample of calcium carbonate is reacted with excess dilute hydrochloric acid.

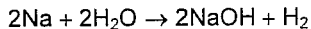


Which volume of CO<sub>2</sub> is produced at r.t.p.?

- A 0.12 dm<sup>3</sup>      B 0.18 dm<sup>3</sup>      C 0.24 dm<sup>3</sup>      D 12 dm<sup>3</sup>

5. O/N 21/P22/Q9

The equation for the reaction of sodium with water is shown.



What is the volume of hydrogen gas, measured at r.t.p., produced when 18.4g of sodium reacts with excess water?

- A 9.6 dm<sup>3</sup>      B 15.0 dm<sup>3</sup>      C 19.2 dm<sup>3</sup>      D 30.0 dm<sup>3</sup>

6. O/N 21/P23/Q2

A student put exactly 25.00 cm<sup>3</sup> of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use?

- A balance, measuring cylinder, thermometer      C balance, pipette, thermometer  
B balance, pipette, stopwatch      D burette, pipette, thermometer

7. O/N 21/P23/Q13

What is the concentration of the solution when 31.8 g of sodium carbonate, Na<sub>2</sub>CO<sub>3</sub>, is dissolved in water to make a solution of 250 cm<sup>3</sup>?

- A 0.075 mol/dm<sup>3</sup>      B 0.30 mol/dm<sup>3</sup>      C 1.2 mol/dm<sup>3</sup>      D 1.5 mol/dm<sup>3</sup>



**8. M/J 21/P23/Q37**

How much hydrogen is needed to react completely with 0.02 moles of butene to make butane?

- A 0.24 dm<sup>3</sup>                      B 0.48 dm<sup>3</sup>                      C 0.96 dm<sup>3</sup>                      D 1.20 dm<sup>3</sup>

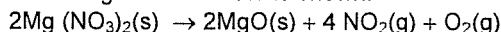
**9. O/N 20/P22/Q13, O/N 20/P21/Q13, O/N 20/P23/Q12**

What is the empirical formula of an oxide of iron, formed by reacting 2.24 g of iron with 0.96 g of oxygen?

- A FeO                      B Fe<sub>2</sub>O                      C Fe<sub>2</sub>O<sub>3</sub>                      D Fe<sub>3</sub>O<sub>4</sub>

**10. MAR 20/P22/Q9**

The equation for the decomposition of magnesium nitrate is shown.



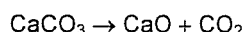
Which volume of gas is produced when 0.1 moles of magnesium nitrate is decomposed completely?

- A 1.2 dm<sup>3</sup>                      B 4.8 dm<sup>3</sup>                      C 6.0 dm<sup>3</sup>                      D 8.4 dm<sup>3</sup>

**11. O/N 19/P22/Q10**

Calcium carbonate is heated. Calcium oxide and carbon dioxide gas are formed.

The equation for the reaction is shown.



225 kg of calcium carbonate is heated until there is no further change in mass.

The yield of calcium oxide is 85 kg.

What is the percentage yield?

- A 37.8 %                      B 47.2 %                      C 67.5 %                      D 85.0 %

**12. O/N 19/P22/Q16**

A sample of dilute nitric acid is added to lumps of limestone in a conical flask. The conical flask is placed on a balance and the loss in mass is measured.

A second sample of nitric acid of a different concentration is separately tested. All other conditions are kept the same.

The loss in mass in 1 minute at each concentration of nitric acid is shown.

| concentration<br>in mol / dm <sup>3</sup> | loss in mass in<br>1 minute / g |
|-------------------------------------------|---------------------------------|
| 0.5                                       | 0.15                            |
| 1.0                                       | 0.25                            |

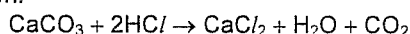
Which row describes and explains the results obtained using 1.0 mol/dm<sup>3</sup> nitric acid compared with 0.5 mol/dm<sup>3</sup> nitric acid?

|          | description               | explanation                                              |
|----------|---------------------------|----------------------------------------------------------|
| <b>A</b> | decrease in reaction rate | decrease in particle collision energy                    |
| <b>B</b> | decrease in reaction rate | increase in particle collision rate                      |
| <b>C</b> | increase in reaction rate | increase in particle collision rate                      |
| <b>D</b> | increase in reaction rate | increase in particle collision rate and collision energy |

**13. O/N 19/P21/Q10**

Calcium carbonate reacts with dilute hydrochloric acid.

The equation for the reaction is shown.



1.00 g of calcium carbonate is added to 50.0 cm<sup>3</sup> of 0.0500 mol / dm<sup>3</sup> hydrochloric acid.

Which volume of carbon dioxide is made in this reaction?

- A 30 cm<sup>3</sup>                      B 60 cm<sup>3</sup>                      C 120 cm<sup>3</sup>                      D 240 cm<sup>3</sup>

**14. O/N 19/P23/Q2**

A student is asked to measure the time taken for 0.4 g of magnesium carbonate to react completely with 25.0 cm<sup>3</sup> of dilute hydrochloric acid.

Which pieces of apparatus does the student need?

- A balance, stop-clock, pipette                      C balance, pipette, thermometer  
B balance, stop-clock, thermometer                      D stop-clock, pipette, thermometer

## 15. O/N 18/P22/Q7

Which gas sample contains the smallest number of molecules?

- A 4 g of helium  
B 16 g of oxygen  
C 28 g of carbon monoxide  
D 28 g of nitrogen

## 16. MAR 19/P22/Q9

Calcium carbonate reacts with dilute hydrochloric acid according to the equation shown.

10 g of calcium carbonate is reacted with 100 cm<sup>3</sup> of 1 mol / dm<sup>3</sup> hydrochloric acid.

The following statements are made.

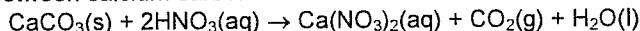
- 1 1.2 dm<sup>3</sup> of carbon dioxide is formed.  
2 5.6 g of calcium chloride is formed.  
3 4.8 g of carbon dioxide is formed.  
4 No calcium carbonate is left when the reaction is completed.

Which statements about the reaction are correct?

- A 1 and 2  
B 1 and 4  
C 2 and 3  
D 3 and 4

## 17. O/N 18/P22/Q8

The equation for the reaction between calcium carbonate and dilute nitric acid is shown.



25 g of calcium carbonate is reacted with an excess of dilute nitric acid.

Which mass of calcium nitrate and which volume of carbon dioxide is produced at room temperature and pressure?

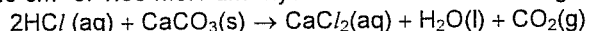
|   | mass of<br>calcium nitrate / g | volume of<br>carbon dioxide / dm <sup>3</sup> |
|---|--------------------------------|-----------------------------------------------|
| A | 29                             | 6                                             |
| B | 29                             | 12                                            |
| C | 41                             | 6                                             |
| D | 41                             | 12                                            |

## 18. O/N 18/P21/Q7

Which gas sample contains the most molecules?

- A 24 dm<sup>3</sup> of carbon dioxide, CO<sub>2</sub>  
B 4 g of hydrogen, H<sub>2</sub>  
C 36 dm<sup>3</sup> of hydrogen chloride, HCl  
D 14 g of nitrogen, N<sub>2</sub>

## 19. O/N 18/P21/Q8

A student mixed together 25.0 cm<sup>3</sup> of 1.00 mol / dm<sup>3</sup> hydrochloric acid and 25.0 g of calcium carbonate.

What is the maximum volume of carbon dioxide gas that could be collected at room temperature and pressure?

- A 300 dm<sup>3</sup>  
B 6.00 dm<sup>3</sup>  
C 0.600 dm<sup>3</sup>  
D 0.300 dm<sup>3</sup>

## 20. O/N 18/P23/Q7

Which gas sample has the greatest mass?

- A 5.0 moles of Cl<sub>2</sub>  
B 10.0 moles of O<sub>2</sub>  
C 15.0 moles of N<sub>2</sub>  
D 20.0 moles of H<sub>2</sub>

## 21. O/N 18/P23/Q8

Which sample of magnesium chloride, MgCl<sub>2</sub>, contains the same number of moles as 69.6 g of potassium sulfate, K<sub>2</sub>SO<sub>4</sub>?

- A 19.0 g  
B 28.5 g  
C 38.0 g  
D 47.5 g

## 22. O/N 18/P23/Q14

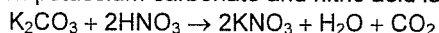
Dilute hydrochloric acid reacts with 1g of limestone.

Which conditions produce the fastest rate of reaction?

- A 2 mol / dm<sup>3</sup> hydrochloric acid and a single lump of limestone  
B 4 mol / dm<sup>3</sup> hydrochloric acid and a single lump of limestone  
C 4 mol / dm<sup>3</sup> hydrochloric acid and small pieces of limestone  
D 4 mol / dm<sup>3</sup> hydrochloric acid and powdered limestone

**23. M/J 18/P22/Q8**

The equation for the reaction between potassium carbonate and nitric acid is shown.



Which volume of carbon dioxide is produced from 69 g of potassium carbonate?

- A 6 dm<sup>3</sup>                      B 12 dm<sup>3</sup>                      C 24 dm<sup>3</sup>                      D 48 dm<sup>3</sup>

**24. M/J 18/P22/Q9**

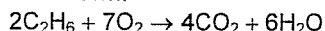
A solution of sodium carbonate, Na<sub>2</sub>CO<sub>3</sub>, has a concentration of 0.03 mol / dm<sup>3</sup>.

Which mass of sodium carbonate is dissolved in 1 dm<sup>3</sup> of this solution?

- A 1.06 g                      B 3.18 g                      C 10.60 g                      D 31.80 g

**25. M/J 18/P21/Q8**

The equation for the combustion of ethane is shown.



Which volume of carbon dioxide, at room temperature and pressure, is formed when 0.5 moles of ethane burn?

- A 48 dm<sup>3</sup>                      B 24 dm<sup>3</sup>                      C 12 dm<sup>3</sup>                      D 6 dm<sup>3</sup>

**26. M/J 18/P21/Q9**

A solution of ethanoic acid, CH<sub>3</sub>COOH, has a concentration of 2 mol / dm<sup>3</sup>.

Which statement about this solution is correct?

- A 20 g of ethanoic acid is dissolved in 10 cm<sup>3</sup> of water.  
 B 30 g of ethanoic acid is dissolved in 250 cm<sup>3</sup> of water.  
 C 60 g of ethanoic acid is dissolved in 1 dm<sup>3</sup> of water.  
 D 120 g of ethanoic acid is dissolved in 2 dm<sup>3</sup> of water.

**27. M/J 18/P23/Q8**

An experiment was done to determine the formula of a hydrocarbon, C<sub>x</sub>H<sub>y</sub>.

10 cm<sup>3</sup> of the gaseous hydrocarbon, C<sub>x</sub>H<sub>y</sub>, was burned in an excess of oxygen to form 20 cm<sup>3</sup> of carbon dioxide and 30 cm<sup>3</sup> of water vapour.

What is C<sub>x</sub>H<sub>y</sub>?

- A CH<sub>4</sub>                      B C<sub>2</sub>H<sub>4</sub>                      C C<sub>2</sub>H<sub>6</sub>                      D C<sub>3</sub>H<sub>8</sub>

**28. M/J 18/P23/Q9**

4.00 g of solid sodium hydroxide is added to water to make a solution with a concentration of 0.200 mol / dm<sup>3</sup>.

What is the volume of water used?

- A 0.5 cm<sup>3</sup>                      B 20 cm<sup>3</sup>                      C 500 cm<sup>3</sup>                      D 2000 cm<sup>3</sup>

**29. MAR 18/P22/Q7**

The concentration of a hydrochloric acid solution is 0.5 mol / dm<sup>3</sup>.

How many moles of hydrochloric acid are present in 25 cm<sup>3</sup> of this solution?

- A 0.0125                      B 0.0200                      C 12.5                      D 20.0

**30. M/J 17/P22/Q7**

Which equations are balanced?

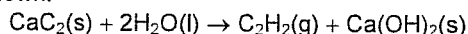
- 1 Fe<sub>2</sub>O<sub>3</sub> + 3CO → 2Fe + 3CO<sub>2</sub>  
 2 ZnCO<sub>3</sub> + 2HCl → ZnCl<sub>2</sub> + CO<sub>2</sub> + 2H<sub>2</sub>O  
 3 Mg(NO<sub>3</sub>)<sub>2</sub> + NaOH → Mg(OH)<sub>2</sub> + 2NaNO<sub>3</sub>  
 4 CaCO<sub>3</sub> + H<sub>2</sub>SO<sub>4</sub> → CaSO<sub>4</sub> + H<sub>2</sub>O + CO<sub>2</sub>

- A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

**31. M/J 17/P22/Q8**

Calcium carbide, CaC<sub>2</sub>, reacts with water to form ethyne, C<sub>2</sub>H<sub>2</sub>, and calcium hydroxide.

The equation for the reaction is shown.

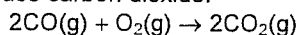


Which volume of ethyne is produced when 6 g of water react completely with calcium carbide?

- A 4 dm<sup>3</sup>                      B 8 dm<sup>3</sup>                      C 36 dm<sup>3</sup>                      D 72 dm<sup>3</sup>

**32. M/J 17/P21/Q8**

Carbon monoxide burns in oxygen to produce carbon dioxide.

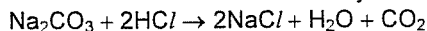


Which mass of carbon dioxide is produced from 14 g of carbon monoxide?

- A 22 g                      B 28 g                      C 44 g                      D 88 g

**33. M/J 17/P23/Q8**

The equation for the reaction between sodium carbonate and dilute hydrochloric acid is shown.



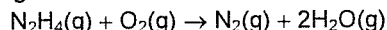
What is the maximum volume of carbon dioxide produced when 26.5 g of sodium carbonate react with dilute hydrochloric acid?

- A 6 dm<sup>3</sup>                      B 12 dm<sup>3</sup>                      C 18 dm<sup>3</sup>                      D 24 dm<sup>3</sup>

**34. MAR 17/P22/Q8**

The gas hydrazine has the molecular formula N<sub>2</sub>H<sub>4</sub>.

Hydrazine burns in air to form nitrogen gas and steam.



Which statements are correct?

- 1 1 mole of hydrazine gives 72 dm<sup>3</sup> of gaseous products when it reacts with oxygen at room temperature and pressure.
- 2 The empirical formula of hydrazine is NH<sub>2</sub>.
- 3 The total number of atoms in 1 mole of hydrazine is 6 × the Avogadro constant.
- 4 The volume of 1 mole of hydrazine at room temperature and pressure is 6 × 24 dm<sup>3</sup>.

- A 1, 2 and 3                      B 1 and 2 only                      C 2, 3 and 4                      D 3 and 4 only

**35. MAR 17/P22/Q9**

Copper(II) carbonate is broken down by heating to form copper(II) oxide and carbon dioxide gas.

The equation for the reaction is shown.



31.0 g of copper(II) carbonate are heated until all of the contents of the test-tube have turned from green to black.

The yield of copper(II) oxide formed is 17.5 g.

What is the percentage yield?

- A 19.02%                      B 21.88%                      C 56.50%                      D 87.50%

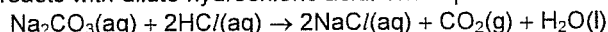
**36. O/N 16/P22/Q8**

Which sample contains the greatest number of molecules?

- A 4 g of hydrogen                      C 24 dm<sup>3</sup> of oxygen  
B 18 g of water                      D 66 g of carbon dioxide

**37. O/N 16/P22/Q9**

Sodium carbonate solution reacts with dilute hydrochloric acid. The equation for the reaction is shown.



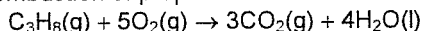
Excess sodium carbonate is added to 10.0 cm<sup>3</sup> of 0.10 mol / dm<sup>3</sup> hydrochloric acid.

Which volume of carbon dioxide gas is made?

- A 12 cm<sup>3</sup>                      B 24 cm<sup>3</sup>                      C 12 000 cm<sup>3</sup>                      D 24 000 cm<sup>3</sup>

**38. O/N 16/P23/Q8**

The equation shows the complete combustion of propane.



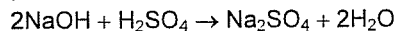
Which statement is correct?

- A 10 cm<sup>3</sup> of propane cannot burn if less than 50 cm<sup>3</sup> of oxygen is present.
- B 10 cm<sup>3</sup> of propane would produce 40 cm<sup>3</sup> of liquid water.
- C 100 cm<sup>3</sup> of oxygen would be sufficient to react completely with 20 cm<sup>3</sup> of propane.
- D This reaction would result in an increase in the volume of gas.

**39. O/N 16/P23/Q9**

Sodium hydroxide reacts with sulfuric acid.

The equation for the reaction is shown.



Which volume of 0.4 mol / dm<sup>3</sup> sodium hydroxide reacts with 50.0 cm<sup>3</sup> of 0.1 mol / dm<sup>3</sup> sulfuric acid?

- A 12.5 cm<sup>3</sup>                      B 25.0 cm<sup>3</sup>                      C 50.0 cm<sup>3</sup>                      D 100.0 cm<sup>3</sup>

**40. M/J 16/P22/Q8**

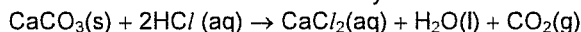
A sample of 16.0 g of a metal oxide, MO, is reduced to 12.8 g of the metal, M.

What is the relative atomic mass, A<sub>r</sub>, of M?

- A 32                      B 64                      C 80                      D 128

**41. M/J 16/P22/Q9**

The equation for the reaction between calcium carbonate and hydrochloric acid is shown.



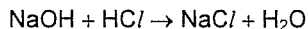
How many moles of calcium carbonate will give 24 cm<sup>3</sup> of carbon dioxide when reacted with an excess of the acid?

- A 1 mol                      B 0.1 mol                      C 0.01 mol                      D 0.001 mol

**42. M/J 16/P21/Q9**

25 cm<sup>3</sup> of 0.1 mol / dm<sup>3</sup> hydrochloric acid exactly neutralise 20 cm<sup>3</sup> of aqueous sodium hydroxide.

The equation for this reaction is:



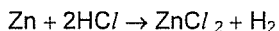
What is the concentration of the sodium hydroxide solution?

- A 0.080 mol / dm<sup>3</sup>      B 0.800 mol / dm<sup>3</sup>      C 0.125 mol / dm<sup>3</sup>      D 1.25 mol / dm<sup>3</sup>

**43. M/J 16/P23/Q9**

An excess of zinc is added to 100 cm<sup>3</sup> of 1.0 mol / dm<sup>3</sup> hydrochloric acid.

The equation for the reaction is:



What is the maximum volume of hydrogen evolved at room temperature and pressure?

- A 1.2 dm<sup>3</sup>                      B 2.0 dm<sup>3</sup>                      C 2.4 dm<sup>3</sup>                      D 24 dm<sup>3</sup>

**44. MAR 16/P22/Q8**

Which quantities of chemicals will react exactly with no reactants left over?

- A 12 g of carbon and 12 g of oxygen                      C 12 g of magnesium and 16 g of oxygen  
B 12 g of carbon and 48 g of oxygen                      D 24 g of magnesium and 16 g of oxygen

**45. M/J 15/P11/Q8**

What is the relative formula mass,  $M_r$ , of CaCO<sub>3</sub>?

- A 50                      B 68                      C 100                      D 204

**46. M/J 15/P13/Q8**

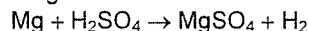
Aluminium oxide has the formula Al<sub>2</sub>O<sub>3</sub>.

Which statement about aluminium oxide is correct?

- A 2 g of aluminium atoms are combined with 3 g of oxygen atoms.  
B 2 g of aluminium atoms are combined with 3 g of oxygen molecules.  
C Aluminium oxide has a relative molecular mass of 102.  
D Pure aluminium oxide contains a higher mass of oxygen than of aluminium.

**47. M/J 13/P12/Q7**

The equation shows the reaction between magnesium and sulfuric acid.



(Mg = 24, H = 1, S = 32, O = 16)

In this reaction, what mass of magnesium sulfate will be formed when 6g of magnesium reacts with excess sulfuric acid?

- A 8                      B 24                      C 30                      D 60



**11. MAR 21/P22/Q10**

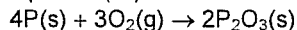
A compound has the formula  $\text{XF}_2$  and has a relative mass of 70.

What is element X?

- A gallium                      B germanium                      C sulfur                      D ytterbium

**12. O/N 19/P22/Q9**

Phosphorus reacts with oxygen to form phosphorus(III) oxide as shown.

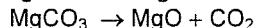


Which mass of phosphorus(III) oxide is produced from 6.2 g of phosphorus?

- A 1.1 g                      B 5.5 g                      C 11.0 g                      D 22.0 g

**13. O/N 19/P23/Q9**

Magnesium carbonate decomposes on heating to form magnesium oxide and carbon dioxide as shown.



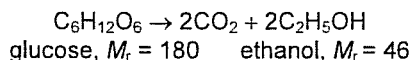
How much magnesium carbonate is needed to make 5.0 g of magnesium oxide?

- A 3.5 g                      B 4.0 g                      C 6.5 g                      D 10.5 g

**14. O/N 19/P23/Q10**

90 g of glucose is dissolved in water.

The glucose solution is fermented.



After the fermentation finishes, 6.8 g of ethanol is obtained from the solution.

What is the percentage yield of ethanol?

- A 7.4                      B 7.6                      C 14.8                      D 29.6

**15. M/J 19/P22/Q8**

25.0 cm<sup>3</sup> of 0.100 mol/dm<sup>3</sup> aqueous sodium hydroxide is neutralised by 24.6 cm<sup>3</sup> of dilute sulfuric acid.

What is the concentration of the dilute sulfuric acid?

- A 0.0508 mol/dm<sup>3</sup>      B 0.0984 mol/dm<sup>3</sup>      C 0.102 mol/dm<sup>3</sup>      D 0.203 mol/dm<sup>3</sup>

**16. M/J 19/P21/Q8**

A tablet contains 0.080 g of ascorbic acid ( $M_r = 176$ ).

What is the concentration of ascorbic acid when one tablet is dissolved in 200 cm<sup>3</sup> of water?

- A  $9.1 \times 10^{-5}$  mol/dm<sup>3</sup>      B  $4.5 \times 10^{-4}$  mol/dm<sup>3</sup>      C  $9.1 \times 10^{-2}$  mol/dm<sup>3</sup>      D  $2.3 \times 10^{-3}$  mol/dm<sup>3</sup>

**17. M/J 19/P23/Q8**

What is the concentration of a solution that contains 25.0 g NaOH in 500 cm<sup>3</sup> of water?

- A 0.125 mol/dm<sup>3</sup>      B 0.800 mol/dm<sup>3</sup>      C 1.25 mol/dm<sup>3</sup>      D 3.20 mol/dm<sup>3</sup>

**18. MAR 19/P22/Q8**

An oxide of nitrogen has the following composition by mass: N, 30.4%; O, 69.6%.

It has a relative molecular mass of 92.

What is the molecular formula of the oxide of nitrogen?

- A NO                      B NO<sub>2</sub>                      C NO<sub>4</sub>                      D N<sub>2</sub>O<sub>4</sub>

**19. O/N 18/P22/Q9**

The formulae of some ions are shown.

| positive ion     | negative ion       |
|------------------|--------------------|
| $\text{Ti}^{4+}$ | $\text{PO}_4^{3-}$ |
| $\text{Al}^{3+}$ | $\text{SO}_4^{2-}$ |
| $\text{Mg}^{2+}$ | $\text{NO}_3^-$    |
| $\text{K}^+$     | $\text{Cl}^-$      |

Which formula is **not** correct?

- A  $\text{Al}_3(\text{SO}_4)_2$                       B  $\text{K}_3\text{PO}_4$                       C  $\text{Mg}(\text{NO}_3)_2$                       D  $\text{TiCl}_4$

**20. O/N 18/P21/Q9**

Iron can react with sulfur to form two ionic compounds.

The iron is present as  $\text{Fe}^{2+}$  in one compound and as  $\text{Fe}^{3+}$  in the other compound.

The sulfur ion is present as  $\text{S}^{2-}$  in both compounds.

What are the formulae of the two compounds?

- A  $\text{FeS}$  and  $\text{Fe}_2\text{S}_3$       B  $\text{FeS}$  and  $\text{Fe}_3\text{S}_2$       C  $\text{FeS}_2$  and  $\text{Fe}_3\text{S}_2$       D  $\text{FeS}_2$  and  $\text{Fe}_2\text{S}_3$

**21. O/N 18/P23/Q9**

Iron(III) chromate is a yellow solid. It contains the ions  $\text{Fe}^{3+}$  and  $\text{CrO}_4^{2-}$ .

What is the formula of iron(III) chromate?

- A  $\text{FeCrO}_4$       B  $\text{Fe}_3(\text{CrO}_4)_2$       C  $\text{Fe}_2\text{CrO}_4$       D  $\text{Fe}_2(\text{CrO}_4)_3$

**22. MAR 18/P22/Q8**

A sample of an iron oxide contains 50.4 g of iron and 21.6 g of oxygen.

What is the empirical formula of the iron oxide?

- A  $\text{FeO}$       B  $\text{FeO}_3$       C  $\text{Fe}_2\text{O}_3$       D  $\text{Fe}_3\text{O}_2$

**23. O/N 17/P22/Q8**

The relative molecular mass of an alcohol is 88.

Its percentage composition by mass is: C, 54.5%; H, 9.1%; O, 36.4%.

Which row shows the empirical formula and molecular formula for this alcohol?

|   | empirical formula                | molecular formula                |
|---|----------------------------------|----------------------------------|
| A | $\text{C}_2\text{H}_4\text{O}$   | $\text{C}_2\text{H}_4\text{O}$   |
| B | $\text{C}_2\text{H}_4\text{O}$   | $\text{C}_4\text{H}_8\text{O}_2$ |
| C | $\text{C}_4\text{H}_8\text{O}_2$ | $\text{C}_4\text{H}_8\text{O}_2$ |
| D | $\text{C}_4\text{H}_8\text{O}_2$ | $\text{C}_2\text{H}_4\text{O}$   |

**24. O/N 17/P23/Q8**

A compound contains 34.5% calcium, 24.1% silicon and 41.4% oxygen by mass.

What is its empirical formula?

- A  $\text{Ca}_2\text{SiO}_3$       B  $\text{CaSiO}_3$       C  $\text{CaSi}_2\text{O}_3$       D  $\text{CaSiO}_5$

**25. MAR 17/P22/Q5**

The formulae of some ions are shown.

| positive ions    | negative ions      |
|------------------|--------------------|
| $\text{Al}^{3+}$ | $\text{Br}^-$      |
| $\text{Ca}^{2+}$ | $\text{CO}_3^{2-}$ |
| $\text{Cu}^{2+}$ | $\text{NO}_3^-$    |
| $\text{Fe}^{3+}$ | $\text{S}^{2-}$    |
| $\text{K}^+$     | $\text{SO}_4^{2-}$ |

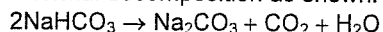
In which row is the formula **not** correct?

|   | compound          | formula                      |
|---|-------------------|------------------------------|
| A | aluminium sulfate | $\text{Al}_2(\text{SO}_4)_3$ |
| B | calcium nitrate   | $\text{Ca}(\text{NO}_3)_2$   |
| C | iron(III) bromide | $\text{Fe}_3\text{Br}$       |
| D | potassium sulfide | $\text{K}_2\text{S}$         |



**26. O/N 16/P21/Q9**

Sodium hydrogencarbonate undergoes thermal decomposition as shown.



What is the maximum mass of sodium carbonate that can be made from 0.100 moles of sodium hydrogencarbonate?

- A 4.15 g                      B 5.30 g                      C 10.6 g                      D 21.2 g

**27. M/J 16/P21/Q8**

A compound, X, contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen by mass.

The relative molecular mass,  $M_r$ , of X is 60.

What is the molecular formula of X?

- A  $\text{CH}_2\text{O}$                       B  $\text{CH}_4\text{O}$                       C  $\text{C}_2\text{H}_4\text{O}$                       D  $\text{C}_2\text{H}_4\text{O}_2$

**28. M/J 16/P23/Q8**

Analysis of a compound formed between magnesium and nitrogen showed it contained 14.4 g of magnesium and 5.6 g of nitrogen.

What is the empirical formula of the compound?

- A  $\text{Mg}_2\text{N}_3$                       B  $\text{Mg}_3\text{N}_2$                       C  $\text{Mg}_4\text{N}_6$                       D  $\text{Mg}_6\text{N}_4$

**29. O/N 15/P12/Q7**

The relative atomic mass of chlorine is 35.5.

When calculating relative atomic mass, which particle is the mass of a chlorine atom compared to?

- A a neutron                      C an atom of carbon-12  
B a proton                      D an atom of hydrogen-1

**30. O/N 14/P12/Q12, O/N 14/P11/Q10**

Iron forms an oxide with the formula  $\text{Fe}_2\text{O}_3$ .

What is the relative formula mass of this compound?

- A 76                      B 100                      C 136                      D 160

**31. O/N 14/P13/Q9**

The table shows the numbers of atoms present in the formula of some compounds.

Which row is **not** correct?

|   | numbers of atoms                     | formula                         |
|---|--------------------------------------|---------------------------------|
| A | 1 × calcium, 1 × carbon, 3 × oxygen  | $\text{CaCO}_3$                 |
| B | 1 × carbon, 5 × hydrogen, 1 × oxygen | $\text{C}_2\text{H}_5\text{OH}$ |
| C | 1 × hydrogen, 1 × oxygen, 1 × sodium | $\text{NaOH}$                   |
| D | 2 × hydrogen, 4 × oxygen, 1 × sulfur | $\text{H}_2\text{SO}_4$         |

**32. M/J 14/P12/Q9**

What is the relative molecular mass,  $M_r$ , of nitrogen dioxide?

- A 15                      B 23                      C 30                      D 46

**33. M/J 14/P11/Q9, M/J 14/P13/Q9**

A compound contains one atom of calcium, two atoms of hydrogen and two atoms of oxygen.

What is the correct chemical formula of the compound?

- A  $\text{CaO}_2\text{H}_2$                       B  $\text{HOCaOH}$                       C  $\text{H}_2\text{CaO}_2$                       D  $\text{Ca(OH)}_2$

**34. M/J 14/P11/Q10, M/J 14/P13/Q8**

In athletics, banned drugs such as nandrolone have been taken illegally to improve performance.

Nandrolone has the molecular formula  $\text{C}_{18}\text{H}_{26}\text{O}_2$ .

What is the relative molecular mass,  $M_r$ , of nandrolone?

(Relative atomic mass: H = 1; C = 12; O = 16)

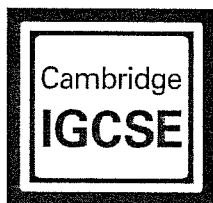
- A 46                      B 150                      C 274                      D 306



## Answers Section

| 4.1   |       | 4.2   |  |
|-------|-------|-------|--|
| 1. A  | 41. D | 1. D  |  |
| 2. D  | 42. C | 2. B  |  |
| 3. C  | 43. A | 3. C  |  |
| 4. A  | 44. D | 4. C  |  |
| 5. A  | 45. C | 5. A  |  |
| 6. C  | 46. C | 6. D  |  |
| 7. C  | 47. C | 7. C  |  |
| 8. B  |       | 8. D  |  |
| 9. C  |       | 9. B  |  |
| 10. C |       | 10. C |  |
| 11. C |       | 11. C |  |
| 12. C |       | 12. C |  |
| 13. A |       | 13. D |  |
| 14. A |       | 14. C |  |
| 15. B |       | 15. A |  |
| 16. A |       | 16. D |  |
| 17. C |       | 17. C |  |
| 18. B |       | 18. D |  |
| 19. D |       | 19. A |  |
| 20. C |       | 20. A |  |
| 21. C |       | 21. D |  |
| 22. D |       | 22. C |  |
| 23. B |       | 23. B |  |
| 24. B |       | 24. B |  |
| 25. B |       | 25. C |  |
| 26. B |       | 26. B |  |
| 27. C |       | 27. D |  |
| 28. C |       | 28. B |  |
| 29. A |       | 29. C |  |
| 30. B |       | 30. D |  |
| 31. A |       | 31. B |  |
| 32. A |       | 32. D |  |
| 33. A |       | 33. D |  |
| 34. A |       | 34. C |  |
| 35. D |       | 35. D |  |
| 36. A |       | 36. D |  |
| 37. A |       | 37. B |  |
| 38. C |       |       |  |
| 39. B |       |       |  |
| 40. B |       |       |  |

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## Unit 5

# ELECTROCHEMISTRY

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IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 5.1 Electrolysis
- 5.2 Fuel cell

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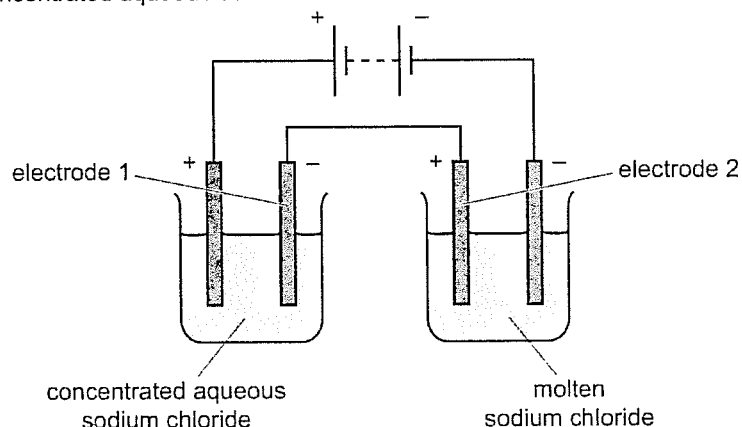
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## Unit 5: Electrochemistry

### 5.1: Electrolysis

1. O/N 22/P22/Q10

The electrolysis of concentrated aqueous sodium chloride and molten sodium chloride is shown.

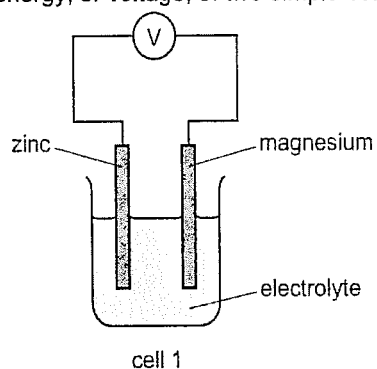


What are the products at electrodes 1 and 2?

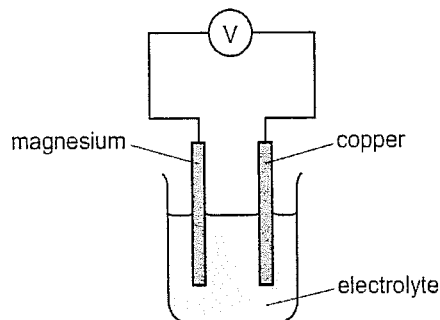
|   | electrode 1 | electrode 2 |
|---|-------------|-------------|
| A | chlorine    | chlorine    |
| B | hydrogen    | chlorine    |
| C | hydrogen    | sodium      |
| D | sodium      | sodium      |

2. O/N 22/P22/Q13

The electrical energy, or voltage, of two simple cells is measured.



cell 1



cell 2

statement 1 The voltage of cell 1 is greater than cell 2.

statement 2 Zinc is more reactive than copper.

statement 3 Magnesium is oxidised in both cells.

statement 4 Magnesium atoms lose electrons to form magnesium ions.

Which option is correct?

A All the statements are correct.

B Only statements 1 and 3 are correct.

C Statement 2 is correct and explains statement 1.

D Statement 4 is correct and explains statement 3.

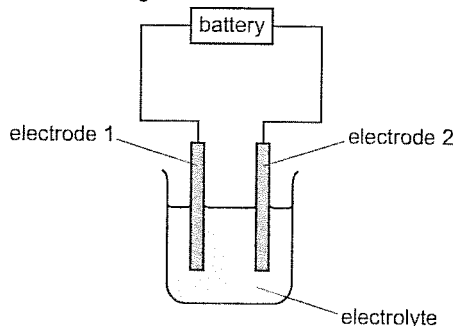
## 3. O/N 22/P22/Q14

Dilute aqueous sodium chloride is electrolysed using carbon electrodes.  
What is the product at the anode?

- A carbon dioxide      B hydrogen      C oxygen      D sodium

## 4. O/N 22/P21/Q10

In the electrolysis diagram, oxidation is occurring at electrode 1 and reduction at electrode 2.

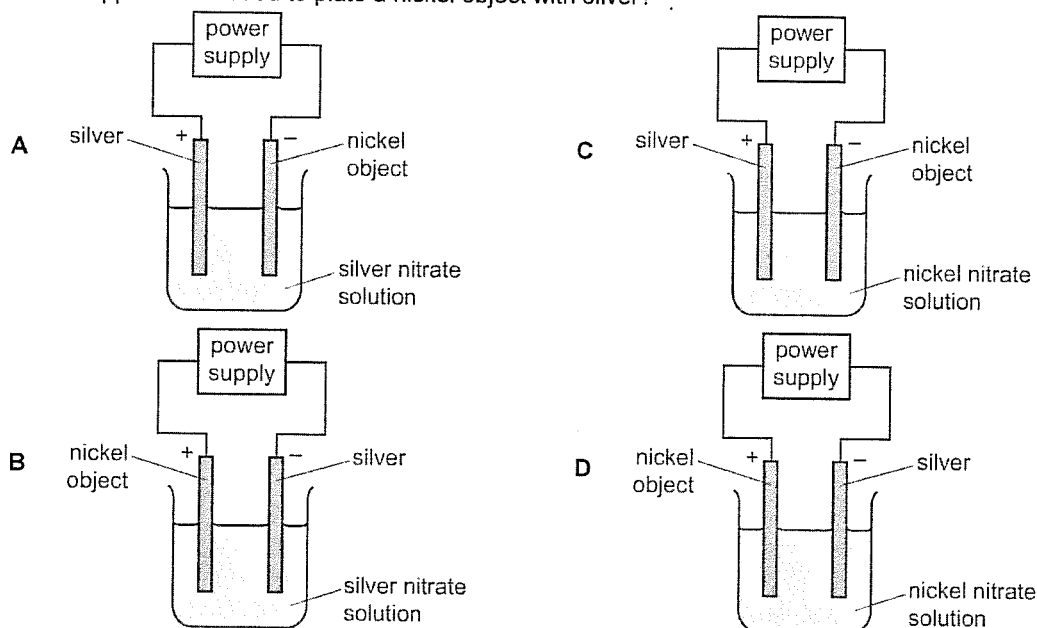


Which row shows the directions of movement of the electrons in the external circuit and of the positive ions in the electrolyte?

|   | direction of movement of electrons in external circuit | direction of movement of positive ions in electrolyte |
|---|--------------------------------------------------------|-------------------------------------------------------|
| A | 1 → 2                                                  | 1 → 2                                                 |
| B | 1 → 2                                                  | 2 → 1                                                 |
| C | 2 → 1                                                  | 1 → 2                                                 |
| D | 2 → 1                                                  | 2 → 1                                                 |

## 5. O/N 22/P23/Q10

Which apparatus is used to plate a nickel object with silver?



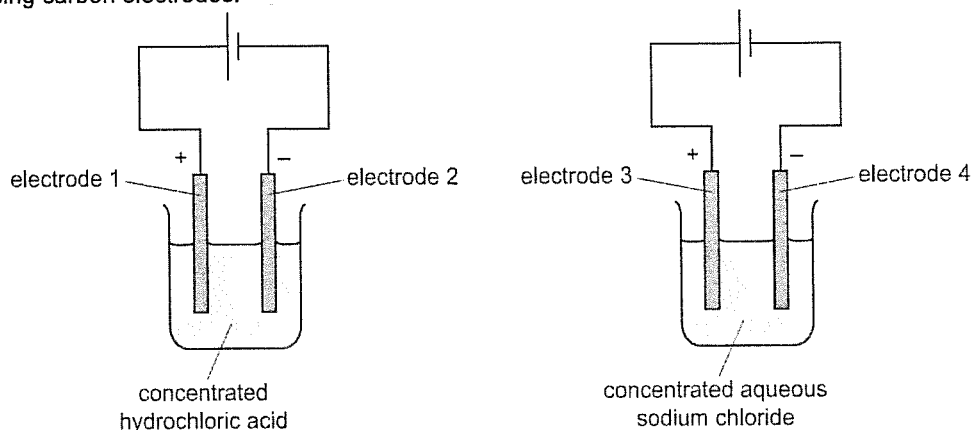
## 6. M/J 22/P21/Q11

Aqueous copper(II) sulfate is electrolysed using copper electrodes.  
What is the ionic half-equation for the reaction at the cathode?

- A  $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$       B  $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$       C  $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$       D  $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$

## 7. M/J 22/P22/Q10, M/J 22/P21/Q10, M/J 22/P23/Q10

The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.

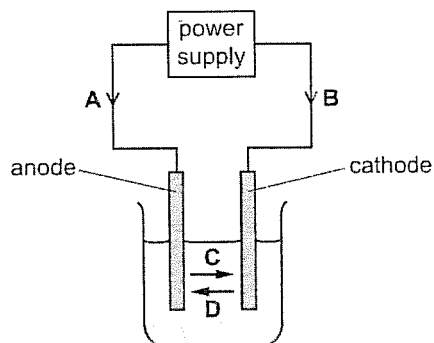


At which electrodes is hydrogen produced?

- A electrode 1 only    B electrodes 1 and 3    C electrode 2 only    D electrodes 2 and 4

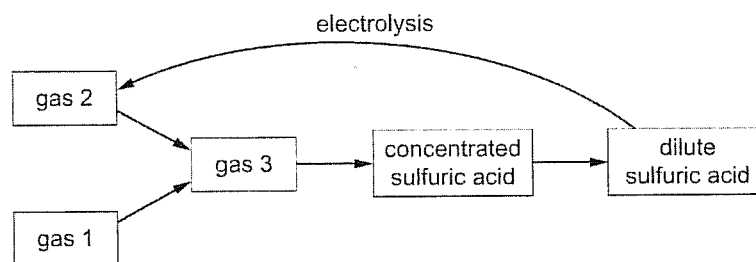
## 8. M/J 22/P22/Q11

The diagram shows the electrolysis of aqueous copper(II) sulfate using inert electrodes. Which arrow shows the movement of electrons in the circuit?



## 9. M/J 22/P21/Q32

The flow chart shows part of the process for the manufacture of sulfuric acid and its electrolysis.



What are gases 1, 2 and 3?

|   | gas 1           | gas 2    | gas 3           |
|---|-----------------|----------|-----------------|
| A | sulfur dioxide  | hydrogen | sulfur trioxide |
| B | sulfur dioxide  | oxygen   | sulfur trioxide |
| C | sulfur trioxide | hydrogen | sulfur dioxide  |
| D | sulfur trioxide | oxygen   | sulfur dioxide  |



**10. M/J 22/P23/Q9**

Aluminium is manufactured from aluminium oxide by electrolysis.

Which row shows the ionic half-equations at each electrode and describes the role of cryolite in the process?

|          | reaction at anode                                     | reaction at cathode                                   | role of cryolite            |
|----------|-------------------------------------------------------|-------------------------------------------------------|-----------------------------|
| <b>A</b> | $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ | $\text{Al}^{3+} + 3\text{e}^- \rightarrow 3\text{Al}$ | catalyst                    |
| <b>B</b> | $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$  | $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ | solvent for aluminium oxide |
| <b>C</b> | $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ | $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$  | solvent for aluminium oxide |
| <b>D</b> | $\text{Al}^{3+} + 3\text{e}^- \rightarrow 3\text{Al}$ | $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ | catalyst                    |

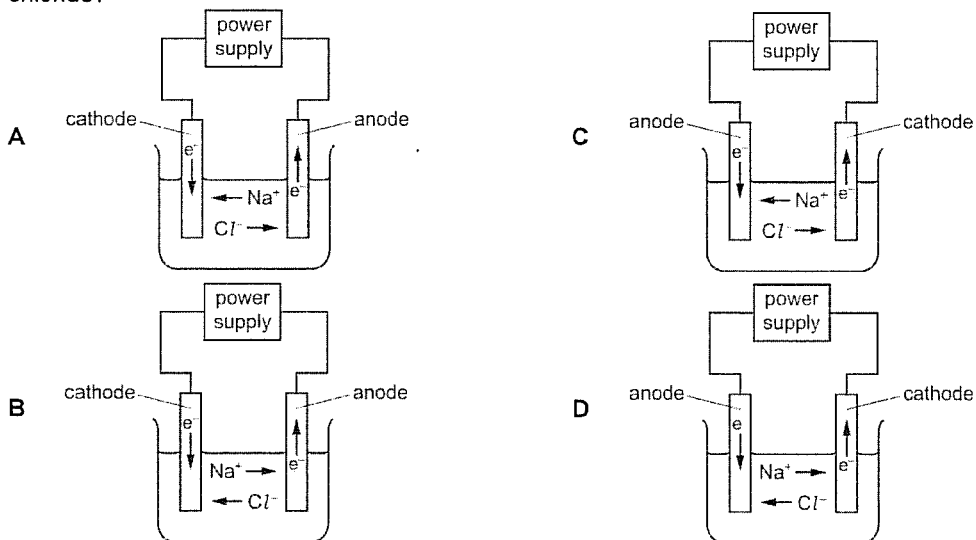
**11. MAR 22/P22/Q6**

Which row explains why copper is a good conductor of electricity at room temperature?

|          | copper ions move freely | electrons move freely |
|----------|-------------------------|-----------------------|
| <b>A</b> | no                      | no                    |
| <b>B</b> | no                      | yes                   |
| <b>C</b> | yes                     | no                    |
| <b>D</b> | yes                     | yes                   |

**12. MAR 22/P22/Q11**

Which diagram shows the direction of movement of ions and electrons during the electrolysis of molten sodium chloride?

**13. MAR 22/P22/Q14**

An object is electroplated with silver using an aqueous silver salt as the electrolyte.

Which row is correct?

|          | the object to be electroplated is the | the other electrode is made from |
|----------|---------------------------------------|----------------------------------|
| <b>A</b> | anode                                 | carbon                           |
| <b>B</b> | anode                                 | silver                           |
| <b>C</b> | cathode                               | carbon                           |
| <b>D</b> | cathode                               | silver                           |

**14. MAR 22/P22/Q13**

Molten sodium chloride and concentrated aqueous sodium chloride are electrolysed using platinum electrodes. What are the products at the negative electrode (cathode) in each electrolysis?

|          | molten sodium chloride | concentrated aqueous sodium chloride |
|----------|------------------------|--------------------------------------|
| <b>A</b> | hydrogen               | hydrogen                             |
| <b>B</b> | hydrogen               | sodium                               |
| <b>C</b> | sodium                 | hydrogen                             |
| <b>D</b> | sodium                 | sodium                               |

**15. O/N 21/P22/Q13**

Concentrated aqueous sodium chloride is electrolysed. Which equation represents the reaction at the cathode?

- A**  $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$       **B**  $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$       **C**  $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$       **D**  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

**16. O/N 21/P21/Q10**

Which statements about the products of electrolysis, using inert electrodes, are correct?

- 1 When molten lead(II) bromide is electrolysed, bromine is formed at the cathode.  
 2 When dilute sulfuric acid is electrolysed, oxygen is formed at the anode.  
 3 When concentrated aqueous sodium chloride is electrolysed, sodium is formed at the cathode.  
 4 When concentrated hydrochloric acid is electrolysed, chlorine is formed at the anode.
- A** 1 and 2      **B** 1 and 3      **C** 2 and 4      **D** 3 and 4

**17. O/N 21/P23/Q10**

Effervescence is observed at the negative electrode (cathode) during the electrolysis of concentrated aqueous sodium chloride.

Which element is produced at the negative electrode (cathode)?

- A** chlorine      **B** hydrogen      **C** oxygen      **D** sodium

**18. M/J 21/P22/Q10, M/J 21/P21/Q10, M/J 21/P23/Q10**

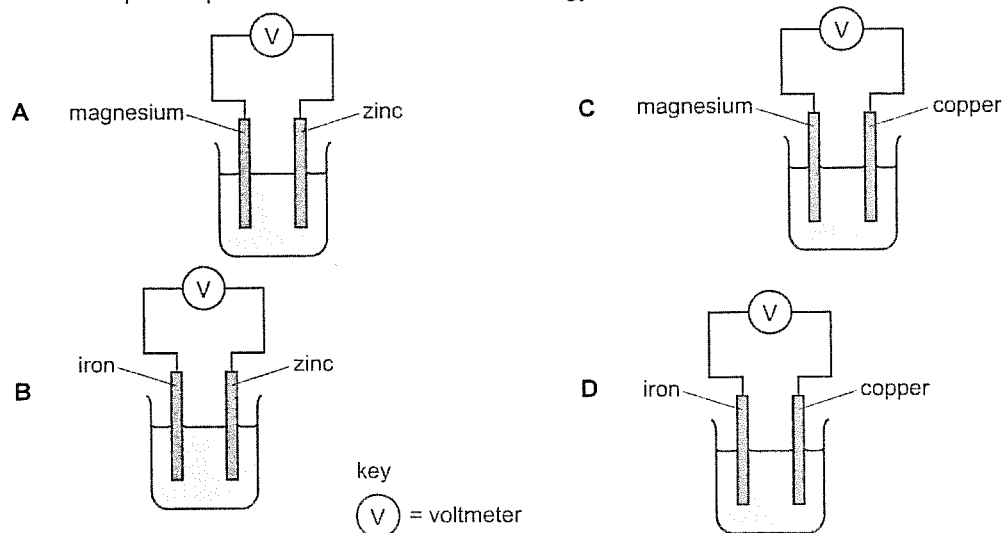
In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.

What would happen in **both** experiments?

- A** A halogen would be formed at the anode.      **C** Hydrogen would be formed at the anode.  
**B** A metal would be formed at the cathode.      **D** Hydrogen would be formed at the cathode.

**19. M/J 21/P23/Q13**

Which simple cell produces the most electrical energy?



**20. M/J 21/P21/Q11**

What is the ionic half-equation for the reaction that occurs at the cathode when molten lead(II) bromide is electrolysed?

- A  $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$     B  $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$     C  $\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$     D  $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$

**21. M/J 21/P23/Q31**

Metal X is a good conductor of electricity and is used for electrical wiring.

Metal Y is used to make an alloy which is resistant to corrosion and is used to make cutlery.

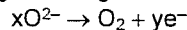
Metal Z is light and strong and is used in the manufacture of aircraft.

What are X, Y and Z?

|   | X         | Y         | Z         |
|---|-----------|-----------|-----------|
| A | aluminium | iron      | copper    |
| B | copper    | iron      | aluminium |
| C | aluminium | copper    | iron      |
| D | copper    | aluminium | iron      |

**22. MAR 21/P22/Q9**

The ionic half-equation for the formation of oxygen during the electrolysis of aluminium oxide is shown.



What are the values of x and y?

|   | x | y |
|---|---|---|
| A | 1 | 2 |
| B | 1 | 4 |
| C | 2 | 2 |
| D | 2 | 4 |

**23. MAR 21/P22/Q12**

During the electrolysis of dilute sulfuric acid, hydrogen is collected at the cathode.

What is the ionic half-equation for this reaction?

- A  $\text{H}^+ + \text{e}^- \rightarrow \text{H}$     B  $\text{H}^+ \rightarrow \text{H} + \text{e}^-$     C  $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$     D  $2\text{H}^+ \rightarrow \text{H}_2 + 2\text{e}^-$

**24. O/N 20/P21/Q14**

Which reaction takes place at the cathode during the electrolysis of molten nickel (II) chloride?

- A  $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$     B  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$     C  $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$     D  $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$

**25. O/N 20/P23/Q13**

Electrolysis is carried out on dilute aqueous potassium bromide.

Which products are formed at the anode and the cathode?

|   | anode    | cathode   |
|---|----------|-----------|
| A | bromine  | hydrogen  |
| B | bromine  | potassium |
| C | hydrogen | bromine   |
| D | hydrogen | potassium |

**26. M/J 20/P21/Q10**

Dilute aqueous sodium chloride is electrolyzed using platinum electrodes.

What is the half-equation for the reaction at the cathode?

- A  $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$     C  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$   
 B  $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$     D  $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$

**27. M/J 20/P22/Q10**

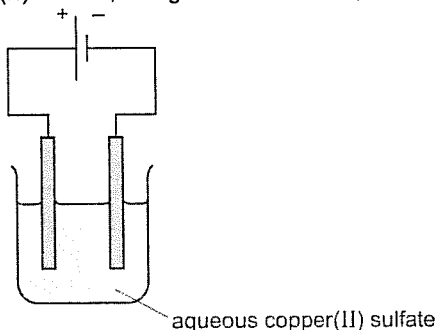
Electrolytes can be broken down by electrolysis.

Which rows are correct for each electrolyte?

|   | electrolyte                         | reaction at cathode                                   | product at anode |
|---|-------------------------------------|-------------------------------------------------------|------------------|
| 1 | dilute aqueous sodium chloride      | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$    | oxygen           |
| 2 | concentrated hydrochloric acid      | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$    | chlorine         |
| 3 | molten aluminium oxide              | $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ | aluminium        |
| 4 | concentrated aqueous sodium bromide | $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$      | bromine          |

**A** 1 and 2**B** 1 and 4**C** 2 and 3**D** 3 and 4**28. M/J 20/P22/Q11, M/J 20/P21/Q11, M/J 20/P23/Q11**

The electrolysis of aqueous copper (II) sulfate, using inert electrodes, is shown.



Which statement about a reaction at an electrode is correct?

- A** Copper ions gain electrons at the negative electrode.  
**B** Copper ions gain electrons at the positive electrode.  
**C** Hydrogen ions gain electrons at the negative electrode.  
**D** Hydrogen ions gain electrons at the positive electrode.

**29. M/J 20/P23/Q10**

Which row describes the reactions during the electrolysis of dilute aqueous sodium chloride?

|          | anode (+) reaction                                                        | cathode (-) reaction                                                      |
|----------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>A</b> | $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$                        | $2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$ |
| <b>B</b> | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$                        | $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ |
| <b>C</b> | $2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$ | $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$                        |
| <b>D</b> | $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$                        |

**30. MAR 20/P22/Q10**

Which statements about the electrolysis of molten lead (II) bromide are correct?

- 1 Lead ions move to the anode and are oxidized.  
 2 Lead ions move to the cathode and are reduced.  
 3 Bromide ions move to the anode and are oxidized.  
 4 Bromide ions move to the cathode and are reduced.

**A** 1 and 3**B** 1 and 4**C** 2 and 3**D** 2 and 4**31. MAR 20/P22/Q11**

Aqueous copper (II) sulfate is electrolyzed using carbon electrodes.

Which statement is correct?

- A** Bubbles of hydrogen are formed at the anode.  
**B** Bubbles of oxygen gas are formed at the cathode.  
**C** Copper is deposited at the anode.  
**D** The blue color of the solution fades.

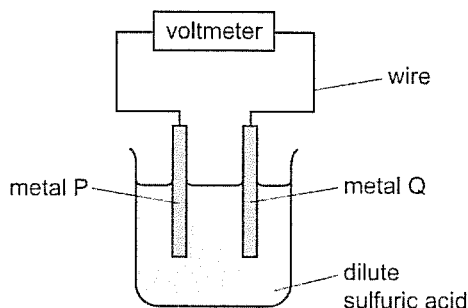
**32. MAR 20/P22/Q27**

Aluminium is extracted from bauxite by electrolysis. Which statement is correct?

- A Aluminium ions are oxidised to form aluminium.
- B The cathode has to be replaced regularly because it reacts with the oxygen which is formed.
- C Cryolite is added to remove impurities.
- D Carbon dioxide is produced at the anode.

**33. O/N 19/P22/Q11**

The diagram shows a simple cell.



Which pair of metals produces the largest voltage?

|   | metal P   | metal Q |
|---|-----------|---------|
| A | magnesium | iron    |
| B | magnesium | copper  |
| C | zinc      | iron    |
| D | zinc      | copper  |

**34. O/N 19/P22/Q12, O/N 19/P21/Q12, O/N 19/P23/Q12**

What are the ionic half-equations for the electrode reactions during the electrolysis of concentrated aqueous sodium chloride?

|   | anode                                                | cathode                                              |
|---|------------------------------------------------------|------------------------------------------------------|
| A | $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ | $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$   |
| B | $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$   |
| C | $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$   | $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ |
| D | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$   | $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ |

**35. O/N 19/P21/Q11**

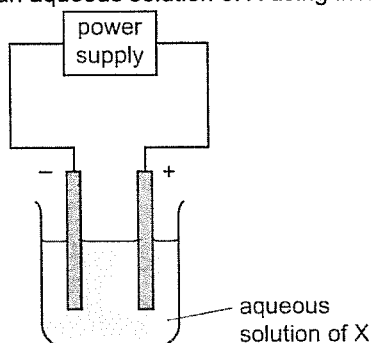
Which rows correctly show cathode and anode products from the electrolysis of the named electrolyte?

|   | electrolyte                                         | cathode product | anode product |
|---|-----------------------------------------------------|-----------------|---------------|
| 1 | copper(II) sulfate solution using copper electrodes | copper          | oxygen        |
| 2 | molten lead(II) bromide                             | lead            | bromine       |
| 3 | dilute sodium bromide solution                      | hydrogen        | oxygen        |
| 4 | copper(II) sulfate solution using carbon electrodes | hydrogen        | oxygen        |

- A 1 and 2 only      B 1 and 4 only      C 2 and 3 only      D 3 and 4 only

## 36. M/J 19/P22/Q9

The diagram shows the electrolysis of an aqueous solution of X using inert electrodes.



Hydrogen is produced at the cathode and chlorine is produced at the anode.  
What is X?

- A concentrated copper(II) chloride solution      C dilute hydrochloric acid  
B concentrated hydrochloric acid      D dilute sodium chloride solution

## 37. M/J 19/P21/Q9

Which statement about the electrolysis of copper(II) sulfate solution using carbon electrodes is correct?

- A A colourless gas is produced at the anode.  
B A colourless gas is produced at the cathode.  
C The colour of the electrolyte remains the same.  
D The mass of both electrodes remains constant.

## 38. M/J 19/P21/Q10

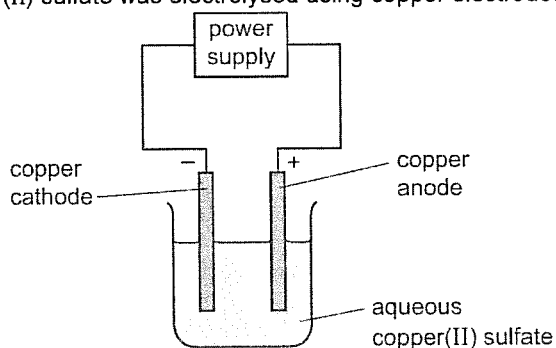
Aluminium metal is extracted from aluminium oxide by electrolysis.

Which ionic half-equation describes a reaction that occurs at the named electrode?

|   | ionic half-equation                                   | electrode |
|---|-------------------------------------------------------|-----------|
| A | $2\text{O}^{2-} \rightarrow \text{O}_2 + 2\text{e}^-$ | anode     |
| B | $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$  | anode     |
| C | $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ | cathode   |
| D | $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$  | cathode   |

## 39. M/J 19/P23/Q9

An aqueous solution of copper(II) sulfate was electrolysed using copper electrodes.



Which equation for the reaction at the anode is correct?

- A  $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$       B  $\text{Cu} + 2\text{e}^- \rightarrow \text{Cu}^{2+}$       C  $\text{Cu}^{2+} \rightarrow \text{Cu} + 2\text{e}^-$       D  $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

## 40. MAR 19/P22/Q10

Which substance is **not** produced during the electrolysis of concentrated aqueous sodium chloride?

- A chlorine      B hydrogen      C sodium      D sodium hydroxide

**41. MAR 19/P22/Q11**

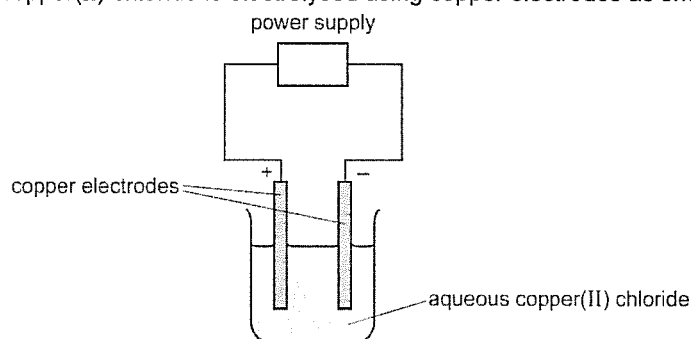
Aqueous copper(II) sulfate is electrolysed using copper electrodes.

What are the ionic half-equations for the reactions that occur at each electrode?

|          | anode                                                                     | cathode                                              |
|----------|---------------------------------------------------------------------------|------------------------------------------------------|
| <b>A</b> | $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$                      | $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ |
| <b>B</b> | $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$                      | $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ |
| <b>C</b> | $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ | $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ |
| <b>D</b> | $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$   |

**42. O/N 18/P22/Q10**

Concentrated aqueous copper(II) chloride is electrolysed using copper electrodes as shown.

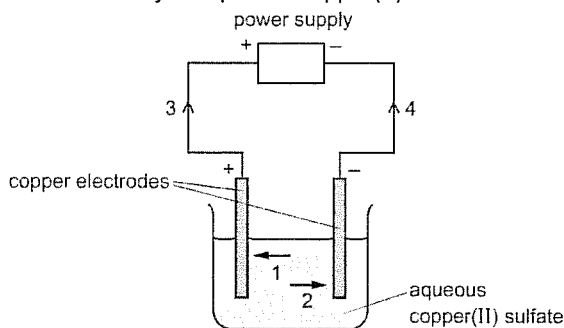


What happens to the mass of each electrode during this process?

|          | positive electrode | negative electrode |
|----------|--------------------|--------------------|
| <b>A</b> | decreases          | decreases          |
| <b>B</b> | decreases          | increases          |
| <b>C</b> | increases          | decreases          |
| <b>D</b> | increases          | increases          |

**43. O/N 18/P22/Q11, O/N 18/P21/Q11, O/N 18/P23/Q11**

The diagram shows a circuit used to electrolyse aqueous copper(II) sulfate.



Which arrows indicate the movement of the copper ions in the electrolyte and of the electrons in the external circuit?

|          | copper ions | electrons |
|----------|-------------|-----------|
| <b>A</b> | 1           | 3         |
| <b>B</b> | 1           | 4         |
| <b>C</b> | 2           | 3         |
| <b>D</b> | 2           | 4         |

**44. O/N 18/P21/Q10**

Aqueous copper(II) sulfate is electrolysed using carbon electrodes.  
What is the product at each electrode?

|   | product at the positive electrode | product at the negative electrode |
|---|-----------------------------------|-----------------------------------|
| A | copper                            | oxygen                            |
| B | hydrogen                          | oxygen                            |
| C | oxygen                            | copper                            |
| D | oxygen                            | hydrogen                          |

**45. O/N 18/P23/Q10**

Electrolysis of copper(II) sulfate can be done using either carbon electrodes or copper electrodes.  
Which statement describes what happens at the positive electrode?

- A Copper is deposited if the electrode is made from carbon.
- B Copper is deposited if the electrode is made from copper.
- C Oxygen gas is produced if the electrode is made from carbon.
- D Oxygen gas is produced if the electrode is made from copper.

**46. M/J 18/P22/Q10**

Aqueous copper(II) sulfate is electrolysed using copper electrodes.  
Which statement about the electrolysis is **not** correct?

- A An oxidation reaction occurs at the positive electrode.
- B The current is carried through the electrolyte by ions.
- C The negative electrode gains mass.
- D The number of copper(II) ions in the electrolyte decreases.

**47. M/J 18/P22/Q11, M/J 18/P21/Q11, M/J 18/P23/Q11**

Dilute sulfuric acid is electrolysed using inert electrodes.  
What are the ionic half-equations for the reactions that take place at each electrode?

|   | positive electrode                                                        | negative electrode                                                        |
|---|---------------------------------------------------------------------------|---------------------------------------------------------------------------|
| A | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$                        | $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ |
| B | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$                        | $4\text{OH}^- + 4\text{H}^+ \rightarrow 4\text{H}_2\text{O}$              |
| C | $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$                        |
| D | $4\text{OH}^- + 4\text{H}^+ \rightarrow 4\text{H}_2\text{O}$              | $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$                        |

**48. M/J 18/P21/Q10**

Aqueous copper(II) sulfate is electrolysed using copper electrodes.  
Which statement is correct?

- A A reduction reaction occurs at the positive electrode.
- B The blue colour of the solution becomes darker.
- C The concentration of copper ions in the solution decreases.
- D The mass of the negative electrode increases.

**49. M/J 18/P23/Q10**

Aqueous copper(II) sulfate is electrolysed using copper electrodes.  
Which statement is correct?

- A Oxygen gas is produced at the positive electrode.
- B The blue colour of the solution gradually fades.
- C The concentration of copper ions in the solution stays the same.
- D The mass of the negative electrode decreases.



**50. MAR 18/P22/Q9**

A solution of copper(II) sulfate can be electrolysed using copper electrodes or carbon electrodes. Which statements are correct?

- 1 Using copper electrodes, oxygen gas forms at the anode.
  - 2 Using copper electrodes, copper atoms lose electrons at the anode.
  - 3 Using carbon electrodes, copper metal forms at the cathode.
  - 4 Using carbon electrodes, copper ions gain electrons at the cathode.
- A 1 and 2                      B 1 and 3                      C 2, 3 and 4                      D 4 only

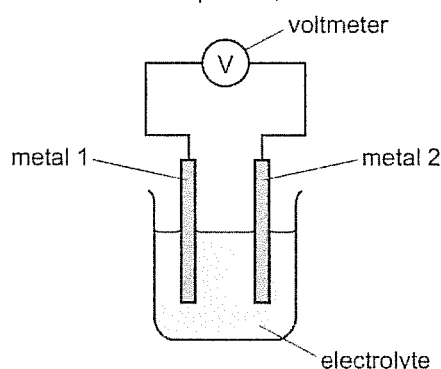
**51. O/N 17/P22/Q9, O/N 17/P21/Q9, O/N 17/P23/Q9**

Which statements about the electrolysis of concentrated copper(II) chloride are correct?

- 1 Electrons are transferred from the cathode to the copper(II) ions.
  - 2 Electrons move round the external circuit from the cathode to the anode.
  - 3 Chloride ions are attracted to the anode.
  - 4 Hydroxide ions transfer electrons to the cathode.
- A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**52. MAR 18/P22/Q10**

Pairs of metals are connected together to make a simple cell, as shown.



The table shows the reading on the voltmeter when different metals are used.

|         |           | metal 2   |         |         |           |
|---------|-----------|-----------|---------|---------|-----------|
|         |           | beryllium | cerium  | cobalt  | manganese |
| metal 1 | beryllium | 0.00 V    | +0.64 V | -1.57 V | -0.67 V   |
|         | cerium    |           | 0.00 V  | -2.21 V | -1.30 V   |
|         | cobalt    |           |         | 0.00 V  | +0.90 V   |
|         | manganese |           |         |         | 0.00 V    |

If metal 2 is more reactive than metal 1, the voltage measured is positive.

The greater the difference in reactivity of the metals, the larger the reading on the voltmeter.

What is the order of reactivity?

|   | <div style="display: flex; align-items: center; justify-content: space-between;"> <span>most reactive</span> <span>→</span> <span>least reactive</span> </div> |           |           |           |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|
| A | cerium                                                                                                                                                         | beryllium | cobalt    | manganese |
| B | cerium                                                                                                                                                         | beryllium | manganese | cobalt    |
| C | cobalt                                                                                                                                                         | manganese | beryllium | cerium    |
| D | cobalt                                                                                                                                                         | manganese | cerium    | beryllium |

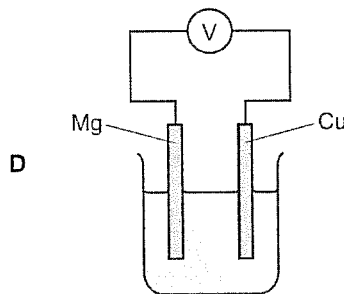
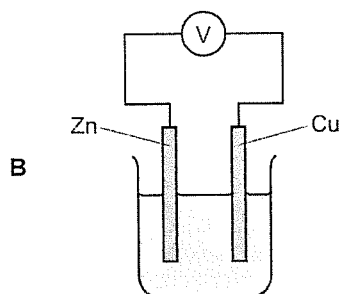
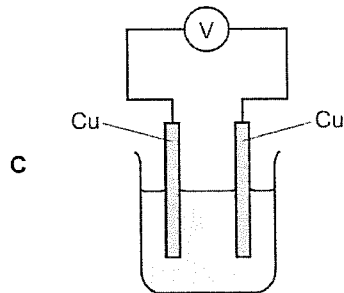
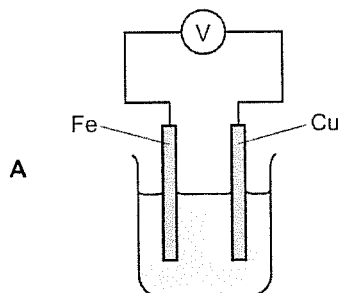
**53. O/N 17/P22/Q26**

Aluminium is extracted from bauxite by electrolysis.  
Which row shows the anode material and the anode reaction?

|   | anode material | anode reaction                     |
|---|----------------|------------------------------------|
| A | carbon         | $Al^{3+} + 3e^{-} \rightarrow Al$  |
| B | carbon         | $2O^{2-} \rightarrow O_2 + 4e^{-}$ |
| C | steel          | $Al^{3+} + 3e^{-} \rightarrow Al$  |
| D | steel          | $2O^{2-} \rightarrow O_2 + 4e^{-}$ |

**54. O/N 17/P22/Q10, O/N 17/P21/Q10, O/N 17/P23/Q10**

Which metal combination produces the highest voltage reading in the cells shown?

**55. M/J 17/P22/Q9, M/J 17/P21/Q9, M/J 17/P23/Q9**

Which statement about electrolysis is correct?

- A Electrons move through the electrolyte from the cathode to the anode.
- B Electrons move towards the cathode in the external circuit.
- C Negative ions move towards the anode in the external circuit.
- D Positive ions move through the electrolyte towards the anode during electrolysis.

**56. MAR 17/P22/Q11**

Four solutions are separately electrolysed.

| experiment | solution                       | electrodes |
|------------|--------------------------------|------------|
| 1          | dilute aqueous sodium chloride | carbon     |
| 2          | aqueous copper(II) sulfate     | copper     |
| 3          | concentrated hydrochloric acid | carbon     |
| 4          | dilute sulfuric acid           | carbon     |

In which two experiments is a colourless gas evolved at the anode?

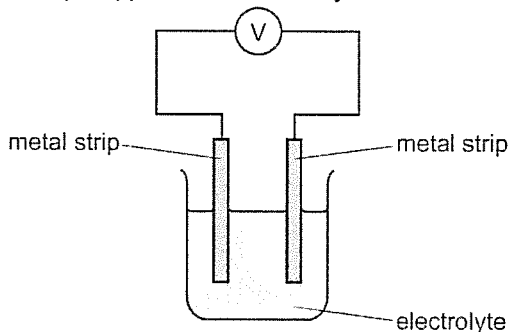
- A 1 and 2      B 1 and 4      C 2 and 3      D 3 and 4

**57. M/J 17/P22/Q10, M/J 17/P21/Q10, M/J 17/P23/Q10**

The reactivity series for a number of different metals is shown.

|               |      |      |        |        |          |                |
|---------------|------|------|--------|--------|----------|----------------|
| most reactive |      | →    |        |        |          | least reactive |
| magnesium     | zinc | iron | copper | silver | platinum |                |

The diagram shows different metal strips dipped into an electrolyte.

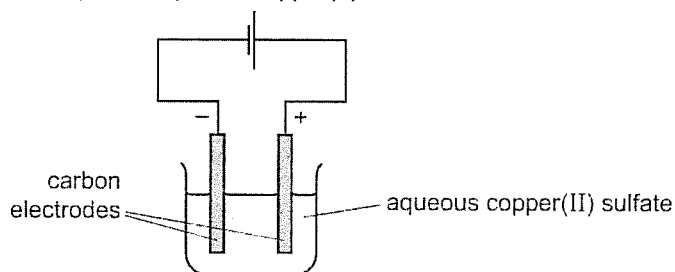


Which pair of metals produces the highest voltage?

- A copper and magnesium  
B magnesium and platinum  
C magnesium and zinc  
D silver and platinum

**58. MAR 17/P22/Q10**

The diagram shows the electrolysis of aqueous copper(II) sulfate.



Which statement is correct?

- A Copper metal is deposited at the positive electrode.  
B In the external circuit the electrons move from positive to negative.  
C In the solution the electrons move from negative to positive.  
D Oxygen gas is produced at the positive electrode.

**59. O/N 16/P21/Q17**

Four ionic half-equations are shown.

- 1  $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$
- 2  $2\text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2\text{e}^-$
- 3  $\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{e}^-$
- 4  $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$

Which statement is correct?

- A In equation 1, copper(II) ions are oxidised to copper.  
B In equation 2, iodide ions are reduced to iodine.  
C In equation 3, iron(II) ions are oxidised to iron(III) ions.  
D In equation 4, chlorine is oxidised to chloride ions.

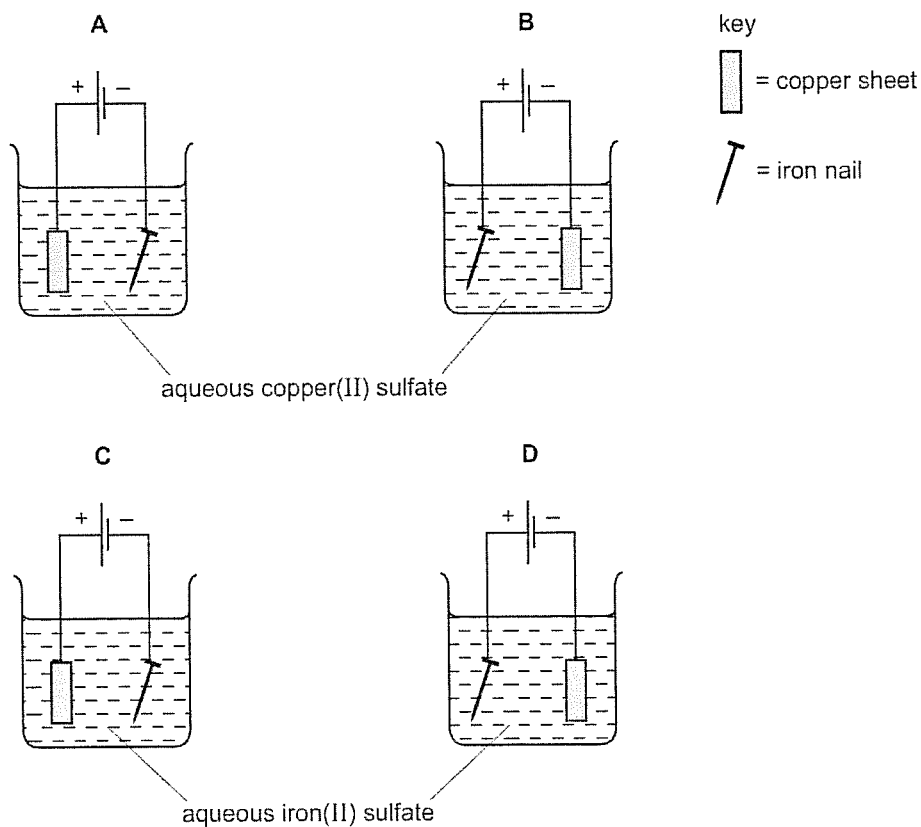
**60. O/N 16/P22/Q11**

A student sets up a number of simple cells by putting strips of two different metals into dilute sulfuric acid. Which cell produces the highest voltage?

- A copper and magnesium  
B copper and zinc  
C iron and copper  
D magnesium and zinc

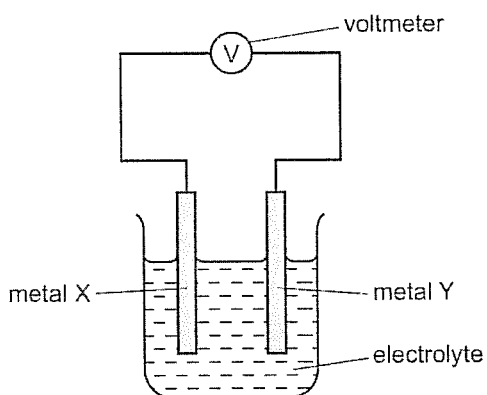
61. O/N 16/P22/Q10, O/N 16/P21/Q10, O/N 16/P23/Q10

Which apparatus could be used to electroplate an iron nail with copper?



62. O/N 16/P21/Q11

The diagram shows a simple cell.

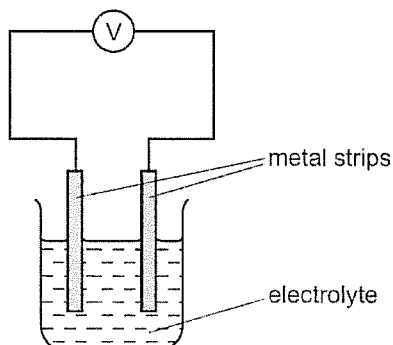


Which two metals produce the highest reading on the voltmeter?

|   | X         | Y      |
|---|-----------|--------|
| A | magnesium | copper |
| B | magnesium | iron   |
| C | zinc      | copper |
| D | zinc      | iron   |

## 63. O/N 16/P23/Q11

The diagram shows two different metal strips dipped into an electrolyte.

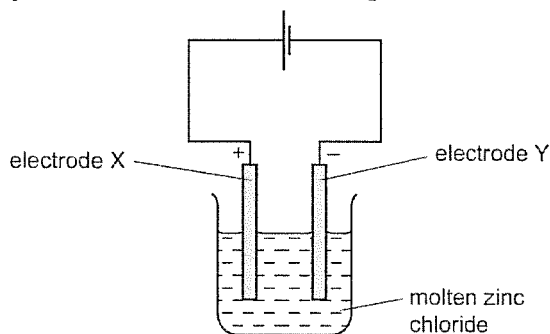


Which pair of metals produces the highest voltage?

- |                        |                      |
|------------------------|----------------------|
| A copper and iron      | C copper and zinc    |
| B copper and magnesium | D magnesium and iron |

## 64. M/J 16/P22/Q10

The diagram shows the electrolysis of molten zinc chloride,  $\text{ZnCl}_2$ .

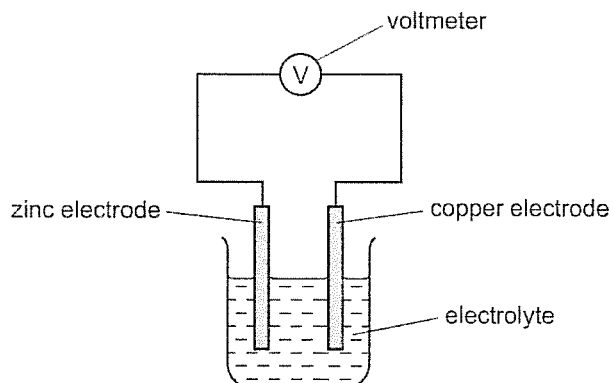


Which statement is correct?

- A Oxidation occurs at electrode X and the equation is:  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ .  
 B Oxidation occurs at electrode Y and the equation is:  $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$ .  
 C Reduction occurs at electrode X and the equation is:  $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$ .  
 D Reduction occurs at electrode Y and the equation is:  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ .

## 65. M/J 16/P21/Q12

The diagram shows a simple cell.

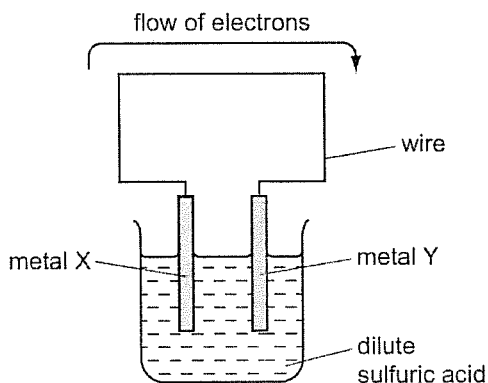


Which statement about the process occurring when the cell is in operation is correct?

- |                                                 |                                                                             |
|-------------------------------------------------|-----------------------------------------------------------------------------|
| A $\text{Cu}^{2+}$ ions are formed in solution. | C The reaction $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ occurs. |
| B Electrons travel through the solution.        | D The zinc electrode increases in mass.                                     |

**66. M/J 16/P22/Q12**

The diagram shows a simple cell.

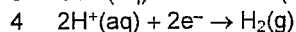
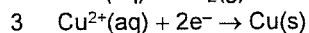
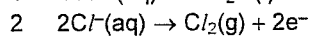
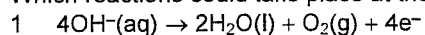


For which pair of metals would electrons flow from metal X to metal Y?

|   | X      | Y    |
|---|--------|------|
| A | copper | iron |
| B | copper | zinc |
| C | iron   | zinc |
| D | zinc   | iron |

**67. M/J 16/P21/Q10**

Which reactions could take place at the anode during electrolysis?



A 1 and 2

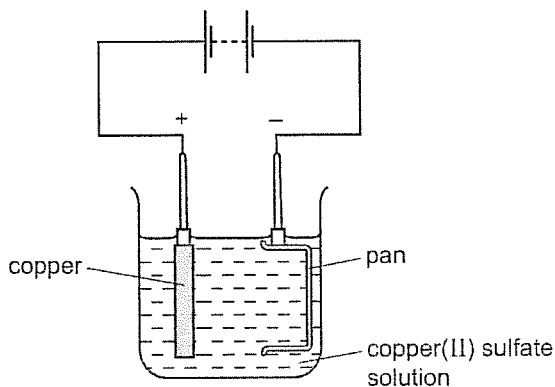
B 1 and 4

C 2 and 4

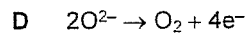
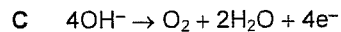
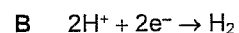
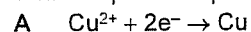
D 3 and 4

**68. M/J 16/P23/Q10**

The diagram shows a method used to copper-plate a pan

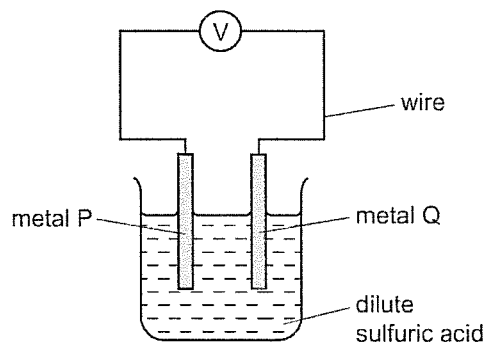


Which equation represents the reaction at the cathode?



## 69. M/J 16/P23/Q12

The diagram shows a simple cell.



Which pair of metals produces the largest voltage?

|   | metal P   | metal Q |
|---|-----------|---------|
| A | iron      | copper  |
| B | magnesium | copper  |
| C | magnesium | zinc    |
| D | zinc      | copper  |

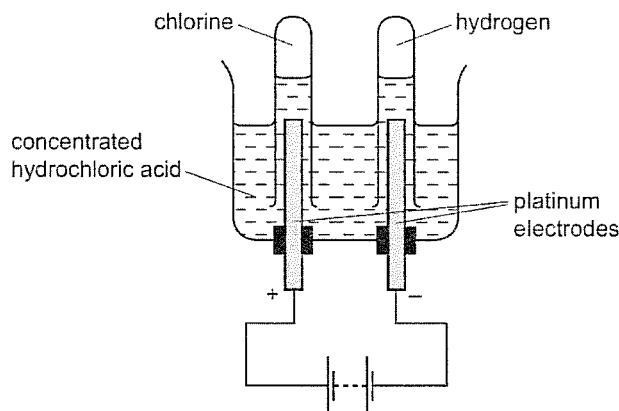
## 70. O/N 15/P11/Q8

Which row describes the electrolysis of molten potassium bromide?

|   | product at anode | product at cathode |
|---|------------------|--------------------|
| A | bromine          | hydrogen           |
| B | bromine          | potassium          |
| C | hydrogen         | bromine            |
| D | potassium        | bromine            |

## 71. MAR 16/P22/Q10

The electrolysis of concentrated hydrochloric acid is shown.

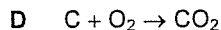
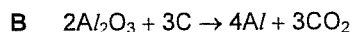
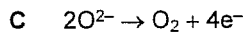
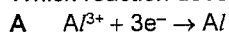


Which statement describes what happens to the electrons during the electrolysis?

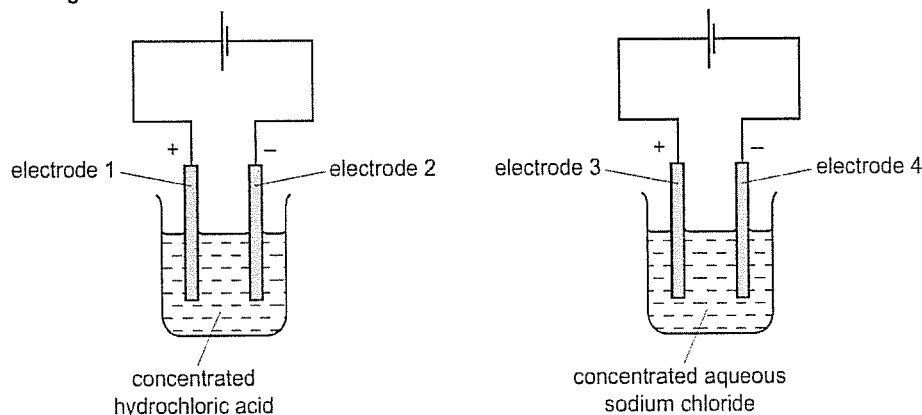
- A They are added to chloride ions.
- B They are added to hydrogen ions.
- C They move through the circuit from positive to negative.
- D They move through the solution from negative to positive.

**72. MAR 16/P22/Q11**

Which reaction does **not** occur in the extraction of aluminium?

**73. O/N 15/P12/Q8**

The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrode(s) is hydrogen produced?

- A** electrode 1 only    **B** electrodes 1 and 3    **C** electrode 2 only    **D** electrodes 2 and 4

**74. O/N 15/P13/Q8**

What are the electrode products when molten silver iodide is electrolysed between inert electrodes?

|          | cathode  | anode  |
|----------|----------|--------|
| <b>A</b> | hydrogen | iodine |
| <b>B</b> | iodine   | silver |
| <b>C</b> | silver   | iodine |
| <b>D</b> | silver   | oxygen |

**75. M/J 15/P12/Q9, M/J 15/P11/Q9, M/J 15/P13/Q9**

Copper and hydrogen can each be formed by electrolysis.

At which electrodes are these elements formed?

|          | copper  | hydrogen |
|----------|---------|----------|
| <b>A</b> | anode   | anode    |
| <b>B</b> | anode   | cathode  |
| <b>C</b> | cathode | anode    |
| <b>D</b> | cathode | cathode  |

**76. M/J 15/P13/Q10**

An object is electroplated with silver using an aqueous silver salt as the electrolyte.

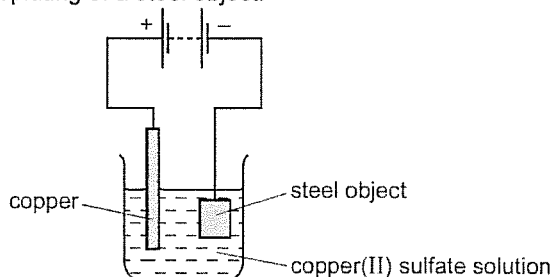
Which set of conditions is used?

|          | the object to be electroplated is the | the other electrode is made from |
|----------|---------------------------------------|----------------------------------|
| <b>A</b> | anode                                 | carbon                           |
| <b>B</b> | anode                                 | silver                           |
| <b>C</b> | cathode                               | carbon                           |
| <b>D</b> | cathode                               | silver                           |



**77. M/J 15/P12/Q10**

The diagram shows the electroplating of a steel object.



A student made the following statements.

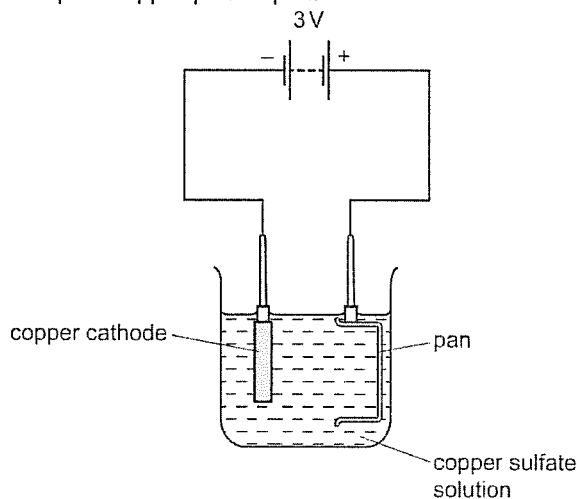
- 1 The object turns a reddish-brown colour.
- 2 The copper sulfate solution changes to a paler blue colour.
- 3 The copper electrode becomes smaller.

Which statements are correct?

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

**78. M/J 15/P11/Q10**

The diagram shows a failed attempt to copper-plate a pan.

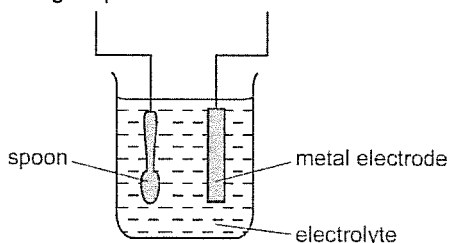


Which action will plate the pan with copper?

- A cooling the copper sulfate solution in an ice bath
- B heating the copper sulfate solution to boiling point
- C increasing the voltage from 3 V to 6 V
- D making the pan the cathode and the copper the anode

**79. MAR 15/P12/Q9, M/J 10/P12/Q8, M/J 10/P11/Q11, M/J 10/P13/Q8**

The diagram shows apparatus for plating a spoon with silver.

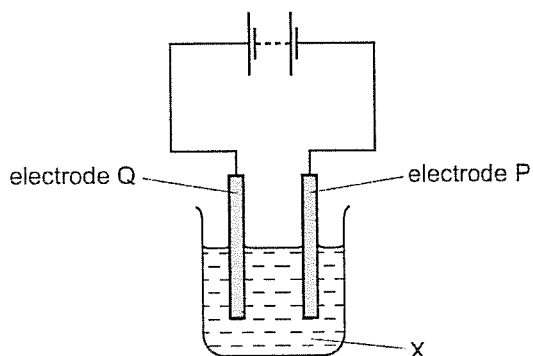


Which statement is **not** correct?

- A Silver would stick to the spoon because it is a very reactive metal.
- B The electrolyte would be a silver salt dissolved in water.
- C The metal electrode would be made from silver.
- D The spoon would be connected to the negative terminal of the power supply.

**80. MAR 15/P12/Q8**

The diagram shows an electrolysis experiment.



During the electrolysis, sodium was formed at electrode P and chlorine at electrode Q. Which row correctly identifies P, Q and X?

|   | P       | Q       | X                                                 |
|---|---------|---------|---------------------------------------------------|
| A | anode   | cathode | concentrated solution of sodium chloride in water |
| B | anode   | cathode | molten sodium chloride                            |
| C | cathode | anode   | concentrated solution of sodium chloride in water |
| D | cathode | anode   | molten sodium chloride                            |

**81. O/N 14/P12/Q10, O/N 14/P11/Q11**

Which metal could **not** be used for electroplating by using an aqueous solution?

- A chromium      B copper      C silver      D sodium

**82. O/N 14/P12/Q11, O/N 14/P11/Q12**

Which products are formed at the electrodes when a concentrated solution of sodium chloride is electrolysed?

|   | cathode (–) | anode (+) |
|---|-------------|-----------|
| A | hydrogen    | chlorine  |
| B | hydrogen    | oxygen    |
| C | sodium      | chlorine  |
| D | sodium      | oxygen    |

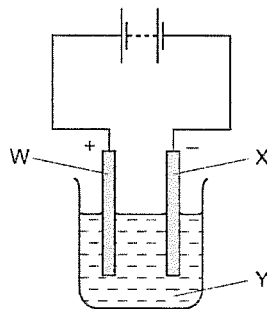
**83. O/N 14/P13/Q11**

A student wishes to electroplate an object with copper. Which row is correct?

|   | object is made the | a suitable electrolyte is |
|---|--------------------|---------------------------|
| A | anode              | CuO(s)                    |
| B | anode              | CuSO <sub>4</sub> (aq)    |
| C | cathode            | CuO(s)                    |
| D | cathode            | CuSO <sub>4</sub> (aq)    |

## 84. O/N 14/P13/Q12

In the electrolysis shown, chlorine is produced at W and sodium at X.



Which labels are correct?

|   | W       | X       | Y                  |
|---|---------|---------|--------------------|
| A | anode   | cathode | $\text{NaCl (l)}$  |
| B | anode   | cathode | $\text{NaCl (aq)}$ |
| C | cathode | anode   | $\text{NaCl (l)}$  |
| D | cathode | anode   | $\text{NaCl (aq)}$ |

## 85. M/J 14/P12/Q11

What will be produced at the anode and at the cathode, if molten potassium chloride is electrolysed?

|   | anode (+) | cathode (-) |
|---|-----------|-------------|
| A | chlorine  | hydrogen    |
| B | chlorine  | potassium   |
| C | hydrogen  | chlorine    |
| D | potassium | chlorine    |

## 86. M/J 14/P11/Q12, M/J 14/P13/Q15

Which products are formed at the anode and cathode when electricity is passed through molten lead(II) bromide?

|   | anode (+)         | cathode (-)       |
|---|-------------------|-------------------|
| A | bromide ions      | lead ions         |
| B | bromine molecules | lead atoms        |
| C | lead atoms        | bromine molecules |
| D | lead ions         | bromide ions      |

## 87. M/J 14/P11/Q14, M/J 14/P13/Q12

Two chemical processes are described below.

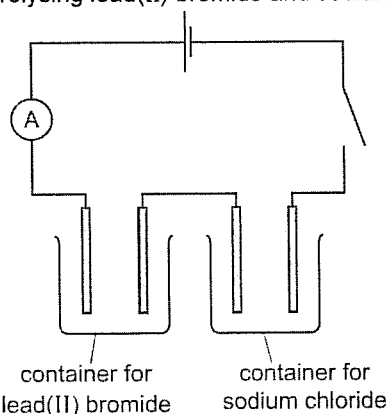
- In the combustion of methane, energy is .....1..... .
- In the electrolysis of molten lead(II) bromide, energy is .....2..... .

Which words correctly complete gaps 1 and 2?

|   | 1         | 2         |
|---|-----------|-----------|
| A | given out | given out |
| B | given out | taken in  |
| C | taken in  | given out |
| D | taken in  | taken in  |

**88. O/N 13/P12/Q10, O/N 13/P11/Q11**

The diagram shows the circuit for electrolysis of lead(II) bromide and sodium chloride to liberate the metal.



In what form are these salts electrolysed for liberating the metal?

|          | lead(II) bromide      | sodium chloride       |
|----------|-----------------------|-----------------------|
| <b>A</b> | concentrated solution | concentrated solution |
| <b>B</b> | concentrated solution | molten                |
| <b>C</b> | molten                | concentrated solution |
| <b>D</b> | molten                | molten                |

**89. O/N 13/P13/Q11**

One molten compound and two aqueous solutions were electrolysed. The table gives the compounds electrolysed and the electrodes used

|   | substance electrolysed         | electrodes |
|---|--------------------------------|------------|
| 1 | concentrated hydrochloric acid | carbon     |
| 2 | concentrated sodium chloride   | platinum   |
| 3 | molten lead bromide            | platinum   |

In which experiments is a gas evolved at the cathode?

**A** 1, 2 and 3

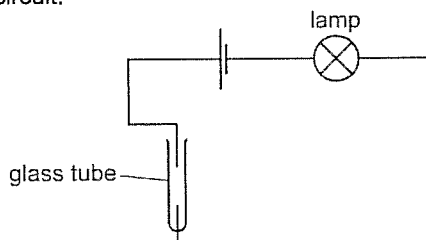
**B** 1 and 2 only

**C** 1 only

**D** 3 only

**90. M/J 13/P12/Q9, M/J 13/P11/Q8, M/J 13/P13/Q9**

The diagram shows an incomplete circuit.



Which substance causes the lamp to light when added to the glass tube?

**A** aqueous sodium chloride

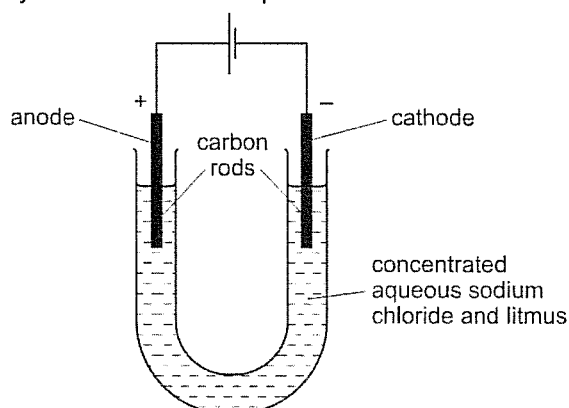
**C** solid sodium chloride

**B** aqueous sugar

**D** solid sugar

## 91. O/N 13/P12/Q13, O/N 13/P11/Q12

The diagram shows the electrolysis of concentrated aqueous sodium chloride.

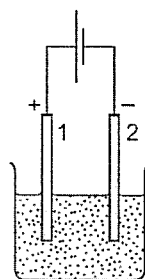


What is the colour of the litmus at each electrode after five minutes?

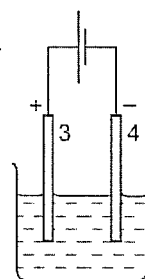
|          | colour at anode | colour at cathode |
|----------|-----------------|-------------------|
| <b>A</b> | blue            | red               |
| <b>B</b> | red             | blue              |
| <b>C</b> | red             | colourless        |
| <b>D</b> | colourless      | blue              |

## 92. O/N 13/P13/Q10

Two electrolysis experiments were carried out as shown in the diagram below. The graphite electrodes are labelled 1-4.



molten  
sodium chloride



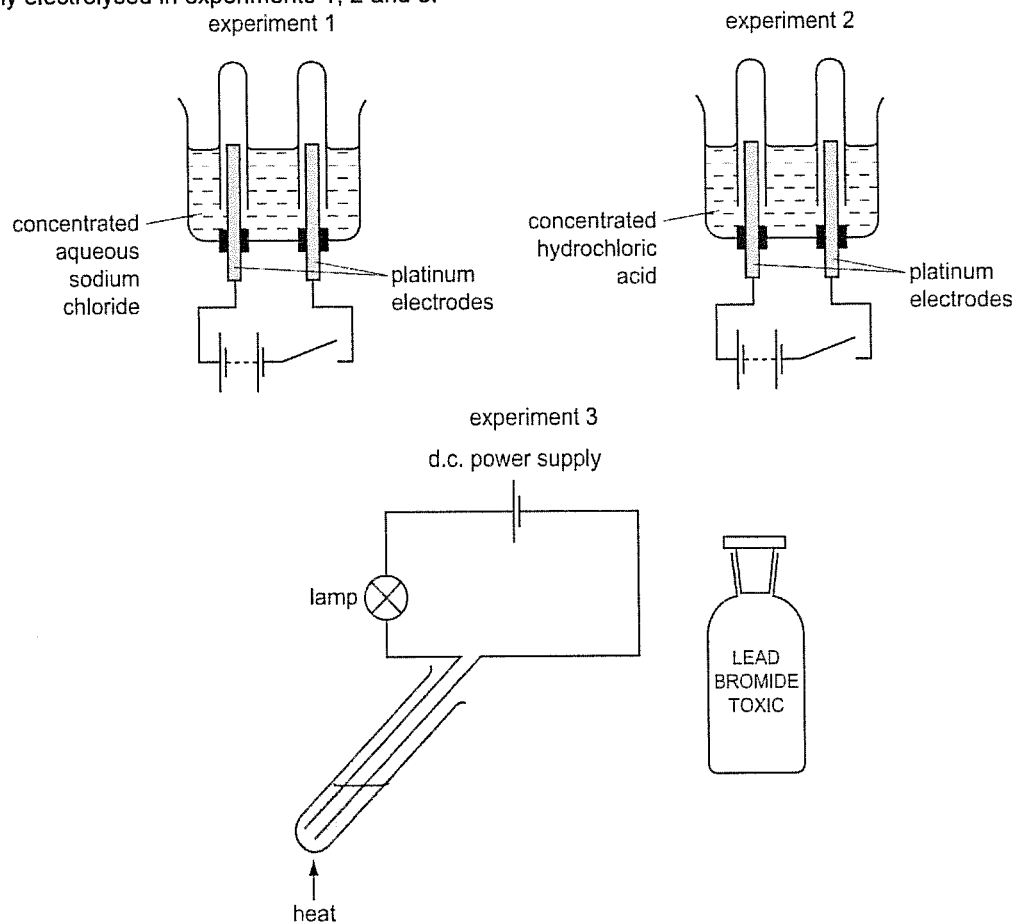
concentrated aqueous  
sodium chloride

Which row describes the products at the electrodes in these experiments?

|          | electrode 1 | electrode 2 | electrode 3 | electrode 4 |
|----------|-------------|-------------|-------------|-------------|
| <b>A</b> | chlorine    | hydrogen    | chlorine    | hydrogen    |
| <b>B</b> | chlorine    | sodium      | chlorine    | hydrogen    |
| <b>C</b> | chlorine    | sodium      | hydrogen    | chlorine    |
| <b>D</b> | sodium      | chlorine    | sodium      | chlorine    |

**93. M/J 13/P12/Q18**

Concentrated aqueous sodium chloride, concentrated hydrochloric acid and molten lead bromide were separately electrolysed in experiments 1, 2 and 3.

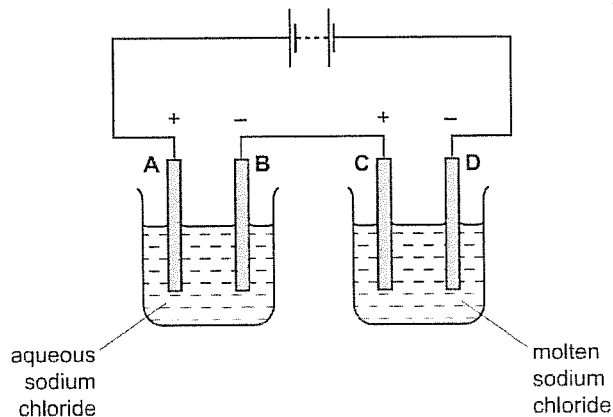


Which statement about the electrode products is correct?

- A Gases were given off at the anode in experiments 2 and 3 only.
- B Gases were given off at the cathode in experiments 1 and 2 only.
- C Metals were formed at the anode in experiments 1 and 3 only.
- D Metals were formed at the cathode in experiments 1 and 3 only.

**94. M/J 13/P11/Q12, M/J 13/P13/Q13**

The diagram shows an electrolysis circuit.  
At which electrode is hydrogen formed?



**5.2: Fuel cell****1. M/J 22/P22/Q17**

When a hydrogen–oxygen fuel cell is in operation, a different reaction happens at each electrode.

at the hydrogen electrode  $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$

at the oxygen electrode  $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

The electrons that are lost at the hydrogen electrode travel through the external circuit to the oxygen electrode, where they are gained by the oxygen and water.

A hydrogen–oxygen fuel cell is operated for a period of time and four moles of oxygen molecule are consumed.

Which mass of hydrogen is consumed?

- A 2.0 g                      B 4.0 g                      C 8.0 g                      D 16.0 g

**2. M/J 22/P23/Q29**

Nitrogen oxide, NO, is formed in the engine of petrol-powered cars.

One constituent of petrol is octane,  $\text{C}_8\text{H}_{18}$ .

Nitrogen oxide is removed from exhaust fumes by catalytic converters.

Which row identifies the reactants that produce nitrogen oxide and a reaction that removes it in a catalytic converter?

|   | reactants that produce NO     | reaction that removes NO                                        |
|---|-------------------------------|-----------------------------------------------------------------|
| A | octane + one gas found in air | $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ |
| B | octane + one gas found in air | $\text{NO} + \text{CO}_2 \rightarrow \text{NO}_2 + \text{CO}$   |
| C | two gases found in air        | $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ |
| D | two gases found in air        | $\text{NO} + \text{CO}_2 \rightarrow \text{NO}_2 + \text{CO}$   |

**3. O/N 21/P21/Q12**

Hydrogen is used as a fuel in rockets and is also used in hydrogen fuel cells.

Which statements are correct?

- Both uses produce water vapour.
- Burning hydrogen produces polluting gases.
- A fuel cell produces electricity.

- A 1, 2 and 3                      B 1 and 3 only                      C 1 only                      D 2 and 3 only

**4. M/J 21/P22/Q21**

Burning fossil fuels releases sulfur dioxide which leads to acid rain.

Which ion in the rain water causes it to be acidic?

- A  $\text{H}^+$                       B  $\text{OH}^-$                       C  $\text{O}^{2-}$                       D  $\text{SO}_4^{2-}$

**5. M/J 21/P21/Q13**

Which equation represents a reaction that takes place in a fuel cell?

- A  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$                       C  $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$   
 B  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$                       D  $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$

**6. MAR 21/P22/Q15**

Hydrogen fuel cells can be used to power cars.

Which statements about a fuel cell are correct?

- The balanced equation for the reaction is  $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ .
- The fuel cell generates electricity.
- In the fuel cell hydrogen is reduced.
- The reactants are gases at room temperature.

- A 1 and 2                      B 1 and 3                      C 2 and 4                      D 3 and 4

**7. O/N 20/P22/Q16**

Which substance does not require oxygen in order to produce energy?

- A coal                      B hydrogen                      C natural gas                      D  $^{235}\text{U}$

**8. M/J 20/P22/Q13**

Which statements about hydrogen fuel cells are correct?

- 1 Water is formed as the only waste product.
- 2 Both water and carbon dioxide are formed as waste products.
- 3 The overall reaction is  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ .
- 4 The overall reaction is endothermic.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**9. MAR 20/P22/Q13**

Which statements about hydrogen fuel cells are correct?

- 1 The reaction between hydrogen and oxygen is endothermic.
- 2 The waste product in a hydrogen fuel cell is water.
- 3 A chemical reaction in the cell produces hydrogen which is used as the fuel.
- 4 A hydrogen fuel cell is used to generate electricity.

A 1 and 2                      B 1 and 3                      C 2 and 4                      D 3 and 4

**10. O/N 19/P22/Q14, O/N 19/P23/Q14**

Which gases are used to generate electricity in a fuel cell?

- A carbon dioxide and oxygen
- B hydrogen and methane
- C hydrogen and oxygen
- D methane and carbon dioxide

**11. M/J 19/P21/Q11**

Which statement about the hydrogen fuel cell is not correct?

- A Chemical energy is converted into electrical energy.
- B Hydrogen is oxidised.
- C The reaction that takes place is endothermic.
- D Water is the only product.

**12. M/J 19/P23/Q11**

Which statement about a fuel cell in a car is correct?

- A The fuel cell produces heat, which powers the car.
- B The fuel cell is supplied with hydrogen directly from the air.
- C The only emission from a fuel cell is nitrogen gas, which is non-polluting.
- D The fuel cell produces electricity, which powers an electric motor.



## Answers Section

| 5.1 |   |     |   |     |   | 5.2 |   |
|-----|---|-----|---|-----|---|-----|---|
| 1.  | B | 41. | A | 81. | D | 1.  | D |
| 2.  | D | 42. | B | 82. | A | 2.  | C |
| 3.  | C | 43. | C | 83. | D | 3.  | B |
| 4.  | A | 44. | C | 84. | A | 4.  | A |
| 5.  | A | 45. | C | 85. | B | 5.  | B |
| 6.  | B | 46. | D | 86. | B | 6.  | C |
| 7.  | D | 47. | C | 87. | B | 7.  | D |
| 8.  | B | 48. | D | 88. | D | 8.  | A |
| 9.  | B | 49. | C | 89. | B | 9.  | C |
| 10. | C | 50. | C | 90. | A | 10. | C |
| 11. | B | 51. | A | 91. | D | 11. | C |
| 12. | A | 52. | B | 92. | B | 12. | D |
| 13. | D | 53. | B | 93. | B |     |   |
| 14. | C | 54. | D | 94. | B |     |   |
| 15. | C | 55. | B |     |   |     |   |
| 16. | C | 56. | B |     |   |     |   |
| 17. | B | 57. | B |     |   |     |   |
| 18. | A | 58. | D |     |   |     |   |
| 19. | C | 59. | C |     |   |     |   |
| 20. | A | 60. | A |     |   |     |   |
| 21. | B | 61. | A |     |   |     |   |
| 22. | D | 62. | A |     |   |     |   |
| 23. | C | 63. | B |     |   |     |   |
| 24. | D | 64. | A |     |   |     |   |
| 25. | A | 65. | C |     |   |     |   |
| 26. | A | 66. | D |     |   |     |   |
| 27. | A | 67. | A |     |   |     |   |
| 28. | A | 68. | A |     |   |     |   |
| 29. | D | 69. | B |     |   |     |   |
| 30. | C | 70. | B |     |   |     |   |
| 31. | D | 71. | B |     |   |     |   |
| 32. | D | 72. | B |     |   |     |   |
| 33. | B | 73. | D |     |   |     |   |
| 34. | B | 74. | C |     |   |     |   |
| 35. | C | 75. | D |     |   |     |   |
| 36. | B | 76. | D |     |   |     |   |
| 37. | A | 77. | C |     |   |     |   |
| 38. | D | 78. | D |     |   |     |   |
| 39. | A | 79. | A |     |   |     |   |
| 40. | C | 80. | D |     |   |     |   |

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## Unit 6

# CHEMICAL ENERGETICS

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IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 6.1 Energetics of a reaction
- 6.2 Energy transfer

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## Unit 6: Chemical Energetics

### 6.1: Energetics of a reaction

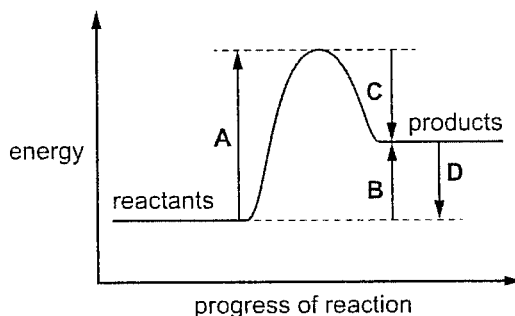
1. O/N 22/P21/Q11, O/N 22/P23/Q11

When an acid is added to an alkali, the temperature of the reaction mixture rises.  
Which words describe this reaction?

- A decomposition and endothermic                      C neutralisation and endothermic  
B decomposition and exothermic                      D neutralisation and exothermic

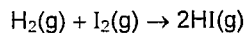
2. M/J 22/P22/Q16

Which arrow on the energy level diagram shows the overall energy change for an endothermic reaction?



3. M/J 22/P21/Q13

The equation for the reaction between gaseous hydrogen and gaseous iodine to form gaseous hydrogen iodide is shown.



The reaction is exothermic.

Which statement explains why the reaction is exothermic?

- A Energy is released when H–H and I–I bonds are broken.  
B The bond energies of the reactants are larger than the bond energies of the products.  
C The products are at a higher energy level than the reactants.  
D More energy is released when two HI bonds are formed than is used when the H–H and I–I bonds are broken.

4. MAR 22/P22/Q15

Which row describes the changes that occur in an endothermic reaction?

|   | energy change                         | temperature |
|---|---------------------------------------|-------------|
| A | energy given out to the surroundings  | decreases   |
| B | energy given out to the surroundings  | increases   |
| C | energy taken in from the surroundings | decreases   |
| D | energy taken in from the surroundings | increases   |

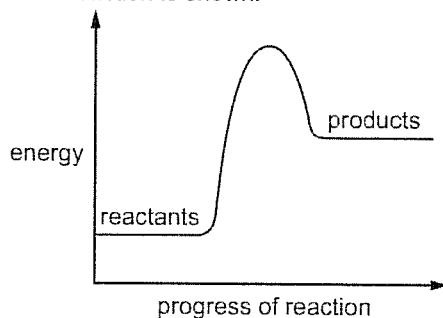
**5. MAR 22/P22/Q17**

Which statement about endothermic and exothermic reactions is correct?

- A In an endothermic reaction, less energy is absorbed in bond breaking than is released in bond forming.
- B In an endothermic reaction, the activation energy is always higher than in an exothermic reaction.
- C In an exothermic reaction, more energy is absorbed in bond breaking than is released in bond forming.
- D In an exothermic reaction, the reactants are higher on an energy level diagram than the products.

**6. O/N 21/P23/Q11**

The energy level diagram for a chemical reaction is shown.

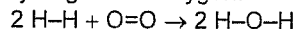


Which statement about this reaction is correct?

- A The reaction is endothermic and energy is given out to the surroundings.
- B The reaction is endothermic and energy is taken in from the surroundings.
- C The reaction is exothermic and energy is given out to the surroundings.
- D The reaction is exothermic and energy is taken in from the surroundings.

**7. M/J 21/P22/Q20**

The equation shows the reaction between hydrogen and oxygen.



The bond energies are shown.

|     | bond energy<br>in kJ/mol |
|-----|--------------------------|
| H-H | 436                      |
| O=O | 495                      |
| O-H | 463                      |

Which row shows the energy change and the type of reaction?

|   | energy change<br>in kJ/mol | type of reaction |
|---|----------------------------|------------------|
| A | 441                        | exothermic       |
| B | 441                        | endothermic      |
| C | 485                        | exothermic       |
| D | 485                        | endothermic      |

**8. M/J 21/P23/Q15**

Four statements about the effect of increasing temperature on a reaction are shown.

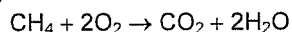
- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A 1, 2 and 3
- B 1, 3 and 4
- C 2, 3 and 4
- D 2 and 3 only

**9. O/N 20/P22/Q14**

The combustion of methane is exothermic.



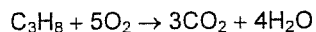
Which statement about this reaction is correct?

- A** The energy needed to break the bonds in methane and oxygen is greater than the energy released in making new bonds in carbon dioxide and water.
- B** The energy needed to break the bonds in methane and oxygen is less than the energy released in making new bonds in carbon dioxide and water.
- C** The energy released in breaking bonds in methane and oxygen is greater than the energy needed to make new bonds in carbon dioxide and water.
- D** The energy released in breaking bonds in methane and oxygen is less than the energy needed to make new bonds in carbon dioxide and water.

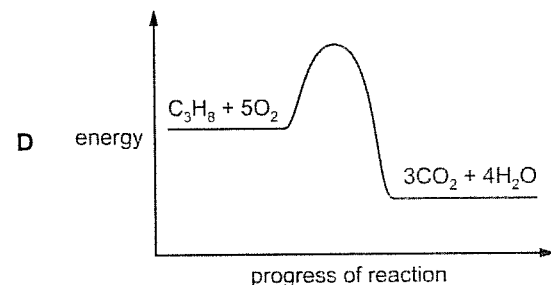
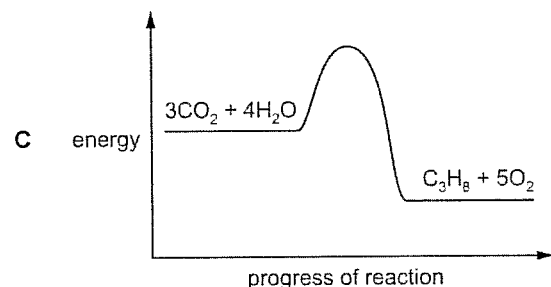
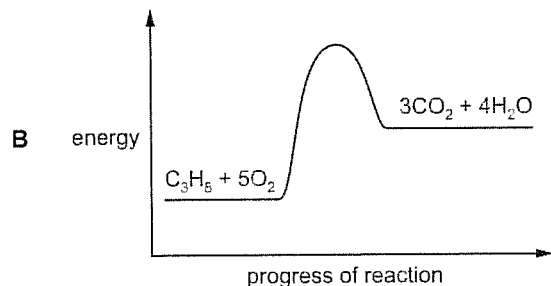
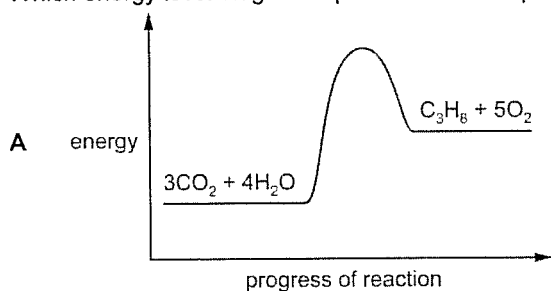
**10. M/J 21/P21/Q12**

The complete combustion of propane is exothermic.

The equation for this reaction is shown.

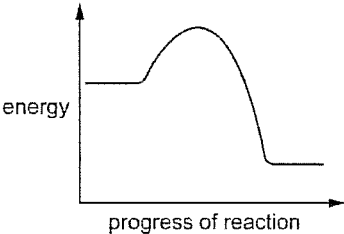
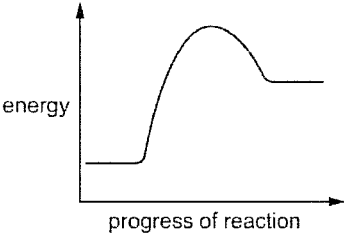
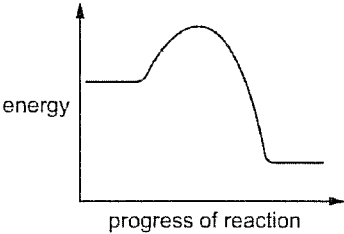
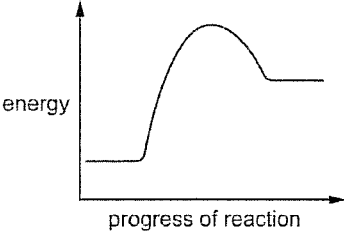


Which energy level diagram represents the complete combustion of propane?



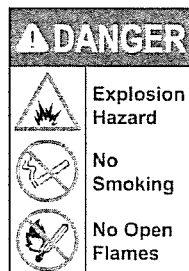
## 11. MAR 21/P22/Q13

Which row describes an endothermic reaction?

|   | energy level diagram                                                                | energy transfer                                             |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| A |    | energy is transferred from the surroundings to the reaction |
| B |    | energy is transferred from the surroundings to the reaction |
| C |   | energy is transferred from the reaction to the surroundings |
| D |  | energy is transferred from the reaction to the surroundings |

## 12. O/N 20/P21/Q18

A sign displayed in a flour mill is shown.



Which statement explains why there is a danger of explosion in a flour mill?

- A Flour burns very quickly because it is a fine powder.
- B Flour is a catalyst for combustion.
- C Flour mills get hot and speed up the rate of combustion.
- D The combustion of flour is exothermic.

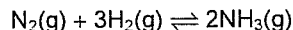
**13. O/N 20/P23/Q18**

Which reaction is an example of a photochemical reaction?

- A glucose forming carbon dioxide and water      C potassium reacting with water  
 B magnesium reacting with oxygen      D silver chloride forming silver metal

**14. O/N 20/P23/Q32**

Ammonia is manufactured in an exothermic reaction.

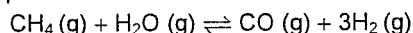


What is the effect of lowering the pressure on the rate of formation of ammonia and percentage yield of ammonia at equilibrium?

|   | rate of formation | percentage yield |
|---|-------------------|------------------|
| A | decreases         | decreases        |
| B | decreases         | increases        |
| C | increases         | decreases        |
| D | increases         | increases        |

**15. M/J 20/P22/Q16**

Methane reacts with steam and an equilibrium is reached.



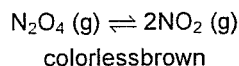
The forward reaction is endothermic.

Which row shows how the amount of hydrogen at equilibrium changes when the pressure or temperature is changed as indicated?

|   | change in temperature | change in pressure | amount of hydrogen |
|---|-----------------------|--------------------|--------------------|
| A | decrease              | no change          | increase           |
| B | increase              | no change          | decrease           |
| C | no change             | increase           | decrease           |
| D | no change             | decrease           | decrease           |

**16. M/J 20/P21/Q16**The equation shows the equilibrium between dinitrogen tetroxide,  $\text{N}_2\text{O}_4$ , and nitrogen dioxide,  $\text{NO}_2$ .

The colours of the reactant and product are also shown.



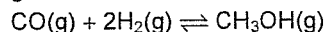
The forward reaction is endothermic.

Which statement is **not** correct?

- A At equilibrium the concentrations of the reactant and the product are constant.  
 B At equilibrium the rate of the forward reaction is equal to the rate of the reverse reaction.  
 C When the pressure is increased a darker brown colour is seen.  
 D When the temperature is increased a darker brown colour is seen.

**17. O/N 19/P22/Q17**

When carbon monoxide reacts with hydrogen, methanol is formed.



The forward reaction is exothermic.

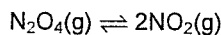
Which statements are correct?

- 1 There are more moles of gas on the left-hand side of the reaction.
  - 2 Increasing the temperature increases the amount of methanol at equilibrium.
  - 3 Increasing the pressure increases the amount of methanol at equilibrium.
  - 4 Increasing the initial amount of hydrogen decreases the amount of methanol at equilibrium.
- A 1 and 2 only      B 1 and 3 only      C 2 and 4 only      D 3 and 4 only



**18. MAR 20/P22/Q16**

An equilibrium reaction is shown.



The forward reaction is endothermic.

What is the effect of changing the temperature and pressure on the equilibrium position?

|          | increasing temperature | increasing pressure |
|----------|------------------------|---------------------|
| <b>A</b> | moves to the left      | moves to the left   |
| <b>B</b> | moves to the left      | moves to the right  |
| <b>C</b> | moves to the right     | moves to the left   |
| <b>D</b> | moves to the right     | moves to the right  |

**19. O/N 19/P22/Q13**

The temperature of the water in two beakers, X and Y, is measured as 21.5 °C.

5g of sodium chloride is dissolved in the water in beaker X. The temperature changes to 18.0 °C.

5g of calcium oxide is dissolved in the water in beaker Y. The temperature changes to 29.4 °C.

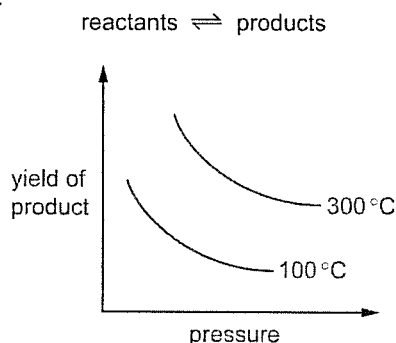
Which types of process are occurring in beakers X and Y?

|          | X           | Y           |
|----------|-------------|-------------|
| <b>A</b> | endothermic | endothermic |
| <b>B</b> | endothermic | exothermic  |
| <b>C</b> | exothermic  | endothermic |
| <b>D</b> | exothermic  | exothermic  |

**20. M/J 19/P21/Q15**

The graph shows how the yield of product in a reversible reaction changes as the temperature and pressure are changed.

All reactants and products are gases.

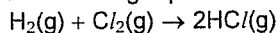


Which row is correct for this reversible reaction?

|          | side of reaction with fewer moles | forward reaction |
|----------|-----------------------------------|------------------|
| <b>A</b> | reactant                          | exothermic       |
| <b>B</b> | reactant                          | endothermic      |
| <b>C</b> | product                           | endothermic      |
| <b>D</b> | product                           | exothermic       |

**21. MAR 19/P22/Q13**

Hydrogen reacts with chlorine according to the following equation.



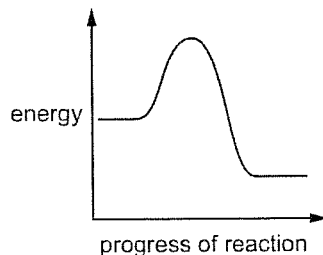
The reaction is exothermic.

Which statement about this reaction is correct?

- A Energy absorbed for bond breaking is greater than the energy released in bond making.
- B Energy absorbed for bond breaking is less than the energy released in bond making.
- C Energy released in bond breaking is greater than the energy absorbed in bond making.
- D Energy released in bond breaking is less than the energy absorbed in bond making.

**22. O/N 19/P23/Q13**

An energy level diagram for a reaction is shown.



Which statement and explanation about this reaction are correct?

|   | statement                   | explanation                                      |
|---|-----------------------------|--------------------------------------------------|
| A | the reaction is endothermic | the products have more energy than the reactants |
| B | the reaction is endothermic | the products have less energy than the reactants |
| C | the reaction is exothermic  | the products have more energy than the reactants |
| D | the reaction is exothermic  | the products have less energy than the reactants |

**23. MAR 19/P22/Q12**

10 g of ammonium nitrate is added to water at 25 °C and the mixture stirred.

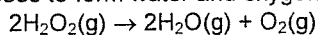
The ammonium nitrate dissolves and, after one minute, the temperature of the solution is 10 °C.

Which word describes this change?

- A endothermic
- B exothermic
- C neutralisation
- D reduction

**24. O/N 18/P22/Q12**

Hydrogen peroxide, H–O–O–H, decomposes to form water and oxygen.



The bond energies are shown in the table. The reaction is exothermic.

| bond | bond energy<br>in kJ / mol |
|------|----------------------------|
| O–H  | +460                       |
| O–O  | +150                       |
| O=O  | +496                       |

What is the energy change for the reaction?

- A –346 kJ / mol
- B –196 kJ / mol
- C +196 kJ / mol
- D +346 kJ / mol

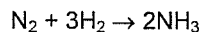
**25. O/N 18/P21/Q34**

Which reaction is endothermic?

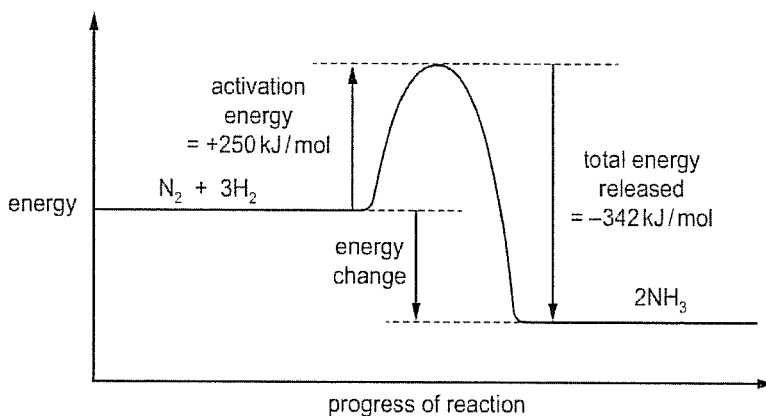
- A  $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- B  $\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$
- C  $2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$
- D  $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$

**26. O/N 18/P22/Q13, O/N 18/P21/Q13, O/N 18/P23/Q13**

The equation for the formation of ammonia is shown.



The energy level diagram for the reaction is shown.

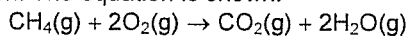


What is the energy change for the reaction?

- A** -592 kJ / mol      **B** -92 kJ / mol      **C** +92 kJ / mol      **D** +592 kJ / mol

**27. O/N 18/P21/Q12**

Methane burns in an excess of oxygen. The equation is shown.



The bond energies are shown in the table.

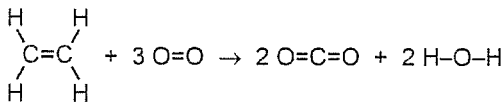
| bond | bond energy<br>in kJ / mol |
|------|----------------------------|
| C-H  | +410                       |
| C=O  | +805                       |
| O-H  | +460                       |
| O=O  | +496                       |

What is the energy change for the reaction?

- A** +818 kJ / mol      **B** +102 kJ / mol      **C** -359 kJ / mol      **D** -818 kJ / mol

**28. O/N 18/P23/Q12**

Ethene burns in oxygen to form carbon dioxide and water vapour.



The bond energies are shown in the table.

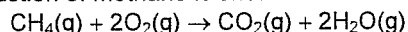
| bond | bond energy<br>in kJ / mol |
|------|----------------------------|
| C=C  | +610                       |
| C-H  | +410                       |
| O=O  | +497                       |
| C=O  | +805                       |
| O-H  | +460                       |

What is the energy change for the reaction?

- A** -2959 kJ/mol      **B** -2313 kJ/mol      **C** -1319 kJ/mol      **D** -399 kJ/mol

## 29. M/J 18/P22/Q13

The equation for the complete combustion of methane is shown.



The bond energies are shown in the table.

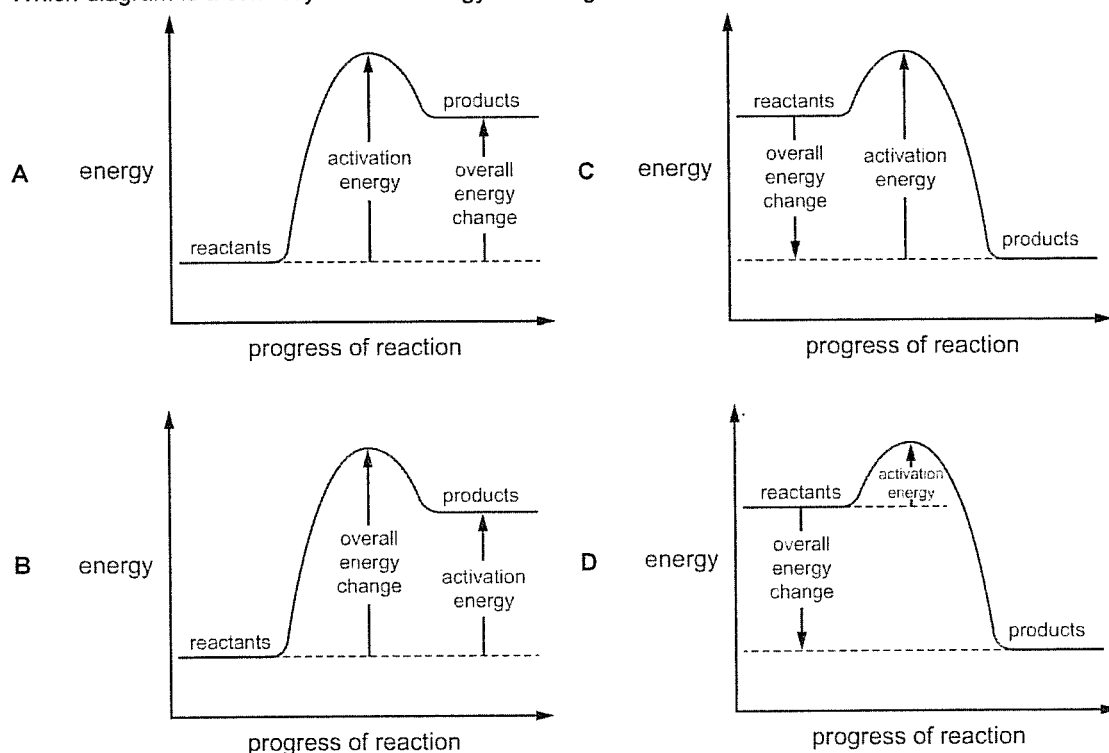
| bond | bond energy<br>in kJ / mol |
|------|----------------------------|
| C–H  | +410                       |
| C=O  | +805                       |
| O–H  | +460                       |
| O=O  | +496                       |

What is the energy change for the reaction?

- A –818 kJ / mol      B –359 kJ / mol      C –323 kJ / mol      D +102 kJ / mol

## 30. M/J 18/P22/Q12

Which diagram is a correctly labelled energy level diagram for an endothermic reaction?



## 31. M/J 18/P21/Q12

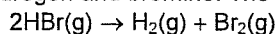
Plant cells use energy from sunlight for photosynthesis.

Which row describes and explains the energy change that occurs?

|   | type of<br>energy change | explanation                                                          |
|---|--------------------------|----------------------------------------------------------------------|
| A | endothermic              | less energy is released making bonds than is absorbed to break bonds |
| B | endothermic              | more energy is released making bonds than is absorbed to break bonds |
| C | exothermic               | less energy is released making bonds than is absorbed to break bonds |
| D | exothermic               | more energy is released making bonds than is absorbed to break bonds |

**32. M/J 18/P21/Q13**

Hydrogen bromide decomposes to form hydrogen and bromine. The equation is shown.



The bond energies are shown in the table. The reaction is endothermic.

| bond  | bond energy<br>in kJ/mol |
|-------|--------------------------|
| Br-Br | +193                     |
| H-Br  | +366                     |
| H-H   | +436                     |

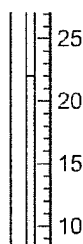
What is the energy change for the reaction?

- A +263 kJ / mol      B +103 kJ / mol      C -103 kJ / mol      D -263 kJ / mol

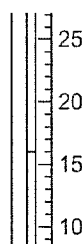
**33. M/J 18/P23/Q3**

Solid R reacted with dilute sulfuric acid.

The initial temperature of the dilute sulfuric acid and the final temperature of the solution are shown.



initial temperature  
of the dilute  
sulfuric acid (°C)



final temperature  
of the solution (°C)

What was the change in temperature in °C?

- A -6      B -4      C 4      D 6

**34. M/J 18/P23/Q12**

Information about two reactions is given.

- The neutralisation reaction between citric acid and sodium hydrogencarbonate is endothermic.
- The displacement reaction between magnesium and carbon dioxide is exothermic.

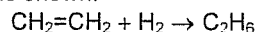
Which statements about the two reactions are correct?

- The energy of the products formed in the neutralisation reaction is greater than the energy of the reactants.
- The energy of magnesium and carbon dioxide is greater than the energy of magnesium oxide and carbon.
- In an exothermic reaction, the energy required to break the bonds is greater than the energy released when the new bonds are formed.

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

**35. M/J 18/P23/Q13**

Ethene reacts with hydrogen. The equation is shown.



The bond energies are shown in the table. The reaction is exothermic.

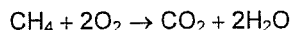
| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| C-C  | +350                     |
| C=C  | +610                     |
| C-H  | +410                     |
| H-H  | +436                     |

What is the energy change for the reaction?

- A -560 kJ / mol      B -124 kJ / mol      C +486 kJ / mol      D +5496 kJ / mol

**36. MAR 18/P22/Q12**

Methane burns in oxygen to form carbon dioxide and water.

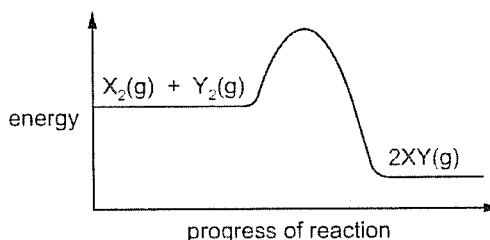


The bond energies are shown in the table.

| bond | bond energy<br>in kJ / mol |
|------|----------------------------|
| C–H  | +410                       |
| C=O  | +805                       |
| O–H  | +460                       |
| O=O  | +496                       |

What is the energy change for the reaction?

- A –818 kJ / mol      B –323 kJ / mol      C +323 kJ / mol      D +818 kJ / mol

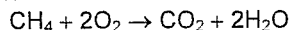
**37. MAR 18/P22/Q11**The energy level diagram for the reaction between  $\text{X}_2$  and  $\text{Y}_2$  to form XY gas is shown.

Which statement is correct?

- A Energy is released when  $\text{X}_2$  and  $\text{Y}_2$  bonds are broken.  
 B Energy is needed to form XY bonds.  
 C The energy change,  $\Delta H$ , for the reaction is negative.  
 D The reaction is endothermic.

**38. O/N 17/P22/Q11**

The equation for the combustion of methane is shown.

The energy change for the combustion of methane is  $-890 \text{ kJ / mol}$ .

The bond energies are shown in the table.

| bond | bond energy<br>in kJ / mol |
|------|----------------------------|
| C–H  | +410                       |
| O=O  | +496                       |
| H–O  | +460                       |

What is the bond energy of the C=O bond?

- A +49 kJ / mol      B +841 kJ / mol      C +1301 kJ / mol      D +1335 kJ / mol

**39. O/N 17/P22/Q12, O/N 17/P21/Q12, O/N 17/P23/Q12**

Which statement describes an exothermic reaction?

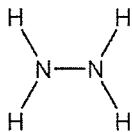
- A The energy absorbed for bond breaking is greater than the energy released by bond formation.  
 B The energy absorbed for bond breaking is less than the energy released by bond formation.  
 C The energy released by bond breaking is greater than the energy absorbed for bond formation.  
 D The energy released by bond breaking is less than the energy absorbed for bond formation.

**40. O/N 17/P23/Q2**25 cm<sup>3</sup> of an alkali are added to 20 cm<sup>3</sup> of an acid. The temperature change is measured.Which apparatus is **not** needed in the experiment?

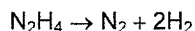
- A 25 cm<sup>3</sup> measuring cylinder      C balance  
 B 100 cm<sup>3</sup> beaker      D thermometer

**41. O/N 17/P21/Q11**

The compound hydrazine is used as a rocket fuel. It has the structural formula shown.



One of the reactions of hydrazine is shown. This reaction is exothermic.



The bond energies are shown in the table.

|     | bond energy<br>in kJ/mol |
|-----|--------------------------|
| H-H | +436                     |
| N-H | +390                     |
| N-N | +160                     |
| N≡N | +945                     |

What is the energy change for this reaction?

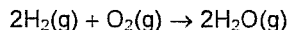
- A -339 kJ/mol      B -97 kJ/mol      C +97 kJ/mol      D +339 kJ/mol

**42. O/N 17/P23/Q11**

Some bond energies are shown in the table.

| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| H-H  | +436                     |
| O=O  | +496                     |
| H-O  | +460                     |

Hydrogen reacts with oxygen. The reaction is exothermic.



What is the energy change for the reaction?

- A -3208 kJ/mol      B -908 kJ/mol      C -472 kJ/mol      D -448 kJ/mol

**43. M/J 17/P22/Q12, M/J 17/P23/Q12, M/J 17/P21/Q12, O/N 14/P12/Q13, O/N 14/P11/Q13**

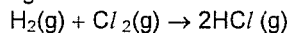
Which statements about exothermic and endothermic reactions are correct?

- During an exothermic reaction, heat is given out.
- The temperature of an endothermic reaction goes up because heat is taken in.
- Burning methane in the air is an exothermic reaction.

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

**44. M/J 17/P22/Q13**

The equation for the reaction between hydrogen and chlorine is shown.



The reaction is exothermic.

The bond energies are shown in the table.

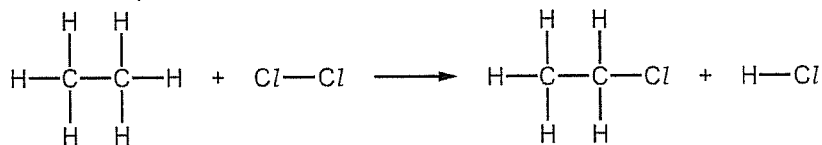
| bond  | bond energy<br>in kJ/mol |
|-------|--------------------------|
| Cl-Cl | +240                     |
| H-Cl  | +430                     |
| H-H   | +436                     |

What is the energy change for the reaction?

- A -1536 kJ/mol      B -184 kJ/mol      C +184 kJ/mol      D +246 kJ/mol

**45. M/J 17/P21/Q13**

Chlorine reacts with ethane to produce chloroethane and hydrogen chloride.



The reaction is exothermic.

The bond energies are shown in the table.

| bond  | bond energy in kJ/mol |
|-------|-----------------------|
| C-Cl  | +340                  |
| C-C   | +350                  |
| C-H   | +410                  |
| Cl-Cl | +240                  |
| H-Cl  | +430                  |

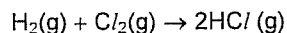
What is the energy change for the reaction?

- A -1420 kJ/mol      B -120 kJ/mol      C +120 kJ/mol      D +1420 kJ/mol

**46. M/J 17/P23/Q13**

Hydrogen and chlorine react to form hydrogen chloride.

The reaction is exothermic.



The overall energy change for this reaction is -184 kJ/mol.

The table gives some of the bond energies involved.

| bond | bond energy in kJ/mol |
|------|-----------------------|
| H-Cl | +430                  |
| H-H  | +436                  |

What is the energy of the Cl-Cl bond?

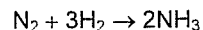
- A -240 kJ/mol      B -190 kJ/mol      C +190 kJ/mol      D +240 kJ/mol

**47. MAR 17/P22/Q12**

Ammonia is made by reacting nitrogen with hydrogen in the presence of an iron catalyst.

The reaction is exothermic.

The equation for the reaction is shown.



The bond energies are shown in the table.

| bond | bond energy in kJ/mol |
|------|-----------------------|
| H-H  | 436                   |
| N-H  | 390                   |
| N≡N  | 945                   |

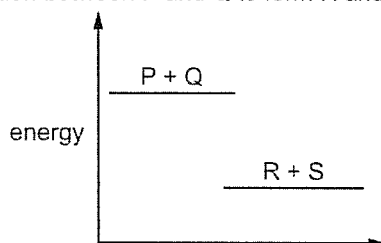
What is the energy given out during this reaction?

- A -4593 kJ/mol      B -1083 kJ/mol      C -959 kJ/mol      D -87 kJ/mol



**48. MAR 17/P22/Q13**

The energy level diagram for the reaction between P and Q to form R and S is shown.



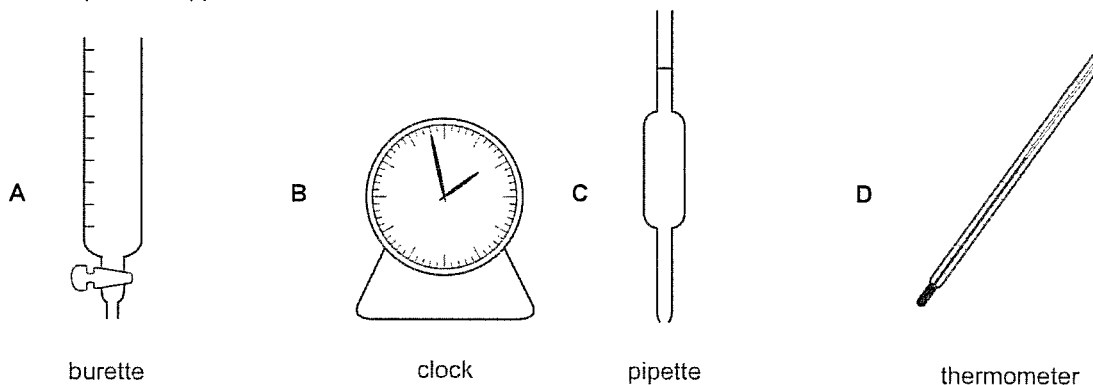
Which row describes the energy changes involved and the type of reaction?

|          | energy changes involved                                                                                               | type of reaction |
|----------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| <b>A</b> | more energy is given out when the bonds in the products are formed than is needed to break the bonds in the reactants | endothermic      |
| <b>B</b> | more energy is given out when the bonds in the products are formed than is needed to break the bonds in the reactants | exothermic       |
| <b>C</b> | more energy is needed to break the bonds in the reactants than is given out when the bonds in the products are formed | endothermic      |
| <b>D</b> | more energy is needed to break the bonds in the reactants than is given out when the bonds in the products are formed | exothermic       |

**49. O/N 16/P22/Q2, O/N 16/P21/Q2, O/N 16/P23/Q2**

A student mixes 25 cm<sup>3</sup> samples of dilute hydrochloric acid with different volumes of aqueous sodium hydroxide. In each case, the student measures the change in temperature to test if the reaction is exothermic.

Which piece of apparatus is **not** needed?

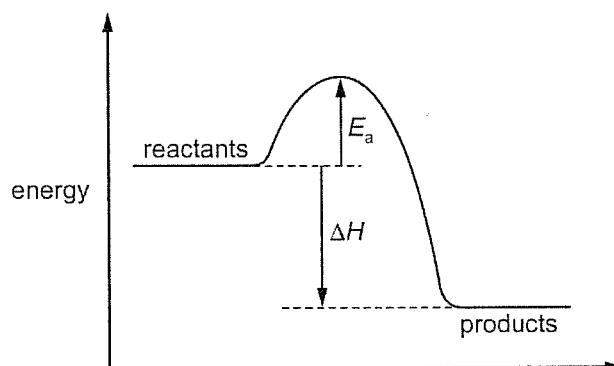
**50. O/N 16/P22/Q12**

Which experiment is the most exothermic?

|          | initial temperature / °C | final temperature / °C |
|----------|--------------------------|------------------------|
| <b>A</b> | 20                       | 5                      |
| <b>B</b> | 20                       | 32                     |
| <b>C</b> | 25                       | 12                     |
| <b>D</b> | 25                       | 34                     |

## 51. O/N 16/P22/Q13

The energy level diagram for a reaction is shown.

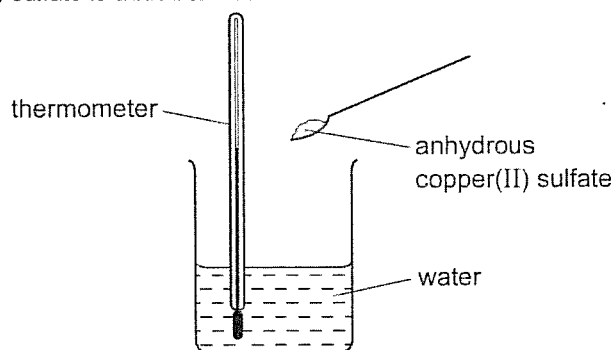


Which row is correct?

|          | sign of $\Delta H$ | overall energy change | sign of $E_a$ |
|----------|--------------------|-----------------------|---------------|
| <b>A</b> | –                  | exothermic            | –             |
| <b>B</b> | +                  | endothermic           | +             |
| <b>C</b> | +                  | endothermic           | –             |
| <b>D</b> | –                  | exothermic            | +             |

## 52. O/N 16/P21/Q12

When anhydrous copper(II) sulfate is added to water a solution is formed and heat is given out.



Which row shows the temperature change and the type of reaction taking place?

|          | temperature change | type of reaction |
|----------|--------------------|------------------|
| <b>A</b> | decrease           | endothermic      |
| <b>B</b> | decrease           | exothermic       |
| <b>C</b> | increase           | endothermic      |
| <b>D</b> | increase           | exothermic       |

## 53. O/N 16/P23/Q12

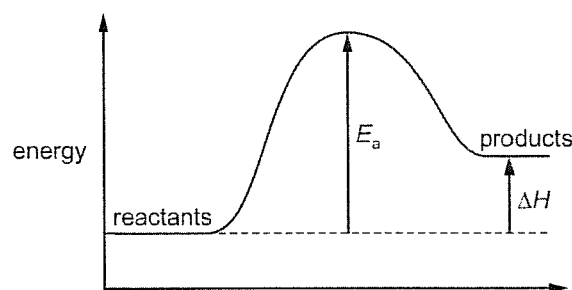
10 g of ammonium nitrate are added to water at 25 °C and the mixture stirred. The ammonium nitrate dissolves and, after one minute, the temperature of the solution is 10 °C.

Which word describes this change?

- A** endothermic      **B** exothermic      **C** neutralisation      **D** reduction

## 54. O/N 16/P21/Q13

The energy level diagram for a reaction is shown.



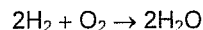
Which statement is **not** correct for this energy level diagram?

- A It could be the energy level diagram for the reaction when petrol is burnt.
- B Less energy is released in bond forming than is needed for bond breaking.
- C The activation energy,  $E_a$ , has a positive value.
- D The energy change,  $\Delta H$ , for the reaction is positive.

## 55. M/J 16/P21/Q13

Hydrogen burns exothermically in oxygen.

The equation for the reaction is:



The table shows the bond energies involved.

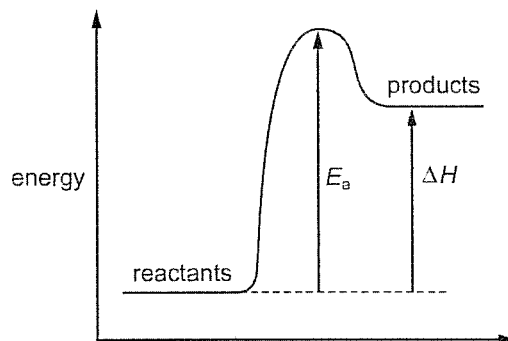
| bond | bond energy in kJ/mol |
|------|-----------------------|
| H–H  | 436                   |
| O=O  | 498                   |
| O–H  | 464                   |

What is the energy given out during the reaction?

- A –3226 kJ/mol      B –884 kJ/mol      C –486 kJ/mol      D –442 kJ/mol

## 56. O/N 16/P23/Q13

The energy level diagram for a reaction is shown.

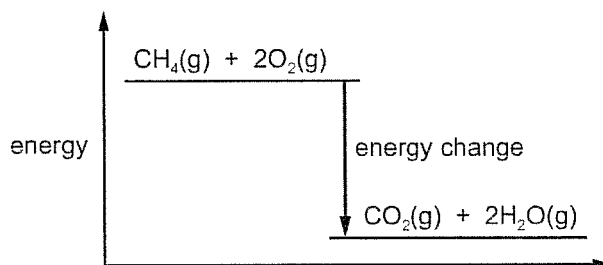


Which row is correct?

|   | sign of $\Delta H$ | overall energy change | sign of $E_a$ |
|---|--------------------|-----------------------|---------------|
| A | –                  | exothermic            | –             |
| B | +                  | endothermic           | +             |
| C | +                  | endothermic           | –             |
| D | +                  | exothermic            | +             |

**57. M/J 16/P22/Q13**

The energy level diagram for the combustion of methane is shown.

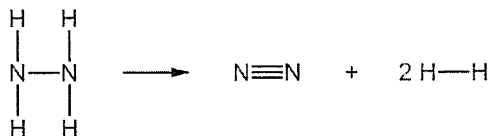


Which row gives the equation and energy change for this reaction?

|          | equation                                                                                                          | energy change in kJ/mol |
|----------|-------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>A</b> | $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ | +891                    |
| <b>B</b> | $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ | -891                    |
| <b>C</b> | $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ | +891                    |
| <b>D</b> | $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ | -891                    |

**58. M/J 16/P23/Q13**

Hydrazine,  $\text{N}_2\text{H}_4$ , decomposes as shown.



The energy change for this reaction is  $-95 \text{ kJ/mol}$ .

The table shows some bond energies involved.

| bond                       | bond energy in kJ/mol |
|----------------------------|-----------------------|
| $\text{N} \equiv \text{N}$ | 945                   |
| $\text{N}-\text{H}$        | 391                   |
| $\text{H}-\text{H}$        | 436                   |

What is the bond energy of the  $\text{N}-\text{N}$  bond?

- A** 158 kJ/mol      **B** 315 kJ/mol      **C** 348 kJ/mol      **D** 895 kJ/mol

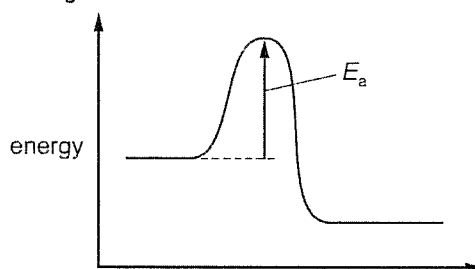
**59. MAR 16/P22/Q13**

Which row describes an endothermic reaction?

|          | energy needed to break bonds/kJ | energy released by forming bonds/kJ | temperature |
|----------|---------------------------------|-------------------------------------|-------------|
| <b>A</b> | 400                             | 200                                 | decreases   |
| <b>B</b> | 400                             | 800                                 | decreases   |
| <b>C</b> | 600                             | 200                                 | increases   |
| <b>D</b> | 600                             | 800                                 | increases   |

## 60. MAR 16/P22/Q17

The diagram shows an energy level diagram for a reaction.



The diagram shows that the reaction is .....1..... .

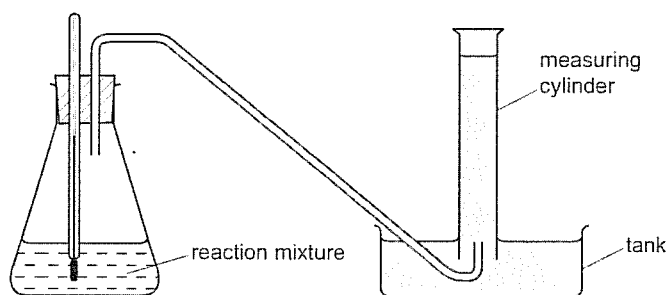
Increasing the temperature increases the rate of reaction. A reason for this is that the .....2..... .

Which words correctly complete gaps 1 and 2?

|   | 1           | 2                           |
|---|-------------|-----------------------------|
| A | endothermic | activation energy decreases |
| B | endothermic | collision rate increases    |
| C | exothermic  | activation energy decreases |
| D | exothermic  | collision rate increases    |

## 61. O/N 15/P12/Q2

The diagram shows apparatus being used to demonstrate how the rate of a chemical reaction changes with temperature.



Which statement must be correct?

- A The reaction is endothermic.                      C The reaction produces a gas.  
B The reaction is exothermic.                      D The reaction produces an acid.

## 62. O/N 15/P12/Q11

Which row correctly describes whether the reaction is exothermic or endothermic?

|   | reaction                                                       | exothermic | endothermic |
|---|----------------------------------------------------------------|------------|-------------|
| A | calcium carbonate $\rightarrow$ calcium oxide + carbon dioxide | ✓          | ✗           |
| B | carbon + oxygen $\rightarrow$ carbon dioxide                   | ✓          | ✗           |
| C | methane + oxygen $\rightarrow$ carbon dioxide + water          | ✗          | ✓           |
| D | sodium + water $\rightarrow$ sodium hydroxide + hydrogen       | ✗          | ✓           |

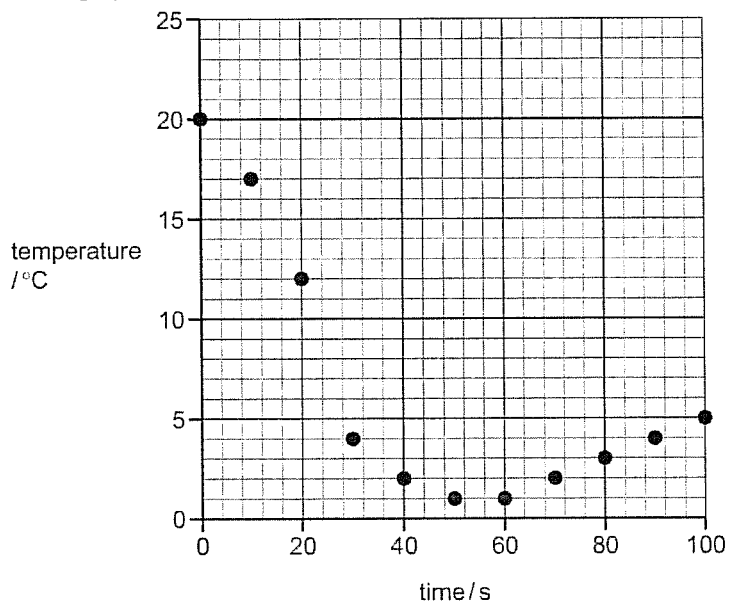
## 63. O/N 15/P11/Q10

Which reaction is endothermic?

- A acid neutralising alkali causing a temperature increase  
B adding magnesium to hydrochloric acid  
C calcium carbonate decomposing when heated  
D combustion of fossil fuels

**64. O/N 15/P11/Q11**

Solid hydrated sodium carbonate was added to solid citric acid.  
The mixture was stirred and the temperature recorded every 10 seconds.  
The results are shown on the graph:

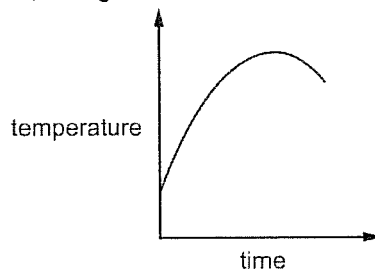


Which row describes the reaction?

|          | reaction type         | energy change |
|----------|-----------------------|---------------|
| <b>A</b> | neutralisation        | endothermic   |
| <b>B</b> | neutralisation        | exothermic    |
| <b>C</b> | thermal decomposition | endothermic   |
| <b>D</b> | thermal decomposition | exothermic    |

**65. O/N 15/P13/Q11**

A metal reacts with an aqueous solution.  
The graph shows the temperature before, during and after the reaction.



Which row describes the reaction?

|          | reaction              | energy change |
|----------|-----------------------|---------------|
| <b>A</b> | combustion            | endothermic   |
| <b>B</b> | combustion            | exothermic    |
| <b>C</b> | thermal decomposition | endothermic   |
| <b>D</b> | thermal decomposition | exothermic    |

**66. O/N 15/P13/Q10**

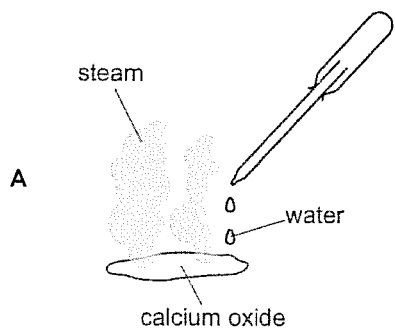
Which reaction is endothermic?

- A** the burning of magnesium ribbon
- B** the combustion of methane
- C** the decomposition of calcium carbonate
- D** the reaction of water with anhydrous copper(II) sulfate

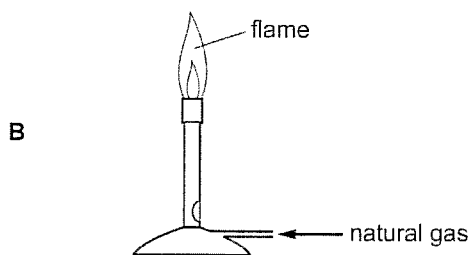
**67. M/J 15/P11/Q11**

The diagrams show four chemical reactions.

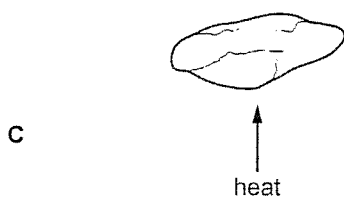
Which reaction is endothermic?



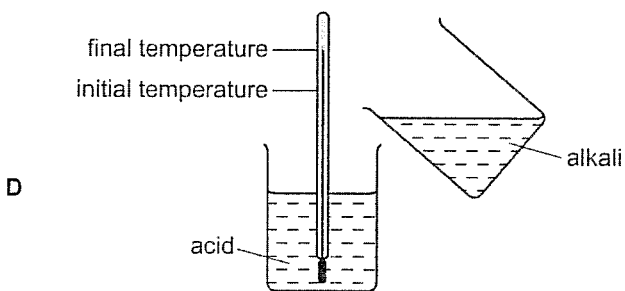
addition of water to calcium oxide



combustion of natural gas



thermal decomposition of limestone



reaction of acid with alkali

**68. MAR 15/P12/Q10**

Limestone can be changed into slaked lime in two chemical reactions.

1 When limestone,  $\text{CaCO}_3$ , is heated it decomposes into lime,  $\text{CaO}$ .

2 Water is slowly dripped onto the cooled lime. The lime appears to expand and steam is produced. Slaked lime,  $\text{Ca(OH)}_2$ , is formed.

Which row shows the correct description of each of the chemical reactions?

|          | reaction 1  | reaction 2  |
|----------|-------------|-------------|
| <b>A</b> | endothermic | endothermic |
| <b>B</b> | endothermic | exothermic  |
| <b>C</b> | exothermic  | endothermic |
| <b>D</b> | exothermic  | exothermic  |

**69. O/N 14/P13/Q13**

What occurs when a fuel burns?

|          | fuel reacts with oxygen | energy change |
|----------|-------------------------|---------------|
| <b>A</b> | no                      | endothermic   |
| <b>B</b> | no                      | exothermic    |
| <b>C</b> | yes                     | endothermic   |
| <b>D</b> | yes                     | exothermic    |

**70. M/J 14/P12/Q12**

Solutions of two chemicals are mixed.

A reaction occurs and the temperature change is measured.

Which statement is correct?

- A** If the reaction is endothermic, the temperature decreases and energy is taken in.
- B** If the reaction is endothermic, the temperature increases and energy is given out.
- C** If the reaction is exothermic, the temperature decreases and energy is given out.
- D** If the reaction is exothermic, the temperature increases and energy is taken in.

**71. M/J 14/P11/Q13, M/J 14/P13/Q14**

Some reactions are endothermic.

How does the temperature and energy change in an endothermic reaction?

|          | temperature change | energy change    |
|----------|--------------------|------------------|
| <b>A</b> | decreases          | energy taken in  |
| <b>B</b> | decreases          | energy given out |
| <b>C</b> | increases          | energy taken in  |
| <b>D</b> | increases          | energy given out |

**72. O/N 13/P13/Q12**

When ammonium nitrate is added to water the temperature of the water decreases.

The ammonium nitrate can be recovered by evaporating the water added.

Which explains these observations?

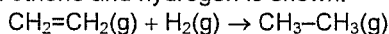
- A** The ammonium nitrate dissolves in the water and the process is endothermic.
- B** The ammonium nitrate reacts with the water and the process is endothermic.
- C** The ammonium nitrate dissolves in the water and the process is exothermic.
- D** The ammonium nitrate reacts with the water and the process is exothermic.



## 6.2: Energy Transfer

### 1. O/N 22/P23/Q17

The equation for the reaction between ethene and hydrogen is shown.



The bond energies are shown.

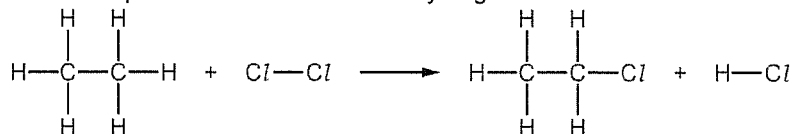
| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| C=C  | 612                      |
| H-H  | 436                      |
| C-C  | 348                      |
| C-H  | 416                      |

What is the overall energy change during this reaction?

- A -284 kJ/mol      B -132 kJ/mol      C +132 kJ/mol      D +284 kJ/mol

### 2. O/N 21/P22/Q11, O/N 21/P21/Q11, O/N 21/P23/Q12

Chlorine reacts with ethane to produce chloroethane and hydrogen chloride.



The reaction is exothermic.

The bond energies are shown in the table.

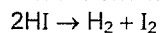
| bond  | bond energy<br>in kJ/mol |
|-------|--------------------------|
| C-Cl  | +340                     |
| C-C   | +350                     |
| C-H   | +410                     |
| Cl-Cl | +240                     |
| H-Cl  | +430                     |

What is the energy change for the reaction?

- A -1420 kJ/mol      B -120 kJ/mol      C +120 kJ/mol      D +1420 kJ/mol

### 3. M/J 21/P23/Q17

The equation for the decomposition of hydrogen iodide is shown.



Some bond energies are shown.

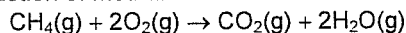
| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| H-H  | 440                      |
| I-I  | 150                      |
| H-I  | 300                      |

What is the energy change for the reaction?

- A -290 kJ/mol      B -10 kJ/mol      C +10 kJ/mol      D +290 kJ/mol

**4. MAR 21/P22/Q14**

The equation for the complete combustion of methane is shown.



The bond energies are shown in the table.

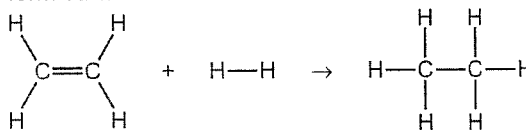
| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| C–H  | +410                     |
| C=O  | +805                     |
| O–H  | +460                     |
| O=O  | +496                     |

What is the energy change for the reaction?

- A –818 kJ/mol      B –359 kJ/mol      C –323 kJ/mol      D +102 kJ/mol

**5. O/N 20/P21/Q17**

Ethene reacts with hydrogen to form ethane.



The bond energies are shown in the table.

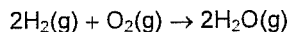
| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| C–C  | +350                     |
| C–H  | +410                     |
| H–H  | +436                     |
| C=C  | +614                     |

What is the energy change for the reaction?

- A –290 kJ/mol      B –120 kJ/mol      C +120 kJ/mol      D +290 kJ/mol

**6. O/N 20/P23/Q16**

The reaction between hydrogen and oxygen releases 486 kJ/mol of energy.



The bond energy of H–H is 436 kJ/mol and that of H–O is 464 kJ/mol.

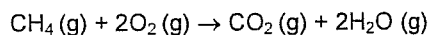
What is the bond energy of O=O?

- A 430 kJ/mol      B 458 kJ/mol      C 498 kJ/mol      D 984 kJ/mol

**7. M/J 20/P22/Q12**

Methane burns in excess oxygen.

The equation is shown.



Bond energies are shown.

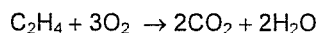
| bond | bond energy<br>/ kJ mol <sup>-1</sup> |
|------|---------------------------------------|
| C=O  | 805                                   |
| C–H  | 410                                   |
| O=O  | 496                                   |
| O–H  | 460                                   |

What is the energy change for the reaction?

- A  $(4 \times 410 + 2 \times 496) - (2 \times 805 + 4 \times 460)$       C  $(410 + 2 \times 496) - (805 + 2 \times 460)$   
 B  $(2 \times 805 + 2 \times 460) - (410 + 2 \times 496)$       D  $(410 + 496) - (805 + 460)$

**8. M/J 20/P23/Q12**

Ethene gas,  $C_2H_4$ , is completely burned in excess oxygen to form carbon dioxide and water. The equation for this exothermic reaction is shown.



The table shows the bond energies involved in the reaction.

| bond | bond energy (kJ/mol) |
|------|----------------------|
| C=C  | 614                  |
| C-H  | 413                  |
| O=O  | 495                  |
| C=O  | 799                  |
| O-H  | 467                  |

What is the total energy change in this reaction?

- A -954 kJ/mol      B -1010 kJ/mol      C -1313 kJ/mol      D -1369 kJ/mol

**9. O/N 19/P21/Q13**

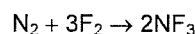
Which statements about endothermic reactions are correct?

- The energy of the products is greater than the energy of the reactants.
- The energy of the reactants is greater than the energy of the products.
- The temperature of the surroundings increases during the reaction.
- The temperature of the surroundings decreases during the reaction.

- A 1 and 3 only      B 1 and 4 only      C 2 and 3 only      D 2 and 4 only

**10. MAR 20/P22/Q12**

Nitrogen trifluoride,  $NF_3$ , is used in the manufacture of certain types of solar panels. The equation for the formation of nitrogen trifluoride is shown.



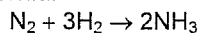
| type of bond | bond energy (kJ mol <sup>-1</sup> ) |
|--------------|-------------------------------------|
| N≡N          | +950                                |
| F-F          | +150                                |
| N-F          | +280                                |

Using the table of bond energies, what is the energy change for this reaction?

- A -560 kJ mol<sup>-1</sup>      B -280 kJ mol<sup>-1</sup>      C +280 kJ mol<sup>-1</sup>      D +3080 kJ mol<sup>-1</sup>

**11. M/J 19/P21/Q12**

Nitrogen reacts with hydrogen to produce ammonia.



The reaction is exothermic. The bond energies are shown in the table.

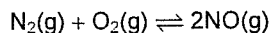
| bond | bond energy in kJ/mol |
|------|-----------------------|
| N≡N  | 945                   |
| H-H  | 436                   |
| N-H  | 390                   |

What is the energy change for this reaction?

- A -1473 kJ/mol      B -87 kJ/mol      C 87 kJ/mol      D 1473 kJ/mol

**12. M/J 19/P23/Q15**

A reaction between nitrogen and oxygen is shown. The forward reaction is endothermic.

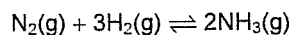


Which change increases the equilibrium yield of nitrogen monoxide, NO?

- A decreasing the pressure                      C increasing the pressure  
B decreasing the temperature                D increasing the temperature

**13. M/J 19/P23/Q32**

Ammonia is manufactured in an exothermic reaction.



What is the effect of lowering the temperature on the rate of formation and equilibrium yield of ammonia?

|   | rate of formation | equilibrium yield |
|---|-------------------|-------------------|
| A | decreases         | decreases         |
| B | decreases         | increases         |
| C | increases         | decreases         |
| D | increases         | increases         |

**14. O/N 17/P21/Q8**

A compound is analysed and found to contain 85.7% carbon and 14.3% hydrogen.

What is its empirical formula?

- A CH                      B CH<sub>2</sub>                      C C<sub>2</sub>H<sub>4</sub>                      D C<sub>6</sub>H

## Answers Section

| 6.1   |       | 6.2   |
|-------|-------|-------|
| 1. D  | 41. B | 1. B  |
| 2. B  | 42. C | 2. B  |
| 3. D  | 43. C | 3. C  |
| 4. C  | 44. B | 4. A  |
| 5. D  | 45. B | 5. B  |
| 6. B  | 46. D | 6. C  |
| 7. C  | 47. D | 7. A  |
| 8. C  | 48. B | 8. C  |
| 9. B  | 49. B | 9. B  |
| 10. D | 50. B | 10. B |
| 11. B | 51. D | 11. B |
| 12. A | 52. D | 12. D |
| 13. D | 53. A | 13. B |
| 14. A | 54. A | 14. B |
| 15. C | 55. C |       |
| 16. C | 56. B |       |
| 17. B | 57. B |       |
| 18. C | 58. A |       |
| 19. B | 59. A |       |
| 20. B | 60. D |       |
| 21. B | 61. C |       |
| 22. D | 62. B |       |
| 23. A | 63. C |       |
| 24. B | 64. A |       |
| 25. A | 65. B |       |
| 26. B | 66. C |       |
| 27. D | 67. C |       |
| 28. C | 68. B |       |
| 29. A | 69. D |       |
| 30. A | 70. A |       |
| 31. A | 71. A |       |
| 32. B | 72. A |       |
| 33. A |       |       |
| 34. B |       |       |
| 35. B |       |       |
| 36. A |       |       |
| 37. C |       |       |
| 38. B |       |       |
| 39. B |       |       |
| 40. C |       |       |

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## Unit 7

# CHEMICAL REACTIONS

IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 7.1 Rate (speed) of reaction
- 7.2 Reversible reactions
- 7.3 Redox

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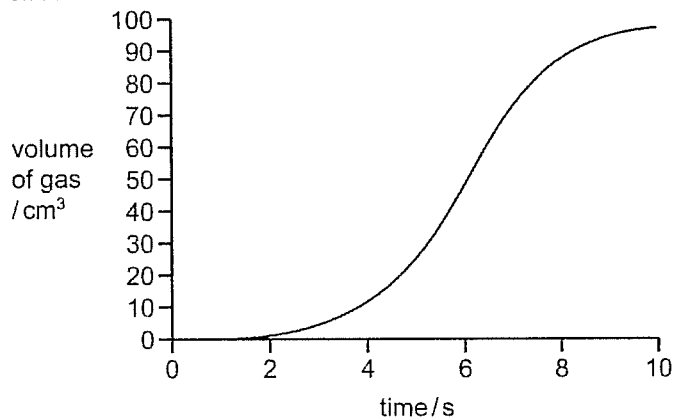
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## Unit 7: Chemical Reactions

### 7.1: Rate (speed) of reaction

1. O/N 22/P22/Q15

The volume of gas given off in a chemical reaction is measured over time. The results are shown.



At which time is the rate of reaction greatest?

- A 0s                      B 4s                      C 6s                      D 10s

2. O/N 22/P21/Q14

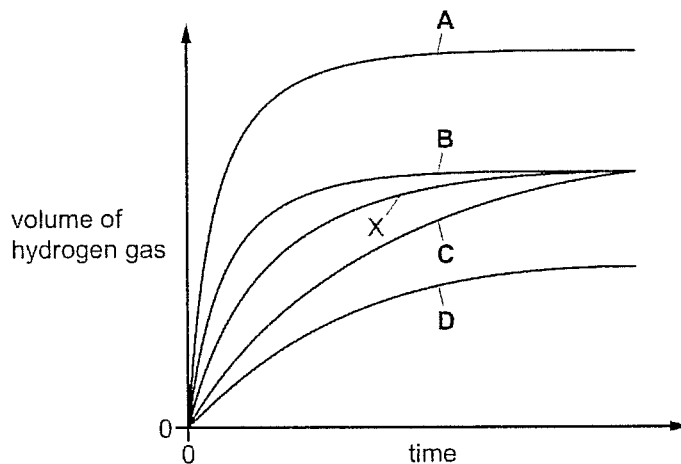
A student adds excess zinc to dilute hydrochloric acid at 25 °C.

The hydrogen gas produced is collected and measured at room temperature and pressure.

The results are plotted and labelled as curve X on the graph.

The experiment is repeated at 50 °C with all other conditions remaining the same.

Which graph shows the results at 50 °C?



3. M/J 22/P22/Q13

Which statements explain why increasing the concentration of a reactant increases the rate of reaction?

- 1 It increases the collision rate of particles.
- 2 It lowers the activation energy.
- 3 A greater proportion of the colliding molecules have the required activation energy.
- 4 There are more particles per unit volume.

- A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4



**4. M/J 22/P23/Q14**

Which statement explains why increasing the concentration of a reactant increases the rate of the reaction?

- A A greater proportion of the particles have the activation energy so there are more successful collisions.
- B Particles have more energy so there are more frequent collisions.
- C There are more particles in the same volume so there are more frequent collisions.
- D The particles move more quickly so there are more frequent collisions.

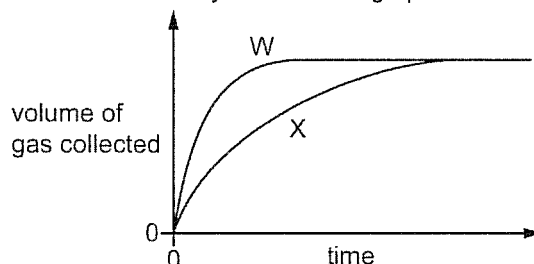
**5. O/N 22/P23/Q14**

Dilute hydrochloric acid is reacted with excess calcium carbonate and the total volume of gas is measured at regular intervals.

The results are shown by line W on the graph.

The experiment is repeated but with one change.

The results of the second experiment are shown by line X on the graph.



Which change is made in the second experiment?

- A A catalyst is added.
- B The calcium carbonate is broken into smaller pieces.
- C The concentration of the dilute hydrochloric acid is increased.
- D The temperature of the dilute hydrochloric acid is decreased.

**6. MAR 22/P22/Q19**

How does increasing the concentration affect the reacting particles in a chemical reaction?

|   | increases the collision rate | increases the proportion of particles with the activation energy |
|---|------------------------------|------------------------------------------------------------------|
| A | ✓                            | x                                                                |
| B | ✓                            | ✓                                                                |
| C | x                            | x                                                                |
| D | x                            | ✓                                                                |

**7. O/N 21/P22/Q19**

Which reaction is a photochemical reaction?

- A addition of bromine to propene
- B esterification of ethanol and ethanoic acid
- C oxidation of ethanol
- D substitution of methane with chlorine

**8. O/N 21/P21/Q13**

Which statements about the effect of increasing the temperature on the rate of a reaction are correct?

- 1 It increases the rate of a reaction.
- 2 It increases the activation energy.
- 3 It increases the frequency of collisions.

- A 1, 2 and 3
- B 1 and 2 only
- C 1 and 3 only
- D 2 and 3 only

**9. M/J 21/P22/Q15, M/J 21/P21/Q15**

Four statements about the effect of increasing temperature on a reaction are shown.

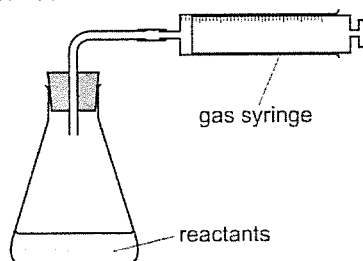
- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A 1, 2 and 3                      B 1, 3 and 4                      C 2, 3 and 4                      D 2 and 3 only

**10. MAR 21/P22/Q16**

The apparatus shown is used to measure the rate of a reaction.



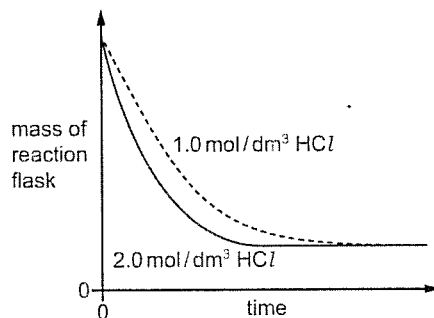
Which equation represents a reaction where the rate can be measured using this apparatus?

- A  $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$                       C  $\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{Cu(s)} + \text{FeSO}_4\text{(aq)}$   
 B  $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$                       D  $2\text{Na(s)} + \text{Br}_2\text{(l)} \rightarrow 2\text{NaBr(s)}$

**11. O/N 21/P23/Q17**

Excess dilute hydrochloric acid is added to equal masses of powdered calcium carbonate in two separate experiments.

Two different concentrations of hydrochloric acid are used. The temperature in both experiments is the same. The results show the change in mass of the reaction flask measured over time.

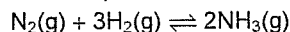


Why is the rate of reaction for the  $1.0 \text{ mol/dm}^3$  hydrochloric acid slower?

|   | collision energy                   | collision rate |
|---|------------------------------------|----------------|
| A | lower                              | higher         |
| B | lower                              | lower          |
| C | same as for $2.0 \text{ mol/dm}^3$ | higher         |
| D | same as for $2.0 \text{ mol/dm}^3$ | lower          |

**12. O/N 20/P22/Q18**

Ammonia is produced by the Haber process. The equation is shown.

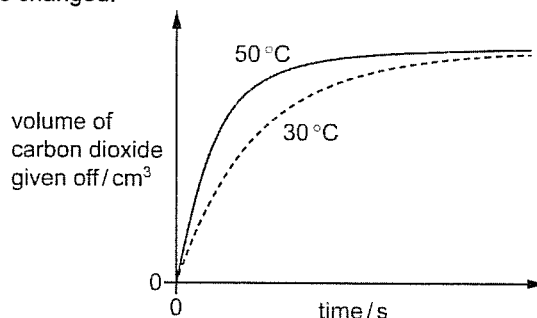


The forward reaction is exothermic. Which statement is correct?

- A Increasing pressure decreases the yield of ammonia, but speeds up the reaction.  
 B Increasing temperature decreases the yield of ammonia, but speeds up the reaction.  
 C Increasing the concentration of hydrogen and nitrogen results in a lower yield of ammonia.  
 D Increasing the temperature increases the yield of ammonia and speeds up the reaction.

**13. M/J 20/P22/Q15**

The results of adding excess marble chips (calcium carbonate) to hydrochloric acid at 50 °C and at 30 °C are shown. Only the temperature is changed.



Which row describes the reacting particles at 30 °C compared to those at 50 °C?

|   | collision rate | collision energy |
|---|----------------|------------------|
| A | higher         | higher           |
| B | higher         | lower            |
| C | lower          | higher           |
| D | lower          | lower            |

**14. O/N 20/P21/Q19**

A student investigates the effect of concentration on the rate of reaction between calcium carbonate and hydrochloric acid. He follows the method shown.

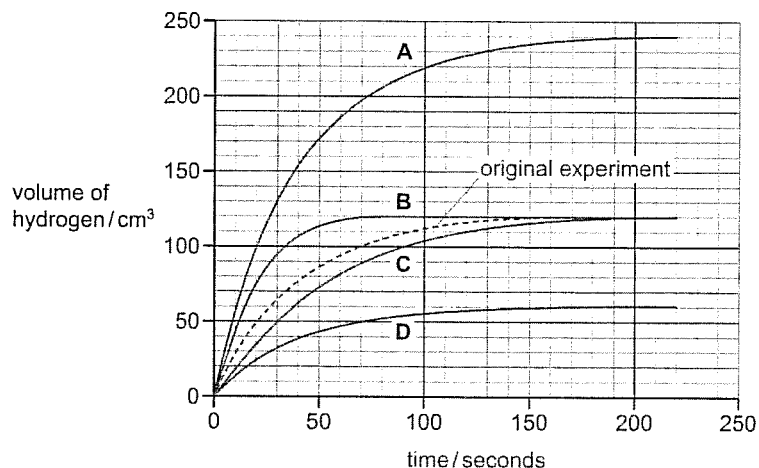
- Place 1g of calcium carbonate in a conical flask.
- Add excess hydrochloric acid.
- Let the reaction continue until no more gas is made.
- Repeat the experiment with different concentrations of hydrochloric acid.

Which essential step has been left out of the method if he is to work out the rate of the reaction?

- |                                |                       |
|--------------------------------|-----------------------|
| A heating the reaction mixture | C timing the reaction |
| B placing a bung in the flask  | D using a catalyst    |

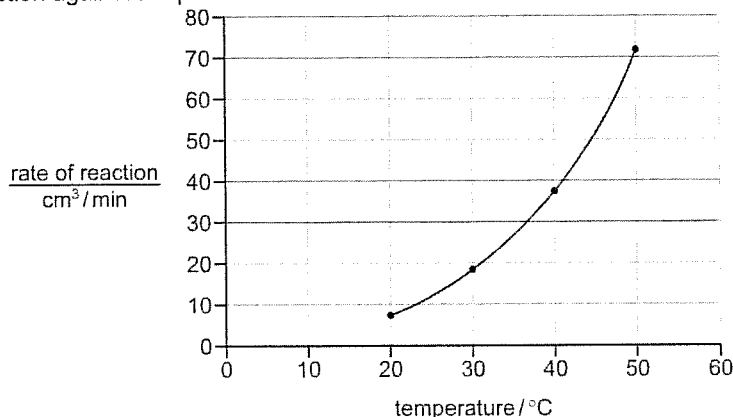
**15. MAR 20/P22/Q15**

A student adds excess magnesium ribbon to 10 cm<sup>3</sup> of 0.5 mol / dm<sup>3</sup> sulfuric acid. The hydrogen gas is collected and its volume measured every 10 seconds. The experiment is repeated using the same mass of magnesium ribbon with 5 cm<sup>3</sup> of 0.5 mol / dm<sup>3</sup> sulfuric acid added to 5 cm<sup>3</sup> of water. Which graph shows the results of the second experiment?



## 16. M/J 20/P21/Q15

The rate of reaction between calcium carbonate chips and hydrochloric acid is studied by collecting the volume of gas released in one minute at different temperatures.  
A graph of rate of reaction against temperature is shown.



Which statement fully explains why increasing the temperature has this effect on the rate?

- A The kinetic energy of the particles increases so the collisions are harder.
- B The number of collisions between particles increases.
- C The activation energy needed for the particles to react is reduced.
- D There are more frequent collisions between particles with enough energy to react.

## 17. O/N 19/P21/Q16

The rate of reaction between magnesium and dilute hydrochloric acid is increased by increasing the concentration of the acid.

How does this affect the reacting particles?

|   | collision rate of particles | proportion of particles with sufficient energy to react |
|---|-----------------------------|---------------------------------------------------------|
| A | increases                   | increases                                               |
| B | increases                   | stays the same                                          |
| C | stays the same              | increases                                               |
| D | stays the same              | stays the same                                          |

## 18. O/N 19/P23/Q16

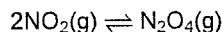
Magnesium reacts with dilute hydrochloric acid.

Which statement about the particles in the reaction is correct?

- A Increasing the concentration of dilute hydrochloric acid increases the collision rate but has no effect on the activation energy.
- B Increasing the concentration of dilute hydrochloric acid increases the collision rate and the activation energy.
- C Increasing the temperature of the reaction increases the activation energy.
- D Increasing the temperature of the reaction causes all collisions to lead to a reaction.

## 19. O/N 19/P23/Q17

Two molecules of nitrogen dioxide combine in a reversible reaction to form dinitrogen tetroxide.  
The forward reaction is exothermic.



Which changes in reaction conditions would **both** increase the amount of dinitrogen tetroxide at equilibrium?

- A decreasing the temperature and decreasing the pressure
- B decreasing the temperature and increasing the pressure
- C increasing the temperature and decreasing the pressure
- D increasing the temperature and increasing the pressure

**20. M/J 19/P22/Q13, M/J 19/P21/Q13, M/J 19/P23/Q13**

Which change in reaction conditions increases both the collision rate and the proportion of molecules with sufficient energy to react?

- A addition of a catalyst  
 B increasing the concentration of a reactant  
 C increasing the surface area of a reactant  
 D increasing the temperature of the reaction

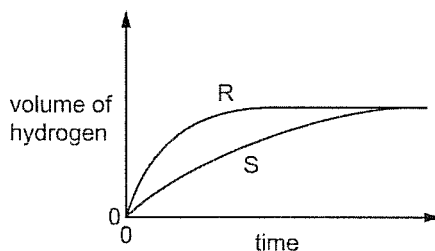
**21. M/J 19/P23/Q19**

Solutions of acid R and acid S have the same concentration.

The same volume of each acid at the same temperature is reacted with the same mass of magnesium ribbon.

The volume of hydrogen produced is measured.

The results are shown.



Which statement about the reactions is correct?

- A Acid S reacts faster than acid R.  
 B The final volume of hydrogen produced in each reaction is different.  
 C Acid R is a stronger acid than acid S.  
 D Acid S is a stronger acid than acid R.

**22. MAR 19/P22/Q15**

A student adds dilute hydrochloric acid at two different temperatures to two different lumps of limestone. The lumps of limestone have the same mass.

The carbon dioxide gas produced is collected in a gas syringe.

The volume of carbon dioxide collected in 1 minute at each temperature is shown.

| temperature / °C | volume of carbon dioxide produced in 1 minute / cm <sup>3</sup> |
|------------------|-----------------------------------------------------------------|
| 25               | 10                                                              |
| 50               | 40                                                              |

Which row describes and explains the results obtained at 50 °C compared with 25 °C?

|   | reaction rate | energy of collisions |
|---|---------------|----------------------|
| A | higher        | lower                |
| B | higher        | higher               |
| C | lower         | lower                |
| D | lower         | higher               |

**23. O/N 18/P22/Q14**

The rate of reaction between magnesium ribbon and 2 mol / dm<sup>3</sup> hydrochloric acid at 25 °C to produce hydrogen gas is measured.

In another experiment, either the concentration of the hydrochloric acid or the temperature is changed. All other conditions are kept the same.

Which conditions increase the rate of reaction?

- A 1 mol / dm<sup>3</sup> hydrochloric acid at 25 °C  
 B 2 mol / dm<sup>3</sup> hydrochloric acid at 10 °C  
 C 2 mol / dm<sup>3</sup> hydrochloric acid at 20 °C  
 D 3 mol / dm<sup>3</sup> hydrochloric acid at 25 °C

**24. O/N 18/P21/Q14**

The effects of a change in conditions on a chemical reaction are listed.

- 1 The total number of collisions per minute increased.
- 2 The number of effective collisions per minute increased.
- 3 The average energy of the particles increased.

Which change in conditions caused all of these effects?

- A addition of a catalyst  
 B increasing the concentration of a solution of a reactant  
 C increasing the surface area of a solid reactant  
 D increasing the temperature

**25. M/J 18/P22/Q14, M/J 18/P21/Q14, M/J 18/P23/Q14**

Which row describes the effects of increasing both concentration and temperature on the collisions between reacting particles?

|   | increasing concentration                                                       | increasing temperature                                                         |
|---|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| A | more collisions per second only                                                | more collisions per second only                                                |
| B | more collisions per second and more collisions with sufficient energy to react | more collisions per second only                                                |
| C | more collisions per second only                                                | more collisions per second and more collisions with sufficient energy to react |
| D | more collisions per second and more collisions with sufficient energy to react | more collisions per second and more collisions with sufficient energy to react |

**26. MAR 18/P22/Q16**

Calcium carbonate reacts with dilute hydrochloric acid to form bubbles of carbon dioxide.

At a higher temperature, the same reaction is faster.

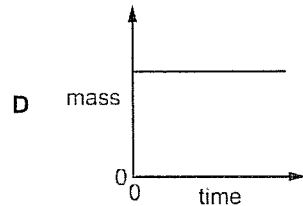
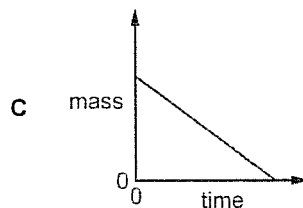
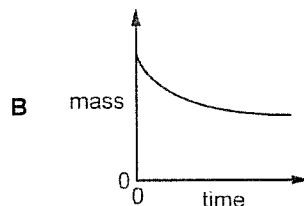
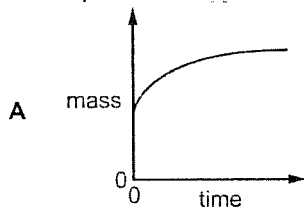
Which row explains this observation?

|   | collision rate | number of molecules with sufficient energy to react |
|---|----------------|-----------------------------------------------------|
| A | increases      | more                                                |
| B | increases      | the same                                            |
| C | stays the same | more                                                |
| D | stays the same | the same                                            |

**27. O/N 17/P22/Q13, O/N 17/P21/Q13, O/N 17/P23/Q13**

The mass of a beaker and its contents is plotted against time.

Which graph represents what happens when sodium carbonate reacts with an excess of dilute hydrochloric acid in an open beaker?



**28. O/N 17/P22/Q15**

Four statements about the effect of increasing temperature on a reaction are shown.

- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A 1, 2 and 3      B 1, 3 and 4      C 2, 3 and 4      D 2 and 3 only

**29. O/N 17/P21/Q2**

student put 25.0 cm<sup>3</sup> of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use to obtain the most accurate results?

- A balance, measuring cylinder, thermometer      C balance, pipette, thermometer  
B balance, pipette, stopwatch      D burette, pipette, thermometer

**30. O/N 17/P21/Q15**

Ethanoic acid reacts slowly with calcium carbonate.

Which statements explain why an increase in temperature increases the rate of the reaction?

- 1 The activation energy of the reaction is decreased.
- 2 There is an increase in collision rate.
- 3 The particles have more energy.
- 4 There will be fewer successful collisions.

- A 1 and 2      B 1 and 3      C 2 and 3      D 2 and 4

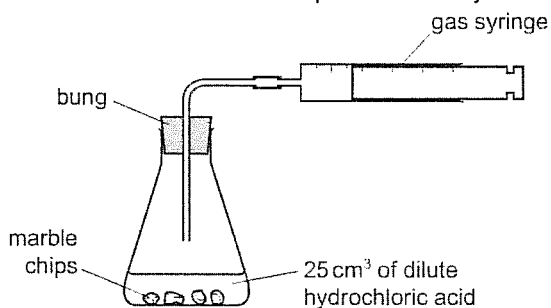
**31. O/N 17/P23/Q15**

Which statement about the effect of concentration and temperature on the rate of a reaction is **not** correct?

- A If the concentration of a reactant is increased, the rate of reaction increases because more particles have sufficient energy to react.  
B If the concentration of a reactant is increased, the rate of reaction increases because there are more collisions between particles per second.  
C If the temperature is increased, the rate of reaction increases because there are more collisions between particles per second.  
D If the temperature is increased, the rate of reaction increases because more particles have sufficient energy to react.

**32. M/J 17/P22/Q15, M/J 17/P21/Q15, M/J 17/P23/Q15, M/J 14/P12/Q14**

A student was investigating the reaction between marble chips and dilute hydrochloric acid.

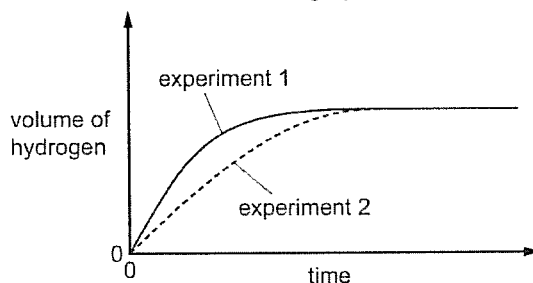


Which changes slow down the rate of reaction?

|   | temperature of acid | concentration of acid | surface area of marble chips |
|---|---------------------|-----------------------|------------------------------|
| A | decrease            | decrease              | decrease                     |
| B | decrease            | decrease              | increase                     |
| C | increase            | decrease              | decrease                     |
| D | increase            | increase              | increase                     |

**33. O/N 16/P22/Q14**

Zinc granules are reacted with excess dilute hydrochloric acid.  
The volume of hydrogen given off is measured at different times.  
The results are shown on the graph, labelled experiment 1.  
The results for a second experiment are also shown on the graph, labelled experiment 2.



Which change to the conditions was made in experiment 2?

- A The concentration of the hydrochloric acid was decreased.
- B The size of the zinc granules was decreased.
- C The surface area of the zinc granules was increased.
- D The temperature was increased.

**34. O/N 16/P22/Q15**

In an experiment nitric acid is added to excess marble chips and the volume of carbon dioxide formed is measured.

The experiment is repeated using smaller marble chips. All other conditions remain the same.

Which statement about the second experiment is correct?

- A The collisions are more frequent and higher energy.
- B The collisions are more frequent and the same energy.
- C The collisions are the same frequency and the same energy.
- D The collisions are the same frequency and higher energy.

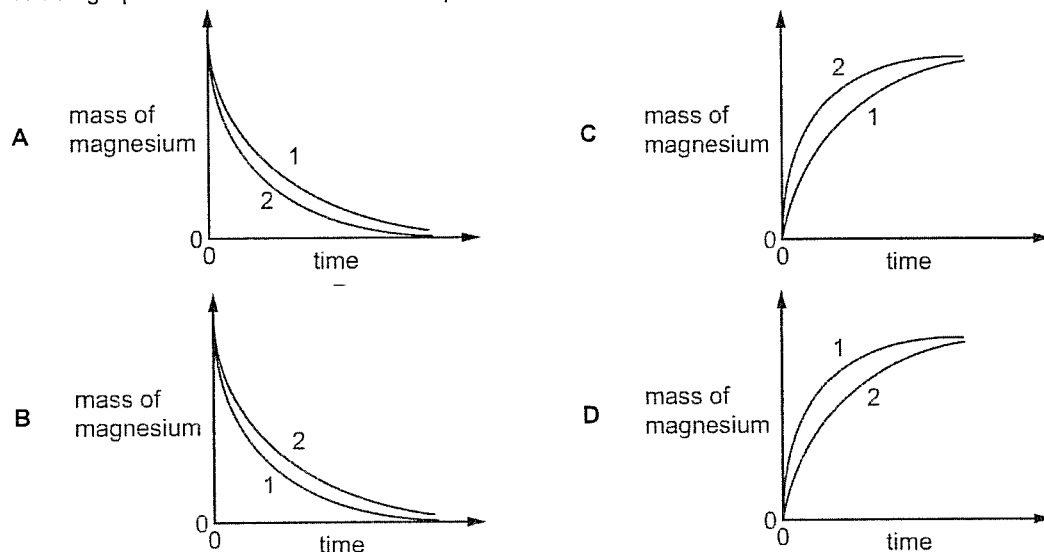
**35. O/N 16/P21/Q14**

The rate of reaction between magnesium and excess dilute hydrochloric acid was followed by measuring the mass of magnesium present at regular time intervals.

Two experiments were performed.

Both experiments used 0.1 g of magnesium ribbon. The acid in experiment 1 was less concentrated than in experiment 2.

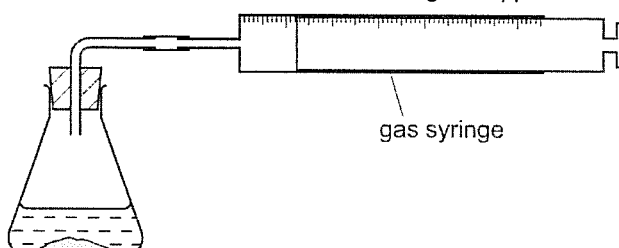
Which graph shows the results of the experiments?



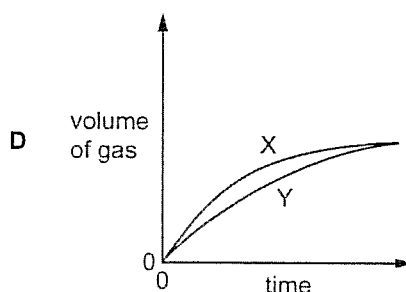
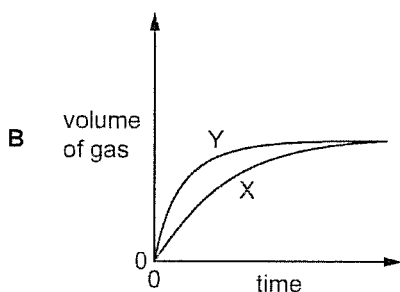
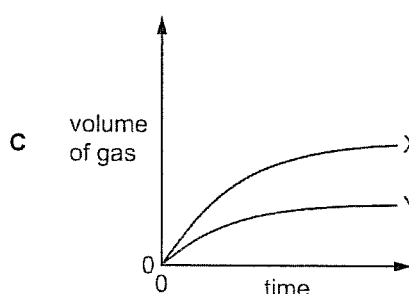
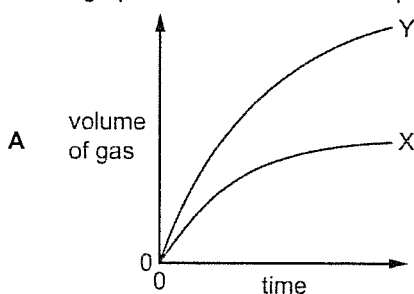


**36. O/N 16/P23/Q14**

An experiment X is carried out between a solid and a solution using the apparatus shown.



The volume of gas given off is measured at different times and the results plotted on a graph. In a second experiment Y, the surface area of the solid is increased but all other factors remain the same. Which graph shows the results of experiments X and Y?

**37. O/N 16/P21/Q15**

Which statement explains why coal dust forms an explosive mixture with air?

- A Coal dust catalyses the explosion.
- B Coal dust has a large surface area.
- C Crushing coal increases the concentration of the coal.
- D Crushing coal increases the temperature of the coal.

**38. O/N 16/P23/Q15**

Which change in conditions increases the energy of the particles in a reaction?

- A addition of a catalyst
- B increase in concentration
- C increase in surface area
- D increase in temperature

**39. M/J 16/P22/Q15**

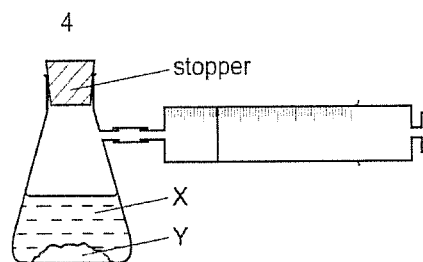
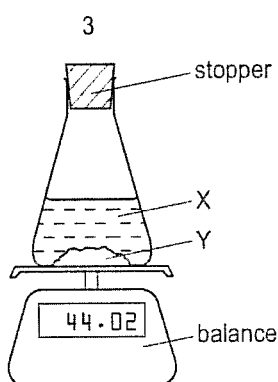
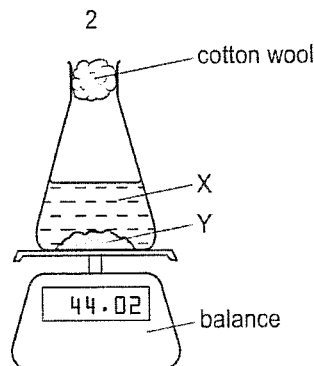
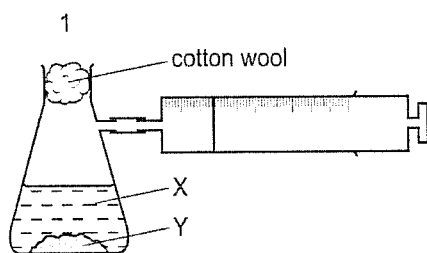
Which row describes how the energy of collision between particles changes when concentration and temperature are increased?

|   | concentration | temperature |
|---|---------------|-------------|
| A | increases     | increases   |
| B | increases     | no change   |
| C | no change     | increases   |
| D | no change     | no change   |

**40. M/J 16/P22/Q14, M/J 16/P21/Q14, M/J 16/P23/Q14**

A liquid X reacts with solid Y to form a gas.

Which two diagrams show suitable methods for investigating the rate (speed) of the reaction?

**A** 1 and 3**B** 1 and 4**C** 2 and 3**D** 2 and 4**41. M/J 16/P21/Q15**

Which statements explain why increasing temperature increases the rate of a chemical reaction?

- 1 Heat makes the molecules move faster and collide more often.
- 2 Heat makes the molecules collide with more energy so they are more likely to react.
- 3 Increasing temperature lowers the activation energy for the reaction.

**A** 1 and 2**B** 1 and 3**C** 1 only**D** 2 only**42. M/J 16/P23/Q15**

Which row explains why increasing temperature increases the rate of reaction?

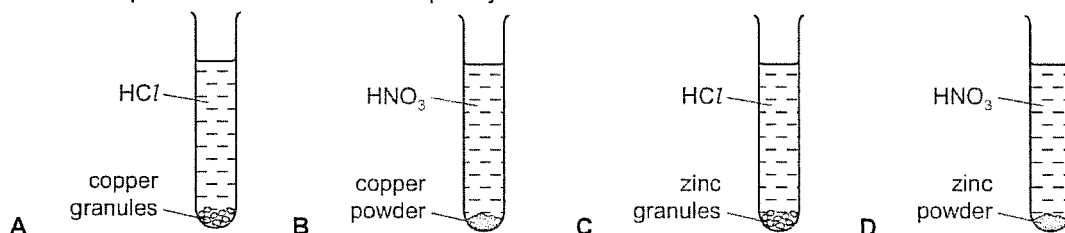
|          | particles collide more often | particles collide with more energy |
|----------|------------------------------|------------------------------------|
| <b>A</b> | ✓                            | ✓                                  |
| <b>B</b> | ✓                            | x                                  |
| <b>C</b> | x                            | ✓                                  |
| <b>D</b> | x                            | x                                  |

**43. MAR 16/P22/Q15**Which statement about catalysts in chemical reactions is **not** correct?

- A** Catalysts are not used up in the reaction.
- B** Catalysts increase the energy of the reacting particles.
- C** Catalysts increase the rate of the reaction.
- D** Catalysts lower the activation energy.

## 44. O/N 15/P12/Q12

The diagram shows four experiments in which equal volumes of aqueous acid (all in an excess) are added to equal masses of metal. Both acids have the same concentration.  
In which experiment has the metal completely reacted in the shortest time?



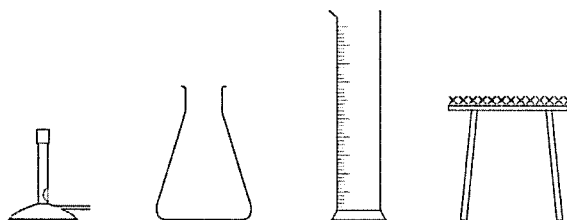
## 45. O/N 15/P11/Q12

The effect of temperature on the rate of the reaction between marble chips and hydrochloric acid can be investigated by measuring the production of carbon dioxide.  
Which item of equipment is **not** required for the investigation?

- A condenser      B gas syringe      C stopclock      D thermometer

## 46. O/N 15/P13/Q2

A student was asked to measure the rate of reaction between dilute hydrochloric acid and marble chips at different temperatures.  
Some of the apparatus used is shown.



Which two other pieces of apparatus would be needed?

- A balance and pipette      C beaker and stopclock  
B balance and stopclock      D burette and pipette

## 47. O/N 15/P13/Q12

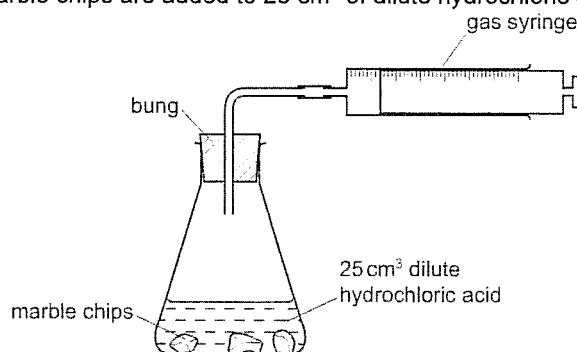
Which of the following changes decreases the rate of the reaction between magnesium and dilute hydrochloric acid?

- 1 diluting the acid  
2 using larger pieces of magnesium  
3 cooling the mixture

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

## 48. M/J 15/P11/Q2

A student uses the apparatus shown in the diagram below to measure the volume of carbon dioxide gas made when different masses of marble chips are added to 25 cm<sup>3</sup> of dilute hydrochloric acid.

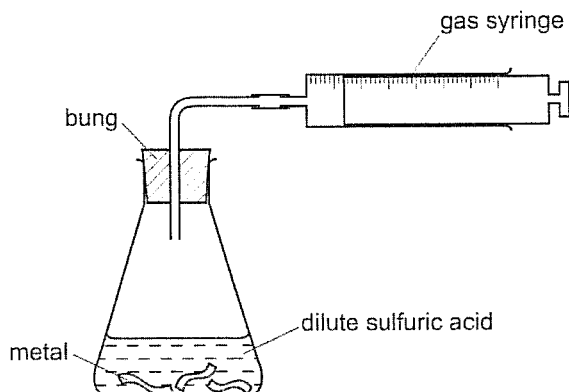


Which other items of apparatus are needed?

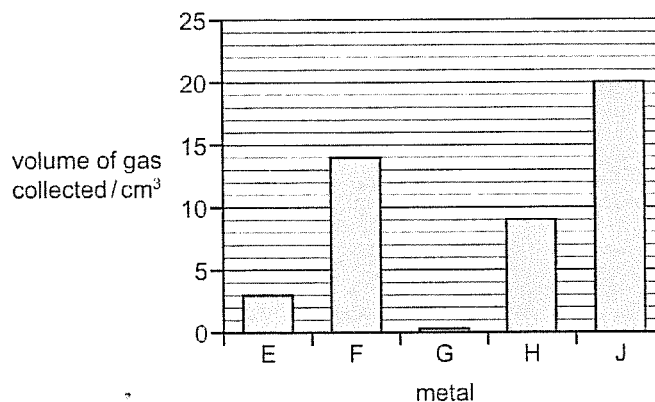
- A funnel and balance      C measuring cylinder and balance  
B funnel and stopwatch      D measuring cylinder and stopwatch

**49. M/J 15/P12/Q26**

Samples of five different metals, E, F, G, H and J were reacted with dilute sulfuric acid using the apparatus shown.



The volume of hydrogen gas collected after one minute was measured.  
The results are shown on the bar chart.

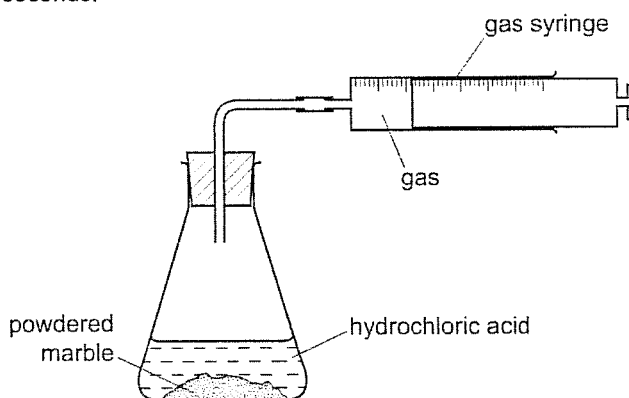


What is the order of reactivity of the metals (most reactive first)?

- A E, F, G, H, J      B G, E, H, F, J      C J, F, H, E, G      D J, H, G, F, E

**50. M/J 15/P11/Q12**

Powdered marble reacts with hydrochloric acid using the apparatus shown.  
The gas syringe fills in 36 seconds.

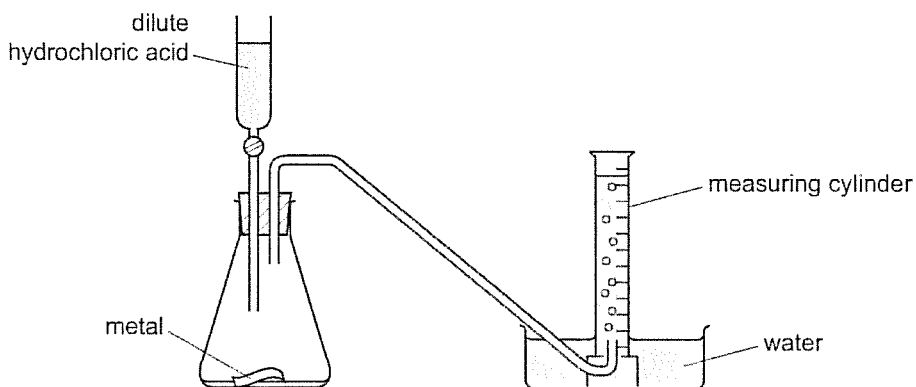


The experiment is repeated using marble chips in place of powdered marble.  
How long does it take to fill the gas syringe in this experiment?

- A 9 seconds      B 18 seconds      C 36 seconds      D 72 seconds

**51. M/J 15/P13/Q14**

The diagram shows an experiment to measure the rate of a chemical reaction.



Which change decreases the rate of reaction?

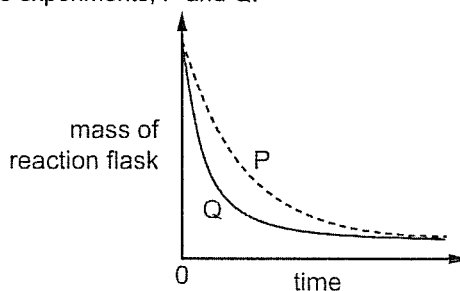
- |                                                |                                       |
|------------------------------------------------|---------------------------------------|
| <b>A</b> adding water to the flask             | <b>C</b> using more concentrated acid |
| <b>B</b> heating the flask during the reaction | <b>D</b> using powdered metal         |

**52. MAR 15/P12/Q11, M/J 10/P12/Q15, M/J 10/P11/Q16, M/J 10/P13/Q15**

A student investigates the rate of reaction between marble chips and hydrochloric acid.

The mass of the reaction flask is measured.

The graph shows the results of two experiments, P and Q.



Which change explains the difference between P and Q?

- |                                             |                                                       |
|---------------------------------------------|-------------------------------------------------------|
| <b>A</b> A catalyst is added in P.          | <b>C</b> Bigger marble chips are used in Q.           |
| <b>B</b> A higher temperature is used in P. | <b>D</b> Hydrochloric acid is more concentrated in Q. |

**53. O/N 14/P12/Q16, O/N 14/P11/Q15**

The rate of a reaction depends on temperature, concentration, particle size and catalysts.

Which statement is **not** correct?

- A** Catalysts can be used to increase the rate of reaction.  
**B** Higher concentration decreases the rate of reaction.  
**C** Higher temperature increases the rate of reaction.  
**D** Larger particle size decreases the rate of reaction.

**54. O/N 14/P12/Q17, O/N 14/P11/Q18, O/N 14/P13/Q18**

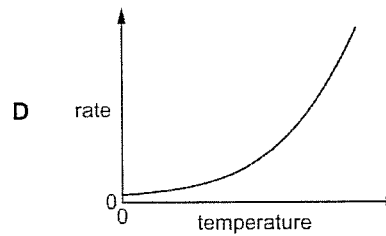
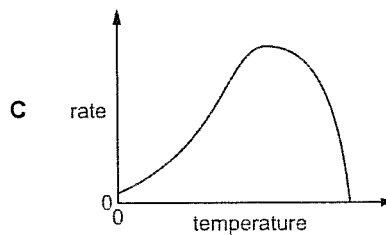
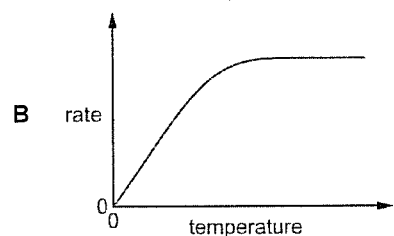
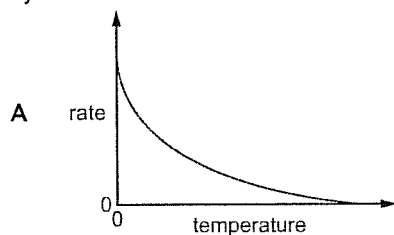
Which changes decrease the rate of reaction between magnesium and air?

- 1 heating the magnesium to a higher temperature
- 2 using a higher proportion of oxygen in the air
- 3 using magnesium ribbon instead of powdered magnesium

- |                     |                 |                 |                 |
|---------------------|-----------------|-----------------|-----------------|
| <b>A</b> 1, 2 and 3 | <b>B</b> 1 only | <b>C</b> 2 only | <b>D</b> 3 only |
|---------------------|-----------------|-----------------|-----------------|

**55. O/N 14/P13/Q15**

Which graph shows the effect of increasing temperature on the rate of reaction of calcium carbonate with dilute hydrochloric acid?

**56. M/J 14/P12/Q16, M/J 14/P11/Q16, M/J 14/P13/Q17**

In separate experiments, a catalyst is added to a reaction mixture and the temperature of the mixture is decreased.

What are the effects of these changes on the rate of the reaction?

|          | catalyst added | temperature decreased |
|----------|----------------|-----------------------|
| <b>A</b> | faster         | faster                |
| <b>B</b> | faster         | slower                |
| <b>C</b> | slower         | faster                |
| <b>D</b> | slower         | slower                |

**57. O/N 13/P12/Q2, O/N 13/P11/Q2**

A student measures the rate of two reactions.

In one reaction, there is a change in mass of the reactants during the reaction.

In the second reaction, there is a change in temperature during the reaction.

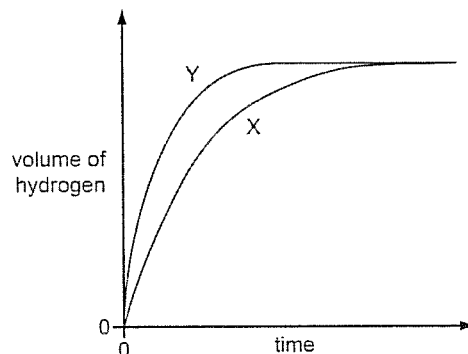
Which piece of apparatus would be essential in **both** experiments?

**A** balance      **B** clock      **C** pipette      **D** thermometer

**58. O/N 13/P12/Q16, O/N 13/P11/Q15, O/N 13/P13/Q15**

A student investigates the rate of reaction between zinc and an excess of sulfuric acid.

The graph shows the results of two experiments, X and Y.



Which change explains the difference between X and Y?

**A** A catalyst is added in Y.

**B** A lower temperature is used in Y.

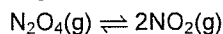
**C** Larger pieces of zinc are used in Y.

**D** Less concentrated acid is used in Y

## 7.2: Reversible reactions

### 1. O/N 22/P21/Q15

Dinitrogen tetroxide,  $\text{N}_2\text{O}_4$ , is converted into nitrogen dioxide,  $\text{NO}_2$ , in a reversible reaction.



The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

|   | pressure<br>/ atmospheres | temperature |
|---|---------------------------|-------------|
| A | 2                         | high        |
| B | 2                         | low         |
| C | 50                        | high        |
| D | 50                        | low         |

### 2. MAR 22/P22/Q18

The reaction used to manufacture ammonia from nitrogen and hydrogen is reversible.

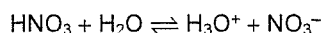
An equilibrium is established between ammonia, nitrogen and hydrogen.

Which statement describes the equilibrium?

- A Both the forward reaction and the backward reaction have the same rate.
- B The rate of the backward reaction is greater than the rate of the forward reaction.
- C The rate of the forward reaction is greater than the rate of the backward reaction.
- D The forward and backward reactions have both stopped.

### 3. MAR 22/P22/Q30

How many species are acting as bases in this reversible reaction?

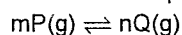


- A 3                                      B 2                                      C 1                                      D 0

### 4. O/N 21/P22/Q20

The equation shown represents a reaction at equilibrium.

m and n represent the balancing numbers for the reactant and product respectively.



A high temperature increases the concentration of Q.

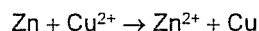
A high pressure increases the concentration of Q.

Which statement about the reaction is correct?

- A The forward reaction is exothermic and m is greater than n.
- B The forward reaction is exothermic and m is less than n.
- C The forward reaction is endothermic and m is greater than n.
- D The forward reaction is endothermic and m is less than n.

### 5. M/J 21/P22/Q16

An example of a redox reaction is shown.



Which statement about the reaction is correct?

- A Zn is the oxidising agent and it oxidises  $\text{Cu}^{2+}$ .
- B Zn is the oxidising agent and it reduces  $\text{Cu}^{2+}$ .
- C Zn is the reducing agent and it oxidises  $\text{Cu}^{2+}$ .
- D Zn is the reducing agent and it reduces  $\text{Cu}^{2+}$ .

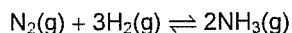
### 6. M/J 21/P21/Q17

Which statement about a reaction in equilibrium is correct?

- A Both the forward and the backward reactions are proceeding at the same rate.
- B Neither the forward nor the backward reaction is proceeding.
- C The amount of product present is no longer affected by changes in temperature or pressure.
- D The amount of product present is only affected by a change in pressure.

**7. M/J 20/P21/Q9, M/J 20/P23/Q9**

The Haber process is a reversible reaction.



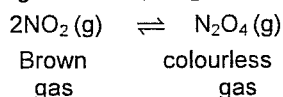
The reaction has a 30% yield of ammonia.

Which volume of ammonia gas,  $\text{NH}_3$ , measured at room temperature and pressure, is obtained by reacting 0.75 moles of hydrogen with excess nitrogen?

- A 3600  $\text{cm}^3$       B 5400  $\text{cm}^3$       C 12 000  $\text{cm}^3$       D 18 000  $\text{cm}^3$

**8. M/J 20/P23/Q16**

At room temperature, the conversion of nitrogen dioxide,  $\text{NO}_2$ , into dinitrogen tetroxide,  $\text{N}_2\text{O}_4$ , is reversible.

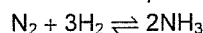


The forward reaction is exothermic. Which changes cause the equilibrium to shift to the left?

|   | pressure | temperature |
|---|----------|-------------|
| A | decrease | decrease    |
| B | decrease | increase    |
| C | increase | decrease    |
| D | increase | increase    |

**9. M/J 20/P23/Q31**

Hydrogen and nitrogen react to form ammonia in the Haber process.



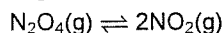
The forward reaction is exothermic. Which statements about the process are correct?

- 1 Nitrogen is obtained from the air.
- 2 Increasing the temperature of the reaction increases the yield of ammonia.
- 3 Increasing the reaction pressure increases the yield of ammonia.
- 4 Vanadium(V) oxide is used as a catalyst.

- A 1 and 2      B 1 and 3      C 2 and 3      D 3 and 4

**10. O/N 19/P21/Q17**

Dinitrogen tetroxide,  $\text{N}_2\text{O}_4$ , is converted into nitrogen dioxide,  $\text{NO}_2$ , in a reversible reaction.



The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

|   | pressure<br>/ atmospheres | temperature |
|---|---------------------------|-------------|
| A | 2                         | high        |
| B | 2                         | low         |
| C | 50                        | high        |
| D | 50                        | low         |

**11. M/J 19/P22/Q14, M/J 19/P23/Q14**

When blue-green crystals of nickel(II) sulfate are heated, water is produced and a yellow solid remains. When water is added to the yellow solid, the blue-green colour returns.

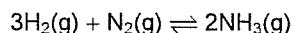
Which process describes these changes?

- A combustion      B corrosion      C neutralisation      D reversible reaction



**12. M/J 19/P22/Q32, M/J 19/P21/Q31**

Ammonia is produced in the Haber process.  
The equation for the reaction is shown.



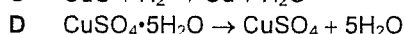
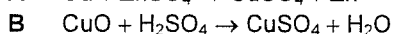
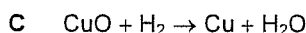
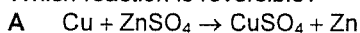
The forward reaction is exothermic.

Which conditions of temperature and pressure produce the highest yield of ammonia?

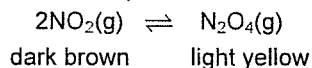
|          | temperature | pressure |
|----------|-------------|----------|
| <b>A</b> | high        | high     |
| <b>B</b> | high        | low      |
| <b>C</b> | low         | high     |
| <b>D</b> | low         | low      |

**13. MAR 19/P22/Q16**

Which reaction is reversible?

**14. MAR 19/P22/Q17**

Some nitrogen dioxide gas was put in a gas syringe. The end of the gas syringe is sealed.  
A reversible reaction occurs. The reaction reaches equilibrium.



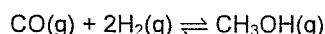
The forward reaction is exothermic.

Which statement about the reaction is correct?

- A** If the gas syringe is placed in a cold water bath, the colour becomes darker.  
**B** If the gas syringe is placed in a hot water bath, the colour becomes lighter.  
**C** If the volume in the gas syringe is increased, the colour becomes lighter.  
**D** If the volume in the gas syringe is decreased, the colour becomes lighter.

**15. O/N 18/P22/Q15**

Methanol is prepared by the reversible reaction shown.



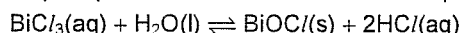
The forward reaction is exothermic.

Which conditions produce the highest equilibrium yield of methanol?

|          | temperature | pressure |
|----------|-------------|----------|
| <b>A</b> | high        | high     |
| <b>B</b> | high        | low      |
| <b>C</b> | low         | high     |
| <b>D</b> | low         | low      |

**16. O/N 18/P21/Q15**

When  $\text{BiCl}_3$  reacts with water, a white precipitate of  $\text{BiOCl}$  is formed. The equation for the reaction is shown.

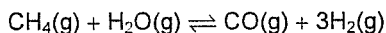


Which statements are correct?

- 1 The reaction is reversible.  
 2 When dilute hydrochloric acid is added to the reaction mixture, more of the white precipitate forms.  
 3 When aqueous sodium hydroxide is added to the reaction mixture, more of the white precipitate forms.
- A** 1, 2 and 3                  **B** 1 and 2 only                  **C** 1 and 3 only                  **D** 2 and 3 only

**17. O/N 18/P23/Q15**

The reversible reaction between methane and steam is shown.



The forward reaction is endothermic.

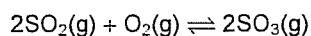
Which changes in pressure and temperature move the equilibrium to the right?

|          | pressure | temperature |
|----------|----------|-------------|
| <b>A</b> | decrease | decrease    |
| <b>B</b> | decrease | increase    |
| <b>C</b> | increase | decrease    |
| <b>D</b> | increase | increase    |

**18. M/J 18/P22/Q15**

Sulfur dioxide reacts with oxygen at 2 atmospheres pressure. The forward reaction is exothermic.

The equation for the reaction is shown.



The reaction reaches equilibrium. The pressure is then doubled.

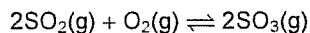
How and why does the amount of sulfur trioxide formed change?

|          | amount of sulfur trioxide | reason                                 |
|----------|---------------------------|----------------------------------------|
| <b>A</b> | decreases                 | the forward reaction is exothermic     |
| <b>B</b> | decreases                 | there are fewer molecules on the right |
| <b>C</b> | increases                 | the forward reaction is exothermic     |
| <b>D</b> | increases                 | there are fewer molecules on the right |

**19. M/J 18/P21/Q15**

The formation of sulfur trioxide is a reversible reaction.

The equation is shown.



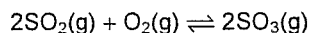
The forward reaction is exothermic.

Which conditions produce the highest equilibrium yield of sulfur trioxide?

|          | pressure | temperature |
|----------|----------|-------------|
| <b>A</b> | high     | high        |
| <b>B</b> | high     | low         |
| <b>C</b> | low      | high        |
| <b>D</b> | low      | low         |

**20. M/J 18/P23/Q15**

In the Contact process, sulfur dioxide is converted into sulfur trioxide in a reversible reaction.



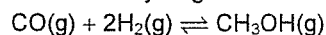
The forward reaction is exothermic.

Which conditions give the highest yield of sulfur trioxide at equilibrium?

|          | pressure<br>/ atmospheres | temperature |
|----------|---------------------------|-------------|
| <b>A</b> | 0.5                       | high        |
| <b>B</b> | 0.5                       | low         |
| <b>C</b> | 1.5                       | high        |
| <b>D</b> | 1.5                       | low         |

**21. MAR 18/P22/Q13**

Methanol is made by reacting carbon monoxide with hydrogen. The reaction is reversible.



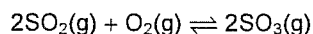
The forward reaction is exothermic.

Which combination of temperature and pressure gives the highest equilibrium yield of methanol?

|          | temperature<br>/°C | pressure<br>/ atmospheres |
|----------|--------------------|---------------------------|
| <b>A</b> | 200                | 10                        |
| <b>B</b> | 200                | 200                       |
| <b>C</b> | 600                | 10                        |
| <b>D</b> | 600                | 200                       |

**22. O/N 17/P22/Q16**

The formation of sulfur trioxide from sulfur dioxide is a reversible reaction.



The forward reaction is exothermic.

Which changes would increase the equilibrium yield of  $\text{SO}_3$ ?

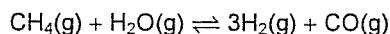
- 1 increasing the pressure
- 2 lowering the temperature
- 3 decreasing the concentration of oxygen

**A** 1, 2 and 3                      **B** 1 and 2 only                      **C** 1 only                      **D** 2 and 3 only

**23. O/N 17/P21/Q16**

Methane reacts with steam to produce hydrogen and carbon monoxide.

The equation for the reaction is shown.



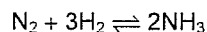
The reaction is reversible. The forward reaction is endothermic.

Which changes in temperature and pressure increase the equilibrium yield of carbon monoxide?

|          | temperature | pressure |
|----------|-------------|----------|
| <b>A</b> | decrease    | decrease |
| <b>B</b> | decrease    | increase |
| <b>C</b> | increase    | decrease |
| <b>D</b> | increase    | increase |

**24. O/N 17/P21/Q31**

Ammonia is made by the Haber process.

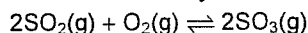


What are the sources of the nitrogen and hydrogen used in the Haber process?

|          | nitrogen    | hydrogen                    |
|----------|-------------|-----------------------------|
| <b>A</b> | fertilisers | reacting methane with steam |
| <b>B</b> | fertilisers | the air                     |
| <b>C</b> | the air     | reacting methane with steam |
| <b>D</b> | the air     | the air                     |

**25. O/N 17/P23/Q16**

The following reaction has reached equilibrium in a closed system.



The forward reaction is exothermic.

Which row shows the effect of increasing the pressure on the equilibrium mixture?

|          | reaction rate | amount of $\text{SO}_2$ | amount of $\text{SO}_3$ |
|----------|---------------|-------------------------|-------------------------|
| <b>A</b> | increases     | decreases               | increases               |
| <b>B</b> | increases     | increases               | decreases               |
| <b>C</b> | unchanged     | decreases               | increases               |
| <b>D</b> | unchanged     | increases               | decreases               |

**26. M/J 17/P22/Q16**

The reaction used to manufacture ammonia from nitrogen and hydrogen is reversible.

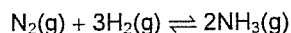
An equilibrium can be established between ammonia, nitrogen and hydrogen.

Which statement describes the equilibrium?

- A** Both the forward reaction and the backward reaction have the same rate.
- B** The rate of the backward reaction is greater than the rate of the forward reaction.
- C** The rate of the forward reaction is greater than the rate of the backward reaction.
- D** The forward and backward reactions have both stopped.

**27. M/J 17/P21/Q16**

Nitrogen, hydrogen and ammonia gases are placed inside a container. The container is then sealed. After some time, an equilibrium forms.



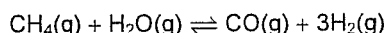
Which statement describes the equilibrium in this container?

- A** The amount of ammonia remains constant from the moment the container is sealed.
- B** The amounts of ammonia, nitrogen and hydrogen in the container are always equal.
- C** The rate of formation of ammonia is equal to the rate of decomposition of ammonia.
- D** The rate of formation of ammonia is faster than the rate of decomposition of ammonia.

**28. M/J 17/P23/Q16**

Hydrogen is produced when methane reacts with steam.

The equation for the reaction is shown.



The forward reaction is endothermic.

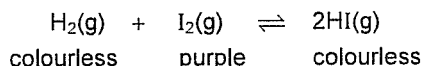
Which conditions produce the highest yield of hydrogen?

|          | pressure | temperature |
|----------|----------|-------------|
| <b>A</b> | high     | high        |
| <b>B</b> | high     | low         |
| <b>C</b> | low      | high        |
| <b>D</b> | low      | low         |

**29. MAR 17/P22/Q16**

The equation for the reversible reaction between hydrogen and iodine to form hydrogen iodide is shown.

The colours of the reactants and products are shown.



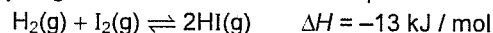
The forward reaction is exothermic.

Which statement is correct?

- A** An increase in pressure has no effect on the equilibrium position.
- B** The purple colour fades when the reaction mixture is heated.
- C** When equilibrium is reached, both forward and reverse reactions stop.
- D** When more hydrogen gas is added, the purple colour increases.

**30. O/N 16/P22/Q16**

At 400 °C the reaction between hydrogen and iodine reaches an equilibrium. The reaction is exothermic.

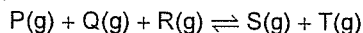


Which change in conditions would increase the percentage of hydrogen iodide in the equilibrium mixture?

- A a decrease in pressure                      C an increase in pressure  
B a decrease in temperature                D an increase in temperature

**31. O/N 16/P21/Q16**

The following reversible reaction takes place in a closed vessel at constant temperature.



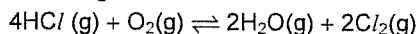
When the system has reached equilibrium, more T is added.

After the addition of T, which substances increase in concentration?

- A P, Q, R and S            B P and Q only            C P, Q and R only            D S only

**32. O/N 16/P23/Q16**

Chlorine can be manufactured by the following reaction. The reaction is exothermic.



Which change increases the yield of chlorine at equilibrium?

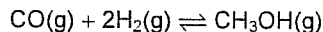
- A adding more HCl (g)                      C decreasing the pressure  
B adding more H<sub>2</sub>O(g)                      D increasing the temperature

**33. M/J 16/P22/Q16**

Methanol is made by reacting carbon monoxide with hydrogen.

The reaction is exothermic and is a chemical equilibrium.

The equation for the reaction is shown.

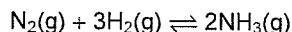


Which changes in temperature and pressure increase the yield of methanol?

|   | temperature | pressure |
|---|-------------|----------|
| A | decrease    | decrease |
| B | decrease    | increase |
| C | increase    | decrease |
| D | increase    | increase |

**34. M/J 16/P22/Q32**

Ammonia is manufactured by a reversible reaction.



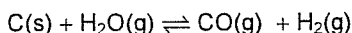
The forward reaction is exothermic.

What is the effect of increasing the pressure on the percentage yield and rate of formation of ammonia?

|   | percentage yield | rate of formation |
|---|------------------|-------------------|
| A | decreases        | decreases         |
| B | decreases        | increases         |
| C | increases        | decreases         |
| D | increases        | increases         |

**35. M/J 16/P21/Q16**

Steam reacts with carbon in an endothermic reaction.

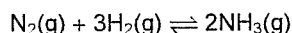


Which conditions of temperature and pressure would give the largest yield of hydrogen?

|          | temperature | pressure |
|----------|-------------|----------|
| <b>A</b> | high        | high     |
| <b>B</b> | high        | low      |
| <b>C</b> | low         | high     |
| <b>D</b> | low         | low      |

**36. M/J 16/P21/Q32**

Ammonia is formed by a reversible reaction.



The forward reaction is exothermic.

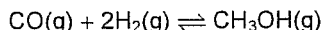
Which changes in conditions would increase the yield of ammonia?

|          | increase in pressure | increase in temperature |
|----------|----------------------|-------------------------|
| <b>A</b> | ✓                    | ✓                       |
| <b>B</b> | ✓                    | ✗                       |
| <b>C</b> | ✗                    | ✓                       |
| <b>D</b> | ✗                    | ✗                       |

**37. M/J 16/P23/Q16**

Methanol is manufactured by reacting carbon monoxide and hydrogen together in the presence of an aluminium oxide catalyst.

The equation for the reaction is shown.



The reaction is a reversible reaction.

The forward reaction is exothermic.

Which change in conditions increases the yield of methanol?

- A** decreasing the concentration of the carbon monoxide
- B** increasing the pressure
- C** increasing the rate of the reaction
- D** increasing the temperature

**38. MAR 16/P22/Q14**

A reversible reaction is shown.



Which statement about an equilibrium mixture of  $\text{NO}_2$  and  $\text{N}_2\text{O}_4$  is correct?

- A** If the pressure is decreased the amount of  $\text{N}_2\text{O}_4$  increases.
- B** If the temperature is increased the amount of  $\text{N}_2\text{O}_4$  increases.
- C** The rates of formation and decomposition of  $\text{N}_2\text{O}_4$  are not the same.
- D** The decomposition of  $\text{N}_2\text{O}_4$  is an endothermic reaction.

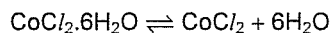
**39. O/N 15/P12/Q14**

Which reaction is **not** a reversible reaction?

- A** combustion of alkanes
- B** hydration of anhydrous copper(II) sulfate
- C** melting lead(II) bromide
- D** thermal decomposition of hydrated cobalt(II) chloride

## 40. M/J 15/P12/Q13, M/J 15/P11/Q13, M/J 15/P13/Q13, O/N 10/P12/Q16, O/N 10/P11/Q17, O/N 10/P13/Q18

When pink crystals of cobalt(II) chloride are heated, steam is given off and the colour of the solid changes to blue.



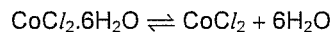
What happens when water is added to the blue solid?

|   | colour          | temperature |
|---|-----------------|-------------|
| A | changes to pink | decreases   |
| B | changes to pink | increases   |
| C | remains blue    | decreases   |
| D | remains blue    | increases   |

## 41. MAR 15/P12/Q12

Hydrated cobalt(II) chloride decomposes on heating.

The equation for the reaction is



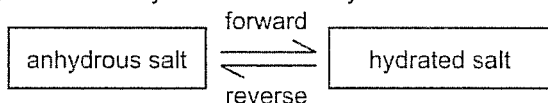
The reaction is reversed by adding water.

Which row describes the colour change and the type of reaction for the **reverse** reaction?

|   | colour change | type of reaction |
|---|---------------|------------------|
| A | blue to pink  | endothermic      |
| B | blue to pink  | exothermic       |
| C | pink to blue  | endothermic      |
| D | pink to blue  | exothermic       |

## 42. O/N 14/P12/Q15, O/N 14/P11/Q16, O/N 14/P13/Q16

The diagram shows the change from an anhydrous salt to its hydrated form.

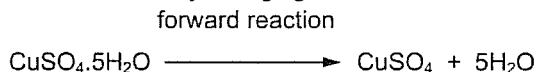


Which statement is correct?

- |                                                                                                 |                                                                                                 |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <p>A forward reaction requires heat and water</p> <p>B forward reaction requires water only</p> | <p>C reverse reaction requires heat and water</p> <p>D reverse reaction requires water only</p> |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## 43. M/J 14/P12/Q18, M/J 14/P11/Q18, M/J 14/P13/Q20, M/J 09/P1/Q17

The equation shows a reaction that is reversed by changing the conditions.

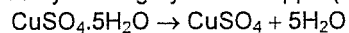


How can the forward reaction be reversed?

|   | by adding water | by heating |
|---|-----------------|------------|
| A | ✓               | ✓          |
| B | ✓               | x          |
| C | x               | ✓          |
| D | x               | x          |

**44. O/N 13/P12/Q14, O/N 13/P11/Q16**

Anhydrous copper(II) sulfate can be made by heating hydrated copper(II) sulfate.

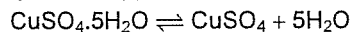


What can be added to anhydrous copper(II) sulfate to turn it into hydrated copper(II) sulfate?

- |                                     |                         |
|-------------------------------------|-------------------------|
| <b>A</b> concentrated sulfuric acid | <b>C</b> sulfur dioxide |
| <b>B</b> sodium hydroxide powder    | <b>D</b> water          |

**45. M/J 13/P11/Q15, M/J 13/P13/Q16**

The equation shows the formation of anhydrous copper(II) sulfate from hydrated copper(II) sulfate.



Statements 1, 2 and 3 refer to this reaction.

- 1 Hydrated copper(II) sulfate is reduced to anhydrous copper(II) sulfate.
- 2 The (II) in the name copper(II) sulfate refers to the oxidation state of the metal.
- 3 The reaction is reversible.

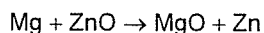
Which statements are correct?

- |                 |                  |                  |                 |
|-----------------|------------------|------------------|-----------------|
| <b>A</b> 1 only | <b>B</b> 1 and 2 | <b>C</b> 2 and 3 | <b>D</b> 3 only |
|-----------------|------------------|------------------|-----------------|



**7.3: Redox****1. O/N 22/P22/Q17**

When magnesium is heated with zinc oxide a reaction occurs.  
The equation is shown.



Which substance is oxidised?

- A magnesium      B magnesium oxide      C zinc      D zinc oxide

**2. O/N 22/P23/Q22**

Which metal forms ions with one oxidation state?

- A aluminium      B chromium      C copper      D iron

**3. M/J 22/P22/Q20**

Substance J takes part in a redox reaction.

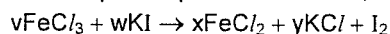
In the reaction, J gains electrons.

Which statement is correct?

- A J is the oxidising agent and it is oxidised in the reaction.  
B J is the oxidising agent and it is reduced in the reaction.  
C J is the reducing agent and it is oxidised in the reaction.  
D J is the reducing agent and it is reduced in the reaction.

**4. M/J 22/P21/Q16**

Aqueous iron(III) chloride,  $\text{FeCl}_3$ , reacts with aqueous potassium iodide, KI.



Which statements are correct?

- 1 In the balanced equation, v, w, x and y have the same value.  
2 Potassium iodide is an oxidising agent.  
3 A dark brown solution is produced in the reaction.

- A 1 and 2      B 1 and 3      C 2 only      D 2 and 3

**5. M/J 22/P23/Q27**

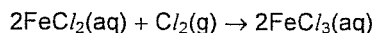
Which substance is used to reduce zinc oxide in the manufacture of zinc?

- A carbon      B carbon dioxide      C hydrogen      D sulfur dioxide

**6. O/N 21/P22/Q16, O/N 21/P21/Q16, O/N 21/P23/Q16**

Iron(II) chloride solution reacts with chlorine gas.

The equation is shown.



Which statements about this reaction are correct?

- 1  $\text{Fe}^{2+}$  ions are reduced to  $\text{Fe}^{3+}$  ions.  
2 Chlorine acts as a reducing agent.  
3  $\text{Fe}^{2+}$  ions each lose an electron.  
4  $\text{Cl}_2$  molecules are reduced to  $\text{Cl}^-$  ions.

- A 1 and 2      B 2 and 3      C 2 and 4      D 3 and 4

**7. O/N 21/P23/Q20**

Which row describes an acid and an oxidising agent?

|   | acid            | oxidising agent   |
|---|-----------------|-------------------|
| A | proton acceptor | electron acceptor |
| B | proton acceptor | electron donor    |
| C | proton donor    | electron acceptor |
| D | proton donor    | electron donor    |

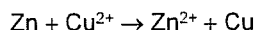
**8. M/J 21/P22/Q24**

When aqueous iodine is added to a solution of vanadium ions,  $V^{2+}$ , the  $V^{2+}$  ions each lose one electron. Which property of transition elements is shown by this reaction?

- A Transition elements have variable oxidation states.  
 B Transition elements form a stable  $1+$  ion.  
 C Transition elements are oxidising agents.  
 D Transition elements can act as catalysts.

**9. M/J 21/P21/Q16, M/J 21/P23/Q16, M/J 17/P22/Q17, M/J 17/P21/Q17, M/J 17/P23/Q17**

An example of a redox reaction is shown.

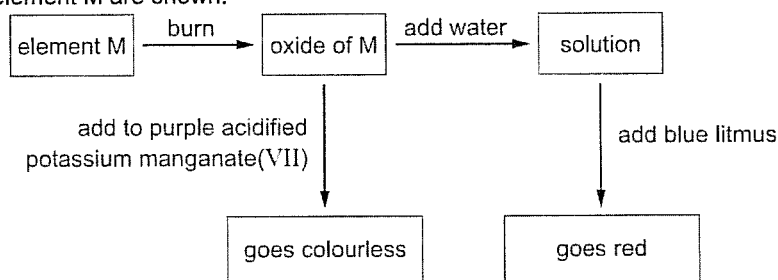


Which statement about the reaction is correct?

- A Zn is the oxidising agent and it oxidises  $Cu^{2+}$ .  
 B Zn is the oxidising agent and it reduces  $Cu^{2+}$ .  
 C Zn is the reducing agent and it oxidises  $Cu^{2+}$ .  
 D Zn is the reducing agent and it reduces  $Cu^{2+}$ .

**10. M/J 21/P21/Q20**

Some reactions of element M are shown.



What is element M?

- A carbon                      B iron                      C magnesium                      D sulfur

**11. M/J 21/P23/Q32**

The formulae of two compounds of manganese are  $MnO_2$  and  $KMnO_4$ .

In these two compounds the oxidation state of potassium is  $+1$  and the oxidation state of oxygen is  $-2$ . What are the oxidation states of manganese in each of these two compounds?

|   | $MnO_2$ | $KMnO_4$ |
|---|---------|----------|
| A | +2      | +3       |
| B | +2      | +7       |
| C | +4      | +3       |
| D | +4      | +7       |

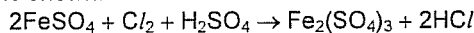
**12. MAR 21/P22/Q18**

Magnesium reacts with copper(II) oxide to give magnesium oxide and copper. Which substance is the oxidising agent in this reaction?

- A copper                      B copper(II) oxide                      C magnesium                      D magnesium oxide

**13. O/N 20/P22/Q20**

The equation for a redox reaction is shown.



Which element is reduced?

- A chlorine                      B iron.                      C oxygen                      D sulfur

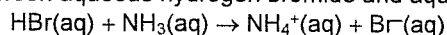
**14. O/N 20/P23/Q17**

Which reaction of hydrochloric acid is a redox reaction?

- A  $2Na + 2HCl \rightarrow 2NaCl + H_2$                       B  $NaOH + HCl \rightarrow NaCl + H_2O$   
 C  $Na_2O + 2HCl \rightarrow 2NaCl + H_2O$                       D  $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

## 15. O/N 20/P22/Q21

The equation shows a reaction between aqueous hydrogen bromide and aqueous ammonia.



Which statement describes the role of aqueous hydrogen bromide?

- A It is a catalyst. C It is a proton acceptor.  
B It is a reducing agent. D It is a proton donor.

## 16. M/J 20/P22/Q17

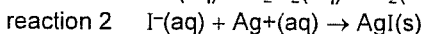
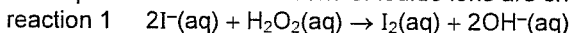
When aqueous iron (III) chloride is added to aqueous potassium iodide a chemical reaction occurs and iodine is formed.

Which statement is correct?

- A Iodide ions are oxidized, they gain electrons in this reaction.  
B Iodide ions are oxidized, they lose electrons in this reaction.  
C Iron (III) chloride is oxidized in this reaction.  
D Neither iodide ions nor iron (III) chloride is oxidized in this reaction.

## 17. M/J 20/P21/Q17

The equations for two reactions of iodide ions are shown.

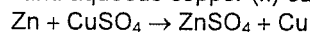


Which statement is correct?

- A Both reactions are redox reactions. C Only reaction 1 is a redox reaction.  
B Neither reaction is a redox reaction. D Only reaction 2 is a redox reaction.

## 18. M/J 20/P23/Q17

The equation for the reaction between zinc and aqueous copper (II) sulfate is shown.



Which statement is correct?

- A The oxidation state of the oxidising agent has changed from 0 to +2.  
B The oxidation state of the reducing agent has changed from 0 to +2.  
C The oxidation state of the reducing agent has changed from +2 to 0.  
D This is not a redox reaction. The solution changes from colourless to blue.

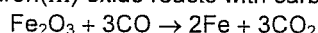
## 19. MAR 20/P22/Q17

In which reaction is the underlined compound acting as a reducing agent?

- A  $\underline{\text{CO}_2} + \text{C} \rightarrow 2\text{CO}$  C  $\underline{\text{Fe}_2\text{O}_3} + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$   
B  $2\text{CuO} + \underline{\text{C}} \rightarrow 2\text{Cu} + \text{CO}_2$  D  $\text{CaCO}_3 + \underline{2\text{HCl}} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$

## 20. O/N 19/P22/Q18

In the blast furnace, iron is formed when iron(III) oxide reacts with carbon monoxide in a redox reaction.



Which substance is the oxidising agent and which substance is the reducing agent?

|   | oxidising agent                | reducing agent                 |
|---|--------------------------------|--------------------------------|
| A | CO                             | Fe <sub>2</sub> O <sub>3</sub> |
| B | CO <sub>2</sub>                | Fe                             |
| C | Fe                             | CO <sub>2</sub>                |
| D | Fe <sub>2</sub> O <sub>3</sub> | CO                             |

## 21. O/N 19/P22/Q19

Which oxide is classified as an amphoteric oxide?

- A aluminium oxide B calcium oxide C copper(II) oxide D nitrogen oxide

**22. M/J 19/P21/Q16**

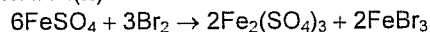
Which changes represent oxidation?

- 1  $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$
- 2  $\text{Cr(VI)} \rightarrow \text{Cr(III)}$
- 3  $\text{Fe(II)} \rightarrow \text{Fe(III)}$

A 1 and 2                      B 1 and 3                      C 1 only                      D 2 only

**23. O/N 19/P21/Q18**

The equation for the reaction between iron(II) sulfate and bromine is shown.



Which row identifies the oxidising agent and the reducing agent?

|   | oxidising agent              | reducing agent               |
|---|------------------------------|------------------------------|
| A | $\text{Br}_2$                | $\text{FeSO}_4$              |
| B | $\text{FeSO}_4$              | $\text{Br}_2$                |
| C | $\text{FeBr}_3$              | $\text{Fe}_2(\text{SO}_4)_3$ |
| D | $\text{Fe}_2(\text{SO}_4)_3$ | $\text{FeBr}_3$              |

**24. M/J 19/P23/Q16**

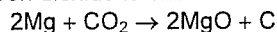
Which changes represent reduction?

- 1  $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$
- 2  $\text{Mn(VII)} \rightarrow \text{Mn(II)}$
- 3 sulfate(IV)  $\rightarrow$  sulfate(VI)

A 1 and 2                      B 1 and 3                      C 1 only                      D 2 only

**25. MAR 19/P22/Q18**

The reaction between magnesium and carbon dioxide is shown in the equation.



Which statement describes what happens in this reaction?

- A Carbon is oxidised.                      C Neither oxidation nor reduction happens.  
 B Magnesium is reduced.                      D The carbon in carbon dioxide is reduced.

**26. MAR 19/P22/Q19**

Which changes involve reduction?

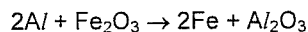
- 1  $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$
- 2  $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$
- 3  $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
- 4  $\text{Pb}^{2+} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4$

A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

**27. O/N 18/P22/Q16**

The thermite reaction can be used to produce iron from iron(III) oxide.

The equation for the reaction is shown.



Which statements about this reaction are correct?

- 1 Aluminium is the oxidising agent.
- 2 Aluminium is less reactive than iron.
- 3 Electrons are transferred from aluminium to iron.
- 4 The iron in the iron(III) oxide is reduced.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 3 and 4

**28. O/N 18/P21/Q16**

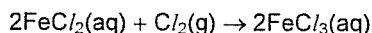
An excess of iron(II) chloride is added to acidified potassium manganate(VII). Which statements are correct?

- 1 The purple colour disappears.
- 2 Iron(II) is reduced to iron(III).
- 3 Manganate(VII) ions are oxidised to manganese(II) ions.
- 4 Potassium manganate(VII) is an oxidising agent.

A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

**29. M/J 18/P22/Q16**

Iron(II) chloride solution reacts with chlorine gas. The equation is shown.



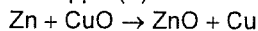
Which statements about this reaction are correct?

- 1  $\text{Fe}^{2+}$  ions are reduced to  $\text{Fe}^{3+}$  ions.
- 2 Chlorine acts as a reducing agent.
- 3  $\text{Fe}^{2+}$  ions each lose an electron.
- 4  $\text{Cl}_2$  molecules are reduced to  $\text{Cl}^-$  ions.

A 1 and 2                      B 2 and 3                      C 2 and 4                      D 3 and 4

**30. O/N 18/P23/Q16**

The equation for the reaction between zinc and copper(II) oxide is shown.



Which row shows the oxidising agent and the reducing agent?

|   | oxidising agent | reducing agent |
|---|-----------------|----------------|
| A | CuO             | Cu             |
| B | CuO             | Zn             |
| C | Zn              | CuO            |
| D | Zn              | ZnO            |

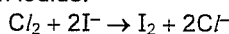
**31. M/J 18/P22/Q17, M/J 18/P21/Q17, M/J 18/P23/Q17, O/N 14/P13/Q20**

Which statement about oxides is correct?

- A A solution of magnesium oxide has a pH less than pH 7.
- B A solution of sulfur dioxide has a pH greater than pH 7.
- C Magnesium oxide reacts with nitric acid to make a salt.
- D Sulfur dioxide reacts with hydrochloric acid to make a salt.

**32. M/J 18/P21/Q16**

Chlorine displaces iodide ions from potassium iodide.

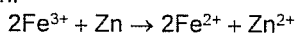


What is the oxidising agent?

A chloride ions                      B chlorine                      C iodide ions                      D iodine

**33. M/J 18/P23/Q16**

The equation for a redox reaction is shown.



Which statements are correct?

- 1  $\text{Fe}^{3+}$  is reduced to form  $\text{Fe}^{2+}$ .
- 2 Zn oxidises the  $\text{Fe}^{3+}$  ions.
- 3  $\text{Fe}^{3+}$  is an oxidising agent.

A 1, 2 and 3                      B 1 and 2 only                      C 1 and 3 only                      D 2 and 3 only

**34. O/N 17/P22/Q14**

Copper metal donates electrons to silver ions.

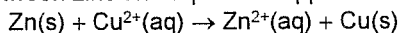
Zinc metal donates electrons to copper ions.

What is the strongest reducing agent?

A copper ions                      B copper metal                      C silver ions                      D zinc metal

**35. MAR 18/P22/Q14**

The ionic equation for the reaction between zinc and aqueous copper ions is shown.

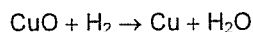


Which statement about this reaction is correct?

- A Copper ions are oxidised and their oxidation state changes.
- B Copper ions are reduced because they lose electrons.
- C Zinc atoms are oxidised and their oxidation state changes.
- D Zinc atoms are reduced because they gain electrons.

**36. O/N 17/P21/Q14**

Copper(II) oxide reacts with hydrogen.



Which row is correct?

|   | oxidising agent  | reducing agent   |
|---|------------------|------------------|
| A | H <sub>2</sub>   | CuO              |
| B | CuO              | H <sub>2</sub>   |
| C | H <sub>2</sub> O | Cu               |
| D | Cu               | H <sub>2</sub> O |

**37. O/N 17/P23/Q14**

Silver chloride reacts when it is exposed to light.

Which row shows what happens to the silver in this process?

|   | half-equation                                    | type of reaction |
|---|--------------------------------------------------|------------------|
| A | $\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$ | oxidation        |
| B | $\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$ | reduction        |
| C | $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ | oxidation        |
| D | $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ | reduction        |

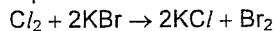
**38. M/J 17/P23/Q25**

Which equation from the zinc extraction process shows the metal being produced by reduction?

- A  $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
- B  $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$
- C  $\text{Zn(g)} \rightarrow \text{Zn(l)}$
- D  $\text{Zn(l)} \rightarrow \text{Zn(s)}$

**39. MAR 17/P22/Q17**

Chlorine displaces bromine from a solution of potassium bromide.



What is the oxidising agent in this reaction?

- A bromide ions
- B bromine
- C chloride ions
- D chlorine

**40. O/N 16/P22/Q17**

Chromium forms the compound chromium(III) sulfate.

What does the (III) represent?

- A the charge on a sulfate ion
- B the number of chromium ions combined with one sulfate ion
- C the number of sulfate ions combined with one chromium ion
- D the oxidation state of chromium

**41. O/N 16/P23/Q17**

Which change represents an oxidation reaction?

- A chlorine changes to chlorate(III) ions
- B chlorine changes to chloride ions
- C copper(II) ions change to copper
- D potassium manganate(VII) ions change to potassium manganate(VI) ions

## 42. M/J 16/P22/Q17, M/J 16/P21/Q17, M/J 16/P23/Q17

Which equation represents a reduction reaction?

- A  $\text{Fe}^{2+} + \text{e}^- \rightarrow \text{Fe}^{3+}$     B  $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$     C  $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$     D  $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+} + \text{e}^-$

## 43. MAR 16/P22/Q30

Nitrogen monoxide is produced in a car engine when petrol is burnt.

The gases from the car engine are passed through a catalytic converter.

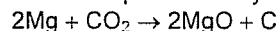
In the catalytic converter the nitrogen monoxide reacts with carbon monoxide to form nitrogen and carbon dioxide.

Which statement is **not** correct?

- A Carbon monoxide is oxidised in the catalytic converter.  
 B Carbon monoxide is produced by the complete combustion of petrol.  
 C Nitrogen monoxide is formed by the reaction of nitrogen and oxygen.  
 D Nitrogen monoxide is reduced in the catalytic converter.

## 44. O/N 15/P12/Q15

The reaction between magnesium and carbon dioxide is represented by the following equation.



Which statement describes what happens in this reaction?

- A Carbon is oxidised.    C Neither oxidation nor reduction happens.  
 B Magnesium is reduced.    D The carbon in carbon dioxide is reduced.

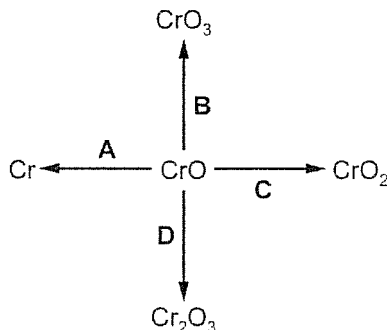
## 45. M/J 15/P12/Q12

In which equation does the oxidation of the underlined compound occur?

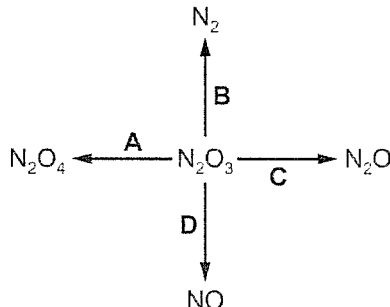
- A  $2\text{CuO} + \text{C} \rightarrow \text{CO}_2 + \underline{\text{Cu}}$     C  $2\text{Mg} + \text{O}_2 \rightarrow \underline{2\text{MgO}}$   
 B  $\text{Fe}_2\text{O}_3 + \underline{3\text{CO}} \rightarrow 2\text{Fe} + 3\text{CO}_2$     D  $\underline{\text{MnO}_2} + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$

## 46. M/J 15/P11/Q14

In which change is chromium(II) oxide, CrO, reduced?



## 47. M/J 15/P13/Q12

In which change is  $\text{N}_2\text{O}_3$  oxidised?

**48. MAR 15/P12/Q2**

During a reaction, the following changes take place.

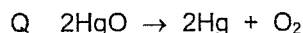
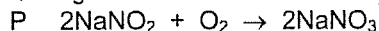
- 1 The temperature rises.
- 2 A gas is given off.

Which apparatus is required to measure the rate of this reaction?

- A balance and burette  
B balance and gas syringe  
C gas syringe and burette  
D gas syringe and stopclock

**49. O/N 14/P12/Q19, O/N 14/P11/Q17, O/N 14/P13/Q17**

The equations for two reactions P and Q are given.



In which of these reactions does oxidation of the underlined substance occur?

|   | P | Q |
|---|---|---|
| A | ✓ | ✓ |
| B | ✓ | x |
| C | x | ✓ |
| D | x | x |

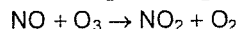
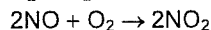
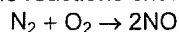
**50. M/J 14/P12/Q15, M/J 14/P11/Q15, M/J 14/P13/Q13**

Which equation shows an oxidation reaction?

- A  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$   
B  $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$   
C  $\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$   
D  $\text{N}_2\text{O}_4 \rightarrow 2\text{NO}_2$

**51. O/N 13/P12/Q19, O/N 13/P11/Q17, O/N 13/P13/Q17, M/J 09/P1/Q18**

The reactions shown may occur in the air during a thunder storm.



Which row shows what happens to the reactant molecules in each of these reactions?

|   | N <sub>2</sub> | NO       | O <sub>3</sub> |
|---|----------------|----------|----------------|
| A | oxidised       | oxidised | oxidised       |
| B | oxidised       | oxidised | reduced        |
| C | reduced        | reduced  | oxidised       |
| D | reduced        | reduced  | reduced        |

**52. O/N 13/P13/Q16**

When green iron(II) sulfate is heated, it turns white and a colourless liquid is produced.

When the liquid is put back into the white solid it changes back to green.

What type of reaction takes place and what is the name of the liquid?

|   | type of reaction | name of liquid |
|---|------------------|----------------|
| A | redox            | sulfuric acid  |
| B | redox            | water          |
| C | reversible       | sulfuric acid  |
| D | reversible       | water          |

**53. M/J 13/P12/Q14, M/J 13/P11/Q17, M/J 13/P13/Q18, O/N 09/P12/Q19, O/N 09/P11/Q18**

In which equation is the underlined substance acting as a reducing agent?

- A  $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + 3\text{CO}_2$   
B  $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$   
C  $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$   
D  $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$



## Answers Section

| 7.1   |       |  |  | 7.2   |       |  |  | 7.3   |       |  |  |
|-------|-------|--|--|-------|-------|--|--|-------|-------|--|--|
| 1. C  | 41. A |  |  | 1. A  | 41. B |  |  | 1. A  | 41. A |  |  |
| 2. B  | 42. A |  |  | 2. A  | 42. B |  |  | 2. A  | 42. C |  |  |
| 3. B  | 43. B |  |  | 3. B  | 43. B |  |  | 3. B  | 43. B |  |  |
| 4. C  | 44. D |  |  | 4. C  | 44. D |  |  | 4. B  | 44. D |  |  |
| 5. D  | 45. A |  |  | 5. D  | 45. C |  |  | 5. A  | 45. B |  |  |
| 6. A  | 46. B |  |  | 6. A  |       |  |  | 6. D  | 46. A |  |  |
| 7. D  | 47. A |  |  | 7. A  |       |  |  | 7. C  | 47. A |  |  |
| 8. C  | 48. C |  |  | 8. B  |       |  |  | 8. A  | 48. D |  |  |
| 9. C  | 49. C |  |  | 9. B  |       |  |  | 9. D  | 49. B |  |  |
| 10. A | 50. D |  |  | 10. A |       |  |  | 10. D | 50. A |  |  |
| 11. D | 51. A |  |  | 11. D |       |  |  | 11. D | 51. B |  |  |
| 12. B | 52. D |  |  | 12. C |       |  |  | 12. B | 52. D |  |  |
| 13. D | 53. B |  |  | 13. D |       |  |  | 13. A | 53. A |  |  |
| 14. C | 54. D |  |  | 14. D |       |  |  | 14. A |       |  |  |
| 15. D | 55. D |  |  | 15. C |       |  |  | 15. D |       |  |  |
| 16. D | 56. B |  |  | 16. C |       |  |  | 16. B |       |  |  |
| 17. B | 57. B |  |  | 17. B |       |  |  | 17. C |       |  |  |
| 18. A | 58. A |  |  | 18. D |       |  |  | 18. B |       |  |  |
| 19. B |       |  |  | 19. B |       |  |  | 19. B |       |  |  |
| 20. D |       |  |  | 20. D |       |  |  | 20. D |       |  |  |
| 21. C |       |  |  | 21. B |       |  |  | 21. A |       |  |  |
| 22. B |       |  |  | 22. B |       |  |  | 22. B |       |  |  |
| 23. D |       |  |  | 23. C |       |  |  | 23. A |       |  |  |
| 24. D |       |  |  | 24. C |       |  |  | 24. A |       |  |  |
| 25. C |       |  |  | 25. A |       |  |  | 25. D |       |  |  |
| 26. A |       |  |  | 26. A |       |  |  | 26. C |       |  |  |
| 27. B |       |  |  | 27. C |       |  |  | 27. D |       |  |  |
| 28. C |       |  |  | 28. C |       |  |  | 28. B |       |  |  |
| 29. C |       |  |  | 29. A |       |  |  | 29. D |       |  |  |
| 30. C |       |  |  | 30. B |       |  |  | 30. B |       |  |  |
| 31. A |       |  |  | 31. C |       |  |  | 31. C |       |  |  |
| 32. A |       |  |  | 32. A |       |  |  | 32. B |       |  |  |
| 33. A |       |  |  | 33. B |       |  |  | 33. C |       |  |  |
| 34. B |       |  |  | 34. D |       |  |  | 34. D |       |  |  |
| 35. A |       |  |  | 35. B |       |  |  | 35. C |       |  |  |
| 36. B |       |  |  | 36. B |       |  |  | 36. B |       |  |  |
| 37. B |       |  |  | 37. B |       |  |  | 37. D |       |  |  |
| 38. D |       |  |  | 38. D |       |  |  | 38. A |       |  |  |
| 39. C |       |  |  | 39. A |       |  |  | 39. D |       |  |  |
| 40. D |       |  |  | 40. B |       |  |  | 40. D |       |  |  |

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## Unit 8

# ACIDS, BASES AND SALTS

IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 8.1 The characteristics properties of acids and bases
- 8.2 Types of oxides and indicators
- 8.3 Preparation of salts
- 8.4 Identification of ions and gases

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## Unit 8: Acids, Bases And Salts

### 8.1: The characteristic properties of acids and bases

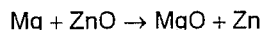
1. O/N 22/P22/Q11

When an acid is added to an alkali, the temperature of the reaction mixture rises.  
Which words describe this reaction?

- A decomposition and endothermic      C neutralisation and endothermic  
B decomposition and exothermic      D neutralisation and exothermic

2. O/N 22/P21/Q16, O/N 22/P23/Q16

When magnesium is heated with zinc oxide a reaction occurs.  
The equation is shown.

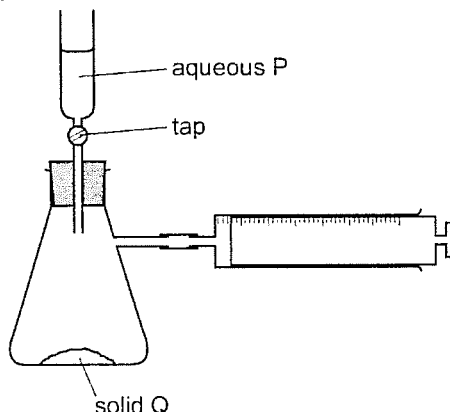


Which substance is oxidised?

- A magnesium      B magnesium oxide      C zinc      D zinc oxide

3. O/N 22/P21/Q17

The diagram shows an experiment.



A small volume of aqueous P is poured on to solid Q and the tap of the funnel closed.  
Which pairs of substances cause the syringe to fill with gas?

|   | $\text{HNO}_3$<br>and<br>Mg | HCl<br>and<br>Cu | $\text{H}_2\text{SO}_4$<br>and<br>$\text{Na}_2\text{CO}_3$ |
|---|-----------------------------|------------------|------------------------------------------------------------|
| A | ✓                           | ✓                | ✓                                                          |
| B | ✓                           | ✓                | x                                                          |
| C | ✓                           | x                | ✓                                                          |
| D | x                           | ✓                | ✓                                                          |

4. M/J 22/P22/Q19, M/J 22/P21/Q19

An acid is neutralised by adding an excess of an insoluble solid base.  
A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A crystallisation → evaporation → filtration      C filtration → crystallisation → evaporation  
B evaporation → crystallisation → filtration      D filtration → evaporation → crystallisation

**5. M/J 22/P22/Q22**

Which statement about acids is correct?

- A Acids are proton acceptors.  
 B Acids transfer electrons to bases in aqueous solution.  
 C Hydrochloric acid reacts with ammonium hydroxide to produce ammonia.  
 D Ethanoic acid partially ionises in aqueous solution.

**6. M/J 22/P22/Q34, M/J 22/P21/Q34, M/J 22/P23/Q34**

What are the products when limestone (calcium carbonate) is heated strongly?

- A calcium hydroxide and carbon dioxide      C calcium oxide and carbon dioxide  
 B calcium hydroxide and carbon monoxide      D calcium oxide and carbon monoxide

**7. M/J 22/P21/Q17**

Which statement about acids is correct?

- A A strong acid has a higher pH than a weak acid of the same concentration.  
 B A strong acid is a proton acceptor.  
 C A weak acid is a proton donor.  
 D A weak acid is fully ionised in aqueous solution.

**8. M/J 22/P23/Q16**

Which statement explains why lime is added to soil?

- A to decrease the pH of acidic soil      C to increase the pH of acidic soil  
 B to decrease the pH of alkaline soil      D to increase the pH of alkaline soil

**9. MAR 22/P22/Q20**

Methyl orange is added to dilute hydrochloric acid and to aqueous sodium hydroxide.

What is the colour of the methyl orange in each solution?

|   | colour in dilute hydrochloric acid | colour in aqueous sodium hydroxide |
|---|------------------------------------|------------------------------------|
| A | orange                             | red                                |
| B | red                                | yellow                             |
| C | red                                | orange                             |
| D | yellow                             | red                                |

**10. MAR 22/P22/Q23**

Aqueous ethanoic acid is a weak acid.

Aqueous sodium hydroxide is a strong base.

Aqueous ethanoic acid is neutralised by aqueous sodium hydroxide.

Which statements are correct?

- 1 Aqueous ethanoic acid accepts protons from hydroxide ions.  
 2 The aqueous ethanoic acid used is fully dissociated into ions.  
 3 The aqueous sodium hydroxide used is fully dissociated into ions.  
 4 The reaction produces a salt and water.

- A 1 and 2      B 1 and 3      C 2 and 4      D 3 and 4

**11. MAR 22/P22/Q34**

Which statements about lime are correct?

- 1 Lime is made by heating calcium carbonate (limestone).  
 2 Lime is used to desulfurise flue gases.  
 3 Lime is used to treat alkaline soil.  
 4 The chemical name for lime is calcium oxide.

- A 1 and 3      B 1, 2 and 4      C 1 and 4 only      D 2, 3 and 4

**12. O/N 21/P22/Q17**

Which statements about acids and bases are correct?

- 1 An acid reacts with a metal to give off hydrogen.
- 2 A base reacts with an ammonium salt to give off ammonia.
- 3 An acid reacts with a carbonate to give off carbon dioxide.
- 4 Alkaline solutions are orange in methyl orange.

A 1, 2 and 3                      B 1, 2 and 4                      C 1, 3 and 4                      D 2, 3 and 4

**13. O/N 21/P22/Q18**

Oxide 1 is a solid that reacts with dilute hydrochloric acid.

Oxide 2 is a gas that reacts with sodium hydroxide solution.

What are the formulae of the oxides?

|   | oxide 1         | oxide 2         |
|---|-----------------|-----------------|
| A | CaO             | MgO             |
| B | MgO             | NO <sub>2</sub> |
| C | NO <sub>2</sub> | SO <sub>2</sub> |
| D | SO <sub>2</sub> | CaO             |

**14. O/N 21/P22/Q32**

Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

A ammonia                      B sodium chloride                      C sodium hydroxide                      D sulfuric acid

**15. O/N 21/P21/Q17**

Which row describes the properties of an acid?

|   | property 1      | property 2     |
|---|-----------------|----------------|
| A | proton acceptor | pH less than 7 |
| B | proton acceptor | pH more than 7 |
| C | proton donor    | pH less than 7 |
| D | proton donor    | pH more than 7 |

**16. O/N 21/P21/Q19**

Copper(II) chloride crystals are made by adding solid copper(II) carbonate to dilute hydrochloric acid until no more dissolves.

Which process is used to obtain pure copper(II) chloride crystals from the mixture?

A distillation of the mixture                      C filtration followed by drying of the residue  
B evaporation of the mixture                      D filtration followed by evaporation of the filtrate

**17. O/N 21/P21/Q32, O/N 21/P23/Q32**

Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

A ammonia                      B sodium chloride                      C sodium hydroxide                      D sulfuric acid

**18. O/N 21/P23/Q19**

Four solid oxides are added to dilute hydrochloric acid and aqueous sodium hydroxide.

Which row describes an amphoteric oxide?

|   | hydrochloric acid | sodium hydroxide |
|---|-------------------|------------------|
| A | ✓                 | ✓                |
| B | x                 | ✓                |
| C | ✓                 | x                |
| D | x                 | x                |

key

✓ = reacts

x = does not react

**19. M/J 21/P22/Q25**

A piece of aluminium is dropped into dilute hydrochloric acid.

No immediate reaction is observed.

Which statement explains this observation?

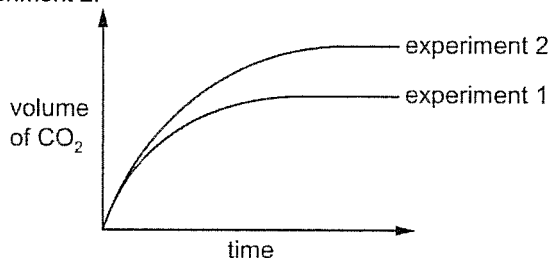
- A Aluminium does not neutralise acids.
- B Aluminium is a non-metal so does not react with acids.
- C Aluminium is below hydrogen in the reactivity series.
- D Aluminium is covered in an unreactive oxide layer.

**20. M/J 21/P22/Q13**

An excess of calcium carbonate reacts with dilute hydrochloric acid. The volume of carbon dioxide produced is measured at regular time intervals. The results are shown as experiment 1.

The experiment is repeated with only one change to the reaction conditions.

The results are shown as experiment 2.



Which change is made in experiment 2?

- A The concentration of the acid is increased.
- B The volume of acid is increased.
- C The mass of calcium carbonate is increased.
- D The calcium carbonate is powdered.

**21. M/J 21/P21/Q18, M/J 21/P23/Q18**

Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

|   | element X | nature of oxide, XO |
|---|-----------|---------------------|
| A | metal     | acidic              |
| B | metal     | basic               |
| C | non-metal | acidic              |
| D | non-metal | basic               |

**22. M/J 21/P21/Q21**

In which equation is the underlined reactant acting as a base?

- A  $\text{CH}_3\text{COO}^- + \underline{\text{H}_3\text{O}^+} \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$
- B  $\underline{\text{NH}_4^+} + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$
- C  $\text{CO}_2 + \underline{2\text{H}_2\text{O}} \rightarrow \text{H}_3\text{O}^+ + \text{HCO}_3^-$
- D  $\underline{\text{H}^+} + \text{OH}^- \rightarrow \text{H}_2\text{O}$

**23. M/J 21/P21/Q32**

A white solid Z reacts with dilute hydrochloric acid to produce a gas.

The same gas is produced when compound Z is heated strongly.

What is Z?

- A calcium
- B calcium carbonate
- C calcium hydroxide
- D calcium oxide

**24. M/J 21/P23/Q33**

Which statement about calcium carbonate is correct?

- A It is made by the thermal decomposition of limestone.
- B It is used to neutralise alkaline soils.
- C It is a reactant in the test for carbon dioxide.
- D It is used to remove impurities in iron extraction.

**25. MAR 21/P22/Q21**

Which statements about strong acids are correct?

- 1 They have a high concentration of OH<sup>-</sup> ions.
- 2 They have a pH value of 1.
- 3 They completely ionise in water.
- 4 They turn red litmus blue.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**26. MAR 21/P22/Q33**

What is an industrial use of calcium carbonate?

- A cracking of hydrocarbons                      C manufacture of cement  
B manufacture of aluminium                      D purification of water

**27. O/N 20/P22/Q12**

Universal indicator solution is added to a neutral solution of concentrated aqueous sodium chloride.

The solution, which contains H<sup>+</sup> (hydrogen), Na<sup>+</sup> (sodium), Cl<sup>-</sup> (chloride) and OH<sup>-</sup> (hydroxide) ions, is electrolysed.

The product at the cathode is hydrogen gas and the product at the anode is chlorine gas. What happens to the colour of the indicator in the solution during electrolysis?

- A The colour changes from blue to green.                      C The colour changes from green to blue.  
B The colour changes from blue to red.                      D The colour changes from green to red.

**28. O/N 20/P21/Q11**

Sodium carbonate reacts with sulfuric acid to form carbon dioxide, water and a sodium salt. An incomplete equation for the reaction is shown.



What is the formula of the sodium salt?

- A Na<sub>2</sub>(SO<sub>4</sub>)<sub>2</sub>                      B Na(SO<sub>4</sub>)<sub>2</sub>                      C Na<sub>2</sub>SO<sub>4</sub>                      D NaSO<sub>4</sub>

**29. O/N 20/P23/Q19**

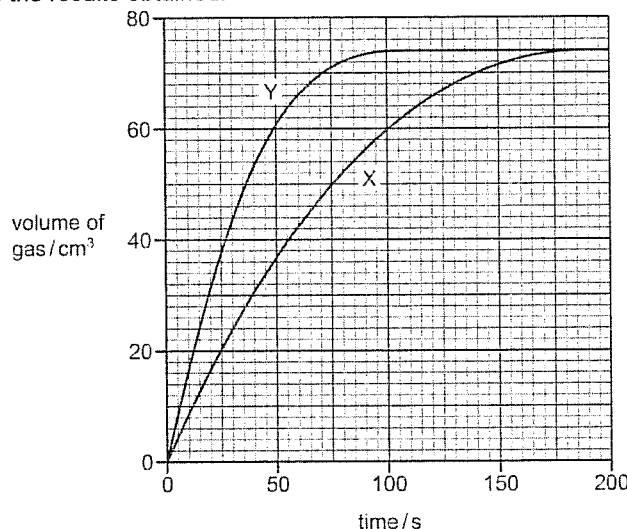
An excess of calcium carbonate is added to dilute hydrochloric acid, X.

The carbon dioxide gas given off is collected and its volume recorded at regular time intervals.

Line X on the graph shows the results obtained.

The experiment is repeated using dilute hydrochloric acid, Y.

Line Y on the graph shows the results obtained.



Which statement about the two hydrochloric acid samples, X and Y, is correct?

- A They had the same volume but Y had higher concentration.  
B They had the same concentration but Y had a larger volume.  
C X had a higher concentration but Y had a larger volume.  
D Y had a higher concentration but X had a larger volume.



**30. O/N 20/P21/Q22**

What is a characteristic of acids?

- A Acids turn methyl orange indicator yellow.  
 B Acids have a high pH value.  
 C Acids react with ammonium salts to give ammonia gas.  
 D Acids react with carbonates to produce salts.

**31. O/N 20/P23/Q7**

Magnesium reacts with sulfuric acid.

What are the formulae of the products formed in this reaction?

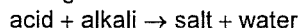
- A  $\text{MgSO}_4$  and  $\text{H}_2$       B  $\text{MgSO}_4$  and  $\text{H}_2\text{O}$       C  $\text{Mg}(\text{SO}_4)_2$  and  $\text{H}_2$       D  $\text{Mg}(\text{SO}_4)_2$  and  $\text{H}_2\text{O}$

**32. O/N 20/P23/Q35**Which calcium compound does **not** neutralise an acid soil?

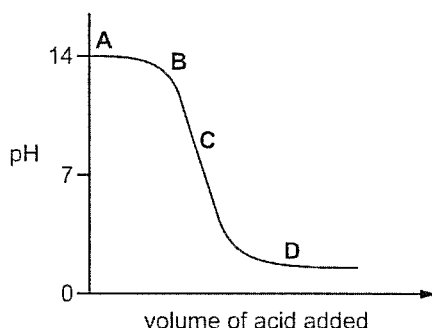
- A calcium oxide      B calcium sulfate      C calcium hydroxide      D calcium carbonate

**33. M/J 20/P22/Q18, M/J 20/P21/Q18, M/J 20/P23/Q18, M/J 12/P11/Q17, M/J 12/P13/Q18**

The graph shows how the pH of a solution changes as an acid is added to an alkali.



Which letter represents the area of the graph where both acid and salt are present?

**34. M/J 20/P22/Q19, M/J 20/P21/Q19, M/J 20/P23/Q19**

Which statement describes a weak acid?

- A It is a proton acceptor and is fully ionised in aqueous solution.  
 B It is a proton acceptor and is partially ionised in aqueous solution.  
 C It is a proton donor and is fully ionised in aqueous solution.  
 D It is a proton donor and is partially ionised in aqueous solution.

**35. M/J 20/P22/Q33, M/J 20/P21/Q33, M/J 20/P23/Q33**

A student suggests three uses of calcium carbonate (limestone).

- 1 manufacture of cement
- 2 manufacture of iron
- 3 treating alkaline soils

Which suggestions are correct?

- A 1 and 2 only      B 1 and 3 only      C 2 and 3 only      D 1, 2 and 3

**36. M/J 20/P21/Q8**

A solution of iron(III) sulfate reacts with aqueous sodium hydroxide to form a red-brown precipitate.

What is the balanced equation, including state symbols, for the reaction?

- A  $\text{FeSO}_4(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_2(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq})$   
 B  $\text{FeSO}_4(\text{l}) + 2\text{NaOH}(\text{l}) \rightarrow \text{Fe}(\text{OH})_2(\text{s}) + \text{Na}_2\text{SO}_4(\text{l})$   
 C  $\text{Fe}_2(\text{SO}_4)_3(\text{aq}) + 6\text{NaOH}(\text{aq}) \rightarrow 2\text{Fe}(\text{OH})_3(\text{s}) + 3\text{Na}_2\text{SO}_4(\text{aq})$   
 D  $\text{Fe}_2(\text{SO}_4)_3(\text{l}) + 6\text{NaOH}(\text{aq}) \rightarrow 2\text{Fe}(\text{OH})_3(\text{s}) + 3\text{Na}_2\text{SO}_4(\text{l})$

**37. O/N 19/P22/Q20**

Which statement describes the properties of hydrochloric acid?

- A Carbon dioxide is produced when limestone reacts with hydrochloric acid.  
 B Hydrogen is produced when sodium hydroxide reacts with hydrochloric acid.  
 C Methyl orange turns yellow in strong hydrochloric acid.  
 D Red litmus paper turns blue when dipped into hydrochloric acid.

**38. O/N 19/P22/Q35**

Which statement about limestone and lime is correct?

- A Limestone combines with water to produce slaked lime.  
 B Lime is obtained from limestone by oxidation.  
 C Lime is used in the desulfurisation of flue gases.  
 D Lime is used in the treatment of alkaline soil.

**39. O/N 19/P21/Q20**

Carbonic acid is a weak acid formed when carbon dioxide dissolves in water.  
 What is the pH of the solution?

- A 1                                  B 5                                  C 7                                  D 9

**40. O/N 19/P23/Q20**

Which statements about dilute sulfuric acid are correct?

- 1 It turns red litmus paper blue.  
 2 It reacts with magnesium(II) oxide to form magnesium(II) sulfate and water.  
 3 It reacts with magnesium to form magnesium(II) sulfate and carbon dioxide.  
 4 Its pH is below pH 7.

- A 1 and 2 only                  B 1 and 3 only                  C 2 and 4 only                  D 3 and 4 only

**41. M/J 19/P22/Q19**

Which row shows the difference between a weak acid and a strong acid?

|   | weak acid         | strong acid       |
|---|-------------------|-------------------|
| A | fully ionised     | partially ionised |
| B | concentrated      | dilute            |
| C | dilute            | concentrated      |
| D | partially ionised | fully ionised     |

**42. M/J 19/P21/Q19**

Ethanoic acid is a weak acid.

Hydrochloric acid is a strong acid.

Which statements are correct?

- 1 Ethanoic acid molecules are partially dissociated into ions.  
 2 1.0 mol/dm<sup>3</sup> ethanoic acid has a higher pH than 1.0 mol/dm<sup>3</sup> hydrochloric acid.  
 3 Ethanoic acid is always more dilute than hydrochloric acid.  
 4 Ethanoic acid is a proton acceptor.

- A 1 and 2                          B 1 and 3                          C 2 and 4                          D 3 and 4

**43. MAR 19/P22/Q20, O/N 12/P12/Q17, O/N 12/P11/Q18, O/N 12/P13/Q18**

Barium hydroxide is an alkali. It reacts with hydrochloric acid.

How does the pH of the hydrochloric acid change as an excess of aqueous barium hydroxide is added?

- A The pH decreases from pH 14 and becomes constant at pH 7.  
 B The pH decreases from pH 14 to about pH 1.  
 C The pH increases from pH 1 and becomes constant at pH 7.  
 D The pH increases from pH 1 to about pH 14.



**51. O/N 17/P23/Q19**

Three solids, P, Q and R, all react with dilute sulfuric acid to produce zinc sulfate.  
P and R produce gases during the reaction.  
The gas produced when P reacts will not burn. The gas produced when R reacts will burn.  
What are P, Q and R?

|   | P              | Q              | R              |
|---|----------------|----------------|----------------|
| A | zinc           | zinc hydroxide | zinc carbonate |
| B | zinc carbonate | zinc           | zinc oxide     |
| C | zinc carbonate | zinc hydroxide | zinc           |
| D | zinc oxide     | zinc carbonate | zinc           |

**52. M/J 17/P22/Q19**

Which statements about a weak acid, such as ethanoic acid, are correct?

- 1 It reacts with a carbonate.
  - 2 It does not neutralise aqueous sodium hydroxide solution.
  - 3 It turns red litmus blue.
  - 4 It is only partially ionised in aqueous solution.
- A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

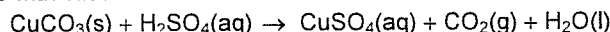
**53. M/J 17/P23/Q19**

Chloric(I) acid,  $\text{HClO}$ , is formed when chlorine dissolves in water. It is a weak acid.  
What is meant by the term *weak acid*?

- A It contains fewer hydrogen atoms than a strong acid.
- B It is easily neutralised by a strong alkali.
- C It is less concentrated than a strong acid.
- D It is only partially ionised in solution.

**54. MAR 17/P22/Q14**

Copper(II) carbonate reacts with dilute sulfuric acid.



The rate of the reaction can be changed by varying the conditions.

Which changes always increase the rate of this chemical reaction?

- 1 increasing the concentration of sulfuric acid
  - 2 increasing the size of the pieces of copper(II) carbonate
  - 3 increasing the temperature
  - 4 increasing the volume of sulfuric acid
- A 1, 3 and 4                      B 1 and 3 only                      C 2 and 3                      D 3 and 4 only

**55. MAR 17/P22/Q19**

A student investigates two acids W and X.

The same volumes of W and X are reacted separately with excess magnesium.

The student makes the following observations.

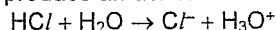
- 1 Hydrogen gas is produced at a faster rate with W than with X.
- 2 The total volume of hydrogen gas produced is the same for both acids.

Which statement explains these observations?

- A The pH of W is higher than the pH of X.
- B W is an organic acid.
- C W is a stronger acid than X.
- D W is more concentrated than X.

**56. O/N 16/P22/Q19, O/N 16/P21/Q19, O/N 16/P23/Q19**

Hydrogen chloride gas reacts with water to produce an acidic solution. The equation for the reaction is shown.



Which statement describes what happens during the reaction?

- A The chloride ion is formed by accepting an electron from the water.
- B The hydrogen chloride loses an electron to form the chloride ion.
- C The water accepts a proton from the hydrogen chloride.
- D The water donates a proton to the hydrogen chloride.

## 57. O/N 16/P22/Q34

Which substance gives off carbon dioxide on heating?

- A lime                      B limestone                      C limewater                      D slaked lime

## 58. M/J 16/P22/Q18, M/J 16/P21/Q18, M/J 16/P23/Q18, O/N 13/P12/Q17, O/N 13/P11/Q18, O/N 13/P13/Q18

Which statements are properties of an acid?

- 1 reacts with ammonium sulfate to form ammonia  
2 turns red litmus blue

|   | 1 | 2 |
|---|---|---|
| A | ✓ | ✓ |
| B | ✓ | x |
| C | x | ✓ |
| D | x | x |

## 59. MAR 16/P22/Q18

Concentrated hydrochloric acid is a *strong* acid.

What is meant by the terms 'strong' and 'acid'?

|   | strong                             | acid            |
|---|------------------------------------|-----------------|
| A | contains a low proportion of water | accepts protons |
| B | contains a low proportion of water | donates protons |
| C | fully ionised                      | accepts protons |
| D | fully ionised                      | donates protons |

## 60. O/N 15/P12/Q17, O/N 15/P11/Q17, O/N 15/P13/Q17, O/N 11/P12/Q23, O/N 11/P11/Q18, O/N 11/P13/Q22

Which property is **not** characteristic of a base?

- A It reacts with a carbonate to form carbon dioxide.  
B It reacts with an acid to form a salt.  
C It reacts with an ammonium salt to form ammonia.  
D It turns universal indicator paper blue.

## 61. M/J 15/P12/Q2

The results of some tests on a colourless liquid X are shown.

- Boiling point = 102 °C
- Universal Indicator turns green

What is X?

- A ethanol                      C pure water  
B hydrochloric acid                      D sodium chloride (salt) solution

## 62. M/J 15/P12/Q15

Which statement is **not** correct?

- A When a base reacts with an ammonium salt, ammonia is given off.  
B When an acid reacts with a base, neutralisation takes place.  
C When an acid reacts with a carbonate, carbon dioxide is given off.  
D When the acidity of a solution increases, the pH increases.

## 63. M/J 15/P13/Q15

Which reaction is **not** characteristic of an acid?

- A It dissolves magnesium oxide.  
B It produces ammonia from ammonium compounds.  
C It produces carbon dioxide from a carbonate.  
D It produces hydrogen from zinc metal.

**64. M/J 15/P11/Q15**

The table shows the pH of four aqueous solutions, W, X, Y and Z.

| substance | pH |
|-----------|----|
| W         | 7  |
| X         | 9  |
| Y         | 2  |
| Z         | 5  |

Universal Indicator is added to each solution.

Which row shows the colour of each solution after the indicator is added?

|          | W      | X      | Y      | Z      |
|----------|--------|--------|--------|--------|
| <b>A</b> | blue   | green  | orange | red    |
| <b>B</b> | green  | blue   | red    | orange |
| <b>C</b> | orange | red    | blue   | green  |
| <b>D</b> | red    | orange | green  | blue   |

**65. MAR 15/P12/Q7**

Which equation for the reaction between sodium carbonate and dilute hydrochloric acid is correct?

- A**  $\text{Na}_2\text{CO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$   
**B**  $\text{Na}_2\text{CO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{Na}_2\text{Cl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$   
**C**  $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$   
**D**  $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$

**66. MAR 15/P12/Q14**

Three chemicals, P, Q and R, were each dissolved in water. The table shows some of the reactions of these solutions.

| solution | reaction when solid sodium carbonate is added | reaction when heated with solid ammonium chloride |
|----------|-----------------------------------------------|---------------------------------------------------|
| P        | gas evolved                                   | no reaction                                       |
| Q        | no reaction                                   | gas evolved                                       |
| R        | no reaction                                   | no reaction                                       |

The pH of the three solutions was also measured.

What are the correct pH values of these solutions?

|          | P  | Q  | R  |
|----------|----|----|----|
| <b>A</b> | 2  | 7  | 13 |
| <b>B</b> | 2  | 13 | 7  |
| <b>C</b> | 7  | 2  | 13 |
| <b>D</b> | 13 | 7  | 2  |

**67. O/N 14/P12/Q32, O/N 14/P11/Q32**

Which method can be used to obtain ammonia from ammonium sulfate?

- A** Heat it with an acid. **C** Heat it with an oxidising agent.  
**B** Heat it with an alkali. **D** Heat it with a reducing agent.

**68. O/N 14/P12/Q18, O/N 14/P11/Q19**

Which substance is the most acidic?

|          | substance         | pH |
|----------|-------------------|----|
| <b>A</b> | calcium hydroxide | 12 |
| <b>B</b> | lemon juice       | 4  |
| <b>C</b> | milk              | 6  |
| <b>D</b> | washing up liquid | 8  |

**69. M/J 14/P11/Q19, M/J 14/P13/Q18**

Which statements about alkalis are correct?

- 1 When reacted with an acid, the pH of the alkali increases.
- 2 When tested with litmus, the litmus turns blue.
- 3 When warmed with an ammonium salt, ammonia gas is given off.

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**70. M/J 14/P11/Q21, M/J 14/P13/Q19**

Which acid reacts with ammonia to produce the salt ammonium sulfate?

**A** hydrochloric      **B** nitric      **C** phosphoric      **D** sulfuric

**71. M/J 13/P12/Q19**

Which statement about the reaction of acids is correct?

- A** They react with ammonium salts to form a salt and ammonia only.  
**B** They react with metal carbonates to give a salt and carbon dioxide only.  
**C** They react with metal hydroxides to give a salt and water only.  
**D** They react with metals to give a salt, hydrogen and water only.

**72. M/J 13/P11/Q16, M/J 13/P13/Q15**

Calcium carbonate reacts with hydrochloric acid to form carbon dioxide.

Which changes would slow this reaction down?

- 1 decreasing the concentration of hydrochloric acid
- 2 decreasing the particle size of calcium carbonate
- 3 decreasing the temperature

**A** 1 and 2 only      **B** 1 and 3 only      **C** 2 and 3 only      **D** 1, 2 and 3

**73. M/J 13/P11/Q18, M/J 13/P13/Q17**

Ant stings hurt because of the methanoic acid produced by the ant.

Which substance could, **most safely**, be used to neutralise the acid?

|          | substance        | pH |
|----------|------------------|----|
| <b>A</b> | baking soda      | 8  |
| <b>B</b> | car battery acid | 1  |
| <b>C</b> | lemon juice      | 3  |
| <b>D</b> | oven cleaner     | 14 |

**8.2: Types of oxides and indicators****1. O/N 22/P22/Q32, O/N 22/P21/Q32, O/N 22/P23/Q32**

Which element has an oxide that is used as a food preservative?

- A helium                      B hydrogen                      C iron                      D sulfur

**2. O/N 22/P23/Q23**

How does the nature of the oxides change across Period 3 from sodium to chlorine?

- A basic → amphoteric → acidic                      C amphoteric → basic → acidic  
B basic → acidic → amphoteric                      D acidic → amphoteric → basic

**3. M/J 22/P22/Q18, M/J 22/P21/Q18, M/J 22/P23/Q18**

The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

| oxide tested | pH of solution |
|--------------|----------------|
| X            | 1              |
| Y            | 13             |

Which information about X and Y is correct?

|   | oxide is acidic | oxide is basic | metal | non-metal |
|---|-----------------|----------------|-------|-----------|
| A | X               | Y              | X     | Y         |
| B | X               | Y              | Y     | X         |
| C | Y               | X              | X     | Y         |
| D | Y               | X              | Y     | X         |

**4. MAR 22/P22/Q21**

Zinc oxide is an amphoteric oxide.

Which types of substances will react with zinc oxide?

- A acids and bases                      C bases only  
B acids only                      D neither acids nor bases

**5. O/N 21/P21/Q18**

Which element forms an amphoteric oxide?

- A aluminium                      B carbon                      C magnesium                      D silicon

**6. O/N 21/P23/Q18**

Basic oxides are neutralised by acidic oxides.

Which element forms an oxide that neutralises calcium oxide?

- A hydrogen                      B magnesium                      C sodium                      D sulfur

**7. M/J 21/P22/Q18**

Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

|   | element X | nature of oxide, XO |
|---|-----------|---------------------|
| A | metal     | acidic              |
| B | metal     | basic               |
| C | non-metal | acidic              |
| D | non-metal | basic               |



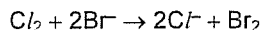
## 8. O/N 20/P22/Q23, O/N 20/P21/Q23, O/N 20/P23/Q23

Zinc oxide is an amphoteric oxide. Which row describes the reactions of zinc oxide?

|   | reaction with alkalis | reaction with acids |
|---|-----------------------|---------------------|
| A | ✓                     | ✓                   |
| B | ✓                     | x                   |
| C | x                     | ✓                   |
| D | x                     | x                   |

## 9. O/N 20/P21/Q21

The reaction between chlorine and bromide ions is a redox reaction.



What is the change in oxidation state of the reducing agent in this reaction?

- A -2 to 0                      B -1 to 0                      C 0 to -1                      D 0 to +1

## 10. M/J 20/P23/Q7

Why does magnesium oxide, MgO, have a very high melting point?

- A There is a very strong double bond between magnesium and oxygen.  
 B There is a very strong attractive force between the magnesium oxide molecules.  
 C The oxide ions are strongly attracted to positive ions.  
 D The magnesium ions are strongly attracted to a sea of electrons.

## 11. M/J 20/P23/Q8

Aluminium metal reacts with iron (III) oxide to form aluminium oxide and iron.

Which chemical equation for the reaction between aluminium and iron (III) oxide is correct?

- A  $\text{FeO} + \text{Al} \rightarrow \text{Al}_2\text{O}_3 + \text{Fe}$                       C  $\text{Fe}_2\text{O}_3 + \text{Al} \rightarrow \text{Al}_2\text{O}_3 + \text{Fe}$   
 B  $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$                       D  $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$

## 12. MAR 20/P22/Q18

X, Y and Z are oxides of elements in the same row of the Periodic Table. Some information about each oxide is shown.

| oxide | solubility in water | ability to neutralise an acid | ability to neutralise an alkali |
|-------|---------------------|-------------------------------|---------------------------------|
| X     | soluble             | x                             | ✓                               |
| Y     | insoluble           | ✓                             | ✓                               |
| Z     | slightly soluble    | ✓                             | x                               |

key

✓ = able

x = not able

Which types of oxides are X, Y and Z?

|   | X          | Y          | Z          |
|---|------------|------------|------------|
| A | acidic     | amphoteric | basic      |
| B | amphoteric | basic      | basic      |
| C | basic      | amphoteric | acidic     |
| D | basic      | acidic     | amphoteric |

## 13. O/N 19/P21/Q19

Which statement about amphoteric oxides is correct?

- A They are made by combining an acidic oxide with a basic oxide.  
 B They react with water to give a solution of pH 7.  
 C They react with both acids and bases.  
 D They do not react with acids or bases.

**14. O/N 19/P23/Q19**

Which substance is a neutral oxide?

A aluminium oxide      B carbon monoxide      C sulfur dioxide      D zinc oxide

15. M/J 19/P22/Q17

Which type of oxide are carbon monoxide and aluminium oxide?

|          | carbon monoxide | aluminium oxide |
|----------|-----------------|-----------------|
| <b>A</b> | acidic          | amphoteric      |
| <b>B</b> | acidic          | basic           |
| <b>C</b> | neutral         | amphoteric      |
| <b>D</b> | neutral         | basic           |

16. M/J 19/P22/Q18, M/J 19/P21/Q18, M/J 19/P23/Q18

The positions of elements W, X, Y and Z in the Periodic Table are shown.

elements W, X, Y, Z placed in the periodic table are shown.

|   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| W |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   |   | Y |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   | X |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Z |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

Which elements form basic oxides?

**A** W, X and Y      **B** W and X only      **C** Y only      **D** Z only

17. M/J 19/P21/Q17

Nitrogen(I) oxide,  $N_2O$ , nitrogen(II) oxide,  $NO$ , and carbon monoxide,  $CO$ , are all non-metal oxides.

They do not react with acids or bases.

Which statement is correct?

**A** They are acidic oxides. **C** They are basic oxides.  
**B** They are amphoteric oxides. **D** They are neutral oxides.

18. MAR 19/P22/Q21

Which statement describes a chemical property of aluminium oxide,  $\text{Al}_2\text{O}_3$ ?

A It reacts with acids but not with bases.  
B It reacts with acids and bases.  
C It reacts with bases but not with acids.  
D It reacts with water.

19. O/N 18/P22/Q17

In which row are the oxides correctly identified?

|          | acidic                          | basic                          |
|----------|---------------------------------|--------------------------------|
| <b>A</b> | magnesium oxide, calcium oxide  | sulfur dioxide, carbon dioxide |
| <b>B</b> | magnesium oxide, sulfur dioxide | carbon dioxide, calcium oxide  |
| <b>C</b> | sulfur dioxide, carbon dioxide  | calcium oxide, magnesium oxide |
| <b>D</b> | sulfur dioxide, magnesium oxide | calcium oxide, carbon dioxide  |

20. O/N 18/P21/Q19

Potassium hydroxide is a base.

Which statement describes a reaction of potassium hydroxide?

**A** Chlorine is formed when it is heated with ammonium chloride.  
**B** It turns Universal Indicator green.  
**C** It reacts with an acid to produce a salt and water.  
**D** It turns methyl orange red.

**21. M/J 18/P21/Q24**

The following statements are made about the metals copper, iron, magnesium and zinc.

- 1 Their oxides are acidic.
- 2 They all conduct electricity in the solid state.
- 3 They all have high melting points.
- 4 They all react with dilute acids to form hydrogen.

Which statements are correct?

- A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

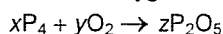
**22. O/N 17/P22/Q4**

Which row describes silicon(IV) oxide?

|   | has a<br>giant structure | is an<br>acidic oxide | conducts<br>electricity |
|---|--------------------------|-----------------------|-------------------------|
| A | ✓                        | ✓                     | ✓                       |
| B | ✓                        | ✓                     | x                       |
| C | ✓                        | x                     | x                       |
| D | x                        | ✓                     | ✓                       |

**23. O/N 17/P22/Q7**

The equation for the reaction between phosphorus and oxygen is shown.



Which values of x, y and z balance the equation?

|   | x | y  | z |
|---|---|----|---|
| A | 1 | 5  | 2 |
| B | 1 | 10 | 2 |
| C | 2 | 5  | 2 |
| D | 2 | 10 | 1 |

**24. O/N 17/P22/Q17, O/N 17/P21/Q17, O/N 17/P23/Q17**

Some properties of four oxides are listed.

Oxide 1 reacts with both acids and alkalis to form salts.

Oxide 2 reacts with acids to form salts but does not react with alkalis.

Oxide 3 reacts with alkalis to form salts but does not react with acids.

Oxide 4 does not react with acids or alkalis.

Which row describes the oxides?

|   | oxide 1    | oxide 2 | oxide 3 | oxide 4    |
|---|------------|---------|---------|------------|
| A | amphoteric | acidic  | basic   | neutral    |
| B | amphoteric | basic   | acidic  | neutral    |
| C | neutral    | acidic  | basic   | amphoteric |
| D | neutral    | basic   | acidic  | amphoteric |

**25. O/N 17/P21/Q26**

Aluminium is extracted by the electrolysis of aluminium oxide.

Which statement is **not** correct?

- A Aluminium ions are oxidised at the cathode.
- B Carbon dioxide is made at the anode.
- C Cryolite is added to lower the melting point of the aluminium oxide.
- D The electrodes are made from graphite.

**26. M/J 17/P22/Q18**

Which type of oxide is aluminium oxide?

- A acidic                      B amphoteric                      C basic                      D neutral

**27. M/J 17/P21/Q18**

Zinc oxide is amphoteric.

Which row describes the reactions of zinc oxide?

|   | reaction with hydrochloric acid | reaction with aqueous sodium hydroxide |
|---|---------------------------------|----------------------------------------|
| A | ✓                               | ✓                                      |
| B | ✓                               | x                                      |
| C | x                               | ✓                                      |
| D | x                               | x                                      |

key

✓ = reaction occurs

x = reaction does not occur

**28. M/J 17/P23/Q18, MAR 16/P22/Q19**

Which oxide is amphoteric?

- A  $Al_2O_3$                       B CaO                      C  $Na_2O$                       D  $SO_2$

**29. MAR 17/P22/Q18**

Beryllium oxide reacts with both sulfuric acid and aqueous sodium hydroxide.

Which type of oxide is beryllium oxide?

- A acidic                      B amphoteric                      C basic                      D neutral

**30. O/N 16/P22/Q18, O/N 16/P21/Q18, O/N 16/P23/Q18**

Germanium oxide is a white powder.

Germanium oxide reacts with concentrated hydrochloric acid.

Germanium oxide reacts with concentrated aqueous sodium hydroxide.

Germanium oxide does not dissolve when added to water.

Which type of oxide is germanium oxide?

- A acidic                      B amphoteric                      C basic                      D neutral

**31. O/N 16/P23/Q25**

Basic oxides and oxygen are used to convert iron into steel.

Which statement is **not** correct?

- A Carbon is converted into carbon dioxide.  
 B Silicon is converted into silicon(IV) oxide.  
 C The basic oxides react with acidic impurities to form slag.  
 D The oxygen reacts with the iron to produce hematite.

**32. M/J 16/P22/Q19, M/J 16/P21/Q19, M/J 16/P23/Q19**

Which row describes whether an amphoteric oxide reacts with acids and bases?

|   | reacts with acids | reacts with bases |
|---|-------------------|-------------------|
| A | no                | no                |
| B | no                | yes               |
| C | yes               | no                |
| D | yes               | yes               |





**43. O/N 13/P12/Q18, O/N 13/P11/Q19, O/N 13/P13/Q19**

Which of the following are properties of the oxides of non-metals?

|          | property 1 | property 2 |
|----------|------------|------------|
| <b>A</b> | acidic     | covalent   |
| <b>B</b> | acidic     | ionic      |
| <b>C</b> | basic      | covalent   |
| <b>D</b> | basic      | ionic      |

**44. M/J 13/P11/Q19, M/J 13/P13/Q21**

The diagram shows one period of the Periodic Table.

|    |    |   |   |   |   |   |    |
|----|----|---|---|---|---|---|----|
| Li | Be | B | C | N | O | F | Ne |
|----|----|---|---|---|---|---|----|

Which two elements form acidic oxides?

- A** carbon and lithium  
**B** carbon and neon

- C** carbon and nitrogen  
**D** nitrogen and neon

**45. M/J 13/P12/Q16**

Two oxides, X and Y, are added separately to dilute sulfuric acid and dilute sodium hydroxide.

X reacts with dilute sulfuric acid but Y does not react.

Y reacts with aqueous sodium hydroxide but X does not react.

Which type of oxide are X and Y?

|          | acidic oxide | basic oxide | metallic oxide |
|----------|--------------|-------------|----------------|
| <b>A</b> | X            | Y           | X              |
| <b>B</b> | X            | Y           | Y              |
| <b>C</b> | Y            | X           | X              |
| <b>D</b> | Y            | X           | Y              |

## 8.3: Preparation of salts

**1. O/N 22/P22/Q20**

Copper(II) sulfate is a soluble salt.

Calcium sulfate is an insoluble salt.

Which row shows suitable reactants for preparing a pure sample of the named salt?

|          | salt               | reactants                                              |
|----------|--------------------|--------------------------------------------------------|
| <b>A</b> | calcium sulfate    | calcium carbonate + dilute sulfuric acid               |
| <b>B</b> | calcium sulfate    | aqueous calcium chloride and aqueous sodium sulfate    |
| <b>C</b> | copper(II) sulfate | copper + dilute sulfuric acid                          |
| <b>D</b> | copper(II) sulfate | aqueous copper(II) chloride and aqueous sodium sulfate |

**2. M/J 22/P23/Q19**

An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A** crystallisation → evaporation → filtration      **C** filtration → crystallisation → evaporation  
**B** evaporation → crystallisation → filtration      **D** filtration → evaporation → crystallisation

**3. MAR 22/P22/Q22**

Information about some silver compounds is shown.

| compound         | formula                  | solubility in water |
|------------------|--------------------------|---------------------|
| silver carbonate | $\text{Ag}_2\text{CO}_3$ | insoluble           |
| silver chloride  | $\text{AgCl}$            | insoluble           |
| silver nitrate   | $\text{AgNO}_3$          | soluble             |
| silver oxide     | $\text{Ag}_2\text{O}$    | insoluble           |

Which equation shows a reaction which cannot be used to make a silver salt?

- A**  $\text{AgNO}_3(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{HNO}_3(\text{aq})$   
**B**  $\text{Ag}_2\text{O}(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow 2\text{AgNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$   
**C**  $\text{Ag}_2\text{CO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow 2\text{AgNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$   
**D**  $2\text{Ag}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{H}_2(\text{g})$

**4. O/N 21/P22/Q15**

Solid X is heated strongly.

The colour of the solid changes from blue to white.

What is solid X?

- A** anhydrous cobalt(II) chloride      **C** hydrated copper(II) sulfate  
**B** calcium carbonate      **D** lead(II) bromide

**5. O/N 21/P22/Q22**

All metal nitrates are soluble in water.

All metal chlorides are soluble except silver and lead.

All metal carbonates are insoluble except sodium and potassium.

Which aqueous solutions produce a precipitate when mixed together?

- 1 silver nitrate + sodium carbonate  
 2 silver nitrate + sodium chloride  
 3 barium nitrate + potassium chloride  
**A** 1 and 2 only      **B** 1 and 3 only      **C** 2 and 3 only      **D** 1, 2 and 3



6. M/J 21/P22/Q19

Information about the solubility of salts is shown.

| salt      | solubility                                                 |
|-----------|------------------------------------------------------------|
| chlorides | soluble (except for lead(II) chloride and silver chloride) |
| nitrates  | soluble                                                    |
| sulfates  | soluble (except for barium sulfate and lead(II) sulfate)   |

Aqueous solutions of which two compounds would produce a precipitate when added together?

- A  $\text{Ba}(\text{NO}_3)_2$  and  $\text{CaCl}_2$                       C  $\text{KCl}$  and  $\text{Na}_2\text{SO}_4$   
B  $\text{CuSO}_4$  and  $\text{Zn}(\text{NO}_3)_2$                       D  $\text{Pb}(\text{NO}_3)_2$  and  $\text{MgSO}_4$

7. O/N 21/P21/Q15

X is a pink solid.

Y is a blue solid.

When X is heated, water is produced and the solid turns blue.

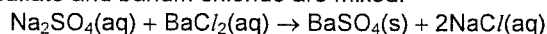
When water is added to Y, the solid turns pink.

What are X and Y?

|          | X                             | Y                             |
|----------|-------------------------------|-------------------------------|
| <b>A</b> | anhydrous cobalt(II) chloride | hydrated cobalt(II) chloride  |
| <b>B</b> | hydrated cobalt(II) chloride  | anhydrous cobalt(II) chloride |
| <b>C</b> | anhydrous copper(II) sulfate  | hydrated copper(II) sulfate   |
| <b>D</b> | hydrated copper(II) sulfate   | anhydrous copper(II) sulfate  |

8. M/J 21/P23/Q19

Aqueous solutions of sodium sulfate and barium chloride are mixed.



Which process is used to separate a sample of barium sulfate from the reaction mixture?

- A** precipitation      **B** filtration      **C** evaporation      **D** distillation

9. MAR 21/P22/Q20

When aqueous sodium hydroxide is added to a solution of a metal ion, a grey-green precipitate forms, which dissolves in excess to form a dark green solution.

What is the identity of the metal ion?

- A** chromium(III)      **B** iron(II)      **C** iron(III)      **D** copper(II)

10. O/N 20/P22/Q22

The equations for three reactions are shown.

- 1  $\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$
- 2  $2\text{AgNO}_3(\text{aq}) + \text{CuI}_2(\text{aq}) \rightarrow \text{Cu}(\text{NO}_3)_2(\text{aq}) + 2\text{AgI}(\text{s})$
- 3  $\text{CuO}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$

Which reactions are suitable for making a salt by precipitation?

- A** 1 and 2 only      **B** 1 and 3 only      **C** 2 and 3 only      **D** 1, 2 and 3

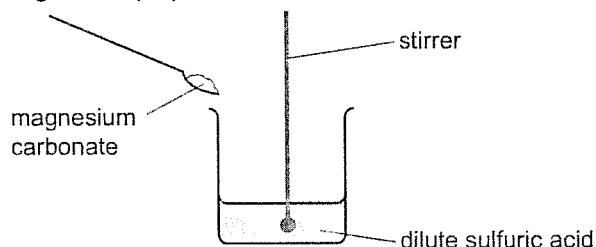
11. O/N 20/P21/Q15

Sodium nitrate is added to water in a beaker and stirred until it dissolves. At the end of the experiment, the beaker feels cold. Which row describes the reaction?

|   | temperature of solution | type of reaction |
|---|-------------------------|------------------|
| A | decreases               | endothermic      |
| B | decreases               | exothermic       |
| C | increases               | endothermic      |
| D | increases               | exothermic       |

**12. O/N 20/P22/Q24, O/N 20/P21/Q24, O/N 20/P23/Q24**

A student carries out an experiment to prepare pure magnesium sulfate crystals.  
The diagram shows the first stage of the preparation.



He adds magnesium carbonate until no more reacts.  
Which process should he use for the next stage?

- A crystallisation      B evaporation      C filtration      D neutralisation

**13. O/N 20/P23/Q3**

Nickel(II) sulfate is a green solid that is soluble in water.

Which method is used to obtain a pure sample of nickel(II) sulfate crystals from a mixture of nickel(II) sulfate and sand?

- A Heat the mixture with water and distil it to give nickel(II) sulfate.  
B Heat the mixture with water and leave it to crystallise.  
C Heat the mixture with water and filter off the nickel(II) sulfate.  
D Heat the mixture with water, filter and allow the solution to crystallise.

**14. M/J 20/P22/Q7**

The bonding, structure and melting point of sodium chloride and sulfur dichloride are shown.

| compound          | bonding  | structure        | melting point/°C |
|-------------------|----------|------------------|------------------|
| sodium chloride   | ionic    | giant lattice    | 801              |
| sulfur dichloride | covalent | simple molecular | -121             |

Why does sulfur dichloride have a lower melting point than sodium chloride?

- A The covalent bonds in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.  
B The covalent bonds in sulfur dichloride are weaker than the ionic bonds in sodium chloride.  
C The attractive forces between molecules in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.  
D The attractive forces between molecules in sulfur dichloride are weaker than the ionic bonds in sodium chloride.

**15. M/J 20/P22/Q8**

Lead (II) nitrate,  $\text{Pb}(\text{NO}_3)_2$ , reacts with potassium iodide, KI, to form a yellow precipitate,  $\text{PbI}_2$ , and a soluble salt,  $\text{KNO}_3$ .

What is the equation for the reaction?

- A  $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$       C  $2\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$   
B  $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$       D  $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$

**16. M/J 20/P21/Q21**

Lead (II) sulfate is an insoluble salt.

Which method is suitable for obtaining solid lead (II) sulfate?

- A Mix aqueous lead (II) nitrate and aqueous potassium sulfate, heat to evaporate all of the water, collect the solid and then wash and dry it.  
B Mix aqueous lead (II) nitrate and aqueous potassium sulfate, filter, collect the filtrate, crystallize, then wash and dry the crystals.  
C Mix aqueous lead (II) nitrate and dilute sulfuric acid, filter, then wash and dry the residue.  
D Titrate aqueous lead (II) hydroxide with dilute sulfuric acid, crystallize, then wash and dry the crystals.

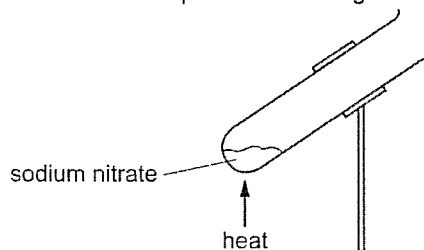
## 17. M/J 20/P22/Q21

Which process is not used in the preparation of an insoluble salt?

- A filtration                      B washing                      C crystallisation                      D drying

## 18. M/J 20/P21/Q26

Sodium nitrate is a white crystalline solid that decomposes on heating.

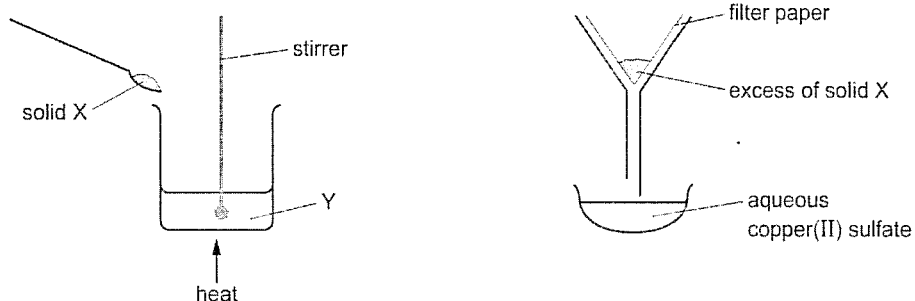


Which row describes the decomposition products formed when sodium nitrate is heated strongly?

|   | solid products | gaseous products               |
|---|----------------|--------------------------------|
| A | sodium nitrite | $\text{NO}_2$ and $\text{O}_2$ |
| B | sodium nitrite | $\text{O}_2$ only              |
| C | sodium oxide   | $\text{NO}_2$ and $\text{O}_2$ |
| D | sodium oxide   | $\text{O}_2$ only              |

## 19. M/J 20/P23/Q20

The apparatus shown is used to prepare aqueous copper (II) sulfate.



What are X and Y?

|   | X                   | Y                           |
|---|---------------------|-----------------------------|
| A | copper              | aqueous iron(II) sulfate    |
| B | copper(II) chloride | dilute sulfuric acid        |
| C | copper(II) oxide    | dilute sulfuric acid        |
| D | sulfur              | aqueous copper(II) chloride |

## 20. M/J 20/P23/Q21

Which two compounds would react together to form the insoluble salt lead (II) chloride?

|   | compound         | solubility in water |
|---|------------------|---------------------|
| 1 | lead(II) nitrate | yes                 |
| 2 | lead(II) sulfate | no                  |
| 3 | silver chloride  | no                  |
| 4 | sodium chloride  | yes                 |

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4



**27. MAR 19/P22/Q22**

The results of two tests on an aqueous solution of X are shown.

| test                                   | observation               |
|----------------------------------------|---------------------------|
| aqueous sodium hydroxide added         | green precipitate formed  |
| acidified aqueous silver nitrate added | yellow precipitate formed |

What is X?

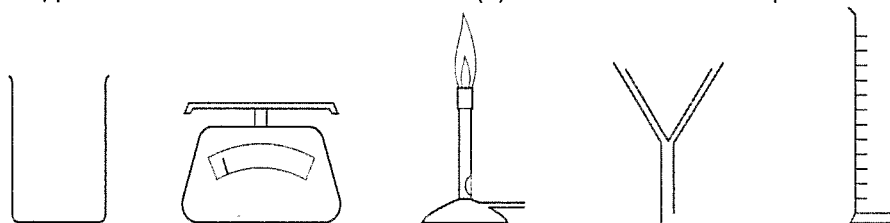
- A copper(II) chloride    B copper(II) iodide    C iron(II) chloride    D iron(II) iodide

**28. MAR 19/P22/Q3, M/J 13/P11/Q3, M/J 13/P13/Q4**

Lead(II) iodide is insoluble in water.

Lead(II) iodide is made by adding aqueous lead(II) nitrate to aqueous potassium iodide.

Which pieces of apparatus are needed to obtain solid lead(II) iodide from 20 cm<sup>3</sup> of aqueous lead(II) nitrate?



- A 1, 2 and 4    B 1, 3 and 5    C 1, 4 and 5    D 2, 4 and 5

**29. MAR 19/P22/Q23**

Four stages used to prepare an insoluble salt are listed.

- 1 drying
- 2 filtration
- 3 precipitation
- 4 washing

In which order are the stages done?

- A 2 → 1 → 3 → 4    B 3 → 2 → 4 → 1    C 3 → 4 → 1 → 2    D 4 → 3 → 2 → 1

**30. O/N 18/P22/Q19**

A few drops of methyl orange are added to a reaction mixture.

During the reaction, a gas is produced and the methyl orange turns from red to orange.

What are the reactants?

- A aqueous sodium hydroxide and ammonium chloride  
 B aqueous sodium hydroxide and calcium carbonate  
 C dilute hydrochloric acid and magnesium  
 D dilute hydrochloric acid and aqueous sodium hydroxide

**31. O/N 18/P22/Q20, O/N 18/P21/Q20, O/N 18/P23/Q20**

Some general rules for the solubility of salts in water are listed.

- Carbonates are insoluble (except ammonium carbonate, potassium carbonate and sodium carbonate).
- Chlorides are soluble (except lead(II) chloride and silver chloride).
- Nitrates are soluble.
- Sulfates are soluble (except barium sulfate, calcium sulfate and lead(II) sulfate).

Which substances produce an insoluble salt when aqueous solutions of them are mixed?

- A barium chloride and magnesium nitrate    C silver nitrate and zinc chloride  
 B calcium chloride and ammonium nitrate    D sodium carbonate and potassium sulfate

**32. M/J 18/P22/Q20, M/J 18/P21/Q20, M/J 18/P23/Q20**

Which methods are suitable for preparing **both** zinc sulfate and copper(II) sulfate?

- 1 reacting the metal oxide with warm dilute aqueous sulfuric acid
  - 2 reacting the metal with dilute aqueous sulfuric acid
  - 3 reacting the metal carbonate with dilute aqueous sulfuric acid
- A 1, 2 and 3    B 1 and 2 only    C 1 and 3 only    D 2 and 3 only

**33. M/J 18/P22/Q19**

The solubility of some salts is shown.

|           | chloride  | nitrate | sulfate   | carbonate |
|-----------|-----------|---------|-----------|-----------|
| barium    | soluble   | soluble | insoluble | insoluble |
| lead(II)  | insoluble | soluble | insoluble | insoluble |
| potassium | soluble   | soluble | soluble   | soluble   |
| zinc      | soluble   | soluble | soluble   | insoluble |

Which two aqueous solutions produce an insoluble salt when mixed together?

- A** barium chloride and zinc nitrate      **C** lead(II) nitrate and potassium carbonate  
**B** barium nitrate and lead(II) nitrate      **D** potassium nitrate and zinc sulfate

**34. M/J 18/P23/Q19**

An excess of aqueous sodium sulfate was added to aqueous barium chloride and the mixture was filtered. Which row shows the identity of the residue and the substances present in the filtrate?

|          | residue         | substances in filtrate              |
|----------|-----------------|-------------------------------------|
| <b>A</b> | barium sulfate  | barium chloride and sodium chloride |
| <b>B</b> | barium sulfate  | sodium chloride and sodium sulfate  |
| <b>C</b> | sodium chloride | barium chloride and sodium sulfate  |
| <b>D</b> | sodium chloride | barium sulfate and sodium sulfate   |

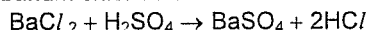
**35. MAR 18/P22/Q20**

Which method is used to make the salt copper(II) sulfate?

- A** dilute acid + alkali      **C** dilute acid + metal  
**B** dilute acid + carbonate      **D** dilute acid + non-metal oxide

**36. O/N 17/P21/Q7**

The equation for the reaction between barium chloride solution and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

|          | $\text{BaCl}_2$ | $\text{H}_2\text{SO}_4$ | $\text{BaSO}_4$ | $2\text{HCl}$ |
|----------|-----------------|-------------------------|-----------------|---------------|
| <b>A</b> | (aq)            | (aq)                    | (s)             | (aq)          |
| <b>B</b> | (aq)            | (l)                     | (s)             | (aq)          |
| <b>C</b> | (l)             | (aq)                    | (s)             | (l)           |
| <b>D</b> | (aq)            | (l)                     | (aq)            | (l)           |

**37. O/N 17/P21/Q19**

Zinc sulfate is made by reacting an excess of zinc oxide with dilute sulfuric acid. The excess zinc oxide is then removed from the solution.

Which process is used to obtain solid zinc sulfate from the solution?

- A** crystallisation      **C** filtration  
**B** dissolving      **D** fractional distillation

**38. M/J 17/P22/Q20**

Silver chloride is a white solid which is insoluble in water.

Which statement describes how a sample of pure silver chloride can be made?

- A** Add aqueous silver nitrate to aqueous sodium chloride and then filter.  
**B** Add aqueous silver nitrate to dilute hydrochloric acid, evaporate and then crystallise.  
**C** Add silver carbonate to dilute hydrochloric acid, evaporate and then crystallise.  
**D** Add silver to dilute hydrochloric acid, filter and then wash the residue.

## 39. O/N 17/P23/Q20

Which ion forms a green precipitate with aqueous sodium hydroxide that dissolves in an excess of aqueous sodium hydroxide?

- A  $\text{Ca}^{2+}$                       B  $\text{Cr}^{3+}$                       C  $\text{Cu}^{2+}$                       D  $\text{Fe}^{2+}$

## 40. M/J 17/P21/Q20

A pure sample of the insoluble salt barium carbonate can be made using the method given.

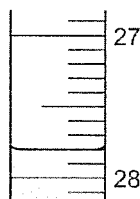
- step 1 Dissolve barium chloride in water.  
 step 2 Separately dissolve sodium carbonate in water.  
 step 3 Mix the two solutions together.  
 step 4 Filter the mixture.  
 step 5  
 step 6 Dry the residue between two sheets of filter paper.

Which instruction is missing from step 5?

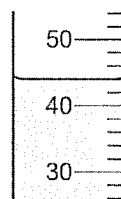
- A Heat the residue to dryness.                      C Place the filtrate in an evaporating basin.  
 B Heat the residue to the point of crystallisation.                      D Wash the residue with water.

## 41. MAR 17/P22/Q2

The diagrams show liquids in a burette and a measuring cylinder.



burette



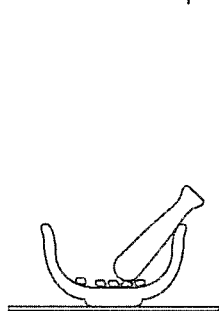
measuring cylinder

Which row shows the correct readings for the burette and the measuring cylinder?

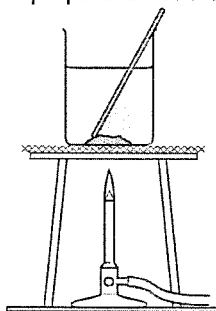
|   | burette | measuring cylinder |
|---|---------|--------------------|
| A | 27.8    | 42                 |
| B | 27.8    | 44                 |
| C | 28.2    | 42                 |
| D | 28.2    | 44                 |

## 42. MAR 17/P22/Q21

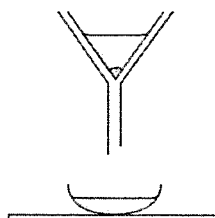
The diagram shows the steps in the preparation of a salt.



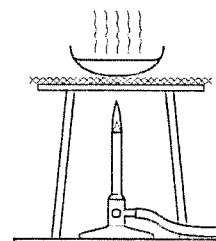
step 1



step 2



step 3



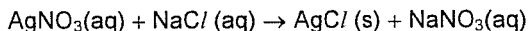
step 4

Which salt is prepared by this method?

- A barium sulfate                      B copper(II) sulfate                      C potassium sulfate                      D sodium sulfate

## 43. M/J 17/P23/Q20

Silver nitrate reacts with sodium chloride to produce silver chloride and sodium nitrate. The equation for the reaction is shown.



How is silver chloride separated from the reaction mixture?

- A crystallisation      B distillation      C evaporation      D filtration

## 44. O/N 16/P22/Q21, O/N 16/P21/Q21, O/N 16/P23/Q21

Information about some silver compounds is shown in the table.

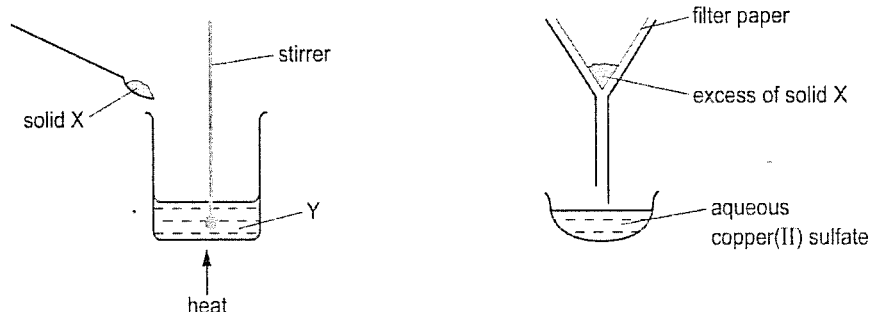
| compound         | formula                  | solubility in water |
|------------------|--------------------------|---------------------|
| silver carbonate | $\text{Ag}_2\text{CO}_3$ | insoluble           |
| silver chloride  | $\text{AgCl}$            | insoluble           |
| silver nitrate   | $\text{AgNO}_3$          | soluble             |
| silver oxide     | $\text{Ag}_2\text{O}$    | insoluble           |

Which equation shows a reaction which **cannot** be used to make a silver salt?

- A  $\text{AgNO}_3(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{HNO}_3(\text{aq})$   
 B  $\text{Ag}_2\text{O}(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow 2\text{AgNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$   
 C  $\text{Ag}_2\text{CO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow 2\text{AgNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$   
 D  $2\text{Ag}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{H}_2(\text{g})$

## 45. O/N 16/P22/Q20, O/N 16/P21/Q20, O/N 16/P23/Q20

The apparatus shown is used to prepare aqueous copper(II) sulfate.



What are X and Y?

|   | X                   | Y                           |
|---|---------------------|-----------------------------|
| A | copper              | aqueous iron(II) sulfate    |
| B | copper(II) chloride | sulfuric acid               |
| C | copper(II) oxide    | sulfuric acid               |
| D | sulfur              | aqueous copper(II) chloride |

## 46. O/N 16/P21/Q3

Information about the solubility of four solids, P, Q, R and S, is given in the table.

|                     | P         | Q         | R         | S         |
|---------------------|-----------|-----------|-----------|-----------|
| solubility in water | dissolves | insoluble | insoluble | dissolves |

A student attempted to separate mixtures of these solids using the following method.

- 1 Add the mixture to a beaker of water and stir.
- 2 Filter the mixture.
- 3 Crystallise one of the solids from the filtrate.

Which of the following mixtures could **not** be separated by this method?

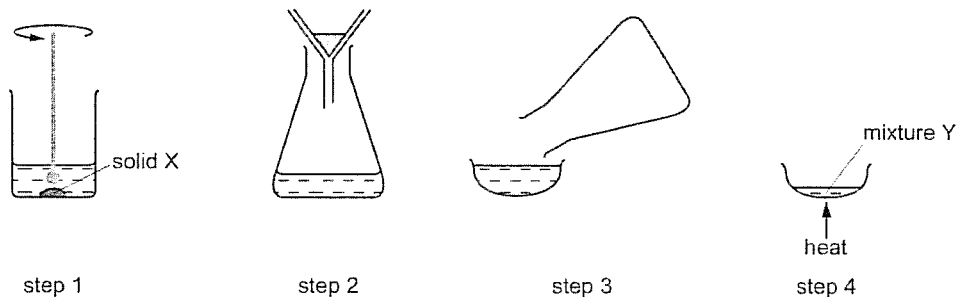
- A a mixture of P and R      C a mixture of Q and R  
 B a mixture of Q and P      D a mixture of R and S



**47. O/N 16/P23/Q3**

A solid X is purified in five steps.

The first four steps of the purification are shown in the diagram.



In **step 5**, how is a pure sample of solid X obtained from mixture Y?

- A** dissolving      **B** distillation      **C** evaporating      **D** filtering

**48. M/J 16/P22/Q20**

Silver chloride is insoluble in water and is prepared by precipitation.

Which two substances can be used to make silver chloride?

- A** barium chloride and silver nitrate      **C** hydrochloric acid and silver bromide  
**B** hydrochloric acid and silver      **D** sodium chloride and silver iodide

**49. M/J 16/P21/Q20**

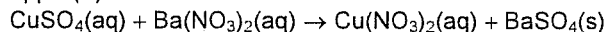
Which substance reacts with dilute sulfuric acid to form a salt that can be removed from the resulting mixture by filtration?

- A** aqueous barium chloride      **C** copper  
**B** aqueous sodium hydroxide      **D** copper(II) carbonate

**50. M/J 16/P23/Q20**

Barium sulfate is an insoluble salt.

It can be made by reacting copper(II) sulfate solution with barium nitrate solution.

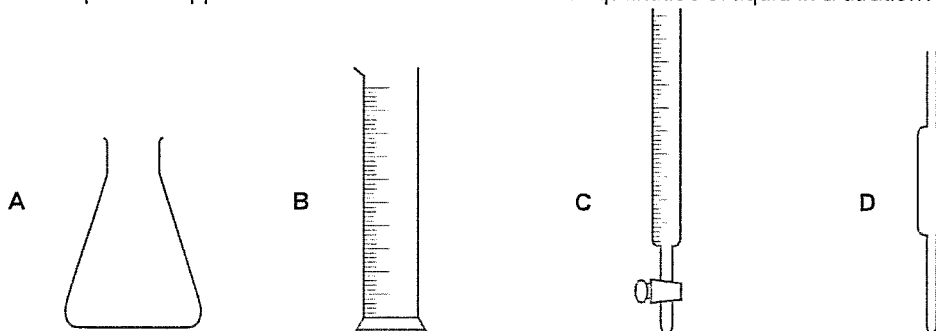


What is the correct order of steps to obtain a pure, dry sample of barium sulfate from the reaction mixture?

|          | step 1 | step 2                                                 | step 3                                   |
|----------|--------|--------------------------------------------------------|------------------------------------------|
| <b>A</b> | filter | evaporate the filtrate to dryness                      | leave the solid formed to cool           |
| <b>B</b> | filter | evaporate the filtrate to the point of crystallisation | leave the filtrate to cool               |
| <b>C</b> | filter | leave the residue in a warm place to dry               | wash the residue with water              |
| <b>D</b> | filter | wash the residue with water                            | leave the residue in a warm place to dry |

**51. MAR 16/P22/Q2**

Which piece of apparatus is used to measure variable quantities of liquid in a titration?



**52. MAR 16/P22/Q20, O/N 10/P12/Q23, O/N 10/P11/Q22, O/N 10/P13/Q23**

A salt is made by adding an excess of an insoluble metal oxide to an acid.

How is the excess metal oxide removed from the mixture?

- A chromatography    B crystallisation    C distillation    D filtration

**53. O/N 15/P12/Q19**

A salt is produced in each of the following reactions.

P magnesium + dilute hydrochloric acid

Q zinc oxide + dilute sulfuric acid

R sodium hydroxide + dilute hydrochloric acid

S copper carbonate + dilute sulfuric acid

Which statements about the products of the reactions are correct?

1 A flammable gas is produced in reaction P.

2 Water is formed in all reactions.

3 All the salts formed are soluble in water.

- A 1, 2 and 3    B 1 and 2 only    C 1 and 3 only    D 2 and 3 only

**54. O/N 15/P11/Q18**

Four stages in the preparation of a salt from an acid and a solid metal oxide are listed.

1 Add excess solid.

2 Evaporate half the solution and leave to cool.

3 Filter to remove unwanted solid.

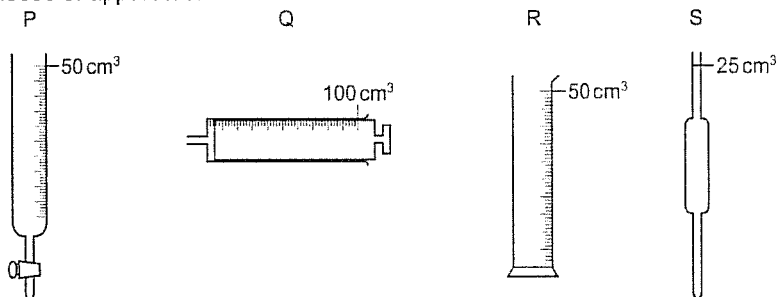
4 Heat the acid.

In which order should the stages be carried out?

- A 1 → 3 → 4 → 2    B 2 → 1 → 3 → 4    C 4 → 1 → 3 → 2    D 4 → 2 → 1 → 3

**55. O/N 15/P11/Q2**

P, Q, R and S are pieces of apparatus.



Which row describes the correct apparatus for the measurement made?

|   | apparatus | measurement made                                                                       |
|---|-----------|----------------------------------------------------------------------------------------|
| A | P         | the volume of acid added to alkali in a titration                                      |
| B | Q         | 1 cm <sup>3</sup> of acid to add to calcium carbonate in a rate-determining experiment |
| C | R         | 75 cm <sup>3</sup> of a gas given off in a rate-determining experiment                 |
| D | S         | 20 cm <sup>3</sup> of alkali for use in a titration                                    |

**56. O/N 15/P13/Q18**

Zinc sulfate is a soluble salt and can be prepared by reacting excess zinc carbonate with dilute sulfuric acid.

Which piece of equipment would **not** be required in the preparation of zinc sulfate crystals?

- A beaker    B condenser    C evaporating dish    D filter funnel

**57. M/J 15/P12/Q17**

Which method is used to make the salt copper sulfate?

- A dilute acid + alkali    B dilute acid + carbonate    C dilute acid + metal    D dilute acid + non-metal oxide

**58. M/J 15/P12/Q18**

Two tests are carried out to identify an aqueous solution of X.

test 1 Aqueous sodium hydroxide is added and a blue precipitate is produced.

test 2 Dilute nitric acid is added followed by aqueous silver nitrate and a white precipitate is produced.  
What is X?

- A copper carbonate    B copper chloride    C iron(III) carbonate    D iron(III) chloride

**59. M/J 15/P11/Q17**

Four steps to prepare a salt from an excess of a solid base and an acid are listed.

- 1 crystallisation
- 2 evaporation
- 3 filtration
- 4 neutralisation

In which order are the steps carried out?

- A 2 → 3 → 4 → 1    B 3 → 1 → 2 → 4    C 4 → 2 → 1 → 3    D 4 → 3 → 2 → 1

**60. M/J 15/P11/Q18**

Which two compounds give a white precipitate when their aqueous solutions are mixed?

- A silver nitrate and sodium chloride    C sodium hydroxide and copper(II) sulfate  
B silver nitrate and sodium iodide    D sodium hydroxide and iron(II) chloride

**61. M/J 15/P13/Q17**

Which of the following methods are suitable for preparing both zinc sulfate and copper sulfate?

- 1 Reacting the metal oxide with warm dilute aqueous sulfuric acid.
- 2 Reacting the metal with dilute aqueous sulfuric acid.
- 3 Reacting the metal carbonate with dilute aqueous sulfuric acid.

- A 1 and 2 only    B 1 and 3 only    C 2 and 3 only    D 1, 2 and 3

**62. MAR 15/P12/Q16**

Which two processes are involved in the preparation of magnesium sulfate from dilute sulfuric acid and an excess of magnesium oxide?

- A neutralisation and filtration    C thermal decomposition and filtration  
B neutralisation and oxidation    D thermal decomposition and oxidation

**63. O/N 14/P12/Q21, O/N 14/P11/Q21**

How many different salts could be made from a supply of dilute sulfuric acid, dilute hydrochloric acid, copper, magnesium oxide and zinc carbonate?

- A 3    B 4    C 5    D 6

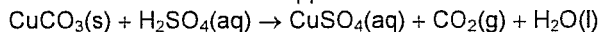
**64. O/N 14/P13/Q21**

Which salt preparation uses a burette and a pipette?

- A calcium nitrate from calcium carbonate and nitric acid  
B copper(II) sulfate from copper(II) hydroxide and sulfuric acid  
C potassium chloride from potassium hydroxide and hydrochloric acid  
D zinc chloride from zinc and hydrochloric acid

**65. M/J 14/P12/Q20**

Copper carbonate reacts with dilute sulfuric acid to make copper sulfate.



Which row gives the correct order of steps for making copper sulfate crystals?

|   | step 1                                  | step 2                                | step 3                                         | step 4        |
|---|-----------------------------------------|---------------------------------------|------------------------------------------------|---------------|
| A | add excess acid to the copper carbonate | filter                                | evaporate filtrate to point of crystallisation | leave to cool |
| B | add excess acid to the copper carbonate | filter                                | evaporate to dryness                           | leave to cool |
| C | add excess copper carbonate to the acid | evaporate to point of crystallisation | leave to cool                                  | filter        |
| D | add excess copper carbonate to the acid | filter                                | evaporate filtrate to point of crystallisation | leave to cool |

**66. M/J 14/P11/Q8, M/J 14/P13/Q7**

Solid F is an element.

Solid G is a compound.

Neither solid conducts electricity but G conducts electricity when dissolved in water.

These properties suggest that F is .....1..... and that G is .....2..... with .....3..... bonds.

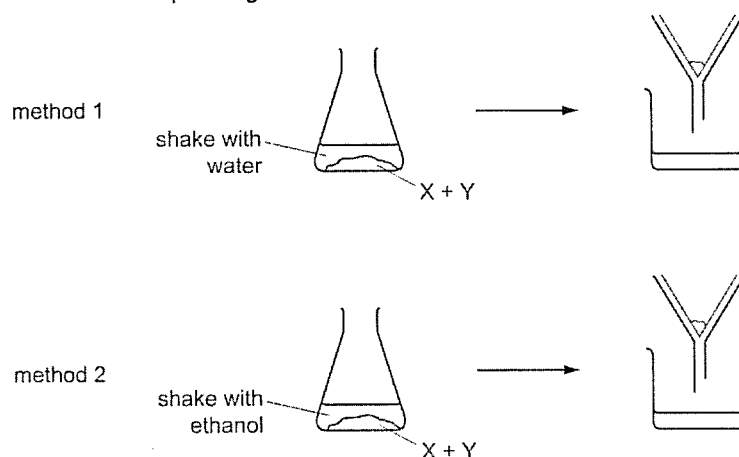
Which words correctly complete gaps 1, 2 and 3?

|          | 1        | 2    | 3        |
|----------|----------|------|----------|
| <b>A</b> | diamond  | AgCl | covalent |
| <b>B</b> | diamond  | NaCl | ionic    |
| <b>C</b> | graphite | AgCl | ionic    |
| <b>D</b> | graphite | NaCl | covalent |

**67. O/N 13/P12/Q9, O/N 13/P11/Q8, O/N 13/P13/Q8**

A solid mixture contains an ionic salt, X, and a covalent organic compound, Y.

Two students suggest methods of separating the mixture as shown.



Which methods of separation are likely to work?

|          | 1 | 2 |
|----------|---|---|
| <b>A</b> | ✓ | ✓ |
| <b>B</b> | ✓ | x |
| <b>C</b> | x | ✓ |
| <b>D</b> | x | x |

**68. M/J 13/P11/Q13, M/J 13/P13/Q12**

Some white anhydrous copper(II) sulfate powder is put into a beaker of water and stirred.

What would show that the process was exothermic?

**A** A blue solution is formed.**C** The beaker feels warmer.**B** The beaker feels cooler.**D** The powder dissolves in the water.**69. O/N 12/P12/Q18, O/N 12/P11/Q16**Which of these pairs of aqueous ions **both** react with dilute sulfuric acid to give a visible result?**A**  $\text{Ba}^{2+}$  and  $\text{Cl}^-$ **B**  $\text{Ba}^{2+}$  and  $\text{CO}_3^{2-}$ **C**  $\text{NH}_4^+$  and  $\text{Cl}^-$ **D**  $\text{NH}_4^+$  and  $\text{CO}_3^{2-}$ **70. O/N 12/P12/Q19, O/N 12/P11/Q19, O/N 12/P13/Q19**

A compound is a salt if it

**A** can neutralise an acid.**C** dissolves in water.**B** contains more than one element.**D** is formed when an acid reacts with a base.

**71. M/J 13/P12/Q21, M/J 13/P11/Q21, M/J 13/P13/Q19**

Two indicators, bromophenol blue and Congo red, show the following colours in acidic solutions and in alkaline solutions.

| indicator        | acid   | alkali |
|------------------|--------|--------|
| bromophenol blue | yellow | blue   |
| Congo red        | violet | red    |

A few drops of each indicator are added to separate samples of a solution of pH 2. What are the colours of the indicators in this solution?

|          | in a solution of pH 2 |              |
|----------|-----------------------|--------------|
|          | bromophenol blue is   | Congo red is |
| <b>A</b> | blue                  | red          |
| <b>B</b> | blue                  | violet       |
| <b>C</b> | yellow                | red          |
| <b>D</b> | yellow                | violet       |

## 8.4: Identification of ions and gases

### 1. O/N 22/P21/Q25

Copper(II) nitrate and zinc carbonate are heated strongly in separate test-tubes. Which row identifies the gases produced?

|   | copper(II) nitrate          | zinc carbonate            |
|---|-----------------------------|---------------------------|
| A | oxygen and nitrogen dioxide | carbon dioxide only       |
| B | oxygen and nitrogen dioxide | carbon dioxide and oxygen |
| C | nitrogen dioxide only       | carbon dioxide and oxygen |
| D | nitrogen dioxide only       | carbon dioxide only       |

### 2. O/N 22/P23/Q19

Tests are done on an aqueous solution.

| test        | a few drops of aqueous sodium hydroxide are added | aqueous sodium hydroxide is added in excess         |
|-------------|---------------------------------------------------|-----------------------------------------------------|
| observation | white precipitate                                 | precipitate dissolves to give a colourless solution |

Which cation produce these observations?

- 1 aluminium,  $\text{Al}^{3+}$
- 2 calcium,  $\text{Ca}^{2+}$
- 3 zinc,  $\text{Zn}^{2+}$

A 1 and 2                      B 1 and 3                      C 1 only                      D 2 and 3

### 3. M/J 22/P22/Q14

When the colourless gas  $\text{N}_2\text{O}_4$  is heated, it forms the brown gas  $\text{NO}_2$ .

When the reaction mixture is cooled, the brown colour fades and turns back to colourless.

Which type of reaction is described by these observations?

A decomposition              B displacement              C reduction              D reversible

### 4. M/J 22/P22/Q28, M/J 22/P21/Q28, M/J 22/P23/Q28

Three metal compounds, J, K and L, are heated using a Bunsen burner.

The results are shown.

J colourless gas produced, which relights a glowing splint

K colourless gas produced, which turns limewater milky

L no reaction

Which row identifies J, K and L?

|   | J                   | K                   | L                   |
|---|---------------------|---------------------|---------------------|
| A | magnesium carbonate | potassium carbonate | potassium nitrate   |
| B | magnesium carbonate | potassium nitrate   | potassium carbonate |
| C | potassium nitrate   | magnesium carbonate | potassium carbonate |
| D | potassium nitrate   | potassium carbonate | magnesium carbonate |

### 5. M/J 22/P23/Q13

Metal M reacts with steam and produces gas G.

Which row identifies gas G and the type of reaction when metal M reacts with steam?

|   | gas G    | type of reaction |
|---|----------|------------------|
| A | hydrogen | redox            |
| B | hydrogen | neutralisation   |
| C | oxygen   | redox            |
| D | oxygen   | neutralisation   |

**6. M/J 22/P21/Q14**

Acidified aqueous silver nitrate is added to a test-tube containing aqueous chloride ions.

The test-tube is then left in direct sunlight.

Which row describes the observations and explains what happens to the reaction mixture?

|          | observation on adding aqueous silver nitrate | observation after leaving in sunlight | explanation             |
|----------|----------------------------------------------|---------------------------------------|-------------------------|
| <b>A</b> | yellow precipitate                           | precipitate dissolves                 | silver chloride forms   |
| <b>B</b> | yellow precipitate                           | precipitate turns grey                | silver ions are reduced |
| <b>C</b> | white precipitate                            | precipitate dissolves                 | silver chloride forms   |
| <b>D</b> | white precipitate                            | precipitate turns grey                | silver ions are reduced |

**7. M/J 22/P23/Q20**

Which ion forms a precipitate that dissolves in excess with both aqueous ammonia and with aqueous sodium hydroxide?

**A** calcium ion,  $\text{Ca}^{2+}$     **B** copper(II) ion,  $\text{Cu}^{2+}$     **C** iron(III) ion,  $\text{Fe}^{3+}$     **D** zinc ion,  $\text{Zn}^{2+}$

**8. O/N 21/P21/Q28**

Which statements about the thermal decomposition of copper(II) nitrate are correct?

- 1 A brown gas is given off.
- 2 A gas which relights a glowing splint is given off.
- 3 The solid residue is an acidic oxide.

**A** 1 only    **B** 1 and 2    **C** 1 and 3    **D** 2 and 3

**9. M/J 21/P22/Q17**

When bismuth(III) chloride,  $\text{BiCl}_3$ , reacts with water, a white precipitate of bismuth(III) oxychloride,  $\text{BiOCl}$ , is formed. The equation for the reaction is shown.



The reaction is in equilibrium.

Which changes cause the white precipitate to dissolve?

- 1 adding acid
- 2 adding water
- 3 adding sodium chloride solution

**A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3

**10. O/N 20/P22/Q32**

The results of tests on solid S and its aqueous solution are shown.

| tests on solid S                                                         | tests on aqueous solution of S                              |                                                             |
|--------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| effect of heat                                                           | effect of aqueous sodium hydroxide                          | effect of aqueous ammonia                                   |
| brown gas given off, together with a gas which relights a glowing splint | white ppt., soluble in excess, giving a colourless solution | white ppt., soluble in excess, giving a colourless solution |

What is S?

**A** aluminium nitrate    **B** aluminium sulfate    **C** zinc sulfate    **D** zinc nitrate

**11. O/N 20/P23/Q22**

An aqueous cation reacts with aqueous sodium hydroxide to form a white precipitate.

The precipitate is insoluble in excess sodium hydroxide.

What is the aqueous cation?

**A** aluminium ion    **B** calcium ion    **C** chromium ion    **D** zinc ion

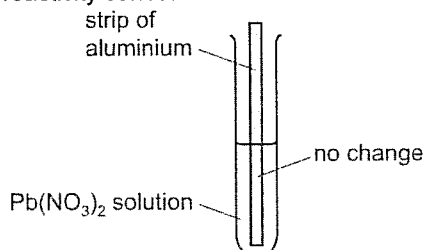
## 12. O/N 20/P23/Q27

A flammable gas needs to be removed from a tank at an industrial plant.  
For safety reasons, an inert gas is used.  
Which gas is suitable?

- A argon                      B hydrogen                      C methane                      D oxygen

## 13. O/N 20/P21/Q28

A strip of aluminium is placed into a test-tube containing aqueous lead(II) nitrate and left for several minutes.  
Aluminium is higher than lead in the reactivity series.



Which statement explains why lead is **not** displaced by this strip of aluminium?

- A A thin insoluble layer of aluminium nitrate forms on the aluminium.  
B Nitrate ions are reduced in aqueous solution.  
C The ionic bonds between lead and nitrate ions are too strong.  
D There is an unreactive oxide layer on the aluminium.

## 14. MAR 20/P22/Q21

Salt S is dissolved in water and three tests are carried out on the solution.

|   | test                                                                  | result                                                         |
|---|-----------------------------------------------------------------------|----------------------------------------------------------------|
| 1 | aqueous sodium hydroxide is added                                     | green precipitate formed, insoluble in excess sodium hydroxide |
| 2 | dilute nitric acid is added                                           | no reaction                                                    |
| 3 | aqueous barium nitrate is added to the acidified solution from test 2 | white precipitate formed                                       |

What is the identity of S?

- A Copper (II) chloride    B Copper (II) sulfate    C Iron (II) chloride    D Iron (II) sulfate

## 15. M/J 19/P22/Q7

Calcium metal reacts with water to form a solution of calcium hydroxide and hydrogen gas.  
Which equation is correct?

- A  $\text{Ca(s)} + \text{H}_2\text{O(aq)} \rightarrow \text{CaOH(aq)} + \text{H(g)}$                       C  $\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$   
B  $\text{Ca(s)} + 2\text{H}_2\text{O(aq)} \rightarrow \text{Ca(OH)}_2\text{(s)} + 2\text{H}_2\text{(g)}$                       D  $\text{Ca(s)} + \text{H}_2\text{O(l)} \rightarrow \text{CaOH(l)} + \text{H(g)}$

## 16. M/J 19/P22/Q24

A student heated the carbonates and nitrates of sodium and copper.  
The results are shown.

|   | compound heated      | gases released              | solid formed     |
|---|----------------------|-----------------------------|------------------|
| 1 | sodium carbonate     | carbon monoxide             | sodium oxide     |
| 2 | copper(II) carbonate | carbon dioxide              | copper           |
| 3 | sodium nitrate       | oxygen only                 | sodium nitrite   |
| 4 | copper(II) nitrate   | nitrogen dioxide and oxygen | copper(II) oxide |

Which rows describe the correct results?

- A 1 and 3                      B 2 and 3                      C 3 and 4                      D 4 only



**17. M/J 19/P21/Q24**

Three metal compounds, P, Q and R, are heated using a Bunsen burner.

The results are shown.

P colourless gas produced, which relights a glowing splint

Q colourless gas produced, which turns limewater milky

R no reaction

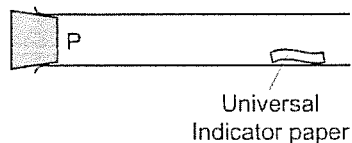
Which row shows the identity of P, Q and R?

|   | P                   | Q                   | R                   |
|---|---------------------|---------------------|---------------------|
| A | magnesium carbonate | potassium carbonate | potassium nitrate   |
| B | magnesium carbonate | potassium nitrate   | potassium carbonate |
| C | potassium nitrate   | magnesium carbonate | potassium carbonate |
| D | potassium nitrate   | potassium carbonate | magnesium carbonate |

**18. M/J 19/P23/Q1**

Hydrogen chloride gas ( $M_r = 36.5$ ) is released at P in the apparatus shown.

The Universal Indicator paper turns red after 38 s.



The experiment is repeated using sulfur dioxide ( $M_r = 64$ ).

What is the result for sulfur dioxide?

|   | Universal Indicator turns | time for Universal Indicator to change colour/s |
|---|---------------------------|-------------------------------------------------|
| A | blue                      | 26                                              |
| B | blue                      | 51                                              |
| C | red                       | 26                                              |
| D | red                       | 51                                              |

**19. O/N 18/P21/Q18**

Aqueous sodium hydroxide is added to solid Q in a test-tube.

A gas is produced which turns damp red litmus blue.

What is Q?

A aluminium

C ammonium chloride

B ammonia

D sodium nitrate

**20. O/N 18/P23/Q17**

The results of some experiments with sulfur dioxide are shown.

| experiment | description                                    | result                           |
|------------|------------------------------------------------|----------------------------------|
| 1          | mix with dilute hydrochloric acid              | does not react                   |
| 2          | mix with concentrated sodium hydroxide         | a salt forms                     |
| 3          | add Universal Indicator                        | Universal Indicator turns purple |
| 4          | add acidified aqueous potassium manganate(VII) | purple solution turns colourless |

Which results are correct?

A 1, 2 and 4

B 2, 3 and 4

C 1 and 2 only

D 3 and 4 only

**21. O/N 18/P23/Q18**

A white precipitate is produced when small amounts of two colourless solutions are mixed together. Which pairs of solutions produce a white precipitate?

- 1 sodium hydroxide and zinc nitrate
- 2 sodium hydroxide and aluminium chloride
- 3 barium chloride and sulfuric acid
- 4 acidified barium nitrate and potassium sulfate

A 1, 2, 3 and 4      B 1, 2 and 4 only      C 1 and 2 only      D 2 only

**22. O/N 18/P23/Q19**

Solution Q is warmed with ammonium chloride.

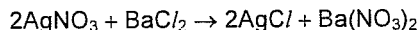
In a separate experiment, solution Q is added to methyl orange.

Which observations show that solution Q is basic?

|   | warmed with ammonium chloride | added to methyl orange |
|---|-------------------------------|------------------------|
| A | gas is produced               | turns red              |
| B | gas is produced               | turns yellow           |
| C | no reaction                   | turns red              |
| D | no reaction                   | turns yellow           |

**23. M/J 18/P21/Q19**

A student mixes silver nitrate and barium chloride to form a white precipitate of silver chloride. The equation is shown.



Which row describes the solubility of the salts?

|   | soluble                                             | insoluble                                           |
|---|-----------------------------------------------------|-----------------------------------------------------|
| A | silver nitrate                                      | barium chloride, barium nitrate and silver chloride |
| B | silver nitrate and barium chloride                  | barium nitrate and silver chloride                  |
| C | silver nitrate, barium chloride and barium nitrate  | silver chloride                                     |
| D | silver nitrate, barium chloride and silver chloride | barium nitrate                                      |

**24. O/N 17/P21/Q28**

Information about the nitrates and carbonates of two metals, Q and R, is shown.

|                | appearance  | solubility in water | effect of heat                                         |
|----------------|-------------|---------------------|--------------------------------------------------------|
| nitrate of Q   | white solid | soluble             | colourless gas evolved which relights a glowing splint |
| carbonate of Q | white solid | soluble             | no reaction                                            |
| nitrate of R   | white solid | soluble             | brown gas evolved                                      |
| carbonate of R | white solid | insoluble           | colourless gas evolved which turns limewater milky     |

Which statement is correct?

- A Q is calcium and R is magnesium.  
B Q is magnesium and R is sodium.

- C Q is potassium and R is copper.  
D Q is sodium and R is calcium.

**25. O/N 17/P22/Q20**

Compound P reacts with hydrochloric acid to produce a gas that turns limewater milky.

What is P?

- A** sodium carbonate    **B** sodium chloride    **C** sodium hydroxide    **D** sodium sulfate

**26. O/N 17/P21/Q20**

What is used to test for chlorine?

- A** a glowing splint    **C** limewater  
**B** damp litmus paper    **D** potassium manganate(VII) solution

**27. M/J 17/P22/Q21**

Dilute sulfuric acid is added to two separate aqueous solutions, X and Y. The observations are shown.

|            |                             |
|------------|-----------------------------|
| solution X | white precipitate           |
| solution Y | bubbles of a colourless gas |

Which row shows the ions present in the solutions?

|          | solution X       | solution Y         |
|----------|------------------|--------------------|
| <b>A</b> | $\text{Ba}^{2+}$ | $\text{CO}_3^{2-}$ |
| <b>B</b> | $\text{Ca}^{2+}$ | $\text{Cl}^-$      |
| <b>C</b> | $\text{Cu}^{2+}$ | $\text{CO}_3^{2-}$ |
| <b>D</b> | $\text{Fe}^{2+}$ | $\text{NO}_3^-$    |

**28. M/J 17/P21/Q21**

Substance X reacts with warm dilute hydrochloric acid to produce a gas which decolourises acidified aqueous potassium manganate(VII).

Substance X gives a yellow flame in a flame test.

What is X?

- A** potassium chloride    **B** potassium sulfite    **C** sodium chloride    **D** sodium sulfite

**29. M/J 17/P23/Q7**

Aqueous iron(III) sulfate and aqueous sodium hydroxide react to give a precipitate of iron(III) hydroxide and a solution of sodium sulfate.

What is the balanced equation for this reaction?

- A**  $\text{Fe}_2(\text{SO}_4)_3(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq})$   
**B**  $\text{Fe}_2(\text{SO}_4)_3(\text{aq}) + 3\text{NaOH}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s}) + 3\text{Na}_2\text{SO}_4(\text{aq})$   
**C**  $\text{Fe}_2(\text{SO}_4)_3(\text{aq}) + 6\text{NaOH}(\text{aq}) \rightarrow 2\text{Fe}(\text{OH})_3(\text{s}) + 3\text{Na}_2\text{SO}_4(\text{aq})$   
**D**  $2\text{Fe}_2(\text{SO}_4)_3(\text{aq}) + 6\text{NaOH}(\text{aq}) \rightarrow 4\text{Fe}(\text{OH})_3(\text{s}) + 6\text{Na}_2\text{SO}_4(\text{aq})$

**30. M/J 17/P23/Q21**

Aqueous sodium hydroxide reacts with an aqueous solution of compound Y to give a green precipitate.

Aqueous ammonia also reacts with an aqueous solution of compound Y to give a green precipitate.

In each case the precipitate is insoluble when an excess of reagent is added.

Which ion is present in Y?

- A** chromium(III)    **B** copper(II)    **C** iron(II)    **D** iron(III)

**31. MAR 17/P22/Q20**

A student is given an unknown solution.

Which two tests provide evidence that the solution is copper(II) sulfate?

- 1 adding dilute hydrochloric acid  
 2 adding aqueous sodium hydroxide  
 3 adding dilute nitric acid, then silver nitrate solution  
 4 adding dilute nitric acid, then barium nitrate solution  
**A** 1 and 3    **B** 1 and 4    **C** 2 and 3    **D** 2 and 4

**32. O/N 16/P21/Q23**

Four substances, P, Q, R and S, are tested as shown.

| test                                                    | substance                                        |                                           |                                                       |             |
|---------------------------------------------------------|--------------------------------------------------|-------------------------------------------|-------------------------------------------------------|-------------|
|                                                         | P                                                | Q                                         | R                                                     | S           |
| dilute hydrochloric acid added                          | gas given off which 'pops' with a lighted splint | gas given off which turns limewater milky | no reaction                                           | no reaction |
| dilute aqueous sodium hydroxide added and warmed gently | no reaction                                      | no reaction                               | gas given off which turns damp, red litmus paper blue | no reaction |

What are P, Q, R and S?

|          | P                        | Q                        | R                        | S                      |
|----------|--------------------------|--------------------------|--------------------------|------------------------|
| <b>A</b> | Mg                       | $\text{Na}_2\text{CO}_3$ | $\text{NH}_4\text{Cl}$   | $\text{NaCl}$          |
| <b>B</b> | Mg                       | $\text{NH}_4\text{Cl}$   | $\text{Na}_2\text{CO}_3$ | $\text{NaCl}$          |
| <b>C</b> | Mg                       | $\text{Na}_2\text{CO}_3$ | $\text{NaCl}$            | $\text{NH}_4\text{Cl}$ |
| <b>D</b> | $\text{Na}_2\text{CO}_3$ | Mg                       | $\text{NaCl}$            | $\text{NH}_4\text{Cl}$ |

**33. O/N 16/P23/Q23**

Compound T is added to dilute hydrochloric acid and warmed gently.

The mixture gives off a gas which turns acidified aqueous potassium manganate(VII) from purple to colourless.

A flame test on compound T gives a lilac flame.

What is compound T?

- A** sodium sulfate      **B** sodium sulfite      **C** potassium sulfate      **D** potassium sulfite

**34. MAR 16/P22/Q21**

A substance is heated with aluminium foil in aqueous sodium hydroxide. A gas is produced which turns damp, red litmus paper blue.

Which anion is present in the substance?

- A** carbonate      **B** iodide      **C** nitrate      **D** sulfate

**35. O/N 15/P12/Q18**

A sting from insect X has a pH of 6 and a sting from insect Y has a pH of 8.

The table shows the pH of four substances.

| substance                 | pH |
|---------------------------|----|
| hydrochloric acid         | 1  |
| sodium hydrogen carbonate | 8  |
| sodium hydroxide          | 14 |
| vinegar                   | 5  |

Which substances are used to treat the two stings?

|          | X                         | Y                         |
|----------|---------------------------|---------------------------|
| <b>A</b> | hydrochloric acid         | sodium hydroxide          |
| <b>B</b> | sodium hydrogen carbonate | vinegar                   |
| <b>C</b> | sodium hydroxide          | hydrochloric acid         |
| <b>D</b> | vinegar                   | sodium hydrogen carbonate |

**36. O/N 15/P11/Q16**

A white solid is insoluble in water.

When it is added to hydrochloric acid, bubbles of gas are formed.

Adding aqueous ammonia to the solution formed gives a white precipitate. Adding excess aqueous ammonia causes the precipitate to re-dissolve.

What is the white solid?

- A** aluminium nitrate    **B** ammonium nitrate    **C** calcium carbonate    **D** zinc carbonate

**37. O/N 15/P13/Q16**

Three liquids, P, Q and R, are added to a mixture of hydrochloric acid and Universal Indicator solution.

The following observations are made.

**P** the colour of the indicator turns purple.

**Q** the colour of the indicator does not change.

**R** there is effervescence and the indicator turns blue.

What are P, Q and R?

|          | P                         | Q                         | R                         |
|----------|---------------------------|---------------------------|---------------------------|
| <b>A</b> | sodium carbonate solution | water                     | sodium hydroxide solution |
| <b>B</b> | sodium hydroxide solution | water                     | sodium carbonate solution |
| <b>C</b> | water                     | sodium carbonate solution | sodium hydroxide solution |
| <b>D</b> | water                     | sodium hydroxide solution | sodium carbonate solution |

**38. M/J 15/P13/Q18**

Which gas relights a glowing splint?

- A** ammonia    **B** carbon dioxide    **C** hydrogen    **D** oxygen

**39. MAR 15/P12/Q17**

Which statement about aqueous sodium hydroxide is correct?

- A** When it is added to a solution containing sulfate ions, a white precipitate is formed.  
**B** When it is added to a solution of copper(II) ions, a blue precipitate is formed which dissolves in excess to give deep blue solution.  
**C** When it is added to a solution of iron(II) ions, a green precipitate is formed which does not dissolve in excess.  
**D** When it is added to ammonium chloride, a gas is produced which turns blue litmus red.

**40. O/N 14/P13/Q19**

A colourless solution is tested by the following reactions.

Which reaction is **not** characteristic of an acid?

- A** A piece of magnesium ribbon is added. Bubbles are seen and the magnesium disappears.  
**B** A pungent smelling gas is produced when ammonium carbonate is added.  
**C** Copper oxide powder is added and the mixture is warmed. The solution turns blue.  
**D** The solution turns blue litmus red.

**41. M/J 14/P12/Q22, M/J 14/P11/Q22, M/J 14/P13/Q25**

Aqueous sodium hydroxide is added to solid X and the mixture is heated.

A green precipitate is formed and an alkaline gas is given off.

Which ions are present in X?

- A**  $\text{NH}_4^+$  and  $\text{Fe}^{2+}$     **B**  $\text{NH}_4^+$  and  $\text{Fe}^{3+}$     **C**  $\text{OH}^-$  and  $\text{Fe}^{2+}$     **D**  $\text{OH}^-$  and  $\text{Fe}^{3+}$

**42. O/N 13/P12/Q21, O/N 13/P11/Q20**

Compound X is tested and the results are shown in the table.

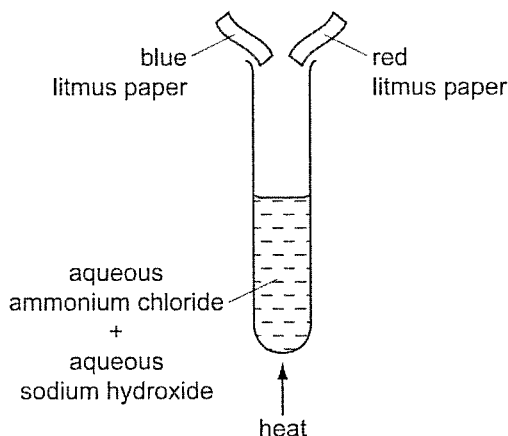
| test                                                  | result                                                   |
|-------------------------------------------------------|----------------------------------------------------------|
| aqueous sodium hydroxide is added, then heated gently | gas given off which turns damp red litmus paper blue     |
| dilute hydrochloric acid is added                     | effervescence, gas given off which turns limewater milky |

Which ions are present in compound X?

- A** ammonium ions and carbonate ions      **C** calcium ions and carbonate ions  
**B** ammonium ions and chloride ions      **D** calcium ions and chloride ions

**43. M/J 13/P12/Q20, M/J 13/P11/Q20, M/J 13/P13/Q20**

The diagram shows an experiment.



What happens to the pieces of litmus paper?

|          | blue litmus paper | red litmus paper |
|----------|-------------------|------------------|
| <b>A</b> | changes colour    | changes colour   |
| <b>B</b> | changes colour    | no colour change |
| <b>C</b> | no colour change  | changes colour   |
| <b>D</b> | no colour change  | no colour change |

**44. O/N 13/P13/Q20**

The cations shown are identified by the colour of the precipitates formed when an excess of an aqueous solution of X is added.

| cations present                 | effect of adding an excess of aqueous X |
|---------------------------------|-----------------------------------------|
| iron(II) ( $\text{Fe}^{2+}$ )   | green precipitate                       |
| copper(II) ( $\text{Cu}^{2+}$ ) | light blue precipitate                  |
| iron(III) ( $\text{Fe}^{3+}$ )  | red-brown precipitate                   |

What is X?

- A** ammonia      **B** limewater      **C** silver nitrate      **D** sodium hydroxide

## Answers Section

| 8.1   |       |  | 8.2   |       |  | 8.3   |       |  | 8.4   |       |  |
|-------|-------|--|-------|-------|--|-------|-------|--|-------|-------|--|
| 1. D  | 41. D |  | 1. D  | 41. B |  | 1. B  | 41. B |  | 1. A  | 41. A |  |
| 2. A  | 42. A |  | 2. A  | 42. A |  | 2. D  | 42. B |  | 2. B  | 42. A |  |
| 3. C  | 43. D |  | 3. B  | 43. A |  | 3. D  | 43. D |  | 3. D  | 43. C |  |
| 4. D  | 44. D |  | 4. A  | 44. C |  | 4. C  | 44. D |  | 4. C  | 44. D |  |
| 5. D  | 45. C |  | 5. A  | 45. C |  | 5. A  | 45. C |  | 5. A  |       |  |
| 6. C  | 46. D |  | 6. D  |       |  | 6. D  | 46. C |  | 6. D  |       |  |
| 7. C  | 47. D |  | 7. B  |       |  | 7. B  | 47. D |  | 7. D  |       |  |
| 8. C  | 48. B |  | 8. A  |       |  | 8. B  | 48. A |  | 8. B  |       |  |
| 9. B  | 49. B |  | 9. B  |       |  | 9. A  | 49. A |  | 9. B  |       |  |
| 10. D | 50. A |  | 10. C |       |  | 10. A | 50. D |  | 10. D |       |  |
| 11. B | 51. C |  | 11. D |       |  | 11. A | 51. C |  | 11. B |       |  |
| 12. A | 52. B |  | 12. A |       |  | 12. C | 52. D |  | 12. A |       |  |
| 13. B | 53. D |  | 13. C |       |  | 13. D | 53. C |  | 13. D |       |  |
| 14. D | 54. B |  | 14. B |       |  | 14. D | 54. C |  | 14. D |       |  |
| 15. C | 55. C |  | 15. C |       |  | 15. D | 55. A |  | 15. C |       |  |
| 16. D | 56. C |  | 16. A |       |  | 16. C | 56. B |  | 16. C |       |  |
| 17. D | 57. B |  | 17. D |       |  | 17. C | 57. B |  | 17. C |       |  |
| 18. A | 58. D |  | 18. B |       |  | 18. B | 58. B |  | 18. D |       |  |
| 19. D | 59. D |  | 19. C |       |  | 19. C | 59. D |  | 19. C |       |  |
| 20. B | 60. A |  | 20. C |       |  | 20. B | 60. A |  | 20. A |       |  |
| 21. B | 61. D |  | 21. C |       |  | 21. D | 61. B |  | 21. A |       |  |
| 22. C | 62. D |  | 22. B |       |  | 22. D | 62. A |  | 22. B |       |  |
| 23. B | 63. B |  | 23. A |       |  | 23. B | 63. B |  | 23. C |       |  |
| 24. D | 64. B |  | 24. B |       |  | 24. D | 64. C |  | 24. D |       |  |
| 25. C | 65. D |  | 25. A |       |  | 25. A | 65. D |  | 25. A |       |  |
| 26. C | 66. B |  | 26. B |       |  | 26. C | 66. B |  | 26. B |       |  |
| 27. C | 67. B |  | 27. A |       |  | 27. D | 67. A |  | 27. A |       |  |
| 28. C | 68. B |  | 28. A |       |  | 28. C | 68. C |  | 28. D |       |  |
| 29. D | 69. D |  | 29. B |       |  | 29. B | 69. B |  | 29. C |       |  |
| 30. D | 70. D |  | 30. B |       |  | 30. C | 70. D |  | 30. C |       |  |
| 31. A | 71. C |  | 31. D |       |  | 31. C | 71. D |  | 31. D |       |  |
| 32. B | 72. B |  | 32. D |       |  | 32. C |       |  | 32. A |       |  |
| 33. D | 73. A |  | 33. D |       |  | 33. C |       |  | 33. D |       |  |
| 34. D |       |  | 34. C |       |  | 34. B |       |  | 34. C |       |  |
| 35. A |       |  | 35. D |       |  | 35. B |       |  | 35. B |       |  |
| 36. C |       |  | 36. C |       |  | 36. A |       |  | 36. D |       |  |
| 37. A |       |  | 37. B |       |  | 37. A |       |  | 37. B |       |  |
| 38. C |       |  | 38. C |       |  | 38. A |       |  | 38. D |       |  |
| 39. B |       |  | 39. A |       |  | 39. B |       |  | 39. C |       |  |
| 40. C |       |  | 40. B |       |  | 40. D |       |  | 40. B |       |  |

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## Unit 9

# THE PERIODIC TABLE

IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 9.1 The periodic table
- 9.2 Alkali metals and halogens
- 9.3 Transition elements
- 9.4 Noble gases

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## Unit 9: The Periodic Table

## 9.1: The periodic table

1. O/N 22/P22/Q21

Strontium displaces magnesium from molten magnesium chloride.

Bromine displaces iodine from aqueous potassium iodide.

Which row describes the change in reactivity down both Group II and Group VII of the Periodic Table?

|          | reactivity down the group |           |
|----------|---------------------------|-----------|
|          | Group II                  | Group VII |
| <b>A</b> | decreases                 | decreases |
| <b>B</b> | decreases                 | increases |
| <b>C</b> | increases                 | decreases |
| <b>D</b> | increases                 | increases |

2. O/N 22/P22/Q22

Elements J and K are in the same period in the Periodic Table.

J reacts with acids to produce a salt and hydrogen.

K reacts with sodium to form an ionic compound.

Which statement about J and K is correct?

- A** An atom of J has more electrons than an atom of K.  
**B** J and K are both metals.  
**C** J and K are both non-metals.  
**D** J is to the left of K in the Periodic Table.

3. O/N 22/P22/Q23

Part of the Periodic Table is shown.

Which element has a high density, a high melting point and forms a brown oxide?

A simplified periodic table with 18 columns and 4 rows. The first column is labeled 'A' in the top cell. The second column is empty. The third column is empty. The fourth column is empty. The fifth column is empty. The sixth column is empty. The seventh column is empty. The eighth column is empty. The ninth column is empty. The tenth column is empty. The eleventh column is empty. The twelfth column is empty. The thirteenth column is empty. The fourteenth column is empty. The fifteenth column is empty. The sixteenth column is empty. The seventeenth column is empty. The eighteenth column is empty. The label 'B' is in the fifth column, third row. The label 'C' is in the second column, fourth row. The label 'D' is in the thirteenth column, second row. There is an empty box above the fifth column, first row.

4. O/N 22/P23/Q21

Elements E and F are in Group I of the Periodic Table.

E has a higher melting point than F.

Elements J and L are in Group VII of the Periodic Table.

J has a higher density than L.

Which elements have the highest atomic numbers in each group?

- A** E and J                      **B** E and L                      **C** F and J                      **D** F and L

5. M/J 22/P21/Q22

Element M forms both  $M^+$  and  $M^{2+}$  ions.

In which part of the Periodic Table is M placed?

- A** Group I                      **B** Group II                      **C** Group III                      **D** transition elements

## 6. M/J 22/P22/Q21, M/J 22/P21/Q21, M/J 22/P23/Q21

Elements in Group IV of the Periodic Table are shown.

carbon  
silicon  
germanium  
tin  
lead

What does **not** occur in Group IV as it is descended?

- A The proton number of the elements increases.  
B The elements become more metallic.  
C The elements have more electrons in their outer shell.  
D The elements have more electron shells.

## 7. M/J 22/P23/Q22

W, X, Y and Z are elements in Period 3 of the Periodic Table.

The numbers of outer-shell electrons in an atom of each element are shown.

| element | number of<br>outer-shell electrons |
|---------|------------------------------------|
| W       | 1                                  |
| X       | 2                                  |
| Y       | 7                                  |
| Z       | 8                                  |

Which elements are non-metals?

- A W, X and Y      B W and X only      C Y and Z      D Z only

## 8. MAR 22/P22/Q24

Three properties of element X are listed.

- It contains atoms with a full outer shell of electrons.
- It is monoatomic.
- It is unreactive.

In which part of the Periodic Table is the element placed?

- A Group I      B Group VII      C Group VIII      D transition elements

## 9. O/N 21/P21/Q22

Group VII elements show trends in their physical properties going down the group.

| element  | X    | Y   | Z     |
|----------|------|-----|-------|
| chlorine | -101 | -34 | 0.003 |
| bromine  | -7   | 59  | 3.1   |
| iodine   | 114  | 184 | 4.9   |

Which row shows the missing headings for the properties in the table?

|   | X                            | Y                            | Z                            |
|---|------------------------------|------------------------------|------------------------------|
| A | density in g/cm <sup>3</sup> | boiling point in °C          | melting point in °C          |
| B | melting point in °C          | boiling point in °C          | density in g/cm <sup>3</sup> |
| C | boiling point in °C          | density in g/cm <sup>3</sup> | melting point in °C          |
| D | boiling point in °C          | melting point in °C          | density in g/cm <sup>3</sup> |

10. O/N 21/P21/Q20

Moving from right to left across the Periodic Table the elements show increasing metallic character. Why does metallic character increase from right to left across a period?

- A The atoms have more electrons in their outer shells.  
B The atoms more readily gain electrons to form negative ions.  
C The atoms more readily lose electrons to form positive ions.  
D The charge on the nucleus of each atom gets larger.

**11. O/N 21/P21/Q23**

Some properties of two metals, G and H, are shown.

| metal G                                                        | metal H                                                 |
|----------------------------------------------------------------|---------------------------------------------------------|
| the formula of the chloride is $GCl$<br>reacts with cold water | high melting point<br>has more than one oxidation state |

Which row about metals G and H is correct?

|          | metal G                           | metal H                           |
|----------|-----------------------------------|-----------------------------------|
| <b>A</b> | in Group I of the Periodic Table  | in Group II of the Periodic Table |
| <b>B</b> | in Group I of the Periodic Table  | transition metal                  |
| <b>C</b> | in Group II of the Periodic Table | in Group I of the Periodic Table  |
| <b>D</b> | in Group II of the Periodic Table | transition metal                  |

12. O/N 21/P22/Q21,O/N 21/P21/Q21,O/N 21/P23/Q21

A period of the Periodic Table is shown.

|         |   |    |     |    |   |    |     |      |
|---------|---|----|-----|----|---|----|-----|------|
| group   | I | II | III | IV | V | VI | VII | VIII |
| element | R | S  | T   | V  | W | X  | Y   | Z    |

The letters are not their chemical symbols.

Which statement is correct?

- A Element R does not conduct electricity.  
B Elements R and Y react together to form an ionic compound.  
C Element Z exists as a diatomic molecule.  
D Element Z reacts with element T.

13. O/N 21/P23/Q22

Part of the Periodic Table is shown.

[illegible]

Which pairs of the elements J, K, L, M and N react together to form a product with a 1 : 1 ratio?

- |          |         |         |          |         |         |
|----------|---------|---------|----------|---------|---------|
| <b>A</b> | J and L | K and M | <b>C</b> | J and N | K and L |
| <b>B</b> | J and M | K and N | <b>D</b> | J and N | K and M |

14. M/J 21/P22/Q23

The diagram shows the positions of elements E, F, G and H in the Periodic Table.

[illegible]

Which statements about elements E, F, G and H are correct?

- 1 E has a higher density than F.
- 2 E has a higher melting point than F.
- 3 G has a darker colour than H.
- 4 G has a lower melting point than H.

A 1 and 3

**B** 1 and 4

C 2 and 3

D 2 and 4

15. MAR 21/P22/Q19

Part of the Periodic Table is shown.

Which element forms an acidic oxide?

[illegible]

16. O/N 20/P21/Q26

The positions of four elements in the Periodic Table are shown. Which element is a gas that displaces iodine from sodium iodide?

A blank periodic table grid is shown. The grid is composed of squares representing elements. The following positions are labeled:

- A**: Located in the first column (Group 1), third row from the top (Period 3).
- B**: Located in the sixth column (Group 6), top row (Period 1).
- C**: Located in the eleventh column (Group 11), second row from the top (Period 2).
- D**: Located in the twelfth column (Group 12), second row from the top (Period 2).

The rest of the grid is empty.

17. O/N 20/P22/Q25

Which row about elements in the Periodic Table is correct?

|          | statement 1                                                      | statement 2                          |
|----------|------------------------------------------------------------------|--------------------------------------|
| <b>A</b> | two elements in the same group have similar chemical properties  | metals are on the left of the table  |
| <b>B</b> | two elements in the same group have similar chemical properties  | metals are on the right of the table |
| <b>C</b> | two elements in the same period have similar chemical properties | metals are on the left of the table  |
| <b>D</b> | two elements in the same period have similar chemical properties | metals are on the right of the table |



**24. O/N 17/P22/Q21**

Which statement about nitrogen and phosphorus is **not** correct?

- A Both are in the same group of the Periodic Table.  
 B Both are in the same period of the Periodic Table.  
 C Both are non-metals.  
 D Both have the same number of electrons in their outer shell.

**25. O/N 17/P21/Q22**

Astatine is an element in Group VII of the Periodic Table.

Astatine is .....1..... reactive than iodine.

The melting point of astatine is .....2..... than the melting point of iodine.

Astatine is .....3..... in colour than bromine.

Which words complete gaps 1, 2 and 3?

|   | 1    | 2      | 3       |
|---|------|--------|---------|
| A | less | higher | darker  |
| B | less | lower  | lighter |
| C | more | higher | darker  |
| D | more | lower  | lighter |

**26. O/N 17/P21/Q21**

Which statements about the trends across a period of the Periodic Table are correct?

- 1 Aluminium is more metallic than sodium.  
 2 Beryllium is more metallic than carbon.  
 3 Boron is more metallic than lithium.  
 4 Magnesium is more metallic than silicon.

- A 1 and 2                      B 1 and 3                      C 2 and 4                      D 3 and 4

**27. M/J 17/P22/Q22, M/J 17/P23/Q22, M/J 17/P21/Q22**

Which element is less reactive than the other members of its group in the Periodic Table?

- A astatine                      B caesium                      C fluorine                      D rubidium

**28. MAR 18/P22/Q21, O/N 15/P11/Q20**

The Periodic Table lists all the known elements.

Elements are arranged in order of ..... 1 ..... number.

The melting points of Group I elements ..... 2 ..... down the group.

The melting points of Group VII elements ..... 3 ..... down the group.

Which words correctly complete gaps 1, 2 and 3?

|   | 1       | 2        | 3        |
|---|---------|----------|----------|
| A | nucleon | decrease | increase |
| B | nucleon | increase | decrease |
| C | proton  | decrease | increase |
| D | proton  | increase | decrease |

**29. MAR 17/P22/Q22, M/J 13/P11/Q22, M/J 13/P13/Q23**

Which property of elements increases across a period of the Periodic Table?

- A metallic character                      C number of outer shell electrons  
 B number of electron shells                      D tendency to form positive ions

30. O/N 16/P22/Q24

Part of the Periodic Table is shown.

Which element has two electrons in its outer shell and three electron shells?

A blank periodic table is shown. Above the noble gases (Group 18), there is a single box. In the top-left corner, there is a 2x2 grid of boxes. The top-left box is labeled 'A', the top-right box is labeled 'B', and the bottom-right box is labeled 'C'. In the top-right corner, there is a 2x2 grid of boxes. The bottom-right box of this grid is labeled 'D'. The rest of the periodic table is empty.

31. M/J 16/P21/Q21, M/J 16/P22/Q21, M/J 16/P23/Q21

Where in the Periodic Table is the metallic character of the elements greatest?

|          | left or right<br>side of a period | at the top or bottom<br>of a group |
|----------|-----------------------------------|------------------------------------|
| <b>A</b> | left                              | bottom                             |
| <b>B</b> | left                              | top                                |
| <b>C</b> | right                             | bottom                             |
| <b>D</b> | right                             | top                                |

32. MAR 16/P22/Q22, M/J 10/P12/Q27, M/J 10/P11/Q25, M/J 10/P13/Q27

An element does not conduct electricity and exists as diatomic molecules.

Where in the Periodic Table is the element found?

The diagram shows a cross-section of a building with four vertical sections labeled A, B, C, and D. A horizontal line connects the top of section A to the top of section C. Section B is the central part of the building, and section D is the rightmost part.

**33. O/N 15/P12/Q22**

The diagram shows elements W, X, Y and Z in a section of the Periodic Table.

Which statement about the reactivity of the elements is correct?

- A X is more reactive than Y, and W is more reactive than Z.  
B X is more reactive than Y, and Z is more reactive than W.  
C Y is more reactive than X, and W is more reactive than Z.  
D Y is more reactive than X, and Z is more reactive than W.



34. O/N 15/P13/Q20

J and K are two elements from the same period in the Periodic Table. The table gives some properties of J and K.

|                                    | J          | K           |
|------------------------------------|------------|-------------|
| appearance                         | shiny grey | dull yellow |
| electrical conductivity when solid | good       | poor        |
| malleability                       | malleable  | brittle     |

Which statement about J and K is correct?

- Which statement about J and K is correct?
- A J forms an acidic oxide.
  - B J is found to the left of K in the Periodic Table.
  - C K forms positive ions when it reacts.
  - D K is more metallic than J.

35. M/J 15/P12/Q19

The positions of four elements in the Periodic Table are shown. Which element does **not** form a compound with chlorine?

A blank periodic table grid is shown. The grid is composed of 18 columns and 7 rows. The columns are numbered 1 through 18 from left to right. The rows are labeled A through G from top to bottom. The grid is divided into four main sections by the noble gases (columns 2, 10, and 18). The sections are:
 

- Section 1: Columns 1 and 2, rows A through G.
- Section 2: Columns 3 through 10, rows A through G.
- Section 3: Columns 11 through 18, rows A through G.
- Section 4: Column 19, rows A through G.

 The labels A, B, C, and D are placed in the following positions:
 

- A: Row A, Column 1.
- B: Row B, Column 2.
- C: Row C, Column 10.
- D: Row D, Column 19.

36. M/J 15/P12/Q22, M/J 15/P11/Q22, M/J 15/P13/Q22

In the outline of the Periodic Table below, some elements are shown as numbers.

[illegible]

Which two numbers are **metals** in the same period?

- A** 1 and 2                      **B** 1 and 7                      **C** 3 and 5                      **D** 5 and 6

37. M/J 15/P11/Q21

The following statements are about elements in the Periodic Table.

- 1 Their atoms have a full outer shell of electrons.
- 2 They form basic oxides.
- 3 They are found in Group 0.
- 4 They are present in small quantities in the air.

Which statements are correct for the noble gases?

- A** 1, 2 and 3      **B** 1, 2 and 4      **C** 1, 3 and 4      **D** 2, 3 and 4

38. MAR 15/P12/Q20

The diagram shows a section of the Periodic Table.

Which element is described below?

'A colourless, unreactive gas that is denser than air.'

A simplified periodic table with 18 columns and 4 rows. The first two columns are on the left, and the last two are on the right. The middle 14 columns are connected by a horizontal line. A box labeled 'A' is in the top right corner. A box labeled 'B' is in the second row, 16th column. A box labeled 'C' is in the third row, 17th column. A box labeled 'D' is in the fourth row, 18th column.

39. M/J 14/P12/Q21

Element X is a non-metal.

In which position of the Periodic Table could element X be found?

- A** at the bottom of Group I  
**B** at the top of Group 0  
**C** at the top of Group I  
**D** in the transition elements

40. M/J 14/P11/Q7, M/J 14/P13/Q10

Element X is in Group I of the Periodic Table. X reacts with element Y to form an ionic compound.

Which equation shows the process that takes place when X forms ions?

- A**  $X + e^- \rightarrow X^+$       **B**  $X - e^- \rightarrow X^-$       **C**  $X + e^- \rightarrow X^-$       **D**  $X - e^- \rightarrow X^+$

41. M/J 14/P11/Q23, M/J 14/P13/Q24

Which statement about the Periodic Table is correct?

- Which statement about the periodic table is correct?
- A** Elements in the same period have the same number of outer electrons.
  - B** The elements on the left are usually gases.
  - C** The most metallic elements are on the left.
  - D** The relative atomic mass of the elements increases from right to left.

42. O/N 13/P12/Q20, O/N 13/P11/Q21, O/N 13/P13/Q21

Calcium, on the left of Period 4 of the Periodic Table, is more metallic than bromine on the right of this period.

Why is this?

Calcium has

- A** fewer electrons.  
**B** fewer protons.  
**C** fewer full shells of electrons.  
**D** fewer outer shell electrons.

43. M/J 13/P12/Q22

W, X, Y and Z are elements in the same period in the Periodic Table.

W and Y are metals. X and Z are non-metals.

Which shows the correct order of these elements across the period?

- A**
- |   |  |   |  |   |  |   |  |
|---|--|---|--|---|--|---|--|
| W |  | X |  | Y |  | Z |  |
|---|--|---|--|---|--|---|--|
- B**
- |   |   |  |  |  |   |  |   |
|---|---|--|--|--|---|--|---|
| X | Z |  |  |  | W |  | Y |
|---|---|--|--|--|---|--|---|
- C**
- |   |  |  |  |  |   |   |   |
|---|--|--|--|--|---|---|---|
| Y |  |  |  |  | W | X | Z |
|---|--|--|--|--|---|---|---|
- D**
- |   |  |   |  |  |  |   |   |
|---|--|---|--|--|--|---|---|
| W |  | Y |  |  |  | X | Z |
|---|--|---|--|--|--|---|---|

## 9.2: Alkali metals and halogens

### 1. O/N 22/P21/Q21

Group II elements show the same trends as Group I elements.  
Which statement about elements in Group II is correct?

- A The melting point of barium is higher than the melting point of calcium.
- B Barium is more reactive than beryllium.
- C Strontium would not react with oxygen.
- D Magnesium is more dense than barium.

### 2. O/N 22/P21/Q22

Some information about properties of Group I elements is shown.

| element   | melting point / °C | density in g/cm <sup>3</sup> |
|-----------|--------------------|------------------------------|
| lithium   | 181                | 0.53                         |
| sodium    | 98                 | 0.97                         |
| potassium | X                  |                              |
| rubidium  | Y                  | Z                            |

What are the values for X, Y and Z?

|   | X  | Y   | Z    |
|---|----|-----|------|
| A | 63 | 252 | 0.26 |
| B | 63 | 39  | 0.26 |
| C | 39 | 63  | 1.53 |
| D | 63 | 39  | 1.53 |

### 3. O/N 21/P22/Q12

Chlorine gas is bubbled into aqueous potassium iodide.  
What is the ionic equation for the reaction that takes place?

- A  $\text{Cl} + \text{I}^- \rightarrow \text{Cl}^- + \text{I}$
- B  $\text{Cl}_2 + 2\text{I}^- \rightarrow \text{Cl}_2^- + \text{I}_2$
- C  $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$
- D  $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + 2\text{I}$

### 4. O/N 21/P23/Q6

A Group I element combines with a Group VII element and forms an ionic bond.  
Which row shows how the electronic structures change?

|   | Group I element |               | Group VII element |               |
|---|-----------------|---------------|-------------------|---------------|
|   | before bonding  | after bonding | before bonding    | after bonding |
| A | 2,8,1           | 2,8,2         | 2,7               | 2,6           |
| B | 2,8             | 2,7           | 2,8               | 2,8,1         |
| C | 2,8,1           | 2,8           | 2,7               | 2,8           |
| D | 2,8             | 2,8,1         | 2,8               | 2,7           |

**5. M/J 21/P22/Q7**

Chemical compounds formed from a Group I element and a Group VII element contain ionic bonds. How are the ionic bonds formed?

- A Electrons are transferred from Group VII atoms to Group I atoms.
- B Electrons are shared between Group I atoms and Group VII atoms.
- C Electrons are lost by Group I atoms and Group VII atoms.
- D Electrons are transferred from Group I atoms to Group VII atoms.

**6. M/J 21/P22/Q22**

Which statement about the trends shown by the elements of Period 3 in the Periodic Table is **not** correct?

- A The elements become less metallic across the period.
- B The group number increases across the period.
- C The number of electron shells increases across the period.
- D The number of outer electrons increases across the period.

**7. M/J 21/P23/Q22**

Helium and neon exist as monoatomic gases at room temperature and pressure.

statement 1 Helium and neon have eight electrons in their outer shell.

statement 2 Helium and neon are unreactive.

Which option is correct?

- A Statement 1 and statement 2 are incorrect.
- B Statement 1 is correct and explains statement 2.
- C Statement 1 is correct, but does not explain statement 2.
- D Statement 1 is incorrect, but statement 2 is correct.

**8. O/N 20/P22/Q9, O/N 20/P21/Q9, O/N 20/P23/Q9**

Rubidium is in Group I of the Periodic Table and bromine is in Group VII. Rubidium reacts with bromine to form an ionic compound. Which row shows the electron change taking place for rubidium and the correct formula of the rubidium ion?

|   | electron change | formula of ion formed |
|---|-----------------|-----------------------|
| A | electron gained | $\text{Rb}^+$         |
| B | electron gained | $\text{Rb}^-$         |
| C | electron lost   | $\text{Rb}^+$         |
| D | electron lost   | $\text{Rb}^-$         |

**9. O/N 20/P21/Q27**

A flammable gas needs to be removed from a tank at an industrial plant.

For safety reasons, an inert gas is used.

Which gas is suitable?

- A argon
- B hydrogen
- C methane
- D oxygen

**10. O/N 20/P23/Q21**

Which metal has variable oxidation states?

- A aluminium
- B calcium
- C copper
- D sodium

**11. O/N 20/P23/Q25**

Which statement about the halogens and their compounds is correct?

- A The colour of the element gets lighter going down Group VII.
- B The elements get less dense going down Group VII.
- C When chlorine is added to sodium iodide solution, iodine is formed.
- D When iodine is added to sodium bromide solution, bromine is formed.

**12. O/N 20/P23/Q26**

Elements in Group II of the Periodic Table show the same trends in their reaction with water and their density as Group I.

Which row shows how the properties of barium compare with calcium?

|          | reaction with water | density |
|----------|---------------------|---------|
| <b>A</b> | faster              | higher  |
| <b>B</b> | faster              | lower   |
| <b>C</b> | slower              | higher  |
| <b>D</b> | slower              | lower   |

**13. M/J 20/P22/Q22**

Which statement about Group I and Group VII elements is correct?

- A** Group VII elements are monoatomic non-metals.
- B** Lithium is more reactive with water than caesium.
- C** The melting points of Group I metals increase down the group.
- D** Potassium bromide reacts with chlorine to produce an orange solution.

**14. M/J 20/P21/Q22**

A Group I metal (lithium, sodium or potassium) is reacted with a Group VII element (chlorine, bromine or iodine). Which compound is formed when the Group I metal of highest density reacts with the Group VII element of lowest density?

- A** lithium chloride
- B** potassium chloride
- C** potassium iodide
- D** lithium iodide

**15. M/J 20/P23/Q6**

Caesium is a metal in Group I of the Periodic Table.

Which description of the bonding in caesium is correct?

- A** electrostatic attraction between oppositely charged ions
- B** electrostatic attraction between positive metal ions and mobile electrons
- C** neighbouring metal atoms sharing pairs of electrons
- D** strong attractive forces between atoms

**16. M/J 20/P23/Q22**

The elements in Group I include lithium, sodium and potassium.

Which statements about these elements are correct?

- 1 Sodium is denser than lithium.
- 2 Lithium has a lower melting point than potassium.
- 3 Potassium is a relatively soft metal.
- 4 Sodium is less reactive than lithium but more reactive than potassium.

- A** 1 and 2
- B** 1 and 3
- C** 2 and 4
- D** 3 and 4

**17. MAR 20/P22/Q4**

Lithium reacts with fluorine to form the compound lithium fluoride. Which statement about this reaction is correct?

- A** Each fluorine atom gains one electron.
- B** Each fluorine atom gains two or more electrons.
- C** Each fluorine atom loses one electron.
- D** Each fluorine atom loses two or more electrons.

18. MAR 20/P22/Q23

The diagram shows elements W, X, Y and Z in a section of the Periodic Table.

[illegible]

Which statement about the reactivity of the elements is correct?

- A** X is more reactive than Y, and W is more reactive than Z.  
**B** X is more reactive than Y, and Z is more reactive than W.  
**C** Y is more reactive than X, and W is more reactive than Z.  
**D** Y is more reactive than X, and Z is more reactive than W.

## 19. MAR 20/P22/Q25

Sodium is a Group I metal. Which property, that is typical of most metals, is **not** shown by sodium?

- A** conductor of heat      **B** high melting point      **C** malleable      **D** shiny

20. O/N 19/P22/Q23

Helium is a noble gas.

Which statement about helium is correct?

- A** It has eight electrons in its outer shell.      **C** It is reactive.
- B** It is a diatomic gas.      **D** It is used for filling balloons.

21. O/N 19/P22/Q24, O/N 19/P21/Q24, O/N 19/P23/Q24

Which pair of elements reacts together most violently?

- A** chlorine and lithium  
**B** chlorine and potassium  
**C** iodine and lithium  
**D** iodine and potassium

22. O/N 19/P22/Q25

Iron(II) ions can be oxidised to iron(III) ions by hydrogen peroxide.

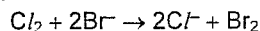
Which statement explains why iron is a transition element?

- A** Iron is a transition element because it can be oxidised.  
**B** Iron is a transition element because it has variable oxidation states.  
**C** Iron is a transition element because it takes part in redox reactions.  
**D** Iron is a transition element because it reacts with chlorine.

23. O/N 19/P23/Q18

Chlorine displaces bromine from aqueous potassium bromide.

The ionic equation for the reaction is shown.



Which statement about this reaction is correct?

- A** Bromide ions act as an oxidising agent.  
**B** Bromide ions are oxidised when electrons are lost.  
**C** Chlorine acts as a reducing agent.  
**D** Chlorine is reduced when electrons are lost.

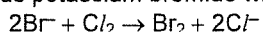
24. O/N 19/P23/Q23

Which statement describes a gas which is in Group VIII of the Periodic Table?

- A A colourless gas that helps substances burn.  
B A pollutant gas present in car exhausts.  
C A gas that is less dense than air and makes a 'pop' sound with a lighted splint.  
D A gas that is used in lamps.

25. M/J 19/P22/Q16

The ionic equation for the reaction of aqueous potassium bromide with chlorine gas is shown.



Which statement is correct?

- A** Bromide ions are oxidised by gaining electrons.  
**B** Bromide ions are oxidised by losing electrons.  
**C** Chlorine is oxidised by gaining electrons.  
**D** Chlorine is oxidised by losing electrons.

**26. M/J 19/P22/Q21**

The melting points and boiling points of the elements of Group I of the Periodic Table are shown.

| element   | melting point<br>/°C | boiling point<br>/°C |
|-----------|----------------------|----------------------|
| lithium   | 181                  | 1330                 |
| sodium    | 98                   | 883                  |
| potassium | 63                   | 759                  |
| rubidium  | 39                   | 688                  |
| caesium   | 28                   | 671                  |

Which pair of elements are liquid at 800 °C?

- A caesium and rubidium  
B potassium and sodium  
C lithium and sodium  
D potassium and caesium

27. M/J 19/P22/Q20

Part of the Periodic Table is shown.

A simplified periodic table with 18 columns and 4 rows. The first column is labeled 'Y' at the bottom. The second column is empty. The third column is labeled 'Z' at the bottom. The fourth column is empty. The fifth column is empty. The sixth column is empty. The seventh column is empty. The eighth column is empty. The ninth column is empty. The tenth column is empty. The eleventh column is empty. The twelfth column is empty. The thirteenth column is empty. The fourteenth column is empty. The fifteenth column is empty. The sixteenth column is empty. The seventeenth column is labeled 'X' at the top. The eighteenth column is empty.

Which row describes the properties of X, Y and Z?

|          | good conductor<br>of electricity | high melting<br>point |
|----------|----------------------------------|-----------------------|
| <b>A</b> | X                                | Z                     |
| <b>B</b> | Y                                | Z and X               |
| <b>C</b> | Y and Z                          | Z                     |
| <b>D</b> | Z and X                          | X                     |

28. M/J 19/P21/Q21

Which statement about elements in Group I and Group VII of the Periodic Table is correct?

- A Bromine reacts with potassium chloride to produce chlorine.  
B Iodine is a monatomic non-metal.  
C Lithium has a higher melting point than potassium.  
D Sodium is more reactive with water than potassium.

**29. M/J 19/P23/Q21**

Which statement about the properties of elements in Group I and in Group VII is correct?

- A Bromine displaces iodine from an aqueous solution of potassium iodide.  
B Chlorine, bromine and iodine are diatomic gases at room temperature.  
C Lithium, sodium and potassium are soft non-metals.  
D Lithium, sodium and potassium have an increasing number of electrons in their outer shells.

## 30. MAR 19/P22/Q24

The elements sodium to argon form Period 3 of the Periodic Table. Which row describes the trend across Period 3 from left to right?

|          | number of outer shell electrons | metallic character | group number |
|----------|---------------------------------|--------------------|--------------|
| <b>A</b> | decreases                       | decreases          | decreases    |
| <b>B</b> | decreases                       | increases          | decreases    |
| <b>C</b> | increases                       | decreases          | increases    |
| <b>D</b> | increases                       | increases          | increases    |

## 31. MAR 19/P22/Q25

Astatine is below iodine in Group VII in the Periodic Table. Which row describes the properties of astatine?

|          | state at room temperature | reactivity                                                 |
|----------|---------------------------|------------------------------------------------------------|
| <b>A</b> | gas                       | displaces chlorine, bromine and iodine                     |
| <b>B</b> | gas                       | displaces iodine but does not displace chlorine or bromine |
| <b>C</b> | solid                     | displaces iodine but does not displace chlorine or bromine |
| <b>D</b> | solid                     | does not displace chlorine, bromine or iodine              |

32. MAR 19/P22/Q26

Which statement explains why elements in Group VIII of the Periodic Table are unreactive?

- Which statement explains why elements in Group 18 are unreactive?
- A They are monatomic gases. C They have a full outer shell of electrons.
- B They form stable diatomic molecules. D They share electrons with each other.

33. O/N 18/P22/Q21, O/N 18/P21/Q21, O/N 18/P23/Q21, O/N 14/P12/Q22, O/N 14/P11/Q23

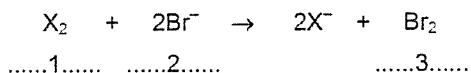
Elements in Group I of the Periodic Table react with water.

Which row describes the products made in the reaction and the trend in reactivity of the elements?

|          | products                     | trend in reactivity          |
|----------|------------------------------|------------------------------|
| <b>A</b> | metal hydroxide and hydrogen | less reactive down the group |
| <b>B</b> | metal hydroxide and hydrogen | more reactive down the group |
| <b>C</b> | metal oxide and hydrogen     | less reactive down the group |
| <b>D</b> | metal oxide and hydrogen     | more reactive down the group |

34. O/N 18/P22/Q22, O/N 18/P21/Q22, O/N 18/P23/Q22, M/J 11/P12/Q18, M/J 11/P11/Q23, M/J 11/P13/Q18

The equation shows the reaction between a halogen and aqueous bromide ions.



Which words complete gaps 1, 2 and 3?

|          | 1        | 2          | 3          |
|----------|----------|------------|------------|
| <b>A</b> | chlorine | brown      | colourless |
| <b>B</b> | chlorine | colourless | brown      |
| <b>C</b> | iodine   | brown      | colourless |
| <b>D</b> | iodine   | colourless | brown      |





**41. MAR 18/P22/Q22**

Metal X reacts with non-metal Y to form an ionic compound with the formula  $X_2Y$ .

Which statements are correct?

- 1 X is in Group I of the Periodic Table.
- 2 X is in Group II of the Periodic Table.
- 3 Y is in Group VI of the Periodic Table.
- 4 Y is in Group VII of the Periodic Table.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**42. MAR 18/P22/Q23, O/N 15/P11/Q19**

Which statements about Group I and Group VII elements are correct?

- 1 In Group I, lithium is more reactive than potassium.
- 2 In Group VII, chlorine is more reactive than fluorine.

|   | statement 1 | statement 2 |
|---|-------------|-------------|
| A | ✓           | ✓           |
| B | ✓           | x           |
| C | x           | ✓           |
| D | x           | x           |

**43. O/N 17/P22/Q22**

Sodium and rubidium are elements in Group I of the Periodic Table.

Which statement is correct?

- A Sodium atoms have more electrons than rubidium atoms.
- B Sodium has a lower density than rubidium.
- C Sodium has a lower melting point than rubidium.
- D Sodium is more reactive than rubidium.

**44. O/N 17/P23/Q21**

A period of the Periodic Table is shown.

| group   | I | II | III | IV | V | VI | VII | VIII |
|---------|---|----|-----|----|---|----|-----|------|
| element | R | S  | T   | V  | W | X  | Y   | Z    |

The letters are not their chemical symbols.

Which statement is correct?

- A Element R does not conduct electricity.
- B Elements R and Y react together to form an ionic compound.
- C Element Z exists as a diatomic molecule.
- D Element Z reacts with element T.

**45. O/N 17/P23/Q22**

Some properties of element X are shown.

|                                  |                             |
|----------------------------------|-----------------------------|
| melting point in °C              | 98                          |
| boiling point in °C              | 883                         |
| reaction with cold water         | gives off $H_2$ gas         |
| reaction when heated with oxygen | burns to give a white solid |

In which part of the Periodic Table is X found?

A Group I                      B Group VII                      C Group VIII                      D transition elements

**46. M/J 17/P22/Q23**

The elements oxygen and sulfur are in the same group of the Periodic Table.

Which statement about oxygen and sulfur is **not** correct?

- A They are non-metals.
- B They have giant covalent structures.
- C They have six electrons in their outer shells.
- D They react together to form an acidic oxide.



**54. M/J 16/P23/Q22**

Which statement about the elements in Group I is correct?

- A Hydrogen is evolved when they react with water.
- B Ions of Group I elements have a  $-1$  charge.
- C Sodium is more reactive than potassium.
- D Solid sodium is a poor electrical conductor.

**55. M/J 16/P23/Q24**

Two statements about noble gases are given.

- 1 Noble gases are reactive, monatomic gases.
- 2 Noble gases all have full outer shells of electrons.

Which is correct?

- A Both statements are correct and statement 2 explains statement 1.
- B Both statements are correct but statement 2 does not explain statement 1.
- C Statement 1 is correct but statement 2 is incorrect.
- D Statement 2 is correct but statement 1 is incorrect.

**56. MAR 16/P22/Q24**

The elements in a group of the Periodic Table show the following trends.

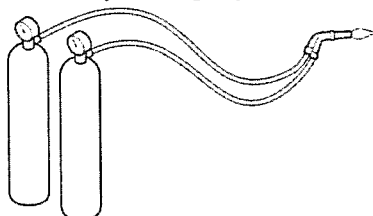
- 1 The element with the lowest proton number has the lowest reactivity.
- 2 All the elements in the group form basic oxides.
- 3 The density of the elements increases down the group.
- 4 The melting point of the elements decreases down the group.

In which group are the elements found?

- A I                                      B IV                                      C VI                                      D VII

**57. O/N 15/P12/Q30**

Metals are welded by using the heat produced by burning a gas in oxygen.



Which gas could **not** be used for this purpose?

- A ethene                                      B hydrogen                                      C helium                                      D methane

**58. O/N 15/P11/Q22**

The Group 0 elements are unreactive.

The gas used to fill balloons is ..... X..... .

This gas is unreactive because it has ..... Y..... electrons in its outermost shell.

Which words correctly complete gaps X and Y?

|   | X      | Y     |
|---|--------|-------|
| A | argon  | eight |
| B | argon  | two   |
| C | helium | eight |
| D | helium | two   |

## 59. O/N 15/P13/Q3

Argon,  $A_r$ , has a higher relative atomic mass than potassium, K, but appears before it in the Periodic Table.

Why is argon listed before potassium in the Periodic Table?

- A Argon has fewer neutrons than potassium. C Argon has more neutrons than potassium.  
B Argon has fewer protons than potassium. D Argon has more protons than potassium.

## 60. O/N 15/P13/Q19

An element, X, is a dark grey crystalline solid at room temperature.

It has a melting point of  $114\text{ }^{\circ}\text{C}$  and a density of  $4.9\text{ g / cm}^3$ .

When heated gently it forms a purple vapour.

Where in the Periodic Table is X found?

## 61. M/J 15/P12/Q21

The table shows some properties of the Group I metals.

| metal     | melting point/ $^{\circ}\text{C}$ | hardness        | reaction with water         |
|-----------|-----------------------------------|-----------------|-----------------------------|
| lithium   | 181                               | moderately soft | steady effervescence        |
| sodium    | 98                                | soft            | vigorous effervescence      |
| potassium | 63                                | very soft       | very vigorous effervescence |
| rubidium  | ?                                 | ?               | ?                           |

What are the properties of rubidium?

- A melts below  $63\text{ }^{\circ}\text{C}$ , very soft, reacts explosively with water  
B melts below  $63\text{ }^{\circ}\text{C}$ , very soft, reacts slowly with water  
C melts above  $181\text{ }^{\circ}\text{C}$ , very soft, reacts explosively with water  
D melts above  $181\text{ }^{\circ}\text{C}$ , very soft, reacts slowly with water

## 62. M/J 15/P11/Q19

Which element is in the same group of the Periodic Table as lithium?

|   | electrical conductivity | density in $\text{g / cm}^3$ |
|---|-------------------------|------------------------------|
| A | high                    | 0.97                         |
| B | high                    | 8.93                         |
| C | low                     | 0.07                         |
| D | low                     | 3.12                         |

63. MAR 15/P12/Q18

Which pair of elements will react together most violently?

- Which pair of elements will react together most readily?
- |                                 |                               |
|---------------------------------|-------------------------------|
| <b>A</b> chlorine and lithium   | <b>C</b> iodine and lithium   |
| <b>B</b> chlorine and potassium | <b>D</b> iodine and potassium |

64. M/J 15/P13/Q19

The noble gases, which are in Group 0 of the Periodic Table, are all very ..... 1.....

..... 2....., one of these gases, is used to provide an inert atmosphere in lamps.

Another, ..... 3....., is used for filling balloons because it is less dense than air.

Which words complete the sentences about noble gases?

|          | 1          | 2      | 3      |
|----------|------------|--------|--------|
| <b>A</b> | reactive   | argon  | helium |
| <b>B</b> | reactive   | helium | argon  |
| <b>C</b> | unreactive | argon  | helium |
| <b>D</b> | unreactive | helium | argon  |

65. M/J 15/P13/Q21

X is a Group I metal.

Y and Z are Group VII elements.

When X reacts with Y a salt is formed. A solution of this salt reacts with Z to form a different salt.

What are X, Y and Z?

|          | X  | Y      | Z      |
|----------|----|--------|--------|
| <b>A</b> | K  | $Cl_2$ | $I_2$  |
| <b>B</b> | Li | $Cl_2$ | $Br_2$ |
| <b>C</b> | Mg | $Br_2$ | $Cl_2$ |
| <b>D</b> | Na | $I_2$  | $Cl_2$ |

66. MAR 15/P12/Q19

The table shows some information about elements in Group VII of the Periodic Table.

| name     | state at room temperature | colour       |
|----------|---------------------------|--------------|
| chlorine | gas                       | yellow-green |
| bromine  | liquid                    | brown        |
| iodine   | ?                         | ?            |
| astatine | solid                     | black        |

Which information about iodine completes the table?

|          | state  | colour |
|----------|--------|--------|
| <b>A</b> | liquid | black  |
| <b>B</b> | liquid | green  |
| <b>C</b> | solid  | grey   |
| <b>D</b> | solid  | yellow |



**73. M/J 13/P12/Q25**

Bromine is in Group VII on the Periodic Table.

Which describes the appearance of bromine at room temperature?

- A grey solid      B purple fumes      C red-brown liquid      D yellow gas

**74. M/J 13/P11/Q25, M/J 13/P13/Q27**

Group I metals are also known as the Alkali Metals.

Which statement about the metals in Group I is **not** correct?

- A In their reactions they lose electrons.  
B Their atoms all have one electron in their outer shell.  
C They form +1 ions in their reactions with non-metals.  
D They form covalent compounds by sharing electrons



## 9.3: Transition elements

### 1. O/N 22/P21/Q23

Which statements describe properties of transition elements?

- 1 They form coloured compounds.
- 2 They have variable oxidation states.
- 3 They have low densities.
- 4 They are volatile.

A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4

### 2. M/J 22/P22/Q23

Which elements have both a high melting point and variable oxidation states?

A alkali metals              B transition elements      C halogens                      D noble gases

### 3. M/J 22/P23/Q24

Which row describes the properties of a typical transition element?

|   | melting point | density | used as catalyst |
|---|---------------|---------|------------------|
| A | high          | high    | yes              |
| B | high          | low     | no               |
| C | low           | high    | yes              |
| D | low           | low     | no               |

### 4. O/N 21/P22/Q23

Which row describes properties of transition elements?

|   | property 1         | property 2         | property 3                |
|---|--------------------|--------------------|---------------------------|
| A | coloured compounds | high density       | variable oxidation states |
| B | high density       | high melting point | one oxidation state       |
| C | high melting point | coloured compounds | one oxidation state       |
| D | low melting point  | high density       | variable oxidation states |

### 5. O/N 21/P23/Q23

Which property is shown by transition metals but **not** shown by Group I metals?

- A good electrical conductivity                      C loss of electrons to form positive ions  
B good thermal conductivity                      D variable oxidation states

### 6. M/J 21/P21/Q23

Which elements in the table are transition elements?

| element | property                      |
|---------|-------------------------------|
| E       | forms $E^{3+}$ ions only      |
| F       | forms $F^+$ and $F^{2+}$ ions |
| G       | forms only white salts        |
| H       | used in catalytic converters  |

A E and G                      B E and H                      C F and G                      D F and G

7. M/J 21/P23/Q21

Which property is shown by transition elements?

- A low density  
B low melting point  
C variable oxidation state  
D white compounds

8. MAR 21/P22/Q28

The properties of an element are listed.

Its melting point is  $3414^{\circ}\text{C}$ .

Some of its compounds are catalysts.

It has variable oxidation states.

Where is the element found in the Periodic Table?

- A alkali metals      B halogens      C noble gases      D transition elements

9. O/N 20/P22/Q28

Transition elements can have variable oxidation states. Which pair of compounds shows a transition element in two different oxidation states?

- A**  $\text{Cr}_2\text{O}_3$  and  $\text{Cr}_2(\text{SO}_4)_3$       **C**  $\text{ZnS}$  and  $\text{ZnSO}_4$   
**B**  $\text{Cu}_2\text{O}$  and  $\text{CuCO}_3$       **D**  $\text{NiO}$  and  $\text{Ni}(\text{NO}_3)_2$

10. M/J 20/P22/Q23, M/J 20/P21/Q23, M/J 20/P23/Q23

The properties of the element titanium, Ti, can be predicted from its position in the Periodic Table.

Which row identifies the properties of titanium?

|          | can be used<br>as a catalyst | conducts electricity<br>when solid | has low density | forms coloured<br>compounds |
|----------|------------------------------|------------------------------------|-----------------|-----------------------------|
| <b>A</b> | ✓                            | ✓                                  | ✓               | x                           |
| <b>B</b> | ✓                            | ✓                                  | x               | ✓                           |
| <b>C</b> | ✓                            | x                                  | ✓               | ✓                           |
| <b>D</b> | x                            | ✓                                  | ✓               | ✓                           |

## 11. MAR 20/P22/Q24

Some properties of substances are listed.

- 1 They conduct electricity.
- 2 They have low densities.
- 3 They have high melting points.
- 4 They are malleable.

Which properties are shown by transition metals?

- A** 1 and 3 only      **B** 1 and 4 only      **C** 1, 2 and 3      **D** 1, 3 and 4

12. O/N 19/P21/Q25

Which pair of compounds shows that transition elements have variable oxidation states?

- A**  $\text{Cr}_2\text{O}_3$  and  $\text{CrBr}_3$       **B**  $\text{CuSO}_4$  and  $\text{CuCl}_2$       **C**  $\text{Fe}_2\text{O}_3$  and  $\text{FeCl}_2$       **D**  $\text{NiO}$  and  $\text{NiCl}_2$

13. O/N 19/P23/Q25

Iron reacts with dilute hydrochloric acid to form iron(II) chloride,  $\text{FeCl}_2$ . Iron reacts with chlorine to form iron(III) chloride,  $\text{FeCl}_3$ .

Which property of transition elements is shown by this information?

- A** Transition elements have high melting points.
- B** Transition elements can act as catalysts.
- C** Transition elements have variable oxidation states.
- D** Transition elements have coloured compounds.

**14. M/J 18/P22/Q23**

Which row describes the properties of a transition element?

|   | property 1                 | property 2                  |
|---|----------------------------|-----------------------------|
| A | forms colourless compounds | acts as a catalyst          |
| B | forms colourless compounds | low electrical conductivity |
| C | high density               | acts as a catalyst          |
| D | high density               | low electrical conductivity |

**15. M/J 18/P21/Q23**

Which row shows the catalytic activity of transition elements and their compounds?

|   | catalytic activity of transition elements | catalytic activity of compounds of transition elements |
|---|-------------------------------------------|--------------------------------------------------------|
| A | good                                      | good                                                   |
| B | good                                      | poor                                                   |
| C | poor                                      | good                                                   |
| D | poor                                      | poor                                                   |

**16. M/J 18/P23/Q23**

Which row describes a typical transition element?

|   | density in g/cm <sup>3</sup> | melting point in °C | boiling point in °C | colour of oxide |
|---|------------------------------|---------------------|---------------------|-----------------|
| A | 0.97                         | 98                  | 883                 | white           |
| B | 2.64                         | 769                 | 1382                | white           |
| C | 3.10                         | -7                  | 59                  | yellow          |
| D | 8.96                         | 1085                | 2562                | red             |

**17. O/N 17/P22/Q23**

Which properties do the elements chromium, iron and vanadium have in common?

- 1 They all conduct electricity.  
 2 They, or their compounds, can act as catalysts.  
 3 They all form coloured compounds.
- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

**18. O/N 17/P22/Q25, O/N 17/P21/Q25, O/N 17/P23/Q25**What is a property of **all** metals?

- A conduct electricity      B hard      C low melting points      D react with water

**19. O/N 17/P21/Q23**

Which row describes the properties of a typical transition element?

|   | melting point | forms coloured compounds | can act as a catalyst |
|---|---------------|--------------------------|-----------------------|
| A | high          | no                       | no                    |
| B | high          | yes                      | yes                   |
| C | low           | no                       | yes                   |
| D | low           | yes                      | no                    |



**24. O/N 16/P22/Q28**

The alloy brass is a mixture of copper and another metal.  
Brass is used to make the pins of electrical plugs.  
Copper is used to make electrical wiring.  
Which row about brass is correct?

|          | hardness           | electrical conductivity | other metal |
|----------|--------------------|-------------------------|-------------|
| <b>A</b> | harder than copper | better than copper      | tin         |
| <b>B</b> | harder than copper | worse than copper       | zinc        |
| <b>C</b> | softer than copper | better than copper      | tin         |
| <b>D</b> | softer than copper | worse than copper       | zinc        |

**25. O/N 16/P21/Q24**

Which statement about transition elements and their compounds is correct?

- A** All the transition elements have an oxidation state of +2 only.  
**B** Aqueous solutions of the salts of transition elements are generally coloured.  
**C** Transition elements change from metal to non-metal across the period.  
**D** Transition elements can act as catalysts but their compounds cannot.

**26. M/J 16/P22/Q23**

The table gives information about four elements, P, Q, R and S.

|   | melting point in °C | electrical conductivity of element when solid | density in g/cm <sup>3</sup> | colour of iodide of element |
|---|---------------------|-----------------------------------------------|------------------------------|-----------------------------|
| P | 98                  | good                                          | 0.97                         | white                       |
| Q | -39                 | good                                          | 13.53                        | red                         |
| R | 1410                | poor                                          | 2.33                         | colourless                  |
| S | 1535                | good                                          | 7.87                         | green                       |

Which elements could be transition elements?

- A** P, Q and S      **B** Q and S only      **C** R and S only      **D** S only

**27. M/J 16/P21/Q23**

Which of the following could be a transition element?

|          | melting point in °C | density in g/cm <sup>3</sup> | colour | electrical conductor |
|----------|---------------------|------------------------------|--------|----------------------|
| <b>A</b> | 114                 | 4.9                          | purple | no                   |
| <b>B</b> | 659                 | 2.7                          | grey   | yes                  |
| <b>C</b> | 1677                | 4.5                          | grey   | yes                  |
| <b>D</b> | 3727                | 2.3                          | black  | yes                  |

**28. M/J 16/P21/Q24**

Two statements about argon are given.

- 1 Argon has a full outer shell of electrons.  
 2 Argon is very reactive and is used in lamps.

Which is correct?

- A** Both statements are correct and statement 2 explains statement 1.  
**B** Both statements are correct but statement 2 does not explain statement 1.  
**C** Statement 1 is correct but statement 2 is incorrect.  
**D** Statement 2 is correct but statement 1 is incorrect.

**29. M/J 16/P23/Q23**

Osmium is a transition element.

Which row gives the expected properties of osmium?

|          | melting point | density | compounds formed |
|----------|---------------|---------|------------------|
| <b>A</b> | high          | high    | coloured         |
| <b>B</b> | high          | high    | white            |
| <b>C</b> | high          | low     | white            |
| <b>D</b> | low           | high    | coloured         |

**30. O/N 15/P12/Q20**

The table shows the symbols of three metals with names that begin with the letter C.

Which row correctly shows the melting point of the metals?

|          | Co   | Cr   | Cs   |
|----------|------|------|------|
| <b>A</b> | high | high | high |
| <b>B</b> | high | high | low  |
| <b>C</b> | low  | low  | high |
| <b>D</b> | low  | low  | low  |

**31. O/N 15/P12/Q13, O/N 15/P11/Q13, O/N 15/P13/Q13, M/J 12/P11/Q14, M/J 12/P13/Q15**

The element vanadium, V, forms several oxides.

In which change is oxidation taking place?

- A**  $\text{VO}_2 \rightarrow \text{V}_2\text{O}_3$       **B**  $\text{V}_2\text{O}_5 \rightarrow \text{VO}_2$       **C**  $\text{V}_2\text{O}_3 \rightarrow \text{VO}$       **D**  $\text{V}_2\text{O}_3 \rightarrow \text{V}_2\text{O}_5$

**32. O/N 15/P12/Q21, O/N 15/P11/Q21, O/N 15/P13/Q21**

The table gives information about four elements.

Which element is a transition metal?

|          | electrical conductivity | density in $\text{g/cm}^3$ | melting point in $^\circ\text{C}$ |
|----------|-------------------------|----------------------------|-----------------------------------|
| <b>A</b> | good                    | 0.97                       | 98                                |
| <b>B</b> | good                    | 7.86                       | 1535                              |
| <b>C</b> | poor                    | 2.33                       | 1410                              |
| <b>D</b> | poor                    | 3.12                       | -7                                |

**33. M/J 15/P12/Q20, M/J 15/P11/Q20, M/J 15/P13/Q20, M/J 12/P11/Q21, M/J 12/P13/Q19**

Which properties of the element titanium, Ti, can be predicted from its position in the Periodic Table?

|          | can be used as a catalyst | conducts electricity when solid | has low density | forms coloured compounds |
|----------|---------------------------|---------------------------------|-----------------|--------------------------|
| <b>A</b> | ✓                         | ✓                               | ✓               | x                        |
| <b>B</b> | ✓                         | ✓                               | x               | ✓                        |
| <b>C</b> | ✓                         | x                               | ✓               | ✓                        |
| <b>D</b> | x                         | ✓                               | ✓               | ✓                        |



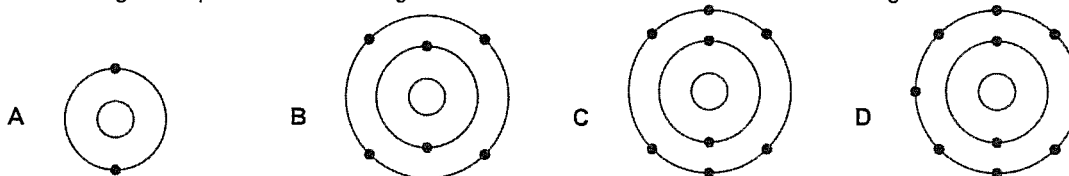




## 9.4: Noble gases

### 1. M/J 22/P21/Q25

Which diagram represents the arrangement of the outer-shell electrons of a noble gas?



### 2. M/J 22/P23/Q25

Which row describes an atom of a noble gas?

|   | number of protons | number of neutrons | number of electrons |
|---|-------------------|--------------------|---------------------|
| A | 2                 | 2                  | 0                   |
| B | 2                 | 2                  | 2                   |
| C | 8                 | 8                  | 8                   |
| D | 8                 | 8                  | 10                  |

### 3. O/N 21/P22/Q24, O/N 21/P21/Q24, O/N 21/P23/Q24

The noble gases are in Group VIII of the Periodic Table.

Which statement explains why noble gases are unreactive?

- A They all have eight electrons in their outer shells.
- B They all have full outer shells.
- C They are all gases.
- D They are all monoatomic.

### 4. M/J 21/P21/Q22

Why is helium used to fill balloons?

- A Helium is monoatomic.
- B Helium is in Group VIII of the Periodic Table.
- C Helium has a full outer electron shell.
- D Helium is less dense than air.

### 5. MAR 21/P22/Q6

Lithium and fluorine react to form lithium fluoride.

A student writes three statements about the reaction.

- 1 Lithium atoms lose an electron when they react.
- 2 Each fluoride ion has one more electron than a fluorine atom.
- 3 Lithium fluoride is a mixture of elements.

Which statements are correct?

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

### 6. O/N 20/P22/Q27

A flammable gas needs to be removed from a tank at an industrial plant. For safety reasons, an inert gas is used.

Which gas is suitable?

- A argon
- B hydrogen
- C methane
- D oxygen

### 7. O/N 20/P23/Q20

Period 3 of the Periodic Table contains the elements sodium to argon.

Element Q is a non-metal from this period.

Which statement about Q is correct?

- A It conducts electricity.
- B It has a lower proton number than sodium.
- C It has electrons in only three shells.
- D It is malleable.

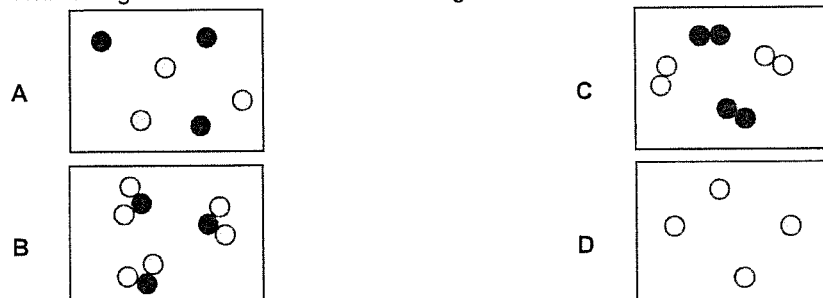
## 8. M/J 20/P23/Q24

Which statement about the noble gases is correct?

- A Argon is used in light bulbs and balloons.      C They all have full outer electron shells.  
 B Helium reacts with oxygen in the air.      D They are all diatomic molecules.

## 9. M/J 20/P22/Q24

Which diagram shows a mixture of noble gases?



## 10. M/J 20/P21/Q24

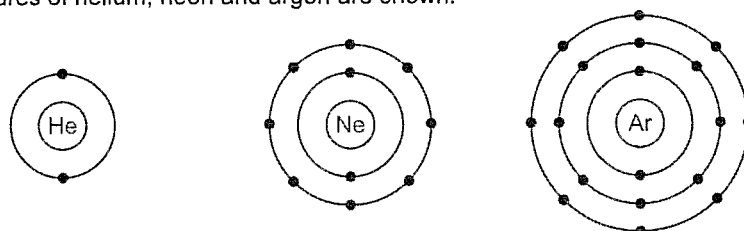
A balloon is filled with helium. Helium is a noble gas and makes the balloon rise up in the air.  
 The density of air is  $1.23 \text{ g/dm}^3$ .

Which gas is helium?

|   | density in $\text{g/dm}^3$ | reaction with oxygen       |
|---|----------------------------|----------------------------|
| A | 0.0899                     | burns rapidly              |
| B | 0.179                      | does not react with oxygen |
| C | 1.78                       | does not react with oxygen |
| D | 3.75                       | does not react with oxygen |

## 11. M/J 19/P22/Q22

The electronic structures of helium, neon and argon are shown.



Which row describes these gases?

|   | reactivity | form of the gas | electronic structure                |
|---|------------|-----------------|-------------------------------------|
| A | reactive   | monoatomic      | incomplete outer shell of electrons |
| B | unreactive | diatomic        | complete outer shell of electrons   |
| C | unreactive | diatomic        | incomplete outer shell of electrons |
| D | unreactive | monoatomic      | complete outer shell of electrons   |

## 12. M/J 19/P21/Q22

Which statement about elements in Group VIII of the Periodic Table is correct?

- A They all have a full outer shell of electrons.  
 B They all react with Group I elements to form ionic compounds.  
 C They are all diatomic molecules.  
 D They are all liquids at room temperature.

**13. M/J 19/P23/Q22**

Gas G has 10 electrons. Gas H has eight more electrons than gas G. Both gases are monoatomic. Which statement about G and H is correct?

- A Both gases are in the same group of the Periodic Table.
- B Both gases are in the same period of the Periodic Table.
- C Both gases are very reactive.
- D Gas G has a higher atomic mass than gas H.

**14. O/N 18/P22/Q23, O/N 18/P21/Q23, O/N 18/P23/Q23**

An inert gas R is used to fill weather balloons. Which descriptions of R are correct?

|   | number of outer shell electrons in atoms of R | structure of gas R |
|---|-----------------------------------------------|--------------------|
| A | 2                                             | diatomic molecules |
| B | 2                                             | single atoms       |
| C | 8                                             | diatomic molecules |
| D | 8                                             | single atoms       |

**15. O/N 17/P22/Q24, O/N 17/P21/Q24, O/N 17/P23/Q24, M/J 14/P12/Q24, M/J 14/P11/Q24, M/J 14/P13/Q23**

Why is argon gas used to fill electric lamps?

- A It conducts electricity.
- B It glows when heated.
- C It is less dense than air.
- D It is not reactive.

**16. M/J 17/P22/Q24, M/J 17/P21/Q24, M/J 17/P23/Q24, O/N 12/P12/Q24, O/N 12/P11/Q23, O/N 12/P13/Q23**

Why are weather balloons sometimes filled with helium rather than hydrogen?

- A Helium is found in air.
- B Helium is less dense than hydrogen.
- C Helium is more dense than hydrogen.
- D Helium is unreactive.

**17. MAR 17/P22/Q24**

The noble gases are in Group VIII of the Periodic Table. Which statement explains why noble gases are unreactive?

- A They all have eight electrons in their outer shells.
- B They all have full outer shells.
- C They are all gases.
- D They are all monoatomic.

**18. O/N 15/P13/Q22**

Hydrogen and helium have both been used to fill balloons.

Which property of helium makes it the preferred choice to hydrogen?

- A easily compressed into a gas cylinder
- B forms monatomic molecules
- C lower density
- D unreactive

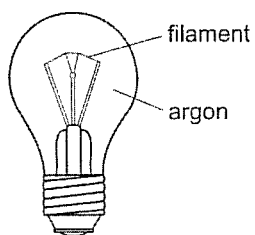
**19. O/N 14/P12/Q25, O/N 14/P11/Q25, O/N 14/P13/Q25**

An inert gas X is used to fill weather balloons. Which descriptions of X are correct?

|   | number of outer electrons in atoms of X | structure of gas X |
|---|-----------------------------------------|--------------------|
| A | 2                                       | single atoms       |
| B | 2                                       | diatomic molecules |
| C | 8                                       | single atoms       |
| D | 8                                       | diatomic molecules |

**20. O/N 13/P12/Q23, O/N 13/P11/Q24, O/N 13/P13/Q24**

The diagram shows a light bulb.



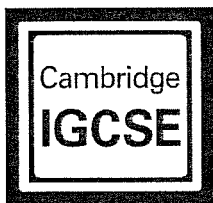
Why is argon used instead of air in the light bulb?

- |                                                    |                                                      |
|----------------------------------------------------|------------------------------------------------------|
| <b>A</b> Argon is a good conductor of electricity. | <b>C</b> The filament glows more brightly.           |
| <b>B</b> Argon is more reactive than air.          | <b>D</b> The filament does not react with the argon. |

## Answers Section

| 9.1   |       |  |  | 9.2   |       |  |  | 9.3   |       |  |  | 9.4   |  |
|-------|-------|--|--|-------|-------|--|--|-------|-------|--|--|-------|--|
| 1. C  | 41. C |  |  | 1. B  | 41. A |  |  | 1. A  | 41. D |  |  | 1. A  |  |
| 2. D  | 42. D |  |  | 2. D  | 42. D |  |  | 2. B  | 42. B |  |  | 2. B  |  |
| 3. B  | 43. D |  |  | 3. C  | 43. B |  |  | 3. A  | 43. B |  |  | 3. B  |  |
| 4. C  |       |  |  | 4. C  | 44. B |  |  | 4. A  | 44. D |  |  | 4. D  |  |
| 5. D  |       |  |  | 5. D  | 45. A |  |  | 5. D  | 45. B |  |  | 5. A  |  |
| 6. C  |       |  |  | 6. C  | 46. B |  |  | 6. D  |       |  |  | 6. A  |  |
| 7. C  |       |  |  | 7. D  | 47. D |  |  | 7. C  |       |  |  | 7. C  |  |
| 8. C  |       |  |  | 8. C  | 48. C |  |  | 8. D  |       |  |  | 8. C  |  |
| 9. B  |       |  |  | 9. A  | 49. C |  |  | 9. B  |       |  |  | 9. A  |  |
| 10. C |       |  |  | 10. C | 50. C |  |  | 10. B |       |  |  | 10. B |  |
| 11. B |       |  |  | 11. C | 51. A |  |  | 11. D |       |  |  | 11. D |  |
| 12. B |       |  |  | 12. A | 52. B |  |  | 12. C |       |  |  | 12. A |  |
| 13. D |       |  |  | 13. D | 53. C |  |  | 13. C |       |  |  | 13. A |  |
| 14. D |       |  |  | 14. B | 54. A |  |  | 14. C |       |  |  | 14. B |  |
| 15. B |       |  |  | 15. B | 55. D |  |  | 15. A |       |  |  | 15. D |  |
| 16. C |       |  |  | 16. B | 56. A |  |  | 16. D |       |  |  | 16. D |  |
| 17. A |       |  |  | 17. A | 57. C |  |  | 17. A |       |  |  | 17. B |  |
| 18. B |       |  |  | 18. C | 58. D |  |  | 18. A |       |  |  | 18. D |  |
| 19. A |       |  |  | 19. B | 59. B |  |  | 19. B |       |  |  | 19. A |  |
| 20. B |       |  |  | 20. D | 60. D |  |  | 20. A |       |  |  | 20. D |  |
| 21. C |       |  |  | 21. B | 61. A |  |  | 21. C |       |  |  |       |  |
| 22. B |       |  |  | 22. B | 62. A |  |  | 22. A |       |  |  |       |  |
| 23. C |       |  |  | 23. B | 63. B |  |  | 23. D |       |  |  |       |  |
| 24. B |       |  |  | 24. D | 64. C |  |  | 24. B |       |  |  |       |  |
| 25. A |       |  |  | 25. B | 65. D |  |  | 25. B |       |  |  |       |  |
| 26. C |       |  |  | 26. C | 66. C |  |  | 26. D |       |  |  |       |  |
| 27. A |       |  |  | 27. C | 67. D |  |  | 27. C |       |  |  |       |  |
| 28. C |       |  |  | 28. C | 68. B |  |  | 28. C |       |  |  |       |  |
| 29. C |       |  |  | 29. A | 69. C |  |  | 29. A |       |  |  |       |  |
| 30. C |       |  |  | 30. C | 70. C |  |  | 30. B |       |  |  |       |  |
| 31. A |       |  |  | 31. D | 71. A |  |  | 31. D |       |  |  |       |  |
| 32. C |       |  |  | 32. C | 72. C |  |  | 32. B |       |  |  |       |  |
| 33. C |       |  |  | 33. B | 73. C |  |  | 33. B |       |  |  |       |  |
| 34. B |       |  |  | 34. B | 74. D |  |  | 34. D |       |  |  |       |  |
| 35. D |       |  |  | 35. D |       |  |  | 35. A |       |  |  |       |  |
| 36. D |       |  |  | 36. C |       |  |  | 36. B |       |  |  |       |  |
| 37. C |       |  |  | 37. B |       |  |  | 37. B |       |  |  |       |  |
| 38. D |       |  |  | 38. B |       |  |  | 38. C |       |  |  |       |  |
| 39. B |       |  |  | 39. A |       |  |  | 39. A |       |  |  |       |  |
| 40. D |       |  |  | 40. C |       |  |  | 40. D |       |  |  |       |  |

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## Unit 10

# METALS

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IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 10.1 Properties of metals
- 10.2 Reactivity series
- 10.3 Extraction of metals
- 10.4 Uses of metals

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## Unit 10: Metals

### 10.1: Properties of metals

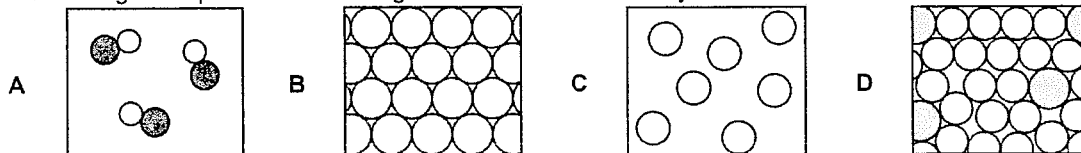
1. O/N 22/P23/Q25

Which statement about the reactions of metals is correct?

- A Iron and carbon dioxide are produced when iron (III) oxide is heated with carbon.
- B Magnesium reacts with dilute hydrochloric acid producing hydrogen and chlorine.
- C Potassium reacts vigorously with water producing hydrogen and an acidic solution.
- D Zinc reacts with dilute sulfuric acid producing sulfur dioxide.

2. M/J 22/P22/Q27

Which diagram represents the arrangement of atoms in an alloy?



3. M/J 22/P21/Q26

Which statements about the general properties of metals are correct?

- 1 They are good conductors of heat and electricity.
- 2 They have low melting points.
- 3 They react with dilute acids to form a salt and water.
- 4 They react with oxygen to form basic oxides.

- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

4. M/J 22/P23/Q6

Which statement about the properties of metals is correct?

- A Metals are malleable because the layers of positive ions can slide over each other.
- B Metals conduct electricity when solid because the positive ions move freely through the metal.
- C Metals conduct electricity because there is a strong force of attraction between the positive ions and the delocalised electrons.
- D Metals have a high melting point because the positive ions attract each other strongly.

5. M/J 22/P23/Q26

Some properties of four elements, P, Q, R and S, are shown.

Solid P reacts with dilute hydrochloric acid to give hydrogen.

Solid Q does not conduct electricity.

Solid R is used to make saucepans because it is a good conductor of heat.

Solid S reacts with oxygen to form a compound where atoms of S share electrons with atoms of oxygen.

Which elements are metals?

- A P and R
- B P and S
- C Q and R
- D Q and S

6. MAR 22/P22/Q27

Metal M is mixed with copper to produce brass.

What is M?

- A chromium
- B nickel
- C vanadium
- D zinc

7. O/N 21/P22/Q5, O/N 21/P21/Q5, O/N 21/P23/Q5

Which description of brass is correct?

- A alloy
- B compound
- C element
- D non-metal



**8. O/N 21/P22/Q25, O/N 21/P21/Q25, O/N 21/P23/Q25**Which statement is correct for **all** metals?

- A They conduct electricity when molten.      C They have a low density.  
 B They gain electrons when they form ions.      D They have a low melting point.

**9. O/N 21/P21/Q4**

Which statement explains why metals conduct electricity when solid?

- A They have atoms which are free to move.  
 B They have electrons which are free to move.  
 C They have molecules which are free to move.  
 D They have positive ions which are free to move.

**10. M/J 21/P22/Q26, M/J 21/P23/Q25**

Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A calcium      B copper      C sodium      D zinc

**11. MAR 21/P22/Q27**

Which row describes the reactions of magnesium hydroxide and magnesium oxide?

|   | effect of heat on hydroxide | effect of heating oxide with carbon |
|---|-----------------------------|-------------------------------------|
| A | forms magnesium oxide       | magnesium and carbon dioxide formed |
| B | forms magnesium oxide       | no reaction                         |
| C | no reaction                 | magnesium and carbon dioxide formed |
| D | no reaction                 | no reaction                         |

**12. M/J 20/P22/Q26**

Many metal carbonates decompose when they are heated.

Which row describes what happens when potassium carbonate, calcium carbonate and copper(II) carbonate are heated using a Bunsen burner?

|   | decomposes easily    | decomposes with difficulty | does not decompose at Bunsen temperatures |
|---|----------------------|----------------------------|-------------------------------------------|
| A | calcium carbonate    | copper(II) carbonate       | potassium carbonate                       |
| B | copper(II) carbonate | calcium carbonate          | potassium carbonate                       |
| C | copper(II) carbonate | potassium carbonate        | calcium carbonate                         |
| D | potassium carbonate  | calcium carbonate          | copper(II) carbonate                      |

**13. M/J 20/P21/Q25, M/J 20/P23/Q25, M/J 12/P11/Q26, M/J 12/P13/Q22**Which property is shown by **all** metals?

- A They are extracted from their ores by heating with carbon.  
 B They conduct electricity.  
 C They form acidic oxides.  
 D They react with hydrochloric acid to form hydrogen.



20. O/N 16/P22/Q26

Y displaces X from its aqueous sulfate.

X does not displace W from its aqueous sulfate.

X displaces Z from its aqueous sulfate.

What is the order of reactivity of elements W, X, Y and Z?

|          | most<br>reactive | $\longrightarrow$ |   |   | least<br>reactive |
|----------|------------------|-------------------|---|---|-------------------|
| <b>A</b> | W                | X                 | Y | Z |                   |
| <b>B</b> | W                | Y                 | X | Z |                   |
| <b>C</b> | Z                | X                 | Y | W |                   |
| <b>D</b> | Z                | W                 | Y | X |                   |

21. O/N 16/P22/Q23

Aqueous sodium hydroxide was added slowly, until in excess, to separate solutions of W, X, Y and Z. The results are shown.

| solution | initial observation with aqueous sodium hydroxide | final observation with excess aqueous sodium hydroxide |
|----------|---------------------------------------------------|--------------------------------------------------------|
| W        | white precipitate formed                          | precipitate dissolves                                  |
| X        | white precipitate formed                          | no change                                              |
| Y        | pale blue precipitate formed                      | no change                                              |
| Z        | green precipitate formed                          | no change                                              |

Which row identifies the metal ions in the solutions?

|   | metal ion in solution W | metal ion in solution X | metal ion in solution Y | metal ion in solution Z |
|---|-------------------------|-------------------------|-------------------------|-------------------------|
| A | aluminium               | calcium                 | copper(II)              | iron(II)                |
| B | aluminium               | calcium                 | iron(II)                | copper(II)              |
| C | aluminium               | iron(II)                | calcium                 | copper(II)              |
| D | calcium                 | aluminium               | copper(II)              | iron(II)                |

22. O/N 16/P23/Q24

Part of the Periodic Table is shown.

[illegible]

Which row correctly describes the properties of elements W, X, Y and Z?

|   | has variable oxidation states | reacts with cold water | very unreactive | has four outer shell electrons |
|---|-------------------------------|------------------------|-----------------|--------------------------------|
| A | W                             | Y                      | Z               | X                              |
| B | X                             | W                      | Y               | Z                              |
| C | Z                             | W                      | Y               | X                              |
| D | Z                             | Y                      | X               | W                              |

**23. M/J 16/P22/Q26**

Which property is **not** considered a typical metallic property?

- A good conductor of heat                      C malleable (can be hammered into shape)  
B low melting point                              D strong

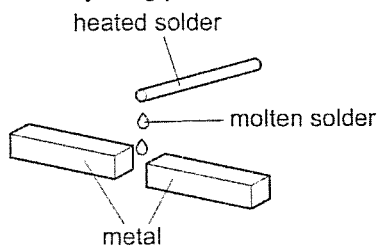
**24. O/N 15/P13/Q23**

Which statement is true for **all** metals?

- A Their atoms lose one or more electrons when they react.  
B They are brittle.  
C They do not conduct electricity when solid.  
D They melt at low temperatures when they are heated.

**25. M/J 15/P11/Q23**

Solder is an alloy of lead and tin. It is used for joining pieces of metal.



Which statement about solder is correct?

- A It can be represented by a chemical formula.                      C It contains lead and tin chemically combined  
B It contains a mixture of lead and tin.                              D It has a higher melting point than lead or tin.

**26. O/N 14/P12/Q29, O/N 14/P11/Q28**

Aluminium is the most common metal in the Earth's crust.

Which is **not** a property of aluminium?

- A low density                                      C good conductor of electricity  
B resistance to corrosion                      D poor conductor of heat

**27. O/N 13/P13/Q25**

Duralumin is an alloy. It contains aluminium, copper and magnesium.

It has many uses including the manufacture of cooking utensils and ships.

Which statement about duralumin and its properties is correct?

- A It is a good conductor of electricity.  
B It is brittle.  
C It is soluble in water.  
D The aluminium, copper and magnesium are chemically combined.

## 10.2: Reactivity Series

### 1. O/N 22/P22/Q24

The reactions of four metals, W, X, Y and Z, are listed.

- Metal W displaces metal X from the oxide of metal X.
- Metal Y has a greater tendency to form positive ions than metal W.
- Aqueous ions of metal Z are reduced by metal X.

What is the order of reactivity of the metals?

|   | least reactive $\longrightarrow$ most reactive |   |   |   |
|---|------------------------------------------------|---|---|---|
| A | Y                                              | W | X | Z |
| B | Y                                              | X | W | Z |
| C | Z                                              | W | X | Y |
| D | Z                                              | X | W | Y |

### 2. O/N 22/P22/Q26

Silver is below copper in the reactivity series.

Which row describes the reactions of silver?

|   | reaction with steam            | reaction with dilute hydrochloric acid |
|---|--------------------------------|----------------------------------------|
| A | no reaction                    | no reaction                            |
| B | no reaction                    | reacts to produce hydrogen gas         |
| C | reacts to produce hydrogen gas | no reaction                            |
| D | reacts to produce hydrogen gas | reacts to produce hydrogen gas         |

### 3. M/J 22/P22/Q25

The reactions of metals P, Q, R and S are shown.

| metal | reaction with water | reaction with hydrochloric acid | reduction of the metal oxide with carbon |
|-------|---------------------|---------------------------------|------------------------------------------|
| P     | no reaction         | no reaction                     | reduced                                  |
| Q     | slow                | vigorous                        | no reaction                              |
| R     | vigorous            | vigorous                        | no reaction                              |
| S     | very slow           | vigorous                        | reduced                                  |

What is the order of reactivity of the metals?

|   | least reactive $\longrightarrow$ most reactive |   |   |   |
|---|------------------------------------------------|---|---|---|
| A | P                                              | S | Q | R |
| B | P                                              | Q | S | R |
| C | R                                              | S | Q | P |
| D | R                                              | Q | S | P |

4. **M/J 22/P21/Q27**

Reactions of three metals and their oxides are shown.

| metal | add dilute hydrochloric acid to metal | heat metal oxide with carbon |
|-------|---------------------------------------|------------------------------|
| 1     | ✓                                     | ✓                            |
| 2     | ✓                                     | x                            |
| 3     | x                                     | ✓                            |

key

✓ = reacts

x = does not react

What is the order of reactivity of these metals, from most reactive to least reactive?

A 1 → 2 → 3

B 1 → 3 → 2

C 2 → 1 → 3

D 2 → 3 → 1

5. **MAR 22/P22/Q25**

Which word equation represents a reaction that occurs?

A sodium oxide + carbon → sodium + carbon dioxide

B sodium oxide + iron → sodium + iron(II) oxide

C iron(II) oxide + copper → iron + copper(II) oxide

D iron(III) oxide + carbon → iron + carbon dioxide

6. **O/N 21/P23/Q26**

Chromium is a more reactive metal than iron but less reactive than zinc.

Which statements are correct?

1 Chromium does not react with dilute hydrochloric acid.

2 Chromium oxide is reduced when it is heated with carbon.

3 Chromium reacts with zinc oxide to form zinc.

4 Chromium reacts with steam to form hydrogen gas.

A 1 and 2

B 1 and 3

C 2 and 4

D 3 and 4

7. **M/J 21/P21/Q25**

Some properties of metal J are listed.

• J does not react with cold water.

• J reacts with dilute hydrochloric acid.

• No reaction occurs when the oxide of J is heated with carbon.

What is J?

A copper

B iron

C magnesium

D sodium

8. **M/J 21/P21/Q26**

Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

A calcium

B copper

C sodium

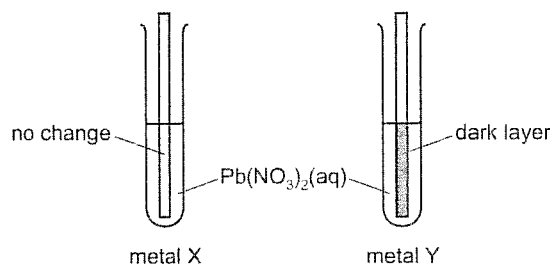
D zinc

9. **O/N 20/P23/Q28**

An experiment is performed to determine the order of reactivity of metals X and Y compared to lead.

Strips of each metal were added to separate test-tubes containing aqueous lead(II) nitrate,  $\text{Pb}(\text{NO}_3)_2$ .

The results are shown.



What is the order of reactivity, least reactive first?

A  $\text{Pb} \rightarrow \text{X} \rightarrow \text{Y}$ B  $\text{X} \rightarrow \text{Y} \rightarrow \text{Pb}$ C  $\text{X} \rightarrow \text{Pb} \rightarrow \text{Y}$ D  $\text{Y} \rightarrow \text{Pb} \rightarrow \text{X}$

**10. M/J 20/P22/Q28**

Four iron nails are added to four different metal sulfate solutions.

In which solution does a displacement reaction occur?

- A copper(II) sulfate    B magnesium sulfate    C sodium sulfate    D zinc sulfate

**11. M/J 20/P21/Q28**

Element Y reacts with copper (II) oxide to form copper.

Element Y will not react with zinc oxide. Copper has no reaction with zinc oxide.

What is the order of reactivity of these three elements, most reactive first?

- A  $\text{Cu} \rightarrow \text{Y} \rightarrow \text{Zn}$     B  $\text{Cu} \rightarrow \text{Zn} \rightarrow \text{Y}$     C  $\text{Zn} \rightarrow \text{Cu} \rightarrow \text{Y}$     D  $\text{Zn} \rightarrow \text{Y} \rightarrow \text{Cu}$

**12. M/J 20/P23/Q28**

P, Q, R and S are four metals. P displaces Q from a solution of its sulfate. Q reacts with hydrochloric acid and can be extracted from its ore using carbon. R does not react with hydrochloric acid. The carbonate of S does not decompose when heated strongly. What is the order of reactivity of the metals, starting with the most reactive?

|   | <div> <div>most reactive</div> <div>→</div> <div>least reactive</div> </div> |   |   |   |
|---|------------------------------------------------------------------------------|---|---|---|
| A | R                                                                            | P | Q | S |
| B | R                                                                            | Q | P | S |
| C | S                                                                            | P | Q | R |
| D | S                                                                            | Q | P | R |

**13. MAR 20/P22/Q26**

Four metals, iron, copper, magnesium and Y, are heated separately with their oxides. The results are shown.

| metal     | magnesium oxide | Y oxide | copper oxide | iron oxide |
|-----------|-----------------|---------|--------------|------------|
| Y         | x               | x       | ✓            | ✓          |
| magnesium | x               | ✓       | ✓            | ✓          |
| copper    | x               | x       | x            | x          |
| iron      | x               | x       | x            | x          |

key

✓ = reacts

x = no reaction

What is the order of reactivity of the metals, least reactive first?

|   | <div> <div>least reactive</div> <div>→</div> <div>most reactive</div> </div> |      |      |           |
|---|------------------------------------------------------------------------------|------|------|-----------|
| A | copper                                                                       | iron | Y    | magnesium |
| B | copper                                                                       | Y    | iron | magnesium |
| C | magnesium                                                                    | iron | Y    | copper    |
| D | magnesium                                                                    | Y    | iron | copper    |

**14. O/N 19/P23/Q28**

Which word equation represents a reaction which occurs?

- A sodium oxide + carbon  $\rightarrow$  sodium + carbon dioxide  
 B sodium oxide + iron  $\rightarrow$  sodium + iron(II) oxide  
 C iron(II) oxide + copper  $\rightarrow$  iron + copper(II) oxide  
 D iron(III) oxide + carbon  $\rightarrow$  iron + carbon dioxide

**15. O/N 19/P22/Q28**

The properties of four metals are listed.

- Metal W does not react with dilute hydrochloric acid.
- Metal X reacts with dilute hydrochloric acid.
- Metal Y displaces metal X from an aqueous solution of its ions.
- Metal Z reacts with water and dilute hydrochloric acid.

What is the order of reactivity of the metals?

|   | most<br>reactive |   | least<br>reactive |
|---|------------------|---|-------------------|
| A | W                | X | Y                 |
| B | W                | Y | X                 |
| C | Z                | X | Y                 |
| D | Z                | Y | X                 |

**16. M/J 19/P22/Q26, M/J 19/P21/Q26, M/J 19/P23/Q26**

Four metals, zinc, M, copper and magnesium, are reacted with aqueous solutions of their nitrates. The results are shown.

| metal     | magnesium<br>nitrate | M<br>nitrate | copper<br>nitrate | zinc<br>nitrate |                 |
|-----------|----------------------|--------------|-------------------|-----------------|-----------------|
| magnesium |                      | ✓            | ✓                 | ✓               | key             |
| zinc      | x                    | ✓            | ✓                 |                 |                 |
| M         | x                    |              | ✓                 | x               | x = no reaction |
| copper    | x                    | x            |                   | x               |                 |

What is the order of reactivity of these four metals starting with the most reactive?

- A copper → zinc → M → magnesium      C magnesium → M → zinc → copper  
B copper → M → zinc → magnesium      D magnesium → zinc → M → copper

**17. MAR 19/P22/Q27**

In which reaction does Fe(s) form ions when the mixture is heated?

- A Fe(s) + CaO(s)      B Fe(s) + MgO(s)      C Fe(s) + ZnO(s)      D Fe(s) + CuO(s)

**18. MAR 19/P22/Q28, O/N 13/P13/Q26**

The list gives the order of some metals and hydrogen in the reactivity series. Metal X is also included.

most reactive      K  
Mg  
Zn  
H  
X  
least reactive      Cu

Which row correctly shows the properties of metal X?

|   | reacts with<br>dilute acids | oxide reduced<br>by carbon |
|---|-----------------------------|----------------------------|
| A | no                          | no                         |
| B | no                          | yes                        |
| C | yes                         | no                         |
| D | yes                         | yes                        |



**19. MAR 19/P22/Q29**

Which metal carbonate does **not** produce carbon dioxide when it is heated?

- A copper(II) carbonate                      C potassium carbonate  
B iron(II) carbonate                         D zinc carbonate

**20. O/N 18/P22/Q24**

Heating copper(II) carbonate produces copper(II) oxide and carbon dioxide.

Heating the copper(II) oxide formed with carbon produces copper.

Which processes are involved in this conversion of copper(II) carbonate to copper?

- A sublimation followed by oxidation                      C thermal decomposition followed by oxidation  
B sublimation followed by reduction                    D thermal decomposition followed by reduction

**21. O/N 18/P21/Q24**

A student heated copper(II) carbonate and copper(II) nitrate in separate test-tubes.

Both compounds decomposed.

Which row shows the gases produced from each reaction?

|   | copper(II) carbonate | copper(II) nitrate          |
|---|----------------------|-----------------------------|
| A | carbon dioxide       | nitrogen dioxide only       |
| B | carbon dioxide       | oxygen only                 |
| C | carbon dioxide       | oxygen and nitrogen dioxide |
| D | oxygen               | oxygen and nitrogen dioxide |

**22. O/N 18/P22/Q25**

Four metals, W, X, Y and Z, are separately reacted with water and dilute hydrochloric acid.

The results are shown.

|                                        | metal  |             |                   |             |
|----------------------------------------|--------|-------------|-------------------|-------------|
|                                        | W      | X           | Y                 | Z           |
| reaction with water                    | fizzes | no reaction | fizzes vigorously | no reaction |
| reaction with dilute hydrochloric acid | fizzes | no reaction | fizzes violently  | fizzes      |

What is the order of reactivity of the four metals starting with the least reactive?

|   | least reactive      →      most reactive |   |   |   |
|---|------------------------------------------|---|---|---|
| A | X                                        | W | Z | Y |
| B | X                                        | Z | W | Y |
| C | Y                                        | W | Z | X |
| D | Y                                        | Z | W | X |

**23. O/N 18/P21/Q25**

Metal X reacts with steam but not with cold water.

What is X?

- A calcium                      B copper                      C sodium                      D zinc

**24. O/N 18/P23/Q24**

Heating copper(II) carbonate produces copper(II) oxide and carbon dioxide.

Heating the copper(II) oxide formed with carbon produces copper.

Which colour changes are observed during these reactions?

- A green → black → brown                      C blue → black → silver  
B green → white → brown                      D blue → white → brown

**25. O/N 18/P23/Q25**

Calcium reacts with cold water to produce hydrogen.

Lead reacts slowly when heated in air to form an oxide but has almost no reaction with steam.

Silver does not react with either air or water.

Zinc reacts when heated with steam to produce hydrogen.

What is the order of reactivity starting with the least reactive?

|   | least reactive |      | → | most reactive |         |
|---|----------------|------|---|---------------|---------|
| A | calcium        | lead |   | zinc          | silver  |
| B | calcium        | zinc |   | lead          | silver  |
| C | silver         | lead |   | zinc          | calcium |
| D | silver         | zinc |   | lead          | calcium |

**26. M/J 18/P22/Q25**

Metal X is more reactive than metal Y. Metal Y is more reactive than metal Z.

Which statement is correct?

- A When metal X is placed in a solution of Y sulfate, there is no reaction.
- B When metal X is placed in a solution of Z sulfate, a reaction occurs.
- C When metal Y is placed in a solution of Z sulfate, there is no reaction.
- D When metal Z is placed in a solution of X sulfate, a reaction occurs.

**27. M/J 18/P21/Q25**

Silver is a less reactive metal than cadmium.

Cadmium is a less reactive metal than barium.

Which statement is correct?

- A Barium does not react when heated with silver oxide.
- B Cadmium displaces barium from a solution of barium chloride.
- C Cadmium displaces silver from a solution of silver nitrate.
- D Cadmium reacts when heated with barium oxide.

**28. M/J 18/P21/Q26**

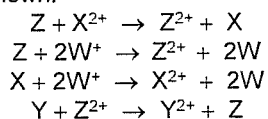
Aluminium metal is extracted from aluminium oxide using electrolysis.

Which statement about the extraction process is **not** correct?

- A A large amount of electricity is required.
- B Molten cryolite is used to dissolve the aluminium oxide.
- C Oxygen gas is released which reacts to form carbon dioxide.
- D The negative electrodes burn away and have to be replaced.

**29. M/J 18/P23/Q25**

The ionic equations for four reactions are shown.

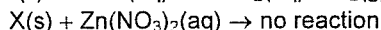
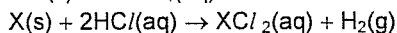
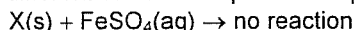


What is the order of reactivity of the four metals, W, X, Y and Z?

|   | most reactive |   | → | least reactive |   |
|---|---------------|---|---|----------------|---|
| A | W             | X |   | Z              | Y |
| B | X             | W |   | Y              | Z |
| C | Y             | Z |   | X              | W |
| D | Z             | W |   | X              | Y |

**30. MAR 18/P22/Q26**

A sample of solid X was added to three different solutions to predict the position of X in the reactivity series.



Which other solution would react with solid X?

- A  $CaSO_4(aq)$       B  $CuSO_4(aq)$       C  $MgSO_4(aq)$       D  $Na_2SO_4(aq)$

**31. O/N 17/P22/Q27**

Which statement about the metal zinc is **not** correct?

- A It forms an oxide more readily than iron.  
B It is manufactured by the electrolysis of zinc blende.  
C It is used to make brass.  
D It is used to prevent iron from rusting.

**32. O/N 17/P22/Q28**

Calcium nitrate decomposes when it is heated.

What is the equation for the thermal decomposition of calcium nitrate?

- A  $2Ca(NO_3)_2 \rightarrow 2CaO + O_2 + 4NO_2$       C  $Ca(NO_3)_2 \rightarrow Ca + O_2 + 2NO_2$   
B  $Ca(NO_3)_2 \rightarrow Ca(NO_2)_2 + O_2$       D  $Ca(NO_3)_2 \rightarrow Ca + 3O_2 + N_2$

**33. O/N 17/P22/Q34, O/N 17/P21/Q34, O/N 17/P23/Q34**

Some marble chips (calcium carbonate) are heated strongly and substances X and Y are formed.

Substance X is a white solid that reacts with water, giving out heat. Substance Y is a colourless gas.

What are substances X and Y?

|   | X                 | Y              |
|---|-------------------|----------------|
| A | calcium chloride  | oxygen         |
| B | calcium hydroxide | carbon dioxide |
| C | calcium oxide     | carbon dioxide |
| D | calcium sulfate   | oxygen         |

**34. O/N 17/P23/Q28**

Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A calcium      B copper      C sodium      D zinc

**35. M/J 17/P22/Q14**

A gas is produced when calcium carbonate is heated.

Which type of change is this?

- A chemical      B exothermic      C physical      D separation

**36. M/J 17/P22/Q26, M/J 17/P21/Q26, M/J 17/P23/Q26**

Element E:

- forms an alloy
- has a basic oxide
- is below hydrogen in the reactivity series.

What is E?

- A carbon      B copper      C sulfur      D zinc

**37. M/J 17/P22/Q27**

A list of metals is shown.

aluminium  
copper  
iron  
magnesium  
silver  
zinc

Which metal will displace all of the other metals from aqueous solutions of their salts?

- A aluminium      B iron      C magnesium      D zinc

**38. M/J 17/P23/Q27**

The section of the reactivity series shown includes a newly discovered element, symbol X.  
The only oxide of X has the formula XO.

Ca  
Mg  
Fe  
X  
H  
Cu

Which equation shows a reaction which occurs?

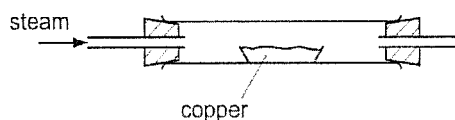
- A  $\text{Cu(s)} + \text{X}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{X(s)}$       C  $\text{X(s)} + \text{Fe}_2\text{O}_3(\text{s}) \rightarrow 2\text{Fe(s)} + 3\text{XO(s)}$   
B  $2\text{X(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{X}^+(\text{aq}) + \text{Cu(s)}$       D  $\text{X(s)} + 2\text{HCl(aq)} \rightarrow \text{XCl}_2(\text{aq}) + \text{H}_2(\text{g})$

**39. MAR 17/P22/Q29**

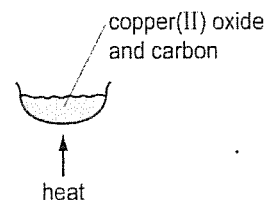
Two experiments are carried out.

In experiment 1, copper is heated with steam.

In experiment 2, copper(II) oxide is heated with carbon.



experiment 1



experiment 2

Which row describes what happens in experiments 1 and 2?

|   | experiment 1 | experiment 2 |
|---|--------------|--------------|
| A | no reaction  | no reaction  |
| B | no reaction  | reaction     |
| C | reaction     | no reaction  |
| D | reaction     | reaction     |

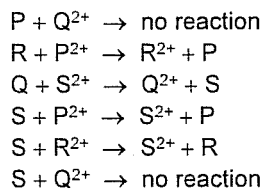
**40. MAR 17/P22/Q28**

Which metal carbonate does **not** produce carbon dioxide when it is heated with a Bunsen burner?

- A copper(II) carbonate      C sodium carbonate  
B magnesium carbonate      D zinc carbonate

**41. O/N 16/P21/Q26**

The ionic equations represent the reactions between four metals, P, Q, R and S, and solutions of the salts of the same metals.



What is the correct order of reactivity of the metals?

|          | <div> <div>most</div> <div>→</div> <div>least</div> </div> |   |   |   |
|----------|------------------------------------------------------------|---|---|---|
| <b>A</b> | P                                                          | R | S | Q |
| <b>B</b> | Q                                                          | R | S | P |
| <b>C</b> | Q                                                          | S | R | P |
| <b>D</b> | S                                                          | Q | P | R |

**42. O/N 16/P21/Q28**

Zinc oxide can be reacted with carbon to produce zinc metal.

Which equation for this reaction is correct?

- A**  $2ZnO + C \rightarrow 2Zn + CO$ 
**C**  $ZnO + C \rightarrow Zn + CO$   
**B**  $2ZnO + 2C \rightarrow 2Zn + 2CO_2$ 
**D**  $ZnO + 2C \rightarrow Zn + 2CO_2$

**43. O/N 16/P23/Q26**

The results of two experiments are given.

- 1 Cobalt displaces manganese from an aqueous solution of a manganese salt.
- 2 Manganese displaces silver from an aqueous solution of a silver salt.

Three more experiments are carried out.

- 3 Cobalt is added to an aqueous solution of a silver salt.
- 4 Manganese is added to an aqueous solution of a cobalt salt.
- 5 Silver is added to an aqueous solution of a cobalt salt.

In which experiments does a reaction take place?

- A** 3 only
 **B** 3 and 4
 **C** 4 and 5
 **D** 5 only

**44. M/J 16/P22/Q25**

Some magnesium compounds undergo thermal decomposition.

What are the products of thermal decomposition of magnesium nitrate,  $Mg(NO_3)_2$ , and magnesium hydroxide,  $Mg(OH)_2$ ?

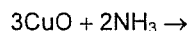
|          | $Mg(NO_3)_2$           | $Mg(OH)_2$     |
|----------|------------------------|----------------|
| <b>A</b> | MgO, $NO_2$ and $O_2$  | MgO and $H_2O$ |
| <b>B</b> | MgO, $NO_2$ and $O_2$  | MgO and $H_2$  |
| <b>C</b> | $Mg(NO_2)_2$ and $O_2$ | MgO and $H_2O$ |
| <b>D</b> | $Mg(NO_2)_2$ and $O_2$ | MgO and $H_2$  |



**49. O/N 15/P13/Q7**

Copper(II) oxide reacts with ammonia.

The left hand side of the balanced equation for this reaction is:



What completes the equation?

- A  $3\text{Cu} + 2\text{HNO}_3$       B  $3\text{Cu} + 2\text{N} + 3\text{H}_2\text{O}$       C  $3\text{Cu} + \text{N}_2 + 3\text{H}_2\text{O}$       D  $3\text{Cu} + 2\text{NO} + 3\text{H}_2\text{O}$

**50. O/N 15/P13/Q31**

In which reaction is carbon dioxide **not** produced?

- A complete combustion of petrol      C respiration  
B hydrochloric acid reacting with magnesium      D thermal decomposition of limestone

**51. M/J 15/P12/Q33**

A zinc compound forms carbon dioxide in two different reactions.

- 1 It is heated strongly.  
2 It is added to hydrochloric acid.

Which type of reaction occurs in 1 and 2?

|   | 1                     | 2              |
|---|-----------------------|----------------|
| A | combustion            | neutralisation |
| B | combustion            | oxidation      |
| C | thermal decomposition | neutralisation |
| D | thermal decomposition | oxidation      |

**52. M/J 15/P13/Q26**

The statements describe how different metals react with cold water.

- Calcium sinks, fizzing and releasing a steady stream of hydrogen.
- Copper does not react.
- Sodium floats, fizzing and rapidly releasing hydrogen.
- Zinc does not react but does react with steam, releasing hydrogen.

Using the information, where should hydrogen be placed in the reactivity series?

- A below copper      C between calcium and zinc  
B between sodium and calcium      D between zinc and copper

**53. MAR 15/P12/Q13**

When copper is heated in air a black coating forms on the copper.

What happens to the copper in this reaction?

- A The copper catches fire.      C The copper gains oxygen.  
B The copper decomposes.      D The copper loses oxygen.

**54. MAR 15/P12/Q23**

Metal X lies between zinc and iron in the reactivity series.

Which statements about metal X are correct?

- 1 It reacts with steam to produce hydrogen gas.  
2 It does not react with steam but will produce hydrogen with dilute acid.  
3 The metal can be obtained from its oxide by heating strongly with charcoal.  
4 The metal oxide cannot be reduced using carbon.
- A 1 and 3      B 1 and 4      C 2 and 3      D 2 and 4

**55. MAR 15/P12/Q32**

The list gives four experiments carried out with calcium carbonate.

- 1 acid added  
2 alkali added  
3 heated strongly  
4 water added

Which experiments produce carbon dioxide?

- A 1 and 2      B 1 and 3      C 2 and 3      D 2 and 4

**56. O/N 14/P12/Q24, O/N 14/P11/Q26**

The table shows the reactions of four different metals with water.

| metal | reaction                                                  |
|-------|-----------------------------------------------------------|
| W     | reacts vigorously with cold water                         |
| X     | no reaction with water                                    |
| Y     | reacts very slowly with water, more vigorously with steam |
| Z     | reacts violently with cold water                          |

What is the correct order of reactivity, from most reactive to least reactive?

- A**  $W \rightarrow X \rightarrow Y \rightarrow Z$     **B**  $W \rightarrow Z \rightarrow Y \rightarrow X$     **C**  $Z \rightarrow W \rightarrow X \rightarrow Y$     **D**  $Z \rightarrow W \rightarrow Y \rightarrow X$

**57. O/N 14/P12/Q27, O/N 14/P11/Q29**

The oxide of element X is reduced by heating with carbon.

Element X does not react with cold water, steam or dilute hydrochloric acid.

What is X?

- A** copper                      **B** iron                      **C** magnesium                      **D** zinc

**58. O/N 14/P13/Q26**

The metal beryllium does not react with cold water.

It reacts with hydrochloric acid but cannot be extracted from its ore by using carbon.

Where should it be placed in the reactivity series?

magnesium

**A**

zinc

**B**

iron

**C**

copper

**D**

**59. M/J 14/P12/Q23**

A student carried out an experiment to find the order of reactivity of five metals.

They were tested with cold water, hot water and steam and the results recorded in a table.

| metal | cold water        | hot water          | steam             |
|-------|-------------------|--------------------|-------------------|
| V     | no reaction       | reacts slowly      | vigorous reaction |
| W     | no reaction       | no reaction        | slow reaction     |
| X     | reacts slowly     | vigorous reaction  | not attempted     |
| Y     | no reaction       | no reaction        | no reaction       |
| Z     | vigorous reaction | explosive reaction | not attempted     |

What is the order of reactivity of these metals?

|          | most reactive $\longrightarrow$ least reactive |   |   |   |   |
|----------|------------------------------------------------|---|---|---|---|
| <b>A</b> | V                                              | W | Y | X | Z |
| <b>B</b> | W                                              | X | Z | V | Y |
| <b>C</b> | Z                                              | X | V | W | Y |
| <b>D</b> | Z                                              | X | Y | W | V |



**60. M/J 14/P11/Q27, M/J 14/P13/Q26**

In an experiment, three test-tubes labelled X, Y and Z were half-filled with dilute hydrochloric acid. A different metal was added to each test-tube. After a few minutes the following observations were made.

In tube X, bubbles slowly rose to the surface.

In tube Y, there was a rapid release of bubbles.

In tube Z, no bubbles were produced.

Which three metals match the observations?

|          | tube X    | tube Y    | tube Z |
|----------|-----------|-----------|--------|
| <b>A</b> | copper    | zinc      | iron   |
| <b>B</b> | magnesium | iron      | copper |
| <b>C</b> | zinc      | magnesium | copper |
| <b>D</b> | zinc      | magnesium | iron   |

**61. M/J 14/P11/Q33, M/J 14/P13/Q34**

When compound X is heated, it changes colour from green to black. Compound Y is formed and a gas is given off which turns limewater milky.

What are X and Y?

|          | X                 | Y             |
|----------|-------------------|---------------|
| <b>A</b> | calcium carbonate | calcium oxide |
| <b>B</b> | copper carbonate  | carbon        |
| <b>C</b> | copper carbonate  | copper oxide  |
| <b>D</b> | copper sulfate    | copper oxide  |

**62. O/N 13/P12/Q29, O/N 13/P11/Q27**

Reactions of three metals and their oxides are listed in the table.

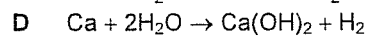
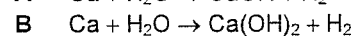
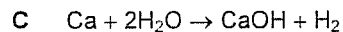
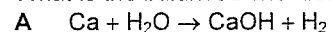
| metal | reacts with cold water | metal oxide reacts with carbon |
|-------|------------------------|--------------------------------|
| W     | no                     | no                             |
| X     | no                     | yes                            |
| Y     | yes                    | no                             |

What is the order of reactivity of the metals?

|          | least reactive | —————→ | most reactive |
|----------|----------------|--------|---------------|
| <b>A</b> | W              | X      | Y             |
| <b>B</b> | X              | W      | Y             |
| <b>C</b> | X              | Y      | W             |
| <b>D</b> | Y              | W      | X             |

**63. M/J 13/P12/Q10, M/J 13/P11/Q10, M/J 13/P13/Q11**

What is the balanced chemical equation for the reaction between calcium and water?



**64. M/J 13/P12/Q30**

Q, R, S and T are four metals.

Q is found naturally as the metal.

R reacts with steam but not with cold water.

S reacts violently with cold water.

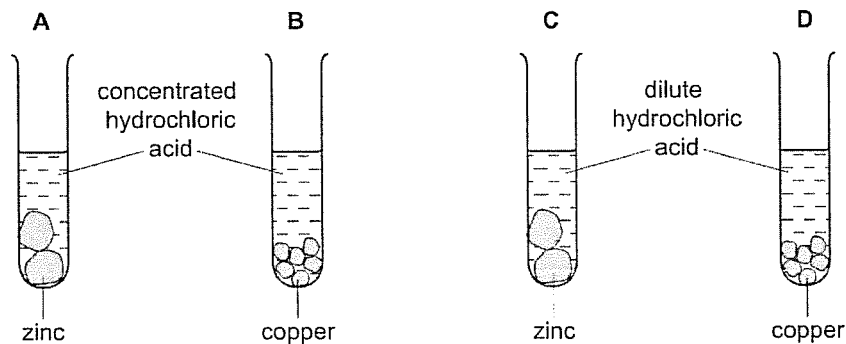
The oxide of T is reduced to T by heating with carbon.

What is the order of reactivity of the four metals, starting with the most reactive first?

A  $Q \rightarrow R \rightarrow T \rightarrow S$     B  $Q \rightarrow T \rightarrow R \rightarrow S$     C  $S \rightarrow R \rightarrow Q \rightarrow T$     D  $S \rightarrow R \rightarrow T \rightarrow Q$

**65. M/J 13/P12/Q15**

The diagram shows an experiment to compare the rate of reaction when a metal is added to hydrochloric acid. In which test-tube is the reaction fastest?



## 10.3: Extraction of metals

### 1. O/N 22/P21/Q24

Which statement about the extraction of aluminium by electrolysis is correct?

- A Aluminium is extracted from its ore, cryolite.
- B Aluminium is formed at the positive electrode.
- C Bauxite is used to lower the temperature of the extraction process.
- D Graphite is used for both the positive and negative electrodes.

### 2. O/N 22/P21/Q26

Iron from a blast furnace can be converted to steel.

Which statements about steel are correct?

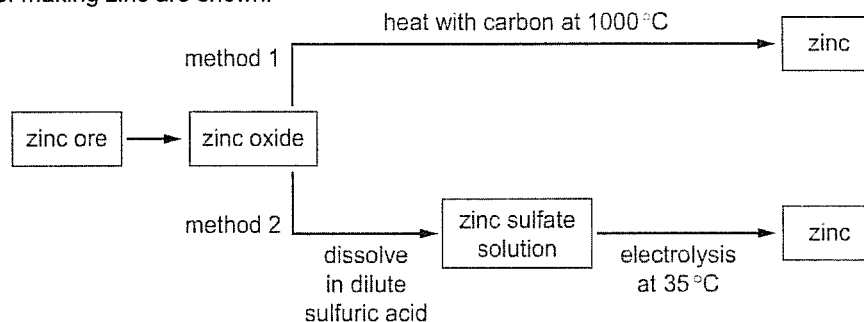
- 1 Steel contains more carbon than the iron obtained from the blast furnace.
- 2 Steel is produced by blowing oxygen through the iron.
- 3 Calcium oxide is added to molten iron to remove basic oxides.

- A 1 and 2                      B 1 and 3                      C 2 and 3                      D 2 only

### 3. O/N 22/P23/Q24

Zinc is a metal with a boiling point of 907 °C.

Two methods of making zinc are shown.



Which statement is correct?

- A Carbon oxidises zinc oxide in method 1.
- B Zinc vapour is produced in both methods.
- C Zinc is produced at the anode in method 2.
- D Zinc compounds are reduced in both methods.

### 4. M/J 22/P22/Q29

Processes involved in the extraction of zinc are listed.

- 1 Heat zinc oxide with carbon.
- 2 Condense zinc vapour.
- 3 Vaporise the zinc.
- 4 Roast zinc ore in air.

In which order are the processes carried out?

- A 1 → 2 → 3 → 4      B 4 → 3 → 1 → 2      C 4 → 1 → 3 → 2      D 1 → 4 → 3 → 2

### 5. M/J 22/P21/Q23

In the extraction of aluminium by electrolysis, cryolite is added to the bauxite ore.

Which row describes the role of cryolite and gives the ionic half-equation at the cathode?

|   | role of cryolite                    | ionic half-equation at the cathode |
|---|-------------------------------------|------------------------------------|
| A | catalyst                            | $Al^{3+} + 3e^{-} \rightarrow Al$  |
| B | catalyst                            | $Al^{3+} + 3e^{-} \rightarrow 3Al$ |
| C | lowers melting point of electrolyte | $Al^{3+} + 3e^{-} \rightarrow Al$  |
| D | lowers melting point of electrolyte | $Al^{3+} + 3e^{-} \rightarrow 3Al$ |

**6. MAR 22/P22/Q26**

Which statement about the extraction of aluminium is correct?

- A Aluminium is formed at the cathode during the electrolysis of aluminium oxide.  
 B Hematite is mainly aluminium oxide.  
 C Molten cryolite is used to raise the melting point of the aluminium oxide.  
 D Oxygen gains electrons at the anode during the electrolysis of aluminium oxide.

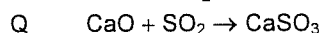
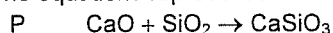
**7. O/N 21/P21/Q26**

Which statement about the extraction of metals is correct?

- A Aluminium is extracted from the ore bauxite by electrolysis.  
 B Aluminium is extracted from the ore hematite by electrolysis.  
 C Iron is extracted from the ore bauxite by electrolysis.  
 D Iron is extracted from the ore hematite by electrolysis.

**8. M/J 21/P22/Q33**

The equations represent two reactions, P and Q, of lime (calcium oxide).



In which processes do the reactions occur?

|   | P                        | Q                        |
|---|--------------------------|--------------------------|
| A | extraction of iron       | extraction of iron       |
| B | extraction of iron       | flue gas desulfurisation |
| C | flue gas desulfurisation | extraction of iron       |
| D | flue gas desulfurisation | flue gas desulfurisation |

**9. M/J 21/P21/Q27**

Which substances are used in the extraction of aluminium?

- A bauxite and cryolite  
 B bauxite and hematite  
 C cryolite and zinc blende  
 D hematite and zinc blende

**10. M/J 21/P23/Q26**

Which compounds are released by the extraction of zinc from zinc blende and by respiration?

|   | extraction of zinc                  | respiration                          |
|---|-------------------------------------|--------------------------------------|
| A | CO <sub>2</sub> and SO <sub>2</sub> | CO <sub>2</sub> only                 |
| B | CO <sub>2</sub> and SO <sub>2</sub> | CO <sub>2</sub> and H <sub>2</sub> O |
| C | CO <sub>2</sub> only                | CO <sub>2</sub> only                 |
| D | CO <sub>2</sub> only                | CO <sub>2</sub> and H <sub>2</sub> O |

**11. M/J 21/P23/Q29**Which reaction does **not** occur during the extraction of iron from hematite in a blast furnace?

- A  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$   
 B  $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$   
 C  $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$   
 D  $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$

**12. MAR 21/P22/Q24**

The metal beryllium does not react with cold water.

It reacts with hydrochloric acid but cannot be extracted from its ore by using carbon.

Where is beryllium placed in the reactivity series?

- A magnesium  
 B zinc  
 C iron  
 D copper

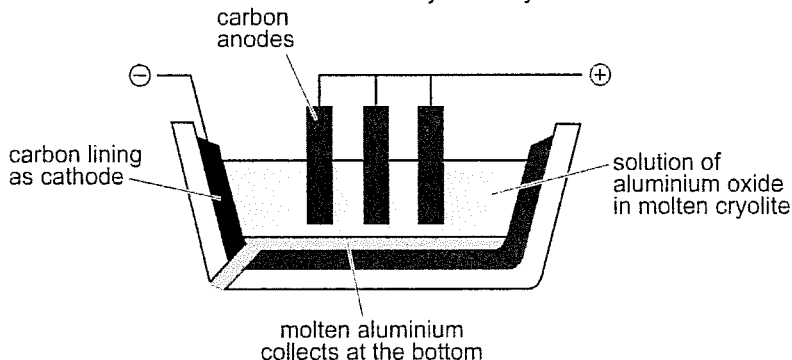
**13. MAR 21/P22/Q25**

Why is cryolite used in the extraction of aluminium from bauxite?

- A as a catalyst for the process  
 B as a solvent for aluminium oxide  
 C it stops the carbon anodes burning away  
 D it reduces aluminium ions in aluminium oxide

**14. O/N 20/P22/Q31**

The apparatus used for the extraction of aluminium oxide by electrolysis is shown.



Which equation represents a reaction taking place at the anode?

- A  $O + 2e^- \rightarrow O^{2-}$   
 B  $2O^{2-} \rightarrow O_2 + 4e^-$   
 C  $Al^{3+} \rightarrow Al + 3e^-$   
 D  $Al^{3+} + 3e^- \rightarrow Al$

**15. O/N 20/P21/Q29**

Which statements about the metal zinc are correct?

- 1 It is extracted from the ore bauxite.  
 2 It is used to galvanise steel.  
 3 It is used to make the alloy brass.  
 4 It reacts with dilute hydrochloric acid to produce hydrogen gas.

- A 2 and 3 only  
 B 1, 2 and 4  
 C 1, 3 and 4  
 D 2, 3 and 4

**16. O/N 20/P21/Q32**

Iron can be protected from rusting by attaching a piece of a more reactive metal, e.g. magnesium, to the iron. Which equation represents the reaction that takes place?

- A  $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$   
 B  $Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$   
 C  $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$   
 D  $Mg^{2+}(aq) + 2e^- \rightarrow Mg(s)$

**17. M/J 20/P22/Q25**

Which property is shown by **all** metals?

- A They are extracted from their ores by heating with carbon.  
 B They conduct electricity.  
 C They form acidic oxides.  
 D They react with hydrochloric acid to form hydrogen.

**18. M/J 20/P22/Q27, M/J 20/P21/Q27, M/J 20/P23/Q27**

Molten iron from the blast furnace contains impurities.

The process of turning the impure iron into steel involves blowing oxygen into the molten iron and adding calcium oxide.

What are the reasons for blowing in oxygen and adding calcium oxide?

|   | blowing in oxygen                         | adding calcium oxide                      |
|---|-------------------------------------------|-------------------------------------------|
| A | carbon is removed by reacting with oxygen | reacts with acidic impurities making slag |
| B | carbon is removed by reacting with oxygen | reacts with slag and so removes it        |
| C | iron reacts with the oxygen               | reacts with acidic impurities making slag |
| D | iron reacts with the oxygen               | reacts with slag and so removes it        |

**19. O/N 19/P22/Q29**

Which statement about the extraction of aluminium from aluminium oxide is correct?

- A Aluminium is formed at the positive electrode during electrolysis.  
 B Pure aluminium oxide is dissolved in molten cryolite.  
 C Pure aluminium oxide is electrolysed using aluminium as the positive electrode.  
 D Pure aluminium oxide is heated with carbon to form carbon dioxide and aluminium.

**20. O/N 19/P21/Q29**

Which statement about the extraction of aluminium is correct?

- A Aluminium is formed at the cathode during the electrolysis of aluminium oxide.  
 B Hematite is mainly aluminium oxide.  
 C Molten cryolite is used to raise the melting point of the aluminium oxide.  
 D Oxygen gains electrons at the anode during the electrolysis of aluminium oxide.

**21. O/N 19/P21/Q35**

Which process is used to obtain lime from limestone?

- A cracking  
 B fractional distillation  
 C neutralisation  
 D thermal decomposition

**22. O/N 19/P23/Q29**

Why is cryolite used in the extraction of aluminium by electrolysis?

- A It changes bauxite to aluminium oxide.  
 B It decreases the melting point of the aluminium.  
 C It dissolves the aluminium oxide.  
 D It protects the anodes from corrosion.

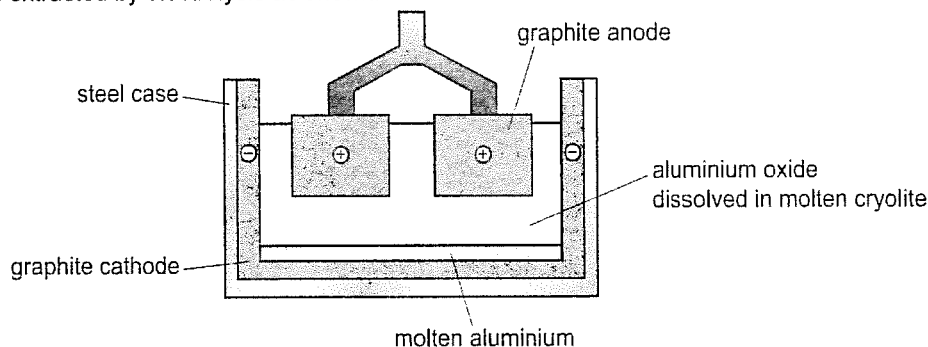
**23. O/N 19/P23/Q35**

Which type of reaction occurs when lime is manufactured from limestone?

- A combustion  
 B neutralisation  
 C redox  
 D thermal decomposition

**24. M/J 19/P22/Q10**

Aluminium is extracted by electrolysis as shown.

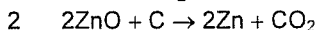
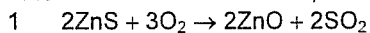


Which row shows the ionic half-equations at the cathode and the anode?

|   | cathode                           | anode                              |
|---|-----------------------------------|------------------------------------|
| A | $Al^{3+} \rightarrow Al + 3e^{-}$ | $2O^{2-} \rightarrow O_2 + 4e^{-}$ |
| B | $Al^{3+} \rightarrow Al + 3e^{-}$ | $2O^{2-} + 4e^{-} \rightarrow O_2$ |
| C | $Al^{3+} + 3e^{-} \rightarrow Al$ | $2O^{2-} \rightarrow O_2 + 4e^{-}$ |
| D | $Al^{3+} + 3e^{-} \rightarrow Al$ | $2O^{2-} + 4e^{-} \rightarrow O_2$ |

**25. M/J 19/P22/Q25, M/J 19/P21/Q25, M/J 19/P23/Q25**

Zinc is extracted from its ore, zinc blende, using two chemical reactions.

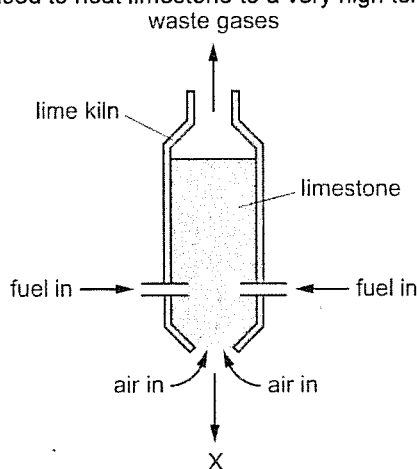


Which substance is reduced in reactions 1 and 2?

|          | reaction 1     | reaction 2 |
|----------|----------------|------------|
| <b>A</b> | O <sub>2</sub> | C          |
| <b>B</b> | O <sub>2</sub> | ZnO        |
| <b>C</b> | ZnS            | C          |
| <b>D</b> | ZnS            | ZnO        |

**26. M/J 19/P22/Q34, M/J 19/P21/Q34, M/J 19/P23/Q34, M/J 15/P11/Q34**

The diagram represents a lime kiln used to heat limestone to a very high temperature.



What leaves the kiln at X?

- |                            |                          |
|----------------------------|--------------------------|
| <b>A</b> calcium carbonate | <b>C</b> calcium oxide   |
| <b>B</b> calcium hydroxide | <b>D</b> calcium sulfate |

**27. M/J 19/P23/Q10**

In the manufacture of aluminium by electrolysis, aluminium oxide is dissolved in molten cryolite. Why is cryolite used?

- A** It lowers the melting point of the aluminium.
- B** It makes the aluminium a better conductor.
- C** It removes impurities from the aluminium.
- D** The mixture has a lower melting point than pure aluminium oxide.

**28. M/J 19/P23/Q17**

Which statement about carbon monoxide and aluminium oxide is correct?

- A** Carbon monoxide and aluminium oxide are both amphoteric.
- B** Carbon monoxide and aluminium oxide are both neutral.
- C** Carbon monoxide is amphoteric but aluminium oxide is neutral.
- D** Carbon monoxide is neutral but aluminium oxide is amphoteric.

**29. MAR 19/P22/Q30**

Which statement about the extraction of metals is correct?

- A** Aluminium is extracted by the electrolysis of hematite.
- B** Cryolite acts as a reducing agent in the extraction of aluminium.
- C** Zinc is extracted by the electrolysis of zinc blende.
- D** Zinc is obtained by heating zinc oxide with coke.

**30. O/N 18/P22/Q27**

Bauxite contains aluminium oxide.

Aluminium is extracted from aluminium oxide by electrolysis.

Why is cryolite added to the electrolytic cell used to extract aluminium?

- A Cryolite prevents the carbon anodes being burned away.
- B Cryolite removes impurities from the bauxite.
- C Cryolite increases the rate at which aluminium ions are discharged.
- D Molten cryolite dissolves the aluminium oxide.

**31. O/N 18/P23/Q27**

Which statement about the manufacture of aluminium by electrolysis is correct?

- A Aluminium ions are oxidised to aluminium by gaining electrons.
- B Aluminium is extracted from its ore hematite.
- C Molten cryolite is used to dissolve the aluminium oxide.
- D Oxygen is formed at the negative electrode.

**32. O/N 18/P23/Q34**

Which equation represents the formation of lime?

- A  $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- B  $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$
- C  $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$
- D  $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

**33. M/J 18/P22/Q24**

Stainless steel is an alloy of iron, carbon and other metals.

Which row is correct?

|   | stainless steel is<br>harder than pure iron | stainless steel resists<br>corrosion better than<br>pure iron |
|---|---------------------------------------------|---------------------------------------------------------------|
| A | ✓                                           | ✓                                                             |
| B | ✓                                           | x                                                             |
| C | x                                           | ✓                                                             |
| D | x                                           | x                                                             |

**34. M/J 18/P22/Q26**

Which statement about the industrial extraction of zinc is correct?

- A Cryolite is added to lower the melting point.
- B Molten zinc oxide is electrolysed.
- C Zinc oxide is heated with coke.
- D Zinc sulfide is heated with coke.

**35. M/J 18/P22/Q34**

Limestone is used in many industrial processes.

In which process is it **not** used?

- A manufacture of alkenes
- B manufacture of cement
- C manufacture of iron
- D manufacture of lime

**36. M/J 18/P21/Q34**

Which process is used to convert limestone (calcium carbonate) into lime?

- A electrolysis
- B fractional distillation
- C incomplete combustion
- D thermal decomposition

**37. M/J 18/P23/Q26**

Which equation represents the first stage in the extraction of zinc from zinc blende?

- A  $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$
- B  $\text{ZnS} + \text{H}_2\text{O} \rightarrow \text{ZnO} + \text{H}_2\text{S}$
- C  $\text{ZnO} + \text{CO} \rightarrow \text{Zn} + \text{CO}_2$
- D  $\text{ZnO} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2\text{O}$



**38. M/J 18/P23/Q34**

Limestone is an important material with many uses.  
Limestone is heated to produce .....1..... and carbon dioxide.  
This reaction is called .....2.....  
Which words correctly complete gaps 1 and 2?

|          | 1           | 2                     |
|----------|-------------|-----------------------|
| <b>A</b> | lime        | neutralisation        |
| <b>B</b> | lime        | thermal decomposition |
| <b>C</b> | slaked lime | neutralisation        |
| <b>D</b> | slaked lime | thermal decomposition |

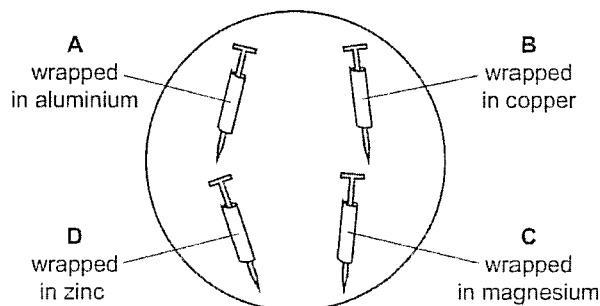
**39. MAR 18/P22/Q25**

Aluminium is extracted from aluminium oxide using electrolysis.  
Carbon dioxide is formed in this process.  
Which equation shows the formation of carbon dioxide during the extraction of aluminium from aluminium oxide by electrolysis?

- A**  $Al_2(CO_3)_3 \rightarrow Al_2O_3 + 3CO_2$                       **C**  $C + O_2 \rightarrow CO_2$   
**B**  $Al_2O_3 + 3CO \rightarrow 2Al + 3CO_2$                       **D**  $C^{4+} + 2O^{2-} \rightarrow CO_2$

**40. O/N 17/P21/Q30**

Four iron nails had different metals wrapped around them.  
The nails were placed in an open dish filled with water and left for a week.  
Which iron nail has no protection against rusting?

**41. O/N 17/P23/Q26**

Aluminium is obtained by the electrolysis of a mixture of aluminium oxide and cryolite.  
Why is cryolite used?

- A** as a catalyst to speed up the process  
**B** as a coolant to prevent the process getting too hot  
**C** as a solvent for aluminium oxide  
**D** as the main source of aluminium ions

**42. O/N 17/P23/Q27**

Metal M is mixed with copper to produce brass.  
What is M?

- A** chromium                      **B** nickel                      **C** vanadium                      **D** zinc

**43. M/J 17/P22/Q25**

Which process is involved in the extraction of zinc from zinc blende?

- A** Cryolite is added to lower the melting point of zinc blende.  
**B** Molten zinc blende is electrolysed.  
**C** Zinc blende is heated with carbon.  
**D** Zinc blende is roasted in air.

**44. M/J 17/P22/Q28, M/J 17/P21/Q28, M/J 17/P23/Q28, M/J 09/P1/Q31**

Stainless steel is an alloy of iron and other metals. It is strong and does not rust but it costs much more than normal steel.

What is **not** made from stainless steel?

- A cutlery  
B pipes in a chemical factory  
C railway lines  
D saucepans

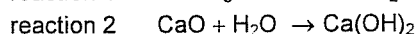
**45. M/J 17/P21/Q27**

Zinc metal is extracted from its ore zinc blende in a similar method to that used to extract iron from hematite. In which way is zinc extraction different from iron extraction?

- A Carbon and carbon monoxide are the main reducing agents.  
B Hot air at the base of the furnace reacts with coke to keep the furnace hot.  
C The metal is removed as a vapour at the top of the furnace.  
D The metal oxide is added into the top of the furnace.

**46. M/J 17/P23/Q34**

Two equations are shown.



Which terms describe reactions 1 and 2?

|   | reaction 1            | reaction 2 |
|---|-----------------------|------------|
| A | reduction             | hydration  |
| B | reduction             | hydrolysis |
| C | thermal decomposition | hydration  |
| D | thermal decomposition | hydrolysis |

**47. O/N 16/P22/Q25**

Impurities in iron obtained from the blast furnace include carbon, phosphorus and silicon. Which impurities are removed from the molten iron as gases when it is made into steel?

- A carbon and phosphorus  
B carbon and silicon  
C carbon only  
D phosphorus and silicon

**48. O/N 16/P22/Q27**

Which statement about the industrial extraction of aluminium from aluminium oxide is correct?

- A Aluminium is extracted by heating its oxide with carbon.  
B Aluminium is extracted using electrolysis and is collected at the anode.  
C Aluminium is extracted using platinum electrodes and direct current.  
D Molten cryolite is used as a solvent for aluminium oxide.

**49. O/N 16/P21/Q27**

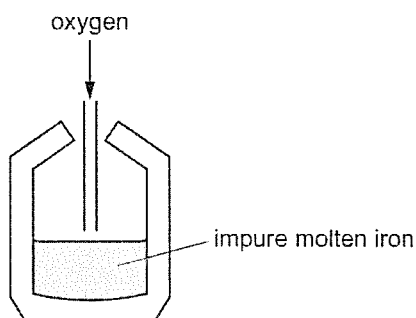
Aluminium is extracted by electrolysis.

From which ore is aluminium extracted and at which electrode is aluminium deposited during electrolysis?

|   | ore      | electrode |
|---|----------|-----------|
| A | bauxite  | negative  |
| B | bauxite  | positive  |
| C | cryolite | negative  |
| D | cryolite | positive  |

**50. O/N 16/P21/Q25**

Impure iron from the blast furnace is converted to steel as shown.

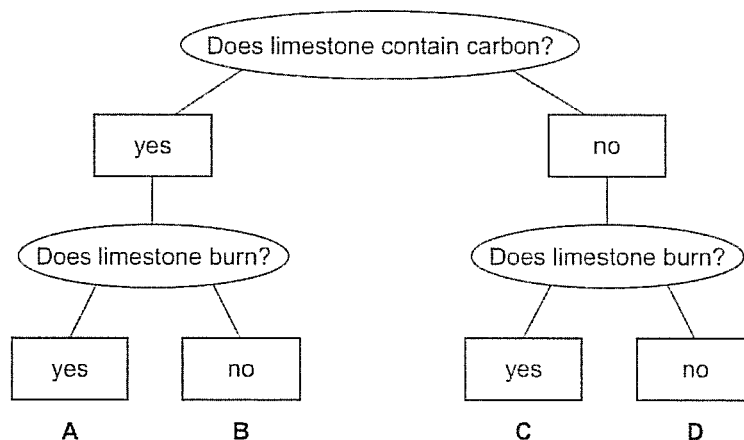


Which statement about the process is correct?

- A Acidic oxides are added to remove alkaline impurities.
- B Coke is added as a reducing agent.
- C Oxygen is blown in to oxidise the impure iron.
- D The steel produced contains less carbon than the impure iron.

**51. O/N 16/P21/Q34**

Which box corresponds to limestone?

**52. O/N 16/P23/Q27**

Cryolite,  $\text{Na}_3\text{AlF}_6$ , is added to aluminium oxide in the electrolytic extraction of aluminium. What is the reason for this?

- A to decrease the melting point of the electrolyte
- B to protect the anodes
- C to produce more aluminium
- D to stop the aluminium reacting with air

**53. O/N 16/P23/Q28**

Different forms of steel contain different proportions of carbon.

Steel P contains a high proportion of carbon.

Steel Q contains a low proportion of carbon.

Which statement is correct?

- |                                          |                                             |
|------------------------------------------|---------------------------------------------|
| A P is stronger and more brittle than Q. | C P is less strong and more brittle than Q. |
| B P is stronger and less brittle than Q. | D P is less strong and less brittle than Q. |

**54. M/J 16/P22/Q28**

Why is cryolite used during the extraction of aluminium by electrolysis?

- A It is a catalyst for the reaction.
- B It lowers the melting point of the electrolyte.
- C It protects the anodes.
- D It separates the aluminium from the electrolyte.

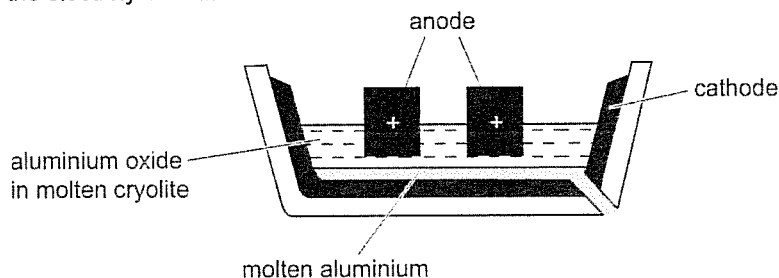
**55. M/J 16/P22/Q27**

Iron from a blast furnace is treated with oxygen and with calcium oxide to make steel.  
Which substances in the iron are removed?

|          | oxygen removes | calcium oxide removes |
|----------|----------------|-----------------------|
| <b>A</b> | carbon         | acidic oxides         |
| <b>B</b> | carbon         | basic oxides          |
| <b>C</b> | iron           | acidic oxides         |
| <b>D</b> | iron           | basic oxides          |

**56. M/J 16/P21/Q28**

Aluminium is manufactured by electrolysis of aluminium oxide.  
The diagram shows the electrolysis cell.



Which statement about the process is **not** correct?

- A** Aluminium ions gain electrons during the electrolysis and are reduced.
- B** Cryolite is added to reduce the melting point of the aluminium oxide.
- C** The anode and cathode are made of graphite.
- D** The cathode has to be replaced regularly because it is burnt away.

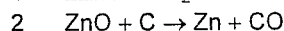
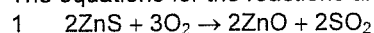
**57. M/J 16/P23/Q34**

Which process is used to make lime (calcium oxide) from limestone (calcium carbonate)?

- A** chromatography
- B** electrolysis
- C** fractional distillation
- D** thermal decomposition

**58. MAR 16/P22/Q16**

Zinc is extracted from zinc blende by roasting it in air to form zinc oxide.  
The zinc oxide is then heated with carbon to form zinc.  
The equations for the reactions are shown.



Which statement about reactions 1 and 2 is **not** correct?

- A** In reaction 1 the oxidation state of sulfur increases and it is oxidised.
- B** In reaction 1 the oxidation state of zinc increases and it is oxidised.
- C** In reaction 2 the carbon acts as a reducing agent and it is oxidised.
- D** In reaction 2 the oxidation state of zinc decreases and it is reduced.

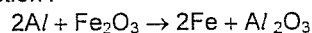
**59. MAR 16/P22/Q26**

Iron is extracted from hematite in the blast furnace.  
The hematite contains silicon(IV) oxide (sand) as an impurity.  
What reacts with this impurity to remove it?

- A** calcium oxide
- B** carbon
- C** carbon dioxide
- D** slag

**60. MAR 16/P22/Q27**

The reaction below is called the 'thermite reaction'.



Which pair of substances reacts in a similar way?

- A** Fe and MgO
- B** Fe and ZnO
- C** Mg and CuO
- D** Zn and  $\text{Al}_2\text{O}_3$

## 61. O/N 15/P12/Q23

Which two elements make up mild steel?

- A aluminium and magnesium  
B copper and zinc  
C iron and carbon  
D tin and lead

## 62. O/N 15/P12/Q26

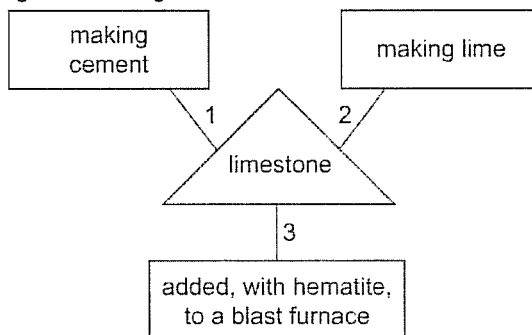
Iron is extracted from hematite in a blast furnace.

Which reaction increases the temperature in the blast furnace to over 1500 °C?

- A calcium carbonate → calcium oxide + carbon dioxide  
B calcium oxide + silicon dioxide → calcium silicate  
C carbon + oxygen → carbon dioxide  
D carbon dioxide + carbon → carbon monoxide

## 63. O/N 15/P12/Q34, O/N 15/P11/Q34, O/N 15/P13/Q34, O/N 12/P12/Q34, O/N 12/P11/Q34

A student is asked to draw a diagram showing the uses of limestone.



Which numbered lines show a correct use of limestone?

- A 1, 2 and 3  
B 1 and 2 only  
C 1 and 3 only  
D 2 and 3 only

## 64. O/N 15/P11/Q26

Iron is extracted from its ore (hematite) in the blast furnace.

Which gas is produced as a waste product?

- A carbon dioxide  
B hydrogen  
C nitrogen  
D oxygen

## 65. O/N 15/P13/Q26

Which conditions are necessary to make mild steel from iron?

- A add calcium oxide and blow oxygen through it  
B heat with calcium oxide  
C heat with carbon and limestone  
D heat with nickel and chromium

## 66. M/J 15/P12/Q25, M/J 15/P11/Q25, M/J 15/P13/Q25, M/J 11/P12/Q28, M/J 11/P11/Q29, M/J 11/P13/Q28

Which row describes the conditions used to make steel from the iron produced by a blast furnace?

|   | calcium oxide<br>(lime) | oxygen | heat |
|---|-------------------------|--------|------|
| A | ✓                       | ✓      | ✓    |
| B | ✓                       | ✓      | x    |
| C | x                       | ✓      | ✓    |
| D | x                       | ✓      | x    |

## 67. M/J 15/P11/Q26

W, X, Y and Z are four metals.

Some properties of these metals are listed below.

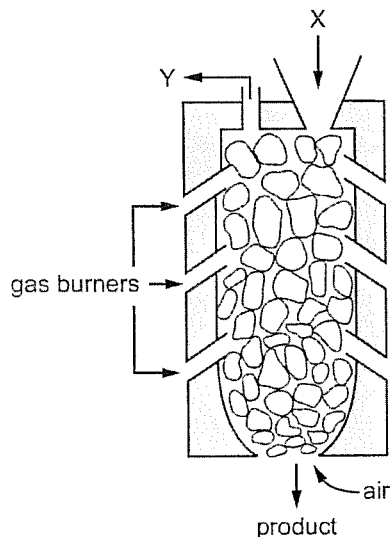
- 1 Only W and Z can be extracted by reduction of their oxides with carbon.  
2 Only X will react with cold water.  
3 Only Z can be found 'native' (not combined with any other element).

What is the correct order of these metals in the reactivity series (most reactive first)?

- A X, W, Y, Z  
B X, Y, W, Z  
C Z, W, Y, X  
D Z, Y, W, X

**68. M/J 15/P13/Q34**

The diagram shows a kiln used to manufacture lime.



Which row identifies X and Y?

|   | X         | Y              |
|---|-----------|----------------|
| A | lime      | carbon dioxide |
| B | lime      | steam          |
| C | limestone | carbon dioxide |
| D | limestone | steam          |

**69. MAR 15/P12/Q25**

Molten iron from the blast furnace contains impurities.

The process of turning the impure iron into steel involves blowing oxygen into the molten iron and adding calcium oxide.

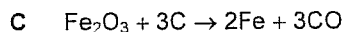
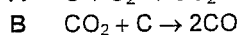
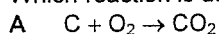
What are the reasons for blowing in oxygen and adding calcium oxide?

|   | blowing in oxygen                         | adding calcium oxide                      |
|---|-------------------------------------------|-------------------------------------------|
| A | carbon is removed by reacting with oxygen | reacts with acidic impurities making slag |
| B | carbon is removed by reacting with oxygen | reacts with slag and so removes it        |
| C | iron reacts with the oxygen               | reacts with acidic impurities making slag |
| D | iron reacts with the oxygen               | reacts with slag and so removes it        |

**70. M/J 14/P12/Q28**

Four reactions that take place in the blast furnace to produce iron are shown.

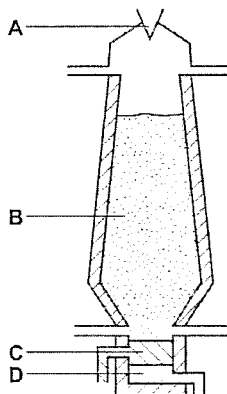
Which reaction is used to keep the furnace hot?



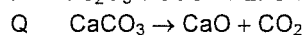
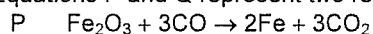
**71. M/J 14/P12/Q29, M/J 14/P11/Q29, M/J 14/P13/Q30**

The diagram shows a blast furnace.

In which part is iron ore changed to iron?

**72. O/N 13/P12/Q26, O/N 13/P11/Q28**

Equations P and Q represent two reactions which occur inside a blast furnace.

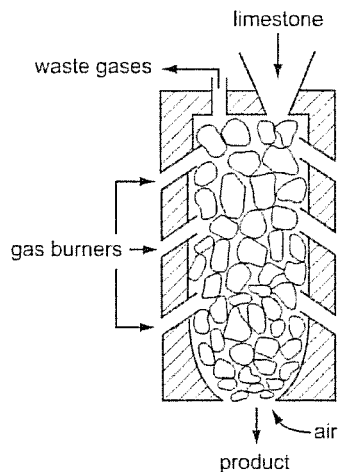


Which type of reactions are P and Q?

|   | P                     | Q                     |
|---|-----------------------|-----------------------|
| A | redox                 | redox                 |
| B | redox                 | thermal decomposition |
| C | thermal decomposition | redox                 |
| D | thermal decomposition | thermal decomposition |

**73. O/N 13/P12/Q34, O/N 13/P11/Q35, O/N 13/P13/Q35**

The diagram shows a kiln used to heat limestone.

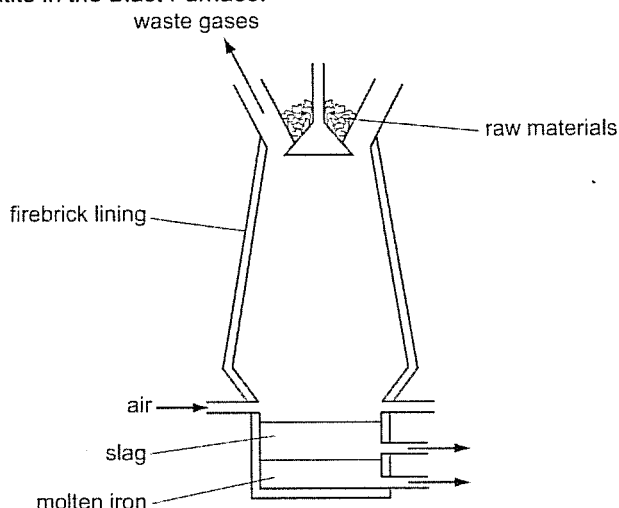


What is the product and what waste gas is formed?

|   | product                          | waste gas       |
|---|----------------------------------|-----------------|
| A | lime, CaO                        | carbon monoxide |
| B | lime, CaO                        | carbon dioxide  |
| C | slaked lime, Ca(OH) <sub>2</sub> | carbon monoxide |
| D | slaked lime, Ca(OH) <sub>2</sub> | carbon dioxide  |

**74. O/N 13/P13/Q28**

Iron is extracted from hematite in the Blast Furnace.



The hematite contains silica as an impurity.  
What reacts with this impurity to remove it?

- A calcium oxide      B carbon      C carbon dioxide      D oxygen

**75. M/J 13/P12/Q28**

Which statement is incorrect?

- A Carbon dioxide is a waste product in the extraction of iron.  
B Carbon monoxide is a reducing agent.  
C The extraction of iron from hematite involves reduction.  
D When iron is converted into steel, oxygen is used to oxidise the iron.

**76. M/J 13/P11/Q28, M/J 13/P13/Q29**

Which substance is **not** involved in the extraction of iron from hematite?

- A carbon      B carbon monoxide      C calcium carbonate      D nitrogen

**77. M/J 13/P11/Q30, M/J 13/P13/Q28**

Below are some metals in decreasing order of reactivity.

magnesium  
zinc  
iron  
copper

Titanium reacts with acid and cannot be extracted from its ore by heating with carbon.

Where should titanium be placed in this list?

- A below copper      C between magnesium and zinc  
B between iron and copper      D between zinc and iron



**10.4: Uses of metals****1. O/N 22/P21/Q27**

Which metal is used to galvanise steel?

- A copper                      B lead                      C tin                      D zinc

**2. O/N 21/P21/Q27, O/N 21/P23/Q27**

Aluminium objects do not need protection from corrosion.

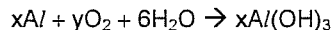
Iron objects must be protected from corrosion.

Which statement explains why aluminium resists corrosion?

- A Aluminium does not form ions easily.  
 B Aluminium does not react with water or air.  
 C Aluminium has a protective oxide layer.  
 D Aluminium is below iron in the reactivity series.

**3. M/J 21/P22/Q11**

A reaction involving aluminium is shown.



Which values of x and y balance the equation?

|   | x | y |
|---|---|---|
| A | 2 | 3 |
| B | 3 | 2 |
| C | 3 | 4 |
| D | 4 | 3 |

**4. M/J 21/P22/Q27**

Aluminium is extracted from its ore by electrolysis.

Which equation represents the reaction that occurs at the anode during the electrolysis?

- A  $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$       B  $\text{Al}^{3+} \rightarrow \text{Al} + 3\text{e}^-$       C  $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$       D  $2\text{O}^{2-} + 2\text{e}^- \rightarrow \text{O}_2$

**5. M/J 21/P21/Q28**

Different types of steel alloys are manufactured by changing the percentage of carbon in the alloy.

The properties of four steel alloys are shown.

| alloy mixture | percentage of carbon in the alloy | strength of the alloy | hardness of the alloy |
|---------------|-----------------------------------|-----------------------|-----------------------|
| 1             | 0.00 to 0.20                      | high                  | low                   |
| 2             | 0.21 to 0.30                      | high                  | medium                |
| 3             | 0.31 to 0.40                      | medium                | high                  |
| 4             | 0.41 to 1.50                      | low                   | high                  |

What are the properties of the steel alloy containing 0.23% of carbon?

|   | strength | hardness |
|---|----------|----------|
| A | high     | low      |
| B | low      | high     |
| C | high     | medium   |
| D | medium   | high     |

**6. M/J 21/P22/Q28**

Mild steel consists mostly of iron. Mild steel can be prevented from rusting by a process called galvanising. Copper is not a very strong metal, however if it is mixed with a suitable metal a strong alloy called brass is produced.

Which statement is correct?

- A Copper corrodes very quickly when wet and brass does not.
- B Copper is mixed with zinc to produce brass.
- C Galvanising mild steel changes it from a pure metal into an alloy.
- D When a steel object is galvanised this means it is coated with a thin layer of tin.

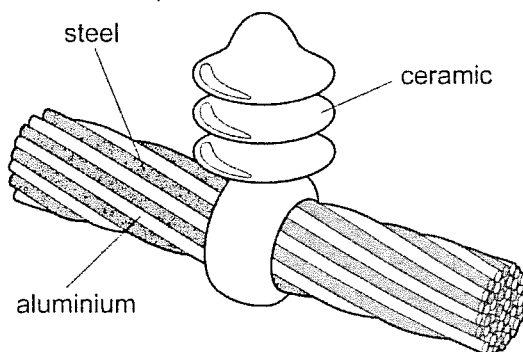
**7. M/J 21/P23/Q24**

Which statement explains why galvanising prevents iron from rusting?

- A Zinc is more reactive than iron and corrodes in preference to iron.
- B Zinc is more reactive than iron and loses electrons less easily than iron.
- C Zinc is less reactive than iron and corrodes in preference to iron.
- D Zinc is less reactive than iron and loses electrons more easily than iron.

**8. MAR 21/P22/Q11, O/N 15/P12/Q9, O/N 15/P11/Q9, O/N 15/P13/Q9, O/N 11/P12/Q10, O/N 11/P11/Q9, O/N 11/P13/Q9**

The diagram shows a section of an overhead power cable.



Which statement explains why a particular substance is used?

- A Aluminium has a low density and is a good conductor of electricity.
- B Ceramic is a good conductor of electricity.
- C Steel can rust in damp air.
- D Steel is more dense than aluminium.

**9. MAR 21/P22/Q26**

Which statements about the uses of metals are correct?

- 1 Iron is used to make aircraft because iron has a low density.
- 2 Copper is used to make electric cables because copper is a good conductor of electricity.
- 3 Aluminium is used to make brass because aluminium is strong and hard.
- 4 Iron is mixed with additives to make an alloy used in chemical plant.

- A 1 and 2                      B 3 and 4                      C 1 and 3                      D 2 and 4

**10. O/N 20/P22/Q30**

Why is aluminium metal unreactive with air?

- A It is covered with a layer of oxide.
- B It is low in the reactivity series.
- C It is produced by electrolysis of its oxide.
- D It melts at a high temperature.

**11. O/N 20/P21/Q30**

What is the symbol of the metal used in the manufacture of aircraft because of its strength and low density?

- A Al                      B Cu                      C Fe                      D Zn

**12. M/J 20/P22/Q32**

Which process, used to prevent iron from rusting, involves sacrificial protection?

- A alloying                      B electroplating                      C galvanising                      D painting

**13. M/J 20/P21/Q32**

Coating iron helps to prevent rusting.

Which coating will continue to protect the iron even when the coating is damaged?

- A copper                      B paint                      C plastic                      D zinc

**14. MAR 20/P22/Q28**

Some properties of aluminium are listed.

- 1 It conducts heat.
- 2 It has a low density.
- 3 It is strong.
- 4 It is resistant to corrosion.

Which of these properties make aluminium suitable for making food containers for chilled food products?

- A 1, 2 and 4                      B 1, 3 and 4                      C 1 only                      D 4 only

**15. O/N 19/P22/Q27**

Which row describes the uses of aluminium, copper and mild steel?

|   | aluminium         | copper            | mild steel        |
|---|-------------------|-------------------|-------------------|
| A | aircraft bodies   | electrical wiring | car bodies        |
| B | car bodies        | cooking utensils  | electrical wiring |
| C | electrical wiring | aircraft bodies   | food containers   |
| D | food containers   | aircraft bodies   | cooking utensils  |

**16. O/N 19/P21/Q28**

Iron rusts but aluminium does not easily corrode.

Which statement explains why aluminium does **not** easily corrode?

- A It is an alloy.                      C It is not a transition element.  
B It is below iron in the reactivity series.                      D Its surface is protected by an oxide layer.

**17. O/N 19/P23/Q27**

Which statement about metals and their uses is correct?

- A Aluminium is used in the manufacture of aircraft because it has a high density.  
B Copper is used to make cooking utensils because it is a poor conductor of heat.  
C Mild steel is used to make car bodies because it is brittle and breaks easily.  
D Stainless steel is used to make cutlery because it is resistant to corrosion.

**18. M/J 19/P22/Q27**

Which property of aluminium makes it useful in the manufacture of aircraft?

- A conducts electricity    B high boiling point    C low density    D silver colour

**19. M/J 19/P21/Q27**

Why is aluminium used to make containers for storing food?

- A It conducts electricity.                      C It is resistant to corrosion.  
B It has a high melting point.                      D It is strong.

**20. M/J 19/P23/Q27**

Aluminium is used to make containers for storing food.

Which property makes it suitable for this use?

- A conducts heat    B low density    C resists corrosion    D shiny surface

**21. O/N 18/P22/Q26**

Which statement about the uses of metals is **not** correct?

- A Aluminium is used in aircraft because of its strength and good electrical conductivity.  
B Copper is used in electrical wiring because of its good electrical conductivity.  
C Stainless steel resists corrosion and is used to make cutlery.  
D Transition elements are often used as catalysts.

**22. O/N 18/P21/Q26**

Which row shows uses of the metals listed?

|   | aluminium            | copper               | mild steel       |
|---|----------------------|----------------------|------------------|
| A | aircraft manufacture | food containers      | cutlery          |
| B | cutlery              | electrical wiring    | chemical plant   |
| C | electrical wiring    | aircraft manufacture | cooking utensils |
| D | food containers      | cooking utensils     | car bodies       |

**23. O/N 18/P21/Q27**

Aluminium objects do not need protection from corrosion.

Iron objects must be protected from corrosion.

Why does aluminium resist corrosion?

- A Aluminium does not form ions easily.  
 B Aluminium does not react with water or air.  
 C Aluminium has a protective oxide layer.  
 D Aluminium is below iron in the reactivity series.

**24. O/N 18/P23/Q26**

Which row describes the use of a metal and the property upon which the use depends?

|   | metal     | use               | property                                  |
|---|-----------|-------------------|-------------------------------------------|
| A | aluminium | aircraft bodies   | aluminium is a heat conductor             |
| B | aluminium | cooking utensils  | aluminium has a low density               |
| C | copper    | cooking utensils  | copper has a high density                 |
| D | copper    | electrical wiring | copper is a good conductor of electricity |

**25. M/J 18/P22/Q27**

Which row describes the use of an alloy and the property upon which the use depends?

|   | alloy           | use       | property                      |
|---|-----------------|-----------|-------------------------------|
| A | mild steel      | cutlery   | resistant to corrosion        |
| B | mild steel      | machinery | strong                        |
| C | stainless steel | cutlery   | low density                   |
| D | stainless steel | machinery | good conductor of electricity |

**26. M/J 18/P22/Q29, M/J 18/P21/Q29, M/J 18/P23/Q29, M/J 15/P12/Q31**

A steel bicycle which had been left outdoors for several months was starting to rust.

What would **not** reduce the rate of corrosion?

- A Remove the rust and paint the bicycle.  
 B Remove the rust and store the bicycle in a dry shed.  
 C Remove the rust and wipe the bicycle with a clean, damp cloth.  
 D Remove the rust and wipe the bicycle with an oily cloth.



**34. MAR 16/P22/Q25**

Brass is an alloy of two metals.

Which row gives a correct use for the two metals from which brass is made?

|          | metal 1                      | metal 2                    |
|----------|------------------------------|----------------------------|
| <b>A</b> | used for electrical wiring   | used for galvanising steel |
| <b>B</b> | used for galvanising steel   | used for making aircraft   |
| <b>C</b> | used for making aircraft     | used for making cutlery    |
| <b>D</b> | used for making cooking pans | used for electrical wiring |

**35. O/N 15/P11/Q25**

Which metal would be suitable for all of the following uses?

- making aircraft bodies
- making food containers
- making overhead power cables

**A** aluminium                      **B** brass                      **C** mild steel                      **D** pure iron

**36. M/J 15/P12/Q24**

Which statement is **not** correct?

- A** Aluminium is used in food containers because of its resistance to corrosion.  
**B** Aluminium is used in the manufacture of aircraft because of its strength and low density.  
**C** Mild steel is used in car bodies because of its strength and low density.  
**D** Stainless steel is used in chemical plant because of its strength and resistance to corrosion.

**37. O/N 15/P13/Q25**

The table show three uses of aluminium and a reason why aluminium is used for that purpose.

|   | use                         | reason                  |
|---|-----------------------------|-------------------------|
| 1 | aircraft manufacture        | high tensile strength   |
| 2 | overhead electricity cables | low density             |
| 3 | food containers             | resistance to corrosion |

Which reasons explain the use?

**A** 1, 2 and 3                      **B** 1 and 2 only                      **C** 1 and 3 only                      **D** 2 and 3 only

**38. M/J 15/P11/Q24**

What is a major use of aluminium?

- A** making brass                      **C** making electrical wiring  
**B** making cutlery                      **D** making food containers

**39. M/J 15/P13/Q24**

Which statement about the uses of aluminium, mild steel and stainless steel is correct?

- A** Aluminium is used for food containers as it has a high density.  
**B** Mild steel is used for car bodies as it is resistant to corrosion.  
**C** Stainless steel is used for aircraft bodies as it is strong.  
**D** Stainless steel is used for cutlery as it is resistant to corrosion.

**40. MAR 15/P12/Q26**

Some properties of aluminium are listed.

- 1 It has mechanical strength.
- 2 It is resistant to corrosion.
- 3 It has a low density.
- 4 It conducts heat.

Which **three** properties make aluminium useful for making the bodies of aircraft?

**A** 1, 2 and 3                      **B** 1, 2 and 4                      **C** 1, 3 and 4                      **D** 2, 3 and 4

41. O/N 14/P12/Q31, O/N 14/P11/Q30, O/N 14/P13/Q30, O/N 09/P12/Q29, O/N 09/P11/Q31

Which object is **least** likely to contain aluminium?

- A a bicycle frame      B a hammer      C a saucepan      D an aeroplane body

42. O/N 14/P13/Q28

A list of properties of aluminium is shown.

- 1 It conducts heat.
- 2 It has a low density.
- 3 It is resistant to corrosion.

Which properties make aluminium useful for making food storage containers?

- A 1, 2 and 3      B 1 and 3 only      C 1 only      D 3 only

43. O/N 14/P13/Q29

Which metal is commonly used to form alloys with a non-metallic element?

- A copper      B iron      C magnesium      D zinc

44. M/J 14/P12/Q10

Electrical cables are made from either .....1....., because it is a very good conductor of electricity, or from.....2....., because it has a low density.

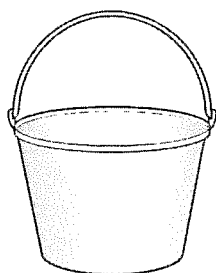
Overhead cables have a .....3..... core in order to give the cable strength.

Which words correctly complete gaps 1, 2 and 3?

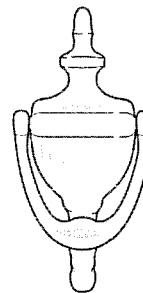
|   | 1         | 2         | 3         |
|---|-----------|-----------|-----------|
| A | aluminium | copper    | magnesium |
| B | copper    | aluminium | magnesium |
| C | copper    | aluminium | steel     |
| D | magnesium | copper    | steel     |

45. M/J 14/P12/Q26, M/J 14/P11/Q26, M/J 14/P13/Q27

The diagrams show two items that may be found in the home. Each item contains zinc.



zinc plated bucket



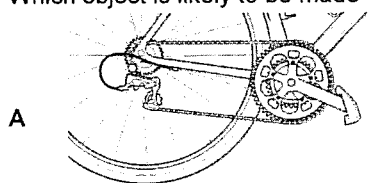
brass door-knocker

In which is zinc used as an alloy?

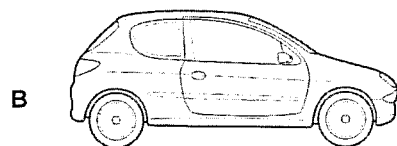
|   | bucket | door-knocker |
|---|--------|--------------|
| A | ✓      | ✓            |
| B | ✓      | x            |
| C | x      | ✓            |
| D | x      | x            |

## 46. M/J 14/P12/Q27

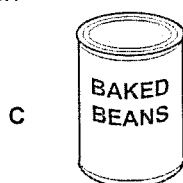
Which object is likely to be made from stainless steel?



bicycle chain



car body



can of beans



teaspoon

## 47. M/J 14/P11/Q28, M/J 14/P13/Q29

The table shows properties of four metals.

Which metal is the most suitable for aircraft construction?

|   | density | strength | resistance to corrosion |
|---|---------|----------|-------------------------|
| A | high    | high     | low                     |
| B | high    | low      | low                     |
| C | low     | high     | high                    |
| D | low     | low      | high                    |

## 48. O/N 13/P12/Q28, O/N 13/P11/Q29

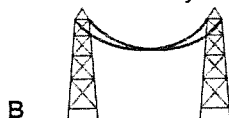
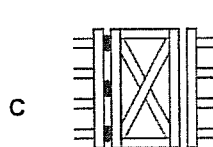
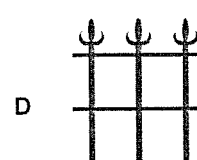
Which row describes the uses of mild steel and stainless steel?

|   | mild steel                | stainless steel           |
|---|---------------------------|---------------------------|
| A | car bodies, cutlery       | chemical plant, machinery |
| B | car bodies, machinery     | chemical plant, cutlery   |
| C | chemical plant, cutlery   | car bodies, machinery     |
| D | chemical plant, machinery | car bodies, cutlery       |

## 49. O/N 13/P12/Q30, O/N 13/P11/Q32

The diagrams show four uses of iron.

In which of these uses is the iron most likely to rust?

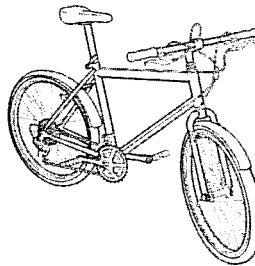
iron bucket  
electroplated  
with zinciron cored  
aluminium  
electricity cablesiron hinges  
on a gatepainted  
iron fence



**50. O/N 13/P13/Q27**

A new bicycle is being developed.

Two different materials are used in its construction, both of which must be corrosion resistant.



Which two metals could be used?

- |                                        |                                        |
|----------------------------------------|----------------------------------------|
| <b>A</b> aluminium and mild steel      | <b>C</b> mild steel and pure iron      |
| <b>B</b> aluminium and stainless steel | <b>D</b> pure iron and stainless steel |

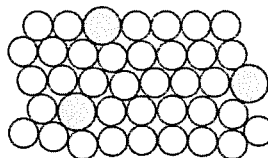
**51. M/J 13/P12/Q27**

Why is aluminium used to make food containers?

- |                                |                                 |
|--------------------------------|---------------------------------|
| <b>A</b> It has a low density. | <b>C</b> It keeps the food hot. |
| <b>B</b> It is strong.         | <b>D</b> It resists corrosion.  |

**52. M/J 13/P12/Q29**

The diagram represents the structure of substance S.



What is S?

- |                   |                         |                          |                       |
|-------------------|-------------------------|--------------------------|-----------------------|
| <b>A</b> an alloy | <b>B</b> an ionic solid | <b>C</b> a macromolecule | <b>D</b> a pure metal |
|-------------------|-------------------------|--------------------------|-----------------------|

**53. M/J 13/P11/Q27, M/J 13/P13/Q24**

Which property makes aluminium ideal for making food containers?

- |                               |                                  |
|-------------------------------|----------------------------------|
| <b>A</b> conducts electricity | <b>C</b> mechanical strength     |
| <b>B</b> conducts heat        | <b>D</b> resistance to corrosion |

**54. M/J 13/P11/Q29, M/J 13/P13/Q30**

Pure metals conduct electricity and can be hammered into different shapes.

Why are metals sometimes used as alloys?

- |                                                                 |
|-----------------------------------------------------------------|
| <b>A</b> Alloys are cheaper than the metals they are made from. |
| <b>B</b> Alloys are easier to hammer into different shapes.     |
| <b>C</b> Alloys are harder and keep their shape better.         |
| <b>D</b> Alloys conduct electricity better.                     |

## Answers Section

| 10.1  | 10.2        | 10.3        | 10.4        |
|-------|-------------|-------------|-------------|
| 1. A  | 1. D 41. C  | 1. D 41. C  | 1. D 41. B  |
| 2. D  | 2. A 42. C  | 2. D 42. D  | 2. C 42. D  |
| 3. B  | 3. A 43. A  | 3. D 43. D  | 3. D 43. B  |
| 4. A  | 4. C 44. A  | 4. C 44. C  | 4. C 44. C  |
| 5. A  | 5. D 45. B  | 5. C 45. C  | 5. C 45. C  |
| 6. D  | 6. C 46. A  | 6. A 46. C  | 6. B 46. D  |
| 7. A  | 7. C 47. A  | 7. A 47. C  | 7. A 47. C  |
| 8. A  | 8. C 48. B  | 8. B 48. D  | 8. A 48. B  |
| 9. B  | 9. C 49. C  | 9. A 49. A  | 9. D 49. C  |
| 10. C | 10. A 50. B | 10. B 50. D | 10. A 50. B |
| 11. B | 11. D 51. C | 11. D 51. B | 11. A 51. D |
| 12. B | 12. C 52. D | 12. A 52. A | 12. C 52. A |
| 13. B | 13. A 53. C | 13. B 53. A | 13. D 53. D |
| 14. B | 14. D 54. A | 14. B 54. B | 14. D 54. C |
| 15. B | 15. D 55. B | 15. D 55. A | 15. A       |
| 16. D | 16. D 56. D | 16. C 56. D | 16. D       |
| 17. B | 17. D 57. A | 17. B 57. D | 17. D       |
| 18. B | 18. B 58. A | 18. A 58. B | 18. C       |
| 19. C | 19. C 59. C | 19. B 59. A | 19. C       |
| 20. B | 20. D 60. C | 20. A 60. C | 20. C       |
| 21. A | 21. C 61. C | 21. D 61. C | 21. A       |
| 22. C | 22. B 62. B | 22. C 62. C | 22. D       |
| 23. B | 23. D 63. D | 23. D 63. A | 23. C       |
| 24. A | 24. A 64. D | 24. C 64. A | 24. D       |
| 25. B | 25. C 65. A | 25. B 65. A | 25. B       |
| 26. D | 26. B       | 26. C 66. A | 26. C       |
| 27. A | 27. C       | 27. D 67. B | 27. C       |
|       | 28. D       | 28. D 68. C | 28. B       |
|       | 29. C       | 29. D 69. A | 29. C       |
|       | 30. B       | 30. D 70. A | 30. D       |
|       | 31. B       | 31. C 71. B | 31. C       |
|       | 32. A       | 32. A 72. B | 32. D       |
|       | 33. C       | 33. A 73. B | 33. C       |
|       | 34. C       | 34. C 74. A | 34. A       |
|       | 35. A       | 35. A 75. D | 35. A       |
|       | 36. B       | 36. D 76. D | 36. C       |
|       | 37. C       | 37. A 77. C | 37. A       |
|       | 38. D       | 38. B       | 38. D       |
|       | 39. B       | 39. C       | 39. D       |
|       | 40. C       | 40. B       | 40. A       |



## Unit 11

# AIR AND WATER

IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 11.1 Water and rusting
- 11.2 Air
- 11.3 Nitrogen and fertilisers
- 11.4 Carbon dioxide and methane

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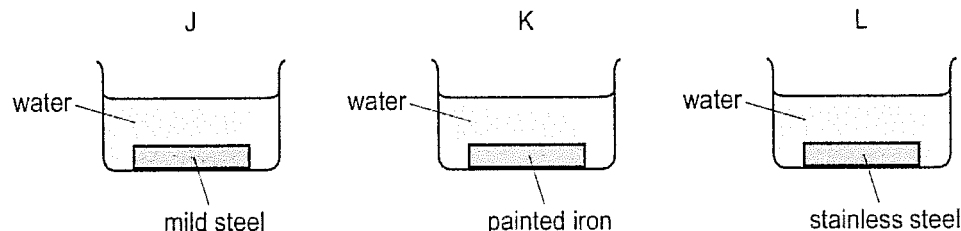
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## Unit 11: Air and Water

### 11.1: Water and rusting

1. O/N 22/P22/Q25

Three experiments, J, K and L, are set up to investigate rusting.



In which experiments does rusting occur?

|   | J | K | L |         |
|---|---|---|---|---------|
| A | x | ✓ | ✓ | key     |
| B | x | ✓ | x | ✓ = yes |
| C | ✓ | x | x | x = no  |
| D | ✓ | x | ✓ |         |

2. O/N 22/P22/Q27

Iron is galvanised by coating it in zinc.

Brass is made by mixing copper with zinc.

Which row gives the reasons for each of these uses of zinc?

|   | reason for galvanising iron | reason for making brass |
|---|-----------------------------|-------------------------|
| A | prevents corrosion          | produces a softer metal |
| B | prevents corrosion          | produces a harder metal |
| C | produces a harder metal     | produces a softer metal |
| D | produces a harder metal     | produces a harder metal |

3. O/N 22/P23/Q15

When hydrated copper(II) sulfate is heated, it produces white copper(II) sulfate. When water is added, the white copper(II) sulfate turns blue.

Which type of reaction is shown by these observations?

- A decomposition      B displacement      C redox      D reversible

4. M/J 22/P22/Q15, M/J 22/P21/Q15, M/J 22/P23/Q15

Water is added to anhydrous copper(II) sulfate.

What happens during the reaction?

- A The copper(II) sulfate turns blue and the solution formed gets colder.  
 B The copper(II) sulfate turns blue and the solution formed gets hotter.  
 C The copper(II) sulfate turns white and the solution formed gets colder.  
 D The copper(II) sulfate turns white and the solution formed gets hotter.

5. M/J 22/P22/Q30

Which process uses sacrificial protection to prevent steel from rusting?

- A galvanising      B oiling      C copper plating      D painting

**6. M/J 22/P21/Q24**

Mild steel is galvanised to prevent corrosion of the iron.

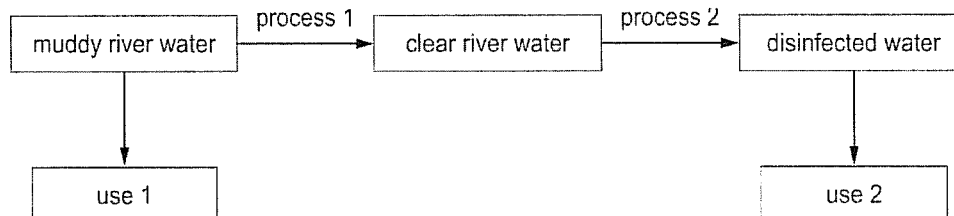
Which statements about galvanising are correct?

- 1 Galvanising prevents corrosion because the zinc forms an alloy.
- 2 If the coating is damaged, water and oxygen do not corrode the iron.
- 3 Zinc is a sacrificial metal and corrodes in preference to iron.

A 1 and 2                      B 1 and 3                      C 2 only                      D 2 and 3

**7. M/J 22/P21/Q29**

The diagram shows the uses and treatment processes of muddy river water.



Which row identifies uses 1 and 2 and processes 1 and 2?

|   | use 1          | use 2          | process 1    | process 2    |
|---|----------------|----------------|--------------|--------------|
| A | drinking       | watering crops | chlorination | filtration   |
| B | drinking       | watering crops | filtration   | chlorination |
| C | watering crops | drinking       | chlorination | filtration   |
| D | watering crops | drinking       | filtration   | chlorination |

**8. M/J 22/P23/Q30**

A magnesium block is attached to iron to prevent it from rusting.

Which statement about this method of rust prevention is correct?

- A Magnesium corrodes instead of iron because it is more reactive.
- B Magnesium prevents oxygen from reaching the iron.
- C The iron does not rust because it has a greater tendency to form ions than magnesium.
- D This method of rust prevention is called galvanising.

**9. O/N 21/P22/Q10**

Iron can be electroplated with zinc to make it resistant to corrosion.

Which row about electroplating iron with zinc is correct?

|   | positive electrode<br>(anode) | negative electrode<br>(cathode) | electrolyte  |
|---|-------------------------------|---------------------------------|--------------|
| A | iron                          | zinc                            | iron nitrate |
| B | iron                          | zinc                            | zinc nitrate |
| C | zinc                          | iron                            | iron nitrate |
| D | zinc                          | iron                            | zinc nitrate |

**10. O/N 21/P22/Q27**

Aluminium objects do not need protection from corrosion.

Iron objects must be protected from corrosion.

Which statement explains why aluminium resists corrosion?

- A Aluminium does not form ions easily.
- B Aluminium does not react with water or air.
- C Aluminium has a protective oxide layer.
- D Aluminium is below iron in the reactivity series.

**11. O/N 21/P22/Q28**

Which statements explain why zinc is used to protect iron from rusting?

- 1 Zinc is more reactive than iron.
- 2 Zinc is less reactive than iron.
- 3 Zinc can form alloys with iron.
- 4 Zinc acts as a sacrificial metal.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**12. O/N 21/P21/Q29**

Covering iron with zinc prevents the iron from rusting even when the zinc is scratched.

Covering iron with tin prevents the iron from rusting, but when the tin is scratched the iron underneath starts to rust.

Which statement is correct?

- A Both tin and zinc prevent iron from rusting by sacrificial protection.
- B Both tin and zinc prevent iron from rusting by stopping water and carbon dioxide reaching the iron.
- C Tin is more reactive than iron and prevents iron from rusting until it is scratched.
- D Zinc loses electrons more easily than iron and prevents iron from rusting by corroding first.

**13. O/N 21/P23/Q29**

Ships are made of steel, an alloy of iron.

Blocks of magnesium are attached to the underside of ships to prevent rusting.

Which statement explains how the magnesium prevents rusting?

- A Magnesium oxidises instead of iron.
- B Magnesium stops air and water getting to the iron.
- C The magnesium forms an alloy with iron which does not corrode.
- D The magnesium reacts with rust as soon as it is formed.

**14. M/J 21/P22/Q29**

Water is used for the irrigation of crops and for drinking water.

For which uses must water be chlorinated?

|   | irrigation | drinking |
|---|------------|----------|
| A | ✓          | ✓        |
| B | ✓          | x        |
| C | x          | ✓        |
| D | x          | x        |

**15. M/J 21/P23/Q23**

What are possible effects of an inadequate water supply during a drought?

- 1 crop failure
- 2 wastage of water
- 3 human disease
- 4 death of farm animals

A 1, 2 and 3                      B 1 and 2 only                      C 1, 3 and 4                      D 3 and 4 only

**16. MAR 21/P22/Q17**

P is a hydrated metal salt with a blue colour. When P is heated, water is given off, leaving solid Q.

R is a hydrated metal salt with a pink colour. When R is heated, water is given off, leaving solid S.

Which row gives the name of P and the colour of S?

|   | name of P                    | colour of S |
|---|------------------------------|-------------|
| A | hydrated cobalt(II) chloride | blue        |
| B | hydrated cobalt(II) chloride | white       |
| C | hydrated copper(II) sulfate  | blue        |
| D | hydrated copper(II) sulfate  | white       |

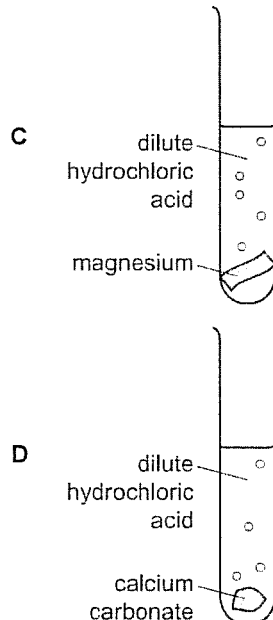
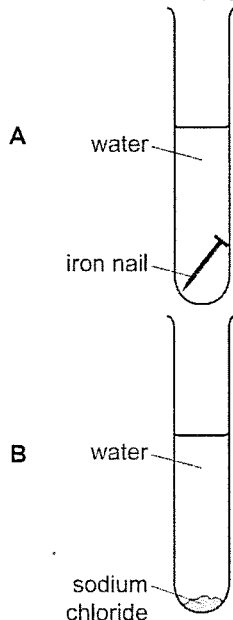
## 17. M/J 20/P22/Q29

Which statement about pure water is not correct?

- A It condenses at 100 °C.  
 B It freezes at 0 °C.  
 C It turns cobalt(II) chloride paper blue.  
 D It turns anhydrous copper(II) sulfate blue.

## 18. M/J 20/P23/Q14

In which tube is a physical change taking place?



## 19. M/J 20/P23/Q29

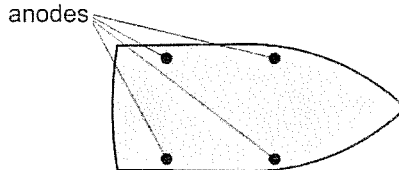
Which substances can be used to detect the presence of water?

- 1 cobalt(II) chloride  
 2 copper(II) sulfate  
 3 litmus  
 4 methyl orange

- A 1 and 2                      B 1 and 3                      C 2 and 4                      D 3 and 4

## 20. M/J 20/P23/Q32

The diagram shows the positions of sacrificial anodes on the steel hull of a yacht.



Which metal is used to make the anodes?

- A calcium                      B copper                      C sodium                      D zinc

## 21. MAR 20/P22/Q29

Water is treated at a waterworks to make it fit to drink. What is present in the water when it leaves the waterworks?

- A bacteria only                      C chlorine compounds only  
 B bacteria and insoluble substances                      D chlorine compounds and soluble substances

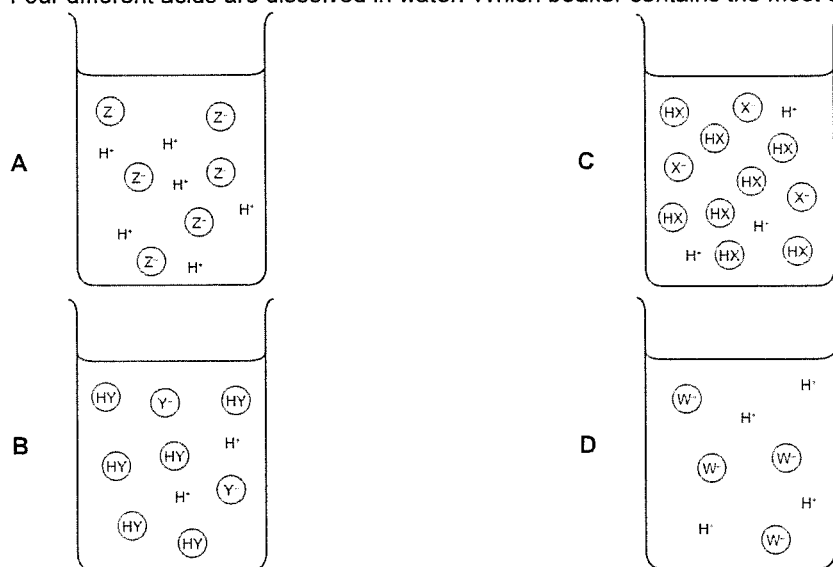
## 22. MAR 20/P22/Q32

Zinc is used to cover iron to prevent it from rusting. Why is zinc a suitable metal to use?

- A Iron is more reactive than zinc.                      C Zinc is more reactive than iron.  
 B Iron atoms are bigger than zinc atoms.                      D Zinc atoms are bigger than iron atoms

**23. MAR 20/P22/Q19**

Four different acids are dissolved in water. Which beaker contains the most concentrated strong acid solution?

**24. O/N 19/P22/Q30, O/N 19/P23/Q30**

River water contains soluble impurities, insoluble impurities and bacteria.

River water is made safe to drink by filtration and chlorination.

Which statement is correct?

- A Filtration removes bacteria and insoluble impurities, and chlorination removes soluble impurities.
- B Filtration removes insoluble impurities, and chlorination kills the bacteria.
- C Filtration removes soluble and insoluble impurities, and chlorination kills the bacteria.
- D Filtration removes soluble impurities and bacteria, and chlorination removes insoluble impurities.

**25. M/J 19/P22/Q29, M/J 19/P21/Q28, M/J 19/P23/Q28**

Water can be treated by filtration then chlorination.

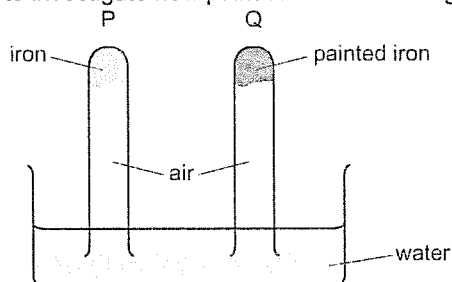
Which uses do **not** need water of this quality?

- 1 water for cooling in industry
- 2 water for washing clothes
- 3 water for drinking

- A 1, 2 and 3
- B 1 and 2 only
- C 1 and 3 only
- D 2 and 3 only

**26. M/J 19/P22/Q31, M/J 19/P21/Q30, M/J 19/P23/Q30**

The diagram shows an experiment to investigate how paint affects the rusting of iron.



What happens to the water level in tubes P and Q?

|   | tube P    | tube Q    |
|---|-----------|-----------|
| A | falls     | rises     |
| B | no change | rises     |
| C | rises     | falls     |
| D | rises     | no change |



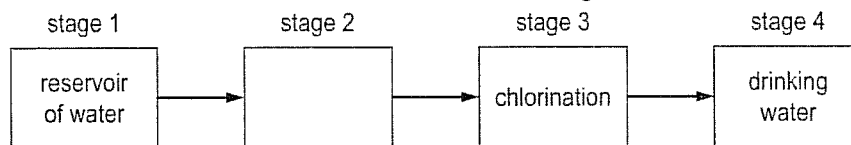
**27. M/J 19/P21/Q14**

When blue-green crystals of nickel(II) sulfate are heated, water is produced and a yellow solid remains. When water is added to the yellow solid, the blue-green colour returns. Which process describes these changes?

- A combustion      B corrosion      C neutralisation      D reversible reaction

**28. MAR 19/P22/Q31**

The diagram shows how water is treated to make it suitable for drinking.

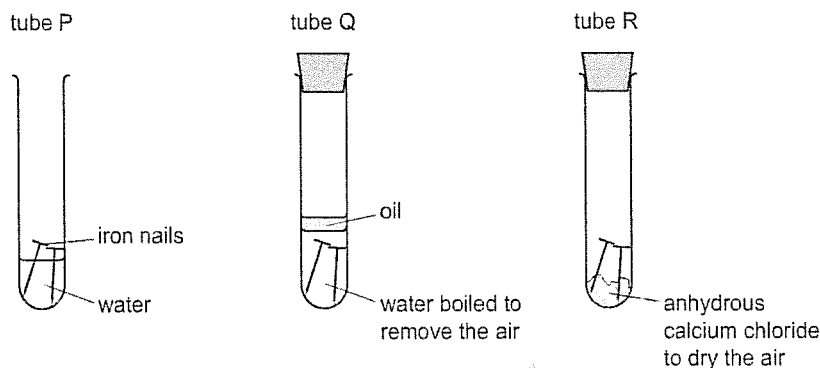


What happens in stage 2?

- A condensation      B sublimation      C evaporation      D filtration

**29. O/N 18/P22/Q31, O/N 18/P21/Q31, O/N 18/P23/Q31, MAR 15/P12/Q30**

The diagrams show experiments involving the rusting of iron.



A student predicted the following results.

- 1 In tube P, the iron nails rust.
- 2 In tube Q, the iron nails do not rust.
- 3 In tube R, the iron nails do not rust.

Which predictions are correct?

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

**30. M/J 18/P22/Q30, M/J 18/P21/Q30, M/J 18/P23/Q30, O/N 15/P12/Q27, O/N 15/P11/Q27, O/N 15/P13/Q27, M/J 10/P12/Q34, M/J 10/P11/Q32, M/J 10/P13/Q34**

Which statements about water are correct?

- 1 Household water contains dissolved salts.
- 2 Water for household use is filtered to remove soluble impurities.
- 3 Water is treated with chlorine to kill bacteria.
- 4 Water is used in industry for cooling.

- A 1, 2, 3 and 4      B 1, 2 and 3 only      C 1, 3 and 4 only      D 2, 3 and 4 only

**31. MAR 18/P22/Q29, O/N 09/P12/Q32, O/N 09/P11/Q35**

Water must be purified before it is suitable for use in the home.

Which processes are used to remove solid impurities and to kill bacteria?

|   | to remove solid impurities | to kill bacteria |
|---|----------------------------|------------------|
| A | chlorination               | chlorination     |
| B | chlorination               | filtration       |
| C | filtration                 | chlorination     |
| D | filtration                 | filtration       |

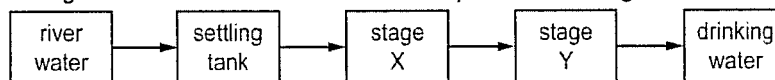
## 32. O/N 17/P23/Q30

A piece of zinc is attached to the hull of a steel boat. Steel is an alloy of iron. Which statement explains why the zinc prevents the iron from rusting?

- A Zinc is less reactive than iron, and iron is less likely to lose electrons than zinc.
- B Zinc is less reactive than iron, and iron is more likely to lose electrons than zinc.
- C Zinc is more reactive than iron, and iron is less likely to lose electrons than zinc.
- D Zinc is more reactive than iron, and iron is more likely to lose electrons than zinc.

## 33. O/N 17/P22/Q29, O/N 17/P21/Q29, O/N 17/P23/Q29

The flow chart shows stages in the treatment of river water to produce drinking water.

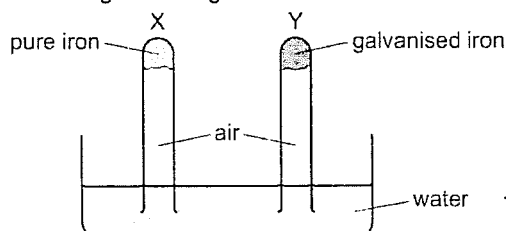


What occurs at stages X and Y?

|   | X            | Y            |
|---|--------------|--------------|
| A | distillation | chlorination |
| B | distillation | filtration   |
| C | filtration   | chlorination |
| D | filtration   | distillation |

## 34. O/N 17/P22/Q30

An experiment to investigate the effect of galvanising iron is shown.



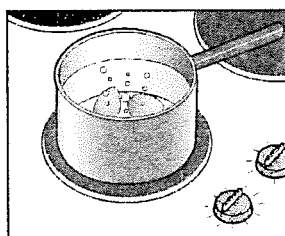
The experiment is left for seven days.

What happens to the water level in tubes X and Y?

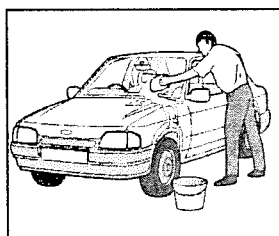
|   | tube X    | tube Y    |
|---|-----------|-----------|
| A | falls     | rises     |
| B | no change | no change |
| C | rises     | falls     |
| D | rises     | no change |

## 35. M/J 17/P22/Q29, M/J 17/P21/Q29, M/J 17/P23/Q29, M/J 14/P12/Q30, M/J 14/P11/Q30, M/J 14/P13/Q28

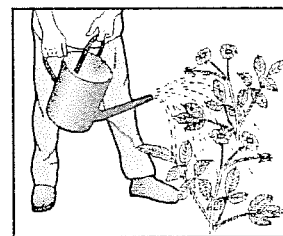
The diagram shows some uses of water in the home.



1



2



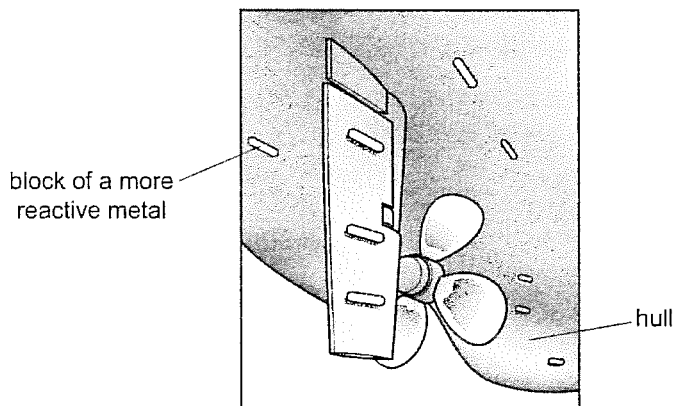
3

For which uses is it important for the water to have been treated?

- A 1 only
- B 2 only
- C 3 only
- D 1, 2 and 3

**36. O/N 16/P22/Q31**

Boats made from steel can be protected from rusting by attaching blocks of a more reactive metal to the hull of the boat.



Which statement is correct?

- A Copper is used for the blocks because it does not react with water.
- B Magnesium is not used for the blocks because it reacts with steel.
- C The metal used for the blocks loses electrons more easily than steel.
- D This form of protection from rusting is called galvanising.

**37. O/N 16/P21/Q31**

Underwater steel pipes can be protected from corrosion by attaching magnesium blocks to them. Which equation represents the reaction that prevents corrosion?

- A  $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$     B  $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$     C  $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$     D  $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$

**38. O/N 16/P23/Q31**

A metal, X, is used to make oil pipelines.

X corrodes in air and water.

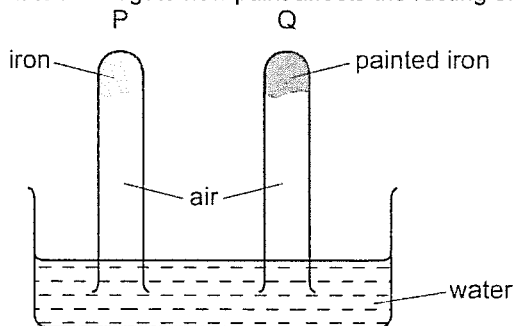
X can be protected from corrosion by attaching blocks of element Y.

Which statement is correct?

- A This process is known as galvanising.
- B Y forms positive ions more readily than X.
- C Y is an unreactive metal.
- D Y is an unreactive non-metal.

**39. M/J 16/P22/Q29, M/J 16/P21/Q29, M/J 16/P23/Q29, O/N 12/P12/Q31, O/N 12/P11/Q31**

The diagram shows an experiment to investigate how paint affects the rusting of iron.



What happens to the water level in tubes P and Q?

|   | tube P    | tube Q    |
|---|-----------|-----------|
| A | falls     | rises     |
| B | no change | rises     |
| C | rises     | falls     |
| D | rises     | no change |

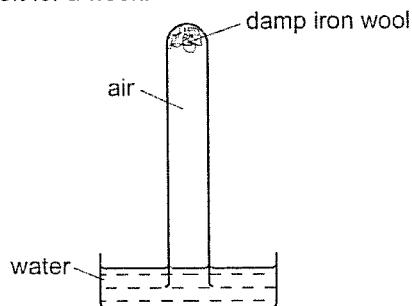
**40. MAR 16/P22/Q28**

One method of preventing the rusting of iron is to keep oxygen away from the surface of the metal. Which way of rust prevention does **not** use this method?

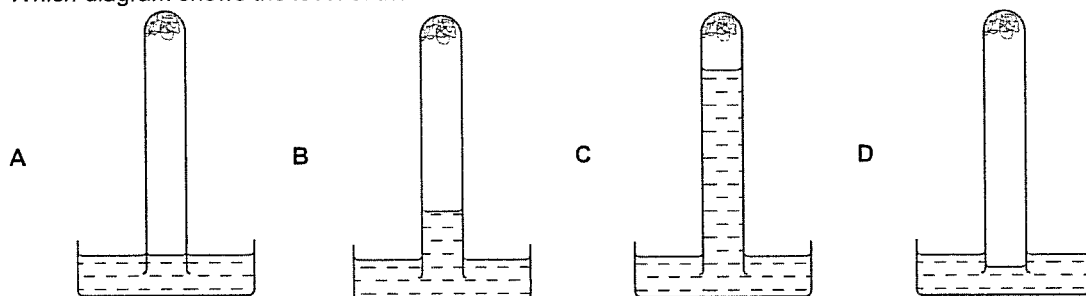
- A coating the iron with grease      C covering the iron with plastic  
B connecting the iron to a more reactive metal      D painting the iron

**41. O/N 15/P12/Q32, O/N 15/P11/Q32, O/N 15/P13/Q32**

The apparatus shown is set up and left for a week.



Which diagram shows the level of the water at the end of the week?

**42. O/N 15/P11/Q14**

Some crystals of hydrated cobalt(II) chloride are heated in a test-tube until no further change is observed. The test-tube is allowed to cool and a few drops of water are then added to the contents. Which colours are observed?

|   | before heating | after heating | after adding water |
|---|----------------|---------------|--------------------|
| A | blue           | pink          | blue               |
| B | blue           | white         | blue               |
| C | pink           | blue          | pink               |
| D | white          | blue          | white              |

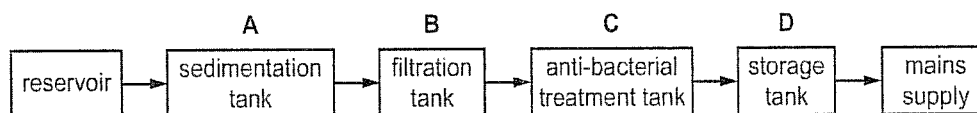
**43. O/N 15/P13/Q14**

If anhydrous copper(II) sulfate is added to water, which colour change is observed?

- A blue to pink      B blue to white      C pink to blue      D white to blue

**44. M/J 15/P12/Q27, M/J 15/P11/Q27, M/J 15/P13/Q27**

The diagram shows stages in producing drinking water. In which tank is chlorine added to the water?





**52. M/J 14/P12/Q31**

A piece of uncoated iron and three pieces of iron with various coatings were left exposed to the air. Which piece of iron would rust?

- A** the painted piece                      **C** the uncoated piece  
**B** the tin-coated piece                  **D** the zinc-coated piece

53. M/J 14/P11/Q31, M/J 14/P13/Q33

Four steel paper clips are treated as described before being placed in a beaker of water. Which paper clip rusts most quickly?

- A coated with grease  
B dipped in paint and allowed to dry  
C electroplated with zinc  
D washed with soap and rinsed

54. O/N 13/P12/Q33, O/N 13/P11/Q34

In many countries river water is used for the washing of clothes. The same water is not considered to be safe for drinking.

Why is it **not** safe for drinking?

- A because river water contains dissolved salts  
B because river water may contain harmful bacteria  
C because river water may contain small particles of sand  
D because river water may contain soap from washing clothes

55. O/N 13/P13/Q32

Iron rusts. This process involves the .....1..... of iron. Rusting can be prevented by covering the iron with grease or paint which stops .....2..... from reaching the surface of the iron.

Which words correctly complete gaps 1 and 2?

|          | 1         | 2        |
|----------|-----------|----------|
| <b>A</b> | oxidation | nitrogen |
| <b>B</b> | oxidation | oxygen   |
| <b>C</b> | reduction | nitrogen |
| <b>D</b> | reduction | oxygen   |

56. O/N 13/P13/Q34

Water is treated at a water works to make it fit to drink.

What is present in the water when it leaves the waterworks?

- A bacteria only  
B bacteria and insoluble substances  
C chlorine only  
D chlorine and soluble substances

57. M/J 13/P12/Q17

Heating pink cobalt(II) chloride crystals forms a blue solid and steam.

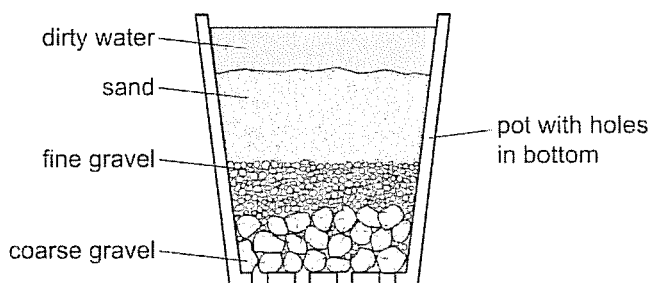
The blue solid turns pink when water is added.

Which terms describe the pink cobalt(II) chloride and the reaction?

|          | pink cobalt(II) chloride is | the reaction is reversible |
|----------|-----------------------------|----------------------------|
| <b>A</b> | anhydrous                   | yes                        |
| <b>B</b> | anhydrous                   | no                         |
| <b>C</b> | hydrated                    | yes                        |
| <b>D</b> | hydrated                    | no                         |

**58. M/J 13/P12/Q31**

The diagram shows a stage in the purification of dirty water.

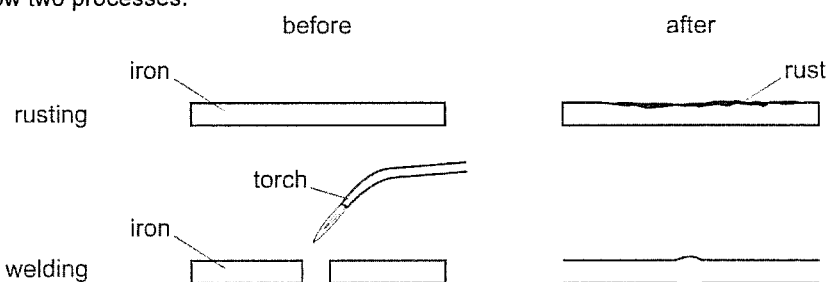


Which process does this apparatus show?

- A** chlorination      **B** condensation      **C** distillation      **D** filtration

**59. M/J 13/P12/Q32**

The diagrams show two processes.



For which processes is oxygen involved?

|          | rusting | welding |
|----------|---------|---------|
| <b>A</b> | ✓       | ✓       |
| <b>B</b> | ✓       | x       |
| <b>C</b> | x       | ✓       |
| <b>D</b> | x       | x       |

**60. M/J 13/P11/Q31, M/J 13/P13/Q31**

Water has been contaminated with sea-water.

Which substances can be removed by chlorination and filtration?

- A** bacteria, sand and sodium chloride      **C** bacteria and sodium chloride only  
**B** bacteria and sand only      **D** sand and sodium chloride only

**61. M/J 13/P11/Q32, M/J 13/P13/Q33**

Iron rusts when it reacts with .....1.....

Rusting can be prevented by covering the iron with a more reactive metal, such as .....2.....

Which words correctly complete gaps 1 and 2?

|          | 1                | 2         |
|----------|------------------|-----------|
| <b>A</b> | oxygen           | copper    |
| <b>B</b> | oxygen           | magnesium |
| <b>C</b> | oxygen and water | copper    |
| <b>D</b> | oxygen and water | magnesium |

**11.2: Air****1. M/J 21/P23/Q27**

Which gas is an air pollutant that causes acid rain?

- A argon                      B carbon monoxide      C methane                      D nitrogen dioxide

**2. MAR 20/P22/Q31**

Oxides of nitrogen, such as NO and NO<sub>2</sub>, are formed in the petrol engines of cars. They are removed from the exhaust gases by reactions in the car's catalytic converter. Which row describes how oxides of nitrogen are formed in a petrol engine, and a reaction that happens in the catalytic converter?

|   | how oxides of nitrogen are formed                                            | a reaction that happens in the catalytic converter                      |
|---|------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| A | by the reaction between nitrogen and oxygen from the air                     | $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$         |
| B | by the reaction between nitrogen and oxygen from the air                     | $2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$ |
| C | by the reaction between nitrogen compounds in petrol and oxygen from the air | $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$         |
| D | by the reaction between nitrogen compounds in petrol and oxygen from the air | $2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$ |

**3. O/N 18/P22/Q30, O/N 18/P21/Q30, O/N 18/P23/Q30**

Argon is a noble gas used to fill light bulbs.

What is the approximate percentage of argon in air?

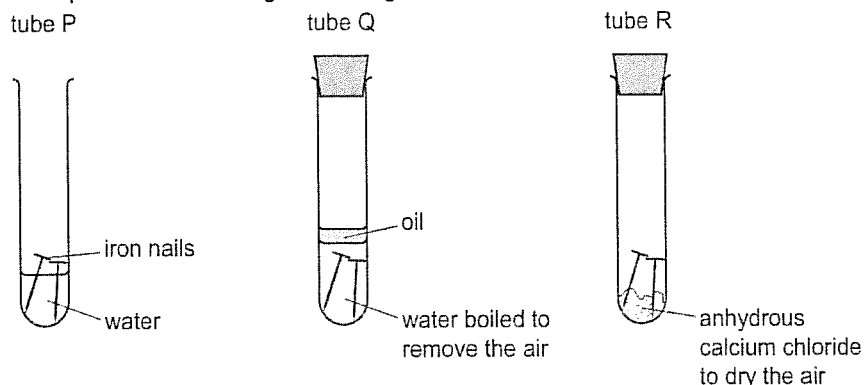
- A 1%                      B 20%                      C 79%                      D 99%

**4. O/N 18/P22/Q34**What is **not** a use of lime?

- A It is used as a bleach in the manufacture of wood pulp.  
 B It is used to desulfurise flue gases.  
 C It is used to neutralise acidic industrial waste.  
 D It is used to treat acidic soil.

**5. O/N 18/P23/Q31**

The diagrams show experiments involving the rusting of iron.



A student predicted the following results.

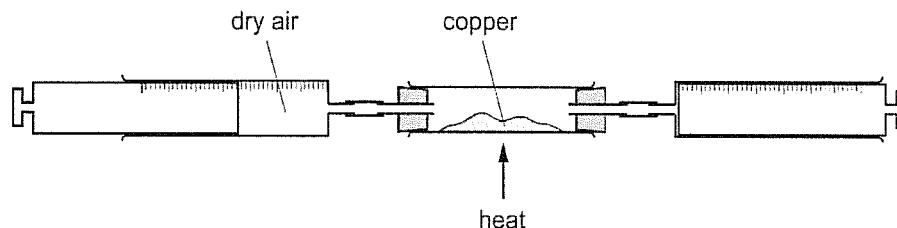
- 1 In tube P, the iron nails rust.
- 2 In tube Q, the iron nails do not rust.
- 3 In tube R, the iron nails do not rust.

Which predictions are correct?

- A 1, 2 and 3                      B 1 and 2 only                      C 1 and 3 only                      D 2 and 3 only



6. **M/J 18/P22/Q28, M/J 18/P21/Q28, M/J 18/P23/Q28, O/N 11/P12/Q32, O/N 11/P11/Q31, O/N 11/P13/Q33**  
Dry air is passed over hot copper until all the oxygen has reacted.



The volume of gas at the end of the reaction is 120 cm<sup>3</sup>.

What is the starting volume of dry air?

- A 132 cm<sup>3</sup>      B 152 cm<sup>3</sup>      C 180 cm<sup>3</sup>      D 570 cm<sup>3</sup>
7. **M/J 18/P23/Q33**  
Element Z forms an oxide, ZO<sub>2</sub>. Three uses of ZO<sub>2</sub> are listed.
- bleaching agent
  - killing bacteria
  - manufacturing an important acid
- What is Z?
- A carbon      B lead      C nitrogen      D sulfur
8. **MAR 18/P22/Q28**  
Air is a mixture of gases.  
The melting and boiling points of some gases present in clean, dry air are shown.  
In the fractional distillation of liquid air, which gas boils first?

|   | gas      | melting point / °C | boiling point / °C |
|---|----------|--------------------|--------------------|
| A | argon    | -189               | -186               |
| B | krypton  | -157               | -153               |
| C | nitrogen | -210               | -196               |
| D | oxygen   | -219               | -183               |

9. **MAR 18/P22/Q32, M/J 14/P12/Q35, M/J14/P11/Q35, M/J 14/P13/Q37, O/N 11/P12/Q34, O/N 11/P11/Q35, O/N 11/P13/Q34**  
Which pollutant gas is produced by the decomposition of vegetation?
- A carbon monoxide      B methane      C nitrogen dioxide      D sulfur dioxide
10. **M/J 17/P22/Q34**  
Statements about methods of manufacture and uses of calcium oxide are shown.
- 1 It is manufactured by reacting acids with calcium carbonate.
  - 2 It is manufactured by heating calcium carbonate.
  - 3 It is used to desulfurise flue gases.
  - 4 It is used to treat alkaline soil.
- Which statements are correct?
- A 1 and 2      B 1 and 4      C 2 and 3      D 3 and 4
11. **M/J 17/P21/Q30**  
Oxides of nitrogen are found in polluted air.  
Which statement about oxides of nitrogen is correct?
- A Oxides of nitrogen are formed by the reaction of nitrogen with oxygen during the fractional distillation of liquid air.
- B Oxides of nitrogen are formed in a car engine by the reaction of petrol with nitrogen from the air.
- C Oxides of nitrogen are removed from exhaust gases by reaction with carbon dioxide in a catalytic converter.
- D Oxides of nitrogen are removed from exhaust gases by reduction in a catalytic converter.

## 12. M/J 17/P21/Q31

Photosynthesis and respiration are important natural processes.  
Which statement is correct?

- A Carbon dioxide is formed by the reaction of glucose with water during photosynthesis.
- B Carbon dioxide is removed from the air by respiration.
- C Glucose reacts with water to form oxygen during respiration.
- D Photosynthesis produces glucose and oxygen.

## 13. M/J 17/P21/Q34

Which statement is **not** correct?

- A Converting limestone into lime is a thermal decomposition reaction.
- B Flue gas desulfurisation is a neutralisation reaction.
- C In the extraction of iron, calcium carbonate is converted into calcium oxide.
- D Slaked lime is added to soil as a fertiliser.

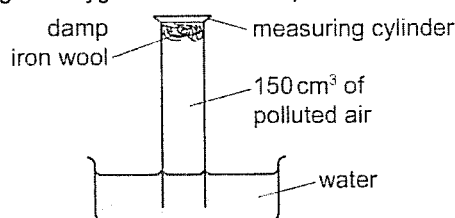
## 14. M/J 17/P23/Q30

The carbon cycle describes how carbon dioxide gas is added to or removed from the atmosphere.  
Which row describes the movement of carbon dioxide during each process?

|   | photosynthesis              | combustion                  | respiration                 |
|---|-----------------------------|-----------------------------|-----------------------------|
| A | added to the atmosphere     | added to the atmosphere     | removed from the atmosphere |
| B | added to the atmosphere     | removed from the atmosphere | added to the atmosphere     |
| C | removed from the atmosphere | added to the atmosphere     | added to the atmosphere     |
| D | removed from the atmosphere | added to the atmosphere     | removed from the atmosphere |

## 15. MAR 17/P22/Q31

An experiment to find the percentage of oxygen in 150 cm<sup>3</sup> of polluted air is shown.



The apparatus is left for one week.

After this time, the volume of gas in the measuring cylinder is 122 cm<sup>3</sup>.

What is the percentage of oxygen, to the nearest whole number, in the polluted air?

- A 19%
- B 21%
- C 28%
- D 81%

## 16. O/N 16/P22/Q30, O/N 16/P21/Q30, O/N 16/P23/Q30, M/J 13/P12/Q34, M/J 13/P11/Q34, M/J 13/P13/Q32, O/N 10/P12/Q35, O/N 10/P11/Q34, O/N 10/P13/Q36

Which information about carbon dioxide and methane is correct?

|   |                                   | carbon dioxide | methane |
|---|-----------------------------------|----------------|---------|
| A | formed when vegetation decomposes | ✓              | x       |
| B | greenhouse gas                    | ✓              | ✓       |
| C | present in unpolluted air         | x              | x       |
| D | produced during respiration       | x              | ✓       |

key

✓ = true

x = false

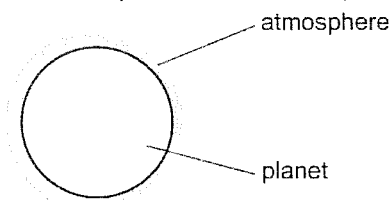
**17. MAR 17/P22/Q30**

Which two gases are obtained from liquid air by fractional distillation?

- A carbon dioxide and oxygen      C nitrogen and oxygen  
B carbon dioxide and water vapour      D nitrogen and water vapour

**18. M/J 16/P22/Q30, M/J 16/P21/Q30, M/J 16/P23/Q30, M/J 12/P12/Q29, M/J 12/P11/Q31, M/J 12/P13/Q32, O/N 09/P12/Q36, O/N 09/P11/Q34**

A new planet has been discovered and its atmosphere has been analysed.



The table shows the composition of its atmosphere.

| gas            | percentage by volume |
|----------------|----------------------|
| carbon dioxide | 4                    |
| nitrogen       | 72                   |
| oxygen         | 24                   |

Which gases are present in the atmosphere of the planet in a higher percentage than they are in the Earth's atmosphere?

- A carbon dioxide and oxygen      C nitrogen and oxygen  
B carbon dioxide only      D nitrogen only

**19. M/J 16/P22/Q31**

The gases coming from a car's engine contain oxides of nitrogen.

How are these oxides formed?

- A Nitrogen reacts with carbon dioxide.      C Nitrogen reacts with oxygen.  
B Nitrogen reacts with carbon monoxide.      D Nitrogen reacts with petrol.

**20. M/J 16/P22/Q34**

Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

- A ammonia      B sodium chloride      C sodium hydroxide      D sulfuric acid

**21. M/J 16/P21/Q31**

Many car exhaust systems contain a catalytic converter.

Which change does **not** occur in a catalytic converter?

- A carbon dioxide → carbon  
B carbon monoxide → carbon dioxide  
C nitrogen oxides → nitrogen  
D unburnt hydrocarbons → carbon dioxide and water

**22. M/J 16/P23/Q31**

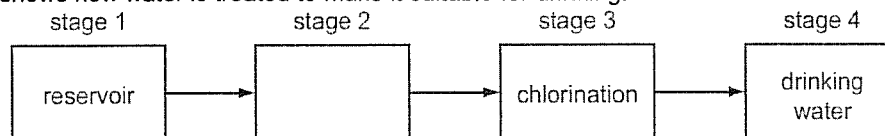
Catalytic converters are used to remove some gaseous pollutants from car exhaust fumes.

Which gas is removed from the fumes by oxidation?

- A carbon dioxide      B carbon monoxide      C nitrogen      D nitrogen oxide

**23. MAR 16/P22/Q29**

The diagram shows how water is treated to make it suitable for drinking.



What happens in stage 2?

- A condensation      B distillation      C evaporation      D filtration

**24. O/N 15/P12/Q28**

Which gas is a pollutant of the air?

- A argon                      B carbon dioxide                      C nitrogen                      D sulfur dioxide

**25. O/N 15/P12/Q29, O/N 15/P11/Q29, O/N 15/P13/Q29**

Carbon monoxide is an air pollutant produced when petrol is burned in a car engine.

Why is carbon monoxide considered to be an air pollutant?

- A It causes climate change.                      C It is a significant greenhouse gas.  
B It causes the corrosion of buildings.                      D It is poisonous.

**26. O/N 15/P12/Q33**

A farmer moves his cows into a concrete shelter for protection.

There is little access for fresh air once the door is closed.

Which gases would increase in amount in the shelter?

- A carbon dioxide and carbon monoxide                      C carbon monoxide and oxygen  
B carbon dioxide and methane                      D methane and oxygen

**27. O/N 15/P11/Q28**

Which is a use of oxygen?

- A as the gas in a lamp                      C to react with methane in a Bunsen burner  
B to react with ethene to form ethanol                      D to react with hematite to form iron

**28. O/N 15/P11/Q33**

Carbon dioxide and methane both contribute to climate change.

Which process produces both gases?

- A complete combustion of natural gas                      C heating calcium carbonate  
B farming cattle                      D respiration

**29. O/N 15/P13/Q33**

Unwanted vegetation is sometimes placed in a bin where it decays to form compost. This compost can be used to fertilise soils.

Which gas is likely to be present in a higher percentage inside the bin than in the air outside the bin?

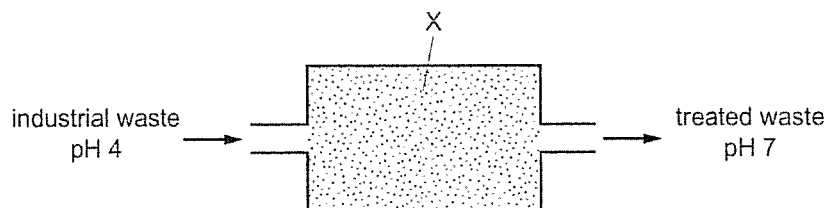
- A carbon monoxide    B methane                      C oxygen                      D sulfur dioxide

**30. M/J 15/P12/Q28**Which gas is **not** present in a sample of clean air?

- A carbon dioxide    B chlorine                      C oxygen                      D water vapour

**31. M/J 15/P12/Q34**

Substance X is used to treat industrial waste.



What is X and which type of reaction occurs during the treatment?

|   | X                    | type of reaction |
|---|----------------------|------------------|
| A | calcium oxide (lime) | neutralisation   |
| B | calcium oxide (lime) | redox            |
| C | carbon               | neutralisation   |
| D | carbon               | redox            |

**32. M/J 15/P12/Q30**

Carbon monoxide is given out from the exhaust of vehicles that burn fossil fuels.  
Which row shows why carbon monoxide is a pollutant?

|          | acidic | toxic |
|----------|--------|-------|
| <b>A</b> | no     | no    |
| <b>B</b> | no     | yes   |
| <b>C</b> | yes    | no    |
| <b>D</b> | yes    | yes   |

**33. M/J 15/P13/Q33**

Four reactions produce carbon dioxide.

- 1 respiration
- 2 fermentation
- 3 combustion of methane
- 4 manufacture of lime

Which reactions do **not** use oxygen from the air?

- A** 1 and 2                      **B** 1 and 3                      **C** 2 and 4                      **D** 3 and 4

**34. MAR 15/P12/Q24**

Which of these gases is an atmospheric pollutant?

- 1 carbon monoxide
- 2 nitrogen dioxide
- 3 sulfur dioxide

- A** 1 only                      **B** 2 only                      **C** 3 only                      **D** 1, 2 and 3

**35. MAR 15/P12/Q28**

A sample of air from a town centre was analysed and found to contain mainly nitrogen and oxygen, but also traces of the four gases below.

Which of these gases is a pollutant?

- A** argon                      **B** carbon dioxide                      **C** sulfur dioxide                      **D** water vapour

**36. MAR 15/P12/Q31, M/J 12/P11/Q32, M/J 12/P13/Q34**

Gas X is a waste gas from digestion in animals.

Gas Y is formed when gas X is burnt with a small amount of oxygen.

Gas Z is formed when gas X is burnt with an excess of oxygen.

What are X, Y and Z?

|          | X               | Y               | Z               |
|----------|-----------------|-----------------|-----------------|
| <b>A</b> | carbon dioxide  | methane         | carbon monoxide |
| <b>B</b> | carbon monoxide | methane         | carbon dioxide  |
| <b>C</b> | methane         | carbon dioxide  | carbon monoxide |
| <b>D</b> | methane         | carbon monoxide | carbon dioxide  |

**37. O/N 14/P13/Q35**

A man blows up a balloon.

What is the approximate composition of his exhaled air in the balloon?

|          | % composition  |        |          |
|----------|----------------|--------|----------|
|          | carbon dioxide | oxygen | nitrogen |
| <b>A</b> | 0.03           | 20     | 79       |
| <b>B</b> | 0.03           | 79     | 20       |
| <b>C</b> | 4              | 16     | 79       |
| <b>D</b> | 4              | 20     | 75       |

**38. O/N 14/P12/Q33, O/N 14/P11/Q35**

Which is an air pollutant that affects a part of the body other than the lungs and blood system?

- A lead compounds    B nitrogen    C oxides of nitrogen    D sulfur dioxide

**39. M/J 14/P12/Q33**

Sulfur dioxide,  $\text{SO}_2$ , nitrogen dioxide,  $\text{NO}_2$ , and carbon monoxide,  $\text{CO}$ , are air pollutants.

Which row correctly shows their major source?

|   | motor car engines             | power stations                |
|---|-------------------------------|-------------------------------|
| A | CO                            | $\text{NO}_2$ , $\text{SO}_2$ |
| B | $\text{NO}_2$ , CO            | $\text{SO}_2$                 |
| C | $\text{SO}_2$ , $\text{NO}_2$ | CO                            |
| D | $\text{SO}_2$                 | $\text{NO}_2$ , CO            |

**40. O/N 13/P12/Q35, O/N 13/P11/Q33**

Which air pollutant is **not** made when coal burns in a power station?

- A carbon monoxide    B lead compounds    C nitrogen oxides    D sulfur dioxide

**41. O/N 13/P13/Q33**

Oxides of nitrogen are given out from car exhausts.

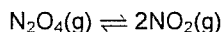
Which row best shows why oxides of nitrogen are unwanted?

|   | acidic | toxic |
|---|--------|-------|
| A | no     | no    |
| B | no     | yes   |
| C | yes    | no    |
| D | yes    | yes   |

## 11.3: Nitrogen and fertilisers

### 1. O/N 22/P22/Q16, O/N 22/P23/Q13

Dinitrogen tetroxide,  $\text{N}_2\text{O}_4$ , is converted into nitrogen dioxide,  $\text{NO}_2$ , in a reversible reaction.



The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

|   | pressure<br>/ atmospheres | temperature |
|---|---------------------------|-------------|
| A | 2                         | high        |
| B | 2                         | low         |
| C | 50                        | high        |
| D | 50                        | low         |

### 2. O/N 22/P22/Q30

Ammonium nitrate,  $\text{NH}_4\text{NO}_3$ , is a fertiliser and is added to fields to help crops grow.

Slaked lime,  $\text{Ca}(\text{OH})_2$ , is an alkali and is added to fields to reduce the acidity of the soil.

Ammonium nitrate and slaked lime should not be added to a field at the same time because they react with each other to form a gas, Z.

What is Z?

- A ammonia                      B hydrogen                      C nitrogen                      D oxygen

### 3. O/N 22/P21/Q30

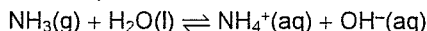
Which row explains why a high temperature and an iron catalyst are used in the manufacture of ammonia by the Haber process?

|   | high temperature                           | iron catalyst                              |
|---|--------------------------------------------|--------------------------------------------|
| A | increases the rate of the reaction         | increases the equilibrium yield of ammonia |
| B | increases the rate of the reaction         | increases the rate of the reaction         |
| C | increases the equilibrium yield of ammonia | increases the equilibrium yield of ammonia |
| D | increases the equilibrium yield of ammonia | increases the rate of the reaction         |

### 4. O/N 22/P23/Q20

Ammonia,  $\text{NH}_3$ , dissolves in water to form a dilute solution of ammonium hydroxide,  $\text{NH}_4\text{OH}$ .

The reaction is reversible and exists as an equilibrium mixture.



Which statement about the mixture is correct?

- A All of the ammonia and water molecules have turned into ions.  
 B The ammonia and water molecules have stopped changing into ions.  
 C The concentrations of the ammonia molecules and ammonium ions are always equal.  
 D The rate of the formation of ammonia molecules is equal to the rate of formation of the ammonium ions.

### 5. O/N 22/P23/Q27

Which statement about the manufacture of ammonia is correct?

- A Ammonia is manufactured by heating hydrogen and nitrogen at  $50^\circ\text{C}$  and 1.0 atm.  
 B Ammonia is obtained by heating hydrogen and nitrogen in the Contact process.  
 C Hydrogen for the manufacture of ammonia is extracted from air.  
 D The reaction between hydrogen and nitrogen to form ammonia is reversible.

**6. M/J 22/P22/Q31, M/J 22/P21/Q31, M/J 22/P23/Q31**

Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

**A**  $\text{Ca}(\text{NO}_3)_2$  and  $(\text{NH}_4)_2\text{SO}_4$

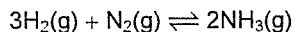
**C**  $\text{KNO}_3$  and  $(\text{NH}_4)_2\text{SO}_4$

**B**  $\text{Ca}(\text{NO}_3)_2$  and  $(\text{NH}_4)_3\text{PO}_4$

**D**  $\text{KNO}_3$  and  $(\text{NH}_4)_3\text{PO}_4$

**7. M/J 22/P21/Q30**

The equation for the manufacture of ammonia in the Haber process is shown.



The forward reaction is exothermic.

Which row describes the effect of the stated change on the reaction rate and the yield of ammonia?

|          | change               | effect on reaction rate | effect on yield of ammonia |
|----------|----------------------|-------------------------|----------------------------|
| <b>A</b> | decrease pressure    | increases               | decreases                  |
| <b>B</b> | decrease temperature | decreases               | increases                  |
| <b>C</b> | increase pressure    | increases               | decreases                  |
| <b>D</b> | increase temperature | increases               | increases                  |

**8. MAR 22/P22/Q29**

Ammonia is produced using the Haber process.

Which row shows the source of the raw materials and the reaction conditions?

|          | source of nitrogen | source of hydrogen | temperature / °C | pressure / atm |
|----------|--------------------|--------------------|------------------|----------------|
| <b>A</b> | air                | hydrocarbons       | 200              | 200            |
| <b>B</b> | hydrocarbons       | air                | 450              | 2              |
| <b>C</b> | air                | hydrocarbons       | 450              | 200            |
| <b>D</b> | air                | hydrocarbons       | 450              | 2              |

**9. O/N 21/P22/Q29**

Which conditions are used in the Haber process?

|          | temperature / °C | pressure / atmospheres |
|----------|------------------|------------------------|
| <b>A</b> | 100              | 10                     |
| <b>B</b> | 450              | 10                     |
| <b>C</b> | 450              | 200                    |
| <b>D</b> | 1000             | 500                    |

**10. O/N 21/P21/Q30**

Which statements about the Haber process are correct?

1 One of the raw materials is extracted from liquid air by fractional distillation.

2 One of the raw materials is produced by the reaction of steam and methane.

3 The catalyst for the Haber process is vanadium(V) oxide.

**A** 1 only

**B** 1 and 2 only

**C** 2 and 3 only

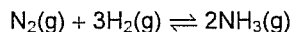
**D** 1, 2 and 3



**11. O/N 21/P21/Q14**

Ammonia is made by reacting nitrogen with hydrogen.

The equation for the reaction is shown.



The forward reaction is exothermic.

Which changes in temperature and pressure decrease the yield of ammonia?

|          | temperature | pressure |
|----------|-------------|----------|
| <b>A</b> | decrease    | decrease |
| <b>B</b> | decrease    | increase |
| <b>C</b> | increase    | decrease |
| <b>D</b> | increase    | increase |

**12. O/N 21/P23/Q28**

Which statement describes how oxides of nitrogen are formed in a car engine?

- A** Nitrogen from the air reacts with oxygen from petrol.
- B** Nitrogen from the air reacts with oxygen from the air.
- C** Nitrogen from petrol reacts with oxygen from petrol.
- D** Nitrogen from petrol reacts with oxygen from the air.

**13. O/N 21/P23/Q30**

Which process is used to produce hydrogen for the Haber process?

- A** electrolysis of water
- C** reacting iron with sulfuric acid
- B** reacting aluminium with sodium hydroxide
- D** reacting methane with steam

**14. M/J 21/P22/Q30**

Which natural resource **cannot** provide a raw material for the manufacture of ammonia?

- A** air
- B** limestone
- C** petroleum
- D** water

**15. M/J 21/P22/Q31**

Ammonia is made in the Haber process.

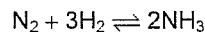
Which conditions are used in the Haber process?

|          | temperature / °C | pressure / atmospheres | catalyst used     |
|----------|------------------|------------------------|-------------------|
| <b>A</b> | 450              | 200                    | iron              |
| <b>B</b> | 450              | 5                      | vanadium(V) oxide |
| <b>C</b> | 200              | 450                    | iron              |
| <b>D</b> | 200              | 5                      | vanadium(V) oxide |

**16. M/J 21/P21/Q29**

Ammonia is made by reacting nitrogen with hydrogen in the Haber process.

The equation for the process is shown.

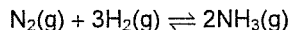


Which changes in reaction conditions would produce a greater yield of ammonia?

- 1 adding more iron catalyst
  - 2 increasing the reaction pressure
  - 3 increasing the particle size of the iron catalyst
- A** 1 only      **B** 2 only      **C** 1 and 2      **D** 2 and 3

**17. M/J 21/P23/Q28**

Ammonia is made from nitrogen and hydrogen. The equation for the reaction is shown.



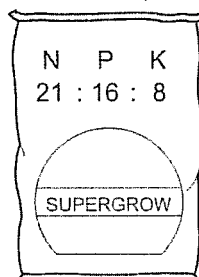
The forward reaction is exothermic.

Which conditions give the greatest equilibrium yield of ammonia?

|          | temperature<br>/°C | pressure<br>/atm |
|----------|--------------------|------------------|
| <b>A</b> | 200                | 15               |
| <b>B</b> | 200                | 150              |
| <b>C</b> | 500                | 15               |
| <b>D</b> | 500                | 150              |

**18. MAR 21/P22/Q30, M/J 15/P12/Q29, M/J 15/P11/Q29, M/J 15/P23/Q29, O/N 11/P12/Q35, O/N 11/P11/Q34, O/N 11/P13/Q35**

Which combination of chemical compounds can be used to produce the fertiliser shown?



- A**  $(\text{NH}_4)_3\text{PO}_4$ ,  $\text{KCl}$   
**B**  $\text{NH}_4\text{NO}_3$ ,  $\text{Ca}_3(\text{PO}_4)_2$

- C**  $\text{NH}_4\text{NO}_3$ ,  $\text{CO}(\text{NH}_2)_2$   
**D**  $\text{NH}_4\text{NO}_3$ ,  $\text{K}_2\text{SO}_4$ ,  $(\text{NH}_4)_2\text{SO}_4$

**19. O/N 20/P22/Q17**

Nitrogen,  $\text{N}_2$ , and hydrogen,  $\text{H}_2$ , can be converted into ammonia,  $\text{NH}_3$ , using a catalyst. What is the purpose of the catalyst?

- A** to increase the amount of ammonia produced  
**B** to increase the rate of reaction  
**C** to reduce the amount of reactants needed  
**D** to reduce the rate of reaction

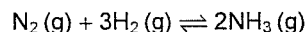
**20. O/N 20/P23/Q31**

What is the catalyst in the Haber process?

- A** Fe                      **B** Ni                      **C** Pt                      **D**  $\text{V}_2\text{O}_5$

**21. M/J 20/P22/Q9**

The Haber process is a reversible reaction.



The reaction has a 30% yield of ammonia.

Which volume of ammonia gas,  $\text{NH}_3$ , measured at room temperature and pressure, is obtained by reacting 0.75 moles of hydrogen with excess nitrogen?

- A**  $3600 \text{ cm}^3$               **B**  $5400 \text{ cm}^3$               **C**  $12\,000 \text{ cm}^3$               **D**  $18\,000 \text{ cm}^3$

**22. MAR 20/P22/Q33**

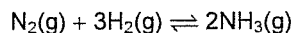
Fertilizers are mixtures of different compounds used to increase the growth of crops.

Which pair of substances contain the three essential elements for plant growth?

- A** ammonium nitrate and calcium phosphate  
**B** ammonium nitrate and potassium chloride  
**C** ammonium phosphate and potassium chloride  
**D** potassium nitrate and calcium carbonate

**23. M/J 20/P22/Q31**

In the Haber process, nitrogen and hydrogen are reacted to make ammonia.



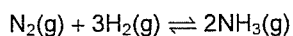
The forward reaction is exothermic.

Which conditions produce the maximum yield of ammonia?

|          | pressure | temperature |
|----------|----------|-------------|
| <b>A</b> | high     | high        |
| <b>B</b> | high     | low         |
| <b>C</b> | low      | high        |
| <b>D</b> | low      | low         |

**24. M/J 20/P21/Q31**

Ammonia is manufactured by the Haber process.



What are the conditions used in the Haber process?

|          | temperature<br>/°C | pressure<br>/atm |
|----------|--------------------|------------------|
| <b>A</b> | 400                | 100              |
| <b>B</b> | 400                | 300              |
| <b>C</b> | 20                 | 300              |
| <b>D</b> | 20                 | 100              |

**25. O/N 19/P22/Q34, O/N 19/P21/Q34, O/N 19/P23/Q34**

Ammonium sulfate is used as a fertiliser.

It is made from ammonia and sulfuric acid.

Which words complete gaps 1, 2 and 3?

The .....1..... is made by the .....2..... process in which .....3..... is used as a catalyst.

|          | 1             | 2       | 3                 |
|----------|---------------|---------|-------------------|
| <b>A</b> | ammonia       | Contact | iron              |
| <b>B</b> | ammonia       | Haber   | vanadium(V) oxide |
| <b>C</b> | sulfuric acid | Contact | vanadium(V) oxide |
| <b>D</b> | sulfuric acid | Haber   | iron              |

**26. O/N 19/P21/Q9**

Four fertilisers are each supplied in 100 kg bags.

Which fertiliser supplies the greatest mass of nitrogen per 100 kg bag?

- |                                                           |                                                         |
|-----------------------------------------------------------|---------------------------------------------------------|
| <b>A</b> ammonium nitrate, $\text{NH}_4\text{NO}_3$       | <b>C</b> ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$ |
| <b>B</b> ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$ | <b>D</b> urea, $\text{CO}(\text{NH}_2)_2$               |

**27. M/J 19/P22/Q28**

The exhaust gases from cars contain oxides of nitrogen.

How are these oxides of nitrogen formed?

- A** Nitrogen and oxygen from the air react together at the high temperatures in the engine.
- B** Nitrogen and oxygen from the petrol react together in the car exhaust.
- C** Nitrogen from the petrol reacts with oxygen at the high temperatures in the engine.
- D** Nitrogen reacts with oxygen from the air in the catalytic converter.

**28. M/J 19/P21/Q29**

Oxides of nitrogen are formed in car engines and are a source of air pollution. To decrease this pollution, catalytic converters are fitted to car exhausts. What happens to the oxides of nitrogen in the catalytic converter?

- A combustion      B cracking      C oxidation      D reduction

**29. M/J 19/P23/Q29**

Catalytic converters in car exhausts change polluting gases into non-polluting gases. Which statements about oxides of nitrogen and car engines are correct?

- 1 The nitrogen in oxides of nitrogen comes from compounds in petrol.  
2 The oxygen in oxides of nitrogen comes from the air in the car engine.  
3 Catalytic converters convert oxides of nitrogen into nitrogen and other gases.

- A 1 and 2      B 2 and 3      C 2 only      D 3 only

**30. MAR 19/P22/Q33**

The raw materials for the Haber process are hydrogen and nitrogen. What are the sources of the hydrogen and nitrogen?

- A hydrogen from ethanol and nitrogen from NPK fertilisers  
B hydrogen from methane and nitrogen from air  
C hydrogen from sulfuric acid and nitrogen from air  
D hydrogen from water and nitrogen from ammonium nitrate

**31. O/N 18/P22/Q28**

Which statement about the Haber process is correct?

- A The hydrogen used is obtained from the air.  
B The nitrogen used is obtained from nitrates in the soil.  
C Nitrogen reacts with hydrogen to make ammonia.  
D The reaction takes place at room temperature and pressure.

**32. O/N 18/P21/Q28**

Which statement describes the role of iron in the Haber process?

- A It is used as a catalyst.  
B It is used as a reducing agent.  
C It is used to condense the ammonia gas into a liquid.  
D It is used to increase the yield of ammonia.

**33. O/N 18/P23/Q28**

Ammonia is manufactured by the Haber process from nitrogen and hydrogen. Which row gives the main sources of these two gases?

|   | hydrogen    | nitrogen    |
|---|-------------|-------------|
| A | air         | air         |
| B | air         | natural gas |
| C | natural gas | air         |
| D | natural gas | natural gas |

**34. M/J 18/P22/Q31, M/J 18/P21/Q31, M/J 18/P23/Q31**

Ammonia is manufactured by reacting hydrogen with nitrogen in the Haber process. Which row describes the sources of hydrogen and nitrogen and the conditions used in the manufacture of ammonia in the Haber process?

|   | source of hydrogen | source of nitrogen | temperature of reaction / °C | pressure of reaction / atm |
|---|--------------------|--------------------|------------------------------|----------------------------|
| A | air                | natural gas        | 250                          | 2                          |
| B | air                | natural gas        | 250                          | 200                        |
| C | natural gas        | air                | 450                          | 2                          |
| D | natural gas        | air                | 450                          | 200                        |

**35. MAR 18/P22/Q18**

A solution of compound Z gives a light blue precipitate with aqueous ammonia. The precipitate dissolves in an excess of ammonia.

A flame test is done on compound Z.

What is the colour of the flame?

- A blue-green      B lilac      C red      D yellow

**36. MAR 18/P22/Q34**

The diagram shows the pH values of the soil in two parts of a garden, X and Y.

|             |             |
|-------------|-------------|
| X<br>pH 7.0 | Y<br>pH 5.5 |
|-------------|-------------|

Lime is used to neutralise the soil in one part of the garden.

To which part of the garden should the lime be added and why?

|   | part of the garden | because lime is |
|---|--------------------|-----------------|
| A | X                  | acidic          |
| B | X                  | basic           |
| C | Y                  | acidic          |
| D | Y                  | basic           |

**37. MAR 18/P22/Q31**

Which pair of compounds would make an NPK fertiliser?

- A ammonium sulfate and potassium phosphate      C calcium phosphate and potassium chloride  
B calcium hydroxide and ammonium nitrate      D potassium nitrate and ammonium sulfate

**38. O/N 17/P22/Q31**

Which metal is used as a catalyst in the Haber process for the manufacture of ammonia?

- A iron      B nickel      C platinum      D vanadium

**39. O/N 17/P23/Q31**

The Haber process for making ammonia is carried out at a temperature of 450 °C and a pressure of 200 atmospheres in the presence of a catalyst.

Which statement is **not** correct?

- A Lowering the pressure increases the rate at which ammonia is produced.  
B Lowering the temperature slows down the rate at which ammonia is produced.  
C Maintaining a very high pressure is very difficult and needs expensive equipment.  
D The reaction is a reversible reaction which can proceed forwards and backwards.

**40. M/J 17/P22/Q31**

Which statement about the conditions used in the Haber process is **not** correct?

- A A high temperature is used because the forward reaction is exothermic.  
B A high pressure is used because there are fewer moles of gas in the products than in the reactants.  
C An iron catalyst is used to increase the rate of the forward reaction.  
D The unreacted hydrogen and nitrogen are recycled to increase the amount of ammonia produced.

**41. M/J 17/P22/Q33, M/J 17/P21/Q33, M/J 17/P23/Q33**

Which statement about sulfuric acid is correct?

- A It is made by the Haber process.  
B It is made in the atmosphere by the action of lightning.  
C It reacts with ammonia to produce a fertiliser.  
D It reacts with copper metal to produce hydrogen gas.



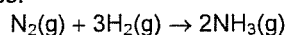
**48. M/J 16/P21/Q34**

A farmer's soil is very low in both nitrogen (N) and phosphorus (P).  
Which fertiliser would improve the quality of this soil most effectively?

|          | percentage   |                |               |
|----------|--------------|----------------|---------------|
|          | nitrogen (N) | phosphorus (P) | potassium (K) |
| <b>A</b> | 11           | 11             | 27            |
| <b>B</b> | 12           | 37             | 10            |
| <b>C</b> | 28           | 10             | 10            |
| <b>D</b> | 31           | 29             | 9             |

**49. M/J 16/P23/Q32**

Ammonia is produced by the Haber process.



Which statement about the Haber process is **not** correct?

- A** An iron catalyst is used to increase the rate of reaction.
- B** The reaction is carried out at high temperature to increase the rate of reaction.
- C** The reaction is carried out at low pressure to increase the yield of ammonia.
- D** The reaction is reversible.

**50. MAR 16/P22/Q32, M/J 12/P12/Q31**

Fertilisers are used to provide three elements needed to increase the yield of crops.

Which two compounds would provide all three of these elements?

- A** ammonium nitrate and calcium phosphate
- C** potassium nitrate and calcium phosphate
- B** ammonium nitrate and potassium sulfate
- D** potassium nitrate and potassium sulfate

**51. MAR 16/P22/Q34**

Why does a farmer put lime (calcium oxide) on the soil?

- A** to act as a fertiliser
- C** to make the soil less acidic
- B** to kill pests
- D** to make the soil less alkaline

**52. O/N 15/P12/Q31**

Which elements are present in NPK fertilisers?

- A** nitrogen, phosphorus, potassium
- C** sodium, phosphorus, potassium
- B** nitrogen, potassium, calcium
- D** sodium, potassium, calcium

**53. O/N 15/P11/Q30**

Fertilisers are mixtures of different compounds used to increase the growth of crops.

Which pair of substances contains the three essential elements for plant growth?

- A** ammonium nitrate and calcium phosphate
- B** ammonium nitrate and potassium chloride
- C** ammonium phosphate and potassium chloride
- D** potassium nitrate and calcium carbonate

**54. O/N 15/P13/Q30**

Which compound is **not** a fertiliser?

- A** ammonium sulfate,  $(\text{NH}_4)_2\text{SO}_4$
- C** potassium chloride,  $\text{KCl}$
- B** calcium hydroxide,  $\text{Ca}(\text{OH})_2$
- D** urea,  $\text{CO}(\text{NH}_2)_2$

**55. MAR 15/P12/Q29**

Which elements does an NPK fertiliser contain?

- A** nickel, phosphorus, potassium
- C** nitrogen, phosphorus, potassium
- B** nickel, potassium, calcium
- D** nitrogen, potassium, calcium

**56. M/J 15/P11/Q30**

The pollutants released into the air from car exhausts and some power stations include oxides of the type XO and YO<sub>2</sub>.

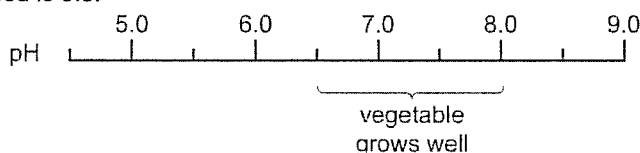
What are X and Y?

|          | X                   | Y                           |
|----------|---------------------|-----------------------------|
| <b>A</b> | carbon only         | nitrogen only               |
| <b>B</b> | carbon only         | nitrogen and sulfur only    |
| <b>C</b> | carbon and nitrogen | carbon and nitrogen only    |
| <b>D</b> | carbon and nitrogen | carbon, nitrogen and sulfur |

**57. MAR 15/P12/Q33**

The diagram shows the soil pH range over which a vegetable grows well.

The pH of the soil to be used is 5.5.



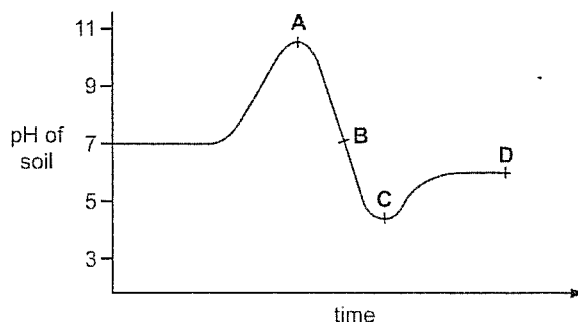
Why is lime added to the soil before planting the vegetable?

- A** The lime acts as a catalyst. **C** The lime is an indicator.  
**B** The lime changes the soil acidity. **D** The lime supplies nitrogen.

**58. O/N 14/P12/Q23, O/N 14/P11/Q22, O/N 14/P13/Q22, M/J 09/P1/Q21**

The graph shows how the pH of soil in a field changes over time.

At which point was the soil neutral?

**59. O/N 14/P13/Q32**

Which pair of compounds would make a N, P, K fertiliser?

- A** ammonium sulfate and potassium phosphate **C** calcium phosphate and potassium chloride  
**B** calcium hydroxide and ammonium nitrate **D** potassium nitrate and ammonium sulfate.

**60. M/J 14/P12/Q17**

Different plants grow best under different pH conditions.

Which plant grows best in alkaline soil?

|          | plant      | grows best in soil at pH |
|----------|------------|--------------------------|
| <b>A</b> | cabbage    | 6-8                      |
| <b>B</b> | potato     | 4-7                      |
| <b>C</b> | strawberry | 5-7                      |
| <b>D</b> | wheat      | 6-7                      |



**61. M/J 14/P12/Q32**Which compound would **not** be an effective fertiliser?

- A ammonium nitrate,  $\text{NH}_4\text{NO}_3$                       C calcium phosphate,  $\text{Ca}_3(\text{PO}_4)_2$   
 B calcium oxide,  $\text{CaO}$                               D potassium nitrate,  $\text{KNO}_3$

**62. M/J 14/P11/Q32, M/J 14/P13/Q32**

Which compound contains two of the three essential elements needed for a complete fertiliser?

- A ammonium chloride                              C ammonium phosphate  
 B ammonium nitrate                                D ammonium sulfate

**63. O/N 13/P12/Q27, O/N 13/P11/Q31**

Farmers add calcium oxide (lime) and ammonium salts to their fields.

The compounds are not added at the same time because they react with each other.

Which gas is produced in this reaction?

- A ammonia                      B carbon dioxide                      C hydrogen                      D nitrogen

**64. O/N 13/P13/Q31**

A sample of fertiliser is tested by warming it with aqueous sodium hydroxide.

A colourless gas is produced which turns red litmus paper blue.

Which element, essential for plant growth, must be present?

- A nitrogen                      B phosphorus                      C potassium                      D sulfur

**65. M/J 13/P12/Q33**

Which substance would make the best general fertiliser?

|   | relative amount |    |    | solubility in water |
|---|-----------------|----|----|---------------------|
|   | P               | K  | N  |                     |
| A | 5               | 0  | 5  | soluble             |
| B | 5               | 5  | 20 | insoluble           |
| C | 5               | 10 | 15 | soluble             |
| D | 10              | 5  | 10 | insoluble           |

**66. M/J 13/P12/Q35**Which process does **not** produce carbon dioxide?

- A fermentation                                      C the production of lime from limestone  
 B respiration                                        D the treatment of acidic soil with lime

**67. M/J 13/P11/Q33, M/J 13/P13/Q34**

Nitrogen, phosphorus and potassium are essential elements for plant growth.

Which mixture provides all three essential elements?

|   | mixture                                       | formula                                                       |
|---|-----------------------------------------------|---------------------------------------------------------------|
| A | ammonium phosphate<br>+<br>potassium chloride | $(\text{NH}_4)_3\text{PO}_4$<br>+<br>$\text{KCl}$             |
| B | ammonium phosphate<br>+<br>ammonium nitrate   | $(\text{NH}_4)_3\text{PO}_4$<br>+<br>$\text{NH}_4\text{NO}_3$ |
| C | ammonium phosphate<br>+<br>ammonium chloride  | $(\text{NH}_4)_3\text{PO}_4$<br>+<br>$\text{NH}_4\text{Cl}$   |
| D | ammonium nitrate<br>+<br>potassium chloride   | $\text{NH}_4\text{NO}_3$<br>+<br>$\text{KCl}$                 |

**11.4: Carbon dioxide and methane****1. O/N 22/P22/Q29, O/N 22/P21/Q29, O/N 22/P23/Q29**

Which substance in polluted air damages stonework and kills trees?

- A carbon dioxide      B carbon monoxide      C lead compounds      D sulfur dioxide

**2. O/N 22/P22/Q33, O/N 22/P21/Q33, O/N 22/P23/Q33**

Which substance gives off carbon dioxide on heating?

- A lime      B limestone      C limewater      D slaked lime

**3. O/N 22/P21/Q18**

Ethanoic acid reacts with water to produce an acidic solution.

Which row describes the roles of ethanoic acid and water in this reaction?

|   | ethanoic acid       | water               |
|---|---------------------|---------------------|
| A | accepts a proton    | donates a proton    |
| B | accepts an electron | donates an electron |
| C | donates a proton    | accepts a proton    |
| D | donates an electron | accepts an electron |

**4. O/N 22/P21/Q20**Carbon forms two oxides: carbon monoxide, CO, and carbon dioxide, CO<sub>2</sub>.

Which row describes these two oxides?

|   | CO      | CO <sub>2</sub> |
|---|---------|-----------------|
| A | acidic  | acidic          |
| B | acidic  | neutral         |
| C | neutral | acidic          |
| D | neutral | neutral         |

**5. M/J 22/P22/Q32**

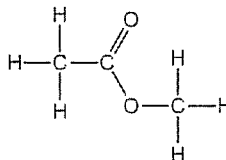
Which processes produce carbon dioxide?

- 1 respiration  
2 photosynthesis  
3 fermentation  
4 combustion of hydrogen

- A 1 and 3      B 1 and 4      C 2 and 3      D 2 and 4

**6. M/J 22/P21/Q35**

The structure of ester W is shown.



Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

|   | name of ester W  | carboxylic acid | alcohol  |
|---|------------------|-----------------|----------|
| A | ethyl methanoate | ethanoic acid   | methanol |
| B | ethyl methanoate | methanoic acid  | ethanol  |
| C | methyl ethanoate | ethanoic acid   | methanol |
| D | methyl ethanoate | methanoic acid  | ethanol  |

**7. M/J 22/P23/Q32**

Which processes increase the amount of carbon dioxide in the air?

- 1 combustion of hydrogen
- 2 combustion of methane
- 3 photosynthesis by plants
- 4 thermal decomposition of limestone

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**8. MAR 22/P22/Q32**

Which information about carbon dioxide and methane is correct?

|   |                                   | carbon dioxide | methane |
|---|-----------------------------------|----------------|---------|
| A | formed when vegetation decomposes | ✓              | ✗       |
| B | greenhouse gas                    | ✓              | ✓       |
| C | present in unpolluted air         | ✗              | ✗       |
| D | produced during respiration       | ✗              | ✓       |

key

✓ = correct

✗ = not correct

**9. O/N 21/P22/Q26**

Carbon dioxide is produced during the extraction of aluminium from bauxite. Which statement describes how this carbon dioxide is made?

- A Carbon monoxide reduces aluminium oxide forming carbon dioxide and aluminium.
- B Carbon is burned in the blast furnace to release heat energy.
- C Oxygen made in the process reacts with the carbon electrode.
- D The ore of aluminium undergoes thermal decomposition.

**10. O/N 21/P22/Q30**Which process does **not** produce a greenhouse gas?

- A acid rain on limestone buildings
- B combustion of wood
- C digestion in cows
- D zinc reacting with sulfuric acid

**11. O/N 21/P21/Q8**

Which statement about carbon is correct?

- A Diamond and graphite both have simple molecular structures.
- B Diamond and graphite are both used to make cutting tools.
- C Each carbon atom in diamond is bonded to three other carbon atoms.
- D Graphite conducts electricity and has a giant covalent structure.

**12. O/N 21/P23/Q35**

What is the main constituent of natural gas?

- A hydrogen
- B carbon monoxide
- C methane
- D nitrogen

**13. M/J 21/P22/Q32**

Which process in the carbon cycle is responsible for removing carbon dioxide from the atmosphere?

- A combustion
- B decomposition
- C photosynthesis
- D respiration

**14. M/J 21/P21/Q30, M/J 20/P21/Q30, O/N 17/P22/Q32**

Which process removes carbon dioxide from the atmosphere?

- A combustion of fossil fuels
- B fermentation
- C photosynthesis
- D respiration

**15. MAR 21/P22/Q31**Which process does **not** produce carbon dioxide?

- A combustion of a hydrocarbon
- B photosynthesis
- C reaction between an acid and a metal carbonate
- D respiration

**16. M/J 21/P23/Q34**

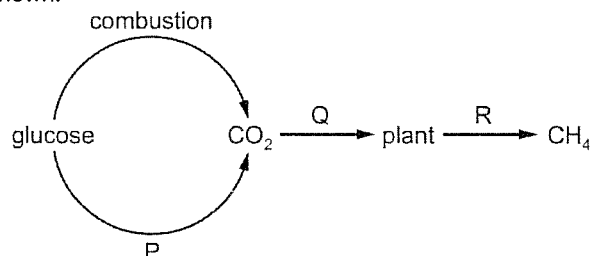
Ethanol is reacted with acidified potassium manganate(VII).

Which row describes the type of reaction and the type of organic compound formed?

|          | type of reaction | organic compound |
|----------|------------------|------------------|
| <b>A</b> | oxidation        | carboxylic acid  |
| <b>B</b> | oxidation        | alkene           |
| <b>C</b> | dehydration      | carboxylic acid  |
| <b>D</b> | dehydration      | alkene           |

**17. O/N 20/P22/Q33, O/N 20/P21/Q34, O/N 20/P23/Q33**

Part of the carbon cycle is shown.



What are processes P, Q and R?

|          | P              | Q              | R              |
|----------|----------------|----------------|----------------|
| <b>A</b> | decomposition  | respiration    | photosynthesis |
| <b>B</b> | respiration    | photosynthesis | decomposition  |
| <b>C</b> | respiration    | decomposition  | photosynthesis |
| <b>D</b> | photosynthesis | respiration    | decomposition  |

**18. M/J 20/P22/Q30**

Three processes in the carbon cycle are shown.

- 1 Methane reacts with oxygen producing carbon dioxide and water.
- 2 Carbon dioxide and water are absorbed and used by plants to make oxygen.
- 3 Oxygen is used by living things to release energy.

Which processes have taken place?

|          | 1              | 2              | 3              |
|----------|----------------|----------------|----------------|
| <b>A</b> | combustion     | photosynthesis | respiration    |
| <b>B</b> | combustion     | respiration    | photosynthesis |
| <b>C</b> | photosynthesis | combustion     | respiration    |
| <b>D</b> | respiration    | photosynthesis | combustion     |

**19. M/J 20/P23/Q30**

Which processes increase the amount of carbon dioxide in the atmosphere?

- 1 burning ethanol
- 2 farming cattle
- 3 growing trees

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**20. O/N 19/P21/Q33**

Which statement about the carbon cycle is correct?

- A Carbon is absorbed from the atmosphere by combustion and released into it by respiration.  
 B Carbon is absorbed from the atmosphere by photosynthesis and released into it by combustion.  
 C Carbon is absorbed from the atmosphere by both respiration and combustion.  
 D Carbon is released into the atmosphere by both photosynthesis and respiration.

**21. MAR 20/P22/Q1**

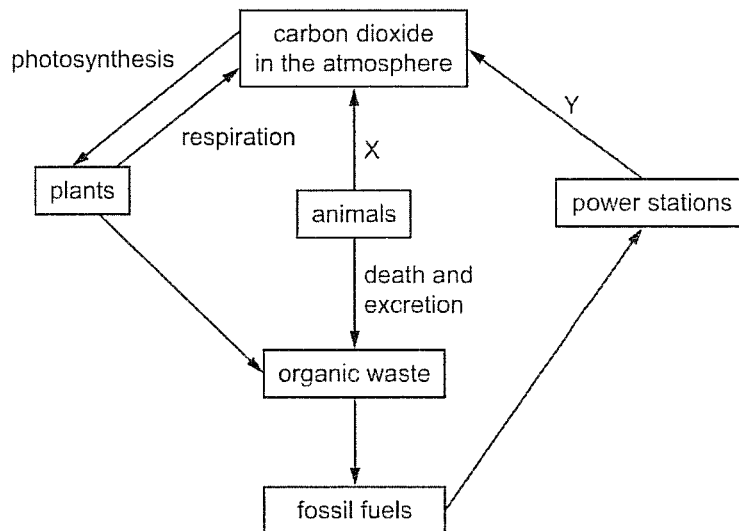
The formula of methane is  $\text{CH}_4$  and the formula of ethane is  $\text{C}_2\text{H}_6$ .

Which row describes diffusion and the relative rates of diffusion of methane and ethane?

|          | description of diffusion                                        | relative rate of diffusion                |
|----------|-----------------------------------------------------------------|-------------------------------------------|
| <b>A</b> | particles move from a high concentration to a low concentration | ethane diffuses more quickly than methane |
| <b>B</b> | particles move from a high concentration to a low concentration | methane diffuses more quickly than ethane |
| <b>C</b> | particles move from a low concentration to a high concentration | ethane diffuses more quickly than methane |
| <b>D</b> | particles move from a low concentration to a high concentration | methane diffuses more quickly than ethane |

**22. O/N 19/P22/Q33**

The diagram represents an outline of the carbon cycle.



Which processes are X and Y?

|          | X              | Y           |
|----------|----------------|-------------|
| <b>A</b> | combustion     | respiration |
| <b>B</b> | decomposition  | respiration |
| <b>C</b> | photosynthesis | combustion  |
| <b>D</b> | respiration    | combustion  |

**23. O/N 19/P23/Q33**

The following processes are part of the carbon cycle.

- 1 photosynthesis
- 2 combustion
- 3 respiration

Which processes decrease the amount of carbon dioxide in the atmosphere?

- A 1 only                      B 1 and 2 only                      C 1 and 3 only                      D 2 and 3 only

**24. M/J 19/P22/Q30**

Some of the processes involved in the carbon cycle are shown.

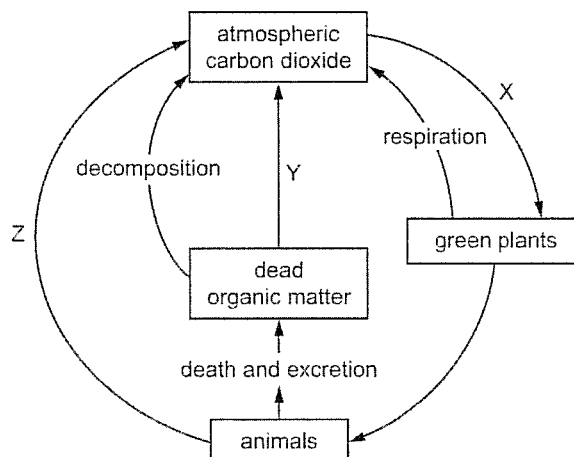
- 1 glucose + oxygen → carbon dioxide + water
- 2 carbon dioxide + water → glucose + oxygen
- 3 methane + oxygen → carbon dioxide + water

What are the names of these processes?

|   | 1              | 2              | 3              |
|---|----------------|----------------|----------------|
| A | combustion     | respiration    | photosynthesis |
| B | photosynthesis | combustion     | respiration    |
| C | respiration    | combustion     | photosynthesis |
| D | respiration    | photosynthesis | combustion     |

**25. M/J 19/P21/Q32**

The carbon cycle is shown.



Which row describes processes X, Y and Z?

|   | X              | Y              | Z              |
|---|----------------|----------------|----------------|
| A | respiration    | combustion     | photosynthesis |
| B | respiration    | photosynthesis | combustion     |
| C | photosynthesis | combustion     | respiration    |
| D | photosynthesis | respiration    | combustion     |

**26. MAR 19/P22/Q34**

Which process removes carbon dioxide from the Earth's atmosphere?

- A combustion                      B heating limestone                      C photosynthesis                      D respiration

**27. O/N 18/P22/Q32**

In the carbon cycle, which two processes add carbon dioxide to the atmosphere?

- A combustion and carbonate formation                      C combustion and respiration  
B combustion and photosynthesis                      D respiration and photosynthesis

**28. M/J 19/P23/Q31**

Which row about the carbon cycle is correct?

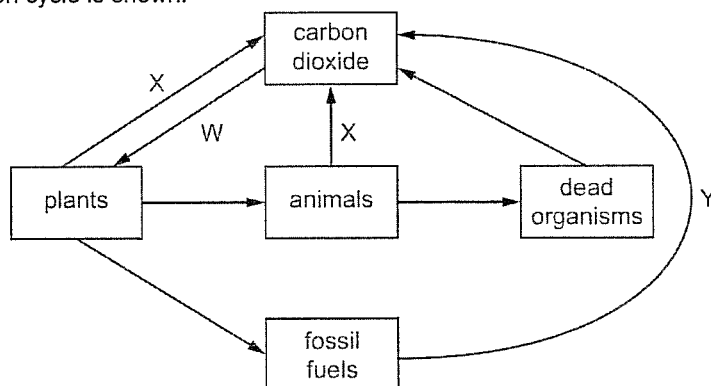
|          | process for removing carbon dioxide from the atmosphere | process for returning carbon dioxide to the atmosphere |
|----------|---------------------------------------------------------|--------------------------------------------------------|
| <b>A</b> | photosynthesis                                          | combustion of hydrocarbons                             |
| <b>B</b> | photosynthesis                                          | cracking of hydrocarbons                               |
| <b>C</b> | respiration                                             | combustion of hydrocarbons                             |
| <b>D</b> | respiration                                             | cracking of hydrocarbons                               |

**29. O/N 18/P21/Q29**Which statement about air pollutants is **not** correct?

- A** Carbon monoxide is formed from the complete combustion of petroleum.  
**B** Lead compounds are formed from some types of petrol.  
**C** Oxides of nitrogen are formed from the combustion reactions inside car engines.  
**D** Sulfur dioxide is formed from the combustion of coal.

**30. O/N 18/P21/Q32**

A diagram of the carbon cycle is shown.



Which processes are represented by the letters W, X and Y?

|          | W              | X              | Y              |
|----------|----------------|----------------|----------------|
| <b>A</b> | photosynthesis | combustion     | respiration    |
| <b>B</b> | photosynthesis | respiration    | combustion     |
| <b>C</b> | respiration    | combustion     | photosynthesis |
| <b>D</b> | respiration    | photosynthesis | combustion     |

**31. O/N 18/P23/Q32**

Which statement about the carbon cycle is correct?

- A** Animals and plants need carbon dioxide for respiration.  
**B** Combustion of plants and natural gas produces carbon dioxide.  
**C** Plants produce glucose from carbon dioxide and oxygen.  
**D** Oxygen is produced by both animals and plants.

**32. M/J 18/P22/Q32, M/J 18/P21/Q32, M/J 18/P23/Q32**

Which statements about the carbon cycle are correct?

- 1 Carbon dioxide is added to the atmosphere by respiration.  
 2 Carbon dioxide is added to the atmosphere by combustion of coal.  
 3 Carbon dioxide is removed from the atmosphere by photosynthesis.

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**33. MAR 18/P22/Q19**

Carbon, copper, magnesium, sodium and sulfur can all form oxides.  
How many of these elements form acidic oxides?

- A 1                      B 2                      C 3                      D 4

**34. MAR 18/P22/Q30**

Which processes do **not** produce carbon dioxide?

- 1 heating limestone
- 2 burning gasoline in car engines
- 3 photosynthesis
- 4 production of nylon

- A 1 and 2                      B 1 and 3                      C 2 and 4                      D 3 and 4

**35. O/N 17/P21/Q32, O/N 17/P23/Q32**

Which process does **not** produce carbon dioxide?

- A combustion of alkanes                      C respiration  
B photosynthesis                      D thermal decomposition of limestone

**36. M/J 17/P22/Q30**

The carbon cycle includes the processes combustion, photosynthesis and respiration.

Which row shows how each process changes the amount of carbon dioxide in the atmosphere?

|   | combustion | photosynthesis | respiration |
|---|------------|----------------|-------------|
| A | decreases  | decreases      | increases   |
| B | decreases  | increases      | decreases   |
| C | increases  | decreases      | increases   |
| D | increases  | increases      | decreases   |

**37. M/J 15/P11/Q33**

These statements are about a gas.

- 1 It is produced by thermal decomposition of a carbonate.
- 2 It is produced by the fermentation of glucose.
- 3 It makes up 1% of unpolluted air.
- 4 It is produced during the production of steel from iron.

Which statements are correct for carbon dioxide?

- A 1 and 2 only                      B 1, 2 and 3                      C 1, 2 and 4                      D 1, 3 and 4

**38. O/N 14/P12/Q34, O/N 14/P11/Q34, O/N 14/P13/Q34, M/J 10/P12/Q35, M/J 10/P11/Q34, M/J 10/P13/Q35**

Which statement about methane is **not** correct?

- A It is a liquid produced by distilling petroleum.                      C It is produced by animals, such as cows.  
B It is produced as vegetation decomposes.                      D It is used as a fuel.

**39. M/J 13/P11/Q35, M/J 13/P13/Q36**

The list shows four methods that were suggested for the formation of carbon dioxide.

- 1 action of an alkali on a carbonate
- 2 action of heat on a carbonate
- 3 complete combustion of methane
- 4 reaction of a carbonate with oxygen

Which methods would result in the production of carbon dioxide?

- A 1 and 2                      B 1 and 4                      C 2 and 3                      D 3 and 4



## Answers Section

| 11.1  |       |       |       | 11.2  |       |       |  | 11.3  |       |       |       | 11.4  |  |       |  |
|-------|-------|-------|-------|-------|-------|-------|--|-------|-------|-------|-------|-------|--|-------|--|
| 1. C  | 41. B | 2. B  | 42. C | 1. D  | 41. D | 2. A  |  | 1. A  | 41. C | 2. A  | 42. C | 1. D  |  | 2. B  |  |
| 3. D  | 43. D | 3. D  | 44. C | 3. A  |       | 3. A  |  | 3. B  | 43. B | 3. B  | 44. B | 3. C  |  | 3. C  |  |
| 4. B  | 45. D | 4. B  | 46. B | 4. A  |       | 4. A  |  | 4. D  | 45. C | 4. D  | 46. C | 4. C  |  | 4. C  |  |
| 5. A  | 47. A | 5. A  | 48. C | 5. A  |       | 5. A  |  | 5. D  | 47. A | 5. D  | 48. D | 5. A  |  | 5. A  |  |
| 6. D  | 49. A | 6. D  | 50. C | 6. B  |       | 6. B  |  | 6. D  | 49. C | 6. D  | 50. C | 6. C  |  | 6. C  |  |
| 7. D  | 51. D | 7. D  | 52. C | 7. D  |       | 7. D  |  | 7. B  | 51. C | 7. B  | 52. A | 7. D  |  | 7. D  |  |
| 8. A  | 53. D | 8. A  | 54. B | 8. C  |       | 8. C  |  | 8. C  | 53. C | 8. C  | 54. B | 8. B  |  | 8. B  |  |
| 9. D  | 55. B | 9. D  | 56. D | 9. B  |       | 9. B  |  | 9. C  | 55. C | 9. C  | 56. D | 9. C  |  | 9. C  |  |
| 10. C | 57. C | 10. C | 58. D | 10. C |       | 10. C |  | 10. B | 57. B | 10. B | 58. B | 10. D |  | 10. D |  |
| 11. B | 59. A | 11. B | 60. B | 11. D |       | 11. D |  | 11. C | 59. A | 11. C | 60. A | 11. D |  | 11. D |  |
| 12. D | 61. D | 12. D |       | 12. D |       | 12. D |  | 12. B | 61. B | 12. B | 62. C | 12. C |  | 12. C |  |
| 13. A |       | 13. A |       | 13. D |       | 13. D |  | 13. D | 62. C | 13. D | 63. A | 13. C |  | 13. C |  |
| 14. C |       | 14. C |       | 14. C |       | 14. C |  | 14. B | 63. A | 14. B | 64. A | 14. C |  | 14. C |  |
| 15. C |       | 15. C |       | 15. A |       | 15. A |  | 15. A | 64. A | 15. A | 65. C | 15. B |  | 15. B |  |
| 16. C |       | 16. C |       | 16. B |       | 16. B |  | 16. B | 65. C | 16. B | 66. D | 16. A |  | 16. A |  |
| 17. C |       | 17. C |       | 17. C |       | 17. C |  | 17. B | 66. D | 17. B | 67. A | 17. B |  | 17. B |  |
| 18. B |       | 18. B |       | 18. A |       | 18. A |  | 18. A |       | 18. A |       | 18. A |  | 18. A |  |
| 19. A |       | 19. A |       | 19. C |       | 19. C |  | 19. B |       | 19. B |       | 19. B |  | 19. B |  |
| 20. D |       | 20. D |       | 20. D |       | 20. D |  | 20. A |       | 20. A |       | 20. B |  | 20. B |  |
| 21. D |       | 21. D |       | 21. A |       | 21. A |  | 21. A |       | 21. A |       | 21. B |  | 21. B |  |
| 22. C |       | 22. C |       | 22. B |       | 22. B |  | 22. C |       | 22. C |       | 22. D |  | 22. D |  |
| 23. A |       | 23. A |       | 23. D |       | 23. D |  | 23. B |       | 23. B |       | 23. A |  | 23. A |  |
| 24. B |       | 24. B |       | 24. D |       | 24. D |  | 24. B |       | 24. B |       | 24. D |  | 24. D |  |
| 25. B |       | 25. B |       | 25. D |       | 25. D |  | 25. C |       | 25. C |       | 25. C |  | 25. C |  |
| 26. D |       | 26. D |       | 26. B |       | 26. B |  | 26. D |       | 26. D |       | 26. C |  | 26. C |  |
| 27. D |       | 27. D |       | 27. C |       | 27. C |  | 27. A |       | 27. A |       | 27. C |  | 27. C |  |
| 28. D |       | 28. D |       | 28. B |       | 28. B |  | 28. D |       | 28. D |       | 28. A |  | 28. A |  |
| 29. A |       | 29. A |       | 29. B |       | 29. B |  | 29. B |       | 29. B |       | 29. A |  | 29. A |  |
| 30. C |       | 30. C |       | 30. B |       | 30. B |  | 30. B |       | 30. B |       | 30. B |  | 30. B |  |
| 31. C |       | 31. C |       | 31. A |       | 31. A |  | 31. C |       | 31. C |       | 31. B |  | 31. B |  |
| 32. C |       | 32. C |       | 32. B |       | 32. B |  | 32. A |       | 32. A |       | 32. A |  | 32. A |  |
| 33. C |       | 33. C |       | 33. C |       | 33. C |  | 33. C |       | 33. C |       | 33. B |  | 33. B |  |
| 34. D |       | 34. D |       | 34. D |       | 34. D |  | 34. D |       | 34. D |       | 34. D |  | 34. D |  |
| 35. A |       | 35. A |       | 35. C |       | 35. C |  | 35. A |       | 35. A |       | 35. B |  | 35. B |  |
| 36. C |       | 36. C |       | 36. D |       | 36. D |  | 36. D |       | 36. D |       | 36. C |  | 36. C |  |
| 37. C |       | 37. C |       | 37. C |       | 37. C |  | 37. A |       | 37. A |       | 37. C |  | 37. C |  |
| 38. B |       | 38. B |       | 38. A |       | 38. A |  | 38. A |       | 38. A |       | 38. A |  | 38. A |  |
| 39. D |       | 39. D |       | 39. B |       | 39. B |  | 39. A |       | 39. A |       | 39. C |  | 39. C |  |
| 40. B |       | 40. B |       | 40. B |       | 40. B |  | 40. A |       | 40. A |       |       |  |       |  |

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## Unit 12

# SULFUR AND CONTACT PROCESS

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IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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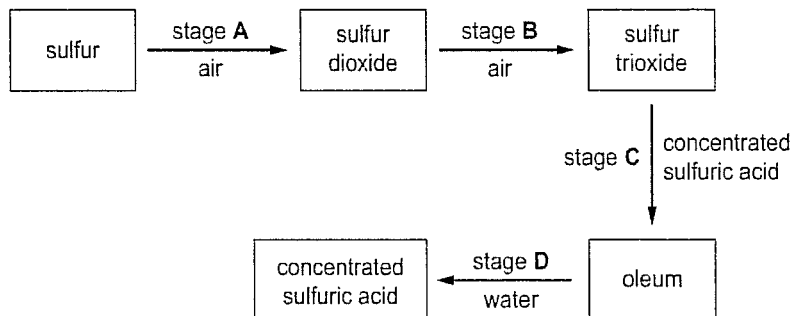
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## Unit 12: Sulfur And Contact Process

### 1. O/N 22/P22/Q31, O/N 22/P21/Q31, O/N 22/P23/Q31, O/N 16/P23/Q33

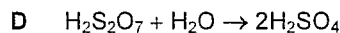
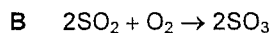
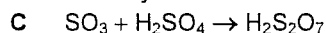
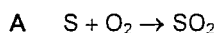
The scheme shows four stages in the conversion of sulfur to sulfuric acid.

In which stage is a catalyst used?



### 2. M/J 22/P22/Q33

Which reaction in the Contact process requires the use of a catalyst?



### 3. M/J 22/P21/Q33

Which statements about sulfur dioxide are correct?

1 Sulfur dioxide decolourises acidified potassium manganate(VII).

2 Sulfur dioxide forms when acids react with carbonates.

3 Sulfur dioxide is used as a bleach.

4 Sulfur dioxide is used to treat acidic soil.

A 1 and 3

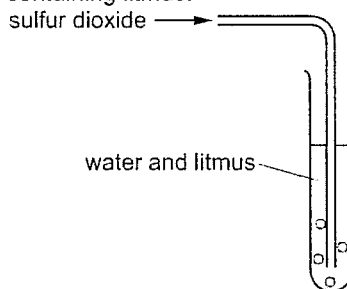
B 1 and 4

C 2 and 3

D 2 and 4

### 4. M/J 22/P23/Q17

Sulfur dioxide is bubbled through water containing litmus.

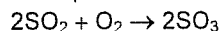


Which row describes and explains what happens to the litmus?

|   | observation   | explanation                       |
|---|---------------|-----------------------------------|
| A | it turns blue | sulfur dioxide is a basic oxide   |
| B | it turns blue | sulfur dioxide is an acidic oxide |
| C | it turns red  | sulfur dioxide is an acidic oxide |
| D | it turns red  | sulfur dioxide is a basic oxide   |

**5. MAR 22/P22/Q31, O/N 21/P23/Q31, M/J 21/P21/31, M/J 21/P23/30, M/J 20/P22/34**

The equation for a reaction occurring in the Contact process is shown.

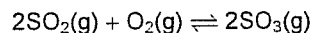


What is the catalyst used in this reaction?

- A iron                      B phosphoric(V) acid    C sulfuric acid                      D vanadium(V) oxide

**6. M/J 22/P23/Q33**

In the Contact process, sulfur dioxide is converted into sulfur trioxide.



What is the effect of lowering the pressure on the rate of formation and percentage yield of sulfur trioxide at equilibrium?

|   | rate of formation | percentage yield |
|---|-------------------|------------------|
| A | decreases         | decreases        |
| B | decreases         | increases        |
| C | increases         | decreases        |
| D | increases         | increases        |

- A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**7. O/N 21/P22/Q31**

Which reaction involving sulfur dioxide is correct?

- A It is produced during the extraction of zinc from zinc blende.  
B It reacts with concentrated sulfuric acid to form oleum.  
C It reacts with sulfur to form sulfur trioxide.  
D It turns an acidified solution of potassium manganate(VII) purple.

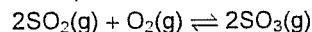
**8. O/N 21/P21/Q31**

Which raw material is used in the Contact process?

- A air                      B ammonia                      C carbon                      D nitrogen

**9. O/N 21/P23/Q15**

Sulfuric acid is manufactured using the Contact process. One of the reactions is shown.



The forward reaction is exothermic.

statement 1 The equation has more molecules on the left-hand side than on the right-hand side.

statement 2 Using a higher pressure shifts the equilibrium to the left.

statement 3 Higher temperatures increase the rate of reaction.

statement 4 Increasing the temperature shifts the equilibrium to the right.

Which alternative is correct?

- A Statement 1 is correct and explains statement 2.  
B Statement 1 and statement 3 are correct.  
C Statement 2 and statement 4 are correct.  
D Statement 3 is correct and explains statement 4.

**10. M/J 21/P22/Q14, M/J 21/P21/Q14, M/J 17/P21/Q14**

When sulfur is heated it undergoes a .....1..... change as it melts.

Further heating causes the sulfur to undergo a .....2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

|   | 1        | 2        |
|---|----------|----------|
| A | chemical | chemical |
| B | chemical | physical |
| C | physical | chemical |
| D | physical | physical |

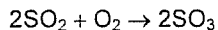
**11. MAR 21/P22/Q32**

Which substance is used as a bleach in the manufacture of paper?

- A carbon dioxide      B nitrogen dioxide      C silicon dioxide      D sulfur dioxide

**12. O/N 20/P22/Q19**

During the manufacture of sulfuric acid, sulfur dioxide is converted to sulfur trioxide.



Which type of reaction is this?

- A displacement      C oxidation  
B neutralisation      D thermal decomposition

**13. O/N 20/P22/Q34**

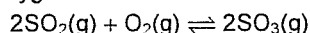
The element sulfur is found in a number of different minerals.

Which mineral contains the greatest percentage by mass of sulfur?

- A barite,  $\text{BaSO}_4$       B galena,  $\text{PbS}$       C gypsum,  $\text{CaSO}_4$       D pyrite,  $\text{FeS}_2$

**14. O/N 20/P21/Q20**

The reaction between sulfur dioxide and oxygen is shown.



The reaction is exothermic.

Which of the changes shifts the position of equilibrium to the right?

- 1 Increase the concentration of oxygen.  
2 Increase the pressure.  
3 Increase the temperature.

- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 only

**15. O/N 20/P21/Q33**

Which row describes two uses of sulfur dioxide?

|   | use 1                         | use 2                                |
|---|-------------------------------|--------------------------------------|
| A | bleaching paper pulp          | neutralising acidic industrial waste |
| B | bleaching paper pulp          | preserving food and drink            |
| C | extracting iron from hematite | neutralising acidic industrial waste |
| D | extracting iron from hematite | preserving food and drink            |

**16. O/N 20/P23/Q34**

Which row shows the conditions used for the manufacture of sulfuric acid in the Contact process?

|   | pressure / atm | temperature / °C | catalyst          |
|---|----------------|------------------|-------------------|
| A | 250            | 200              | vanadium(V) oxide |
| B | 2              | 450              | vanadium(V) oxide |
| C | 250            | 200              | iron              |
| D | 2              | 450              | iron              |

**17. M/J 20/P21/Q34**

The Contact process is used to manufacture concentrated sulfuric acid and consists of four steps. Which step involves a catalyst?

- A production of sulfur dioxide gas      C production of oleum  
B production of sulfur trioxide gas      D production of concentrated sulfuric acid

**18. M/J 20/P23/Q34**

Which reaction in the Contact process is catalysed by vanadium (V) oxide?

- A  $\text{S(s)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)}$                       C  $\text{SO}_3\text{(g)} + \text{H}_2\text{SO}_4\text{(l)} \rightarrow \text{H}_2\text{S}_2\text{O}_7\text{(l)}$   
 B  $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{SO}_3\text{(g)}$                       D  $\text{H}_2\text{S}_2\text{O}_7\text{(l)} + \text{H}_2\text{O(l)} \rightarrow 2\text{H}_2\text{SO}_4\text{(l)}$

**19. MAR 20/P22/Q30**

Sulfur dioxide, carbon monoxide and oxides of nitrogen are common gaseous pollutants found in the air. Which pollutants contribute to acid rain?

- A carbon monoxide and sulfur dioxide                      C oxides of nitrogen only  
 B oxides of nitrogen and sulfur dioxide                      D sulfur dioxide only

**20. MAR 20/P22/Q34**

Which row describes the conditions used in the manufacture of sulfuric acid by the Contact process?

|   | catalyst          | pressure | temperature |
|---|-------------------|----------|-------------|
| A | iron              | high     | high        |
| B | iron              | low      | low         |
| C | vanadium(V) oxide | high     | low         |
| D | vanadium(V) oxide | low      | high        |

**21. M/J 19/P22/Q33, M/J 19/P21/Q33, M/J 19/P23/Q33**

Which row shows the conditions used in the Contact process?

|   | temperature / °C | pressure / atm | catalyst          |
|---|------------------|----------------|-------------------|
| A | 25               | 2              | iron              |
| B | 25               | 200            | iron              |
| C | 450              | 2              | vanadium(V) oxide |
| D | 450              | 200            | vanadium(V) oxide |

**22. MAR 19/P22/Q35**

The Contact process is used to make sulfuric acid.

The steps in the process are listed.

- 1 Dissolve sulfur trioxide in 98% concentrated sulfuric acid.
- 2 Heat sulfur strongly in air.
- 3 Add oleum to water.
- 4 Pass sulfur dioxide over a vanadium(V) oxide catalyst.

Which sequence of steps is correct?

- A  $4 \rightarrow 1 \rightarrow 2 \rightarrow 3$       B  $4 \rightarrow 2 \rightarrow 3 \rightarrow 1$       C  $2 \rightarrow 1 \rightarrow 4 \rightarrow 3$       D  $2 \rightarrow 4 \rightarrow 1 \rightarrow 3$

**23. O/N 18/P22/Q29**

Which statements about sulfur dioxide pollution are correct?

- 1 It increases the pH of rivers.
- 2 It damages limestone buildings.
- 3 It causes respiratory problems.

- A 1 only                      B 2 only                      C 1 and 3                      D 2 and 3

**24. O/N 18/P22/Q33, O/N 18/P21/Q33, O/N 18/P23/Q33**

Which statement about sulfur or one of its compounds is correct?

- A Sulfur occurs naturally as the element sulfur.  
 B Sulfur dioxide is used to kill bacteria in drinking water.  
 C Sulfuric acid is a weak acid.  
 D Dilute sulfuric acid is a dehydrating agent.

25. M/J 18/P21/Q33

Which statement about sulfur and its compounds is **not** correct?

- A Sulfur dioxide is used as a food preservative.  
B Sulfur dioxide turns acidified aqueous potassium manganate(VII) from purple to colourless.  
C Sulfur forms a basic oxide.  
D Sulfur is used in the manufacture of sulfuric acid.

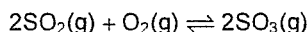
26. O/N 17/P22/Q33, O/N 17/P21/Q33, O/N 17/P23/Q33

Which row shows the conditions used in the manufacture of sulfuric acid by the Contact process?

|          | temperature<br>/ °C | pressure<br>/ atm | catalyst                      |
|----------|---------------------|-------------------|-------------------------------|
| <b>A</b> | 40                  | 200               | Fe                            |
| <b>B</b> | 40                  | 200               | V <sub>2</sub> O <sub>5</sub> |
| <b>C</b> | 400                 | 2                 | Fe                            |
| <b>D</b> | 400                 | 2                 | V <sub>2</sub> O <sub>5</sub> |

27. MAR 18/P22/Q33

The equation for the formation of sulfur trioxide from sulfur dioxide is shown.



The forward reaction is exothermic.

Which combination of pressure and temperature gives the highest equilibrium yield of sulfur trioxide?

|   | pressure | temperature |
|---|----------|-------------|
| A | high     | high        |
| B | high     | low         |
| C | low      | high        |
| D | low      | low         |

28. M/J 17/P22/Q32

Which chemical reaction decreases pollution in the air?

- |          |                                                  |          |                                                                           |
|----------|--------------------------------------------------|----------|---------------------------------------------------------------------------|
| <b>A</b> | $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$  | <b>C</b> | $2\text{CH}_4 + 3\text{O}_2 \rightarrow 2\text{CO} + 4\text{H}_2\text{O}$ |
| <b>B</b> | $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$ | <b>D</b> | $2\text{NO} + 2\text{CO} \rightarrow 2\text{CO}_2 + \text{N}_2$           |

29. MAR 17/P22/Q27

Which substance produces sulfur dioxide when roasted in air?

- A bauxite                  B cryolite                  C hematite                  D zinc blende

30. MAR 17/P22/Q33

The ions present in ammonium sulfate are formed from the products of the Contact and Haber processes. Both of these processes involve the use of a catalyst.

Which row is correct?

|          | ion      | formed from   | process | catalyst          |
|----------|----------|---------------|---------|-------------------|
| <b>A</b> | ammonium | ammonia       | Contact | iron              |
| <b>B</b> | ammonium | ammonia       | Haber   | vanadium(V) oxide |
| <b>C</b> | sulfate  | sulfuric acid | Contact | vanadium(V) oxide |
| <b>D</b> | sulfate  | sulfuric acid | Haber   | iron              |

31. O/N 16/P22/Q33

Sulfuric acid is manufactured by a series of chemical reactions, one of which is catalysed by vanadium(V) oxide. What is the equation for the reaction catalysed by vanadium(V) oxide?

- A  $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$                       C  $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$   
B  $2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_3$                   D  $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$



**32. O/N 16/P21/Q33**

Sulfuric acid is manufactured by the Contact process.  
The most important reaction takes place in the presence of a catalyst.  
What are the reactants and the catalyst for this reaction?

|          | reactants                 | catalyst          |
|----------|---------------------------|-------------------|
| <b>A</b> | sulfur and oxygen         | vanadium(V) oxide |
| <b>B</b> | sulfur dioxide and oxygen | vanadium(V) oxide |
| <b>C</b> | sulfur dioxide and steam  | iron              |
| <b>D</b> | sulfur trioxide and water | platinum          |

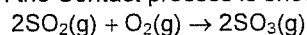
**33. M/J 16/P22/Q33**

The Contact process is used for the manufacture of sulfuric acid.  
Which statement about this process is **not** correct?

- A** A catalyst of iron is used.
- B** Oxygen from the air is used to react with sulfur dioxide.
- C** Sulfur trioxide dissolves in sulfuric acid to form oleum.
- D** The temperature used is around 450 °C.

**34. M/J 16/P21/Q33**

The equation for an exothermic reaction in the Contact process is shown.



Which effects do increasing the temperature and using a catalyst have on the rate of formation of sulfur trioxide,  $\text{SO}_3$ ?

|          | increasing the temperature | using a catalyst |
|----------|----------------------------|------------------|
| <b>A</b> | rate decreases             | rate decreases   |
| <b>B</b> | rate decreases             | rate increases   |
| <b>C</b> | rate increases             | rate decreases   |
| <b>D</b> | rate increases             | rate increases   |

**35. M/J 16/P23/Q33**

One step in the manufacture of sulfuric acid is the oxidation of sulfur dioxide to sulfur trioxide.  
Which conditions are used for this step?

|          | temperature / °C | pressure / atmospheres | catalyst          |
|----------|------------------|------------------------|-------------------|
| <b>A</b> | 450              | 1.5                    | iron              |
| <b>B</b> | 450              | 1.5                    | vanadium(V) oxide |
| <b>C</b> | 450              | 200                    | iron              |
| <b>D</b> | 450              | 200                    | vanadium(V) oxide |

**36. M/J 15/P13/Q30**

Below are two statements about sulfur dioxide.

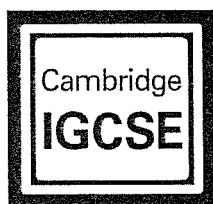
- 1 Sulfur dioxide is formed when fossil fuels burn and it is an acidic oxide.
- 2 Sulfur dioxide is one of the gases in the air which is responsible for 'acid rain'.

Which is correct?

- A** Both statements are correct and statement 1 explains statement 2.
- B** Both statements are correct but statement 1 does not explain statement 2.
- C** Statement 1 is correct but statement 2 is incorrect.
- D** Statement 2 is correct but statement 1 is incorrect.

**Answers Section****Unit 12**

- |     |   |
|-----|---|
| 1.  | B |
| 2.  | B |
| 3.  | A |
| 4.  | C |
| 5.  | D |
| 6.  | A |
| 7.  | A |
| 8.  | A |
| 9.  | B |
| 10. | C |
| 11. | D |
| 12. | C |
| 13. | D |
| 14. | B |
| 15. | B |
| 16. | B |
| 17. | B |
| 18. | B |
| 19. | B |
| 20. | D |
| 21. | C |
| 22. | D |
| 23. | D |
| 24. | A |
| 25. | C |
| 26. | D |
| 27. | B |
| 28. | D |
| 29. | D |
| 30. | C |
| 31. | C |
| 32. | B |
| 33. | A |
| 34. | D |
| 35. | B |
| 36. | A |



## Unit 13

# ORGANIC CHEMISTRY

IGCSE Chemistry P-2  
Classified/Topical MCQ'S  
Article Number: 712

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Editorial Board

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### Syllabus

- 13.1 Fuels
- 13.2 Alkanes
- 13.3 Alkenes and alcohols
- 13.4 Carboxylic acids and esters
- 13.5 Polymers

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## Unit 13: Organic Chemistry

### 13.1: Fuels

1. O/N 22/P22/Q12, O/N 22/P21/Q12, O/N 22/P23/Q12

Some properties of four fuels are shown.

Which fuel is a gas at room temperature and makes two products when it burns in a plentiful supply of air?

|   | fuel     | formula                         | melting point<br>/ °C | boiling point<br>/ °C |
|---|----------|---------------------------------|-----------------------|-----------------------|
| A | hydrogen | H <sub>2</sub>                  | -259                  | -253                  |
| B | methane  | CH <sub>4</sub>                 | -182                  | -164                  |
| C | octane   | C <sub>8</sub> H <sub>18</sub>  | -57                   | 126                   |
| D | wax      | C <sub>31</sub> H <sub>64</sub> | 60                    | 400                   |

2. O/N 22/P23/Q30

Petrol-fuelled cars produce oxides of nitrogen.

Which statement explains how oxides of nitrogen are formed?

- A In the catalytic converter, the elements nitrogen and oxygen combine.
- B Oxygen and nitrogen compounds in petrol combine in the car engine.
- C The high temperatures in the engine provide oxygen and nitrogen with the activation energy needed to react.
- D In the car engine, nitrogen compounds in petrol combine with oxygen.

3. M/J 22/P23/Q11

Which statement about fuels is correct?

- A Coal and ethanol are examples of non-renewable energy sources.
- B Hydrogen and oxygen can be reacted to produce an electric current.
- C Large amounts of energy are taken in by a fuel when it burns.
- D Radioactive isotopes are burned to produce heat.

4. MAR 22/P22/Q16

Which statement about fuels is correct?

- A Heat energy is only produced by burning fuels.
- B Hydrogen is used as a fuel although it is difficult to store.
- C Methane is a good fuel because it produces only water when burned.
- D Uranium is burned in air to produce energy.

5. O/N 21/P22/Q14

Which statements about hydrogen are correct?

- 1 When hydrogen is burned, heat energy is released.
- 2 When hydrogen is used in a fuel cell, electrical energy is generated.
- 3 When hydrogen is used as a fuel, water is the only product.

- A 1, 2 and 3      B 1 and 2 only      C 1 only      D 3 only

6. O/N 21/P23/Q14

A fuel cell is used to generate electricity.

Which chemicals are used in a fuel cell?

- A hydrogen and methane      C nitrogen and methane
- B hydrogen and oxygen      D nitrogen and oxygen

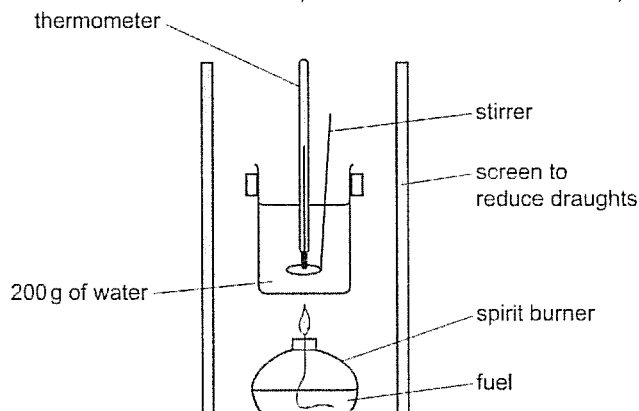
**7. O/N 21/P22/Q34, O/N 21/P21/Q34, O/N 21/P23/Q34**

Fuel X produces carbon dioxide and water when it is burned in air. So does fuel Y.  
What could X and Y be?

|   | X               | Y                              |
|---|-----------------|--------------------------------|
| A | C               | H <sub>2</sub>                 |
| B | C               | C <sub>8</sub> H <sub>18</sub> |
| C | CH <sub>4</sub> | H <sub>2</sub>                 |
| D | CH <sub>4</sub> | C <sub>8</sub> H <sub>18</sub> |

**8. M/J 21/P22/Q12**

Four different fuels are used to heat a beaker of water, for the same amount of time, using the apparatus shown.

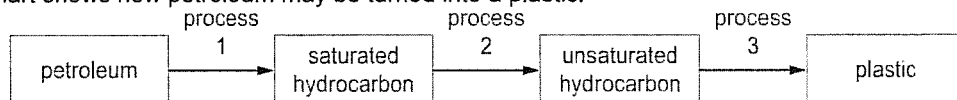


The initial temperature of the water and the temperature after heating by the fuel are recorded.  
Which fuel releases the most heat energy?

|   | initial temperature / °C | temperature after heating / °C |
|---|--------------------------|--------------------------------|
| A | 17                       | 46                             |
| B | 24                       | 52                             |
| C | 26                       | 61                             |
| D | 30                       | 62                             |

**9. M/J 21/P22/Q38**

The flow chart shows how petroleum may be turned into a plastic.



What are processes 1, 2 and 3?

|   | process 1               | process 2               | process 3               |
|---|-------------------------|-------------------------|-------------------------|
| A | cracking                | fractional distillation | polymerisation          |
| B | cracking                | polymerisation          | fractional distillation |
| C | fractional distillation | cracking                | polymerisation          |
| D | fractional distillation | polymerisation          | cracking                |

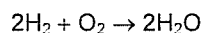
**10. MAR 21/P22/Q29**

Petrol burns in a car engine to produce waste gases which leave through the car exhaust. One of these waste gases is an oxide of nitrogen. Which statement describes how this oxide of nitrogen is formed?

- A Carbon dioxide reacts with nitrogen in the catalytic converter.
- B Nitrogen reacts with oxygen in the car engine.
- C Nitrogen reacts with oxygen in the catalytic converter.
- D Petrol combines with nitrogen in the car engine.

**11. O/N 20/P22/Q15**

Hydrogen reacts with oxygen in a fuel cell.



The reaction is exothermic.

286 kJ of energy is released for every mole of water formed. Which volume of hydrogen gas, measured at room temperature and pressure, would react with oxygen with the release of 7000 J of energy?

- A 587 cm<sup>3</sup>
- B 1175 cm<sup>3</sup>
- C 587 dm<sup>3</sup>
- D 1175 dm<sup>3</sup>

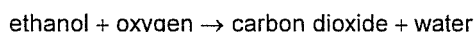
**12. O/N 20/P21/Q16, O/N 20/P23/Q14, M/J 13/P12/Q13, M/J 13/P11/Q14, M/J 13/P13/Q14**

Which substance does not require oxygen in order to produce energy?

- A coal
- B hydrogen
- C natural gas
- D <sup>235</sup>U

**13. O/N 20/P23/Q15**

Ethanol is used as a fuel.



Which statements are correct?

- 1 The reaction is endothermic.
- 2 The products have more energy than the reactants.
- 3 The oxygen for this reaction comes from the air.
- 4 The temperature of the reaction mixture rises during this reaction.

- A 1 and 2
- B 1 and 3
- C 2 and 4
- D 3 and 4

**14. M/J 20/P21/Q13, M/J 20/P23/Q13**

Which statements about hydrogen fuel cells are correct?

- 1 Water is formed as the only waste product.
- 2 Both water and carbon dioxide are formed as waste products.
- 3 The overall reaction is  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ .
- 4 The overall reaction is endothermic.

- A 1 and 3
- B 1 and 4
- C 2 and 3
- D 2 and 4

**15. MAR 20/P22/Q35, O/N 11/P12/Q40, O/N 11/P11/Q37, O/N 11/P13/Q39**

Petroleum is an important raw material that is separated into useful products. Which terms describe petroleum and the method used to separate it?

|   | description | separation method       |
|---|-------------|-------------------------|
| A | compound    | cracking                |
| B | compound    | fractional distillation |
| C | mixture     | cracking                |
| D | mixture     | fractional distillation |

**16. O/N 19/P21/Q36**

Petroleum is separated by fractional distillation.

Which statement about the fractions produced is correct?

- A Bottled gas for heating and cooking is obtained from the naphtha fraction.
- B Diesel oil is used as a fuel for jet aircraft.
- C Substances used to make polishes are obtained from the lubricating fraction.
- D The kerosene fraction contains many useful waxes.

## 17. O/N 19/P22/Q36

Some fractions obtained from petroleum are listed.

|   | fraction     | use                 | position collected in the fractionating column |
|---|--------------|---------------------|------------------------------------------------|
| 1 | gasoline     | waxes and polishes  | below refinery gas                             |
| 2 | bitumen      | making roads        | above kerosene                                 |
| 3 | kerosene     | jet fuel            | below gasoline                                 |
| 4 | refinery gas | heating and cooking | above gasoline                                 |

Which rows are correct?

- A 1, 3 and 4      B 2, 3 and 4      C 3 and 4 only      D 4 only

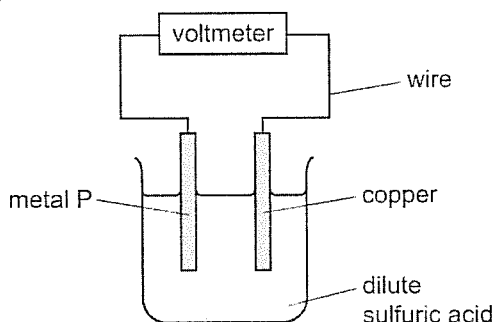
## 18. O/N 19/P21/Q14

Which gases are used to generate electricity in a fuel cell?

- A carbon dioxide and oxygen      C hydrogen and oxygen  
B hydrogen and methane      D methane and carbon dioxide

## 19. O/N 19/P23/Q11

The diagram shows a simple cell.



Which metal P produces the smallest voltage?

- A calcium      B iron      C magnesium      D zinc

## 20. O/N 19/P23/Q36

Which statement is correct?

- A Bitumen is used as a fuel for ships.  
B Coal, natural gas and oxygen are all fuels.  
C Hydrogen is the main constituent of natural gas.  
D Petroleum is separated into useful substances by fractional distillation.

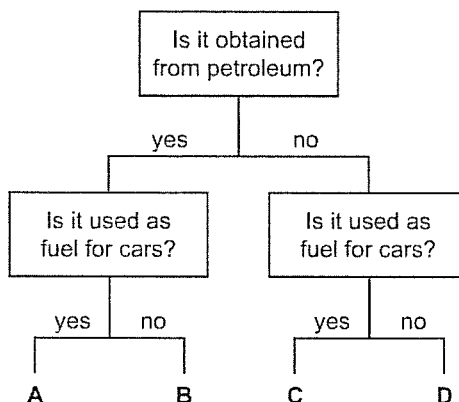
## 21. M/J 19/P22/Q11

Fuel cells are used as energy sources in cars.

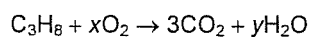
Which row gives a fuel used in a fuel cell and the products formed?

|   | fuel in a fuel cell | products formed          |
|---|---------------------|--------------------------|
| A | hydrogen            | carbon dioxide and water |
| B | hydrogen            | water only               |
| C | petrol              | carbon dioxide and water |
| D | petrol              | water only               |

22. **M/J 19/P22/Q35, M/J 19/P21/Q35, M/J 19/P23/Q35, M/J 13/P12/Q37**  
Which fuel could be gasoline?



23. **M/J 19/P21/Q7**  
Propane burns in oxygen.



Which values of  $x$  and  $y$  balance the equation?

|          | $x$ | $y$ |
|----------|-----|-----|
| <b>A</b> | 5   | 4   |
| <b>B</b> | 7   | 4   |
| <b>C</b> | 10  | 8   |
| <b>D</b> | 13  | 8   |

24. **MAR 19/P22/Q14**  
Hydrogen-oxygen fuel cells can be used to power cars. Platinum is used as a catalyst.  
The amount of energy produced per gram is shown for three fuels.

| fuel     | energy produced per g of fuel / kJ |
|----------|------------------------------------|
| hydrogen | 143                                |
| methane  | 55                                 |
| petrol   | 44                                 |

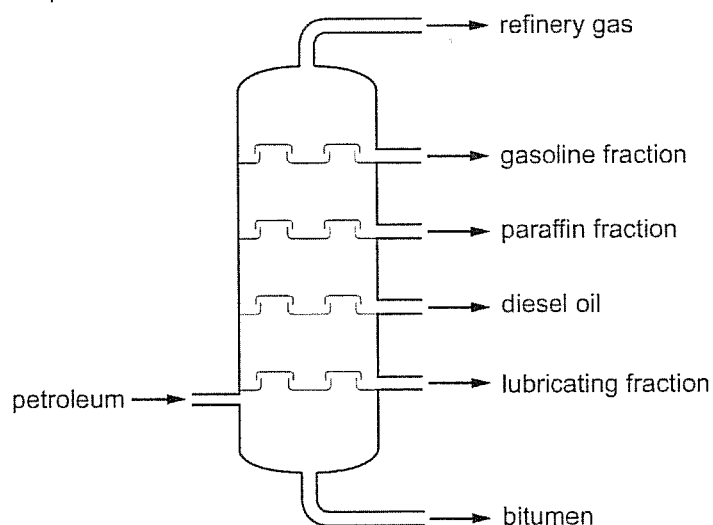
Which statement is correct and is an advantage of a hydrogen-oxygen fuel cell?

- A** Hydrogen is difficult to store.  
**B** Hydrogen produces less energy per gram than methane or petrol.  
**C** Platinum is rare and expensive.  
**D** The only product is water.
25. **MAR 19/P22/Q32**  
What are the main substances produced by the fractional distillation of liquid air?
- A** oxygen and carbon dioxide      **C** helium and nitrogen  
**B** oxygen and nitrogen              **D** hydrogen and oxygen



**26. MAR 19/P22/Q36**

The fractional distillation of petroleum is shown.

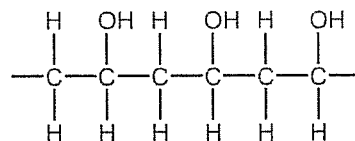


Which fraction is the least volatile?

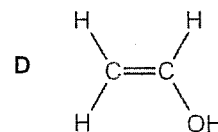
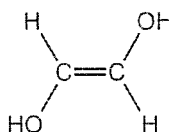
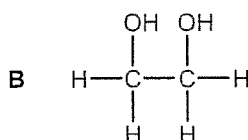
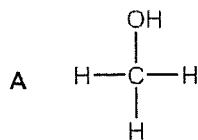
- A bitumen      B diesel oil      C gasoline fraction      D refinery gas

**27. O/N 18/P22/Q40**

The structure of an addition polymer is shown.



Which monomer is used to make this polymer?

**28. M/J 18/P22/Q35, M/J 18/P21/Q35, M/J 18/P23/Q35**

What is **not** the correct use of the fraction named?

|   | name of fraction | use                    |
|---|------------------|------------------------|
| A | fuel oil         | making waxes           |
| B | gas oil          | fuel in diesel engines |
| C | kerosene         | jet fuel               |
| D | naphtha          | making chemicals       |

**29. MAR 18/P22/Q35, M/J 13/P12/Q12**

Statement 1 Hydrogen is used as a fuel.

Statement 2 When hydrogen burns in the air to form water, heat energy is produced.

Which is correct?

- A Both statements are correct and statement 2 explains statement 1.  
 B Both statements are correct but statement 2 does not explain statement 1.  
 C Statement 1 is correct but statement 2 is incorrect.  
 D Statement 2 is correct but statement 1 is incorrect.

## 30. O/N 17/P22/Q36

Which statement is **not** correct?

- A Petroleum is a mixture of hydrocarbons.
- B The main constituent of natural gas is ethane.
- C The naphtha fraction of petroleum is used for making chemicals.
- D When natural gas burns in air, carbon dioxide and water are formed.

## 31. O/N 17/P21/Q36

Some of the fractions obtained from the fractional distillation of petroleum are used as fuels for vehicles.  
Which two fractions are used as fuels for vehicles?

- A bitumen fraction and gasoline fraction
- B bitumen fraction and naphtha fraction
- C gasoline fraction and kerosene fraction
- D kerosene fraction and lubricating fraction

## 32. O/N 17/P23/Q36

Fuel oil and naphtha are two fractions obtained from petroleum.

What are the major uses of these fractions?

|   | fuel oil  | naphtha          |
|---|-----------|------------------|
| A | jet fuel  | making chemicals |
| B | jet fuel  | making roads     |
| C | ship fuel | making chemicals |
| D | ship fuel | making roads     |

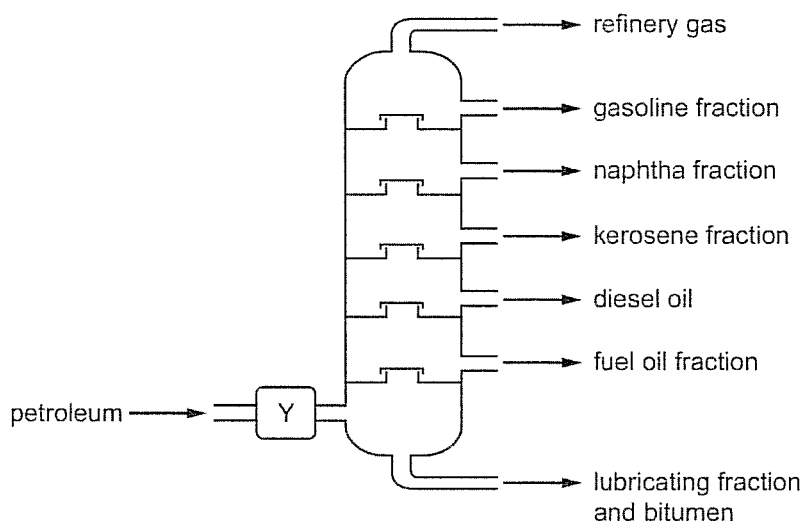
## 33. M/J 17/P22/Q11

Which statement about fuels is correct?

- A Heat energy can only be produced by burning fuels.
- B Hydrogen is used as a fuel although it is difficult to store.
- C Methane is a good fuel because it produces only water when burned.
- D Uranium is burned in air to produce energy.

## 34. M/J 17/P22/Q35

The industrial fractional distillation of petroleum is shown.



Which process happens at Y?

- A burning
- B condensation
- C cracking
- D evaporation

**35. M/J 17/P21/Q35**

Which fraction of petroleum is **not** matched to its correct use?

|          | fraction | use               |
|----------|----------|-------------------|
| <b>A</b> | bitumen  | making roads      |
| <b>B</b> | gasoline | fuel for cars     |
| <b>C</b> | kerosene | fuel for ships    |
| <b>D</b> | naphtha  | chemical industry |

**36. M/J 17/P23/Q32**

Petrol burns in a car engine to produce waste gases which leave through the car exhaust.

One of these waste gases is an oxide of nitrogen.

Which statement describes how this oxide of nitrogen is formed?

- A** Carbon dioxide reacts with nitrogen in the catalytic converter.  
**B** Nitrogen reacts with oxygen in the car engine.  
**C** Nitrogen reacts with oxygen in the catalytic converter.  
**D** Petrol combines with nitrogen in the car engine.

**37. M/J 17/P23/Q35**

Fuel oil, gasoline, kerosene and naphtha are four fractions obtained from the fractional distillation of petroleum.

What is the order of the boiling points of these fractions?

|          | highest boiling point → lowest boiling point |
|----------|----------------------------------------------|
| <b>A</b> | fuel oil → kerosene → gasoline → naphtha     |
| <b>B</b> | fuel oil → kerosene → naphtha → gasoline     |
| <b>C</b> | gasoline → naphtha → kerosene → fuel oil     |
| <b>D</b> | naphtha → gasoline → kerosene → fuel oil     |

**38. MAR 17/P22/Q34**

The table shows the composition of four different types of petroleum.

| fraction   | Arabian Heavy<br>/ % | Arabian Light<br>/ % | Iranian Heavy<br>/ % | North Sea<br>/ % |
|------------|----------------------|----------------------|----------------------|------------------|
| gasoline   | 18                   | 21                   | 21                   | 23               |
| kerosene   | 11                   | 15                   | 13                   | 15               |
| diesel oil | 18                   | 21                   | 20                   | 24               |
| fuel oil   | 53                   | 43                   | 46                   | 38               |

Which type of petroleum is best for the motor vehicle industry?

- A** Arabian Heavy      **B** Arabian Light      **C** Iranian Heavy      **D** North Sea

**39. O/N 16/P22/Q35**

Petroleum is separated into fractions.

Which statement is **not** correct?

- A** Each fraction contains a mixture of hydrocarbon molecules.  
**B** Fuel oil burns easily and is used as fuel in cars.  
**C** Refinery gas is the fraction containing the smallest molecules.  
**D** The fractions are separated depending on their boiling point range.

**40. O/N 16/P23/Q35**

Which list shows the fractions obtained from distilling petroleum, in order of increasing boiling point?

- A** bitumen → diesel oil → fuel oil → lubricating oil  
**B** diesel oil → gasoline → naphtha → kerosene  
**C** gasoline → naphtha → kerosene → diesel oil  
**D** kerosene → lubricating oil → naphtha → refinery gas

**41. O/N 16/P21/Q35**

Petroleum is an important fossil fuel.

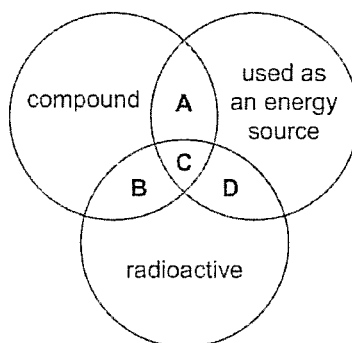
Which row correctly describes petroleum?

|          | type of substance | composition              |
|----------|-------------------|--------------------------|
| <b>A</b> | compound          | mainly hydrocarbons      |
| <b>B</b> | compound          | only hydrogen and carbon |
| <b>C</b> | mixture           | mainly hydrocarbons      |
| <b>D</b> | mixture           | only hydrogen and carbon |

**42. M/J 16/P22/Q11, M/J 16/P21/Q11, M/J 16/P23/Q11, M/J 09/P1/Q15**

The diagram shows some properties that substances may have.

To which labelled part of the diagram does  $^{235}\text{U}$  belong?

**43. MAR 16/P22/Q12, O/N 13/P13/Q13**

Which substance could not be used as a fuel to heat water in a boiler?

**A** ethanol

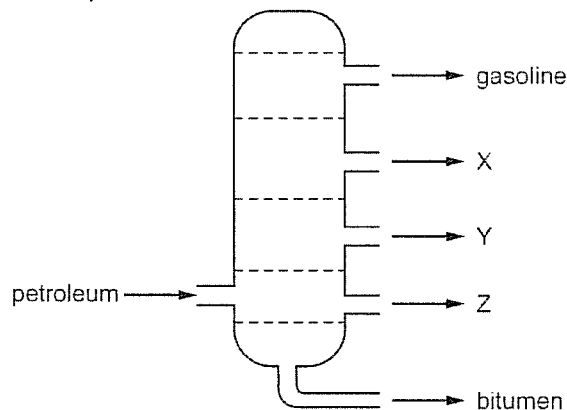
**B** hydrogen

**C** methane

**D** oxygen

**44. M/J 16/P22/Q35, M/J 16/P21/Q35, M/J 16/P23/Q35**

The diagram shows the separation of petroleum into fractions.

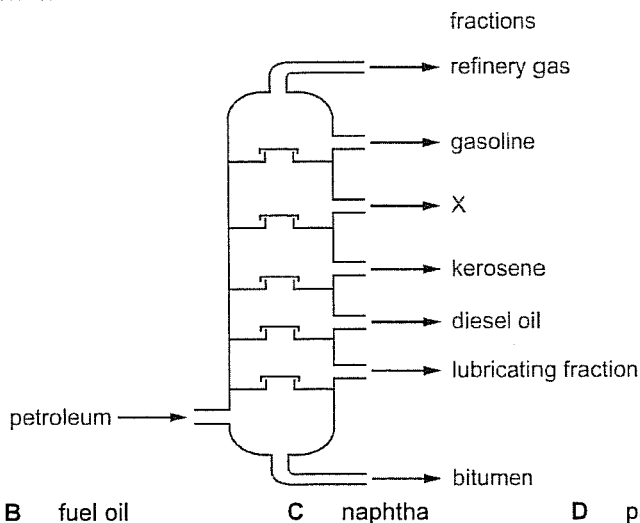


What could X, Y and Z represent?

|          | X                    | Y                    | Z                    |
|----------|----------------------|----------------------|----------------------|
| <b>A</b> | diesel oil           | lubricating fraction | paraffin             |
| <b>B</b> | lubricating fraction | diesel oil           | paraffin             |
| <b>C</b> | paraffin             | lubricating fraction | diesel oil           |
| <b>D</b> | paraffin             | diesel oil           | lubricating fraction |

**45. MAR 16/P22/Q35**

What is the name of fraction X?

**A** alcohol**B** fuel oil**C** naphtha**D** paraffin**46. O/N 15/P12/Q10**

Hydrogen can be used as a fuel.

Which properties of hydrogen would be advantages and which would be disadvantages?

- 1 Hydrogen is expensive to produce.
- 2 Hydrogen reacts exothermically with oxygen.
- 3 When hydrogen burns, a greenhouse gas is not formed.

|          | advantages | disadvantages |
|----------|------------|---------------|
| <b>A</b> | 1          | 2 and 3       |
| <b>B</b> | 1 and 2    | 3             |
| <b>C</b> | 1 and 3    | 2             |
| <b>D</b> | 2 and 3    | 1             |

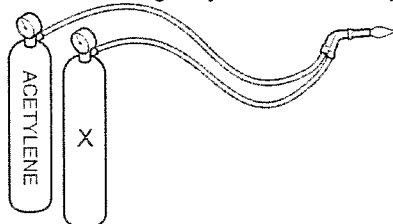
**47. O/N 15/P11/Q37**

What is not the correct use for the fraction named?

|          | name of fraction | use              |
|----------|------------------|------------------|
| <b>A</b> | fuel oil         | making waxes     |
| <b>B</b> | gas oil          | diesel engines   |
| <b>C</b> | kerosene         | jet fuel         |
| <b>D</b> | naphtha fraction | making chemicals |

**48. O/N 15/P13/Q28**

The diagram shows the flame produced from burning a hydrocarbon, acetylene, in a welding torch.



Which gas is X?

**A** hydrogen**B** methane**C** nitrogen**D** oxygen

**49. O/N 15/P13/Q37**

Which row shows the correct use of a fraction obtained by the fractional distillation of petroleum?

|          | fraction | use                       |
|----------|----------|---------------------------|
| <b>A</b> | bitumen  | making waxes and polishes |
| <b>B</b> | fuel oil | aircraft fuel             |
| <b>C</b> | kerosene | fuel for ships            |
| <b>D</b> | naphtha  | making chemicals          |

**50. M/J 15/P12/Q11**

The elements hydrogen and uranium are used as fuels.

The compound ethanol is also used as a fuel.

Which fuels require oxygen in order to produce heat energy?

|          | hydrogen | uranium | ethanol |
|----------|----------|---------|---------|
| <b>A</b> | ✓        | ✓       | ✓       |
| <b>B</b> | ✓        | ✓       | x       |
| <b>C</b> | ✓        | x       | ✓       |
| <b>D</b> | x        | ✓       | ✓       |

**51. M/J 15/P12/Q36, M/J 15/P11/Q36, M/J 15/P13/Q36, O/N 09/P12/Q38, O/N 09/P11/Q38**

Which statement about petroleum is **not** correct?

- A** It can be separated into useful substances by fractional distillation.  
**B** It consists mainly of hydrocarbons.  
**C** It is found underground in many parts of the world.  
**D** Its main use is for making lubricants and polishes.

**52. M/J 15/P13/Q11**

Which substance does **not** use oxygen to produce energy?

- A** coal                      **B** hydrogen                      **C** natural gas                      **D** uranium

**53. M/J 15/P13/Q28**

Oxygen is a reactive element.

Which row shows which of oxygen's reactions are useful?

|          | fuel combustion | rusting | steel manufacture |
|----------|-----------------|---------|-------------------|
| <b>A</b> | no              | no      | yes               |
| <b>B</b> | no              | yes     | no                |
| <b>C</b> | yes             | no      | yes               |
| <b>D</b> | yes             | yes     | no                |

**54. O/N 14/P12/Q14, O/N 14/P11/Q14**

A power station was designed to burn gaseous fuels only.

Which two substances could be used?

- A** carbon dioxide and hydrogen                      **C** hydrogen and methane  
**B** carbon dioxide and  $^{235}\text{U}$                       **D** methane and  $^{235}\text{U}$

**55. O/N 14/P13/Q14**

Which fuel does **not** produce air pollution when it burns?

- A** coal                      **B** diesel oil                      **C** hydrogen                      **D** gasoline (petrol)

## 56. O/N 14/P13/Q31

Which process does **not** involve oxidation?

- A burning a fossil fuel  
 B conversion of iron from the blast furnace into steel  
 C distillation of crude oil  
 D rusting of iron

## 57. M/J 14/P12/Q13

Power stations produce electrical energy from different fuels.

Which fuel causes least pollution to the atmosphere?

- A coal  
 B fuel oil  
 C natural gas  
 D radioactive isotopes

## 58. M/J 14/P12/Q37, M/J 14/P11/Q37, M/J 14/P13/Q36, O/N 11/P12/Q38, O/N 11/P11/Q39, O/N 11/P13/Q37

The table shows the composition of four different types of petroleum (crude oil).

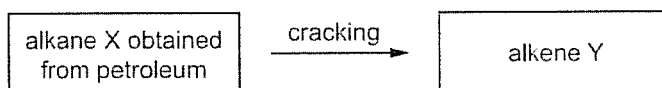
| fraction   | Arabian Heavy<br>/% | Arabian Light<br>/% | Iranian Heavy<br>/% | North Sea<br>/% |
|------------|---------------------|---------------------|---------------------|-----------------|
| gasoline   | 18                  | 21                  | 21                  | 23              |
| kerosene   | 11.5                | 13                  | 13                  | 15              |
| diesel oil | 18                  | 20                  | 20                  | 24              |
| fuel oil   | 52.5                | 46                  | 46                  | 38              |

Which type of petroleum is best for the motor vehicle industry?

- A Arabian Heavy    B Arabian Light    C Iranian Heavy    D North Sea

## 59. M/J 14/P11/Q38, M/J 14/P13/Q39

Alkenes are manufactured by cracking hydrocarbons obtained from petroleum.



Which row describes the process of cracking?

|   | size of X<br>molecules | size of Y<br>molecules | catalyst<br>required | temperature<br>required |
|---|------------------------|------------------------|----------------------|-------------------------|
| A | large                  | small                  | no                   | low                     |
| B | large                  | small                  | yes                  | high                    |
| C | small                  | large                  | no                   | low                     |
| D | small                  | large                  | yes                  | high                    |

## 60. O/N 13/P12/Q15, O/N 13/P11/Q14

Which fuel does **not** produce carbon dioxide when it burns?

- A coal    B hydrogen    C methane    D petrol

## 61. O/N 13/P12/Q31, O/N 13/P11/Q30

In which process is carbon dioxide **not** formed?

- A burning of natural gas  
 B fermentation  
 C heating lime  
 D respiration

## 62. O/N 13/P12/Q39, O/N 13/P11/Q38

Petroleum is a mixture of hydrocarbons which can be separated into fractions using fractional distillation.

Which fraction is used as fuel in jet engines?

- A bitumen    B gasoline    C kerosene    D naphtha

## 63. O/N 13/P13/Q13

Which substance could **not** be used as a fuel to heat water in a boiler?

- A ethanol    B hydrogen    C methane    D oxygen

**64. O/N 13/P13/Q14**

Which substance is not a fossil fuel?

- A coal                      B kerosene                      C gasoline                      D wood

**65. O/N 13/P13/Q38**

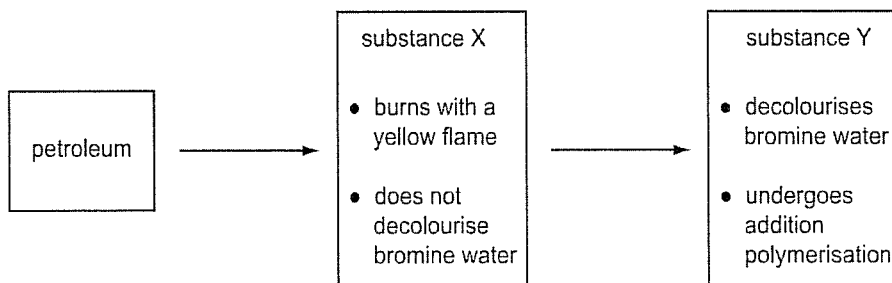
Petroleum is a mixture of different hydrocarbons.

Which process is used to separate the petroleum into groups of similar hydrocarbons?

- A combustion                      C fractional distillation  
B cracking                      D reduction

**66. O/N 13/P13/Q40**

The diagram shows a flow diagram.



Which type of organic compounds are X and Y?

|   | substance X | substance Y |
|---|-------------|-------------|
| A | alcohol     | alkane      |
| B | alkane      | alkene      |
| C | alkene      | alkane      |
| D | alkane      | alcohol     |

**67. M/J 13/P11/Q38, M/J 13/P13/Q39**

The table shows some fractions that are obtained from petroleum by fractional distillation, together with some of their uses.

| fraction     | use              |
|--------------|------------------|
| refinery gas | cooking          |
| gasoline     | fuel for cars    |
| 1            | making chemicals |
| 2            | jet fuel         |
| 3            | fuel for ships   |
| bitumen      | making roads     |

Which row correctly identifies fractions 1, 2 and 3?

|   | 1          | 2          | 3                    |
|---|------------|------------|----------------------|
| A | diesel oil | fuel oil   | lubricating fraction |
| B | fuel oil   | diesel oil | kerosene             |
| C | kerosene   | naphtha    | diesel oil           |
| D | naphtha    | kerosene   | fuel oil             |



**13.2: Alkanes****1. O/N 22/P22/Q34**Which compound has the most  $-\text{CH}_2-$  groups in one molecule?

- A butane                      B butanoic acid                      C butan-1-ol                      D but-1-ene

**2. O/N 22/P22/Q35**

Methane reacts with chlorine in the presence of ultraviolet light.

Which substance is **not** produced in this reaction?

- A  $\text{H}-\text{H}$                       B  $\text{H}-\text{Cl}$                       C  $\begin{array}{c} \text{H} \\ | \\ \text{H}-\text{C}-\text{Cl} \\ | \\ \text{H} \end{array}$                       D  $\begin{array}{c} \text{Cl} \\ | \\ \text{H}-\text{C}-\text{Cl} \\ | \\ \text{H} \end{array}$

**3. O/N 22/P21/Q35**

Which statement about structural isomers is correct?

- A They have the same structure but different reactivity.  
 B They have the same general formula but a different number of carbon atoms in their molecules.  
 C They have the same structure but different relative molecular masses.  
 D They have different structures but the same numbers of each type of atom.

**4. O/N 22/P21/Q37**

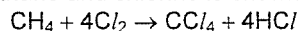
Ethene reacts with water under suitable conditions.

Which statement about this reaction is correct?

- A The product of this reaction has an  $M_r$  of 46.  
 B The reaction produces two different products.  
 C The reaction occurs when ethene gas is bubbled into cold water in the presence of an acid catalyst.  
 D The reaction is a redox reaction.

**5. O/N 22/P23/Q37**

The equation for the reaction between methane and chlorine is shown.



Which type of reaction does methane undergo?

- A substitution                      B reduction                      C condensation                      D addition

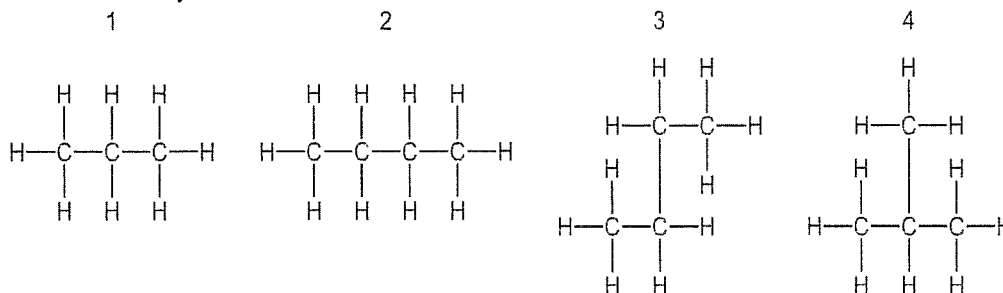
**6. O/N 22/P23/Q38**

Which functional groups form an amide linkage?

- A  $\text{H}_2\text{N}-$  and  $-\text{COOH}$                       B  $\text{H}_2\text{N}-$  and  $\text{H}_2\text{N}-$                       C  $-\text{OH}$  and  $-\text{COOH}$                       D  $-\text{OH}$  and  $\text{H}_2\text{N}-$

**7. M/J 22/P22/Q38**

The structures of some hydrocarbons are shown.

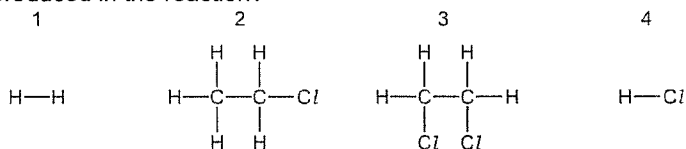


Which statement about the hydrocarbons is correct?

- A 1 and 2 have a different general formula.                      C 2 and 3 are structural isomers.  
 B 1 and 4 are in different homologous series.                      D 3 and 4 have the same empirical formula.

## 8. M/J 22/P22/Q39

Ethane reacts with chlorine in the presence of ultraviolet light.  
Which substances are produced in the reaction?



- A 1, 2 and 3      B 1 and 3 only      C 2, 3 and 4      D 2 and 4 only

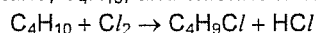
## 9. M/J 22/P21/Q36

Ethanol is made industrially by the fermentation of glucose or by the catalytic addition of steam to ethene.  
Which statement describes an advantage of fermentation compared to catalytic addition?

- A Ethanol is the only product of fermentation.  
B Fermentation uses a batch process but catalytic addition is continuous.  
C Fermentation uses a higher temperature than catalytic addition.  
D Fermentation uses a renewable resource.

## 10. M/J 22/P23/Q36

The equation for the reaction between butane,  $\text{C}_4\text{H}_{10}$ , and chlorine is shown.

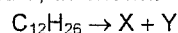


Which type of reaction does butane undergo when it reacts with chlorine?

- A addition      B reduction      C acid-base      D substitution

## 11. M/J 22/P23/Q38

The hydrocarbon  $\text{C}_{12}\text{H}_{26}$  is cracked to give X and Y, as shown.

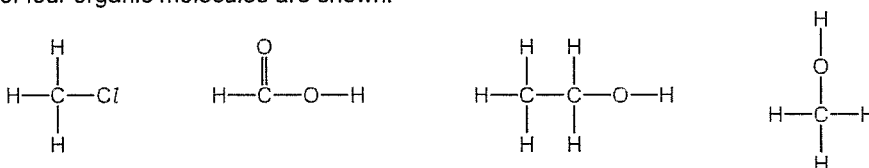


Which statement is correct?

- A If X is  $\text{C}_6\text{H}_{12}$  then Y will react with aqueous bromine.  
B If X is  $\text{C}_{10}\text{H}_{22}$  then Y can be used to make a polymer.  
C If X is a hydrogen molecule then Y is an alkane.  
D X and Y could be structural isomers.

## 12. O/N 21/P22/Q35

The structures of four organic molecules are shown.



How many different homologous series are represented by these molecules?

- A 1      B 2      C 3      D 4

## 13. O/N 21/P21/Q1

Decane has a freezing point of  $-30^\circ\text{C}$  and a boiling point of  $174^\circ\text{C}$ .

A small sample of decane is placed in an open beaker in an oven at a temperature of  $120^\circ\text{C}$  and at atmospheric pressure for 24 hours.

What happens to the sample of decane?

- A It boils.      B It evaporates.      C It melts.      D It sublimes.

## 14. O/N 21/P21/Q36

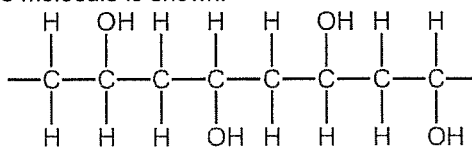
Alkanes react with chlorine to form chloroalkanes.

Which statement about the reactions of alkanes with chlorine is correct?

- A Alkanes react with chlorine by addition.  
B The gaseous product turns red litmus blue.  
C The chlorine atom in chloroethane is covalently bonded.  
D The general formula of the chloroalkanes is  $\text{C}_n\text{H}_{2n}\text{Cl}$ .

**15. O/N 21/P21/Q37**

Part of the structure of a very large molecule is shown.



Which term describes the small unit used to make this molecule?

- A hydrocarbon      B monomer      C polymer      D saturated

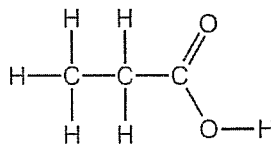
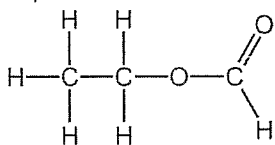
**16. O/N 21/P23/Q36**

Which statement describes the members of a homologous series?

- A compounds with the same physical properties  
B compounds containing the same functional group  
C compounds containing the same number and type of bonds  
D compounds obtained from the same raw material

**17. O/N 21/P23/Q37**

The structures of two compounds are shown.



Which statements about these compounds are correct?

- 1 They have the same molecular formula.  
2 They have similar chemical properties.  
3 They are structural isomers.

- A 1 only      B 1 and 2      C 2 and 3      D 1 and 3

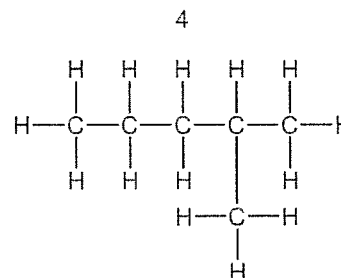
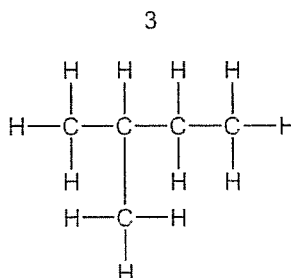
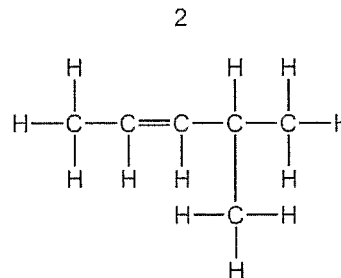
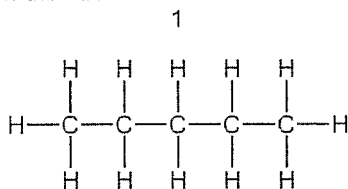
**18. M/J 21/P22/Q35**

Which pair of formulae represents two alkanes?

- A  $\text{CH}_4$  and  $\text{C}_8\text{H}_{18}$       B  $\text{C}_2\text{H}_6$  and  $\text{C}_5\text{H}_8$       C  $\text{C}_3\text{H}_6$  and  $\text{C}_5\text{H}_{12}$       D  $\text{C}_{10}\text{H}_8$  and  $\text{C}_4\text{H}_8$

**19. M/J 21/P23/Q35**

The diagrams show the structural formulae of four compounds.



Which two compounds are structural isomers?

- A 1 and 3      B 1 and 4      C 2 and 3      D 2 and 4

**20. M/J 21/P22/Q36, M/J 21/P21/Q36, M/J 21/P23/Q36**

Which statement about alkanes is correct?

- A They burn in oxygen.
- B They contain carbon, hydrogen and oxygen atoms.
- C They contain double bonds.
- D They contain ionic bonds.

**21. M/J 21/P21/Q35**

Which statement about homologous series and isomerism is correct?

- A Butane and butene are structural isomers.
- B Compounds in the same homologous series have the same general formula.
- C Compounds in the same homologous series have the same molecular formula.
- D Structural isomers have different molecular formulae.

**22. MAR 21/P22/Q34**

Propane reacts with chlorine.

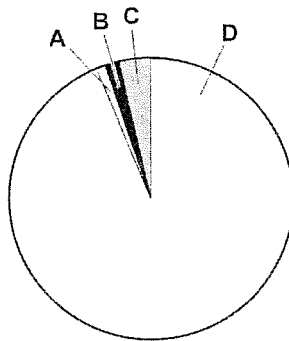
Which row shows a condition required for this reaction and identifies the type of reaction?

|   | condition                | type of reaction |
|---|--------------------------|------------------|
| A | phosphoric acid catalyst | addition         |
| B | phosphoric acid catalyst | substitution     |
| C | ultraviolet light        | addition         |
| D | ultraviolet light        | substitution     |

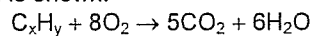
**23. MAR 21/P22/Q35**

The pie chart represents the composition of natural gas.

Which sector represents methane?

**24. MAR 21/P22/Q37**The formula of a hydrocarbon is  $C_xH_y$ .

The equation for its complete combustion is shown.

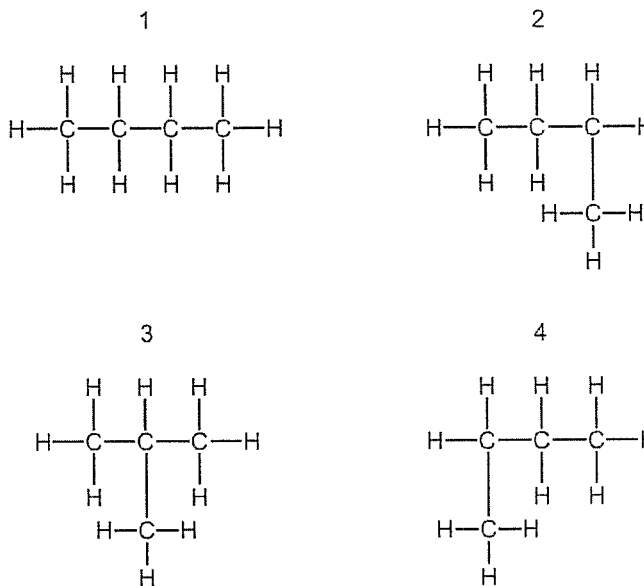


What are the values of x and y?

|   | x  | y  |
|---|----|----|
| A | 5  | 6  |
| B | 5  | 12 |
| C | 6  | 5  |
| D | 12 | 5  |

**25. O/N 20/P22/Q36**

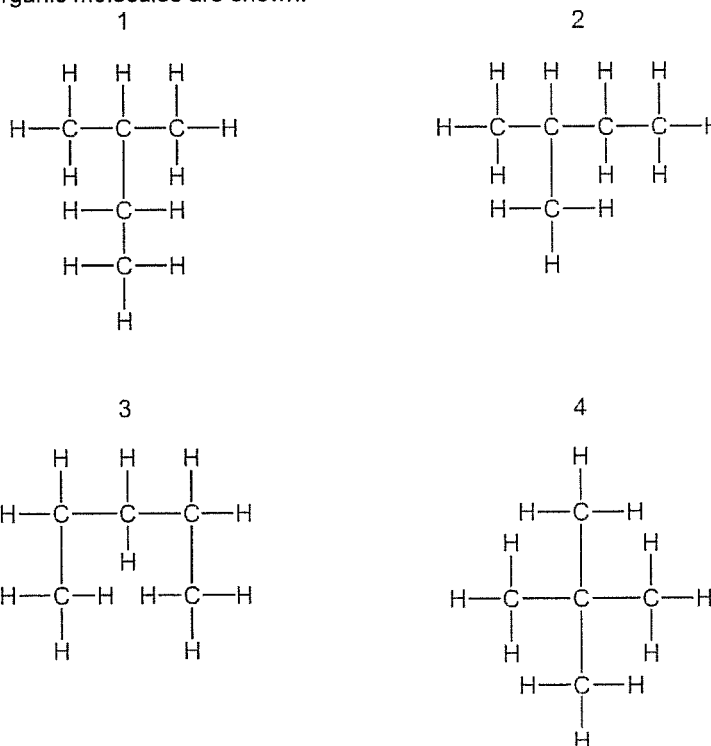
Which structures are structural isomers of each other?



- A** 1, 2, 3 and 4      **B** 1, 2 and 4 only      **C** 1 and 3 only      **D** 2 and 4 only

**26. O/N 20/P21/Q35**

The structures of four organic molecules are shown.



Which molecules are structural isomers of structure 1?

- A** 2 and 4      **B** 2 only      **C** 3 and 4      **D** 3 only

**27. O/N 20/P21/Q36**

Which chemical equation for the substitution of an alkane with chlorine is correct?

- A**  $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$       **C**  $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_6\text{Cl}_2 + \text{H}_2$   
**B**  $\text{C}_3\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_6\text{Cl}_2$       **D**  $\text{C}_3\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_5\text{Cl} + \text{HCl}$

**28. M/J 20/P22/Q36**

Which statement about compounds in the same homologous series is correct?

- A They have the same chemical properties because they have the same number of carbon atoms.
- B They have the same physical properties because they have the same number of carbon atoms.
- C They have different chemical properties because they have different numbers of carbon atoms.
- D They have different physical properties because they have different numbers of carbon atoms.

**29. M/J 20/P22/Q37, M/J 20/P21/Q37, M/J 20/P23/Q37, O/N 14/P12/Q37, O/N 14/P11/Q36, O/N 14/P13/Q36**

Increasing the number of atoms in one molecule of a hydrocarbon increases the amount of energy released when it burns.

What is the correct order?

|   | less energy released | →       | more energy released |
|---|----------------------|---------|----------------------|
| A | ethene               | ethane  | methane              |
| B | ethene               | methane | ethane               |
| C | methane              | ethane  | ethene               |
| D | methane              | ethene  | ethane               |

**30. M/J 20/P22/Q39**

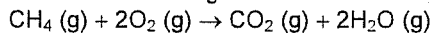
Alkanes undergo substitution reactions in the presence of UV light.

Which equation represents a substitution reaction of ethane?

- A  $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4 + 2\text{HCl}$
- B  $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$
- C  $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{H}_2$
- D  $\text{C}_2\text{H}_6 + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{H}_2$

**31. M/J 20/P21/Q12**

The equation for the complete combustion of methane gas is shown.



Bond energies are shown.

| bond | bond energy in kJ/mol |
|------|-----------------------|
| C—H  | 412                   |
| H—O  | 463                   |
| C=O  | 743                   |
| O=O  | 498                   |

What is the overall energy change, in kJ/mol, for the above reaction?

- A -1192
- B -694
- C +694
- D +1192

**32. M/J 20/P21/Q36**

Which statement about homologous series is correct?

- A Members of a homologous series have the same structural formula.
- B Members of a homologous series all have similar chemical properties.
- C Members of a homologous series all have similar physical properties.
- D Members of all homologous series are hydrocarbons.

**33. M/J 20/P21/Q39**

Ethane,  $\text{C}_2\text{H}_6$ , reacts with chlorine in a substitution reaction.

What are the products of this reaction?

- A chloroethane and hydrogen
- B chloroethane and hydrogen chloride
- C chloroethene and hydrogen
- D chloroethene and hydrogen chloride

**34. M/J 20/P23/Q36**

Which statement about a homologous series is correct?

- A All members have the same general formula.
- B All members have the same molecular formula.
- C All members have similar physical properties.
- D Members show a trend in their chemical properties

**35. M/J 20/P23/Q39**

Alkanes undergo substitution reactions with chlorine in the presence of ultraviolet light. Which equation shows a reaction of this type?

- A  $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$
- B  $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl}_2 + \text{H}_2$
- C  $\text{C}_3\text{H}_8 + 2\text{Cl}_2 \rightarrow \text{C}_3\text{H}_6\text{Cl}_2 + 2\text{HCl}$
- D  $\text{C}_3\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_5\text{Cl} + \text{HCl}$

**36. MAR 20/P22/Q14**

Which change is a physical change?

- A Copper (II) carbonate changes colour from green to black when it is heated, and stays black when it cools.
- B Ethanol reacts with oxygen to form carbon dioxide and water.
- C Hydrogen peroxide decomposes into water and oxygen when it is boiled.
- D Ice forms liquid water when it is heated.

**37. O/N 19/P22/Q37, O/N 19/P21/Q37, O/N 19/P23/Q37**

Which products are obtained by the cracking of an alkane?

|   | alkene | hydrogen | water |
|---|--------|----------|-------|
| A | ✓      | ✓        | ✓     |
| B | ✓      | ✓        | x     |
| C | ✓      | x        | ✓     |
| D | x      | ✓        | ✓     |

**38. O/N 19/P21/Q38**

Ethanol is manufactured by the catalytic addition of steam to ethene and by fermentation.

Which statement describes an advantage of fermentation compared to the catalytic addition of steam to ethene?

- A Fermentation is a more rapid reaction.
- B Fermentation produces a purer product.
- C Fermentation uses a higher temperature.
- D Fermentation uses renewable resources.

**39. M/J 19/P22/Q36**

Which statement about homologous series is not correct?

- A All homologous series are hydrocarbons.
- B Members of a homologous series have the same functional group.
- C Members of a homologous series have similar chemical properties.
- D The alkanes are an example of a homologous series.

**40. M/J 19/P22/Q37**

In bright sunlight, ethane and chlorine combine in substitution reactions.

Which compound is **not** formed in these reactions?

- A  $\text{C}_2\text{H}_3\text{Cl}$
- B  $\text{C}_2\text{H}_5\text{Cl}$
- C  $\text{C}_2\text{H}_4\text{Cl}_2$
- D  $\text{HCl}$

**41. M/J 19/P23/Q7**

When propane burns in air, carbon dioxide and water are formed.

What is the chemical equation for this reaction?

- A  $\text{C}_3\text{H}_8 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- B  $\text{C}_3\text{H}_8 + 3\text{O}_2 \rightarrow 3\text{CO}_2 + \text{H}_2\text{O}$
- C  $\text{C}_3\text{H}_8 + 4\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
- D  $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$

**42. M/J 19/P21/Q37**

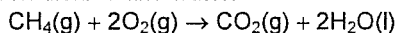
Chlorine reacts with methane.  
Which statements are correct?

- 1 The reaction takes place in the dark.
- 2 The reaction of chlorine with methane forms chloromethane.
- 3 Chloromethane reacts with chlorine to produce dichloromethane.
- 4 The reaction of chlorine with methane is an addition reaction.

A 1 and 2                      B 1 and 3                      C 2 and 3                      D 3 and 4

**43. M/J 19/P23/Q12**

Methane burns in oxygen to form carbon dioxide and water.



The bond energies are shown in the table.

| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| C–H  | 410                      |
| C–O  | 360                      |
| C=O  | 805                      |
| O–H  | 460                      |
| O–O  | 146                      |
| O=O  | 496                      |

What is the energy change for this reaction?

A –818 kJ/mol                      B –102 kJ/mol                      C +102 kJ/mol                      D +818 kJ/mol

**44. M/J 19/P23/Q36**

Which statements about homologous series are correct?

- 1 All members have similar chemical properties.
- 2 All members have the same molecular mass.
- 3 Ethane and ethene are members of the same homologous series.
- 4 Ethane and propane are members of the same homologous series.

A 1 and 3                      B 1 and 4                      C 2 and 3                      D 2 and 4

**45. M/J 19/P23/Q37**

Which type of reaction takes place when methane reacts with chlorine in the presence of ultraviolet light?

A addition                      B cracking                      C polymerisation                      D substitution

**46. O/N 18/P22/Q35, O/N 18/P21/Q35, O/N 18/P23/Q35**

Which equation representing a reaction of methane is correct?

- A  $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$                       C  $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{H}_2$   
 B  $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_4\text{Cl}_2$                       D  $2\text{CH}_4 + 2\text{Cl}_2 \rightarrow 2\text{CH}_3\text{Cl} + \text{Cl}_2 + \text{H}_2$

**47. O/N 18/P23/Q29**

Which equation represents the incomplete combustion of propane,  $\text{C}_3\text{H}_8$ ?

- A  $2\text{C}_3\text{H}_8 + 7\text{O}_2 \rightarrow 6\text{CO} + 8\text{H}_2\text{O}$                       C  $2\text{C}_3\text{H}_8 + 11\text{O}_2 \rightarrow 6\text{CO} + 16\text{H}_2\text{O}$   
 B  $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$                       D  $\text{C}_3\text{H}_8 + 7\text{O}_2 \rightarrow 3\text{CO}_2 + 8\text{H}_2\text{O}$

**48. M/J 18/P23/Q36**

Methane, ethane and propane belong to a family of hydrocarbons called alkanes.

What is the general formula of an alkane?

A  $\text{C}_n\text{H}_{2n}$                       B  $\text{C}_n\text{H}_{2n+1}$                       C  $\text{C}_n\text{H}_{2n-1}$                       D  $\text{C}_n\text{H}_{2n+2}$

**49. M/J 17/P23/Q36**

Butane and methylpropane are isomers with molecular formula  $\text{C}_4\text{H}_{10}$ .

Which statements are correct?

- 1 They have similar chemical properties.
- 2 They have the same general formula.
- 3 They have the same structural formula.

A 1, 2 and 3                      B 1 and 2 only                      C 1 and 3 only                      D 2 and 3 only





**57. M/J 15/P12/Q39, M/J 15/P11/Q39, M/J 15/P13/Q39, O/N 12/P13/Q39**

A hydrocarbon A is cracked to make B and hydrogen.

Compound C is formed by the addition polymerisation of B.

To which homologous series do A, B and C belong?

|          | alkene | alkane  |
|----------|--------|---------|
| <b>A</b> | A      | B and C |
| <b>B</b> | B      | A and C |
| <b>C</b> | C      | A and B |
| <b>D</b> | –      | A and C |

**58. M/J 15/P13/Q37**

Ethene, propene and butene are all members of the same homologous series.

Which statement explains why ethene, propene and butene have similar chemical properties?

- A** They all have the same functional group.      **C** They are all hydrocarbons.  
**B** They are all gases at room temperature.      **D** They are all organic.

**59. M/J 15/P13/Q40**

Ethanol is manufactured from petroleum by reacting ethene with steam.

Which statements about this process are correct?

- 1 Ethene is obtained from the cracking of alkanes.  
 2 The process is carried out in the presence of yeast.  
 3 The reaction is an addition reaction.  
 4 The rate of reaction is increased by a catalyst.
- A** 1 and 3 only      **B** 1 and 4 only      **C** 1, 2 and 3      **D** 1, 3 and 4

**60. MAR 15/P12/Q35**

Petroleum is separated into useful fractions by fractional distillation.

Separation occurs in a fractionating column.

Some properties of three of these fractions are shown.

| fraction | boiling point range/°C | number of carbon atoms in the molecules |
|----------|------------------------|-----------------------------------------|
| 1        |                        | 5–10                                    |
| 2        | 320–350                | 16–24                                   |
| 3        | 120–210                |                                         |

Which statement is correct?

- A** Fraction 1 has a higher boiling point range than fraction 2.  
**B** Fraction 2 is removed from a higher point in the fractioning tower than fraction 1.  
**C** Molecules in fraction 3 have shorter chains than those in fraction 2.  
**D** None of the fractions is liquid at room temperature.

**61. MAR 15/P12/Q37**

Which statement about alkanes is correct?

- A** Ethane has one more carbon atom and one more hydrogen atom than methane.  
**B** They are converted to alcohols by reaction with steam.  
**C** They contain carbon-carbon double bonds.  
**D** They form carbon dioxide and water on combustion.

**62. MAR 15/P12/Q38**

Which statement about alkenes is **not** correct?

- A** They are hydrocarbons.      **C** They contain a C=C bond.  
**B** They are saturated.      **D** They form polymers.

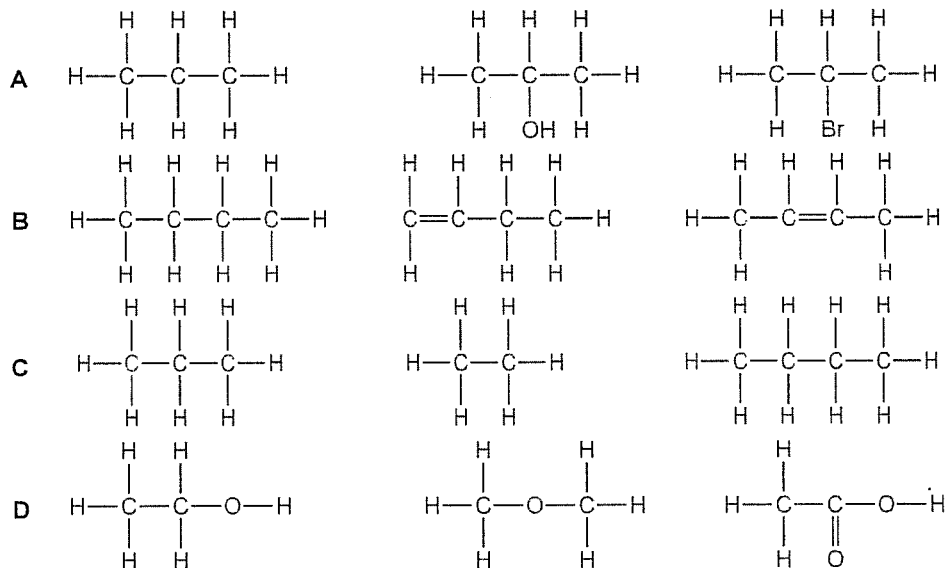
## 63. O/N 14/P13/Q8

Which substance is methane?

|   | volatility | electrical conductivity at room temperature | solubility in water |
|---|------------|---------------------------------------------|---------------------|
| A | high       | good                                        | soluble             |
| B | high       | poor                                        | insoluble           |
| C | low        | good                                        | soluble             |
| D | low        | poor                                        | insoluble           |

## 64. O/N 13/P13/Q39

Which row represents compounds in the same homologous series?



## 13.3: Alkenes and alcohols

### 1. O/N 22/P22/Q36

Ethene reacts with both hydrogen and steam.  
Which row about these reactions is correct?

|   | reactant with ethene | type of reaction | catalyst used   |
|---|----------------------|------------------|-----------------|
| A | hydrogen             | substitution     | phosphoric acid |
| B | hydrogen             | addition         | nickel          |
| C | steam                | substitution     | phosphoric acid |
| D | steam                | addition         | nickel          |

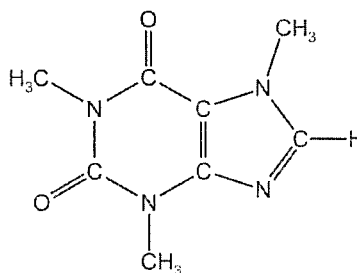
### 2. O/N 22/P22/Q39

Which statement about unsaturated hydrocarbons is correct?

- A  $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$  is an unsaturated hydrocarbon.  
 B Ethene has more hydrogen atoms per molecule than ethane.  
 C Unsaturated hydrocarbons have double bonds between carbon and hydrogen atoms.  
 D Unsaturated hydrocarbons turn aqueous bromine from colourless to brown.

### 3. O/N 22/P22/Q8, O/N 22/P21/Q8, O/N 22/P23/Q8

Caffeine is a stimulant found in coffee.



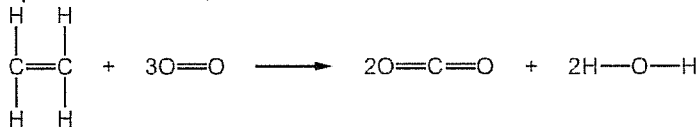
caffeine

Which formula represents caffeine?

- A  $\text{C}_7\text{H}_{10}\text{N}_4\text{O}_2$       B  $\text{C}_8\text{H}_{10}\text{N}_3\text{O}_2$       C  $\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$       D  $\text{C}_8\text{H}_{11}\text{N}_4\text{O}_2$

### 4. O/N 22/P21/Q13

Ethene can undergo complete combustion, as shown.



Some bond energies are given in the table.

| bond | bond energy in kJ/mol |
|------|-----------------------|
| C=C  | 612                   |
| C-H  | 412                   |
| O-H  | 463                   |
| O=O  | 496                   |

The energy change of the reaction is  $-1408$  kJ/mol.

What is the bond energy of the C=O bond in  $\text{CO}_2$ ?

- A 454 kJ/mol      B 673 kJ/mol      C 826 kJ/mol      D 1619 kJ/mol

**5. O/N 22/P23/Q35**

Methanol,  $\text{CH}_3\text{OH}$ , is a member of the homologous series of alcohols.

What is the formula of the alcohol in the same homologous series which contains three carbon atoms?

- A  $\text{C}_3\text{H}_5\text{OH}$       B  $\text{C}_3\text{H}_6\text{OH}$       C  $\text{C}_3\text{H}_7\text{OH}$       D  $\text{C}_3\text{H}_8\text{OH}$

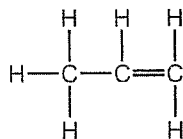
**6. O/N 22/P23/Q36**

Which type of compound reacts with hydrogen in an addition reaction?

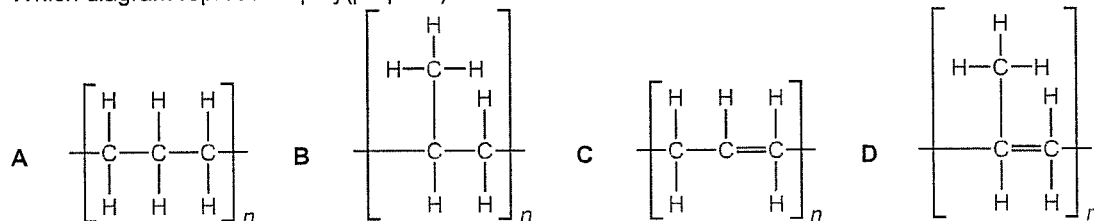
- A alkanes      B alkenes      C alcohols      D carboxylic acids

**7. O/N 22/P23/Q39**

The structure of propene is shown.



Which diagram represents poly(propene)?

**8. M/J 22/P22/Q36**

Ethene reacts with substance X to form ethanol.

What is X?

- A ethanoic acid      B glucose      C hydrogen      D steam

**9. M/J 22/P22/Q37**

Alkenes can be produced by cracking large hydrocarbon molecules to form smaller hydrocarbon molecules.

Which equations represent possible reactions when tetradecane,  $\text{C}_{14}\text{H}_{30}$ , is cracked?

- 1  $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_6 + \text{C}_3\text{H}_6 + \text{C}_4\text{H}_8 + \text{C}_5\text{H}_{10}$   
 2  $\text{C}_{14}\text{H}_{30} \rightarrow \text{H}_2 + \text{C}_2\text{H}_4 + \text{C}_3\text{H}_6 + \text{C}_4\text{H}_8 + \text{C}_5\text{H}_{10}$   
 3  $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_6 + 4\text{C}_3\text{H}_6$   
 4  $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_6 + \text{C}_3\text{H}_8 + \text{C}_9\text{H}_{18}$

- A 1 only      B 1 and 4      C 1, 2 and 3      D 3 and 4

**10. M/J 22/P21/Q37**

Some properties of colourless liquid L are listed.

- It boils at  $65^\circ\text{C}$ .
- When added to water, two layers form which do not mix.
- It does not react with sodium carbonate.
- It has no effect on bromine water.

What is L?

- A ethanol      B hexane      C hexene      D ethanoic acid

**11. M/J 22/P21/Q38**

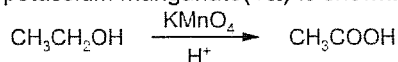
A molecule of compound P contains two carbon atoms and four hydrogen atoms.

Which row represents P?

|   | name of compound | $M_r$ | reacts with aqueous bromine |
|---|------------------|-------|-----------------------------|
| A | ethane           | 30    | x                           |
| B | ethene           | 16    | ✓                           |
| C | ethene           | 28    | ✓                           |
| D | ethene           | 28    | x                           |

**12. M/J 22/P21/Q39**

The reaction of ethanol with acidified potassium manganate(VII) is shown.



Which type of reaction is taking place?

- A addition                      B condensation                      C hydrolysis                      D oxidation

**13. M/J 22/P23/Q37**

Butene has three structural isomers which are alkenes.

Which statements about these isomers are correct?

- 1 They have the same molecular formula.
- 2 They have different numbers of bonds in the molecule.
- 3 They have a C=C bond in the structure.

- A 1 and 2                      B 1 and 3                      C 2 only                      D 3 only

**14. MAR 22/P22/Q5**

Fermentation of sugar produces a mixture of ethanol solution and solid yeast.

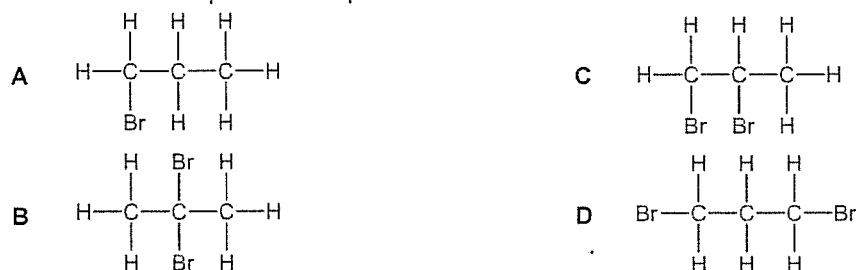
How is the solid yeast removed from the mixture?

- A crystallisation                      C filtration  
B distillation                      D fractional distillation

**15. MAR 22/P22/Q38**

Propene, C<sub>3</sub>H<sub>6</sub>, reacts with bromine, Br<sub>2</sub>, in an addition reaction.

Which structure represents the product of this reaction?

**16. O/N 21/P22/Q36**

Which statement about ethene is correct?

- A It has the chemical formula C<sub>2</sub>H<sub>6</sub>.  
B It burns in excess oxygen producing carbon dioxide and water.  
C It reacts with Br<sub>2</sub> to produce an orange solution.  
D It reacts with oxygen to form ethanol.

**17. O/N 21/P22/Q37**

Ethanol is manufactured by fermentation of sugars or by catalytic hydration of ethene.

Which row states an advantage of each method?

|   | fermentation              | hydration               |
|---|---------------------------|-------------------------|
| A | produces purer ethanol    | is a batch process      |
| B | produces purer ethanol    | is a continuous process |
| C | uses a renewable resource | is a batch process      |
| D | uses a renewable resource | is a continuous process |

**18. O/N 21/P22/Q38**

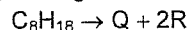
Which statements about unsaturated hydrocarbons are correct?

- 1 They contain both single and double bonds.
- 2 They turn aqueous bromine from colourless to brown.
- 3 They can be manufactured by cracking.

- A 1 and 2 only                      B 1 and 3 only                      C 2 and 3 only                      D 1, 2 and 3

**19. O/N 21/P21/Q33**

An alkane molecule of molecular formula  $C_8H_{18}$  undergoes cracking. The equation for the reaction is shown.

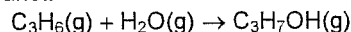


Substance R has two carbon atoms per molecule and decolourises aqueous bromine.  
What is substance Q?

- A butane                      B butene                      C ethane                      D ethene

**20. O/N 21/P21/Q38**

Propene reacts with steam to form propanol.



Which type of reaction takes place?

- A addition                      B condensation                      C oxidation                      D substitution

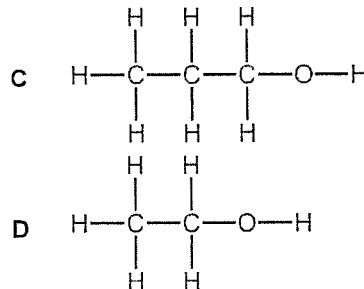
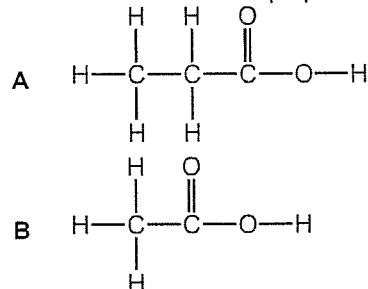
**21. O/N 21/P21/Q39**

Which statement about aqueous ethanoic acid is correct?

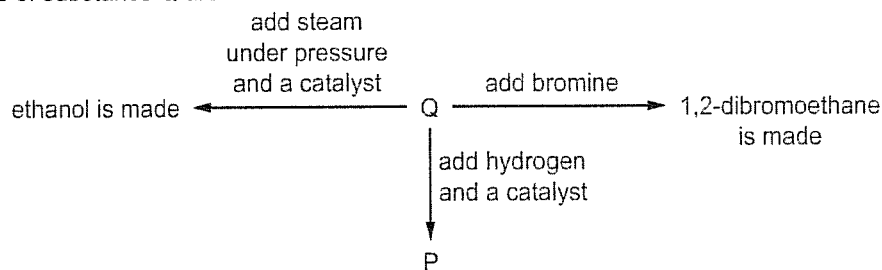
- A It reacts with magnesium to produce a salt and hydrogen.  
B It reacts with sodium hydroxide to produce a salt and hydrogen.  
C It reacts with ammonium salts to produce ammonia.  
D It turns red litmus blue.

**22. O/N 21/P23/Q33**

What is the structure of propanol?

**23. O/N 21/P23/Q38**

Some reactions of substance Q are shown.



What is P?

- A ethane                      B ethanoic acid                      C ethene                      D poly(ethene)

**24. M/J 21/P22/Q34**

Which statement about ethanol is **not** correct?

- A Ethanol can be made by fermentation.  
B Ethanol is oxidised to make ethanoic acid.  
C Ethanol reacts with oxygen exothermically, making it a good fuel.  
D Ethanol reacts with propanoic acid to make propyl ethanoate.

**25. M/J 21/P21/Q37**

What is an advantage of manufacturing ethanol by fermentation?

- A The process is very fast.  
 B The ethanol requires no separation.  
 C The raw materials used are renewable.  
 D There are no other products formed.

**26. M/J 21/P21/Q38**

P, Q, R and S are four organic compounds.

P is an unsaturated hydrocarbon.

Q burns but otherwise is unreactive.

R contains a C–C single bond and a C=C double bond.

S undergoes addition polymerisation.

Which compounds are alkenes?

- A P and R only      B P, R and S      C P, Q and S      D Q, R and S

**27. M/J 21/P23/Q38**

What is an advantage of the fermentation process for producing ethanol compared with the catalytic addition of steam to ethene?

- A Fermentation requires less heat energy.  
 B Ethanol from fermentation needs to be distilled.  
 C Raw materials for fermentation are non-renewable.  
 D The fermentation process is carried out in batches rather than continuously.

**28. MAR 21/P22/Q7**

How many electrons are used to form covalent bonds in a molecule of methanol, CH<sub>3</sub>OH?

- A 5      B 6      C 8      D 10

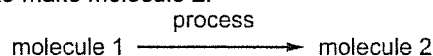
**29. MAR 21/P22/Q36**

Which statement describes the reaction between ethene and steam?

- A a cracking reaction which produces ethane and hydrogen gas as products  
 B an addition reaction which produces ethanol as the only product  
 C an oxidation reaction which produces ethanoic acid as the only product  
 D a slow reaction producing ethanol and carbon dioxide

**30. MAR 21/P22/Q39**

Molecule 1 undergoes a process to make molecule 2.



Which row describes the molecules and the process?

|   | molecule 1     | process        | molecule 2 |
|---|----------------|----------------|------------|
| A | monomer        | cracking       | polymer    |
| B | monomer        | polymerisation | polymer    |
| C | small molecule | polymerisation | monomer    |
| D | small molecule | cracking       | monomer    |

**31. O/N 20/P22/Q37**

Which molecule is **not** produced by an addition reaction of ethene?

- A CH<sub>3</sub>CH<sub>3</sub>      B CH<sub>2</sub>BrCH<sub>2</sub>Br      C CH<sub>3</sub>CH<sub>2</sub>OH      D CH<sub>3</sub>CH<sub>2</sub>CH<sub>3</sub>



**32. O/N 20/P22/Q38, O/N 20/P21/Q38, O/N 20/P23/Q38**

The flow chart shows the preparation of ethanol and some important chemistry of ethanol.

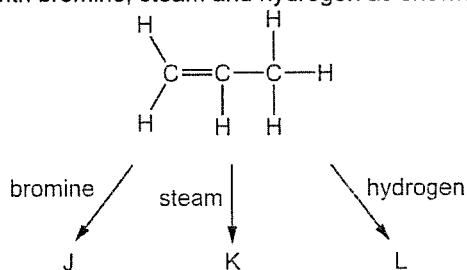
substance X  $\xrightarrow{\text{fermentation}}$  ethanol  $\xrightarrow{\text{process Y}}$  carbon dioxide + substance Z

What are X, Y and Z?

|   | X       | Y              | Z       |
|---|---------|----------------|---------|
| A | yeast   | combustion     | oxygen  |
| B | glucose | combustion     | steam   |
| C | glucose | polymerisation | water   |
| D | yeast   | fermentation   | glucose |

**33. O/N 20/P21/Q37**

Propene is an alkene that reacts with bromine, steam and hydrogen as shown.



What are the products of these reactions?

|   | J              | K              | L       |
|---|----------------|----------------|---------|
| A | bromopropane   | propanol       | butane  |
| B | dibromopropane | propanoic acid | propane |
| C | dibromopropane | propanol       | propane |
| D | bromopropane   | propanoic acid | butane  |

**34. O/N 20/P21/Q31**

Ammonia is manufactured using the Haber process.

Which statement about this process is correct?

- A The catalyst used for this reaction is vanadium pentoxide.  
 B The hydrogen used is extracted from air.  
 C Using a high pressure increases the yield of ammonia.  
 D Using a high temperature increases the yield of ammonia.

**35. O/N 20/P23/Q36**

Which product is obtained when bromine reacts with propene,  $\text{CH}_3\text{CH}=\text{CH}_2$ ?

- A  $\text{CH}_3\text{CH}=\text{CHBr}$       B  $\text{CH}_3\text{CBr}=\text{CHBr}$       C  $\text{CH}_3\text{CH}_2\text{CHBr}_2$       D  $\text{CH}_3\text{CHBrCH}_2\text{Br}$

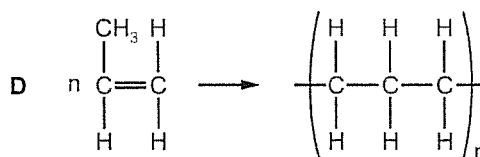
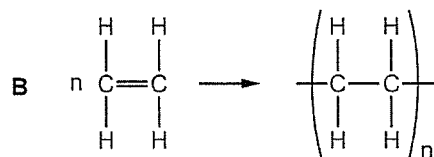
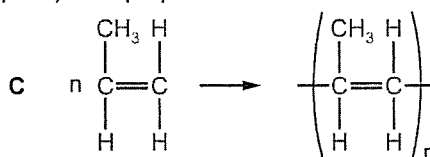
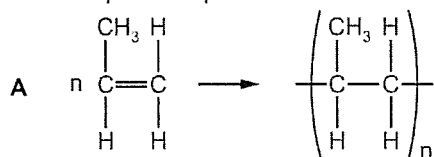
**36. M/J 20/P21/Q35**

Which row about the production of ethanol by fermentation is correct?

|   | raw materials | energy requirement | rate of reaction |
|---|---------------|--------------------|------------------|
| A | non-renewable | high               | slow             |
| B | renewable     | low                | slow             |
| C | non-renewable | low                | fast             |
| D | renewable     | high               | fast             |

**37. O/N 20/P23/Q39**

Which equation represents the formation of poly(propene) from propene?

**38. M/J 20/P23/Q35**

Ethanol is produced by:

- 1 the catalytic addition of steam to ethene
- 2 fermentation.

Which statement is correct?

- A Both processes require similar amounts of energy.  
 B Both processes use a catalyst.  
 C Process 1 uses a renewable resource.  
 D Process 2 produces the purest ethanol

**39. MAR 20/P22/Q36**

Which statements about propene are correct?

- 1 Propene contains only single bonds.
- 2 Propene decolourises bromine water.
- 3 Propene is obtained by cracking.
- 4 Propene is a hydrocarbon.

A 1 and 4

B 2, 3 and 4

C 2 and 4 only

D 4 only

**40. MAR 20/P22/Q37**

Which row describes the production of ethanol and its properties?

|   | can be made from glucose | can be made from ethene | is used as a fuel | is used as a solvent |
|---|--------------------------|-------------------------|-------------------|----------------------|
| A | ✓                        | ✓                       | ✓                 | ✓                    |
| B | ✓                        | x                       | ✓                 | ✓                    |
| C | x                        | ✓                       | ✓                 | x                    |
| D | x                        | ✓                       | x                 | ✓                    |

key

✓ = yes

x = no

**41. O/N 19/P22/Q38**

Ethanol is produced by fermentation or by the reaction of ethene with steam.

Which row is correct?

|   | by fermentation              | from ethene                  |
|---|------------------------------|------------------------------|
| A | uses a temperature of 100 °C | uses a temperature of 350 °C |
| B | needs yeast as a catalyst    | does not need a catalyst     |
| C | very slow reaction           | very fast reaction           |
| D | high yield of ethanol        | low yield of ethanol         |

**42. O/N 19/P23/Q38**

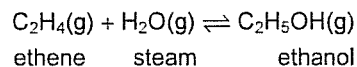
Ethanol is made by fermentation of sugars and by the catalytic addition of steam to ethene.

What are two advantages of making ethanol by the catalytic addition of steam to ethene rather than by fermentation of sugars?

- A faster reaction and renewable raw materials
- B purer product and faster reaction
- C renewable raw materials and continuous process
- D uses more energy and forms a purer product

**43. M/J 19/P22/Q15**

The equation for the manufacture of ethanol is shown.

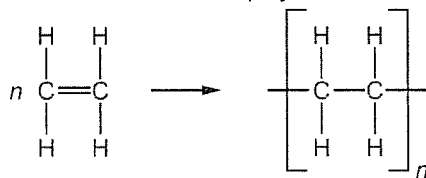


What is the effect of doubling the pressure on this reaction?

- A decreases the rate of formation of ethanol
- B increases the yield of ethene
- C decreases the rate of formation of ethene
- D increases the yield of ethanol

**44. O/N 19/P23/Q39**

The diagram shows the structure of a monomer and of the polymer made from it.



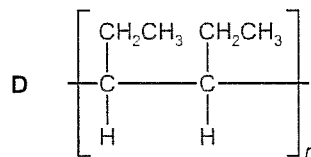
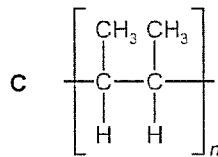
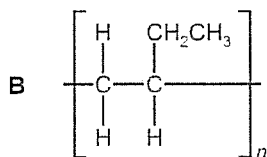
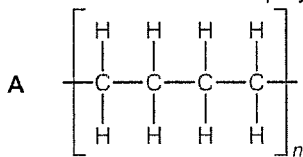
What are the monomer and polymer?

|   | monomer | polymer      |
|---|---------|--------------|
| A | ethane  | poly(ethane) |
| B | ethane  | poly(ethene) |
| C | ethene  | poly(ethane) |
| D | ethene  | poly(ethene) |

**45. M/J 19/P22/Q40**

But-1-ene has the structure  $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ .

What is the structure of poly(but-1-ene)?

**46. M/J 19/P21/Q36**

Why is ethanol a member of the homologous series of alcohols but propane is not?

- A Ethanol has two carbon atoms per molecule but propane has three.
- B Ethanol can be made from ethene but propane is obtained from petroleum.
- C Ethanol is a liquid but propane is a gas.
- D Ethanol contains the same functional group as other alcohols but propane does not.

## 47. MAR 19/P22/Q37

Which statement about members of a homologous series is correct?

- A Successive members differ by  $\text{CH}_3$ .  
 B Successive members have a molecular mass that differs by 14.  
 C They have the same molecular formula.  
 D They have identical physical properties.

## 48. O/N 18/P22/Q36, O/N 18/P21/Q36, O/N 18/P23/Q36

Which two compounds are molecules which both contain a double bond?

- A ethane and ethanoic acid  
 B ethane and ethanol  
 C ethene and ethanoic acid  
 D ethene and ethanol

## 49. M/J 18/P22/Q36

Which statement about alkenes is **not** correct?

- A They decolourise aqueous bromine.  
 B They only contain the elements carbon and hydrogen.  
 C They react with hydrogen to form alkanes.  
 D They react with steam to produce carboxylic acids.

## 50. M/J 18/P21/Q36

Which reaction is **not** a reaction which alkenes undergo?

- A bromination      B hydration      C hydrogenation      D hydrolysis

## 51. M/J 18/P21/Q37, M/J 18/P23/Q37

Which substances can be obtained by cracking hydrocarbons?

- A ethanol and ethene  
 B ethanol and hydrogen  
 C ethene and hydrogen  
 D ethene and poly(ethene)

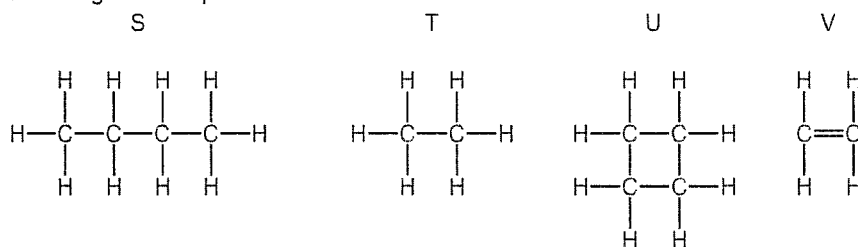
## 52. MAR 18/P22/Q15

In which reaction is the rate of reaction **not** affected by light?

- A the conversion of carbon dioxide and water to glucose and oxygen in green plants  
 B the reaction of bromine with ethene  
 C the reaction of chlorine with methane  
 D the reduction of silver ions to silver

## 53. O/N 17/P22/Q35

The structures of four organic compounds are shown.

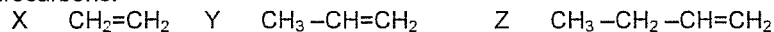


Which compounds are unsaturated?

- A S only      B T and U      C U only      D V only

## 54. O/N 17/P22/Q37, O/N 17/P21/Q37, O/N 17/P23/Q37, M/J 14/P11/Q39, M/J 14/P13/Q38

X, Y and Z are three hydrocarbons.

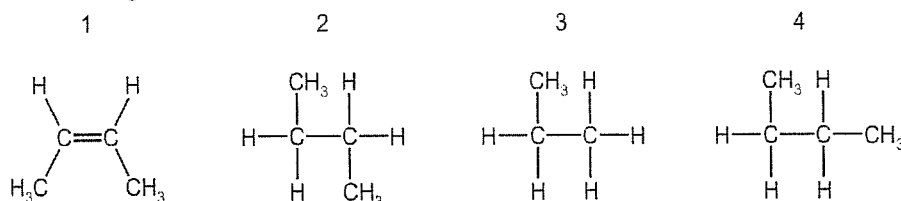


What do compounds X, Y and Z have in common?

- 1 They are all alkenes.  
 2 They are all part of the same homologous series.  
 3 They all have the same boiling point.
- A 1, 2 and 3      B 1 and 2 only      C 1 and 3 only      D 2 and 3 only

## 55. O/N 17/P21/Q35

The structures of some organic molecules are shown.

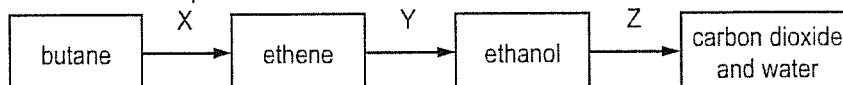


Which structures represent an alkane with four carbon atoms?

- A 1 only                                  B 2 and 3                                  C 2 and 4                                  D 3 and 4

## 56. O/N 17/P21/Q38, O/N 17/P23/Q38, O/N 14/P13/Q38

The diagram shows a reaction sequence.



Which row names the processes X, Y and Z?

|   | X            | Y            | Z           |
|---|--------------|--------------|-------------|
| A | cracking     | fermentation | respiration |
| B | cracking     | hydration    | combustion  |
| C | distillation | fermentation | respiration |
| D | distillation | hydration    | combustion  |

## 57. M/J 17/P22/Q36

Which statement about homologous series is **not** correct?

- A Alkenes have the same general formula,  $\text{C}_n\text{H}_{2n+2}$ .  
 B Each member of the homologous series of alkanes differs from the next by  $\text{CH}_2$ .  
 C The members of a homologous series all have similar chemical properties.  
 D The members of a homologous series all have the same functional group.

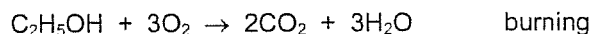
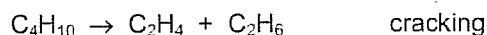
## 58. MAR 17/P22/Q35

Which reaction of ethene is **not** an addition reaction?

- A reaction with bromine                                  C reaction with oxygen  
 B reaction with hydrogen                                  D reaction with steam

## 59. MAR 17/P22/Q36

Ethanol is a fuel used in cars. It can be made from petroleum.

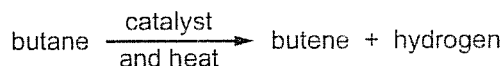


Compounds of how many homologous series appear in these equations?

- A 1                                  B 2                                  C 3                                  D 4

## 60. O/N 16/P22/Q36, O/N 16/P21/Q36, O/N 16/P23/Q36

Butane reacts as shown.



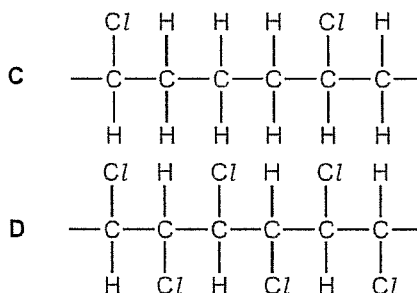
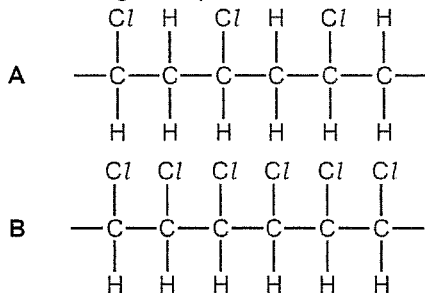
What is this type of reaction?

- A combustion                                  B cracking                                  C polymerisation                                  D reduction

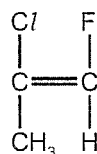
**61. O/N 16/P22/Q39**

Chloroethene,  $\text{CH}_2=\text{CHCl}$ , can be polymerised.

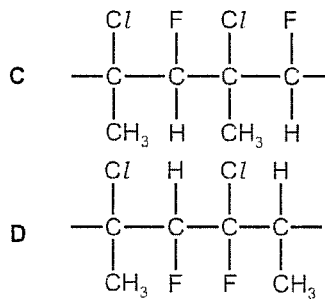
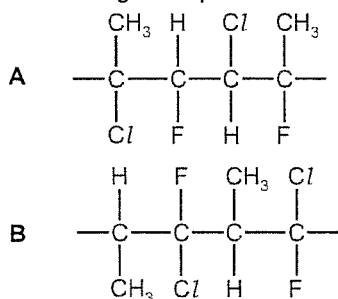
Which diagram represents a section of the polymer?

**62. O/N 16/P23/Q39**

The organic compound shown can be polymerised.



Which diagram represents a section of the polymer?

**63. O/N 16/P21/Q8**

Benzene is a liquid with molecular formula  $\text{C}_6\text{H}_6$ .

Ethene is a gas with molecular formula  $\text{C}_2\text{H}_4$ .

Which statement is correct?

- A** 1 mole of benzene and 1 mole of ethene contain the same number of atoms.  
**B** 1 mole of benzene and 1 mole of ethene both have a volume of  $24 \text{ dm}^3$  at room temperature and pressure.  
**C** Both benzene and ethene have the same empirical formula.  
**D** The number of carbon atoms in 0.5 moles of ethene is equal to the Avogadro constant.

**64. M/J 16/P22/Q39**

In which row are the monomer and polymer chain correctly matched?

|          | monomer                                | part of the polymer chain                                                                       |
|----------|----------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>A</b> | $\text{CH}_3\text{CH}=\text{CHCH}_3$   | $-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-$ |
| <b>B</b> | $\text{CH}_2=\text{CHCl}$              | $-\text{CHCl}-\text{CHCl}-\text{CHCl}-\text{CHCl}-$                                             |
| <b>C</b> | $\text{CH}_3\text{CH}=\text{CH}_2$     | $-\text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_3-\text{CH}-\text{CH}_2-$                         |
| <b>D</b> | $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ | $-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_2\text{CH}_3)-$                       |



70. O/N 15/P12/Q36, O/N 15/P11/Q36, O/N 15/P13/Q36  
Which row describes the formation of a polymer?

|   | monomer | polymer      |
|---|---------|--------------|
| A | ethane  | poly(ethane) |
| B | ethane  | poly(ethene) |
| C | ethene  | poly(ethane) |
| D | ethene  | poly(ethene) |

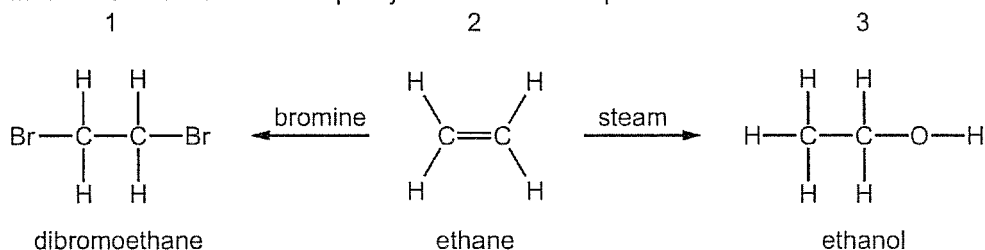
71. O/N 15/P12/Q37

Hydrocarbons obtained by fractional distillation of petroleum can be cracked to make useful products. Which substance could **not** be obtained by cracking propane,  $M_r$  44?

- A  $C_2H_4$       B  $C_3H_6$       C  $C_4H_8$       D  $H_2$

72. O/N 15/P11/Q35

The diagram shows the structure of a simple hydrocarbon and the products of two of its reactions.

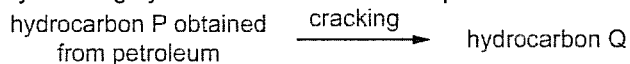


Which structures are named correctly?

|   | structure |   |   |
|---|-----------|---|---|
|   | 1         | 2 | 3 |
| A | ✓         | ✓ | x |
| B | ✓         | x | ✓ |
| C | x         | ✓ | ✓ |
| D | x         | ✓ | x |

73. O/N 15/P11/Q40

Alkenes are manufactured by cracking hydrocarbons obtained from petroleum.



Which row describes the size of the molecules in hydrocarbons P and Q and the effect of Q on aqueous bromine?

|   | size of P molecules | size of Q molecules | effect of Q on aqueous bromine |
|---|---------------------|---------------------|--------------------------------|
| A | large               | small               | decolourises                   |
| B | large               | small               | no effect                      |
| C | small               | large               | decolourises                   |
| D | small               | large               | no effect                      |



## 74. O/N 15/P13/Q38

Ethanol can be formed by

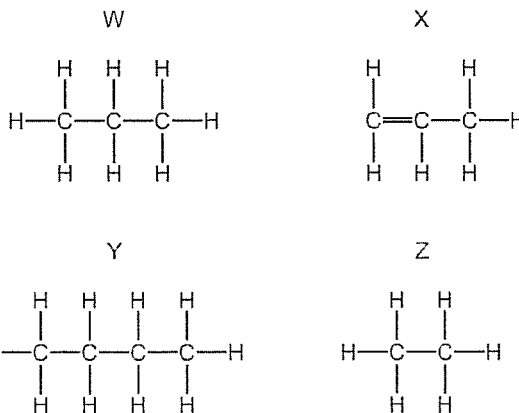
- 1 fermentation
- 2 reaction between steam and ethene

Which of these processes uses a catalyst?

|   | 1 | 2 |
|---|---|---|
| A | ✓ | ✓ |
| B | ✓ | x |
| C | x | ✓ |
| D | x | x |

## 75. O/N 15/P13/Q39

The structures of four compounds are shown.



Which are members of the same homologous series?

- A W, X, Y and Z      B W and X only      C W, Y and Z only      D X and Z only

## 76. O/N 15/P13/Q40

During the process of cracking hydrocarbons, an ..... 1 ..... is converted into an ..... 2 .....

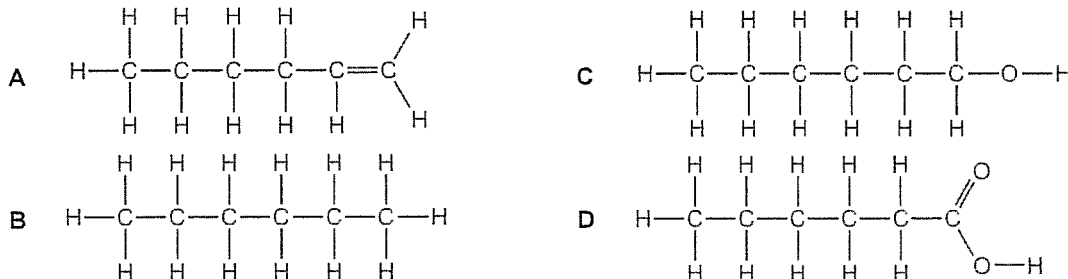
The presence of an ..... 3 ..... can be shown by a visible reaction with ..... 4 .....

Which words complete gaps 1, 2, 3 and 4?

|   | 1      | 2      | 3      | 4       |
|---|--------|--------|--------|---------|
| A | alkane | alkene | alkene | bromine |
| B | alkane | alkene | alkene | steam   |
| C | alkene | alkane | alkane | bromine |
| D | alkene | alkane | alkane | steam   |

## 77. O/N 14/P12/Q36, O/N 14/P11/Q37

Which molecular structure shows hexene?







**88. M/J 13/P12/Q39, M/J 13/P11/Q39, M/J 13/P13/Q38**

Which columns describe the hydrocarbons ethane and ethene?

|                               | 1           | 2                    | 3           | 4                    |
|-------------------------------|-------------|----------------------|-------------|----------------------|
| state at room temperature     | gas         | gas                  | liquid      | liquid               |
| reaction with oxygen          | burns       | burns                | burns       | burns                |
| reaction with aqueous bromine | no reaction | decolourises bromine | no reaction | decolourises bromine |

**A** 1 (ethane) and 2 (ethene)**C** 2 (ethene) and 3 (ethane)**B** 1 (ethane) and 4 (ethene)**D** 3 (ethane) and 4 (ethene)

## 13.4: Carboxylic acids and esters

1. O/N 22/P22/Q19, O/N 22/P23/Q18, MAR 18/P22/Q17

Ethanoic acid reacts with water to produce an acidic solution.

Which row describes the roles of ethanoic acid and water in this reaction?

|   | ethanoic acid       | water               |
|---|---------------------|---------------------|
| A | accepts a proton    | donates a proton    |
| B | accepts an electron | donates an electron |
| C | donates a proton    | accepts a proton    |
| D | donates an electron | accepts an electron |

2. O/N 22/P22/Q37

Which type of reaction occurs when ethanol is converted to ethanoic acid?

- A combustion      B decomposition      C neutralisation      D oxidation

3. O/N 22/P21/Q34

Which formula represents ethanol?

- A  $\text{CH}_3\text{CH}_3$       B  $\text{CH}_2\text{CH}_2$       C  $\text{CH}_3\text{CH}_2\text{OH}$       D  $\text{CH}_3\text{COOH}$

4. O/N 22/P21/Q38

Ethanoic acid is made by reacting ethanol with acidified potassium manganate(VII).

Which type of reaction occurs when ethanol reacts with acidified potassium manganate(VII)?

- A displacement      B fermentation      C oxidation      D neutralisation

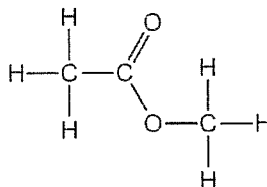
5. O/N 22/P23/Q34

Which formula represents ethyl butanoate?

- A  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$       C  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$   
 B  $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$       D  $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

6. M/J 22/P22/Q35, M/J 22/P21/Q35, M/J 22/P23/Q35, M/J 19/P23/Q39

The structure of ester W is shown.

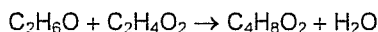


Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

|   | name of ester W  | carboxylic acid | alcohol  |
|---|------------------|-----------------|----------|
| A | ethyl methanoate | ethanoic acid   | methanol |
| B | ethyl methanoate | methanoic acid  | ethanol  |
| C | methyl ethanoate | ethanoic acid   | methanol |
| D | methyl ethanoate | methanoic acid  | ethanol  |

## 7. M/J 22/P23/Q39

An ester,  $C_4H_8O_2$ , is made by reacting 0.06 mol of ethanol,  $C_2H_6O$ , and 0.05 mol of ethanoic acid,  $C_2H_4O_2$ .



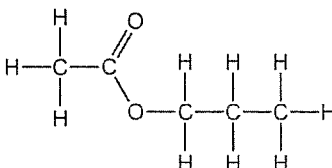
0.0375 mol of the ester was made.

What is the percentage yield and the  $M_r$  of the ester?

|   | percentage yield / % | $M_r$ |
|---|----------------------|-------|
| A | 62.5                 | 48    |
| B | 75.0                 | 48    |
| C | 62.5                 | 88    |
| D | 75.0                 | 88    |

## 8. MAR 22/P22/Q33

The structure of an ester is shown.

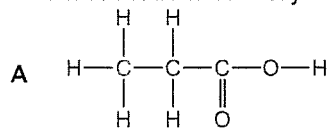


What are the names of the carboxylic acid and the alcohol that react together to form this ester?

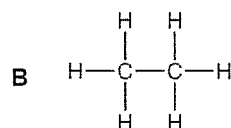
|   | carboxylic acid | alcohol     |
|---|-----------------|-------------|
| A | ethanoic acid   | ethanol     |
| B | ethanoic acid   | propan-1-ol |
| C | propanoic acid  | ethanol     |
| D | propanoic acid  | propan-1-ol |

## 9. MAR 22/P22/Q35

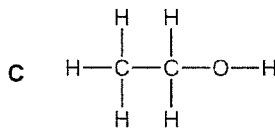
Which structure is correctly named?



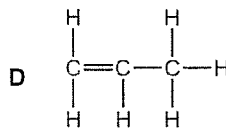
ethanoic acid



ethene



ethanol



propane

## 10. MAR 22/P22/Q37

Carboxylic acids are made by the oxidation of alcohols.

Which carboxylic acid is produced from  $CH_3CH_2OH$ ?

- A butanoic acid      B ethanoic acid      C methanoic acid      D propanoic acid

## 11. MAR 22/P22/Q39

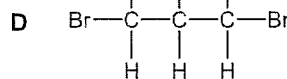
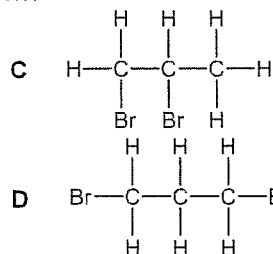
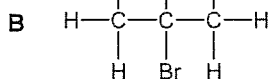
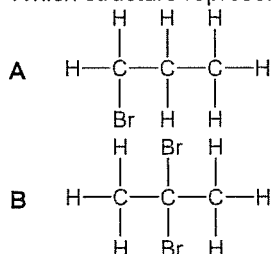
Which statements about ethanol are correct?

- 1 Ethanol is used as a solvent.
- 2 Ethanol can be made directly from ethane.
- 3 Ethanol is a covalent compound.

- A 1 only      B 1 and 2      C 1 and 3      D 2 and 3

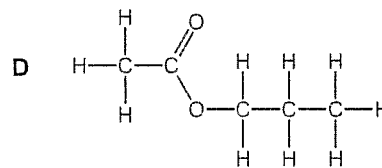
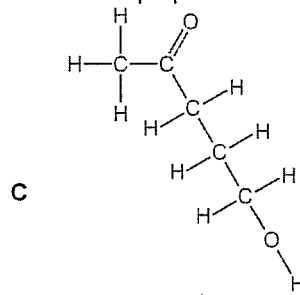
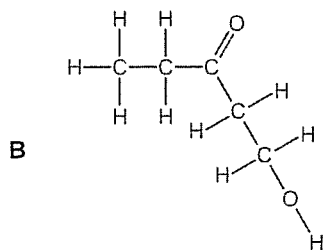
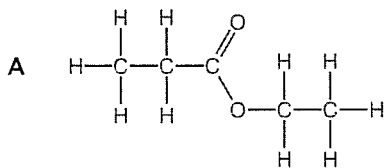
## 12. MAR 22/P22/Q38

Propene,  $C_3H_6$ , reacts with bromine,  $Br_2$ , in an addition reaction.  
Which structure represents the product of this reaction?



## 13. O/N 21/P22/Q33

What is the structure of the ester formed from ethanoic acid and propanol?



## 14. M/J 21/P22/Q37

Which statements about ethanoic acid are correct?

- 1 It is a strong acid.
- 2 It reacts with ethanol to form an ester.
- 3 It has the formula  $CH_3COOH$ .

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

## 15. M/J 21/P21/Q33

What is the structure of butanoic acid?

A  $CH_3CH_2CO_2H$ C  $CH_3CH_2CH_2CH_2CO_2H$ B  $CH_3CH_2CH_2CO_2H$ D  $CH_3CH_2CH_2CO_2CH_3$ 

## 16. M/J 21/P21/Q34

Compound Z contains carbon, hydrogen and oxygen.

Molecules of compound Z have four hydrogen atoms and two carbon atoms.

Compound Z can be made by oxidation of an alcohol.

What is compound Z?

A ethene

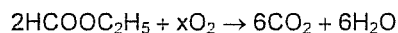
B ethanol

C ethanoic acid

D methyl methanoate

## 17. O/N 20/P22/Q8

Ethyl methanoate,  $HCOOC_2H_5$ , burns in excess oxygen to produce carbon dioxide and water. The equation is shown.



What is the value of x?

A 2

B 7

C 9

D 18

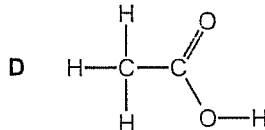
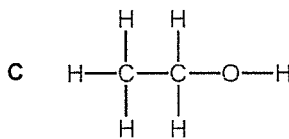
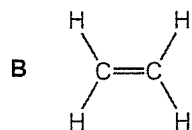
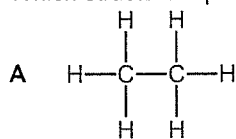
**18. MAR 21/P22/Q38**

The formula of an ester is  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$ .  
Which acid and alcohol react together to make the ester?

|          | acid           | alcohol  |
|----------|----------------|----------|
| <b>A</b> | butanoic acid  | butanol  |
| <b>B</b> | butanoic acid  | propanol |
| <b>C</b> | propanoic acid | butanol  |
| <b>D</b> | propanoic acid | propanol |

**19. O/N 20/P22/Q35**

Which structure represents a molecule of ethanol?

**20. O/N 20/P21/Q39**

Which statements about aqueous ethanoic acid are correct?

- 1 It is an alkane.
- 2 It reacts with sodium carbonate to form carbon dioxide.
- 3 It changes the colour of litmus solution from blue to red.
- 4 It is a hydrocarbon.

**A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

**21. O/N 20/P23/Q37**

Propanol is oxidised by acidified potassium manganate(VII) in a similar way to ethanol.  
Which compound is produced by the oxidation of propanol with acidified potassium manganate(VII)?

**A**  $\text{CH}_3\text{CH}_2\text{OH}$       **B**  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$       **C**  $\text{CH}_3\text{COOH}$       **D**  $\text{CH}_3\text{CH}_2\text{COOH}$

**22. M/J 20/P22/Q38**

An organic compound, P, reacts with zinc to produce a gas, Q.  
What are P and Q?

|          | P             | Q              |
|----------|---------------|----------------|
| <b>A</b> | ethanoic acid | carbon dioxide |
| <b>B</b> | ethanoic acid | hydrogen       |
| <b>C</b> | ethanol       | carbon dioxide |
| <b>D</b> | ethanol       | hydrogen       |

**23. M/J 20/P21/Q38**

Some properties of an organic compound J are listed.

- It is a liquid at room temperature.
- It is soluble in water.
- A solution of J reacts with calcium carbonate to form carbon dioxide.
- A solution of J has a pH of 3.

In which homologous series does J belong?

**A** alkane      **B** alkene      **C** alcohol      **D** carboxylic acid



**24. M/J 20/P23/Q38**

A small quantity of a solid chemical is added to a large excess of aqueous ethanoic acid. No bubbles of gas are seen and the solid dissolves to give a colourless solution. What was the solid chemical?

- A calcium hydroxide    B copper(II) oxide    C magnesium    D sodium carbonate

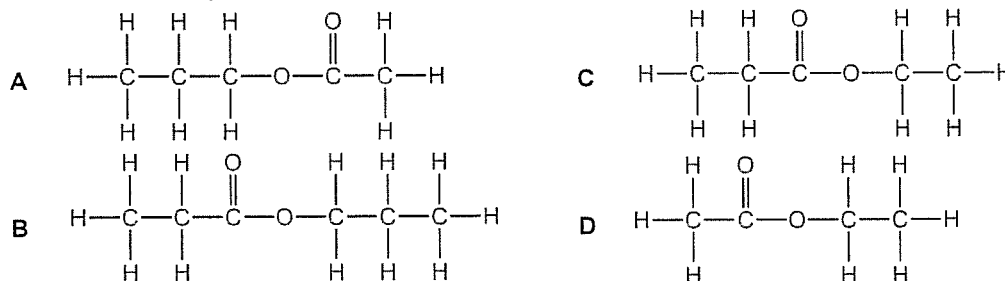
**25. MAR 20/P22/Q38**

Ethanoic acid is a typical carboxylic acid. Which statement about ethanoic acid is correct?

- A It can be oxidized to produce ethanol.  
 B It is a proton acceptor.  
 C It is fully dissociated in water.  
 D It reacts with ethanol to produce ethyl ethanoate and water.

**26. MAR 20/P22/Q39**

Which structure represents the ester made from ethanoic acid and propanol?

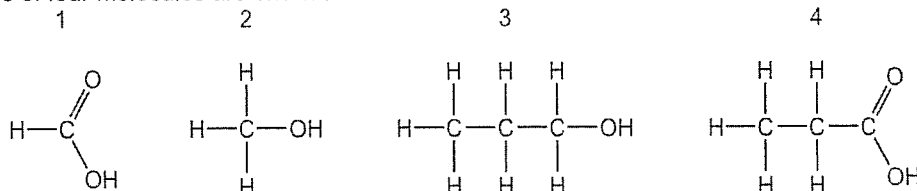
**27. M/J 19/P22/Q38**

What are the properties of aqueous ethanoic acid?

|   | decolourises<br>bromine water | reacts with<br>calcium carbonate to<br>make carbon dioxide | turns damp red<br>litmus blue |
|---|-------------------------------|------------------------------------------------------------|-------------------------------|
| A | ✓                             | ✓                                                          | x                             |
| B | ✓                             | x                                                          | ✓                             |
| C | x                             | ✓                                                          | x                             |
| D | x                             | x                                                          | ✓                             |

**28. M/J 19/P22/Q39**

The structures of four molecules are shown.



Which molecules react together to form the ester propyl methanoate?

- A 1 and 2    B 1 and 3    C 2 and 4    D 3 and 4

**29. M/J 19/P21/Q38**

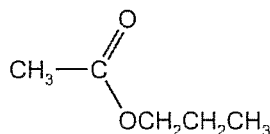
Which statements about aqueous ethanoic acid are correct?

- 1 Ethanoic acid contains the functional group  $\text{-COOH}$ .  
 2 Ethanoic acid reacts with carbonates to produce hydrogen.  
 3 Ethanoic acid turns Universal Indicator paper blue.  
 4 Ethanoic acid has a pH lower than pH 7.

- A 1 and 2    B 1 and 3    C 1 and 4    D 2 and 4

## 30. M/J 19/P21/Q39

The structure of an ester is shown.



What is the name of the ester?

- A ethyl propanoate    B methyl propanoate    C propyl ethanoate    D propyl methanoate

## 31. M/J 19/P23/Q38

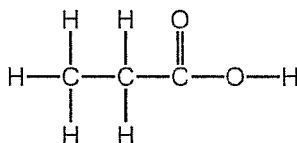
Which statement about aqueous ethanoic acid is correct?

- A It reacts with metal carbonates to form salts, hydrogen and water.  
B It reacts with metal oxides to form salts and oxygen.  
C It reacts with reactive metals to form salts and hydrogen.  
D It turns damp red litmus paper blue.

## 32. MAR 19/P22/Q39

The structure of a compound, G, is shown.

G is in the same homologous series as ethanoic acid.



Which row describes some of the properties of an aqueous solution of G?

|   | produces a gas with magnesium | turns methyl orange yellow |
|---|-------------------------------|----------------------------|
| A | no                            | yes                        |
| B | no                            | no                         |
| C | yes                           | no                         |
| D | yes                           | yes                        |

## 33. O/N 18/P22/Q39

Which reaction can be used to make ethanoic acid?

- A oxidation of ethanol    C reduction of ethanol  
B oxidation of ethene    D reduction of ethene

## 34. O/N 18/P21/Q38

When the alcohol  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$  reacts with the carboxylic acid  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$  an ester is formed.

What is the name and structural formula of this ester?

|   | name             | structural formula                                                      |
|---|------------------|-------------------------------------------------------------------------|
| A | butyl propanoate | $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ |
| B | butyl propanoate | $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$ |
| C | propyl butanoate | $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ |
| D | propyl butanoate | $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$ |

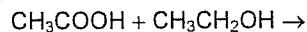
## 35. O/N 18/P23/Q39

Which statement about ethanoic acid is correct?

- A It contains a  $-\text{C}_2\text{H}_5$  group.    C It is formed by the reduction of ethanol.  
B It is a strong acid.    D It reacts with alcohols to form esters.

**36. M/J 18/P22/Q39**

What is the name of the organic product of the reaction shown?



- A ethyl ethanoate      B ethyl methanoate      C methyl ethanoate      D methyl propanoate

**37. M/J 18/P21/Q39**

Which structural formula represents methyl propanoate?

- A  $\text{CH}_3\text{CH}_2\text{COOCH}_3$       C  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_3$   
 B  $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3$       D  $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$

**38. M/J 18/P23/Q39**Which esters have the molecular formula  $\text{C}_5\text{H}_{10}\text{O}_2$ ?

- 1 ethyl propanoate  
 2 propyl ethanoate  
 3 butyl methanoate  
 4 methyl butanoate  
 A 1, 2, 3 and 4      B 1, 2 and 3 only      C 1 and 2 only      D 3 and 4 only

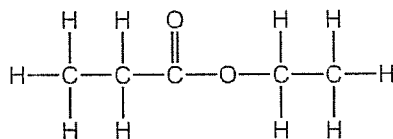
**39. MAR 18/P22/Q39**

Which substances react together to form ethyl propanoate?

- A ethanoic acid and propanol      C ethene and propanol  
 B ethanol and propene      D propanoic acid and ethanol

**40. O/N 17/P22/Q39**

The structure of an ester is shown.



Which substances react to form this ester?

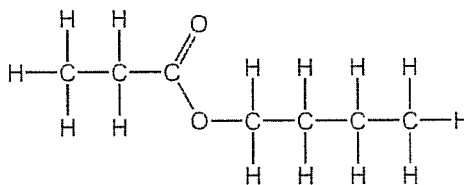
- A ethanol and ethanoic acid      C propanol and ethanoic acid  
 B ethanol and propanoic acid      D propanol and propanoic acid

**41. O/N 17/P21/Q39**Which pair of compounds can be used to prepare  $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$ ?

- A ethanoic acid and ethanol      C propanoic acid and ethanol  
 B ethanoic acid and propanol      D propanoic acid and propanol

**42. O/N 17/P23/Q39**

The structure of an ester is shown.



Which combination of carboxylic acid and alcohol produces this ester?

|   | carboxylic acid | alcohol  |
|---|-----------------|----------|
| A | butanoic acid   | ethanol  |
| B | butanoic acid   | propanol |
| C | ethanoic acid   | butanol  |
| D | propanoic acid  | butanol  |

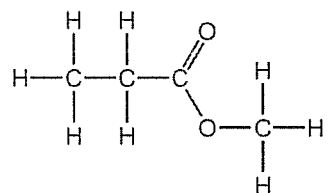
**43. M/J 17/P21/Q19**

Which row shows how the hydrogen ion concentration and pH of ethanoic acid compare to those of hydrochloric acid of the same concentration?

|          | ethanoic acid compared to hydrochloric acid |        |
|----------|---------------------------------------------|--------|
|          | hydrogen ion concentration                  | pH     |
| <b>A</b> | higher                                      | higher |
| <b>B</b> | higher                                      | lower  |
| <b>C</b> | lower                                       | higher |
| <b>D</b> | lower                                       | lower  |

**44. M/J 17/P21/Q39**

The structure of an ester is shown.

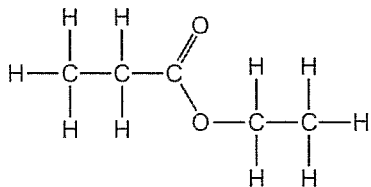


Which row is correct?

|          | name of ester     | names of the carboxylic acid and the alcohol used to form the ester |
|----------|-------------------|---------------------------------------------------------------------|
| <b>A</b> | methyl propanoate | methanoic acid and propanol                                         |
| <b>B</b> | methyl propanoate | methanol and propanoic acid                                         |
| <b>C</b> | propyl methanoate | methanoic acid and propanol                                         |
| <b>D</b> | propyl methanoate | methanol and propanoic acid                                         |

**45. M/J 17/P23/Q39**

The structure of an ester is shown.

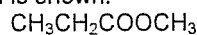


Which alcohol and carboxylic acid produce this ester?

|          | alcohol  | carboxylic acid |
|----------|----------|-----------------|
| <b>A</b> | ethanol  | ethanoic acid   |
| <b>B</b> | ethanol  | propanoic acid  |
| <b>C</b> | propanol | ethanoic acid   |
| <b>D</b> | propanol | propanoic acid  |

**46. MAR 17/P22/Q38**

The structural formula of an organic compound is shown.



What is the name of this compound?

- A butanoic acid      B ethyl ethanoate      C methyl propanoate      D propyl methanoate

**47. O/N 16/P22/Q37, O/N 16/P21/Q37, O/N 16/P23/Q37**

Substance Z has the following characteristics.

- 1 It burns in an excess of oxygen to form carbon dioxide and water.
- 2 It is oxidised by air to form a liquid smelling of vinegar.
- 3 It reacts with carboxylic acids to form esters.

What is substance Z?

- A ethane      B ethanoic acid      C ethanol      D ethyl ethanoate

**48. M/J 16/P22/Q37**

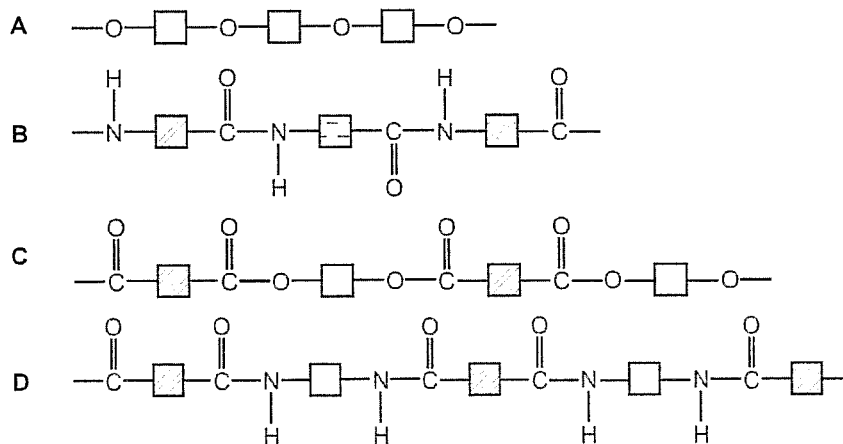
An ester is formed when a carboxylic acid reacts with an alcohol.

Which ester is formed when propanoic acid and ethanol react?

- A  $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_3$       C  $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$   
B  $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$       D  $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$

**49. M/J 16/P21/Q40**

Which formula represents a polyester?

**50. M/J 16/P21/Q37**

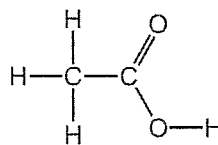
Which compounds contain the same number of carbon, hydrogen and oxygen atoms?

| W                | X                | Y                 | Z               |
|------------------|------------------|-------------------|-----------------|
| ethyl methanoate | methyl ethanoate | methyl methanoate | ethyl ethanoate |

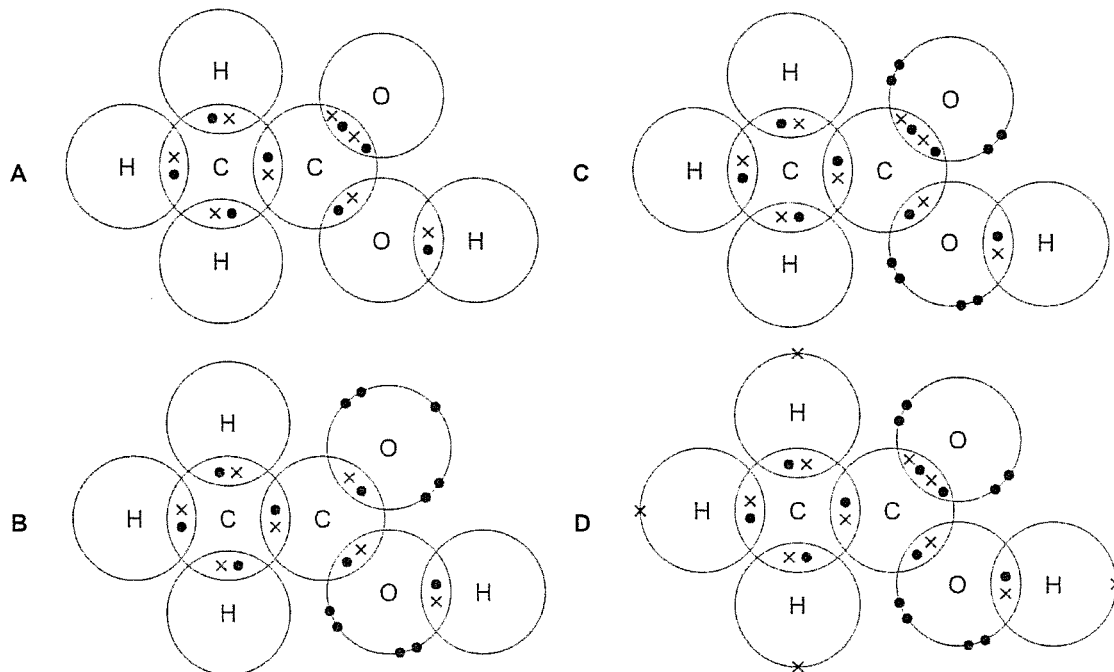
- A W and X      B W and Y      C X and Z      D Y and Z

## 51. O/N 16/P21/Q6

The structure of ethanoic acid is shown.

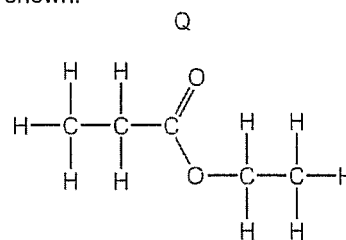
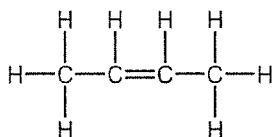


Which diagram shows the arrangement of outer shell electrons in a molecule of ethanoic acid?



## 52. M/J 16/P23/Q37

The structure of an alkene and the structure of an ester are shown.

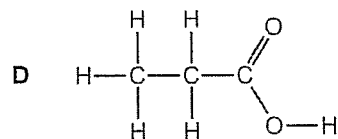
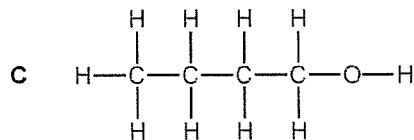
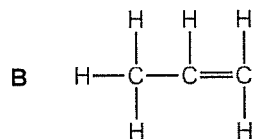
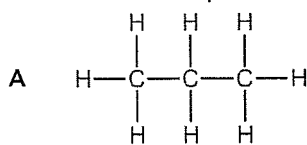
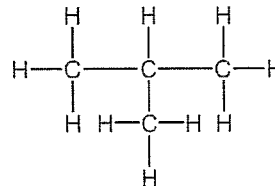
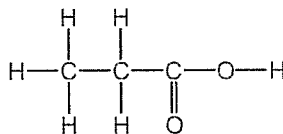
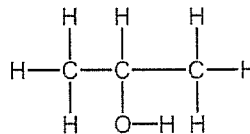
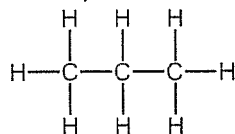


What are the names of P and Q?

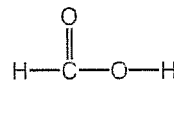
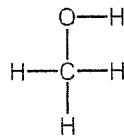
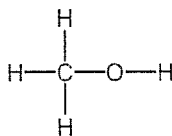
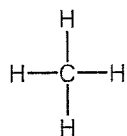
|   | P         | Q                |
|---|-----------|------------------|
| A | but-1-ene | ethyl propanoate |
| B | but-1-ene | propyl ethanoate |
| C | but-2-ene | ethyl propanoate |
| D | but-2-ene | propyl ethanoate |

**53. MAR 16/P22/Q38**

Which structure represents a compound that dissolves in water to form an acidic solution?

**54. O/N 15/P11/Q39**Which homologous series is **not** represented in the compounds shown below?**A** alcohols**B** alkanes**C** alkenes**D** carboxylic acids**55. M/J 15/P12/Q37**

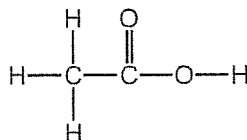
The structures of four different organic compounds are shown.



How many different homologous series are represented by these compounds?

**A** 1**B** 2**C** 3**D** 4**56. M/J 15/P12/Q38**

The structure of a compound, Y, is shown.



Which row describes some of the physical properties of Y?

|          | colourless | characteristic smell | dissolves in water |
|----------|------------|----------------------|--------------------|
| <b>A</b> | no         | no                   | no                 |
| <b>B</b> | no         | no                   | yes                |
| <b>C</b> | yes        | yes                  | no                 |
| <b>D</b> | yes        | yes                  | yes                |

**57. M/J 15/P11/Q38**

Which statements are correct for ethanoic acid?

- 1 It contains a carbon-oxygen double bond.
- 2 It contains two carbon atoms.
- 3 It decolourises bromine water.
- 4 It contains an  $\text{-OH}$  group.

A 1 and 2 only      B 1 and 3      C 1, 2 and 4      D 2, 3 and 4

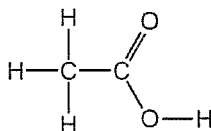
**58. M/J 15/P13/Q35**

Which statement about the names of organic compounds is correct?

- A Compounds containing  $\text{C}=\text{C}$  double bonds are alkanes.
- B The compound of formula  $\text{CH}_3\text{CO}_2\text{H}$  is methanoic acid.
- C The compound of formula  $\text{C}_2\text{H}_4$  is ethane.
- D The compound of formula  $\text{C}_2\text{H}_5\text{OH}$  is an alcohol.

**59. M/J 15/P13/Q38**

Which statement describes the compound shown below?



- A It is a colourless flammable gas.
- B It is a liquid which decolourises bromine water.
- C It is a liquid with a characteristic smell.
- D It is formed when ethane reacts with steam.

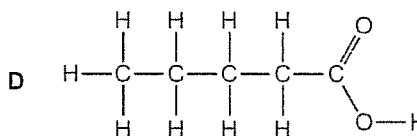
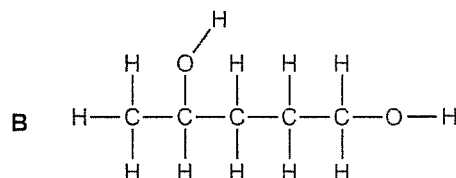
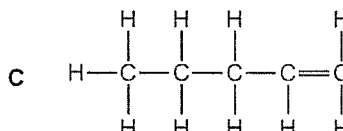
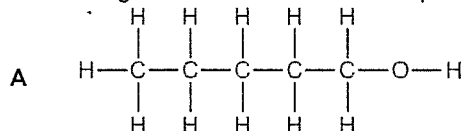
**60. MAR 15/P12/Q40**

Which description of ethanoic acid is correct?

- A a clear, colourless and odourless liquid
- B a colourless liquid with a distinctive odour
- C a soft white solid with a distinctive odour
- D a transparent solid with a low melting point

**61. M/J 14/P12/Q36**

Which diagram shows the structure of pentanoic acid?

**62. O/N 13/P13/Q36**Molecule X is both an alkene and a carboxylic acid.  
Which row describes X?

|   | saturated | $\text{-COOH}$ present |
|---|-----------|------------------------|
| A | no        | no                     |
| B | no        | yes                    |
| C | yes       | no                     |
| D | yes       | yes                    |

**63. M/J 13/P12/Q36, M/J 13/P11/Q36, M/J 13/P13/Q35**

Organic compounds may have names ending in -ane, -ene, -ol or -oic acid.

How many of these endings indicate the compounds contain double bonds in their molecules?

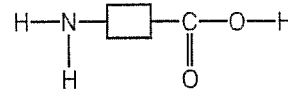
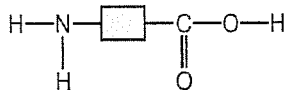
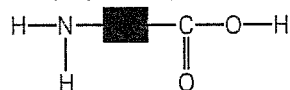
A 1      B 2      C 3      D 4



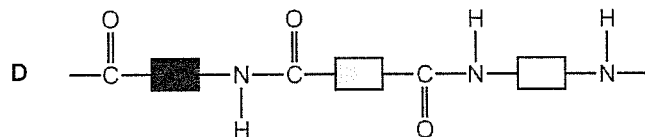
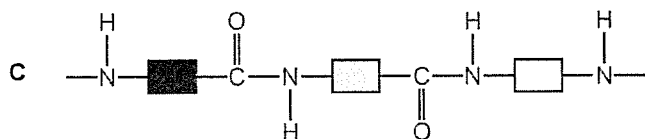
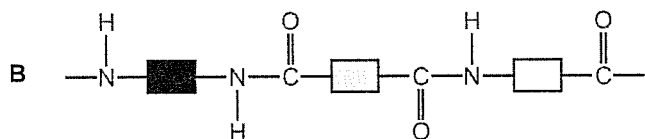
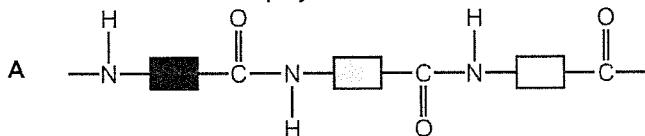
## 13.5: Polymers

### 1. O/N 22/P22/Q38

Hydrolysis of polymer P produces the three compounds shown.

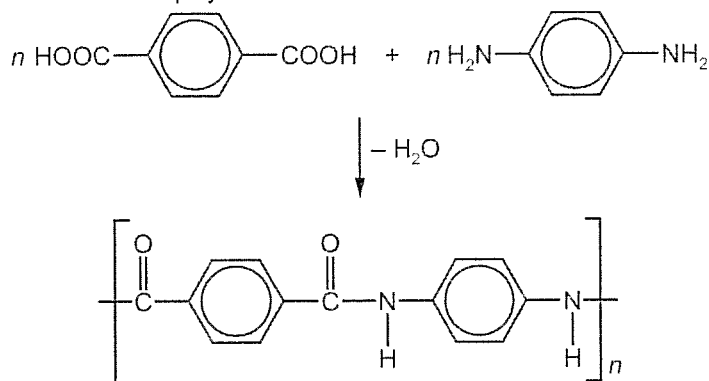


What is the structure of polymer P?



### 2. O/N 22/P22/Q40, O/N 22/P21/Q40, O/N 22/P23/Q40, O/N 17/P23/Q40

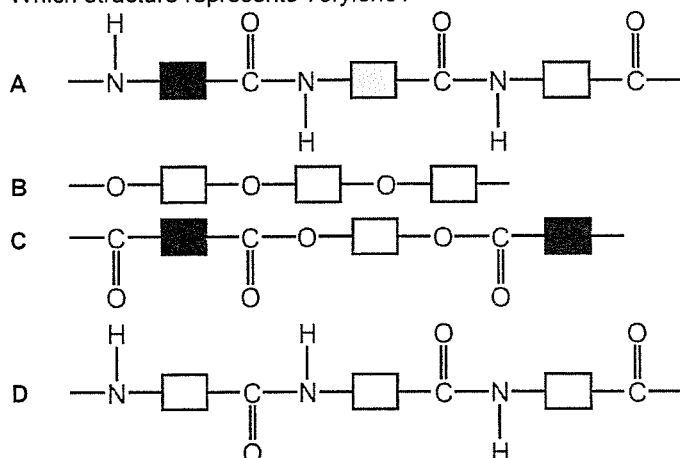
The equation shows the formation of a polymer called *Kevlar*.



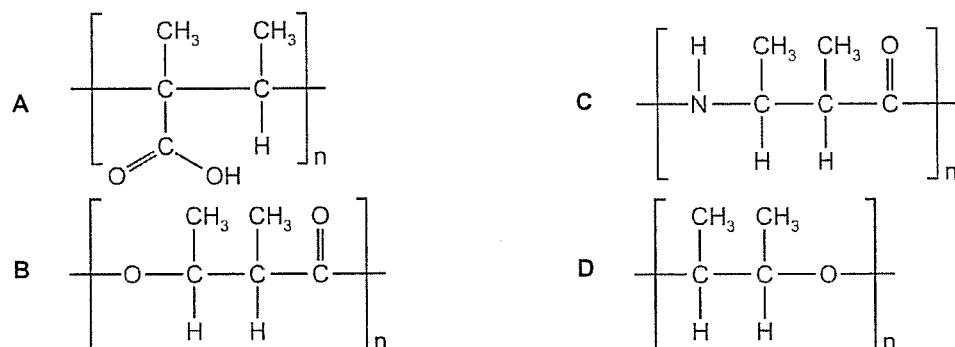
Which row describes *Kevlar*?

|   | how the polymer is formed   | type of polymer |
|---|-----------------------------|-----------------|
| A | addition polymerisation     | polyamide       |
| B | addition polymerisation     | polyester       |
| C | condensation polymerisation | polyamide       |
| D | condensation polymerisation | polyester       |

## 3. O/N 22/P21/Q39

Which structure represents *Terylene*?

## 4. M/J 22/P22/Q40

Which polymer structure has the same linkages as *Terylene*?

## 5. M/J 22/P21/Q40

Which polymer is a synthetic polyamide?

- A nylon                      B poly(ethene)                      C protein                      D *Terylene*

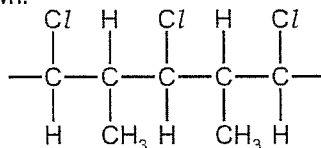
## 6. M/J 22/P23/Q40

Which type of compound is made when a protein is hydrolysed?

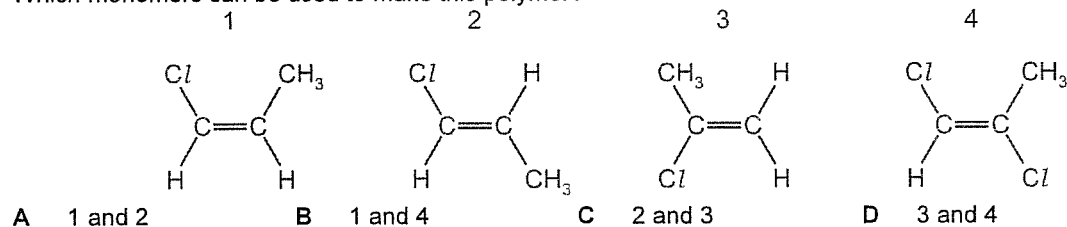
- A alkene                      B amino acid                      C carboxylic acid                      D sugar

## 7. MAR 22/P22/Q36

The structure of part of a polymer is shown.

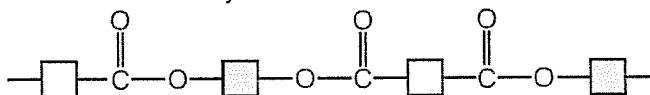


Which monomers can be used to make this polymer?



**8. O/N 21/P22/Q40, O/N 21/P21/Q40, O/N 21/P23/Q40, O/N 16/P21/Q40**

The diagram shows the partial structure of Terylene.



From which pair of compounds is it made?

- A +
- B +
- C +
- D +

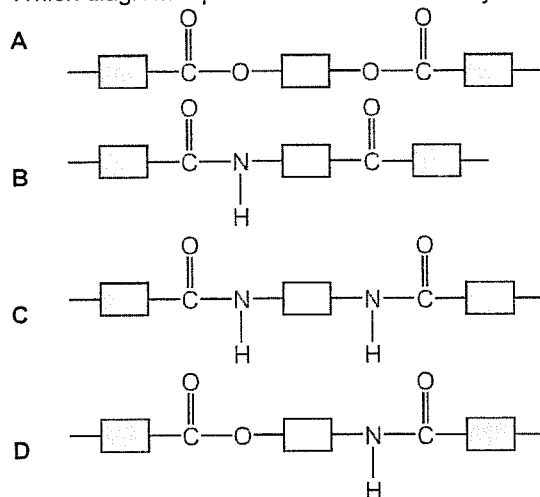
**9. O/N 21/P22/Q39**

Which polymers have the same linkage between monomer units?

- A carbohydrate and polyamide                      C protein and polyamide  
B carbohydrate and polyester                      D protein and polyester

**10. MAR 22/P22/Q40**

Which diagram represents the structure of nylon?

**11. O/N 21/P23/Q39**

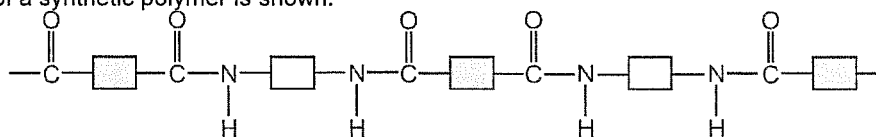
Proteins and starch are natural polymers.

Which row identifies the method of polymerisation of proteins and starch?

|   | proteins     | starch       |
|---|--------------|--------------|
| A | addition     | addition     |
| B | condensation | condensation |
| C | addition     | condensation |
| D | condensation | addition     |

## 12. M/J 21/P22/Q39, M/J 21/P21/Q39, M/J 21/P23/Q39, O/N 17/P21/Q40

The structure of a synthetic polymer is shown.



The structure shows that it is a .....1..... . It is formed by .....2..... polymerisation.

Which words complete gaps 1 and 2?

|   | 1         | 2            |
|---|-----------|--------------|
| A | polyamide | addition     |
| B | polyamide | condensation |
| C | polyester | addition     |
| D | polyester | condensation |

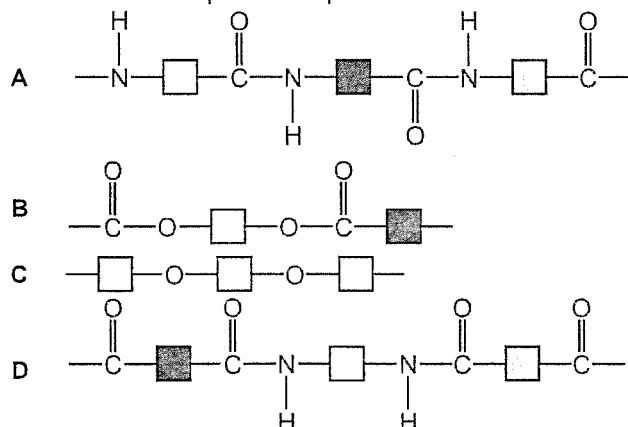
## 13. M/J 21/P22/Q40, M/J 21/P21/Q40, M/J 21/P23/Q40

Which substance is a natural polymer?

- A ethene                      B Terylene                      C nylon                      D protein

## 14. MAR 21/P22/Q40

Which structure represents a protein?



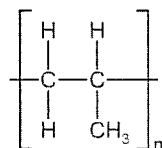
## 15. O/N 20/P22/Q39

Which statement about nylon and *Terylene* is correct?

- A Nylon and *Terylene* are made from monomers with C=C bonds.  
 B Nylon and *Terylene* contain the same linkage.  
 C Nylon is a polyester.  
 D *Terylene* is made from two different monomers.

## 16. O/N 20/P21/Q40

The structure of a polymer is shown.



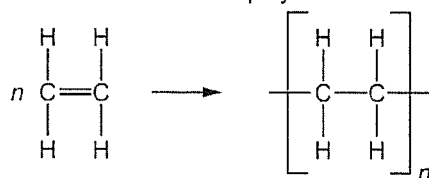
Which monomer forms this polymer?

- A ethane                      B ethene                      C propane                      D propene



## 23. O/N 19/P22/Q39, O/N 19/P21/Q39, O/N 19/P23/Q39

The diagram shows the structure of a monomer and of the polymer made from it.

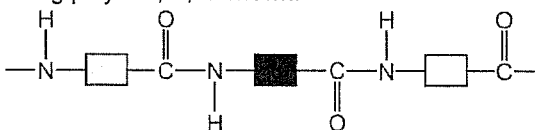


What are the monomer and polymer?

|   | monomer | polymer      |
|---|---------|--------------|
| A | ethane  | poly(ethane) |
| B | ethane  | poly(ethene) |
| C | ethene  | poly(ethane) |
| D | ethene  | poly(ethene) |

## 24. O/N 19/P22/Q40

The structure of a naturally occurring polymer, X, is shown.



What is X?

- A an amino acid      B a carbohydrate      C a protein      D a sugar

## 25. O/N 19/P21/Q40

Which polymers possess the same linkage?

- A nylon and protein      B protein and starch      C starch and nylon      D nylon and Terylene

## 26. O/N 19/P23/Q40

Proteins and starch are both natural polymers.

Both proteins and starch are hydrolysed by dilute acids.

What are the products of hydrolysis of proteins and of starch?

|   | products of hydrolysis of proteins | products of hydrolysis of starch |
|---|------------------------------------|----------------------------------|
| A | amines and carboxylic acids        | simple sugars                    |
| B | amines and carboxylic acids        | alcohols and carboxylic acids    |
| C | amino acids                        | simple sugars                    |
| D | amino acids                        | alcohols and carboxylic acids    |

## 27. M/J 19/P21/Q40

The structure of a polymer is shown.

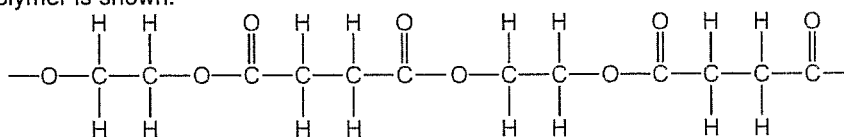


Which type of polymer is shown and by which process is it formed?

|   | type of polymer | formed by                   |
|---|-----------------|-----------------------------|
| A | carbohydrate    | addition polymerisation     |
| B | carbohydrate    | condensation polymerisation |
| C | polyester       | addition polymerisation     |
| D | polyester       | condensation polymerisation |

## 28. M/J 19/P23/Q40

A section of a polymer is shown.

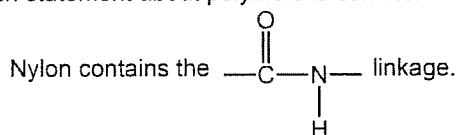


How many different types of monomer units formed this section of polymer?

- A 1                      B 2                      C 3                      D 4

## 29. MAR 19/P22/Q40

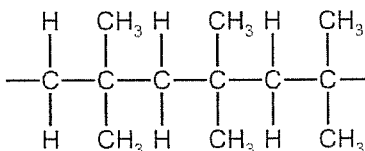
Which statement about polymers is correct?



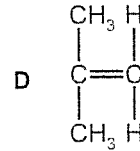
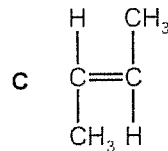
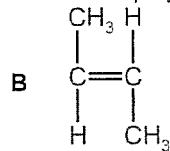
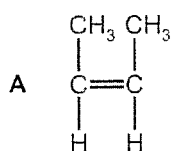
- A                      B Nylon is a polyester.  
C Propane can be polymerised by addition polymerisation.  
D The linkage in *Terylene* contains a carbon-carbon double bond.

## 30. O/N 18/P21/Q40

The structure of a polymer is shown.

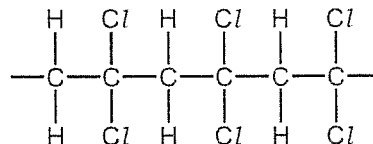


Which monomer is used to make this polymer?

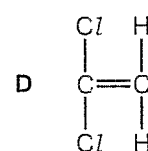
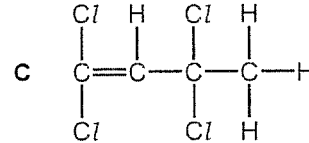
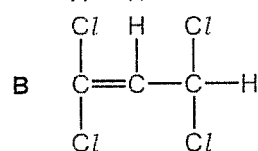
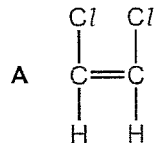


## 31. O/N 18/P23/Q40

The structure of a polymer is shown.



Which monomer is used to make this polymer?



## 32. M/J 18/P22/Q40

Which two compounds react together to form a condensation polymer?

- A  $\text{HOCH}_2\text{CH}_2\text{OH}$  and  $\text{CH}_3\text{COOH}$                       C  $\text{HOCH}_2\text{CH}_2\text{OH}$  and  $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$   
B  $\text{HOCH}_2\text{CH}_2\text{OH}$  and  $\text{CH}_3\text{NH}_2$                       D  $\text{HOCH}_2\text{CH}_2\text{OH}$  and  $\text{HOOCCH}_2\text{CH}_2\text{COOH}$

**33. M/J 18/P21/Q40**

Which row describes addition polymerisation and condensation polymerisation?

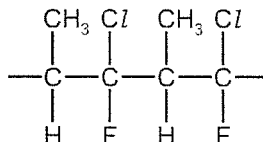
|          | addition polymerisation                                             | condensation polymerisation                                         |
|----------|---------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>A</b> | monomers have a C=C double bond and the polymer is the only product | monomers have a C=C double bond and the polymer is the only product |
| <b>B</b> | monomers have a C=C double bond and the polymer is the only product | the monomers react to form the polymer and a small molecule         |
| <b>C</b> | the monomers react to form the polymer and a small molecule         | monomers have a C=C double bond and the polymer is the only product |
| <b>D</b> | the monomers react to form the polymer and a small molecule         | the monomers react to form the polymer and a small molecule         |

**34. M/J 18/P23/Q40**A polymer linkage contains carbon, hydrogen, nitrogen and oxygen atoms.  
Which row about the polymer is correct?

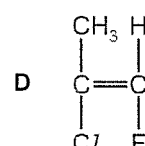
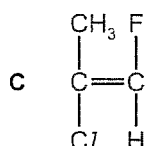
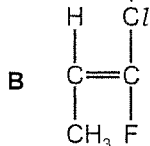
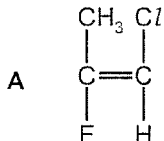
|          | type of polymer | formed by                   |
|----------|-----------------|-----------------------------|
| <b>A</b> | polyamide       | addition polymerisation     |
| <b>B</b> | polyamide       | condensation polymerisation |
| <b>C</b> | polyester       | addition polymerisation     |
| <b>D</b> | polyester       | condensation polymerisation |

**35. MAR 18/P22/Q40**

The structure of a chlorofluorocarbon polymer is shown.



Which monomer is used to make this polymer?

**36. M/J 17/P22/Q39**The formula of an ester is  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3$ .

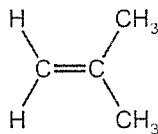
Which acid and alcohol react together to make the ester?

|          | acid           | alcohol  |
|----------|----------------|----------|
| <b>A</b> | butanoic acid  | butanol  |
| <b>B</b> | butanoic acid  | propanol |
| <b>C</b> | propanoic acid | butanol  |
| <b>D</b> | propanoic acid | propanol |

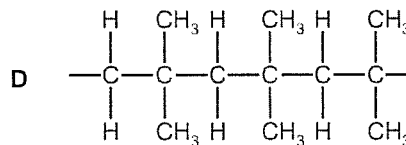
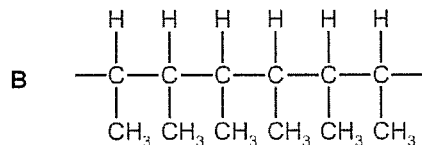
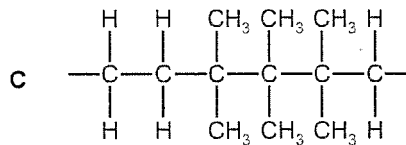
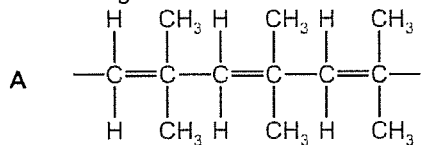


**37. O/N 17/P22/Q40**

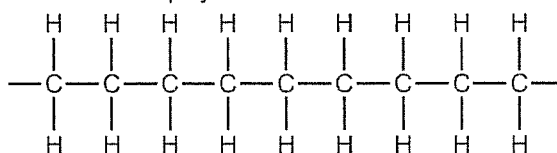
A polymer can be made from methyl propene.



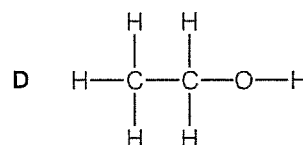
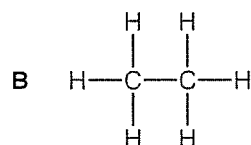
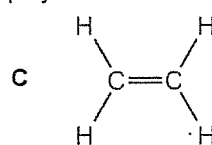
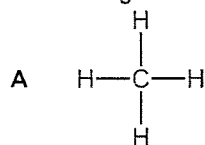
Which diagram shows the structure of the polymer?

**38. M/J 17/P22/Q37, M/J 17/P23/Q37**

The diagram shows part of the molecule of a polymer.



Which diagram shows the monomer from which this polymer could be manufactured?

**39. M/J 17/P22/Q40**

Polyesters and polyamides are types of synthetic polymer.

Which statements are correct?

- 1 They are made by addition polymerisation.
  - 2 They are made by condensation polymerisation.
  - 3 The monomers from which they are made are unsaturated hydrocarbons.
  - 4 The monomers from which they are made contain reactive functional groups at their ends.
- A** 1 and 3                      **B** 1 and 4                      **C** 2 and 3                      **D** 2 and 4

**40. M/J 17/P21/Q40**

Keratin is a protein that is found in human hair.

Keratin is chemically broken down to produce amino acids.

What is the name of this chemical process?

- A** catalysis                      **B** hydration                      **C** hydrolysis                      **D** polymerisation

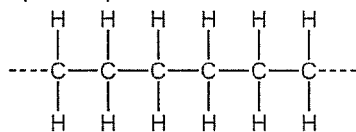
**41. M/J 17/P23/Q40**

How can the amino acids in a protein be separated and identified?

- A** Add a locating agent to the protein.
- B** Hydrolyse the protein and then use chromatography.
- C** Polymerise the protein and then add a locating agent.
- D** Use chromatography on a solution of the protein.

## 42. MAR 17/P22/Q39

The diagram shows the structure of an important product.



This product is formed by ..... 1 ..... of an ..... 2 .....  
Which words complete gaps 1 and 2?

|   | 1                       | 2      |
|---|-------------------------|--------|
| A | addition polymerisation | alkane |
| B | addition polymerisation | alkene |
| C | cracking                | alkane |
| D | cracking                | alkene |

## 43. MAR 17/P22/Q40

Which pair of compounds reacts to form a condensation polymer?

- A  $\text{CH}_3\text{COOH}$  and  $\text{C}_2\text{H}_5\text{NH}_2$       C  $\text{HOC}_6\text{H}_{12}\text{OH}$  and  $\text{HOOC}_3\text{H}_6\text{COOH}$   
 B  $\text{HCOOH}$  and  $\text{HOC}_2\text{H}_4\text{OH}$       D  $\text{H}_2\text{NC}_2\text{H}_4\text{NH}_2$  and  $\text{HOC}_3\text{H}_6\text{OH}$

## 44. O/N 16/P22/Q40

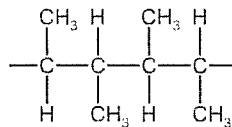
*Terylene* is a synthetic polymer.

Which statement about *Terylene* is **not** correct?

- A It contains amide linkages.      C It is made from small units called monomers.  
 B It contains carbon and oxygen atoms.      D It is formed by condensation polymerisation.

## 45. O/N 16/P21/Q39

The partial structure of addition polymer X is shown.

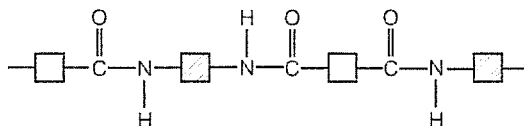


Which monomer is used to form polymer X?

- A  $\text{CH}_2=\text{CH}_2$       B  $\text{CH}_3\text{CH}=\text{CH}_2$       C  $\text{CH}_3\text{CH}=\text{CHCH}_3$       D  $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$

## 46. O/N 16/P23/Q40

The partial structure of a polymer is shown.



Which type of polymer is represented?

- A a carbohydrate      C a polyester  
 B a polyamide      D an addition polymer

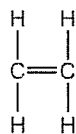
## 47. M/J 16/P23/Q40

Which row shows a natural polymer with the same linkages as a synthetic polymer?

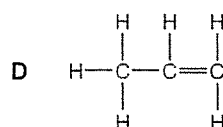
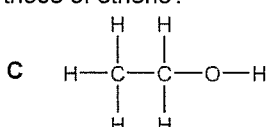
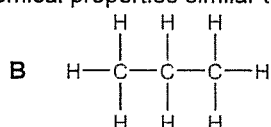
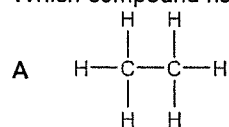
|   | natural polymer      | synthetic polymer |
|---|----------------------|-------------------|
| A | complex carbohydrate | nylon             |
| B | complex carbohydrate | <i>Terylene</i>   |
| C | protein              | nylon             |
| D | protein              | <i>Terylene</i>   |



53. O/N 13/P12/Q38, O/N 13/P11/Q39  
The diagram represents ethene.



Which compound has chemical properties similar to those of ethene?



## Answers Section

| 13.1  |       |  |  | 13.2  |       |  |  | 13.3  |       |       |  |
|-------|-------|--|--|-------|-------|--|--|-------|-------|-------|--|
| 1. B  | 41. C |  |  | 1. C  | 41. D |  |  | 1. B  | 41. C | 81. B |  |
| 2. C  | 42. D |  |  | 2. A  | 42. C |  |  | 2. A  | 42. B | 82. D |  |
| 3. B  | 43. D |  |  | 3. D  | 43. A |  |  | 3. C  | 43. D | 83. C |  |
| 4. B  | 44. D |  |  | 4. A  | 44. B |  |  | 4. C  | 44. D | 84. C |  |
| 5. A  | 45. C |  |  | 5. A  | 45. D |  |  | 5. C  | 45. B | 85. D |  |
| 6. B  | 46. D |  |  | 6. A  | 46. A |  |  | 6. B  | 46. D | 86. C |  |
| 7. D  | 47. A |  |  | 7. D  | 47. A |  |  | 7. B  | 47. B | 87. C |  |
| 8. C  | 48. D |  |  | 8. C  | 48. D |  |  | 8. D  | 48. C | 88. A |  |
| 9. C  | 49. D |  |  | 9. D  | 49. B |  |  | 9. C  | 49. D |       |  |
| 10. B | 50. C |  |  | 10. D | 50. D |  |  | 10. B | 50. D |       |  |
| 11. A | 51. D |  |  | 11. B | 51. B |  |  | 11. C | 51. C |       |  |
| 12. D | 52. D |  |  | 12. C | 52. A |  |  | 12. D | 52. B |       |  |
| 13. D | 53. C |  |  | 13. B | 53. B |  |  | 13. B | 53. D |       |  |
| 14. A | 54. C |  |  | 14. C | 54. C |  |  | 14. C | 54. B |       |  |
| 15. D | 55. C |  |  | 15. B | 55. C |  |  | 15. C | 55. C |       |  |
| 16. C | 56. C |  |  | 16. B | 56. A |  |  | 16. B | 56. B |       |  |
| 17. C | 57. D |  |  | 17. D | 57. B |  |  | 17. D | 57. A |       |  |
| 18. C | 58. D |  |  | 18. A | 58. A |  |  | 18. B | 58. C |       |  |
| 19. B | 59. B |  |  | 19. A | 59. D |  |  | 19. A | 59. C |       |  |
| 20. D | 60. B |  |  | 20. A | 60. C |  |  | 20. A | 60. B |       |  |
| 21. B | 61. C |  |  | 21. B | 61. D |  |  | 21. A | 61. A |       |  |
| 22. A | 62. C |  |  | 22. D | 62. B |  |  | 22. C | 62. C |       |  |
| 23. A | 63. D |  |  | 23. D | 63. B |  |  | 23. A | 63. D |       |  |
| 24. D | 64. D |  |  | 24. B | 64. C |  |  | 24. D | 64. A |       |  |
| 25. B | 65. C |  |  | 25. C |       |  |  | 25. C | 65. C |       |  |
| 26. A | 66. B |  |  | 26. C |       |  |  | 26. B | 66. B |       |  |
| 27. D | 67. D |  |  | 27. A |       |  |  | 27. A | 67. D |       |  |
| 28. A |       |  |  | 28. D |       |  |  | 28. D | 68. B |       |  |
| 29. A |       |  |  | 29. D |       |  |  | 29. B | 69. B |       |  |
| 30. B |       |  |  | 30. B |       |  |  | 30. B | 70. D |       |  |
| 31. C |       |  |  | 31. B |       |  |  | 31. D | 71. C |       |  |
| 32. C |       |  |  | 32. B |       |  |  | 32. B | 72. B |       |  |
| 33. B |       |  |  | 33. B |       |  |  | 33. C | 73. A |       |  |
| 34. D |       |  |  | 34. A |       |  |  | 34. C | 74. A |       |  |
| 35. C |       |  |  | 35. C |       |  |  | 35. D | 75. C |       |  |
| 36. B |       |  |  | 36. D |       |  |  | 36. B | 76. A |       |  |
| 37. B |       |  |  | 37. B |       |  |  | 37. A | 77. A |       |  |
| 38. D |       |  |  | 38. D |       |  |  | 38. B | 78. A |       |  |
| 39. B |       |  |  | 39. A |       |  |  | 39. B | 79. C |       |  |
| 40. C |       |  |  | 40. A |       |  |  | 40. A | 80. B |       |  |

| 13.4 |   |     |   | 13.5 |   |     |   |
|------|---|-----|---|------|---|-----|---|
| 1.   | C | 41. | C | 1.   | A | 41. | B |
| 2.   | D | 42. | D | 2.   | C | 42. | B |
| 3.   | C | 43. | C | 3.   | C | 43. | C |
| 4.   | C | 44. | B | 4.   | B | 44. | A |
| 5.   | A | 45. | B | 5.   | A | 45. | C |
| 6.   | C | 46. | C | 6.   | B | 46. | B |
| 7.   | D | 47. | C | 7.   | A | 47. | C |
| 8.   | B | 48. | D | 8.   | A | 48. | B |
| 9.   | C | 49. | C | 9.   | C | 49. | B |
| 10.  | B | 50. | A | 10.  | C | 50. | B |
| 11.  | C | 51. | C | 11.  | B | 51. | D |
| 12.  | C | 52. | C | 12.  | B | 52. | C |
| 13.  | D | 53. | D | 13.  | D | 53. | D |
| 14.  | C | 54. | C | 14.  | A |     |   |
| 15.  | B | 55. | C | 15.  | D |     |   |
| 16.  | C | 56. | D | 16.  | D |     |   |
| 17.  | B | 57. | C | 17.  | D |     |   |
| 18.  | B | 58. | D | 18.  | A |     |   |
| 19.  | C | 59. | C | 19.  | A |     |   |
| 20.  | C | 60. | B | 20.  | D |     |   |
| 21.  | D | 61. | D | 21.  | A |     |   |
| 22.  | B | 62. | B | 22.  | C |     |   |
| 23.  | D | 63. | B | 23.  | D |     |   |
| 24.  | A |     |   | 24.  | C |     |   |
| 25.  | D |     |   | 25.  | A |     |   |
| 26.  | A |     |   | 26.  | C |     |   |
| 27.  | C |     |   | 27.  | B |     |   |
| 28.  | B |     |   | 28.  | B |     |   |
| 29.  | C |     |   | 29.  | A |     |   |
| 30.  | C |     |   | 30.  | D |     |   |
| 31.  | C |     |   | 31.  | D |     |   |
| 32.  | C |     |   | 32.  | D |     |   |
| 33.  | A |     |   | 33.  | B |     |   |
| 34.  | D |     |   | 34.  | B |     |   |
| 35.  | D |     |   | 35.  | B |     |   |
| 36.  | A |     |   | 36.  | B |     |   |
| 37.  | A |     |   | 37.  | D |     |   |
| 38.  | A |     |   | 38.  | C |     |   |
| 39.  | D |     |   | 39.  | D |     |   |
| 40.  | B |     |   | 40.  | C |     |   |